# A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference

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In Partial Fulfillment
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Bachelor of Science in Computer Science

By

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Philippines

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## Recommendation for Oral Defense

In partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE**, this **UNDERGRADUATE THESIS** entitled:

**A COARSE-TO-FINE SEMANTIC CONFLICT DETECTION SYSTEM FOR EX-ANTE DAVAO CITY ORDINANCES USING INFORMATION RETRIEVAL AND NATURAL LANGUAGE INFERENCE**

has been prepared and submitted by **RALPH PAOLO DULCE** and **YAHYAH ODIN** and is recommended for **ORAL FINAL DEFENSE**.

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## Recommendation for Acceptance

The faculty of the Computer Studies Cluster of the Ateneo de Davao University accepts the Undergraduate Thesis entitled:

**A COARSE-TO-FINE SEMANTIC CONFLICT DETECTION SYSTEM FOR EX-ANTE DAVAO CITY ORDINANCES USING INFORMATION RETRIEVAL AND NATURAL LANGUAGE INFERENCE**

which has been prepared and submitted by **RALPH PAOLO DULCE** and **YAHYAH ODIN** in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science.

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# Abstract

Under constitutional statutory preemption established in *Magtajas v. Pryce Properties* and Republic Act No. 7160, local ordinances cannot contradict superior national statutes. In the Davao City Sangguniang Panlungsod, manual ex-ante consistency reviews impose heavy cognitive burdens on legislative researchers due to statutory volume and archival scanning noise. To automate screening within municipal computing constraints, this study presents an offline coarse-to-fine semantic conflict detection architecture decoupling candidate retrieval from cross-attention inference. In Stage 1, hybrid retrieval fusing Okapi BM25 and dense bi-encoder embeddings via Reciprocal Rank Fusion isolates candidate provisions across 27,357 enactments; this stage attains 100.0% Recall@50 on a controlled benchmark and 65.14% across the unassisted open corpus. In Stage 2, candidate pairs are evaluated through a fine-tuned `DeBERTa-v3` cross-encoder with calibrated decision thresholds. Under zero-leakage premise-grouped cross-validation across 22 premise provisions from 17 statutes on a 350-pair faculty adviser-reviewed benchmark comprising 198 unique instances, the model achieves 86.44% accuracy and a 0.8945 Conflict $F_1$-score. In an open-corpus screening pilot across six draft ordinances, the pipeline surfaced two of seven planted conflicts with six false alarms while clearing 39 of 45 compliant sections at 86.67% specificity. In contrast, oracle candidate retrieval detected all seven planted conflicts, which confirms that Stage 1 candidate retrieval forms the operational constraint. Algorithmic optimizations reduce full-document audit latency by 54.9% to 12.69 seconds on an entry-level CPU with a 1,416.4 MB memory footprint. An offline web prototype delivers explainable preemption alerts and establishes an ex-ante decision-support baseline for Philippine local legislative review.

**Keywords:** Semantic Conflict Detection, Ex-Ante Legislative Review, Statutory Preemption, Natural Language Inference, Information Retrieval, Cross-Encoder, Local Ordinances, Sangguniang Panlungsod, Davao City.

# Chapter 1

# Introduction

## 1.1 Background of the Study

A functional legal system requires strict consistency so that citizens can clearly understand their rights and obligations [75]. However, the sheer volume of modern legislation has grown so rapidly that human practitioners can no longer track every active rule [181, 33]. As governing bodies continuously pass new regulations to address emerging issues, they frequently enact laws that inadvertently contradict older ones [191, 114]. Because the total number of documents is overwhelming, identifying these hidden conflicts through manual review is practically infeasible and cognitively overwhelming for legal staff. This issue is particularly severe in local government units, where historical ordinances are often scattered across fragmented files rather than maintained in a structured digital database [43].

Within the Philippine legal framework, local governments like Davao City operate under delegated authority and can only enact laws permitted by the national government [85]. Due to this structural limitation, the Magtajas v. Pryce Doctrine strictly prohibits any local ordinance from contradicting established national statutes [207]. Consequently, in the local context, city councils must rigorously audit every newly proposed draft ordinance to ensure it logically aligns with existing laws before final approval. If reviewers overlook a normative conflict, such as a local ordinance prohibiting an activity that a national law explicitly permits, the resulting local legislation is legally void. Passing invalid laws triggers severe jurisdictional disputes and wastes administrative resources, while compromising the constitutional rights of the public.

To address these legal risks, the legislative process requires an ex-ante evaluation, drawn from the Latin for 'before the event.' Unlike an 'ex-post' review, which occurs after an enactment takes effect and causes legal harm, an ex-ante audit provides a preventative first line of defense. By checking a proposed ordinance for conflicts while it is still in the drafting phase, the Sangguniang Panlungsod can proactively align local rules with national statutes. This preventative approach is consistent with modern Philippine standards like Republic Act No. 11032 or the Ease of Doing Business and Efficient Government Service Delivery Act of 2018, which encourages government agencies to perform regulatory

assessments to ensure that new rules do not create unnecessary or conflicting burdens on the public.

The decentralized drafting process of local ordinances further complicates manual review because the resulting texts are often highly unstructured [186]. City councils frequently splice and amend different drafts, producing verbose documents that blend formal legal English with regional dialects like Cebuano and administrative "Taglish" [66]. Furthermore, because local archives heavily rely on digitized PDF scans, the text suffers from severe Optical Character Recognition (OCR) errors. While OCR is typically 98–99 percent accurate for clean English text, studies demonstrate that recognition accuracy degrades substantially for low-resource languages [104], with rapid lexicon extraction for Cebuano specifically encountering word error rates of 6.9 percent [157]. These errors cause an "error cascade" that shatters standard vocabulary, making it difficult for traditional search engines to semantically differentiate between a statutory obligation and a strict prohibition.

In light of these challenges, the proponents seek to develop a specialized application capable of retrieving relevant legal documents and comparing them against proposed ordinances to detect potential conflicts.

## 1.2 Problem Statement

This study is conducted to investigate the use of a multi-stage architecture to automate the manual auditing of proposed local legislation in Davao City.

Specifically, the study aims to answer the following research questions:

- What are the specific linguistic parameters and structural markers that define a "semantic conflict" between a proposed local ordinance and existing local or national laws?
- How can a Coarse-to-Fine system be designed to efficiently retrieve relevant national and local statutes before performing deep logical inference on constrained LGU hardware?
- Which combination of hybrid Information Retrieval (IR) algorithms and Natural Language Inference (NLI) models provides the highest accuracy when processing the specific hierarchy of Philippine legal texts?
- What is the empirical performance of the proposed pipeline when evaluated against a manually curated Ground Truth dataset of local and national conflict pairs?
- How usable and viable is the final system interface for key stakeholders within the Davao City Sangguniang Panlungsod?

## 1.3 Objectives of the Study

The primary goal of this research is to develop a computationally efficient, two-stage semantic conflict detection system for proposed Davao City ordinances.

Specifically, this study aims to:

- Identify the specific linguistic characteristics and structural constraints of Davao City's legislative archive that contribute to semantic ambiguity and retrieval failure;
- Design a Coarse-to-Fine computational pipeline that decouples document retrieval from logical inference to optimize resource usage for local deployment;
- Implement a hybrid Information Retrieval (IR) algorithm and Natural Language Inference (NLI) system specifically calibrated for the Philippine legal hierarchy;
- Evaluate the system's reliability by comparing its automated detections against a verified set of known legal conflicts (the Faculty Adviser-Verified Ground Truth Benchmark) to measure how accurately it identifies contradictions; and
- Develop an offline, user-friendly web interface for the Davao City Sangguniang Panlungsod designed to accept uploaded draft documents and provide explainable conflict alerts, establishing a computationally profiled prototype and an institutional post-deployment evaluation roadmap.

## 1.4 Significance of the Study

The Davao City Sangguniang Panlungsod is the primary institutional beneficiary of this automated consistency checking system. By transitioning from manual statutory review to computational auditing, the tool is projected to save legislative staff substantial time in manual cross-referencing across multi-volume legislative digests. This computational intervention actively prevents the city from enacting legally vulnerable measures that invite judicial invalidation, public injunctions, or administrative deadlocks under the doctrine of statutory preemption. It equips the local legislative council with the operational capacity to draft regulatory measures with a heightened degree of legal certainty and procedural speed. Ultimately, the system modernizes the local legislative lifecycle by embedding rigorous logical safeguards directly into the pre-plenary committee review and Second Reading preparation phase, equipping drafters to resolve statutory conflicts during the floor amendment period.

The study provides substantial operational value to legislative staff, particularly legal researchers and committee bill drafters. In contemporary local legislative practice, legal researchers must examine extensive regulatory provisions across disparate national codes without centralized semantic indexing tools. The proposed system provides an automated cross-statutory audit trail, pinpointing candidate national statutes and fellow city ordinances that govern the same regulatory sphere. By surfacing relevant provisions and generating token-level self-attention attribution heatmaps, the architecture is designed to mitigate cognitive fatigue and reading oversight during high-volume legislative sessions.

Consequently, legal officers can conduct evidence-grounded preliminary reviews and focus their analytical expertise on high-level statutory interpretation.

Beyond the immediate city council, this research establishes an edge-deployable computational blueprint for Local Government Units (LGUs) nationwide and oversight agencies such as the Department of the Interior and Local Government (DILG). In compliance with the Ease of Doing Business and Efficient Government Service Delivery Act of 2018 (Republic Act No. 11032), local governments are mandated to perform systematic Regulatory Impact Assessments to ensure local regulations do not create redundant or conflicting business burdens. Demonstrating that an automated consistency checking pipeline operates efficiently on standard consumer-grade office desktop computers demonstrates that advanced artificial intelligence can be deployed within budget-constrained local government environments without recurring cloud subscriptions or specialized computing clusters.

The research also contributes foundational advancements to the Philippine Natural Language Processing and legal informatics community. Prior domestic legal technology research concentrated predominantly on national appellate case summarization and judicial outcome forecasting; local statutory conflict detection remained largely unaddressed. This investigation establishes an empirical baseline for processing localized Philippine statutory corpora marked by historical scanning noise and regional drafting syntax. Furthermore, the empirical findings demonstrate that lightweight, fine-tuned cross-encoders can effectively overcome affirmative entailment bias and resolve directional statutory contradictions on resource-constrained local hardware.

Finally, the general public and the local business community benefit directly through the promulgation of coherent, constitutionally sound public policy. When local ordinances clash with superior national statutes, citizens and local enterprises bear the brunt of legal uncertainty through conflicting compliance directives or unauthorized penalties. As an ex-ante procedural safety net, the system helps ensure that enacted city ordinances harmonize with national standards and existing local legislation. This statutory coherence strengthens institutional transparency and reinforces the local rule of law.

## 1.5 Scope and Limitations

The scope of this study focuses on developing an automated semantic conflict detection system to evaluate draft ordinances from Davao City against a dual statutory search space. This collection encompasses national statutory enactments compiled to the extent accessible from public legal repositories, paired alongside the historical archive of enacted Davao City local ordinances collected to the fullest extent possible from local legislative records. Operationally, the system audits draft measures across two complementary dimensions: vertical statutory preemption, which verifies that local provisions conform to superior national statutes under the landmark *Magtajas* doctrine and the delegated limits of Republic Act No. 7160 (specifically Sections 16 and 458); and horizontal consistency, which checks that proposed clauses do not contradict, duplicate, or inadvertently cause implied

repeals of active local ordinances. While the dual-corpus retrieval index and prototype architecture natively ingest and index both statutory tiers to support horizontal as well as vertical searches, formal empirical benchmark evaluations in this study focus strictly on vertical statutory preemption against superior national statutes, leaving systematic horizontal benchmark construction as an operational recommendation for future work.

To establish an administratively grounded and computationally feasible engineering boundary, several explicit delimitations govern this research. The statutory knowledge base is strictly bounded to codified national legislative instruments and enacted local ordinances. To preserve retrieval index compactness and prevent ambiguous litigation records from distorting similarity rankings, judicial case law from Supreme Court decisions and departmental administrative circulars are excluded from the primary index. Furthermore, the local ordinance corpus reflects the practical realities of local legislative record-keeping, where recent council terms benefit from digital database records while earlier decades rely on curated landmark ordinances due to physical archive gaps. The system evaluates drafts strictly against the combined corpus gathered through public portals and direct institutional coordination with the Sangguniang Panlungsod, leaving the comprehensive optical character recovery of older paper files as an ongoing archival task.

The system operates exclusively on English-language enactments pursuant to Section 20, Chapter 5, Book I of Executive Order No. 292, where the English text controls in formal statutory interpretation. While local drafts occasionally incorporate regional Cebuano phrases or administrative Taglish, restricting the primary reasoning pipeline to English prevents semantic drift and avoids translation distortion. At the inference tier, the system evaluates proposed ordinance sections against retrieved candidate provisions using asymmetric span-level pairing (1-vs-$N$). The pipeline excludes multi-hop reasoning across multiple independent statutes simultaneously. Detecting intricate legal contradictions that emerge only when synthesizing two or more disparate national statutes exceeds the computational capacity of standard workstation hardware and is designated as a recommendation for future research.

The computational architecture is intentionally restricted to an open-weight transformer encoder envelope capable of local deployment on standard Local Government Unit (LGU) desktop computers, specifically targeting sub-600M parameter models requiring less than 2 GB of inference VRAM. Massive autoregressive large language models and proprietary cloud APIs are excluded to maintain institutional data privacy, avoid recurring cloud API expenditures, and guarantee reliable offline operation. Within this architecture, the system operates strictly as a diagnostic decision-support tool providing transparent white-box explainability, such as attention heatmaps and calibrated confidence scores, without rewriting or generating statutory text. Finally, the operational scope of the inference engine is delimited to direct normative and modal contradictions arising from explicit textual collisions, such as local prohibitions of federally authorized activities, unauthorized procedural preconditions, or penalty ceiling infractions. The framework does not model longitudinal legislative histories, historical tax rate compounding, or multi-year financial calculations across unprovided statutory baselines, as observed in complex judicial tax disputes [211]. Modeling such multi-hop temporal construction and arithmetic accounting exceeds the ex-ante diagnostic scope of cross-encoder architectures and is preserved

as an avenue for specialized fiscal analytics.

## 1.6 Definition of Terms

**Bi-Encoder.** A dual-tower transformer architecture that maps queries and candidate statutory provisions independently into a shared dense vector embedding space, enabling rapid semantic similarity search via vector dot products or cosine distance.

**Coarse-to-Fine Architecture.** A two-stage decoupled computational framework that pairs a broad high-throughput candidate retrieval module (coarse stage) with a compute-intensive pairwise sequence classification model (fine stage) to manage extensive textual collections on bounded hardware.

**Contradiction.** In Natural Language Inference, a directional semantic relationship where the truth of the hypothesis is incompatible with the regulatory mandate of the premise, which signals a direct normative conflict or statutory collision.

**Cross-Encoder.** A single-tower transformer architecture that ingests concatenated premise-hypothesis pairs into a unified self-attention matrix, enabling full bidirectional cross-attention across all tokens for granular semantic classification.

**Entailment.** In Natural Language Inference, a relationship where the substantive directive of the hypothesis naturally follows from, conforms with, or is explicitly authorized by the governing statutory premise.

**Ex-Ante Legislative Review.** The proactive legal and technical evaluation of proposed legislative drafts, ordinances, or administrative rules prior to their formal enactment, designed to detect statutory collisions and constitutional defects while the text remains amenable to revision.

**Explainable Artificial Intelligence (XAI).** Computational methods that provide human-interpretable rationales for machine learning decisions; in this system, operationalized through token-level self-attention attribution heatmaps highlighting specific conflicting text spans.

**Fleiss' Kappa ($\kappa$).** A statistical measure of inter-annotator agreement among three or more raters assessing categorical data, utilized to evaluate consensus reliability across domain annotators, established as the evaluation metric for the post-deployment roadmap with the fifteen Sangguniang Panlungsod legal researchers.

**Horizontal Inconsistency.** A normative clash, redundancy, or unintended implied repeal occurring between a newly drafted local ordinance and an existing, fellow local ordinance enacted within the same legislative jurisdiction.

**Information Retrieval (IR).** The computational discipline and algorithmic process of searching and ranking relevant candidate documents or legal provisions from extensive unstructured text collections in response to a query.

**Local Ordinances.** Enacted legislative measures promulgated by the Davao City Sangguniang Panlungsod under devolved legislative powers that carry the force and effect

of law within the territorial jurisdiction of the city.

**Magtajas v. Pryce Doctrine.** The foundational Philippine constitutional principle establishing that local government ordinances are strictly subordinate to national statutes, rendering any ordinance that permits what an Act of Congress forbids, or forbids what an Act of Congress permits, *ultra vires* and void.

**Natural Language Inference (NLI).** A core natural language processing task that evaluates the directional semantic relationship between a premise text and a hypothesis text across three categorical classes: Entailment, Contradiction, or Neutral.

**Neutral.** In Natural Language Inference, a relationship where the truth or enforcement of the hypothesis neither follows from nor contradicts the regulatory command of the statutory premise; this relationship signals thematic and regulatory independence.

**Normative Conflict.** In legal theory and formal deontic logic, a condition where two or more legally valid enactments impose mutually incompatible obligations, prohibitions, or permissions upon the same subject under identical circumstances.

**Optical Character Recognition (OCR) Noise.** Textual distortions such as typographical errors and broken tokens introduced when physical paper documents are scanned and converted into machine-readable text.

**Reciprocal Rank Fusion (RRF).** An unsupervised rank aggregation algorithm that merges ranked candidate lists from disparate retrieval systems (such as lexical BM25 and dense bi-encoders) based on reciprocal rank positions without requiring calibrated score normalization.

**Sangguniang Panlungsod (SP).** The unicameral legislative council of Davao City, vested by Republic Act No. 7160 with the corporate and police authority to enact local ordinances and approve resolutions.

**Semantic Conflict Detection.** The computational formulation and automated identification of directional normative contradictions between legislative provisions using hybrid information retrieval and natural language inference.

**Statutory Preemption.** The constitutional and administrative principle under which superior national legislation invalidates, overrides, or limits the regulatory scope of local enactments governing the same subject matter.

**Vision-Language Model (VLM).** A multimodal deep learning architecture that combines visual processing backbones with transformer language decoders to transcribe document page images into structured text while preserving layout and alphanumeric precision.

# Chapter 2

# Literature Review

## 2.1 Semantic Conflict Detection

In jurisprudence and legal philosophy, the fundamental requirement for a functional legal order is internal consistency, where positive enactments provide unambiguous guidance without subjecting citizens to contradictory commands [75]. In legal theory and formal deontic logic, this phenomenon is strictly defined as a *normative conflict*—a structural clash between legal norms wherein an action is simultaneously commanded, prohibited, or permitted across distinct enactments, violating the principle of non-contradiction [129, 3]. Specifically, normative conflict occurs when two or more valid laws impose incompatible legal duties on the same subject under the same conditions, as when one law mandates an act that another law forbids [3, 105]. When statutory systems generate conflicting obligations, citizens cannot conform their conduct to the law without incurring liability under an opposing mandate; this contradiction destabilizes legal predictability and public trust. Consequently, establishing systematic mechanisms to identify and resolve normative collisions represents a vital safeguard for the integrity of any rule-of-law framework.

However, in real-world legislative governance and local government archives, statutory enactments do not exist as pre-formalized formulas in symbolic logic. Rather, statutes are drafted and promulgated as unstructured, multi-generational natural language texts characterized by high syntactic variance and implicit legal exceptions [33]. Consequently, from a computational perspective, this research formulates and operationalizes the challenge as *Semantic Conflict Detection*. By translating abstract deontic clashes into a computable natural language processing problem, semantic conflict detection leverages continuous vector representation spaces and cross-encoder Natural Language Inference (NLI) to identify directional contradictions between local drafts and governing statutory provisions without relying on brittle, hand-crafted legal ontologies [196, 105]. As Ruhl and Katz [181] established, the volume of positive law expands far faster than human auditing capacity, making automated semantic auditing an operational necessity for modern legislative bodies.

### 2.1.1 Unstructured Legal Corpora

Managing unstructured legal text—text not stored in structured databases—represents a fundamental maintenance challenge for statutory systems. Practical automated auditing requires addressing rapid textual accumulation alongside the linguistic complexity of legislative drafting. These structural realities drive the research transition toward neural retrieval and inference tools for statutory verification.

#### 2.1.1.1 Legal Inflation

The term *legal inflation* describes the continuous exponential expansion in the sheer volume and interconnected complexity of statutory enactments over time. Ruhl and Katz [181] observe that as modern societies grow more complex, legislative bodies enact new laws at a pace that vastly outpaces the capacity of human legal staff to manually review them for consistency. Within legal informatics and administrative science, this rapid accumulation generates *regulatory debt*—formally defined as the compounding institutional burden of uncoordinated, obsolete, or overlapping legal enactments that accumulate when new legislative measures are promulgated without systematically reconciling or harmonizing them against pre-existing statutory codes [181]. In local government administration, this phenomenon is particularly acute because municipal ordinances are rarely compiled into centralized, cross-referenced statutory codes [43]. Conchas and Salapa [43] emphasize that transforming historical municipal enactments into machine-readable data presents severe engineering hurdles because local records are scattered across disparate administrative formats, spanning formal codified enactments and transitory executive orders.

Consequently, the statutory search space for identifying normative conflicts constitutes a fragmented, multi-generational archive rather than a structured relational database. Zhong et al. [249] characterize this challenge as a dynamic consistency problem, wherein governing facts and regulatory obligations continually shift across successive council terms. Because local statutory data lacks centralized indexing, municipal agencies are forced into a reactive posture, discovering statutory preemption defects only after an invalid measure is challenged in court. This systemic backlog motivates the transition in legal informatics from brittle, rule-based systems toward neural language models capable of learning semantic representations directly from unstructured text [33]. Such automated systems are required to process extensive legislative corpora that exceed the working memory and analytical throughput of human researchers.

#### 2.1.1.2 Limits of Manual Review

In contemporary legal informatics, digitizing physical documents into electronic archives no longer represents the primary operational bottleneck; rather, the central hurdle is auditing whether newly introduced provisions logically align with governing positive law. As legislative throughput accelerates, manual review workflows become overwhelmed by the cognitive burden of cross-referencing multi-volume statutory digests. A primary challenge for human reviewers is distinguishing between beneficial regulatory overlap—where multiple enactments address adjacent dimensions of a public policy issue—and destruc-

tive statutory conflict, where an enactment inadvertently contradicts a superior mandate. Recent surveys confirm that conventional keyword-based search platforms consistently fail to capture these subtle legal distinctions [10]. Consequently, latent statutory contradictions frequently remain undetected throughout the drafting lifecycle, shifting the burden of conflict discovery from legislative staff during pre-enactment review to judicial authorities during adversarial litigation [10, 207].

#### 2.1.1.3 Semantic Ambiguity

Traditional search systems rely on exact character matching, operating on the premise that legal concepts can be retrieved by searching for specific vocabulary strings. However, natural legal language exhibits pervasive *semantic ambiguity* driven by two primary linguistic phenomena: *synonymy* (where different words express the identical legal concept, such as *prohibited* versus *unlawful*) and *polysemy* (where the same word assumes drastically different meanings depending on context, such as a *party* referring to a litigant rather than a social gathering) [33, 88]. In statutory drafting, normative directives are governed by deontic modal verbs—such as *shall*, *must*, or *may*—whose binding legal force cannot be determined by simple string matching. Chalkidis et al. [33] demonstrate that this syntactic ambiguity causes exact-match techniques to fail when processing complex legal tasks. For example, a local ordinance regulating "motor vehicles" and a national statute governing "two-wheeled conveyances" might address the exact same civic activity, yet a Boolean query for "automobile" will fail to retrieve either provision due to vocabulary divergence.

Furthermore, legal consistency frequently depends on implied normative boundaries rather than explicit textual declarations. Katz et al. [113] establish that identifying these subtle contradictions requires computational systems that evaluate logical relationships and contextual semantics rather than surface character overlap. Transformer-based language models address this limitation by projecting text sequences into high-dimensional vector spaces that capture deep contextual semantics [247]. However, deploying neural architectures introduces an explainability trade-off. As Sovrano et al. [201] argue, while deep neural models achieve high classification accuracy, their uninspected internal activations function as opaque black boxes that fail to articulate the underlying legal reasoning behind a flagged conflict. In administrative and legislative settings, articulating the verifiable *ratio decidendi* behind an alert is just as vital as the detection itself [193].

### 2.1.2 Local Conflict Detection

While international legal informatics benchmarks such as LexGLUE advance automated conflict detection across Western legal corpora, corresponding research within post-colonial developing jurisdictions like the Philippines remains in its nascent stages [33]. Since the passage of the Local Government Code of 1991, Philippine local government units have enacted thousands of local ordinances that remain largely unindexed in machine-readable formats [85]. Although theoretical algorithms for conflict detection are established in mainstream artificial intelligence literature, their empirical application to localized Philippine statutory text remains largely unaddressed.

#### 2.1.2.1 Philippine Legal NLP

Within domestic computer science, research applying Natural Language Processing to Philippine law has historically focused on administrative classification and text summarization rather than deductive logical reasoning. Early investigations utilized basic supervised classifiers to forecast appellate outcomes in Supreme Court decisions [230] or applied extractive algorithms to condense extensive jurisprudence [183]. More recently, domestic legal informatics has explored transformer-based dense embedding architectures, which map text sequences into continuous mathematical vectors where geometric distance corresponds to topical relatedness [178]. While dense embeddings significantly outperform traditional keyword searches in retrieving domain-specific statutes (such as tax regulations), they remain fundamentally correlational: they identify that two provisions address the same subject, but lack the deductive reasoning layers required to determine whether the two provisions contradict each other.

Analyzing these research patterns reveals a pronounced structural gap in local literature. The Philippine NLP community has successfully automated the categorization and retrieval of national statutory text, while also proving that inference models can resolve semantic contradictions in local dialects for general-domain tasks such as online fact-checking [230, 183, 178, 155]. However, extending these advanced inference models to detect normative conflicts within the decentralized, multi-generational ordinances of a Highly Urbanized City like Davao remains entirely unexplored.

#### 2.1.2.2 Logic and Conflict Research

Within Philippine computational linguistics, the modeling techniques required for automated conflict detection—specifically Natural Language Inference (NLI)—have been successfully deployed across adjacent domains, most notably in detecting misinformation and rumors in low-resource vernacular text. Nuñez et al. [155] demonstrated that transformer architectures effectively resolve directional logical contradictions across code-switched Tagalog and administrative Taglish.

However, when domestic researchers analyze local government records, methodology regresses to legacy information retrieval frameworks. Document indexing across Philippine local government units relies predominantly on Boolean keyword matching and rigid rule-based heuristics. As Naguio et al. [148] demonstrate, these traditional systems suffer from severe *lexical fragility*: they fail when synonymous terms are used across enactments (such as "firecrackers" versus "pyrotechnic devices") and collapse entirely when encountering optical character recognition (OCR) noise in historical scans. Relying on manual rules to isolate statutory contradictions across thousands of localized city ordinances is mathematically unviable for human researchers and rule-based software alike.

#### 2.1.2.3 Research Gaps

While the Philippine AI community has successfully automated the search and summarization of national laws, a significant gap remains in using deep logic models to audit local

ordinances. Research indicates this gap persists because local legislation is substantially more challenging for automated systems to process than national statutory compilations.

The primary structural barrier to local legislative auditing is the critical dataset gap, characterized by the complete absence of curated ground-truth datasets for Philippine local enactments. Unlike national statutes that benefit from centralized repositories, local ordinances represent low-resource textual environments [169]. Standard legal NLP models are trained on massive Western corpora like LexGLUE, but there are no equivalent standardized benchmarks providing annotated pairs of conflicting local ordinances [76, 151]. Without a standardized benchmark to measure progress, local legal informatics research has historically struggled to move beyond superficial keyword search to deductive logical reasoning [76].

Compounding the lack of training data is a pronounced computational resource gap within Local Government Units (LGUs). Deductive reasoning models, particularly deep cross-encoders, demand substantial compute and memory resources that local government offices typically do not possess. Studies on Philippine public administration indicate that local digital adoption remains constrained by limited funding, scarce machine learning personnel, and legacy office desktop hardware [76]. In many local legislative councils, fewer than twenty percent of legislative personnel have received formal ICT training, with most offices still relying on paper records and basic text editors rather than structured databases [76]. This lack of digital infrastructure makes it difficult to justify complex models that would be too computationally demanding to execute on actual LGU workstations [101].

This computational disparity is directly reflected in the hardware profiles of municipal administrative offices. Under the Government Accounting Manual and Commission on Audit guidelines (COA Circular No. 2022-004), the estimated useful life of government information and communications technology equipment is fixed at five years, resulting in municipal workstation fleets that are routinely four to seven years old before replacement cycles occur [42, 76]. Furthermore, technical procurement specifications established by the Department of Information and Communications Technology (DICT) and the Procurement Service of the Department of Budget and Management (DBM-PS) for common-use clerical and legislative office computers specify mid-range quad-core or hexa-core processors (such as Intel Core i3 or i5 and AMD Ryzen 3 or 5 series), an 8 GB system memory ceiling, integrated graphics sharing system RAM without discrete CUDA acceleration, and standard solid-state drives [57]. Because standard local government computers lack dedicated tensor-processing hardware, deploying multi-billion-parameter autoregressive large language models locally is mathematically and financially unviable, while cloud-hosted commercial APIs introduce recurring subscription costs that exceed municipal operating budgets. Consequently, developing practical ex-ante conflict detection for local governance requires engineering lightweight, retrieve-then-entail pipelines capable of executing completely offline on consumer-grade 8 GB CPU-only workstations.

At the textual level, local legislative conflict detection faces a complex sociolinguistic gap. Local ordinances exhibit higher linguistic heterogeneity than national statutes because local drafting frequently blends formal legal English with regional dialects like Cebuano and administrative Taglish [66]. Standard NLP tokenizers designed for mono-

lingual English often break down when encountering localized legal idioms or archaic Spanish loanwords embedded in Philippine local codes [142]. While recent studies confirm that neural inference models can parse regional dialects in informal social media text, extending this capacity to dense, multi-generational local enactments requires specialized domain adaptation that remains largely unexplored in local literature [37, 88].

Finally, the slow adoption of computational consistency checking is driven by an institutional trust gap across the legal profession. As emphasized by Supreme Court Senior Associate Justice Marvic Leonen, the legal community remains cautious regarding artificial intelligence because conventional deep learning models function as opaque black boxes that cannot articulate their underlying reasoning [126]. Legal researchers and legislative counsel express concern that automated tools could introduce subtle statutory errors and lead to the enactment of void legislation [43]. Consequently, academic research has historically favored low-risk document indexing and retrieval systems, leaving deep semantic conflict detection unaddressed in local governance [12].

## 2.2 Philippine Legislative Domain

The Philippine legal system operates under a unitary constitutional framework, wherein Local Government Units (LGUs) possess purely delegated legislative powers conferred by Congress [85]. This jurisdictional structure establishes a strict hierarchical subordination that bounds municipal lawmaking beneath national legislative authority. Any computational system engineered to audit municipal enactments must operationalize these jurisdictional constraints. This section analyzes the statutory and structural conditions governing Philippine local law. First, it details the constitutional and statutory hierarchy subordinating local ordinances beneath national laws. Second, it traces the historical and stylistic roots of Philippine legal syntax. Third, it examines the parliamentary mechanics of the Three Readings legislative procedure during plenary sessions and the challenge of bilingual translation. Fourth, it contrasts ex-ante legislative auditing with ex-post judicial review, and finally reviews existing consistency checking methods to demonstrate the necessity of automated semantic auditing.

### 2.2.1 Hierarchical Constraints

The legal system in the Philippines depends on the fundamental rule that national laws always override local enactments. To understand the legal constraints governing local legislative drafting, automated conflict detection systems must reflect the structural principles that subordinate local ordinances beneath national statutes.

The Philippine legal system operates under a unitary constitutional framework, establishing a rigid hierarchy where national laws strictly control subordinate enactments [85]. At the apex stands the 1987 Constitution, followed by national statutes enacted by Congress, international treaties, presidential executive issuances, and local ordinances. Under this constitutional structure, no Local Government Unit can enact measures that contradict an Act of Congress. The doctrine of national preemption establishes an absolute boundary: whenever the national legislature enacts comprehensive

statutory standards governing a given regulatory domain, local governments are legally precluded from enacting conflicting or diluting local rules [85].

The primary organic charter governing local lawmaking is the Local Government Code of 1991 (Republic Act No. 7160). Section 16 of the Code, known as the General Welfare Clause, delegates police powers to local legislative councils to enact ordinances promoting the health, safety, and general prosperity of their constituents, provided such enactments remain strictly within statutory boundaries [85]. Because Davao City is classified as a Highly Urbanized City (HUC), it operates autonomously from provincial administrative supervision; its ordinances are therefore not reviewed by a provincial board. This administrative independence places the entire burden of ex-ante consistency auditing directly upon the Davao City Sangguniang Panlungsod before measures proceed to final enactment [85].

The definitive judicial test governing the validity of local legislation is established by the Supreme Court in the landmark ruling *Magtajas v. Pryce Properties Corporation, Inc.* [207]. To withstand judicial scrutiny, an ordinance must satisfy six foundational substantive requirements: it must not contravene the Constitution or any national statute, it must not be unfair or oppressive, it must not be partial or discriminatory, it must not prohibit but may regulate trade, it must be general and consistent with public policy, and it must not be unreasonable. Because local legislative bodies possess purely delegated authority, local councils cannot prohibit activities that national legislation has expressly authorized or regulated [85]. This principle of statutory subordination forms the primary legal benchmark against which automated conflict detection systems must evaluate candidate local drafts.

When a local ordinance impermissibly enters an area governed by national legislation, Philippine courts consistently apply the *Magtajas* doctrine to declare the local measure invalid. In *Magtajas v. Pryce Properties Corporation, Inc.* [207], the City Council of Cagayan de Oro enacted ordinances prohibiting casino operations to protect public morals. The Supreme Court struck down the local enactments as *ultra vires*, ruling that a local council cannot prohibit commercial gambling activities that a national statute expressly authorized through the Philippine Amusement and Gaming Corporation (PAGCOR) charter. Similar preemption disputes have directly affected Davao City legislation. In *Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc.* [210], the Supreme Court invalidated Davao City Ordinance No. 0309-07, which imposed a local ban on aerial pesticide spraying across agricultural plantations, holding that the ordinance exceeded local police powers and collided with the regulatory authority granted to the national Fertilizer and Pesticide Authority under Presidential Decree No. 1144. In *City of Manila v. Laguio, Jr.* [209], the Court likewise struck down an ordinance prohibiting motels and massage parlors, holding that local police powers cannot arbitrarily prohibit lawful commercial activities under the guise of zoning regulation.

In contrast to vertical statutory preemption, local lawmaking also involves horizontal consistency, where newly proposed ordinances must align with existing, fellow local enactments. When drafting local legislation, the Sangguniang Panlungsod uses the Three-Reading legislative process, particularly the committee debates and line-by-line amendments during the Second Reading, to harmonize new proposals with older local en-

actments [186]. Unlike vertical conflicts that represent fatal legal defects, inconsistencies between a new draft and an older local ordinance frequently reflect intentional legislative updates, such as revising local tax rates or adjusting traffic routes. To manage these horizontal modifications systematically, legislative drafters rely on standardized legislative saving devices recommended by the Department of the Interior and Local Government (DILG) [58]. Every formal ordinance incorporates an express or general repealing clause, declaring that all prior local enactments inconsistent with the newly approved measure are amended or repealed accordingly. Furthermore, ordinances feature a separability clause, acting as a structural legal safeguard ensuring that if any single section is declared invalid by judicial authorities, the unoffending sections remain in full legal force [58]. However, because local legal researchers must review hundreds of active local enactments under strict legislative session schedules, subtle inconsistencies often go unnoticed during the Second Reading and proceed to final enactment.

A fundamental design decision in legal informatics is delineating which positive legal instruments establish binding preemption standards. Under Article VII, Section 17 of the 1987 Philippine Constitution and Book III, Title I, Chapter 2, Section 2 of the Administrative Code of 1987 (Executive Order No. 292) [172], Executive Orders (EOs) are acts of the President providing rules of a general or permanent character in implementation or execution of constitutional or statutory mandates. Unlike Supreme Court decisions—which function as *ex-post* judicial dispute resolutions settling adversarial controversies under the doctrine of *stare decisis* (Article 8, Civil Code)—Executive Orders operate as *ex-ante*, forward-looking statutory instruments possessing the full force and effect of law. Furthermore, during both the Martial Law period (1972–1986) and the Revolutionary Government under the Freedom Constitution (Proclamation No. 3, 1986–1987), the President exercised direct legislative powers through Executive Orders and Presidential Decrees to establish substantive national enactments that remain operative law today. Modern Executive Orders frequently establish binding nationwide regulatory standards—such as Executive Order No. 26 (s. 2017) establishing nationwide public smoking prohibitions, or Executive Order No. 168 (s. 2014) creating the Inter-Agency Task Force for the Management of Emerging Infectious Diseases (IATF)—that legally preempt conflicting local legislation. Excluding Executive Orders would therefore omit critical nationwide regulatory mandates that local ordinances are legally bound to respect.

### 2.2.2 Historical and Stylistic Heritage

The structural complexity of Philippine legislative text stems directly from the nation's historical experience as a mixed legal jurisdiction [4, 50]. Philippine jurisprudence combines continental civil law traditions with Anglo-American statutory common law conventions. Understanding why Philippine statutes and local ordinances are drafted in their characteristic syntax requires examining how these colonial and post-colonial traditions fused over more than a century of codification.

During more than three centuries of Spanish rule, the Philippine legal system was rooted firmly in the Roman-Germanic civil law tradition [4]. This continental tradition emphasized systematic codification, deductive legal reasoning, and broad normative principles organized into structured books, titles, and chapters. Foundational enactments,

such as the Spanish Civil Code of 1889 and the Penal Code of 1870, relied on generalized legal mandates rather than granular factual definitions. Furthermore, this civilian heritage embedded a rich vocabulary of Latin doctrinal maxims into Philippine jurisprudence, including *in pari materia* (construing related laws together), *generalia specialibus non derogant* (a general law does not override a specific one), and *ultra vires* (acts beyond legal authority) [4]. Modern Philippine codes retain this civilian structure, expecting adjudicators to derive specific legal duties from comprehensive normative commands.

Following the Treaty of Paris and the establishment of the American Insular Government (1900–1935), American legal administrators introduced common law statutory drafting habits without fully displacing the civilian codification base [50]. Enacted by the Philippine Commission and later the Philippine Legislature, early statutes introduced formal drafting conventions borrowed directly from United States federal and state legislation. These conventions included explicit statutory title preambles, exhaustive definitions sections, severability clauses, and general repealing provisions [4]. Significantly, American administrators introduced the practice of drafting extensive conditional provisos (introduced by the traditional formula "*Provided, that...*" and "*Provided, further, that...*"), designed to restrict, qualify, or create exceptions to preceding operative clauses within the same section. The Revised Administrative Code of 1917 (Act No. 2711) and the contemporary Administrative Code of 1987 (Executive Order No. 292) solidified this mixed tradition, confirming English as the controlling statutory language while maintaining civil law institutional categories [172].

This dual heritage produces a legislative drafting style marked by high syntactic density and multi-clause complexity. Unlike standard conversational English, a single statutory provision frequently spans one hundred to two hundred words contained within a solitary grammatical sentence [20]. These sentences embed multiple dependent clauses, qualifying prepositional phrases, and recursive proviso exceptions. At the local level, legislative staff adopt this formal national drafting template while adding procedural boilerplate, resulting in verbose ordinances where the operative regulatory command is separated from its governing subject by dozens of intervening words [58, 186]. From an analytical and document processing perspective, this hybrid syntax presents substantial structural hurdles: archaic legal terms resist straightforward vocabulary normalization, long-distance dependencies obscure the core legal directive, and qualifying provisos invert the practical meaning of preceding clauses unless the text is examined in its full structural context.

### 2.2.3 Plenary Legislative Procedure

Besides the hierarchy of laws, the actual parliamentary mechanism through which laws are drafted, debated, and approved determines how statutory text evolves over time. In the Philippine legal architecture, positive law enactment is governed by a mandatory three-reading procedure conducted during formal plenary sessions of the legislative body. At the national level, Article VI, Section 26, Paragraph 2 of the 1987 Philippine Constitution explicitly dictates that no bill passed by either the House of Representatives or the Senate shall become law unless it has passed three readings on separate days, with printed copies in final form distributed to members at least three days prior to passage [45, 99]. At

the local government level, Sections 53 and 54 of the Local Government Code of 1991 (Republic Act No. 7160) [85], the Local Legislative Drafting Manual of the Department of the Interior and Local Government (DILG) [58], and the Internal Rules of Procedure of the 20th Sangguniang Panlungsod of Davao City (Resolution No. 001-22) [186] replicate this parliamentary doctrine. Under these statutory rules, every proposed municipal or city ordinance must undergo three distinct readings before the full assembly in plenary session. This parliamentary mechanism is designed to guarantee institutional transparency and deliberate democratic accountability before any regulatory constraint attains the force of law.

The legislative progression begins on the plenary floor with the First Reading. In both national congressional chambers and the Davao City Sangguniang Panlungsod, the First Reading is strictly ministerial and introductory: the presiding officer directs the Secretary to read solely the assigned measure number, title, and the name of the author or proponent before the assembled plenary body [99, 186]. Crucially, parliamentary rules prohibit any substantive debate, interpellation, or floor amendments during First Reading. Instead, the presiding officer refers the proposed measure immediately to the appropriate standing committee—such as the Committee on Rules, Laws, and Ordinances—initiating the inter-session committee review phase. In this committee stage, technical working groups, legal counsel, and committee researchers examine the draft, hold public hearings, and prepare the formal committee report. Because local ordinance drafting is decentralized—with individual council members authoring distinct proposals rather than relying on a centralized drafting bureau [85, 186]—drafting conventions and legal terminology vary widely. Legal researchers bear the heavy burden of manually cross-referencing proposed clauses against thousands of national statutes and prior city enactments under strict legislative session schedules. When staff lack automated semantic search tools, potential preemption defects and horizontal contradictions frequently survive committee scrutiny undetected.

Following committee review, the proposed ordinance returns to the plenary floor for the Second Reading, which represents the primary substantive and deliberative phase of the legislative lifecycle. The Second Reading commences with a formal sponsorship speech delivered by the committee chair or measure proponent to summarize the policy objectives and legal basis of the enactment. Immediately following sponsorship, the plenary assembly enters the Period of Interpellation and General Debate. During this period, colleagues interrogate the sponsor, probe statutory ambiguities, question fiscal or regulatory feasibility, and raise substantive legal objections regarding whether the measure contravenes superior national legislation. Once general debate is formally terminated, the body transitions into the Period of Amendments, where committee amendments and individual floor amendments are presented, debated, and voted upon section by section [99, 186]. This period of amendments is the decisive procedural window where proposed clauses are actively reshaped, modified, or deleted to align with governing law.

The Period of Interpellation and Amendments on Second Reading constitutes the exact procedural nexus where an automated semantic conflict detection system delivers its highest institutional utility. Because the Second Reading is the sole phase on the plenary floor where substantive textual alterations, statutory reconciliations, and corrective

rewrites can be introduced, legal researchers and sponsors require granular, evidence-backed conflict diagnostics before and during floor deliberations. If a draft ordinance clause encroaches upon a national statutory domain—violating the *Magtajas* doctrine by attempting to prohibit what national law permits, or by exceeding statutory penalty caps under Section 458 of Republic Act No. 7160—the automated system pinpoints the conflicting statutory premise, flags the contradictory deontic modalities, and provides an explainable legal audit trail. Equipping Sangguniang Panlungsod legal researchers with these automated findings allows them to draft precise, legally sound floor amendments during the Period of Amendments, rectifying latent preemption flaws before the measure advances past the amendable phase.

After the assembly adopts all floor amendments and finalizes the draft on Second Reading, the measure is engrossed, printed in its final form, and scheduled for Third Reading in plenary session. In stark contrast to the dynamic debates of Second Reading, the Third Reading is procedurally rigid and unmodifiable. Under Article VI, Section 26, Paragraph 2 of the 1987 Constitution and corresponding local Sanggunian rules, no amendment whatsoever is permitted upon the Third Reading of a bill or ordinance [45, 99, 186]. Plenary action is confined strictly to a nominal roll-call vote, where each member registers an affirmative, negative, or abstaining vote, accompanied by a brief oral explanation. This procedural finality creates severe institutional risk: if a statutory collision is not discovered and rectified prior to or during the Second Reading, the flawed provision is permanently locked into positive law upon Third Reading approval. The enacted ordinance becomes vulnerable to immediate judicial nullification under the *Magtajas* doctrine, administrative disputes before national executive departments, or citizen litigation. This structural irreversibility underscores why ex-ante computational conflict detection is essential during committee review and Second Reading preparation.

Compounding the procedural complexities of the Three Readings is the statutory mandate requiring local legislation to be accessible across languages. Under Section 469 of the Local Government Code, the Secretary to the Sanggunian is mandated to translate approved ordinances into local dialects, such as Cebuano for Davao City [85, 186, 66]. However, this creates legal friction because the Administrative Code of 1987 establishes that the English statutory text strictly controls in cases of conflict [172]. If an error occurs in regional translation, a citizen relying on the Cebuano text may inadvertently violate the authoritative English version; this disparity conflicts with the public notice doctrine in *Tañada v. Tuvera* [205]. Despite policy initiatives such as the proposed *Batas sa Sariling Wika* legislative measure (House Bill No. 3863) [170], subtle dialect differences often produce semantic drift where the exact legal obligation in English is distorted in regional vernacular text [142]. Consequently, consistency checking systems must operate primarily on the authoritative English statutory provisions while accounting for regional linguistic artifacts generated during local committee debates and drafting iterations.

### 2.2.4 Ex-Ante Legislative Review

The term "ex-ante" derives from Latin meaning "before the event." In legislative informatics and administrative jurisprudence, it denotes the rigorous evaluation of a proposed law or local regulation prior to its official enactment [73]. This prospective auditing process

aims to identify and remediate potential statutory collisions, constitutional infirmities, and unintended regulatory burdens while the draft remains amenable to revision.

In the Philippines, ex-ante regulatory review has achieved heightened institutional prominence following the enactment of Republic Act No. 11032, known as the Ease of Doing Business and Efficient Government Service Delivery Act of 2018 [86]. Under this statutory framework, the Anti-Red Tape Authority (ARTA) alongside the Department of the Interior and Local Government (DILG) mandates that Local Government Units conduct a formal Regulatory Impact Assessment (RIA) for proposed regulations impacting commerce and public welfare [86]. The objective is to verify that proposed ordinances possess valid statutory authority and do not impose duplicative or contradictory requirements on citizens and businesses.

The fundamental distinction between ex-ante and ex-post evaluation lies in timing, methodology, and legal impact across the hierarchy of legal instruments. Ex-ante evaluation focuses strictly on prospective, forward-looking legislative texts—such as draft local ordinances, pending congressional bills, and proposed executive issuances—prior to their passage into positive law. In contrast, ex-post evaluation operates retrospectively through judicial dispute resolution, encompassing adversarial litigation, appellate court rulings, and Supreme Court jurisprudence governed by *stare decisis*. While ex-post judicial adjudication rectifies constitutional injuries after harm has occurred, it incurs substantial societal and litigation expenses, as repealing an active ordinance requires protracted court proceedings [223]. Ex-ante auditing is therefore substantially more efficient, protecting local governments from administrative deadlock and judicial invalidation.

Deploying automated computational tools for ex-ante conflict detection empowers local legislative researchers to identify normative collisions during the drafting stage. Because legislative bodies generate extensive statutory volumes, manual review frequently fails to catch subtle inconsistencies between new drafts and existing laws [181]. By operating as an automated conflict-screening tool, computational systems flag potential preemption conflicts while the measure remains in committee deliberation. This facilitates alignment with superior national statutes, mitigating the risk of enacting local measures that are legally void *ab initio* [207].

### 2.2.5 Legislative Consistency Checking

Because local governments hold subordinate legislative authority within a decentralized drafting environment, systematic consistency checking represents an essential institutional safeguard. However, verifying that an ordinance does not contradict superior statutes requires far more than title matching; it demands deep comprehension of legal semantics and subtle jurisdictional boundaries. Current consistency checking methods remain encumbered by syntactic ambiguity and reliance on basic tracking platforms.

Discourse analysis of Philippine statutory language reveals pervasive ambiguity in how normative keywords and modalities are deployed [206]. In particular, the modal auxiliary “shall” introduces frequent interpretive friction, as it may denote an absolute command, an administrative directive, or a directory permission depending on statutory

context. Empirical studies confirm that such syntactic ambiguities complicate legal interpretation even when enactments adhere to formal drafting guidelines [20].

Philippine jurisprudence recognizes that statutory conflicts rarely present themselves through explicit textual declarations [50]. A local ordinance virtually never declares an overt intent to contradict an Act of Congress; rather, conflicts typically arise as implied preemption. This occurs when a national statute occupies a regulatory field, and a local council introduces unauthorized preconditions or local fees that restrict activities already licensed by the state [206]. In such cases, the Supreme Court has consistently held that the local measure impermissibly modifies national law under the guise of police power regulation. For manual reviewers and keyword-based search tools, detecting implied preemption is exceptionally difficult because the conflicting enactments employ disparate vocabularies; uncovering the contradictory regulatory effect demands deep semantic analysis.

In the local context, the Davao City Sangguniang Panlungsod processes extensive legislative volume, passing nearly 400 ordinances and over 2,000 resolutions within a single three-year council term [160]. To manage this throughput, the city deploys administrative platforms: the Resolution Ordinance Tracking System (ROTS) manages document routing and metadata, while the Legislative Information Support System Program (LISSP) enables council members to access digitized enactments [37]. However, while these digital systems streamline document logistics, substantive consistency auditing across provisions remains entirely manual. Furthermore, archival scans frequently contain optical character recognition (OCR) noise, causing keyword search queries to miss critical governing provisions [133].

## 2.3 Digital Legislative Archives

Building a functional Semantic Conflict Detection system requires access to comprehensive and machine-readable legal corpora. In the Philippines, the digital landscape of legislative data is highly fragmented, split between national statutory repositories and localized city archives. This structural division presents significant technical hurdles for statutory data ingestion. Before deep logical inference can occur, automated auditing pipelines must first secure reliable data streams from these disparate legal environments. Consequently, understanding the specific characteristics of these sources informs the design of statutory retrieval and preprocessing methodologies.

### 2.3.1 National Sources

National legal repositories represent the authoritative apex of the Philippine legislative hierarchy and house laws that universally preempt local ordinances. These databases are maintained by centralized government agencies and contain highly formalized texts such as Republic Acts, Commonwealth Acts, and Batas Pambansa. They form the primary knowledge base for external legal context, providing the authoritative baseline rules against which local drafts are evaluated. Because national legislation undergoes a standardized formal drafting process, the vocabulary within these repositories is generally consistent and adheres to strict legal drafting standards. Accessing these national databases provides

computational systems with the necessary macroscopic view of state-level mandates and prohibitions.

The Official Gazette is the constitutionally mandated repository for Philippine statutes and the primary journal of record for national legislative text [87]. It publishes all newly enacted national laws, presidential executive orders, and administrative issuances, as required by Republic Act No. 453 [163] and Commonwealth Act No. 638 [150]. Because it contains the definitive, legally binding version of a statute, any computational system evaluating local statutory preemption must anchor its baseline rules to this repository. This archive forms the authoritative historical ledger for state-level mandates; it preserves the exact phrasing approved by the legislative and executive branches. Consequently, it remains the most critical target for establishing an authoritative statutory knowledge base.

Complementing the official journal, developers utilize the Open Congress API provided by the House of Representatives Legislative Information System (LEGIS) to access structured, machine-readable metadata regarding legislative routing and statutory amendments [100]. This programmatic interface tracks a statute's formal lifecycle, charting its progression from initial committee filing to final executive approval. By providing standardized output formats, the API allows computational pipelines to establish relational dependencies between different legislative acts without manually parsing unstructured text.

Finally, the Lawphil Project functions as an extensive, privately maintained digital repository of statutory records and jurisprudential decisions [9]. Unlike official government channels that frequently change web templates, this repository provides a comprehensive, pre-digitized plaintext corpus of historical and contemporary Philippine law. This accessibility makes it mathematically ideal for the unsupervised pre-training and vectorization of foundational language models tailored to the local legal domain. By ingesting this vast corpus of clean text, NLP architectures learn the highly specific syntactic structures and deontic operators inherent in Philippine jurisprudence, establishing the core vocabulary required for transformer models to process specialized statutory drafting [247].

### 2.3.2 Local Sources

In contrast to centralized national databases, local legal repositories are maintained independently by individual Local Government Units (LGUs). These archives house local ordinances, resolutions, and executive orders that govern specific geographic jurisdictions. The legislative output at this tier is entirely decentralized, resulting in highly variable archiving practices depending on the technological capacity of the specific city council. These local databases form the immediate target corpus for local conflict detection and contain the actual regulatory constraints imposed on city residents. Consequently, they represent a high-volume, dynamic data environment that continuously expands as the local council holds regular drafting sessions.

The local corpus is anchored upon the Davao City Sangguniang Panlungsod portal, the official digital repository for local legislative outputs [38]. This portal preserves the primary historical record for city regulations, resolutions, and zoning policies enacted

within the city's jurisdiction. It represents a dynamic and continuously expanding text environment that directly reflects the decentralized nature of local lawmaking. The texts housed here encapsulate the specific regulatory constraints and policy directions of the city government, providing the exact ordinance text required to evaluate new drafts for logical consistency against the existing legislative framework.

To manage the extensive volume of local legislation, the city introduced the Legislative Information Support System Program (LISSP) as its primary administrative backbone [37]. This platform is a digital tracking system designed to maintain metadata regarding ordinance routing, committee assignments, and plenary approval statuses. The system provides transparency for monitoring ordinances and resolutions in real time [161]. By digitizing the legislative workflow, LISSP operates as a centralized database of ordinances, providing the structured tracking and contextual data necessary for evaluating the lifecycle of local laws, while addressing the interoperability challenges identified in recent regional e-governance assessments [43].

### 2.3.3 Data Extraction Challenges

While the aforementioned repositories contain the necessary legal knowledge, extracting the needed data for computational analysis presents severe technical challenges. Scraping and formatting these diverse sources exposes the pipeline to significant data degradation and structural inconsistencies across multiple dimensions.

#### 2.3.3.1 Archival Scanning and OCR Degradation

A primary barrier to automated statutory auditing is the lack of native, machine-readable text across archival collections. Both the Official Gazette and the Davao City Sangguniang Panlungsod public archive predominantly host scanned, image-based PDFs rather than selectable digital documents. During manual inspection of these collections, enactments exhibited visual noise such as slanted scans, faded ink, low contrast, and compression artifacts. As Lopresti [133] documents, such noisy visual data results in severe Optical Character Recognition (OCR) degradation, where standard extraction tools produce broken alphanumeric strings and nonsensical tokens. These irregularities suggest that a direct ingestion pipeline suffers from semantic fragmentation, requiring an additional high-precision extraction layer and extensive text-cleaning heuristics to ensure data viability.

The resulting character and word-level corruptions severely hinder automated semantic analysis. In statutory conflict detection, a single corrupted character can alter the legal interpretation of a provision, such as transforming a negative operator into an affirmative assertion or scrambling a penalty amount. Standard tokenizers struggle to process these degraded sequences, fragmenting broken words into out-of-vocabulary sub-tokens that dilute contextual representations [182]. Furthermore, scanning artifacts frequently obscure section demarcations and legislative numbering; this prevents standard parsers from isolating self-contained statutory units. Consequently, automated preemption pipelines must incorporate noise-tolerant document normalization to restore structural integrity before downstream retrieval.

#### 2.3.3.2 Repository Scarcity and Digital Isolation

Another critical challenge is archival fragmentation, wherein legislative repositories suffer from systemic gaps in their digital records. This poses a significant problem for high-fidelity auditing because the absence of even a single related ordinance can lead to false negatives in conflict detection, where a system declares a draft valid simply because it cannot access the contradicting record. A prominent example is the Davao City Sangguniang Panlungsod public archive, which features intermittently missing ordinance files and unorganized chronological listings that hinder systematic data collection [188]. Even relatively complete national databases like Lawphil can lack specific Republic Acts or historical amendments. As Westermann et al. [92] observe, fragmented legal databases force developers to move beyond simple scraping toward fault-tolerant data engineering to ensure an exhaustive corpus.

A further obstacle in statutory data acquisition is the discrepancy between the theoretical and actual accessibility of local legislative databases. For example, while administrative systems like Davao City's LISSP are promoted in institutional reports as public transparency portals [37, 161], empirical verification reveals strict authentication barriers requiring authorized credentials. Technical inspection indicates that local legislative portals often remain computationally isolated behind authentication interfaces; this isolation restricts automated extraction by public web scrapers. Correspondence with local legislative administrators confirms that full-text enactments require administrative clearance and manual provisioning. As Conchas and Salapa [43] observe, isolated administrative data often exists as digitally orphaned artifacts, where metadata about an enactment is disconnected from its full-text searchable content, requiring specialized ingestion pipelines to reconcile sparse public metadata with offline full-text documents.

## 2.4 Exact and Symbolic Methods

Before the adoption of neural networks, computational legal analysis relied heavily on deterministic systems to process text. Deterministic systems operate on predefined rule matrices, wherein a given input produces reproducible outputs without probabilistic inference or statistical estimation. Traditional methods for searching and evaluating legal information assumed that legislative ideas could be cleanly mapped either to specific vocabulary strings or to rigid, hand-coded logic formulas. While highly transparent and easy to debug, these foundational approaches frequently fail when forced to detect hidden conflicts in complex, unstructured legal texts. The core issue is that human language is inherently messy, and rigid systems break down when faced with the ambiguous phrasing typical of local ordinances.

### 2.4.1 Boolean Information Retrieval

Evaluating why legacy legal repositories struggle with semantic conflict detection requires examining how deterministic retrieval architectures process text at the machine level. Boolean Information Retrieval (BIR) represents the foundational search architecture underlying traditional legal databases. BIR operates on the bag-of-words assumption, ab-

stracting statutory enactments into unordered multisets of terms while discarding grammatical syntax, clause order, and relational context. At its core, the architecture relies on an *inverted index*—a mapping that pairs every distinct vocabulary term with a postings list recording document identifiers and term frequencies. Because an inverted index registers solely the discrete presence or absence of literal character strings, the retrieval engine cannot infer latent semantic relationships, grammatical qualifications, or underlying regulatory intent.

At its core, the retrieval mechanism in BIR is strictly based on set theory and formal Boolean logic. A user constructs a query using logical operators like `AND` ($\wedge$), `OR` ($\vee$), and `NOT` ($\neg$) to create intersecting sets of documents. For example, a search for a specific ordinance regarding explosives is formulated mathematically as:

*Boolean Query Set Intersection Formulation:*

$$Q = (\text{fireworks} \vee \text{pyrotechnics}) \wedge \neg\text{exempt} \tag{2.1}$$

Where the compound query $Q$ combines disjunctive keyword sets under conjunction with a negated exemption condition.

The system then retrieves documents by calculating the exact mathematical intersection of these sets to pull only the files that satisfy every condition simultaneously. If a document contains the specific string of characters requested, it is retrieved and scored as a 1; if it misses even a single required term, it is completely ignored and scored as a 0. It evaluates the raw presence of strings, entirely ignoring the conceptual meaning or legal intent behind them [23].

#### 2.4.1.1 Challenges with BIR

The primary limitation of BIR in legal auditing is its absolute, uncompromising reliance on exact string matching. This rigidity creates a severe "vocabulary mismatch" problem when processing natural language. Human language naturally features synonymy, where entirely different words possess the exact same meaning, and polysemy, where the identical word takes on drastically different meanings depending on the context. In statutory drafting, one author might use "prohibited" while another uses "unlawful," referring to the exact same legal concept. Because BIR operates on character-level matching rather than conceptual mapping, it fundamentally cannot connect these two terms; this failure leads to massive blind spots in the search results.

A landmark empirical evaluation of these systems demonstrated this vocabulary gap in a highly practical setting. Legal researchers conducting queries using standard BIR systems confidently estimated they were finding over 75 percent of the relevant documents for a given case. They experienced an illusion of effectiveness because the retrieved documents looked highly accurate, which masked the massive volume of missed relevant files. In reality, the system retrieved less than 20 percent of the actual relevant files contained within the database [23]. The researchers missed the remaining 80 percent simply because the authors of those hidden documents used slightly different, completely valid phrasing that did not perfectly align with the strict search query [225].

In a local government context like Davao City, this specific margin of error is administratively unacceptable. Missing even one existing ordinance during an ex-ante review due to a simple vocabulary mismatch could directly result in the passage of an invalid, conflicting law. Furthermore, legislative vocabulary inherently evolves over time as drafting conventions change. An ordinance passed in the 1990s and a modern draft written in 2026 might strictly regulate the exact same local regulatory subject but share absolutely zero common keywords. Recent studies emphasize that resolving this requires the implementation of "Transformer" models, which abandon exact matching entirely in favor of continuous vector spaces that recognize when different vocabulary points to the same underlying concept [116].

### 2.4.2 Symbolic AI

Because BIR fundamentally fails to capture underlying meaning, early artificial intelligence researchers attempted to automate legal reasoning by directly modeling the rules of law themselves. This approach, widely known as Symbolic AI or "Good Old-Fashioned AI" (GOFAI), operates on the premise that human cognition can be replicated by manipulating formal symbols [228]. Symbolic AI formalizes legal reasoning by modeling statutory constraints into explicit, hand-coded production rules. Rather than searching unstructured text strings, Symbolic AI translates legislative commands into formal deontic and first-order logic representations, allowing computational engines to execute deductive reasoning over regulatory relationships.

To execute this logic, these systems specifically utilize propositional or first-order logic combined with deontic operators. Deontic logic is a specialized branch of mathematical philosophy that formally evaluates normative modalities governing conduct: obligations ($\mathcal{O}$, representing mandatory actions that must be performed), permissions ($\mathcal{P}$, representing discretionary actions that may be undertaken), and prohibitions ($\mathcal{F}$, representing unlawful actions that must not be performed). For example, a system represents a local zoning prohibition mathematically as:

*First-Order Deontic Prohibition Formulation:*

$$\forall x(\text{Business}(x) \land \text{ResidentialZone}(x) \rightarrow \text{Prohibited}(x)) \tag{2.2}$$

Where the universal quantifier ($\forall$) asserts that for any entity $x$, if it is classified as a business ($\land$) and located in a residential zone, it is strictly prohibited ($\rightarrow$).

By converting ordinances into these rigid formulas, a system can theoretically use deductive reasoning to spot a conflict. If Rule A dictates $\mathcal{O}(X)$ (an absolute obligation to do X) and Rule B dictates $\mathcal{F}(X)$ (an absolute prohibition against doing X), the system successfully flags a formal logical contradiction ($\perp$) [13].

#### 2.4.2.1 Challenges with Symbolic AI

While Symbolic AI provides perfectly transparent and mathematically sound reasoning, its absolute reliance on manual programming makes it practically impossible to scale for real-world governance. The primary architectural barrier is the classical "knowledge

acquisition barrier" in expert systems: human domain experts and software engineers must manually read, interpret, and translate thousands of unstructured city ordinances into rigid computer rules. This manual translation process is unacceptably slow, extremely expensive, and highly prone to human error during the coding phase [137]. Furthermore, every time the city council amends an existing law, the underlying symbolic formula must be manually rewritten; this manual requirement makes the system difficult to maintain over time.

Beyond the logistical challenge of manual coding, the rigid binary architecture of Symbolic AI fundamentally clashes with what legal philosopher H.L.A. Hart famously termed the *open texture of law* [94]. As Hart demonstrated in *The Concept of Law*, statutory language cannot anticipate every future factual scenario; consequently, lawmakers deliberately employ open-ended, discretionary standards—such as mandating "reasonable care," "due diligence," or "good moral conduct"—to allow adjudicators flexibility across changing societal contexts. These subjective legal standards cannot be reduced to binary IF-THEN rules because their meaning shifts depending on circumstance and judicial interpretation [12]. While symbolic systems achieve deterministic precision in tightly constrained, domain-restricted formal micro-environments, they fail when exposed to the linguistic variance, unformalized exceptions, and semantic ambiguities characteristic of municipal legislative archives [249].

## 2.5 Graph-Based Architectures

Because exact word matching can be unreliable, computational legal studies frequently explore graph-based architectures to model relationships between statutory provisions. Instead of treating legal texts as isolated character sequences, graph-based frameworks represent statutes as interconnected nodes within a structural network. By analyzing how enactments cite, amend, and conceptually reference one another, these architectures capture latent legal context that bag-of-words models invariably miss.

### 2.5.1 Knowledge Graphs and GNNs

Legal Knowledge Graphs (LKGs) model statutory ecosystems by representing legal entities, provisions, and administrative bodies as structured vertices linked by directional, typed edges. In formal graph theory, a Legal Knowledge Graph models a statutory corpus as a directed, multi-relational graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, wherein vertices $v \in \mathcal{V}$ represent discrete statutory enactments, administrative agencies, or substantive regulatory provisions, while edges $(u, r, v) \in \mathcal{E}$ designate formal juridical relationships $r \in \mathcal{R}$ such as statutory delegation, cross-reference citation, explicit amendment, or regulatory enforcement. Unlike flat text indices, these relational graphs enable Graph Neural Networks (GNNs) to learn node embeddings by aggregating feature representations across local structural neighborhoods [21]. Researchers have developed automated frameworks like GNN-ERE to extract legal entities and relations directly from unstructured statutory text, establishing structured taxonomies without manual relational coding [139]. Furthermore, empirical evaluations demonstrate that graph-based representations improve legal

document retrieval by mapping semantically related court judgments even in the absence of verbatim keyword overlap [93].

#### 2.5.1.1 Graph Modeling and Multi-Hop Reasoning

The foundational strength of Graph Neural Networks lies in their ability to propagate structural context across multi-hop relational paths. In statutory interpretation, normative conflicts rarely occur through isolated direct collisions; rather, they frequently manifest indirectly across chains of administrative definitions and delegated authorities [242]. Formally, Graph Neural Networks execute iterative neighborhood aggregation (message passing), wherein the representation $\mathbf{h}_v^{(l)}$ of node $v$ at layer $l$ is recursively updated by aggregating the latent feature vectors of its adjacent 1-hop topological neighbors. This inductive propagation allows the network to incorporate multi-hop structural context, evaluating whether an ordinance provision conflicts with a superior statutory mandate through intermediate administrative definitions or delegated authority chains. This topological awareness enables systems to distinguish between different legislative functions, such as separating procedural requirements from substantive prohibitions [21]. Consequently, graph neural architectures provide a principled mechanism for structural reasoning across complex, multi-tiered statutory hierarchies.

Furthermore, graph representations capture institutional relationships that remain hidden during sequential text reading. In decentralized governance frameworks like the Philippines, local legislative acts operate within a matrix of national agencies, provincial guidelines, and local zoning boundaries. Graph neural models learn relational node embeddings that reflect these jurisdictional hierarchies, allowing algorithms to evaluate legislative preemption within its broader administrative network [139]. By mapping shared regulatory targets across different enactments, the graph structure guides inference models toward relevant statutory clusters that standard search engines fail to connect. This relational capacity makes graph modeling an appealing conceptual paradigm for analyzing legislative coherence across decentralized jurisdictions.

#### 2.5.1.2 Structural Vulnerabilities in Unstructured Archives

While Graph Neural Networks offer substantial theoretical utility, their practical deployment depends strictly on the completeness and fidelity of the underlying graph representation. A primary vulnerability is their reliance on explicit statutory cross-references to establish graph edges [103]. In Philippine local archives, local ordinances frequently omit formal citations to the governing Republic Acts they inadvertently contradict; drafts rely instead on localized colloquial descriptions. When explicit relational links are missing from the initial graph construction, the GNN remains fundamentally blind to the underlying normative collision, unable to propagate message vectors across disconnected nodes.

This limitation is exacerbated by systemic disconnection and physical degradation across local government records. City archives like Davao City's LISSP store enactments as isolated scanned documents rather than interconnected, machine-readable data nodes [43]. When automated relation extraction pipelines encounter noisy optical charac-

ter recognition outputs, they generate hallucinated edges and misattributed entity linkages [133]. These erroneous connections propagate through successive graph convolution layers, contaminating node representations and introducing false conflict signals throughout the network. Consequently, without expensive, manual human curation to verify every edge, graph-based systems prove too brittle for noisy, real-world local archives.

### 2.5.2 Abstract Meaning Representation

While Graph Neural Networks capture macro-level relationships between distinct legal documents, Abstract Meaning Representation (AMR) evaluates the semantic and logical structure inside individual sentences. AMR represents the meaning of a sentence as a rooted, directed acyclic graph where nodes correspond to semantic concepts and edges represent semantic roles [15]. In legal informatics, AMR provides a formal representation of predicate-argument semantics, abstracting away superficial syntactic variations such as active versus passive voice [232]. For example, whether a statute declares that "The City Mayor shall enforce the penalty" or that "The penalty shall be enforced by the City Mayor," AMR generates the identical semantic rooted graph, identifying the City Mayor as the agent (`:ARG0`) and the penalty as the patient (`:ARG1`). This semantic grounding allows computational systems to evaluate legal statements based on their core logical assertions rather than surface grammar.

#### 2.5.2.1 Semantic Graph Representations and Role Labeling

A principal advantage of Abstract Meaning Representation in statutory analysis is its ability to isolate predicate-argument structures and deontic operators within complex legislative commands. Statutory provisions rely heavily on deontic modalities—such as obligations, permissions, and prohibitions—to govern citizen behavior and administrative authority [219]. Standard grammar parsers often conflate these modal verbs with general descriptive predicates; this conflation obscures the primary regulatory command. In contrast, AMR maps deontic operators directly to core predicates, explicitly identifying whether an action is mandatory, permissible, or prohibited [232]. This structural clarity allows systems to identify subtle policy collisions, such as when a local ordinance authorizes a commercial practice that a national statute expressly forbids.

Furthermore, AMR graphs facilitate semantic normalization across diverse legislative drafting styles. Because local ordinances are drafted by different local committees across multiple decades, identical regulatory concepts are phrased using disparate vocabulary and sentence structures. AMR normalizes these surface differences into standardized semantic frames, allowing different legislative phrasings to map to identical concept graphs [219]. Recent legal NLP benchmarks demonstrate that evaluating semantic graph similarity outperforms raw token comparisons when identifying core regulatory requirements [232]. This structural consistency positions AMR as a valuable framework for extracting precise semantic representations from dense legislative text.

#### 2.5.2.2 Sentence Boundary Constraints and Parser Degradation

Despite its structural precision, Abstract Meaning Representation faces severe operational limitations when applied to Philippine local legislation. The primary architectural constraint is AMR's restriction to single-sentence parsing [219]. In statutory drafting, preemption conflicts rarely manifest within a single isolated sentence; rather, normative contradictions typically emerge across distinct sections, such as a general mandate in one clause negated by an unprovided exception several paragraphs later. Because standard AMR parsers evaluate individual sentences independently, they cannot perform cross-sentence coreference resolution or integrate distant provisos; this boundary limitation creates significant logical blindspots in statutory conflict detection.

Compounding this sentence-level restriction is AMR's high sensitivity to complex statutory syntax and optical character recognition noise. Philippine statutory provisions, such as sections within Republic Act No. 7160, frequently exceed one hundred words, featuring deeply nested conditional clauses and multiple provisos within a single period [221]. When processing these complex syntactic structures alongside scanning artifacts or localized administrative phrasing, standard AMR parsers frequently fail; they produce fragmented or malformed logic graphs [224]. Without extensive domain-specific parser retraining tailored to Philippine legal drafting, AMR cannot provide a dependable, production-ready foundation for ex-ante local statutory auditing.

### 2.5.3 Statutory Citation Networks and Legislative Lineages

Beyond abstract sentence semantics, computational legal informatics models positive law through statutory citation networks and legislative lineage graphs. In democratic legal systems, statutory enactments do not exist in isolation; rather, they form an evolving web of cross-references and amendatory sequences governed by delegating authorities. Frameworks such as Akoma Ntoso [231] and LegalRuleML [158, 13] formalize these relational structures, establishing standardized metadata schemas for legislative drafting and normative rule representation. By mapping statutory relationships into explicit topological networks, researchers can trace how delegated authority flows from national parent codes down to local administrative bodies, as well as how legislative modifications propagate across successive legislative assemblies.

#### 2.5.3.1 Statutory Network Deployments and Operational Granularity

Empirical applications of graph architectures in legal informatics span diverse statutory ecosystems and public sector repositories. In the European Union, the European Legislation Identifier (ELI) within the EUR-Lex database establishes a multi-relational metadata network connecting thousands of Directives and Regulations through explicit legal relations, such as legislative amendments and domestic transpositions. Similarly, computational legal researchers examining the United States Code have constructed large-scale federal statutory networks across all 54 titles, modeling tens of thousands of federal statutory provisions to compute network centrality and analyze structural complexity across federal law [24, 113]. In judicial and statutory benchmark research, datasets such as LeCNet [93] and Chinese Legal Knowledge Graphs [128] evaluate Graph Neural Networks

(GNNs) across judicial judgments and positive law provisions, demonstrating that graph representations improve candidate retrieval by identifying semantically related precedents even in the absence of verbatim keyword overlap [21].

A central architectural consideration in constructing legal knowledge graphs is selecting the operational unit of legal granularity. In legal text modeling, researchers have evaluated coarse document-level representations against fine-grained token and provision-level alternatives. Modeling statutory collections exclusively at the document level obscures the operative commands of positive law, aggregating complex enactments spanning hundreds of diverse articles into single undifferentiated nodes. A document-level graph can indicate that two enactments reference each other, but remains fundamentally blind to which specific administrative fee or licensing standard was modified. Conversely, attempting to construct graphs at the individual token or word level generates tens of millions of sparse, polysemous edges that lack substantive legal meaning, creating intractable computational graphs without isolating regulatory obligations.

To balance computational tractability with normative precision, established legal knowledge engineering establishes the discrete statutory provision or section as the canonical network node [93, 103]. In positive law, an individual numbered section represents the fundamental, self-contained normative command of a legislative act. A section connects the regulated actor and governing deontic operator with the substantive regulatory command, while preserving accompanying administrative exceptions. Organizing statutory networks at provision-level granularity isolates operative legal rules, enabling structural graph algorithms to evaluate legislative compliance with high legal precision.

#### 2.5.3.2 Relational Edge Typologies and Extraction Workflows

The construction of a statutory knowledge graph from unstructured legislative corpora requires a systematic extraction and entity resolution workflow. Because legislative drafting manuals enforce standardized statutory formulas, automated pipelines connect legal provisions through a multi-stage procedural sequence. In the initial stage, entity normalization algorithms scan legal texts to map surface variations—such as "Republic Act No. 7160," "R.A. 7160," and "The Local Government Code of 1991"—to a single canonical entity identifier via standardized dictionary aliasing. Following normalization, high-precision regular expression patterns and structural parsers scan individual sections to extract explicit cross-references, instantiating directed vertical authority edges linking municipal provisions to governing national statutes.

Following citation resolution, the pipeline parses amendatory directives to establish horizontal legislative lineages under the doctrine of *lex posterior derogat legi priori*. Philippine legislative drafting manuals require statutory amendments to adopt standardized linguistic formulas, such as declaring that a specific prior section "is hereby amended to read as follows" or that an older enactment "is hereby repealed in its entirety." Automated relation extraction algorithms identify these explicit amendatory formulas, generating directed modification edges linking amendatory enactments back to their historical parent sections. Assigning enactment dates to network vertices enables the graph engine to track point-in-time statutory validity, determining whether a referenced provision

represents active positive law or an obsolete historical fragment. When scanning noise produces truncated numerical citations, the pipeline resolves candidates through dictionary matching against the national statutory gazette or isolates ambiguous fragments in diagnostic quarantine pools.

Unlike homogeneous graphs that contain a single edge type, statutory networks function as heterogeneous multi-relational graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. Within this multi-relational topology, edge relations $r \in \mathcal{R}$ represent distinct juridical relationships, including structural containment connecting acts to sections, alongside authority delegations and amendatory sequences that modify positive law. Formalizing these diverse legal connections into typed edges enables graph traversal algorithms to evaluate legislative authority paths without conflating structural references with substantive amendments.

#### 2.5.3.3 Neuro-Symbolic and Graph-Augmented Transformers

While topological networks capture formal statutory relationships, pure graph representations exhibit known limitations when detecting preemption conflicts across unlinked enactments. In local legislative practice, draft ordinances frequently collide with superior national statutes without explicitly citing the governing legislation; drafts rely instead on localized terminology or administrative phrasing. Under these conditions, deterministic graph traversal fails because no explicit citation edge connects the conflicting provisions; this absence leaves the network topology blind to the underlying normative contradiction.

To overcome this disconnection, contemporary legal NLP literature investigates neuro-symbolic and graph-augmented architectures that combine topological knowledge graphs with dense transformer models [128, 139]. In these hybrid frameworks, the statutory knowledge graph provides structured structural priors, such as mapping governing legislative hierarchies and pruning obsolete repealed enactments. Concurrently, deep transformer encoders evaluate continuous cross-attention across candidate text spans, resolving semantic equivalence and deontic collisions even in the complete absence of formal citation edges. This synthesis establishes a complementary division of labor: symbolic graph topology enforces structural authority and legislative lineage, while continuous neural inference resolves unstructured textual contradictions.

Within this neuro-symbolic interface, legal informatics literature explores two primary paradigms for integrating knowledge graph structure with transformer cross-encoders: *Prompt-Conditioned Metadata Conditioning* and *Graph-Guided Decision Thresholding*. Under prompt-conditioned metadata injection, structured graph node properties—such as jurisdictional level, governing macro-domain, and formal statutory titles—are serialized into standardized textual prefixes prepended directly to the input sequence (e.g., `[JURISDICTION: NATIONAL | DOMAIN: ...]` and `[JURISDICTION: LOCAL | DOMAIN: ...]`). Studies in legal document classification and natural language inference demonstrate that prepending structured metadata provides explicit contextual framing, enabling transformer self-attention layers to condition their token representations on institutional hierarchies without requiring custom model architectures [33, 128]. At the same time, research examining prompt formatting and representation stability [64, 5] observes that introducing structured prefix tokens alters cross-attention weight distri-

butions. When cross-encoders encounter novel prefix schemas that differ from their primary training data distribution, attention heads can exhibit sensitivity to contrasting lexical tokens in the prompt, highlighting a recognized methodological trade-off between contextual guidance and input representation consistency.

In parallel, an alternative line of research investigates decoupled, decision-level graph integration through *Topologically Calibrated Decision Thresholding* [139, 71]. Rather than modifying the input text string, this approach preserves the original verbatim statutory sequence for neural inference and uses relational graph metrics—such as shortest-path geodesic distance $d_G(u, v)$ and authority hierarchy $H(u, v)$—to inform the post-softmax decision boundary $\tau^*(u, v)$. In statutory knowledge graphs containing typed authority and domain links, the topological distance between two legal provisions reflects institutional proximity: enactments connected by direct citation links ($d_G = 1$) or common macro-domains ($d_G = 2$) operate within a shared regulatory framework, providing an empirical basis for dynamic threshold calibration. Literature exploring this decoupled paradigm notes that separating linguistic cross-attention from relational graph traversal allows language models to evaluate pure textual semantics while graph topology dynamically modulates decision thresholds according to relational proximity. These two paradigms represent distinct methodological approaches to neuro-symbolic integration, testing whether knowledge graph structure is more effectively captured within the model's linguistic input representation or along its operational decision boundary.

## 2.6 Deep Learning and Transformers

The application of Natural Language Processing (NLP) in the legal domain has shifted from rigid, rule-based systems and basic word-matching toward Deep Learning [113]. Legacy algorithms treat legal texts as simple character strings, which frequently fail to capture the dense and interconnected nature of statutory drafting. Deep Learning, conversely, utilizes multi-layer artificial neural networks to map out the actual semantic meaning of legal language. This allows systems to process complex syntax and perform the deductive reasoning required to spot conflicts. This section breaks down the Transformer architecture, explaining its underlying mechanics, computational challenges, and why it remains the most practical framework for auditing unstructured local archives.

### 2.6.1 Transformers and Self-Attention

A major breakthrough was the Transformer architecture, introduced by Vaswani et al. [227] in 2017. Prior neural sequence models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, processed text sequentially from left to right, one token at a time. This sequential dependency created severe computational bottlenecks during training and suffered from vanishing gradient dynamics over long sequences, wherein contextual information from earlier tokens attenuated as sequence length expanded, hindering the model's capacity to resolve dependencies across multi-clause statutory provisions.

Transformers eliminate this sequential bottleneck entirely through the *Self-*

*Attention* mechanism. Rather than reading words one by one, self-attention evaluates all tokens in an input sequence simultaneously in parallel. For every word in a provision, the mechanism computes a set of attention weights measuring how much every other word in the text contributes to its contextual meaning, regardless of their physical distance in the sentence. For example, in a provision stating that "Any licensee who operates without a valid health clearance shall be penalized," self-attention directly connects "licensee" to "penalized" across the intervening relative clause in a single computational step. Because it processes data in parallel, Transformers train substantially faster on modern hardware, directly enabling foundational language models. Modern iterations have introduced refinements like Disentangled Attention to represent token content and relative positional offsets separately, alongside Windowed Attention to process long-context legal documents efficiently. However, while generative Large Language Models (LLMs) excel at text summarization, they are prone to "hallucinations"—fabricating plausible but non-existent statutory citations or misinterpreting statutory cross-references [59]. To ensure absolute legal reliability, consistency auditing requires a discriminative approach: specifically classifying whether two statutory provisions stand in a relationship of Entailment, Neutrality, or Contradiction, rather than probabilistically generating unverified text.

### 2.6.2 Transformer Challenges

Despite their architectural versatility, deploying transformer models within the computational constraints of Philippine Local Government Units (LGUs) introduces substantial operational hurdles.

The foremost engineering bottleneck arises from quadratic memory scaling and context length limits. The standard self-attention mechanism scales quadratically ($O(n^2)$) with input sequence length $n$. Because computing pairwise attention scores requires constructing an explicit $n \times n$ attention matrix across all query-key token pairs, memory consumption scales quadratically with sequence length, such that doubling sequence length quadruples the memory footprint required for attention computation. Consequently, standard foundational architectures like BERT are constrained to a rigid 512-token context window [62], whereas Philippine ordinances frequently span multiple pages. While practitioners employ sliding-window techniques to segment documents into overlapping spans [8, 97], this segmentation severs the semantic dependency between general regulatory mandates and distant provisos [193]. Consequently, sequence truncation creates structural blind spots that can lead to false conflict classifications.

A secondary challenge involves morphological noise and domain adaptation limits. Pretrained transformers require massive corpora of clean text, and while Western legal models like Legal-BERT perform well on foreign statutory benchmarks [31], no dedicated foundation model exists for Philippine local legislation. Applying general tokenizers to local archives—saturated with scanning artifacts, archaic statutory phrases, and administrative Taglish—triggers severe token fragmentation, where domain-specific terms are broken into meaningless sub-tokens that degrade contextual comprehension.

Finally, edge deployment is heavily restricted by local government hardware realities. Executing deep logical inference across thousands of document pairs demands

substantial GPU compute and memory bandwidth [171]. Because most local government units operate standard desktop workstations with modest hardware specifications, deploying unoptimized neural models leads to system latency spikes or memory exhaustion; these resource constraints necessitate decoupled, multi-stage architectures optimized for local inference.

### 2.6.3 Suitability of Transformers

Notwithstanding these hardware limitations, Transformer models remain the most viable approach for Semantic Conflict Detection in local ordinances. General-domain language models typically struggle with legal text because common words frequently take on entirely different legal definitions. For instance, "party" refers to a litigant rather than a celebration, and "suit" signifies a legal action rather than clothing. Greco et al. [88] emphasize that this high degree of polysemy is exactly why domain-specific models are required. While older static systems assign a single, fixed meaning to a word [88], a Transformer dynamically adjusts its reading of a word based on the surrounding sentence context [31]. Chalkidis et al. [32] demonstrated that this contextual awareness allows a model to distinguish between amending a law and violating it—a distinction that keyword searches miss.

The transition to Deep Learning necessitates explicitly moving away from legacy methodologies. Boolean keyword searches fundamentally fail because legislative drafting vocabulary constantly evolves over the decades. Conversely, manually mapping thousands of unlinked Republic Acts and local ordinances into deterministic graph structures creates an intractable administrative burden. As Li et al. [128] observed, hand-crafting these legal knowledge graphs simply does not scale for volatile, unstructured local records. Ultimately, Transformers represent the ideal method: they possess the capacity to map latent semantic relationships in unstructured legal text without requiring an intractable amount of manual data entry.

## 2.7 Single-Stage and Multi-Stage Architectures

In mainstream Natural Language Processing (NLP), end-to-end, single-stage neural architectures are the conventional baseline for document classification, sentiment detection, and question answering. These systems funnel raw input sequences directly into a single neural network that outputs the final classification without intermediate filtering. However, in legal informatics, detecting statutory conflicts requires a level of deductive precision across vast corpora that a monolithic model cannot provide. Forcing a single architecture to simultaneously scan tens of thousands of statutory records and evaluate fine-grained deontic logic creates severe mathematical and operational trade-offs.

### 2.7.1 Single-Stage Monolithic Pipelines

The appeal of a single-stage neural architecture is its conceptual simplicity: a single model handles the entire input-output mapping without requiring intermediate data handoffs. In standard NLP tasks operating over small document collections, this approach minimizes

engineering complexity and eliminates pipeline latency bottlenecks. However, deploying a monolithic model to audit draft local legislation against expansive national repositories introduces severe operational barriers, inevitably sacrificing either candidate recall or deductive precision.

One common monolithic approach involves prompting generative Large Language Models to directly declare legal preemption. However, autoregressive language models operate as probabilistic token predictors rather than deterministic logical discriminators, rendering them vulnerable to legal hallucinations where non-existent statutory citations are fabricated [131]. Furthermore, their massive parameter sizes make offline deployment on standard local government hardware infeasible without heavy quantization, which severely degrades deductive reasoning fidelity in specialized statutory domains [60].

Conversely, relying exclusively on single-stage bi-encoders introduces severe semantic compression limits. In a bi-encoder (or dual-tower) architecture, the search query and the candidate statute are processed through separate neural networks into independent vector coordinates. Because the two texts never interact directly during encoding, the model can pre-compute and store embeddings for all 27,000+ laws in advance, enabling sub-second nearest-neighbor search across extensive archives [171, 35]. However, compressing an entire multi-clause provision into a single fixed-dimensional vector inevitably discards granular syntactic nuance [111, 217]. This architecture optimizes for broad topical similarity rather than logical consistency, causing bi-encoders to conflate permission and prohibition provisions that share identical topical vocabulary [116].

Furthermore, executing a single-stage cross-encoder across an entire statutory repository is computationally prohibitive. In contrast to bi-encoders, a cross-encoder concatenates the draft query and the candidate statute into a single unified sequence, allowing every token in the draft to interact directly with every token in the statute through full bidirectional cross-attention. While this joint comparison achieves high logical sensitivity, full cross-attention scales quadratically ($O(n^2)$) with input length. Comparing a newly proposed local draft against 25,432 national statutes simultaneously would require hundreds of thousands of complex forward passes, instantly exhausting workstation memory and causing intolerable system latency [18]. Forcing the model to process fragmented sliding windows across thousands of enactments severs inter-clause dependencies. This computational ceiling confirms that isolating a compact candidate shortlist via coarse retrieval is essential prior to executing fine-grained sequence classification.

### 2.7.2 Two-Stage Retrieve-then-Entail Architectures

To resolve the mathematical and computational limitations of monolithic architectures, contemporary legal informatics adopts decoupled, two-stage systems. In the retrieve-then-entail paradigm, the workflow is split into two specialized stages: an initial coarse Information Retrieval (IR) phase that scans the entire statutory corpus to extract a top-$k$ candidate shortlist, followed by a fine Natural Language Inference (NLI) phase that performs deep cross-attention on the candidate pairs [118, 79]. Prior investigations in legal contract verification and statutory entailment demonstrate that this decoupled two-tier design dramatically contracts the combinatorial search space of pairwise comparisons

while preserving high inferential accuracy on candidate provisions [118].

The primary engineering advantage of this decoupled architecture is its alignment with edge computing realities. By offloading broad corpus search to pre-computed vector indices that require minimal runtime memory, the computationally intensive $O(n^2)$ matrix operations of the cross-encoder are restricted exclusively to the top-$k$ candidate pairs. Khattab and Zaharia [115] established that this decoupled structure enables advanced neural language processing to execute locally on consumer-grade hardware without requiring distributed GPU clusters. Furthermore, decoupling retrieval from inference provides a pragmatic resolution to the vector compression trade-off. The initial retrieval stage accepts the loss of granular syntactic nuance to maximize search throughput across extensive statutory archives. In contrast, the subsequent inference stage bypasses vector pooling entirely, performing joint token-level cross-attention over uncompressed raw text spans to achieve high-precision deontic reasoning [102].

### 2.7.3 Comparative Architectural Trade-offs

Comparing single-stage and dual-stage architectures highlights a fundamental division of labor in statutory preemption analysis. A single-stage retriever maximizes search speed but remains blind to subtle modal negators, whereas a single-stage cross-encoder possesses high logical sensitivity but cannot scale beyond a few dozen provisions without memory failure. By chaining both mechanisms in a coarse-to-fine hierarchy, the system leverages the complementary strengths of each model family: Stage One absorbs corpus volume and context length, while Stage Two resolves logical nuance and morphological noise [135].

The empirical superiority of this decoupled design is reinforced by over a decade of findings from the Competition on Legal Information Extraction and Entailment (COLIEE). As Goebel et al. [79] document, COLIEE divides legal reasoning into statutory information retrieval (Task 3) and statutory textual entailment (Task 4). Year after year, top-performing teams reject single-stage monoliths in favor of pipelined approaches [6]. While early benchmarks in COLIEE 2021 established this separation, modern iterations through COLIEE 2025 [80] and COLIEE 2026 [81] confirm that decoupled retrieval and inference remain the gold standard. For statutory retrieval, hybrid dense-sparse systems achieve near-perfect candidate recall [152], while cross-encoder entailment systems reach over 95% accuracy by isolating granular reasoning from broad corpus retrieval [5, 220]. This sustained global validation confirms that separating Information Retrieval from Natural Language Inference is the established operational blueprint for resolving semantic conflict across statutory corpora.

## 2.8 Information Retrieval

Information Retrieval (IR) is the process of extracting relevant documents from large unstructured datasets based on a specific query. In legal informatics, IR acts as the foundational filtering mechanism. It serves as the "coarse" stage of a dual-pipeline system, isolating candidate statutes from thousands of irrelevant local records before any intensive logical analysis takes place.

### 2.8.1 Information Retrieval Challenges in Statutory Corpora

Applying information retrieval techniques to statutory collections introduces domain-specific hurdles that general-domain search engines are not engineered to handle. A primary challenge is severe lexical divergence and vocabulary mismatch across legislative eras [23]. In statutory law, statutes governing identical civic activities may be enacted decades apart, using drastically different terminology, archaic phrasing, or specialized administrative acronyms. When newly drafted local ordinances employ modern regulatory expressions, exact keyword matching fails to retrieve governing historical statutes; this omission creates dangerous blindspots in preemption review [225].

Furthermore, statutory corpora exhibit extreme length asymmetry and structural complexity that disrupt standard retrieval scoring. Statutory enactments frequently contain verbose, multi-clause provisions exceeding one thousand tokens, where general commands are qualified by distant provisos and statutory exceptions [33]. Standard passage retrievers struggle to balance term saturation across these extended provisions, frequently penalizing lengthy, comprehensive enactments. Compounding this structural challenge is administrative hierarchy noise: in national legal archives, subordinate issuances (such as executive orders, circulars, and departmental memos) outnumber primary legislative acts, frequently diluting candidate shortlists unless protected by statutory priority safeguards [85].

### 2.8.2 Relevance in Retrieval

To understand how computational systems extract documents, we must first examine the human cognitive process that defines relevance in the physical world. In cognitive science, relevance is not a static property but a dynamic relationship between information and a specific task. According to Relevance Theory, humans naturally filter information by seeking the greatest cognitive benefit for the least mental effort [202]. When a legal researcher checks for conflicts, they do not simply count matching words; they instinctively evaluate the "intent" and "applicability" of a rule based on their background knowledge of the legal system. This human-centric view suggests that a document is relevant only if it provides a new piece of information that changes the researcher's understanding of the legal landscape [190].

Beyond the cognitive baseline, the theoretical definition of relevance in legal information retrieval is anchored in the philosophical requirements of a functional legal system. According to the jurisprudence established by Lon Fuller in *The Morality of Law*, a primary requirement for the internal morality of law is consistency; a system that issues contradictory rules fails to be a legal system at all [75]. Within this philosophical framework, relevance is not simply about finding documents that talk about the same subject matter; it is about identifying provisions that share a "normative overlap." This means that a retrieved document is relevant if it occupies the same jurisdictional or behavioral space as a draft local ordinance, potentially creating a situation where a citizen cannot obey one law without violating another. Grounding the retrieval stage in this quest for consistency treats relevance as the systematic identification of potential normative collisions and threats to the rule of law [75].

However, traditional computational methods struggle to replicate this human judgment because they rely on a rigid mathematical definition of relevance. In standard deterministic systems, relevance is treated as a binary mathematical property based entirely on set intersection [23]. If we denote a candidate document as $D$ and a user query as $Q$, this lexical relevance is formally evaluated as a binary function $R(D, Q) \in \{0, 1\}$. Under this strict lexical definition, a document is deemed fully relevant (i.e. 1) if and only if it contains the exact sequence of characters requested by the user [225]. Consequently, this mechanism establishes a purely topical baseline, effectively grouping documents together simply because they share a surface-level vocabulary [23]. While this mathematical simplicity allows for rapid indexing across massive databases, it completely strips away the legal context required for complex legislative analysis.

This stark discrepancy between superficial lexical matching and actual legal consequence represents a major stumbling block for legacy search architectures. In statutory law, two documents discussing the exact same topic might be lexically relevant to one another, but they are only *legally* relevant if one provision actively modifies, permits, restricts, or preempts the other [225]. For instance, a retrieved document might use identical vocabulary but apply to a completely different jurisdiction or regulatory scope; this jurisdictional divergence renders it unhelpful for an ex-ante conflict check. Conversely, a document regulating the same civic behavior but using entirely different phrasing—such as substituting "motor vehicle" for "automobile"—receives a relevance score of zero in a strict Boolean system [23]. Therefore, to accurately audit local ordinances against superior law, statutory retrieval systems must move beyond binary, keyword-driven matching in favor of dense semantic representations capable of capturing actual jurisprudential relationships [116].

To resolve these limitations, modern computational pipelines fundamentally redefine relevance spatially rather than lexically. Instead of scanning a query as a raw string of text, dense retrieval architectures utilize Bi-Encoder models to compress entire legal provisions into high-dimensional mathematical coordinates, commonly referred to as dense vectors [171]. Under this advanced semantic framework, relevance is no longer determined by counting shared characters, but by calculating the geometric distance—such as cosine similarity—between these coordinate points [116]. If a newly drafted local ordinance and an existing national statute govern the same underlying civic behavior, the neural network learns to map their respective vectors closely together in the mathematical space, completely bypassing the vocabulary mismatch problem [239]. Consequently, the retrieval mechanism evaluates relevance as a continuous proximity score rather than a rigid binary output, successfully surfacing conceptually intertwined laws that basic keyword searches otherwise miss.

### 2.8.3 Long-Context Retrieval

Relying on standard BERT-based architectures introduces an artificial constraint on legal analysis: the 512-token limit. Because standard Self-Attention mechanisms scale quadratically with sequence length, processing long texts requires severe truncation. To get around this, researchers like Chalkidis et al. [33] use Sliding Window techniques to segment documents into overlapping chunks. However, applying this to Philippine ordinances—which

are historically wordy and often bury exceptions deep within 1,000-token provisions—disrupts statutory interpretation. It severs the substantive semantic link between a primary rule and its distant exception.

Consequently, the model operates with a logical blindspot. Beltagy et al. [18] note that this causes logical hallucinations, where a rule might be flagged as an absolute prohibition simply because its exempting clause was pushed into a separate, unconnected processing window. Furthermore, these systems must navigate the trade-off between Sparse Retrieval (e.g., BM25) for exact keyword matching, and Dense Retrieval for capturing conceptual meaning even without direct word overlap.

Resolving this fragmentation means moving beyond short sliding-window chunking where possible. The architectural solution requires an initial retrieval mechanism capable of native long-context retention. Recent Transformer models incorporate advanced positional encoding, such as Rotary Positional Embeddings (RoPE), alongside sparse or optimized attention matrices. Warner et al. [237] demonstrated that these upgrades extend context windows up to 8,192 tokens without triggering memory failures. Deploying this class of architecture allows the system to read entire local ordinances as cohesive semantic units, preserving the law's internal structure and eliminating the Sliding Window fallacy.

### 2.8.4 Retrieval and Fusion Strategies

In computational legal informatics, isolating candidate enactments from extensive statutory collections requires understanding the theoretical foundations, failure modes, and complementary trade-offs of diverse retrieval strategies. Recent findings from the COLIEE benchmarks and legislative consistency literature identify five core methodological dimensions governing coarse statutory search.

The foundational retrieval strategies in computational law encompass sparse probabilistic scoring and dense semantic embeddings. Each relevance scoring metric is to be computed as follows:

*Okapi BM25 Probabilistic Relevance Scoring Function:*

$$S_{\text{BM25}}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \cdot \frac{f(q, D) \cdot (k_1 + 1)}{f(q, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \tag{2.3}$$

Where $f(q, D)$ denotes term frequency of query token $q$ in document $D$, $|D|/\text{avgdl}$ normalizes for provision length, and parameters $k_1 = 1.5$ and $b = 0.75$ govern term saturation and length penalization within Robertson and Zaragoza's probabilistic relevance framework [176].

The Okapi BM25 formulation balances three core retrieval properties within a unified probabilistic relevance framework. First, Inverse Document Frequency ($\text{IDF}(q)$) assigns higher numerical weight to rare, discriminative terms (such as specific statutory numbers, formal administrative acronyms, or penal sanctions) while suppressing ubiquitous statutory stopwords that appear across all enactments. Second, Term Frequency Saturation ($k_1 = 1.5$) enforces sub-linear term weighting, reflecting diminishing marginal

relevance so that repeating a keyword multiple times does not disproportionately inflate provision relevance over texts with concise occurrences. Third, Document Length Normalization ($b = 0.75$) penalizes exceptionally verbose provisions, ensuring that lengthy multi-page legislative digests do not dominate search rankings simply because their word volume increases the incidental probability of keyword matches. In statutory retrieval competitions such as COLIEE Task 3, BM25 is an indispensable baseline (as demonstrated by Team AIIR [239]) because legislative drafting relies on exact statutory citations, formal administrative acronyms, and verbatim penal monetary amounts. However, exact lexical matching suffers from the vocabulary mismatch problem when local ordinances paraphrase statutory commands [23, 140].

*Dense Semantic Cosine Proximity Relevance Function:*

$$S_{\text{dense}}(D, Q) = \frac{\mathbf{e}_Q \cdot \mathbf{e}_D}{\|\mathbf{e}_Q\| \|\mathbf{e}_D\|} \tag{2.4}$$

Where $\mathbf{e}_Q$ and $\mathbf{e}_D$ represent pooled contextual embeddings generated by dense bi-encoders such as `all-MiniLM-L6-v2` [171]. Rather than evaluating token overlap, dense retrievers score relevance via cosine proximity in latent semantic space, capturing conceptual legal intent and retrieving governing statutes even when local drafts employ disparate phrasing.

In heterogeneous statutory databases spanning tens of thousands of enactments, researchers frequently investigate topic classification or domain partitioning to reduce search space complexity. However, empirical literature highlights the severe danger of hard categorical filtering in legal domains. Foundational statutory instruments, such as the Philippine Local Government Code of 1991 (RA 7160) or comprehensive penal codes, function as omnibus legislation that simultaneously govern local administration, taxation, public safety, and environmental sanitation. Hard filtering by a single predicted regulatory category discards the governing omnibus code whenever a query touches an adjacent regulatory area; this exclusion causes catastrophic candidate recall collapse. In contrast, information retrieval theory establishes soft domain priors [176]. Instead of binary candidate exclusion, soft priors apply a calibrated multiplicative score boost (e.g., +20%) to candidate documents matching the predicted topic or latent semantic cluster, preserving access to multi-domain statutory codes while gently elevating topically congruent provisions.

Because lexical and semantic retrievers exhibit orthogonal failure modes, leading computational law frameworks combine both streams through hybrid fusion. In score-level linear combination, popularized by Team JNLP in the COLIEE competitions [79], weighted interpolation combines normalized sparse and dense scores according to $S_{\text{hybrid}}(D, Q) = \alpha \hat{S}_{\text{dense}}(D, Q) + (1 - \alpha) \hat{S}_{\text{BM25}}(D, Q)$, where $\alpha \in [0, 1]$ balances semantic and lexical bias. However, because BM25 scores are unbounded while cosine similarities span $[-1, 1]$, linear combination requires min-max score normalization; this interpolation introduces sensitivity across heterogeneous query lengths. To eliminate score normalization artifacts, Cormack et al. [46] formulated Reciprocal Rank Fusion (RRF), defined as $\text{RRF}(D) = \sum_{m \in \mathcal{M}} (k_{\text{rrf}} + \text{rank}_m(D))^{-1}$, where $\mathcal{M} = \{\text{BM25}, \text{Dense}\}$ and $k_{\text{rrf}} = 60$.

The mathematical advantage of Reciprocal Rank Fusion lies in its scale-invariant rank aggregation. Because BM25 produces unbounded positive relevance scores while

dense bi-encoders output bounded cosine similarities within $[-1, 1]$, score-level linear combination requires sensitive calibration and min-max normalization. In contrast, RRF abstracts away disparate score distributions entirely by operating strictly on the discrete ordinal rank positions assigned to candidate provisions within each individual retrieval stream. The smoothing constant $k_{\text{rrf}} = 60$ stabilizes reciprocal scoring against extreme rank disparities while systematically rewarding candidates that achieve high concordance across both lexical keyword matching and dense vector projections.

Finally, statutory retrieval systems must respect constitutional hierarchies of authority. Anchored in the doctrine of legislative supremacy and the hierarchy of legal norms [114, 85], statutory systems operate under strict formal ordering. In national digital archives, subordinate administrative issuances (such as executive orders, administrative circulars, and departmental memos) frequently constitute nearly half of all indexed records. Because administrative issuances share broad governmental vocabulary with local drafts, unconstrained retrieval often allows routine circulars to crowd out superior national statutes. Literature in legal informatics emphasizes structural safeguards that prioritize primary legislative acts (Republic Acts) over subordinate executive issuances. Implementing a hierarchical statutory priority safeguard shields governing statutes from administrative noise, preventing dilution of candidate shortlists and significantly improving Mean Reciprocal Rank (MRR).

### 2.8.5 Retrieval Benchmarks

Selecting effective retrieval architectures from the extensive landscape of open-weight models requires objective empirical benchmarking. In contemporary legal information retrieval, candidate models are evaluated against standardized empirical leaderboards that assess zero-shot generalization across dense statutory text. The Benchmarking Information Retrieval (BEIR) suite evaluates a dense retriever's capacity for out-of-domain transfer across diverse, specialized datasets [217]. Because Philippine local legislation lacks extensive annotated training corpora, candidate retrievers must possess strong zero-shot generalization. Dominant performance on the BEIR leaderboard serves as a primary indicator of a model's capacity to map novel, localized legal terminology into a structured latent space without prior fine-tuning [217].

Expanding on BEIR, the Massive Text Embedding Benchmark (MTEB) establishes an international standard for ranking embedding systems across more than one hundred datasets and diverse linguistic tasks [147]. In information retrieval literature, candidate architectures are frequently filtered using MTEB's retrieval and clustering leaderboards. Models excelling on these benchmarks demonstrate an empirical capacity to cluster conceptually related doctrines and retrieve relevant provisions based on continuous semantic proximity. Complementing general embedding benchmarks, the annual COLIEE Task 3 evaluation provides a specialized testbed for statute law retrieval [80, 81]. Systems evaluated on Task 3 must extract precise governing articles from civil code collections in response to legal queries. Success in this competition requires balancing broad candidate recall with computational throughput, providing a proven testing foundation for leading statutory retrieval models [239, 79].

### 2.8.6 Candidate Retrieval Models

Surveying leading open-weight architectures documented across recent retrieval literature and standardized benchmarks reveals eight prominent bi-encoder models relevant to statutory candidate discovery. Among distilled edge baselines, Reimers and Gurevych [171] established the Sentence-BERT framework, mapping text into fixed-dimensional dense vector spaces optimized for cosine proximity. The `all-MiniLM-L6-v2` model compresses a 6-layer BERT architecture down to 22.7 million parameters through knowledge distillation to produce 384-dimensional dense vectors. Despite its ultra-compact footprint, empirical benchmarks demonstrate that it retains strong semantic clustering capabilities with sub-two-millisecond latency on consumer CPUs. Complementing this, Xiao et al. [241] introduced `bge-small-en-v1.5` (33.4M parameters), which optimizes contrastive representation learning for high rank sensitivity on MTEB. In public-sector edge deployment, these distilled bi-encoders provide high-throughput candidate filtering without requiring dedicated GPU accelerators.

Scaling bi-encoder capacity to standard 110M-parameter base dimensions enables more expressive semantic representations across general retrieval benchmarks. Xiao et al. [241] developed `bge-base-en-v1.5` (109M parameters, 768 dimensions), utilizing retrofitted contrastive pretraining and instruction-tuned query prefixes to excel across multi-domain retrieval leaderboards. Similarly, Wang et al. [235] introduced `e5-base-v2` (109M parameters), trained on over one billion weakly-supervised text pairs via contrastive objectives; it demonstrated high generalization on unseen technical and legal search tasks. Furthermore, Song et al. [200] developed MPNet, combining masked language modeling with permuted language pretraining to eliminate positional dependence, which serves as the foundational architecture for `all-mpnet-base-v2` (109M parameters).

Beyond general-domain models, legal-domain adapted encoders capture specialized statutory taxonomies. Chalkidis et al. [31] demonstrated that pretraining bidirectional encoders on specialized legal repositories enables dense representations to capture statutory taxonomy and regulatory jargon that general-domain embedders conflate. Mean-pooled sentence embeddings derived from `legal-bert-base-uncased` (110M parameters) map statutory provisions into vector spaces that reflect legislative hierarchy and administrative codification rather than colloquial word associations. This domain-specific pretraining provides a structured baseline to evaluate whether legal pre-training improves candidate provision ranking over general embeddings.

To overcome the quadratic attention complexity of traditional 512-token encoders, modern architectures incorporate native long-context and multilingual designs. Warner et al. [237] developed `ModernBERT-base` (149M parameters), which integrates FlashAttention-2, Rotary Positional Embeddings (RoPE), GeGLU activations, and sequence unpadding to natively process up to 8,192 tokens. In statutory retrieval, this allows direct encoding of verbose multi-paragraph enactments without destructive sliding-window chunking. Expanding to the multilingual domain, Chen et al. [35] introduced `bge-m3` (568M parameters, 1,024 dimensions), supporting 8,192-token contexts and multi-functionality across dense, sparse, and multi-vector representations. In statutory consistency checking, its multi-vector representations provide resilience across

code-switched local records and bilingual administrative issuances.

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Because lexical and semantic retrievers exhibit orthogonal failure modes, leading computational law frameworks combine both streams through hybrid fusion. In score-level linear combination, popularized by Team JNLP in the COLIEE competitions [79], weighted interpolation combines normalized sparse and dense scores according to $S_{\text{hybrid}}(D,Q) = \alpha\hat{S}_{\text{dense}}(D,Q) + (1-\alpha)\hat{S}_{\text{BM25}}(D,Q)$, where $\alpha \in [0,1]$ balances semantic and lexical bias. However, because BM25 scores are unbounded while cosine similarities span $[-1,1]$, linear combination requires min-max score normalization; this interpolation introduces sensitivity across heterogeneous query lengths. To eliminate score normalization artifacts, Cormack et al. [46] formulated Reciprocal Rank Fusion (RRF), defined as $\text{RRF}(D) = \sum_{m\in\mathcal{M}}(k_{\text{rrf}} + \text{rank}_m(D))^{-1}$, where $\mathcal{M} = \{\text{BM25}, \text{Dense}\}$ and $k_{\text{rrf}} = 60$. Because RRF operates strictly on ordinal ranking positions, it provides scale invariance, eliminates distribution sensitivity, and ensures that provisions performing well across both lexical and semantic modalities are prioritized.

Finally, statutory retrieval systems must respect constitutional hierarchies of authority. Anchored in the doctrine of legislative supremacy and the hierarchy of legal norms [114, 85], statutory systems operate under strict formal ordering. In national digital archives, subordinate administrative issuances (such as executive orders, administrative circulars, and departmental memos) frequently constitute nearly half of all indexed records. Because administrative issuances share broad governmental vocabulary with local drafts, unconstrained retrieval often allows routine circulars to crowd out superior national statutes. Literature in legal informatics emphasizes structural safeguards that prioritize primary legislative acts (Republic Acts) over subordinate executive issuances. Implementing a hierarchical statutory priority safeguard shields governing statutes from administrative noise, preventing dilution of candidate shortlists and significantly improving Mean Reciprocal Rank (MRR).

## 2.9 Natural Language Processing

Natural Language Processing (NLP) provides computational techniques to analyze and represent text across linguistic levels toward human-like comprehension [130]. As Jurafsky and Martin [108] explain, it bridges human language and machine logic, enabling algorithms to parse and interpret unstructured text. In the legal field, NLP shifts document auditing from slow manual reviews to scalable, automated systems. However, applying standard NLP to law requires more than basic grammar checks; it demands systems that can navigate dense sentence structures, deontic operators (obligations, permissions, prohibitions), and the highly specific vocabulary found in statutory law.

Early Legal NLP relied on Symbolic Artificial Intelligence (AI) [113]. These systems used manually built ontologies and strict Boolean logic to extract legal information. However, because local legislative drafting language changes constantly over decades, these exact-match frameworks proved far too brittle. Contemporary NLP architectures solve this by moving to continuous vector representations. By mapping textual provisions into high-dimensional semantic spaces, modern Transformers bypass superficial vocabulary mismatches to mathematically quantify the deeper concepts connecting different legal terms. This allows the system to act as an active diagnostic engine that maps legal conflicts, rather than just a passive search index.

This section breaks down how NLP operates within legislative analysis. It covers the systemic challenges of processing localized legal text, the structural need for Natural Language Inference (NLI), the benchmarks used for model selection, and the specific architectures built for these deductive tasks.

### 2.9.1 NLP Challenges

Deploying NLP frameworks against local archives quickly exposes the mechanical limits of standard Language Models. A primary hurdle is the pronounced Global North bias in pre-training data and legal benchmarking. As Ranathunga et al. [169] point out, the vast majority of Legal NLP benchmarks, like LexGLUE [33] and CaseHOLD, are trained exclusively on pristine, digitally native Western legal text. These highly optimized models degrade when applied to Philippine Local Government Unit (LGU) archives, which are filled with scanned PDFs suffering from Optical Character Recognition (OCR) errors, archaic American colonial phrasing, untranslated Spanish legal terms, and local Taglish. Rust et al. [182] demonstrated that standard WordPiece tokenizers shatter unrecognized words into sub-tokens; this fragmentation destroys a model's ability to infer legal intent. Addressing localized morphological noise requires deploying models equipped with massive Byte-Pair Encoding (BPE) vocabularies that absorb linguistic variants without triggering out-of-vocabulary errors, paired with Disentangled Attention architectures that separate word content from corrupted positions [95].

Beyond geographic and morphological bias, traditional NLP methods suffer from structural shortcomings in lexical similarity. Early NLP systems excel at identifying topical overlap by measuring surface-level token intersection using algorithms like TF-IDF or shallow word embeddings. However, Bowman et al. [25] explain that pure lexical similar-

ity is fundamentally unsuited for detecting legal contradictions because lexical metrics are entirely blind to deontic logic and syntactic negation. For example, an ordinance stating that the city permits an activity and a statute stating that the state strictly prohibits it share near-perfect lexical similarity scores. A standard similarity model clusters them as highly related, completely missing the absolute policy collision occurring beneath the shared vocabulary. Consequently, detecting normative conflicts requires advancing from lexical proximity to directional logical inference.

### 2.9.2 From NLP to NLI

Historically, checking legislative consistency relied on Symbolic NLP (hand-written rules) and, later, Statistical NLP. In Statistical NLP, models learned probabilistic weights for specific inputs, like Term Frequency-Inverse Document Frequency (TF-IDF) or N-grams [113]. These models depended heavily on human-engineered features to define grammar rules and categories. While they allowed for soft, probabilistic decisions, they consistently struggled with the deep lexical and semantic ambiguities inherent in legal drafting.

The field largely transitioned to Neural NLP models in the mid-2010s [166]. Unlike statistical systems that treat features as separate entities, neural networks operate end-to-end; they map raw text directly to output predictions through hidden layers. These architectures capture deeper linguistic patterns than traditional statistical methods by learning complex relationships during optimization. However, while machine learning excels at finding patterns, it can overly mimic its training data, making domain-specific pre-training essential for high performance in rigorous legal tasks.

Within modern neural approaches, Natural Language Inference (NLI)—originally formalized under the Recognizing Textual Entailment (RTE) challenges initiated by Dagan et al. [52]—has emerged as the definitive benchmark for evaluating whether an algorithm can perform genuine logical deduction. Yang et al. [243] define NLI as determining the directional logical relationship between two structured text sequences: a governing *Premise* and a candidate *Hypothesis*. In statutory preemption auditing, this formulation maps directly to constitutional hierarchy: the Premise represents the established positive legal standard (the superior national statute), while the Hypothesis represents the newly proposed local claim (the draft city ordinance provision).

The neural network must classify the relationship into one of three mutually exclusive logical states. In an Entailment relationship, the draft ordinance directly carries out, derives its authority from, or compliantly adheres to the governing national statute. In a Contradiction relationship, the draft ordinance directly collides with the national statute (for example, permitting what national law prohibits, or exceeding statutory penalty ceilings under the *Magtajas* doctrine). Finally, in a Neutral relationship, the two provisions govern complementary or non-overlapping subject matter without normative conflict. This directional formulation exposes the fatal flaw of earlier statistical NLP, which mistakenly equated high word overlap with logical entailment [243]. Modern neural approaches fix this by capturing deep compositional semantics. As Le Hong et al. [125] point out, this allows modern models to resolve complex linguistic phenomena—like coreferences and modality—that rigid rule-based systems historically failed to grasp.

### 2.9.3 NLI Benchmarks

Identifying effective Cross-Encoder architectures for the inference stage requires isolating models with robust, pre-trained deductive logic. In natural language processing, research literature relies on established benchmarks to evaluate candidate reasoning models. Wang et al. introduced the General Language Understanding Evaluation (GLUE) [234] and its successor SuperGLUE [233] as foundational baselines for model reasoning. These frameworks evaluate a Transformer's ability to handle diverse linguistic challenges, such as Recognizing Textual Entailment (RTE) and resolving complex coreferences. While high performance on SuperGLUE confirms baseline deductive capacity, it does not independently guarantee success in the legal domain due to the specialized vocabulary shifts present in statutory drafting.

To evaluate specialized legal logic, legal NLP research relies on the Legal General Language Understanding Evaluation (LexGLUE) benchmark. Chalkidis et al. [33] designed LexGLUE specifically to test a model's ability to navigate legalese, where common words assume highly context-dependent meanings. High performance on LexGLUE entailment tasks indicates that an architecture possesses the foundational deductive capacity required to navigate complex statutory drafting and regulatory provisions.

Complementing domain-specific suites, contemporary Hugging Face NLI leaderboards evaluate models on diverse, adversarial textual entailment tasks aggregating MultiNLI (MNLI), FEVER, Adversarial NLI (ANLI R1–R3) [153], and WANLI [132]. Benchmarking candidate models across these suites tests their resilience against spurious lexical heuristics, premise-hypothesis artifacts, and negation traps. High-performing architectures on these leaderboards, such as the pre-aligned DeBERTa-v3 variants curated by Laurer et al. [124], demonstrate strong zero-shot deductive capacity and establish leading candidate foundations for statutory conflict detection. Finally, within legal informatics, the definitive international competition benchmark for statutory inference is Task 4 of the Competition on Legal Information Extraction and Entailment (COLIEE) [167, 80, 81]. In Task 4, systems evaluate whether a natural language query is legally entailed by or contradicts a governing civil code article. Evaluating architectures against COLIEE Task 4 methodologies provides direct empirical evidence of a model's capacity to navigate statutory conditions, exceptions, and deontic modalities.

### 2.9.4 Candidate NLI Models

Evaluating deontic relationships across statutory provisions demands specialized inductive biases across candidate sequence encoders. Standard language models trained exclusively on general corpora frequently fail to capture the specialized semantics of legal drafting, where everyday terms carry specialized juridical meanings. To address this vocabulary divergence, researchers develop domain-adapted encoders initialized with custom legal vocabularies or continuously pretrained on expansive legal repositories. Chalkidis et al. [31] introduced `LEGAL-BERT`, pretrained on diverse English legislation, court judgments, and contracts; this domain pretraining demonstrated superior term sensitivity across legal classification tasks. Henderson et al. [96] expanded this approach with `PoL-BERT` (Pile of Law), pretraining encoders on a 256 GB corpus encompassing administrative regulations,

municipal codes, and statutory provisions. Complementary efforts by Paul et al. [162] (`InLegalBERT`) and Zheng et al. [247] (`CaseLaw-BERT`) investigate the transferability of common-law statutory pretraining, while Xiao et al. [240] evaluate long-document legal encoders. In literature, this family provides a critical testbed for assessing whether domain pretraining provides intrinsic protection against legal misinterpretation.

A pervasive vulnerability of unadapted sequence encoders on deontic tasks is *affirmative entailment bias*, where models exhibit an innate psychological tendency to default toward predicting Entailment or Neutral when reading complex clauses containing subtle prohibitions [5, 220]. In standard language corpora, true logical contradictions are exceedingly rare; text passages typically elaborate, confirm, or describe unrelated facts. Consequently, unadapted neural models learn a strong statistical prior that pairs generally agree. Devlin et al. [62] established `bert-base-uncased` (110M parameters) with masked language modeling, serving as the canonical supervised baseline against which improvements are measured. To overcome heuristic reliance on lexical overlap and break affirmative bias, contemporary literature explores models pre-aligned on multi-task natural language inference datasets (such as MNLI, FEVER, ANLI, and WANLI). Laurer et al. [124] demonstrate that initializing sequence classifiers from warm NLI checkpoints equips models with robust directional contradiction sensitivity before domain adaptation. Structurally, He et al. [95] developed `DeBERTa-v3`, introducing Disentangled Attention to represent word content and relative positions with separate vectors alongside adversarial replaced token generation. In parallel, Clark et al. [39] developed `ELECTRA` with Replaced Token Detection to optimize contextual representations across all input positions. These architectures excel at distinguishing subtle deontic shifts that standard unidirectional or masked encoders frequently miss.

Statutory drafting is inherently verbose, characterized by complex preamble clauses, extensive definitions, and multi-sentence penal qualifications. Traditional transformer encoders are constrained by quadratic self-attention complexity, imposing a rigid 512-token ceiling that necessitates destructive document chunking or sliding window heuristics [33]. Early long-context architectures attempted to resolve this constraint using localized or sparse attention approximations, such as the sliding-window attention in `Longformer` [18] and the block-sparse random attention in `BigBird` [245]. However, empirical legal informatics evaluations note that sparse attention patterns can dilute or sever semantic dependencies between distant provisos and governing commands. To overcome these limitations, modern architectures integrate architectural optimizations that support extended contexts natively. Warner et al. [237] introduced `ModernBERT`, an 8,192-token encoder incorporating FlashAttention-2, Rotary Positional Embeddings (RoPE), GeGLU activation functions, and sequence unpadding. Complementing this, Nussbaum et al. [156] developed `Nomic-BERT` to extend context retention. These modern long-context designs enable joint encoding of complete multi-sentence statutory provisions without sparse approximation artifacts.

Unlike bi-encoders that process queries and documents independently into pooled vector coordinates, cross-encoders execute full bidirectional self-attention across the concatenated pair simultaneously. In retrieval and inference benchmarks, joint cross-attention captures deep inter-token interactions that vector similarity cannot represent. In COLIEE

statute retrieval benchmarks, researchers deploy multilingual cross-encoder rerankers, notably `BGE-Reranker-v2-m3` [241], to score pairwise relevance across multi-thousand token contexts. Similarly, sequence classifiers adapted from MS-MARCO reranking frameworks employ deep cross-attentive scoring to evaluate pair compatibility. In multi-stage neural retrieval paradigms, Nogueira and Cho [154] demonstrated that candidate passages retrieved by coarse search can be evaluated in parallel batches via cross-attentive models (such as MonoBERT), aggregating candidate pairs into structured mini-batches to utilize hardware vectorization. In literature, native cross-encoders represent the conceptual bridge between Stage 1 candidate filtering and Stage 2 logical validation.

Evaluating candidate sequence pairs through multi-layer cross-encoders introduces computational throughput trade-offs during live document review. In standard transformer implementations, mini-batch tensors are padded with dummy zero tokens to match a static global sequence limit, such as 256 or 512 tokens [62]. Because self-attention complexity scales quadratically ($O(L^2)$) with input sequence length $L$, allocating static global padding causes significant floating-point operations to be expended on empty padding positions when input clauses are short. In neural language processing literature, dynamic sequence padding addresses this inefficiency by clamping batch tensor dimensions strictly to the length of the longest clause within each mini-batch, substantially reducing redundant self-attention calculations while producing mathematically identical hidden states. Furthermore, in multi-document and multi-clause document auditing, batch collation literature establishes that flattening candidate pairs across multiple queries into a single continuous tensor pass amortizes runtime call overhead and utilizes multi-threaded central processing unit vectorization.

In post-colonial statutory jurisdictions like the Philippines, statutory archives exhibit unique sociolinguistic complexities. While statutory enactments are drafted in legal English, local ordinances and administrative records frequently feature localized terminology, regional administrative phrasing, Latin legal maxims, and English-Tagalog code-switching. Conneau et al. [44] introduced `XLM-RoBERTa`, trained across 100 languages; this model established strong cross-lingual transfer on semantic benchmarks. He et al. [95] extended disentangled attention to the multilingual domain with `mDeBERTa-v3`; this design provides resilience to mixed-language syntax. Furthermore, Cruz and Cheng [51] developed Tagalog-adapted transformers (`roberta-tagalog-base`) to capture Philippine syntactic structures. In statutory consistency checking, multilingual encoders evaluate whether localized linguistic variances can be processed without catastrophic comprehension loss.

Deploying deep learning in public-sector local government environments requires strict adherence to available edge computing infrastructure. Because local government units typically operate without high-performance computing clusters or enterprise cloud APIs, deploying billion-parameter language models is computationally unfeasible. To establish realistic performance baselines under constrained hardware, the literature explores knowledge distillation and parameter sharing. Sanh et al. [189] introduced `DistilBERT`, compressing transformer depth to reduce model size by 40% while retaining 97% of BERT's language comprehension capabilities and running 60% faster. Lan et al. [120] introduced `ALBERT`, utilizing factorized embedding parameterization and cross-layer

parameter sharing to reduce parameter size down to 12 million in its base configuration (an 89% reduction compared to BERT-base). In dense semantic representation, Reimers and Gurevych [171] established the Sentence-BERT (SBERT) siamese embedding framework, providing the structural foundation for compact distilled bi-encoders such as `all-MiniLM-L6-v2` (22.7M parameters) to deliver sub-two-millisecond latency on commodity CPU hardware. Evaluating distilled edge models allows researchers to quantify the exact accuracy penalty incurred when prioritizing sub-second CPU latency over multi-layer neural capacity.

## 2.10 Ground Truth

In supervised machine learning, a Ground Truth (or Gold Standard) dataset provides the verified target labels used to optimize objective functions and evaluate model parameters. Formally, the ground truth establishes the empirical reference standard against which supervised learning models optimize loss functions [82]. During training, the optimization algorithm computes empirical risk between model predictions and verified target labels; it derives gradients via backpropagation to iteratively adjust model weights. In domain-specific legal NLP, expert-adjudicated ground truth is essential to prevent models from learning superficial statistical artifacts or spurious annotation heuristics that fail to generalize to actual statutory logic.

In model evaluation methodology, the ground truth provides the objective benchmark for measuring generalization capacity across held-out partitions. Datasets are partitioned into independent training, validation, and test subsets. The test partition remains strictly unobserved during hyperparameter tuning and model checkpoint selection; it constitutes a clean out-of-sample evaluation set. Evaluating the model on this held-out data enables objective computation of classification metrics—including precision, recall, Macro $F_1$, and class-specific $F_2$-scores—validating whether the architecture generalizes reliably to unseen legislative drafts [25].

### 2.10.1 Ground Truth in NLI

In Natural Language Inference (NLI), ground truth datasets define the logical relationship between two pieces of text. Annotators read a source text (the Premise) and a claim (the Hypothesis), then categorize their relationship into one of three distinct labels. If the premise guarantees the hypothesis is true, it is labeled as Entailment. If they contradict each other, it is labeled as Contradiction. If there is not enough information to prove either, the pair is labeled Neutral [243].

Most traditional NLI datasets, such as the widely used SNLI (Stanford Natural Language Inference) corpus, are built using crowdsourced workers who write simple variations of everyday sentences [25]. While this approach is cheap and easily generates hundreds of thousands of training pairs, the resulting datasets are incredibly basic and often embed unintended annotation artifacts [91]. They usually feature short, single-sentence premises with obvious logical connections. Training a model exclusively on these simple datasets leaves it entirely unprepared for the dense, multi-layered logic required in

specialized fields.

### 2.10.2 Annotation and Reliability Metrics

Turning raw text into a useful NLI dataset requires strict guidelines to minimize human bias and prevent annotators from relying on subjective legal interpretations. In standard NLI, guidelines dictate simple factual agreement. However, legal conflict detection requires evaluating normative relationships based on deontic logic. Drawing from the frameworks established by Agotnes et al. [3] and Li et al. [129], the annotation guidelines strictly instruct experts to map the text into obligations (must do), permissions (may do), and prohibitions (cannot do). A "Contradiction" is only flagged if the local ordinance mandates an action the national statute prohibits, or prohibits an action the national statute explicitly permits. If a local ordinance imposes an administrative fee on an activity strictly within devolved municipal regulatory competence, the guidelines dictate a "Neutral" classification; however, if the local measure attempts to condition, restrict, or penalize an activity already authorized or licensed under superior national legislation (such as imposing an unauthorized local permit or additional fee on a nationally licensed activity), the provision triggers a "Contradiction" under the doctrine of statutory preemption. This refined deontic framework ensures that the dataset measures actual legal preemption rather than superficial topical friction.

To evaluate the effectiveness of these strict guidelines, researchers must utilize Inter-Rater Reliability (IRR) metrics [144] rather than raw percentage agreements. Percentage agreement is inadequate for evaluating multi-annotator consistency because it fails to correct for expected chance concordance ($p_e$). Under a three-class discrete annotation schema ($\mathcal{C} = 3$), independent stochastic labeling yields a baseline expected agreement of approximately 33.3% purely through chance alignment. Reporting unadjusted percentage concordance confounds random stochastic concordance with genuine evaluative consensus; this metric yields an inflated assessment of inter-annotator reliability.

The two primary statistical formulas used to subtract this random chance are Cohen's Kappa [40] and Fleiss' Kappa [72]. Each inter-rater reliability metric is to be computed as follows:

*Cohen's Pairwise Kappa Reliability Metric ($\kappa$):*

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2.5}$$

Where $p_o$ represents the relative observed agreement between the two individual raters, and $p_e$ represents the hypothetical probability of chance agreement. Cohen's Kappa evaluates agreement between exactly two raters.

*Fleiss' Generalized Multi-Rater Kappa Reliability Metric ($\kappa$):*

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \tag{2.6}$$

Where $\bar{P}$ represents the mean observed agreement across all pairs of raters, while $\bar{P}_e$ represents the expected chance agreement across the entire pool. Fleiss' Kappa generalizes

Cohen's formula to accommodate three or more domain experts establishing a majority consensus.

The resulting $\kappa$ score is interpreted using the universally accepted statistical matrix established by Landis and Koch [122]:

Table 2.1: Interpretation Ranges for Fleiss' Kappa Reliability Metric

| **Kappa ($\kappa$) Score** | **Interpretation** |
|---|---|
| $< 0.00$ | Poor agreement |
| $0.01 - 0.20$ | Slight agreement |
| $0.21 - 0.40$ | Fair agreement |
| $0.41 - 0.60$ | Moderate agreement |
| $0.61 - 0.80$ | Substantial agreement |
| $0.81 - 1.00$ | Almost perfect agreement |

Table 2.1 outlines the standard categorical intervals for evaluating inter-rater agreement across multiple independent judges. It consists of two columns mapping numerical $\kappa$ ranges to their corresponding qualitative interpretations. This scale establishes an objective statistical benchmark for assessing annotation consensus. This statistical scale grounds the evaluation of legal researcher annotation consensus within an authoritative psychometric standard. As the score progresses from negative values toward unity, it reflects a transition from chance-level alignment to robust consensus across evaluators. For this research, the matrix establishes the operational target of $\kappa \geq 0.61$ ("Substantial Agreement"), ensuring that the resulting ground truth benchmark represents dependable legal consistency rather than subjective individual interpretation.

While objective tasks routinely achieve $\kappa$ scores above 0.90, such an expectation is mathematically unrealistic in the legal domain. Bowman et al. [25], in their foundational study for the Stanford Natural Language Inference (SNLI) corpus, established that evaluating semantic relationships is inherently subjective. Their benchmark study hit an approximate performance ceiling with an overall Fleiss' Kappa of 0.70. They explicitly demonstrated that achieving near-perfect agreement ($> 0.80$) in textual inference usually indicates overly simplistic or fabricated test data.

#### 2.10.2.1 Sub-Threshold Reliability

While achieving a "Substantial" Inter-Rater Reliability score ($\kappa \geq 0.61$) validates the clarity of the annotation guidelines, failing to meet this threshold does not inherently invalidate the dataset or halt the experimental pipeline. Klebanov and Beigman [117] establish that a sub-threshold $\kappa$ score typically indicates systematic ambiguity within the text itself—categorized as "hard" instances where seasoned legal professionals can reasonably disagree on statutory scope—rather than carelessness by the annotators. Their research demonstrates that it is still entirely possible to extract a reliable, low-noise subset from the instances where annotators do achieve consensus.

When researchers encounter this sub-threshold reliability, the mitigation strategy

depends entirely on the size of the dataset. For large-scale studies, including the disaster-tweet classification by Ablazo [1], the standard procedure is data omission. Beigman et al. [117] explicitly advocate for filtering out difficult, borderline cases from the gold standard, arguing that evaluating algorithms on inherently ambiguous text introduces stochastic label noise that degrades model optimization. In massive general-domain corpora, items that fail to achieve majority consensus are omitted because the extensive volume of unambiguous data absorbs the sample reduction without compromising statistical power.

However, applying this standard omission strategy to a specialized, few-shot legal dataset mathematically destroys the test set's Minimum Detectable Effect. Furthermore, in local legislation, these "hard" cases often represent the exact jurisdictional friction the logic engine needs to learn. To preserve dataset volume without introducing benchmarking noise, researchers in specialized domains employ Expert Adjudication rather than omission. Gao et al. [78] establish that in these scenarios, a hierarchical adjudication procedure is used where a senior domain expert reviews the conflicting annotations. The senior expert resolves the ambiguity based on governing jurisprudential principles and manually establishes the Gold Standard label, treating the initial low $\kappa$ metric as a measure of text difficulty rather than a rigid experimental blocker.

### 2.10.3 Ground Truth in Legal NLP

When applying NLI to the legal domain, the standard approach of comparing short, isolated sentences quickly breaks down. Statutory phrasing is notoriously complex, filled with conditional clauses, exceptions, and domain-specific vocabulary. Consequently, a hypothesis in a legal context rarely aligns perfectly with a single sentence in the premise. Instead, proving an entailment or contradiction often requires the model to connect evidence scattered across multiple paragraphs or pages.

To address this, researchers have developed specialized legal datasets that reflect real-world document structures. For instance, the Competition on Legal Information Extraction/Entailment (COLIEE) dataset constructs its ground truth by pairing civil code statutes with legal bar exam queries, requiring models to identify specific legal provisions before determining entailment [167]. Similarly, the ContractNLI dataset by Koreeda and Manning [118] engineers its gold standard around non-disclosure agreements. Rather than giving the model pre-cut sentence pairs, ContractNLI forces the system to evaluate hypotheses against entire legal documents, requiring annotators to both classify the logical relationship and extract the specific supporting text span.

Furthermore, unlike general-domain datasets crowdsourced through platforms like Amazon Mechanical Turk, legal ground truth relies heavily on domain experts. Constructing datasets like CaseHOLD [247], which evaluates legal holding statements, requires law students and legal professionals to ensure the labels accurately reflect actual jurisprudence. The push towards comprehensive legal benchmarks, such as LexGLUE [33], demonstrates that evaluating legal NLP models demands rigorously structured, expert-annotated data rather than automated or amateur labeling.

This difficulty is magnified when dealing with local legislation. While national or

federal laws generally follow standardized drafting conventions, local ordinance texts are highly heterogeneous, exhibiting significant variations in terminology, structure, and overall organization across different jurisdictions [27]. Because of this localized variance, local legal documents have historically been neglected in NLP research, hampered primarily by a severe lack of annotated ground truth datasets [28]. Standard legal corpora fail to capture the specific legislative templates, localized governance terminology, or idiosyncratic cross-referencing styles used in local legislation, such as Philippine city ordinances. Without existing resources that reflect these structural and linguistic complexities, evaluating models on local legislation requires building highly specialized, expert-annotated datasets from the ground up.

A critical architectural decision in dataset construction is determining the exact granularity of the extracted legal premise. While the hypothesis (the local draft) is evaluated as a multi-sentence span to capture context, the premise (the national statute) must be strictly isolated at the sentence level. In Philippine statutory drafting, a single paragraph frequently contains a general rule, multiple conditional modifiers, and distinct exceptions. Holzenberger et al. [98] demonstrate that feeding an entire statutory paragraph into an NLI model as a premise dilutes the model's attention mechanism; under this dilution, the model loses track of specific variable constraints. By forcing human annotators to isolate the national rule down to its exact sentence boundary, the resulting Ground Truth prevents the candidate logic engines from getting distracted by surrounding, non-governing text, ensuring the classification mathematically targets the exact point of semantic collision.

The difficulty of dataset construction is magnified by the scarcity of legal domain experts available for annotation. As Zhong et al. [249] explicitly note, the complex terminology required for legal texts makes manual labeling highly time-consuming and difficult to scale, as it strictly requires domain experts rather than crowdsourced workers. Because constructing massive datasets for localized city law is practically unfeasible, researchers must rely on constrained, highly curated Gold Standards. To compensate for this limited data volume, the current NLP standard mandates treating legal conflict detection as a few-shot learning scenario. Gao et al. [77] validate this approach, proving that transformer models can be successfully fine-tuned on specialized datasets containing hundreds of examples, provided the evaluation framework rigorously controls for overfitting.

### 2.10.4 Multi-Sentence and Cross-Document Limits

Traditional Natural Language Inference (NLI) evaluates text using a strict 1-to-1 pairing, comparing a single premise sentence directly against a single hypothesis sentence. However, this highly granular approach fundamentally breaks down in the legal domain due to two specific edge cases. First, intra-document context: a primary rule in one section of an ordinance is frequently modified or negated by an exception buried several paragraphs later (the "1v2" or multi-sentence problem). Second, cross-document dependencies: a contradiction may only emerge when synthesizing provisions from two entirely separate statutes simultaneously.

To address the first edge case, modern legal frameworks utilize Asymmetric Span-

Level Pairing. Yin et al. [244] critique traditional sentence-level datasets, proving that capturing contextual exceptions requires evaluating multi-sentence blocks rather than isolated lines. As demonstrated by Koreeda and Manning in the ContractNLI dataset [118], the premise remains a single, focused legal rule, while the corresponding hypothesis is extracted as a multi-sentence chunk. This asymmetric structure allows the model's attention mechanism to accurately evaluate a primary rule against a distant exception contained within the same local provision.

While asymmetric spans successfully capture intra-document context, resolving the second edge case—cross-document multi-hop reasoning—remains computationally prohibitive. Identifying semantic preemption that spans multiple independent laws represents the current research frontier in legal natural language processing and remains unverified for local governance applications. Recent frameworks have attempted to solve this, but they introduce severe architectural overhead. For instance, Lu et al. [136] address cross-document retrieval by programmatically mapping Entity Graphs to mine evidence paths between laws, a process that is highly resource-intensive and unscalable for unstructured LGU archives. Alternatively, Shen et al. [197] proposed the HopWeaver framework, which relies on massive generative Large Language Models (LLMs) to synthesize complex reasoning paths across documents.

Both of these state-of-the-art solutions immediately violate the strict hardware constraints of consumer-grade LGU workstations. Consequently, practical, edge-deployable systems must intentionally bound their theoretical scope to single-premise, asymmetric span evaluations to avoid the quadratic memory scaling required for multi-document fusion.

### 2.10.5 Dataset Partitions and Grouped Leakage Prevention

When evaluating machine learning models on constrained datasets, researchers often deploy standard random $k$-fold cross-validation to maximize testing coverage. However, Vabalas et al. [222] demonstrated that applying naive random cross-validation to datasets under 1,000 items without strict group isolation introduces severe optimistic bias, causing models to appear artificially accurate due to sample overlap or information leakage across folds. In structured legal and paired-text scenarios where multiple hypothesis statements share an identical statutory premise, random sample splitting permits premise-side semantic features to bleed across training and evaluation partitions. To prevent optimistic bias in few-shot legal scenarios, contemporary evaluation frameworks require both strict, static Train/Validation/Test partitions (e.g., 70/15/15) for hyperparameter tuning and early stopping, alongside strict premise-grouped cross-validation (`GroupKFold`) where entire statutory premise clusters are strictly quarantined into either the training or evaluation partition to guarantee zero cross-fold leakage.

Beyond preventing optimistic bias, utilizing a static partition allows for rigorous validation monitoring during the fine-tuning phase. In datasets with limited volume, neural networks are highly susceptible to memorizing the training data rather than learning underlying logical patterns. Goodfellow et al. [82] establish that permanently isolating a validation set allows architectures to employ early stopping criteria, halting the training

process the moment the model's accuracy on unseen data begins to degrade. Furthermore, as Zhang et al. [246] note regarding few-sample fine-tuning, this static separation ensures strict document-level boundaries, preventing clauses from a specific draft ordinance from accidentally bleeding across the training and testing sets to artificially inflate performance metrics.

### 2.10.6 Minimum Detectable Effect

A standard critique of static splits on small datasets is that the resulting test set lacks the statistical power to reliably compare candidate architectures. However, Card et al. [29] established that a test set's validity is fundamentally determined by its Minimum Detectable Effect (MDE) rather than arbitrary size thresholds.

Formally, statistical power $(1 - \beta)$ quantifies the probability of correctly rejecting the null hypothesis $(H_0)$ when a true performance disparity exists between competing model architectures. In hypothesis testing, the required sample size $N$ is inversely proportional to the square of the standardized effect size ($\delta = |\mu_1 - \mu_2|/\sigma$). When comparing architectures with marginal performance differentials (e.g., 1%–2%), large evaluation corpora ($N > 10^4$) are statistically required to detect the variation at a significance level of $\alpha = 0.05$. Conversely, when evaluating fine-tuned legal models against unadapted zero-shot baselines where the performance delta is substantial ($\delta > 0.80$), a curated holdout partition of approximately 50 instances provides sufficient statistical power to confirm that observed improvements represent genuine domain adaptation rather than random sampling variance.

The practical application of MDE fundamentally shifts how model performance is evaluated in specialized domains. In general NLP research, massive test sets are required because researchers are often evaluating models against benchmarks like GLUE to prove marginal, incremental gains of one percent to two percent over previous state-of-the-art architectures [234]. However, applying a base foundation model—which possesses no prior knowledge of Philippine local governance for example—to a highly specialized task generally results in baseline performance barely above random chance [31]. Zheng et al. [247] demonstrate that in the legal domain, fine-tuning foundational architectures on expertly curated pairs triggers a massive, double-digit jump in classification accuracy compared to zero-shot baselines. Because this expected performance delta is substantial, legal NLP frameworks do not require thousands of annotated examples to demonstrate statistically valid improvements, proving that an evaluation set of approximately 50 curated pairs possesses sufficient statistical power to mathematically capture this distinct leap in model reasoning capacity.

## 2.11 Explainable AI

In legal informatics, a "black box" prediction is fundamentally unacceptable, regardless of nominal classification accuracy. Automated systems that flag normative conflicts between statutory enactments must provide verifiable reasoning (*ratio decidendi*) behind their classifications to genuinely assist legislative analysis [65]. Without transparent jus-

tification, legal practitioners and legislative staff cannot trust, audit, or formally defend algorithmic determinations. Consequently, contemporary legal NLP literature emphasizes Explainable AI (XAI) not as an optional secondary layer, but as an intrinsic structural requirement of high-stakes legal decision support. This stance prioritizes faithful mathematical traces over ungrounded generative summaries.

### 2.11.1 Intrinsic Explainability

Within the explainability literature, researchers distinguish between Post-Hoc Surrogate Models and Intrinsic Explainability Mechanisms. Surrogate methods, such as Local Interpretable Model-agnostic Explanations (LIME) [174] and SHapley Additive exPlanations (SHAP) [138], attempt to approximate the behavior of complex deep networks by fitting a simplified, interpretable linear model around individual predictions. Formally, intrinsic explainability extracts feature attributions directly from the internal mathematical parameters and representations of the model during inference. In contrast, post-hoc surrogate methods treat the deep neural network as an unobservable black box, approximating local decision boundaries by iteratively perturbing input tokens thousands of times and fitting auxiliary linear models over the resulting output shifts [180]. In text classification benchmarks, these external explainers are computationally demanding and inherently prone to approximation errors, as they explain an external proxy rather than the underlying network itself [180]. In contrast, intrinsic approaches extract internal model representations—such as self-attention weights and contextualized hidden states—directly from the transformer layers during the forward pass. Because native attention extraction reflects the mathematical weights computed during inference without fitting secondary approximations, it offers an intrinsically faithful and computationally lightweight mechanism for structural transparency.

While utilizing attention weights as a proxy for feature importance has sparked significant debate, empirical findings demonstrate its utility for targeted textual inspection. Jain and Wallace [107] argued that raw attention distributions do not always correlate consistently with gradient-based feature importance. However, Wiegreffe and Pinter [238] provided critical counter-evidence, demonstrating that attention mechanisms provide faithful representations of internal prioritization under rigorous diagnostic testing. In statutory conflict detection and deontic reasoning, extracting token-level attention spans allows reviewers to identify the specific conflicting provisions (for example, isolating an "absolute prohibition" in a superior statute against an "unauthorized permit" in a local draft), establishing a direct visual link between textual evidence and classification decisions without incurring the computational overhead of auxiliary surrogate models.

Complementing structural attention heatmaps in modern explainability frameworks is Confidence Calibration. Deep learning models, particularly multi-layer transformers, frequently exhibit severe overconfidence in erroneous predictions when evaluated on out-of-domain or specialized text. As Guo et al. [90] demonstrated, modern neural networks trained with standard cross-entropy loss are systematically miscalibrated, exhibiting pronounced overconfidence where output probability vectors assign extreme values that exceed the true empirical likelihood of correct classification. In high-stakes regulatory auditing, uncalibrated softmax scores generate distorted risk assessments. To restore prob-

abilistic reliability, calibration literature advocates post-hoc temperature scaling applied directly to pre-softmax logit vectors. Temperature scaling introduces a learned strictly positive scalar parameter $\tau > 0$ that rescales logit magnitudes before softmax normalization without modifying the rank-order of class predictions. This formulation directly minimizes Expected Calibration Error (ECE), ensuring that predicted probability scores correspond to observed empirical accuracy rates across decision thresholds.

### 2.11.2 Edge Deployment Feasibility

In edge and on-premise computing environments, operational feasibility requires strict adherence to memory and latency boundaries. Post-hoc explainability methods operate by repeatedly perturbing input tokens, requiring hundreds or thousands of forward passes per document to approximate Shapley values or linear boundary slopes [174, 138]. Empirical evaluations by Kumar et al. [119] confirm that permutation-based explainers inflate per-document inference latency by several orders of magnitude, which makes them computationally prohibitive for lengthy legislative texts. Furthermore, Wang et al. [236] demonstrate that running complex post-hoc explainers on edge workstations severely degrades throughput and makes interactive review impractical. In resource-constrained deployment settings, intrinsic extraction of native attention weights and calibrated logits provides a computationally viable pathway to transparency without requiring specialized high-performance computing clusters.

### 2.11.3 Explanation Quality Metrics

Evaluating explanation quality in natural language processing requires distinguishing between two fundamental evaluation criteria: Plausibility and Faithfulness. Formally, *plausibility* measures the extrinsic degree to which an explanation appears convincing, fluent, and domain-intuitive to human legal practitioners, whereas *faithfulness* measures the intrinsic fidelity with which an explanation reflects the actual internal computational mechanics and feature weights utilized by the model to produce its output [106, 65]. Jacovi and Goldberg [106] demonstrate that optimizing explanations purely for plausibility frequently produces deceptive explanations that mask internal reasoning failures and biases. In legal domains, legal scholarship emphasizes that faithfulness must take precedence, as an explanation that sounds plausible but misrepresents statutory text can lead to catastrophic compliance failures.

This trade-off is particularly evident in generative explainability approaches that deploy auto-regressive large language models to draft plain-language explanations of legal predictions. While generative summaries achieve high human plausibility scores, empirical legal informatics studies demonstrate that generative models frequently suffer from legal hallucinations, fabricating non-existent statutory citations or altering legal interpretations during text generation [131, 112]. Because generated natural language explanations are decoupled from the classifier's internal mathematical representations, they remain inherently unfaithful [106]. Consequently, legal informatics literature increasingly favors extractive and attention-grounded visualizations that display the exact mathematical spans influencing classification to eliminate generative distortions.

### 2.11.4 Explainability in Multi-Stage Pipelines

In multi-stage NLP pipelines, literature on Explainable AI emphasizes situating interpretability mechanisms at the fine-grained inference stage rather than the coarse retrieval phase. In retrieve-then-entail systems, the initial retrieval phase functions as a high-throughput candidate selector across extensive document repositories. As Anand et al. [8] observe, computing dense feature attributions during coarse document filtering provides negligible analytical utility while incurring prohibitive computational latency. In contrast, the subsequent cross-encoder inference stage conducts fine-grained semantic entailment across a constrained candidate set; this focus makes it the mathematically appropriate locus for interpretability analysis.

At this second stage, legal NLP frameworks utilize token-level saliency and attention visualizations to bridge model internals with human inspection [229]. By mapping cross-attention weights and calibrated class probabilities back to corresponding spans in the premise and hypothesis texts, systems provide structural verification of detected contradictions without introducing generative paraphrasing risks. This dual mechanism of intrinsic attention mapping and post-hoc confidence calibration allows legislative researchers to trace classification outputs directly back to source statutory phrases.

## 2.12 Theoretical Framework

As established in legal informatics literature, the Retrieve-then-Entail architecture represents the leading theoretical framework for statutory conflict detection and legislative consistency checking. Originating in the international benchmarks of the Competition on Legal Information Extraction and Entailment (COLIEE 2025 and 2026) [80, 81], this framework integrates two complementary theoretical pillars: a scalable Information Retrieval (IR) architecture to isolate relevant statutory context from expansive corpora, followed by a fine-grained Natural Language Inference (NLI) architecture for rapid deontic conflict detection. Specifically, this theoretical framework synthesizes the dense statutory retrieval methodology of Team AIIR Lab [239, 123], the multi-stage ensemble retrieval principles of Team JNLP [80, 79], and the encoder-based textual entailment framework of Team KIS [109].

### 2.12.1 Statutory Retrieval (Team AIIR)

In statute law retrieval, dense bi-encoder architectures offer an effective theoretical foundation for bridging the semantic gap between newly drafted provisions and expansive statutory corpora without relying on computationally demanding generative text decoding or manual taxonomy engineering [239, 123]. Drawing from the empirical methodology developed by the AIIR Lab in recent COLIEE benchmarks, this retrieval pipeline operates across four theoretical dimensions: domain-adapted data augmentation, semantic vectorization, corpus pre-computation, and nearest-neighbor distance evaluation. Figure 2.1 illustrates this dense bi-encoder workflow.

Figure 2.1 details the structural dataflow of the dense statutory retrieval archi-

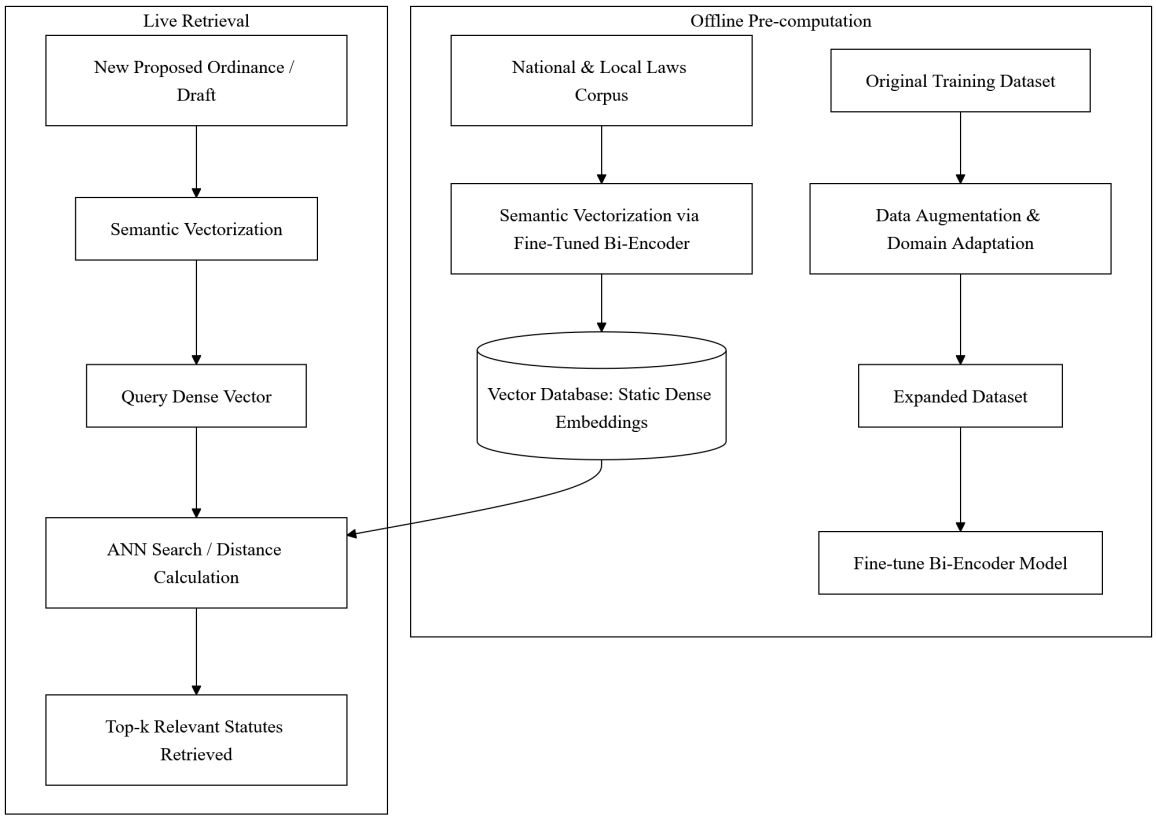



Figure 2.1: Statute Law Retrieval Pipeline Using Dense Bi-Encoder Architecture

tecture adapted from the AIIR Lab methodology. The diagram illustrates four primary operational stages: synthetic legal query augmentation, offline statutory embedding pre-computation, dense vector indexing, and approximate nearest neighbor similarity matching. This visual is presented to demonstrate how modern legal search pipelines computationally decouple corpus-wide indexing from runtime query execution. As depicted, the incoming query provision is projected into a shared semantic coordinate space, allowing its vector representation to be compared directly against pre-computed statutory embeddings via cosine distance. This architectural separation ensures that high-volume statutory collections spanning tens of thousands of provisions can be searched in sub-second latency on consumer-grade hardware, providing the high-recall candidate shortlist necessary to feed the downstream entailment engine.

To operationalize this dense bi-encoder architecture, the pipeline executes across four theoretical dimensions. First, targeted data augmentation and domain adaptation expose dense retrieval models to synthetically augmented legal variants, training the neural network to recognize identical statutory concepts across archaic terminology and modern drafting syntax to enhance robustness against lexical divergence. Second, semantic vectorization via bi-encoders (such as `all-mpnet-base-v2`) projects statutory provisions into a dense, continuous vector space where queries and corpus documents are encoded independently, optimizing embedding coordinates such that logically relevant statutory

pairs exhibit high geometric proximity under cosine distance. Third, offline corpus pre-computation and vector indexing pre-calculate static embeddings across the statutory repository within an indexed vector database, decoupling computationally intensive encoding from live runtime queries and allowing dense retrieval to scale across extensive archives without real-time GPU re-encoding. Fourth, live query vectorization and proximity scoring project incoming draft provisions into the shared embedding space, enabling approximate nearest-neighbor search to calculate continuous proximity scores in sub-second latency and delivering a high-recall candidate shortlist for downstream inference.

### 2.12.2 Multi-Stage Retrieval (Team JNLP)

Complementing pure bi-encoder retrieval, literature in computational law emphasizes multi-stage retrieval pipelines to balance candidate recall against ranking precision. Exemplified by Team JNLP's high-performing architecture in COLIEE benchmarks [80, 79], this multi-stage pipeline progressively filters statutory search spaces across three structured phases: Pre-Retrieval Candidate Pruning, Model Fine-Tuning, and Result Ensembling. Figure 2.2 illustrates this multi-stage workflow.

Figure 2.2 outlines the progressive candidate filtering architecture developed by Team JNLP. The diagram delineates the modular progression from initial corpus-level candidate pruning, through parameter-efficient neural re-ranking, to multi-model score ensembling. This cascaded framework provides theoretical justification for combining sparse lexical matching with dense semantic embeddings in statutory search. In the initial phase, heuristic distance thresholds and lexical filters rapidly eliminate irrelevant enactments, after which fine-tuned language models compute calibrated relevance scores across the remaining shortlist. The resulting predictions are synthesized through rank fusion, ensuring that the candidate set benefits from both keyword specificity and conceptual semantic coverage.

#### 2.12.2.1 Pre-Retrieval

The primary objective of the pre-retrieval phase is to drastically reduce the expansive statutory search space while maintaining high recall across governing provisions. Team JNLP achieves this through a multi-stage candidate pruning procedure combining distance-based heuristic filtering with neural re-ranking [80, 79]. Dense embeddings for statutory queries and corpus provisions are extracted using high-capacity multilingual representations. Rather than relying solely on raw cosine similarity, the methodology calculates multi-dimensional distance metrics (such as L1 distance bins), training a gradient-boosted decision tree to predict relevance and counter severe sample imbalance. Following initial filtering, a specialized passage re-ranker evaluates candidate texts and eliminates non-relevant provisions; this step narrows the candidate pool to top-ranked candidates while preserving recall.

#### 2.12.2.2 Model Fine-Tuning

Once the candidate space is constrained, the framework applies targeted natural language models to perform fine-grained relevance scoring. Because fine-tuning large models across

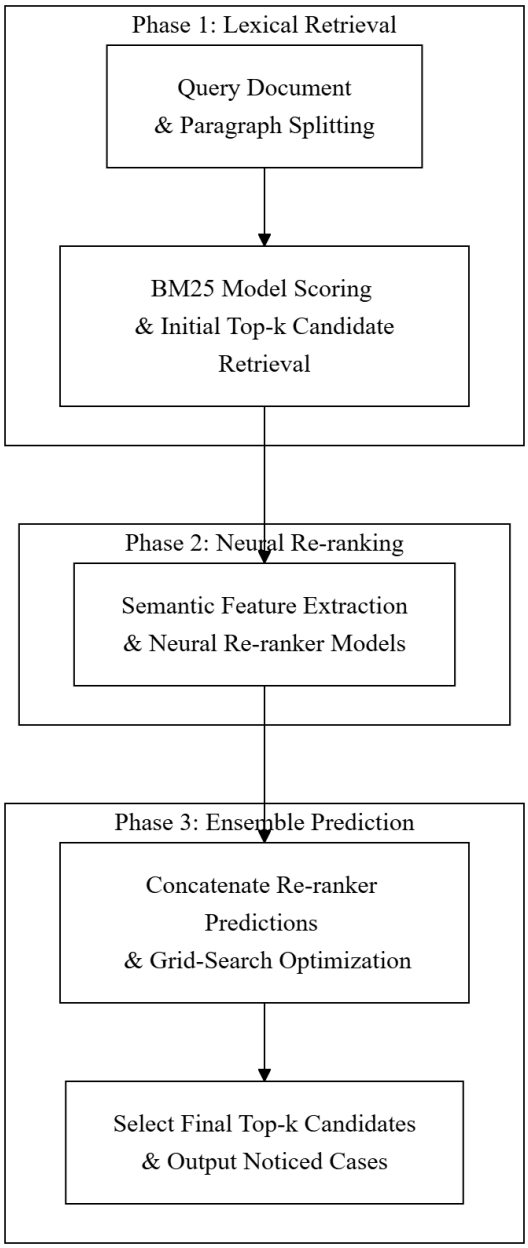



Figure 2.2: Multi-Stage Statutory Retrieval Pipeline With Candidate Pruning and Ensembling

full parameter sets is computationally intensive, legal NLP frameworks employ Parameter-Efficient Fine-Tuning (PEFT) techniques, such as LoRA and QLoRA. These techniques enable foundational language models to adapt to specialized statutory semantics with minimal memory overhead. Furthermore, because statutory candidate sets exhibit severe class imbalance where only a small minority of retrieved provisions contain true preemption conflicts, sampling strategies and asymmetric loss functions are applied during training to preserve sensitivity to positive conflict signals.

#### 2.12.2.3 Result Ensemble

To maximize retrieval robustness and mitigate individual model variance, multi-stage retrieval pipelines aggregate relevance scores across distinct retrieval architectures. Relevance predictions are aggregated through weighted score aggregation; this process combines calibrated probabilities from complementary lexical and neural models. The weighting coefficients assigned to individual scoring components are optimized via systematic hyperparameter search algorithms such as Optuna. This optimization ensures balanced contributions between lexical keyword precision and dense semantic recall to deliver a stable candidate shortlist for downstream reasoning.

### 2.12.3 Textual Entailment (Team KIS)

In the second phase of legal retrieve-then-entail architectures, the theoretical objective transitions from broad candidate discovery to fine-grained deontic reasoning. To ensure high computational throughput and deterministic classification, modern legal NLP frameworks deploy non-generative cross-encoder architectures, exemplified by the ModernBERT-based rationale extraction methodology developed by Team KIS [109]. Figure 2.3 shows this cross-encoder classification structure.

Figure 2.3 illustrates the sequence classification pipeline utilized for statutory textual entailment. The visual depicts the concatenation of the retrieved statutory premise and draft hypothesis into a unified input sequence demarcated by delimiter tokens, which is then processed through bidirectional self-attention layers to produce calibrated class probabilities. This cross-encoder structure illustrates how fine-grained deontic reasoning operates without the hallucination risks inherent in generative text decoders. By evaluating full cross-attention between premise and hypothesis tokens simultaneously, the cross-encoder identifies subtle modal contradictions that independent vector projections fail to capture. The calibrated classification head then maps the combined representation into deterministic Entailment, Neutral, or Contradiction outputs with formal statistical reliability.

### 2.12.4 Theoretical Synthesis

This synthesis integrates the foundational theoretical models reviewed throughout this chapter into a unified framework for ex-ante legislative conflict detection. While each individual methodology addresses an isolated phase of legal document processing, their conceptual integration establishes a principled retrieve-then-entail architecture. The theoretical foundation for statutory candidate discovery draws from the dense retrieval methodology established by the AIIR Lab [239, 123]. Their approach demonstrates that targeted domain augmentation mitigates OCR noise in archival legal text, while continuous semantic vectorization enables high-throughput candidate identification across large statutory collections without relying on rigid keyword indices.

To balance candidate recall against precision, the framework incorporates the multi-stage ranking principles demonstrated by Team JNLP [80, 79]. This methodology utilizes lexical filtering (BM25) to prune extraneous documents, followed by neural scoring and

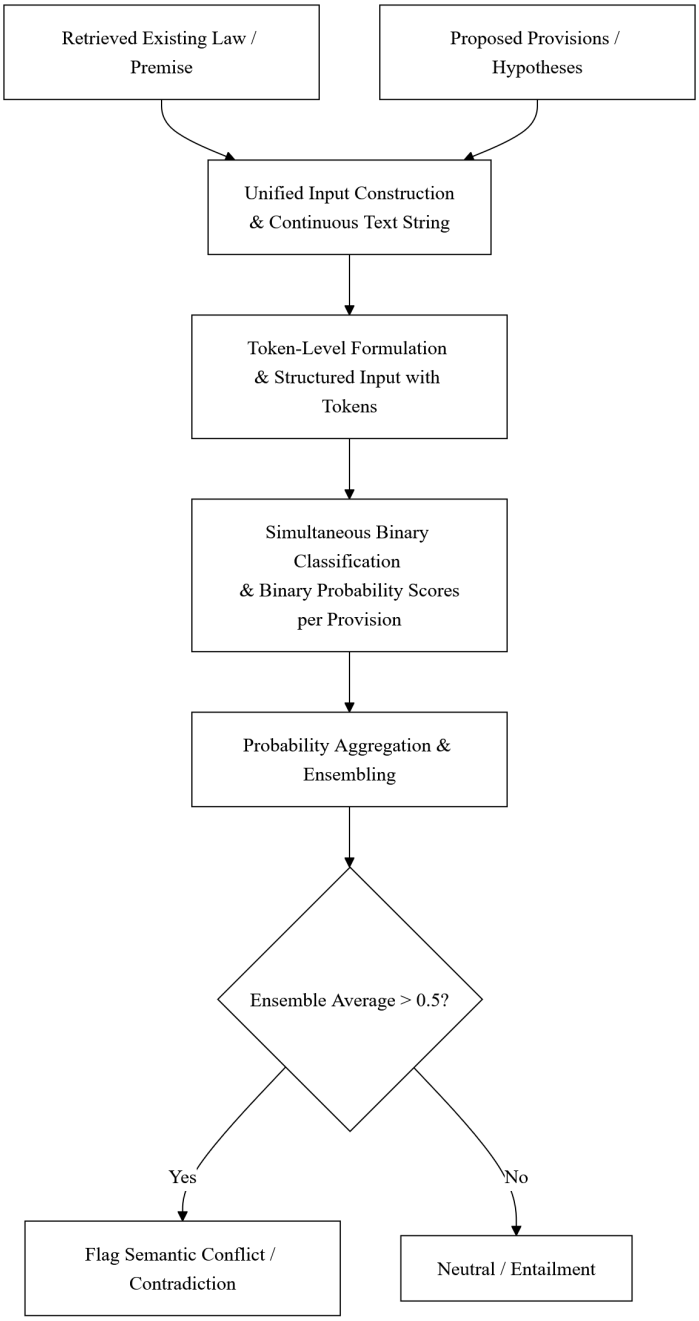



Figure 2.3: Statutory Textual Entailment Pipeline Using Bidirectional Cross-Encoder

rank fusion to isolate authoritative candidate enactments on standard computing hardware. For granular deontic conflict evaluation, the framework adopts the Team KIS framework for textual entailment [109]. Their methodology formats premise-hypothesis pairs into a continuous sequence evaluated via lightweight cross-encoders, utilizing token-level separation boundaries to classify logical relationships without generative hallucination risks.

Synthesizing these foundational frameworks establishes an end-to-end theoretical foundation for automated preemption auditing. The retrieval stage narrows extensive statutory search spaces down to high-probability candidate provisions, while the textual entailment stage conducts deterministic normative conflict checks. The latest findings from the COLIEE 2026 proceedings [81, 152] reinforce this architectural consensus, validating that pipelined retrieval and textual entailment provide the most reliable foundation for automated legal consistency analysis.

# Chapter 3

# Methodology

This chapter explains the methodology used to build and evaluate the Coarse-to-Fine Semantic Conflict Detection System. It describes how Philippine legal texts are collected and cleaned, and how the two-stage Retrieve-then-Entail architecture is organized. The chapter also details the creation of the ground truth dataset and the selection of candidate models. Finally, it outlines the evaluation metrics and hardware constraints, which are chosen to match the computing capacity of an average Local Government Unit (LGU).

## 3.1 Conceptual Framework

This section maps how data flows through the proposed system. The architecture follows an Input-Process-Output (IPO) design organized into a two-stage "Coarse-to-Fine" retrieve-then-entail pipeline (Figure 3.1). In information retrieval and natural language processing, a coarse-to-fine retrieve-then-entail architecture is a cascaded processing design that couples a fast, broad candidate search phase with a computationally intensive, fine-grained semantic classification phase. In this research, Stage 1 is the coarse retrieval stage: it rapidly scans the comprehensive dual statutory corpus across national statutes and local city ordinances using hybrid sparse-dense search to filter the vast document collection down to a compact shortlist of top-$k$ ($k = 50$) candidate provisions. Stage 2 is the fine inference stage: it pairs the draft ordinance section with each retrieved candidate statute to perform deep bidirectional cross-attention, computing the calibrated probability of a substantive legal contradiction. By decoupling high-throughput statutory candidate retrieval from compute-intensive cross-encoder inference, this two-stage design circumvents the quadratic memory limits ($O(n^2)$) and runtime overhead that otherwise prohibit whole-corpus cross-attention, enabling deep semantic conflict detection to execute on standard local government office computers without requiring multi-million-peso cloud clusters or dedicated data centers. The workflow is divided into two operational phases: Phase 1 (Offline Foundation and Optimization) and Phase 2 (Live Online Retrieve-then-Entail Pipeline).

Figure 3.1 illustrates the complete end-to-end information flow of the conflict detection architecture, detailing the functional separation between offline foundation

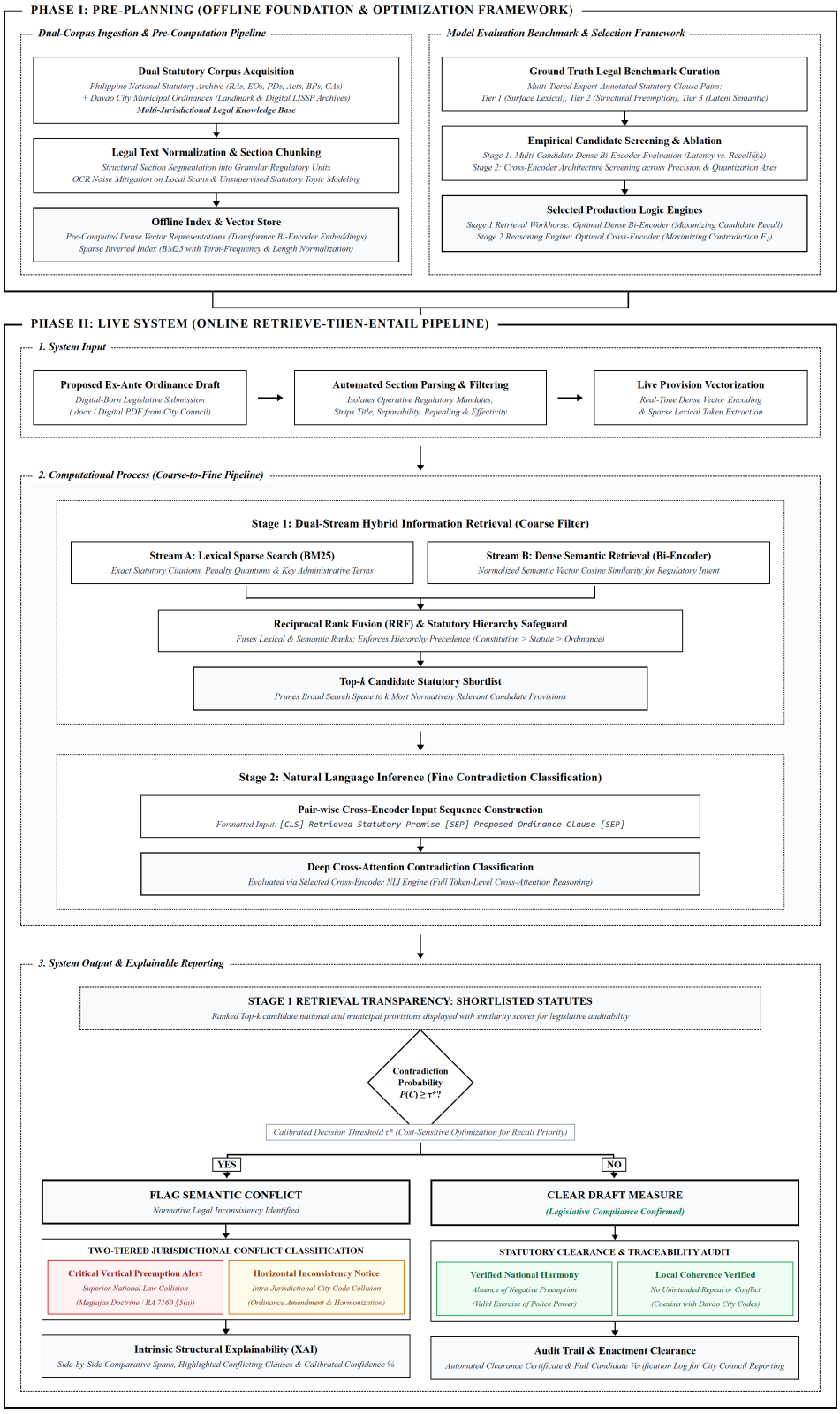



Figure 3.1: Detailed Information Flow of the Coarse-to-Fine Semantic Conflict Detection Architecture.

pre-computation (Phase 1) and the live pre-enactment review pipeline (Phase 2). The flowchart depicts an Input-Process-Output workflow organized into modular processing layers; it shows how heterogeneous statutory archives are ingested, normalized, indexed, and logically verified under local edge compute constraints. In Phase 1, the system compiles the dual statutory knowledge base (combining 25,432 national statutes and 1,925 local ordinances), builds pre-computed BM25 and dense embedding indexes, and screens candidate neural architectures against an adviser-reviewed 350-pair ground truth benchmark (comprising 198 unique instances across 22 statutory acts). In Phase 2, an ex-ante ordinance draft undergoes structural boundary parsing and procedural boilerplate filtering before being routed through Stage 1 hybrid retrieval (fusing lexical BM25 and dense bi-encoder streams via Reciprocal Rank Fusion) and Stage 2 Cross-Encoder inference. This coarse-to-fine design restricts compute-intensive cross-attention to a focused candidate shortlist ($k = 50$), allowing deep semantic conflict detection to execute efficiently on standard local government desktop workstations while producing transparent, explainable audit reports.

### 3.1.1 Phase 1: Offline Foundation and Optimization

This preliminary phase establishes the empirical foundation of the system by compiling the statutory corpus, building pre-computed retrieval indexes, and selecting candidate transformer logic engines through a structured benchmark screening process.

During dual-corpus ingestion and pre-computation, the system compiles a comprehensive dual-jurisdiction statutory archive combining the Philippine national statutory collection (Republic Acts, Executive Orders, Presidential Decrees, Acts, Batas Pambansa, and Commonwealth Acts) with Davao City local legislation (historical Landmark Ordinances and digital LISSP ordinance records). The compiled corpus undergoes legal text normalization, optical character recognition (OCR) noise mitigation on historical scans, and structural section segmentation into granular, self-contained regulatory units ($\sim$ 200–300 tokens). Cleaned statutory provisions are then indexed offline: dense semantic representations are pre-computed using a lightweight transformer bi-encoder, while sparse inverted lexical indexes are constructed using BM25 ($k_1 = 1.5, b = 0.75$) alongside unsupervised statutory topic modeling. In parallel, the system constructs an offline Statutory Knowledge Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. Deterministic statutory parsers extract cross-statute citation networks and amendatory legislative lineages across national acts and city ordinances, mapping structural relationships (CONTAINS, CITES_AUTHORITY, AMENDS, REPEALS, and GOVERNS_DOMAIN) to establish topological validity and authority baselines prior to live query execution.

Following corpus indexing, model benchmarking and selection evaluates candidate neural architectures against a curated Ground Truth Legal Benchmark stratified across three granular difficulty tiers: Tier 1 (Surface Lexical), Tier 2 (Structural Preemption), and Tier 3 (Latent Semantic). A multi-stage empirical screening and ablation process evaluates candidate bi-encoders by balancing inference latency against candidate Recall@$k$, alongside candidate cross-encoders evaluated across precision, latency, and quantization axes. Based on these empirical evaluations, the highest-performing models are selected as the production engines: an optimal dense bi-encoder for Stage 1 candidate retrieval, and

an optimal fine-tuned cross-encoder for Stage 2 contradiction reasoning.

### 3.1.2 Phase 2: Live System Workflow

The live system executes the automated conflict detection pipeline when a newly drafted local ordinance is submitted for pre-enactment review, operating through an Input-Process-Output structure.

The workflow begins with system input ingestion, where a legislative researcher uploads a newly proposed ex-ante ordinance draft. Because draft measures originate within local government offices as digital-born documents (.docx or digital PDF), they are free from optical scanning noise. The ingestion pipeline segments the draft into individual substantive provisions, filtering out non-regulatory procedural boilerplate such as the Title, Separability, Repealing, and Effectivity clauses. The isolated substantive provisions are then vectorized in real time via the selected dense bi-encoder alongside sparse lexical token extraction.

The computational process core then executes two complementary stages: hybrid candidate retrieval followed by natural language inference. In Stage 1 (Dual-Stream Hybrid Information Retrieval), the system narrows the comprehensive statutory search space down to a focused candidate pool. Stream A executes lexical sparse search via BM25 to match exact statutory citations, numerical penalty amounts, and administrative agency terms, while Stream B executes dense semantic search via the selected bi-encoder, measuring cosine similarity between the draft provision and statutory embeddings to capture abstract regulatory intent. The ranked outputs are combined using Reciprocal Rank Fusion (RRF) integrated with a Statutory Hierarchy Safeguard that prioritizes higher-ranking national statutes over local provisions. To ensure candidate safety when draft clauses reference unindexed statutory families, Stage 1 integrates a graph-guided neighborhood expansion: explicit citations in the draft provision trigger localized multi-hop traversals over the Statutory Knowledge Graph, pulling governing national acts and amendatory predecessors directly into the top-$k$ candidate pool without exhausting local LGU memory.

In Stage 2 (Natural Language Inference), following the standard sequence-pair classification formulation established by Devlin et al. [62] and adapted for transformer cross-encoders by Reimers and Gurevych [171], each retrieved statutory candidate is paired with the draft ordinance provision into a single cross-encoder sequence:

*Cross-Encoder Joint Sequence Input Representation:*

$$\mathbf{x} = [\text{CLS}]\,\text{Statutory Premise}\,[\text{SEP}]\,\text{Ordinance Hypothesis}\,[\text{SEP}] \tag{3.1}$$

Where [CLS] marks the classification sequence start and [SEP] separates jurisdictional text boundaries.

Equation 3.1 combines the retrieved statutory section and the draft ordinance provision into a single token sequence separated by delimiter tokens. This allows every word in the statutory premise to interact directly with every word in the draft hypothesis through bidirectional cross-attention. This direct comparison is necessary because

legal conflicts often depend on small details—such as specific conditions, penalty ceilings, or exceptions—that independent bi-encoder vectors can easily miss. In addition, the Stage 2 logic pipeline cross-references candidate provisions against the Statutory Knowledge Graph to confirm amendatory status under *lex posterior derogat legi priori*, ensuring that superseded statutory text is flagged before cross-attention scoring. The model's classification head then calculates calibrated probabilities for Entailment, Neutral, and Contradiction, outputting a conflict probability $P(\text{Contradiction})$.

Finally, the system presents an audit trail displaying the top-$k$ shortlisted laws for human review. It compares the contradiction probability against a calibrated decision threshold $\tau^*$, tuned with an $F_2$-score weighting to prioritize catching true conflicts. When the probability meets or exceeds this threshold ($P(\text{Contradiction}) \geq \tau^*$), the system flags a potential legal conflict and classifies it into one of two alert types: a Vertical Preemption Alert for conflicts with superior national statutes under the *Magtajas* doctrine, or a Horizontal Inconsistency Notice for contradictions or overlaps with existing Davao City ordinances. Confirmed preemption contradictions are dynamically registered as prospective CONFLICTS_WITH relations within the knowledge graph. To help legislative researchers interpret the result, the user interface displays side-by-side text comparisons, highlighted conflicting phrases, and model confidence scores. If the probability remains below the threshold ($P(\text{Contradiction}) < \tau^*$), the system reports that no direct conflicts were found, confirming consistency with both national statutes and prior local enactments.

## 3.2 Research Design

This study uses an applied research design to build a practical software tool for local governance. The researchers use an experimental systems development approach to solve the delays and difficulties of manual ordinance review through automated detection. This setup connects principles from legal informatics and natural language processing to address a real local need. By focusing on practical transformer-based models, the design connects theoretical computer science with everyday government office work. The experiment directly tests whether a retrieve-then-entail pipeline can operate reliably in a local legislative setting.

The experimental setup tests the retrieval and classification modules separately to measure search speed and accuracy independently. Building the system step-by-step allows the researchers to adjust the information retrieval and natural language inference stages using local legal texts. This engineering process ensures that the finished tool accounts for the writing styles found in both city ordinances and national statutes. In this way, applied research methods help turn modern NLP concepts into a usable tool for the Davao City Sangguniang Panlungsod. As a result, the findings establish a clear benchmark for automating pre-enactment legal checks on local legislation.

The experimental design incorporates an incremental ablation methodology to evaluate both isolated components and composite system interactions. The pipeline is evaluated under two distinct structural configurations: a baseline coarse-to-fine retrieve-then-entail architecture coupling lexical BM25 and dense bi-encoders with cross-encoder NLI,

and a neuro-symbolic graph-augmented pipeline incorporating the Statutory Knowledge Graph for candidate expansion, topological verification, and lineage tracking. By benchmarking both architectures across the multi-tier legal dataset, the experimental design isolates the individual contributions of sparse keyword matching, dense latent semantics, cross-attention contradiction reasoning, and relational statutory network traversal. This comparative setup confirms whether graph topological augmentation reliably resolves retrieval omissions on unmentioned statutory preemption authorities while maintaining execution within edge hardware constraints.

## 3.3 Corpus Acquisition and Exploratory Analysis

Building the legal database requires collecting text from different public repositories so that the system includes as many relevant laws as possible. These sources provide the national statutes and city ordinances needed to check draft legislation for consistency. To keep the language consistent and make vector processing simpler, the system gathers only the English-language versions of all legal texts. This choice matches Philippine statutory rules under Section 20, Chapter 5, Book I of Executive Order No. 292 (Administrative Code of 1987) [172], which explicitly mandates that in the interpretation of laws promulgated in official languages, the English text shall control unless otherwise specifically provided. National statutes and local ordinances are gathered using separate automated scripts designed for the format of each source.

### 3.3.1 Corpus Composition

The Lawphil Project (Arellano Law Foundation) is the primary digitized extraction source for national statutory text because its documents are provided in clean HTML, follow a consistent layout, and offer broad historical coverage [9]. Automated scraping scripts browse through chronological and alphabetical indexes to collect Republic Acts, Batas Pambansa, Commonwealth Acts, Acts, Executive Orders, and Presidential Decrees. Using a custom web extraction pipeline built by the proponents, statutory extraction was conducted as of **March 2026**, collecting 25,432 unique legal records enacted between 1900 and 2026. The Official Gazette of the Republic of the Philippines [87] and the Chan Robles Virtual Law Library were used as authoritative secondary cross-references to verify enactment dates, chronological numbering, and historical amendment status. All collected national laws are converted into standardized JSON records with metadata, such as dates of approval, titles, and section numbers. As illustrated in Figure 3.2, the Lawphil repository presents statutory text in clean, unstyled HTML typography with a predictable hierarchical URL schema (e.g., `/statutes/repacts/ra2018/ra_11035_2018.html` for Republic Act No. 11035). This standardized structure separates introductory legislative headers from operative provisions and eliminates the need for dynamic JavaScript browser rendering, facilitating rapid, fault-tolerant web scraping.

Figure 3.2 displays a representative national statutory webpage from the Lawphil Project repository, featuring the complete legislative record of Republic Act No. 11035 (*Balik Scientist Act*). The document interface exhibits a clean, unstyled HTML layout that cleanly isolates the formal congress heading, approval date, official title, and individual

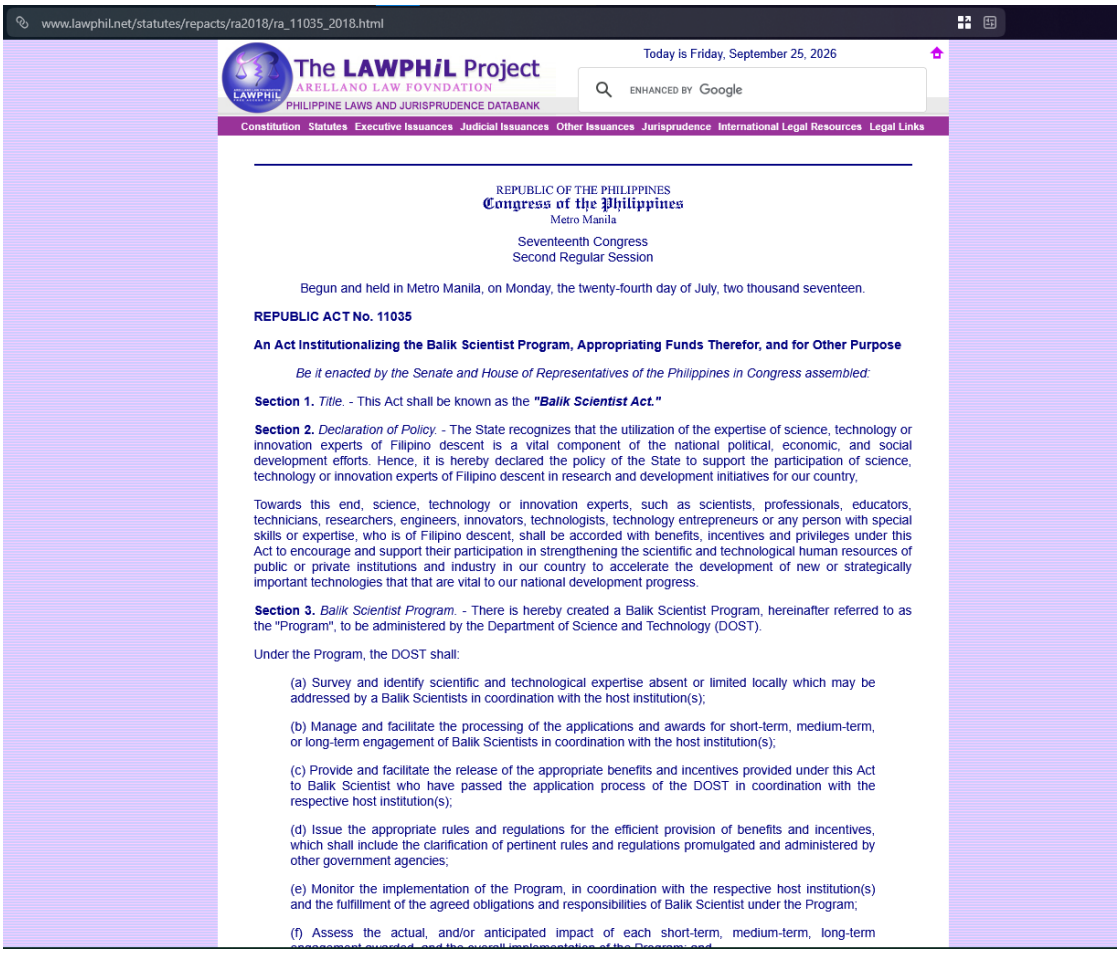



Figure 3.2: Sample Statutory Enactment Webpage from the Lawphil Project Repository Illustrating Clean Document Hierarchy and Structured Legislative Metadata (Republic Act No. 11035).

numbered provisions into predictable DOM elements. This predictable visual structure illustrates why the national statutory repository facilitates automated web crawling without requiring dynamic JavaScript execution or complex layout segmentation. In the displayed layout, the operative provisions appear in sequential chronological order, flanked by transparent URL parameters (`/statutes/repacts/ra2018/ra_11035_2018.html`) that enable automated validation against congressional records. By capturing these structured HTML pages directly, the ingestion preprocessor compiles over 25,000 national enactments into standardized text records while completely avoiding the visual noise and character degradation present in scanned archival paper.

Collecting national laws from Lawphil helps avoid the disorganization common to public legal websites. Although the Official Gazette and congressional websites also host statutory records, their web pages frequently change layout and often provide older laws only as image-only PDF scans. In contrast, Lawphil keeps a standard HTML structure across all historical eras, clearly separating introductory preambles from operative legal sections. Scraping from this repository ensured uniform text formatting across the

national collection (over 25,000 enactments), preventing formatting differences from skewing vector embeddings during Stage 1 retrieval. Listing 3.1 illustrates the web scraping and provision extraction logic implemented to retrieve and structure the statutory archive, establishing a consistent text foundation for semantic analysis and conflict detection.

```
def extract_statutory_corpus(base_url, start_year=1900, end_year=2026):
    """Extracts, sanitizes, and structures national statutes from the online archive."""
    statute_records = []
    for year in range(start_year, end_year + 1):
        year_index = fetch_page(f"{base_url}/year/{year}")
        statute_links = parse_index_links(year_index)
        for link in statute_links:
            # Fetch raw web page and parse HTML DOM tree
            html_content = fetch_page_with_retry(link, delay_seconds=1.0)
            soup = BeautifulSoup(html_content, "html.parser")

            # Extract statutory metadata and clean document body
            metadata = extract_header_metadata(soup)
            body_text = clean_document_body(soup)
            sections = segment_into_sections(body_text)

            statute_records.append({
                "statute_id": metadata.id,
                "title": metadata.title,
                "instrument": metadata.type,
                "date_enacted": metadata.date,
                "sections": sections
            })
    return statute_records

```

Listing 3.1: Code Snippet of the Statutory Corpus Acquisition and Provision Extraction Pipeline

Listing 3.1 details the programmatic execution logic of the statutory web extraction script, outlining the sequential functions for annual index traversal, DOM parsing, metadata extraction, and provision boundary segmentation. The script is structured to crawl legislative archives spanning from 1900 through 2026, parsing HTML document trees via BeautifulSoup while implementing rate-limiting delays to preserve source server availability. During execution, the pipeline extracts key statutory metadata—including enactment identifiers, approval dates, and legislative titles—before passing the document body to a regex-based section segmenter that isolates individual statutory provisions into distinct JSON fields. Methodologically, this automated extraction guarantees uniform data serialization across all 25,432 national enactments, eliminating web scraping artifacts and ensuring that downstream embedding models evaluate pure statutory syntax.

To capture the complete statutory search space, the national collection is paired with Davao City local legislation gathered across two complementary acquisition streams. The first channel consists of the official Davao City Sangguniang Panlungsod Public Web Portal, which provides selective historical landmark ordinances and city uploads spanning approximately fifteen years, capturing prominent regulatory codes in public safety, zoning, and health [188]. The second and primary channel comprises the local Legislative Information Support System Program (LISSP), acquired directly through formal administrative coordination with the Sangguniang Panlungsod Secretariat and Information Technology division [37]. Through this institutional coordination, local authorities granted

the proponents direct access to digitized enactments via a shared repository, encompassing enactments passed primarily by the 19th and 20th City Councils between 2021 and 2025. Together, these two channels yielded an archive of exactly 1,925 local enactments spanning from 1950 to 2025. These 1,925 documents represent the complete accessible digital corpus gathered within the proponents' institutional capabilities. In practice, the true cumulative historical total of ordinances enacted since Davao City was chartered under Commonwealth Act No. 51 in 1936 remains unrecorded on the internet and unknown to local authorities. In institutional consultations, Sangguniang Panlungsod personnel confirmed that an official cumulative ledger does not exist because local legislation is numbered sequentially per calendar year and council term rather than in an unbroken cumulative sequence. Furthermore, the vast majority of historical enactments, particularly those prior to the 2021 LISSP digital initiative, exist exclusively as physical paper documents preserved in archive rooms and filing cabinets at the Sangguniang Panlungsod building on San Pedro Street, requiring manual page-by-page physical inventorying to tally. When combined with the 25,432 national statutes, the total statutory search space comprises **27,357 legal enactments** ($N_{\text{sections}} = 178,494$).

### 3.3.2 Corpus Characteristics and Inherent Biases

A close look at the collected national laws reveals a large imbalance across legal topics. Over 40% of the national statutes consist of routine administrative actions, such as converting local schools, renaming roads, adjusting municipal boundaries, or granting broadcast franchises. In contrast, major regulatory areas that interact with city powers, such as public health, land zoning, and local taxation, make up less than 15% of the total national collection. Searching through this raw distribution without filtering would cause search algorithms to return irrelevant administrative laws instead of actual regulatory rules. To keep search results useful, the system must separate these routine administrative laws from substantive regulatory provisions.

The local ordinance archive also shows an uneven time distribution and gaps in older records. For ordinances passed before 2021, city record-keeping mainly focused on digitizing major "Landmark Ordinances," leaving ordinary regulations from earlier decades in physical paper storage [188]. As a result, the pre-2021 local records represent a selective collection of prominent laws rather than a complete legislative record. While post-2021 records from the digital LISSP database are much more complete, some numerical gaps remain due to missing committee drafts and unindexed attachments. These temporal biases mean that the available local archive contains substantially more recent ordinances than older enactments.

A fundamental technical distinction must be maintained regarding the document media of the legislative corpus. All 1,925 archival ordinance documents received from the Sangguniang Panlungsod are historical scanned physical paper PDFs, rather than digital-born documents exported from word processing software. Older ordinances scanned at low resolution suffer from faded print, ink bleed-through, staple shadows, and rotational skew, as illustrated in Figure 3.3. The scanned copies also contain non-text items such as government stamps, official signatures, and handwritten marginal notes that confuse standard sentence splitting tools. Clean, digital-born text (documents exported directly from mod-

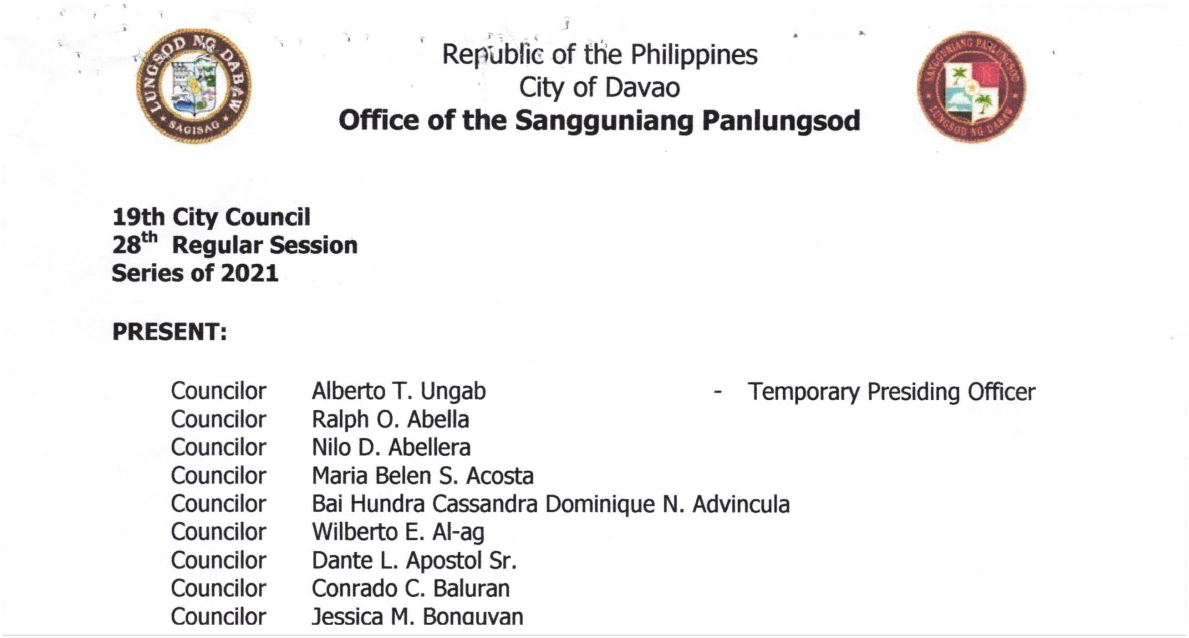
ern word processors or Google Docs without scanning noise) appears exclusively during Phase 2 live system testing when legislative staff submit newly drafted ex-ante measures for pre-enactment conflict review. At the language level, local ordinances sometimes include regional terms and administrative Taglish in their introductory sections [66]. These informal drafting habits differ noticeably from the standardized formal English found in national laws.

Figure 3.3 displays a representative physical document scan from the Davao City Sangguniang Panlungsod legislative archive (Ordinance No. 0667-21, enacted in 2021). Notably, this document represents one of the tamest and cleanest historical scans across the local collection, having been produced using modern electronic typesetting and printed on clean bond paper. Even on this mild scan, off-the-shelf optical character recognition tools still fail: standard OCR engines suffer from character corruptions (misreading "2021" as "202L" and "TITLE" as "TITTE") and erroneously fold the non-regulatory council session attendance roster from Panel (a) into substantive regulatory commands. While contemporary enactments present manageable surface noise, older historical enactments in the repository exhibit far more severe physical degradation. Figure 3.4 illustrates an authentic historical scan from the pre-LGC archive (Ordinance No. 123, Series of 1960), representing one of the most intense and severely degraded documents preserved in the local collection.

Figure 3.4 displays the extreme physical degradation present in mid-century local legislation, examining two pages of Davao City Ordinance No. 123 from the 1960 legislative term. Panel (a) illustrates a severe diagonal carbon smudge and ink smear that obliterates nearly half of the opening legislative preamble, ordinance title, and Section 1 permit mandates, alongside three physical binder punch holes that cut directly through operative characters along the left margin. Panel (b) reveals mechanical typewriter degradation characterized by uneven ink ribbon transfer, misaligned baseline characters, and handwritten pen strike-throughs and amendments (such as "As amended by Ord. No. 131, S. 1961"). When standard optical character recognition is applied to documents with this level of physical deterioration, traditional rule-based pipelines suffer severe character degradation, producing elevated transcription error rates and misinterpreting or completely dropping substantive penal provisions. This visual evidence justifies why the proponents implemented rigorous layout perimeter cropping, noise whitelisting, and local vision-language model (VLM) extraction to recover authoritative legal text from historical city archives.

In positive legislation, direct contradictions between laws occur rarely. Lawmakers generally draft provisions to operate concurrently with superior legislation rather than in direct opposition [75]. As a result, actual legal conflicts make up less than 1% of random pairs formed between local ordinances and national statutes. Taking random pairs of text from the unedited collections would lead to an overwhelming majority of unrelated pairs. A machine learning model trained on such unbalanced data could score 99% nominal accuracy simply by always predicting "Neutral," while remaining unable to detect real legal conflicts.

(a) Modern Electronic Typesetting with Header Scan Noise & Council Roster (Page 1)



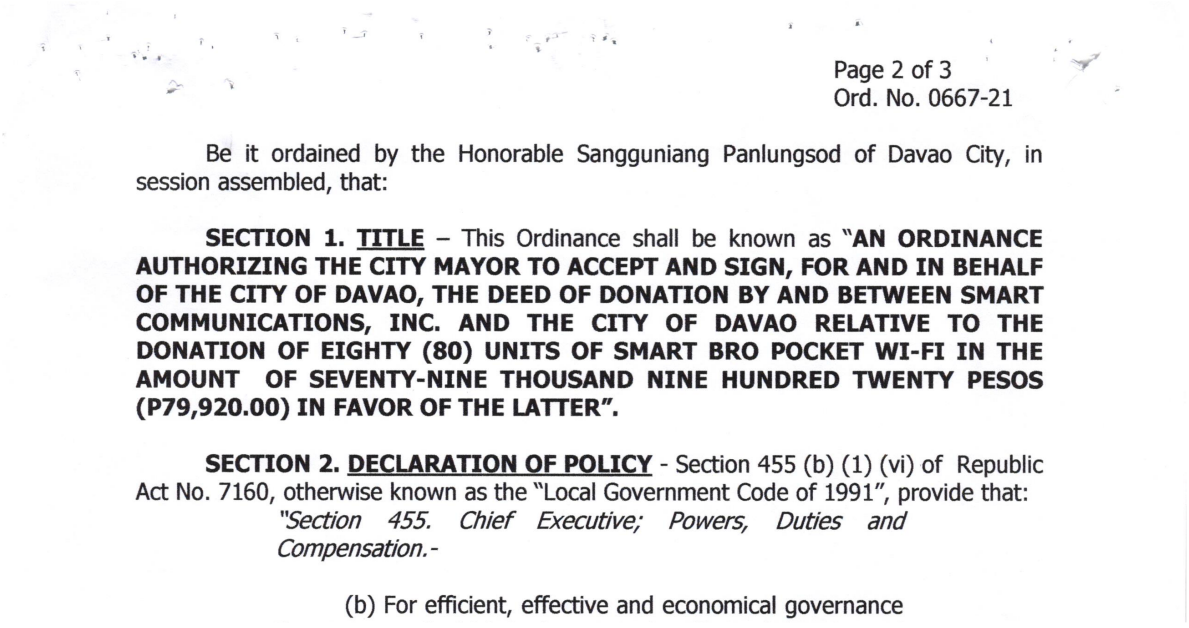
Republic of the Philippines
City of Davao
**Office of the Sangguniang Panlungsod**

**19th City Council**
**28th Regular Session**
**Series of 2021**

**PRESENT:**

| | | |
|---|---|---|
| Councilor | Alberto T. Ungab | - Temporary Presiding Officer |
| Councilor | Ralph O. Abella | |
| Councilor | Nilo D. Abellera | |
| Councilor | Maria Belen S. Acosta | |
| Councilor | Bai Hundra Cassandra Dominique N. Advincula | |
| Councilor | Wilberto E. Al-ag | |
| Councilor | Dante L. Apostol Sr. | |
| Councilor | Conrado C. Baluran | |
| Councilor | Jessica M. Bonguyan | |

(b) Substantive Enacting Sections with OCR Character Corruption & Margin Skew (Page 2)



Page 2 of 3
Ord. No. 0667-21

Be it ordained by the Honorable Sangguniang Panlungsod of Davao City, in session assembled, that:

**SECTION 1. TITLE** – This Ordinance shall be known as "**AN ORDINANCE AUTHORIZING THE CITY MAYOR TO ACCEPT AND SIGN, FOR AND IN BEHALF OF THE CITY OF DAVAO, THE DEED OF DONATION BY AND BETWEEN SMART COMMUNICATIONS, INC. AND THE CITY OF DAVAO RELATIVE TO THE DONATION OF EIGHTY (80) UNITS OF SMART BRO POCKET WI-FI IN THE AMOUNT OF SEVENTY-NINE THOUSAND NINE HUNDRED TWENTY PESOS (P79,920.00) IN FAVOR OF THE LATTER".**

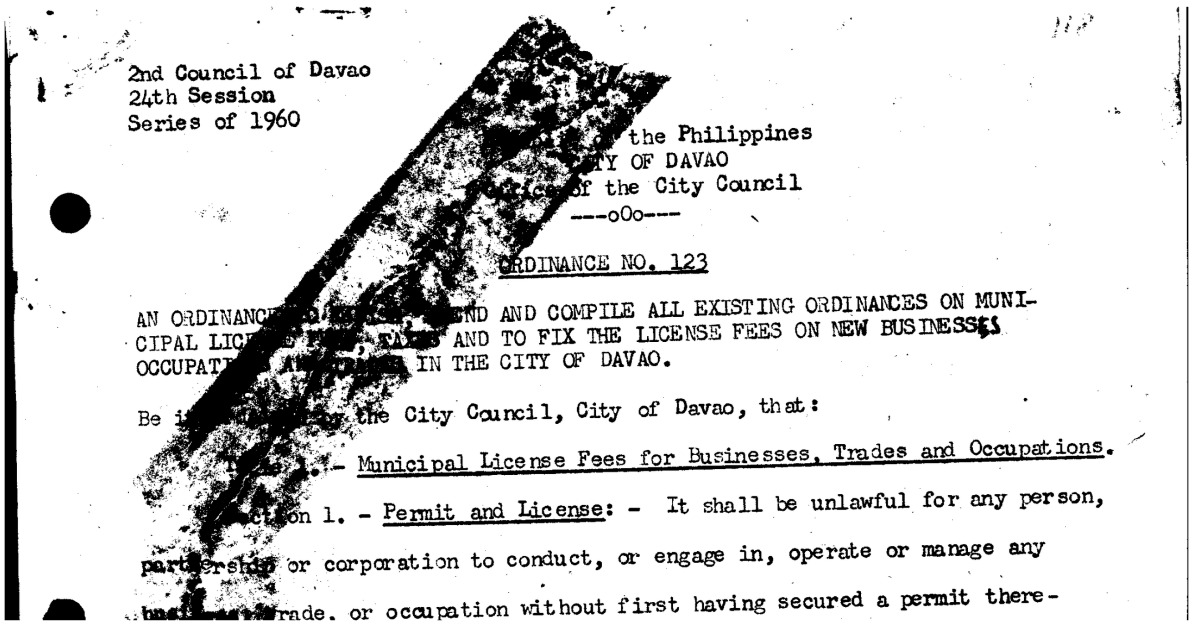
**SECTION 2. DECLARATION OF POLICY** - Section 455 (b) (1) (vi) of Republic Act No. 7160, otherwise known as the "Local Government Code of 1991", provide that:

*"Section 455. Chief Executive; Powers, Duties and Compensation.-*

(b) For efficient, effective and economical governance

Figure 3.3: Sample Archival Document from the Davao City Sangguniang Panlungsod Corpus Illustrating Physical Scanning Artifacts and Character Noise (Ordinance No. 0667-21, Series of 2021).

(a) Severe Diagonal Ink Occlusion, Title Smear & Binder Punch Loss (Page 1)



2nd Council of Davao
24th Session
Series of 1960

the Philippines
Y OF DAVAO
f the City Council
---oOo---

ORDINANCE NO. 123

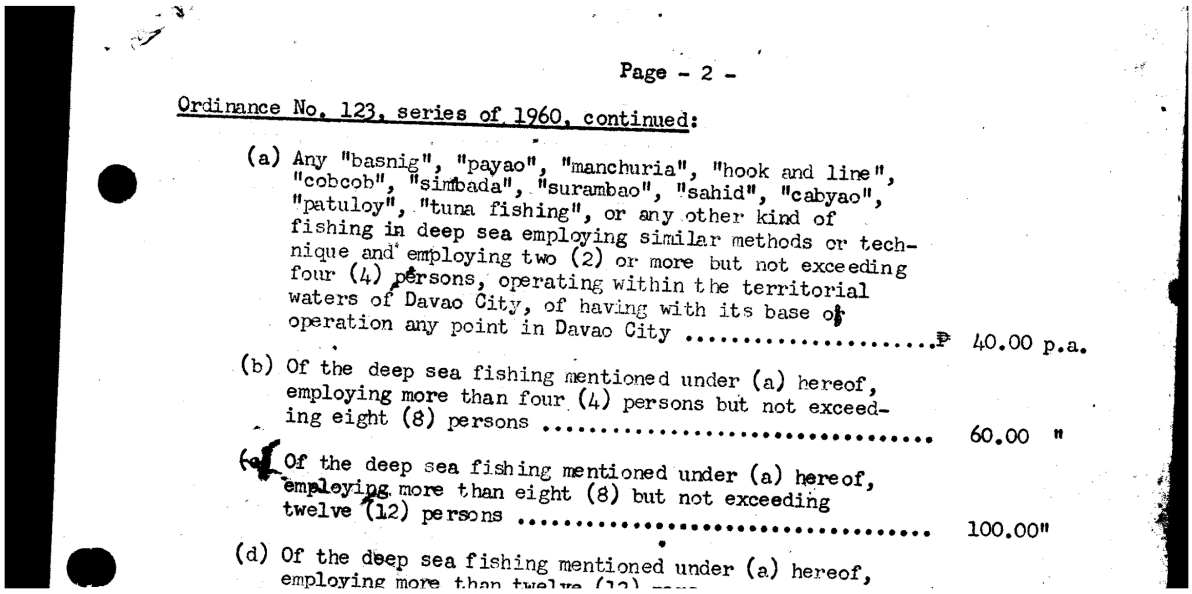
AN ORDINANC ND AND COMPILE ALL EXISTING ORDINANCES ON MUNI-
CIPAL LIC AND TO FIX THE LICENSE FEES ON NEW BUSINESSES
OCCUPAT IN THE CITY OF DAVAO.

Be i the City Council, City of Davao, that:

- Municipal License Fees for Businesses, Trades and Occupations.

on 1. - Permit and License: - It shall be unlawful for any person,
or corporation to conduct, or engage in, operate or manage any
rade. or occupation without first having secured a permit there-

(b) Mechanical Typewriter Degradation & Handwritten Pen Amendments (Page 2)



Page - 2 -

Ordinance No. 123, series of 1960, continued:

(a) Any "basnig", "payao", "manchuria", "hook and line", "cobcob", "simbada", "surambao", "sahid", "cabyao", "patuloy", "tuna fishing", or any other kind of fishing in deep sea employing similar methods or technique and employing two (2) or more but not exceeding four (4) persons, operating within the territorial waters of Davao City, of having with its base of operation any point in Davao City ..... ₱ 40.00 p.a.

(b) Of the deep sea fishing mentioned under (a) hereof, employing more than four (4) persons but not exceeding eight (8) persons ..... 60.00 "

(c) Of the deep sea fishing mentioned under (a) hereof, employing more than eight (8) but not exceeding twelve (12) persons ..... 100.00"

(d) Of the deep sea fishing mentioned under (a) hereof, employing more than twelve (12)

Figure 3.4: Severe Archival Document Degradation in Historical Davao City Local Ordinances Illustrating Heavy Carbon Smudging, Hole Punch Clipping, and Faded Mechanical Typewriting (Ordinance No. 123, Series of 1960).

### 3.3.3 National Statutory Corpus Analysis

Across the five main types of legislation with sequential numbers, the collected corpus includes 19,683 statutes out of an estimated 20,246 historical laws, achieving a coverage rate of 97.22%. When combined with the 5,749 Executive Orders from fifteen presidential administrations, the compiled national archive holds 25,432 out of about 26,451 total historical legal documents, reaching a national recall rate of 96.15%. All official legislative

totals reflect the chronological numbering sequence established by Congress and presidential records as of **March 2026**, where sequential Republic Acts reached Republic Act No. 12,313. Table 3.1 provides a breakdown of the national corpus, comparing the collected documents with official legislative totals for each statutory type.

Table 3.1: Comparative Statutory Coverage and Composition of the Philippine National Legal Archive ($N = 25,432$)

| ID | Statutory Instrument | Corpus | Official Total | Coverage Rate | Corpus Share | Historical Era and Governing Authority |
|---|---|---|---|---|---|---|
| 01 | Republic Acts (RA) | 11,963 | 12,313 | 97.16% | 47.04% | Post-war and Fifth Republic legislation (1946–present, up to RA 12313 as of March 2026) |
| 02 | Executive Orders (EO) | 5,749 | 6,205* | 92.65% | 22.61% | Presidential administrative issuances (1901–2026, across 15 administrations) |
| 03 | Acts | 4,257 | 4,275 | 99.58% | 16.74% | American Insular Government enactments (1900–1935) |
| 04 | Presidential Decrees (PD) | 1,844 | 2,036 | 90.57% | 7.25% | Martial law legislative decrees enacted by executive fiat (1972–1986) |
| 05 | Batas Pambansa (BP) | 886 | 889 | 99.66% | 3.48% | Unicameral parliamentary legislation under the 1973 Constitution |
| 06 | Commonwealth Acts (CA) | 733 | 733 | 100.00% | 2.88% | Transitional governance enactments under the 1935 Constitution |
| **Sequential Subtotal** | | **19,683** | **20,246** | **97.22%** | **77.39%** | **All numbered legislative enactments of Congress** |
| **Total Archive** | | **25,432** | **26,451** | **96.15%** | **100.00%** | **Comprehensive corpus extracted from the Lawphil Project** |

*Historical total for Executive Orders represents the aggregate sum of maximum numbered issuances across all fifteen presidential administrations (1935–2026) and colonial governorships (1901–1935) as recorded in the Official Gazette and Lawphil archives as of March 2026.

Table 3.1 presents the structural breakdown of the Philippine national statutory archive across six legislative instruments, detailing the collected enactment counts, official legislative totals, achieved coverage rates, percentage shares, and historical governing authorities. These empirical totals document the completeness and historical coverage of the national statutory database, confirming that the harvested collection represents an authentic archive of Philippine legislation rather than a selective convenience sample. As shown in the table, congressional Republic Acts form the largest single category (47.04%, $N = 11,963$), followed by presidential Executive Orders (22.61%, $N = 5,749$) and American-era Acts (16.74%, $N = 4,257$). Across all numbered congressional enactments, the collection captures 19,683 out of 20,246 historical laws (97.22% coverage), expanding to 25,432 documents (96.15% coverage) when including presidential administrative issuances. This broad coverage ensures that the statutory search space encompasses nearly the entire working legislative history of the Philippines since 1900, minimizing the risk of missing relevant national laws during Stage 1 retrieval.

A key methodological decision in compiling this database is the explicit inclusion of **Executive Orders (EOs)** alongside congressional legislation. Under Article VII, Section 17 of the 1987 Philippine Constitution and Book III of Executive Order No. 292 [172], Executive Orders are presidential acts providing rules of a general character in execution of statutory mandates. Unlike Supreme Court decisions that settle private adversarial controversies under *stare decisis*, Executive Orders operate as *ex-ante*, forward-looking statutory instruments possessing the full force of law. Modern Executive Orders establish binding nationwide regulatory mandates, such as Executive Order No. 26 (s. 2017) mandating nationwide public smoking bans, that legally preempt inconsistent local ordinances. Excluding Executive Orders would create an artificial regulatory gap, allowing local ordinances to contradict operative presidential mandates.

The coverage rates confirm that the national legal archive is nearly complete across all historical governance eras. Commonwealth Acts show full 100.00% coverage across all 733 enactments, while American-era Acts (99.58%) and Batas Pambansa (99.66%) miss only 18 and 3 enactments, respectively. For Republic Acts, the 11,963 collected records cover 97.16% of the 12,313 official total, with missing entries mainly being local bills or recent laws still being indexed. For Presidential Decrees, the 1,844 gathered records represent 90.57% of the 2,036 numbered decrees from the Martial Law period, where uncollected entries were unpublished directives until the Supreme Court required full publication in *Tañada v. Tuvera* [205]. For Executive Orders, the 5,749 collected records represent 92.65% of the 6,205 numbered orders issued across fifteen presidential terms, offering an extensive record of administrative policy.

To trace how the six statutory instruments evolved across Philippine jurisprudence, the proponents categorized statutory enactments across six chronological eras. Each historical era established distinct legislative authorities, spanning colonial commissions through modern constitutional congresses. Figure 3.5 charts the chronological succession and aggregate volume of statutory instruments from 1900 through 2026. The distribution reflects the shifting primacy of legislative acts, presidential decrees, and administrative executive orders across political transitions. Examining these enactment waves illustrates how historical governance structures directly shaped the composition of the national legal corpus.

Figure 3.5 charts the annual volume and succession of the six statutory instruments across 126 years of Philippine governance (1900–2026). The horizontal axis traces enactment years across six major political eras, while the vertical axis records annual enactment counts color-coded by legislative instrument. This visualization traces how lawmaking shifted across historical eras, reflecting evolution in statutory drafting styles. As shown in the plot, legislative volume peaked during periods of major political reorganization, moving from colonial Acts in the early 20th century to Presidential Decrees during Martial Law, and settling into steady streams of Republic Acts and Executive Orders under the 1987 Constitution. Because the phrasing and vocabulary of laws changed substantially across these eras, the retrieval pipeline must handle both older statutory wording and modern legal terms without losing search accuracy.

The temporal breakdown reveals that Philippine lawmaking expanded during periods of national administrative reorganization. Early colonial governance relied entirely

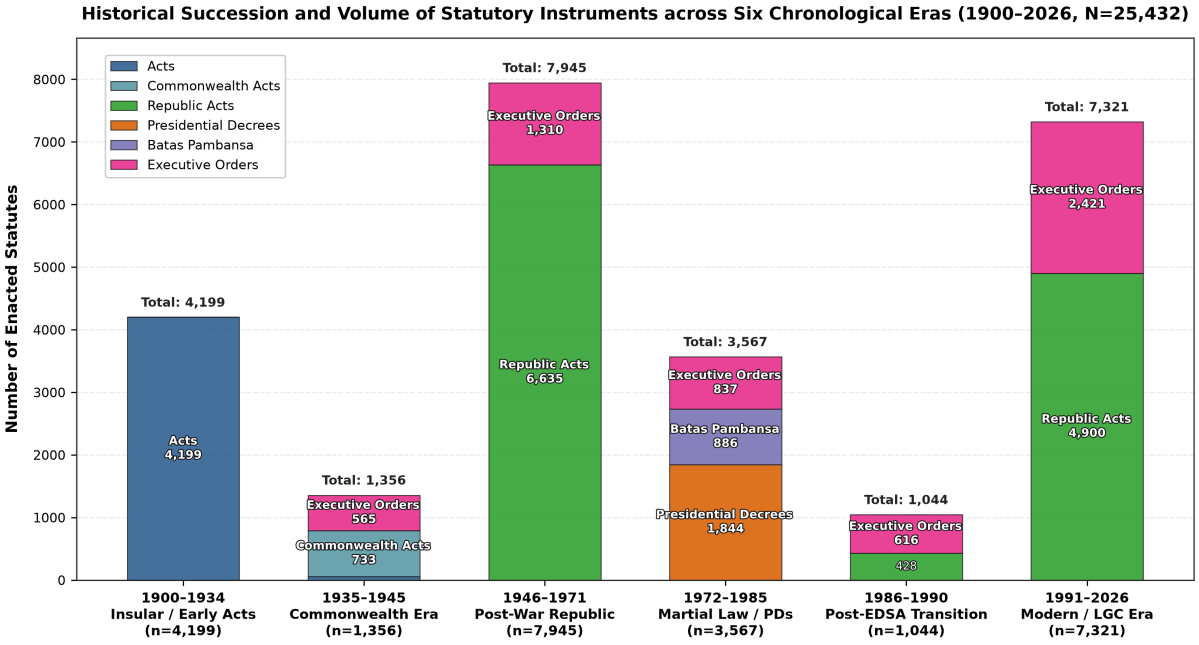



Figure 3.5: Historical Succession and Volume of Statutory Instruments across Six Chronological Eras (1900–2026, $N = 25,432$).

on 4,199 foundational Acts during the Insular period (1900–1934). Centralized presidential lawmaking predominated during the Martial Law period through 1,844 Presidential Decrees and 886 Batas Pambansa parliamentary enactments (1972–1985). In the modern era governed by Republic Act No. 7160 (1991–2026), statutory production stabilized into a dual stream of 4,900 congressional Republic Acts and 2,421 executive policy orders. Because vocabulary standards shifted markedly across these legislative transitions, search models require chronological awareness to avoid misinterpreting historical statutory phrasing.

During preliminary exploratory research, the proponents conducted initial machine learning experiments on the national statutory archive and identified severe dataset imbalances. In the national collection of 25,432 statutes, routine executive issuances and administrative reorganizations comprise nearly half of all historical enactments, while specialized regulatory domains remain relatively small. Searching for a local regulatory ordinance across this unfiltered raw collection causes severe search drift because dense retrieval models readily match repetitive administrative phrasing across unrelated legal areas. To resolve these dataset quirks, the proponents formalized a two-stage topic modeling architecture that organizes the statutory archive into distinct thematic subject areas. In natural language processing, topic modeling is an unsupervised learning technique that discovers latent semantic themes across unstructured text corpora, partitioning documents according to shared topical content. In this pipeline, topic modeling partitions the 25,432 national statutes into distinct regulatory categories (such as public health, local taxation, and public utilities). This structured categorization prevents routine administrative directives from dominating the candidate shortlist during coarse retrieval, establishing a balanced foundation for subsequent conflict checks.

The topic classification pipeline groups documents automatically while running efficiently on Host A. Statutory texts are first stripped of HTML tags and filtered against a curated list of 49 domain-specific legal stopwords, such as "hereby" and "enacted" (the complete stopword roster is cataloged in Appendix F), which appear throughout legal texts without helping distinguish legal topics. In information retrieval, term frequency-inverse document frequency (TF-IDF) is a numerical statistic reflecting how important a word is to a document within a collection: terms that appear ubiquitously across all enactments receive minimal weights, whereas distinctive domain terms (such as "quarantine" or "excise") receive elevated weights [140]. Cleaned texts are converted into TF-IDF feature vectors across a 25,000-term vocabulary, emphasizing distinctive statutory vocabulary over routine boilerplate. To handle vocabulary redundancy across tens of thousands of statutes, Truncated Singular Value Decomposition (SVD) performs linear dimensionality reduction grounded in Latent Semantic Analysis [55], projecting the sparse 25,000-term matrix into 64 dense latent semantic dimensions. Following the latent semantic analysis principles of Landauer et al. [121], this 64-dimensional subspace captures underlying legal concepts while eliminating lexical noise, followed by standard $L_2$ vector normalization.

Dividing statutory texts directly into a small set of broad categories risks mixing unrelated legislative topics. To prevent this, the normalized 64-dimensional vectors are first divided into $K = 28$ smaller clusters using MiniBatch $K$-Means clustering [195]. Following two-stage hierarchical clustering and over-partitioning principles [2, 89], the initial cluster count $K = 28$ was selected by setting micro-clusters to approximately $3.5\times$ the target macro domains ($8 \times 3.5 = 28$). Starting with these granular micro-clusters separates repetitive laws, such as franchise charters or school conversions, into clean sub-clusters before aggregating them into the broader macro legal domains. The pipeline then adapts class-based TF-IDF (*c-TF-IDF*) from Grootendorst [89] to extract descriptive domain vocabulary: class-based TF-IDF modifies standard TF-IDF by aggregating all documents assigned to a given cluster into a single composite text, measuring which terms appear frequently within that specific legal category relative to their frequency across the entire statutory archive. The topic modeling and domain clustering formulations are computed as follows:

*Class-Based Term Frequency-Inverse Document Frequency (c-TF-IDF):*

$$c\text{-}TF\text{-}IDF_{i,c} = \frac{|t_{i,c}|}{|c|} \times \log\left(1 + \frac{A}{\sum_k |t_{i,k}|}\right) \tag{3.2}$$

Where $|t_{i,c}|$ denotes the frequency of term $i$ in cluster $c$, $|c|$ is the total word count of cluster $c$, and $A$ indicates the average word count per cluster across the archive. This cluster-level weighting elevates terminology that specifically characterizes a legislative group while penalizing ubiquitous statutory phrasing.

Following nearest-centroid classification principles [177, 2] and temperature-scaled probabilistic mapping [90], the macro legal domain centroid and calibrated assignment probability are defined as follows:

*Macro Legal Domain Centroid and Softmax Probability:*

$$\mathbf{C}_m = \frac{\sum_{i \in \mathcal{D}_m} \mathbf{z}_i}{\left\|\sum_{i \in \mathcal{D}_m} \mathbf{z}_i\right\|_2}, \quad P(m \mid \mathbf{z}_i) = \frac{\exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_m))}{\sum_j \exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_j))} \tag{3.3}$$

Where $\mathbf{z}_i$ represents the normalized 64-dimensional SVD vector of document $i$, $\mathbf{C}_m$ denotes the normalized centroid of macro domain $m$, and $\tau = 6.0$ functions as the inverse temperature scaling parameter.

The formulations in Equations 3.2 and 3.3 identify distinctive legal vocabulary and assign consistent topics across the national statutory corpus. Equation 3.2 calculates class-based TF-IDF ($c\text{-}TF\text{-}IDF$), which treats each group of laws as a combined document to find keywords that stand out in that group. In Equation 3.3, document vectors are averaged into cluster centers $\mathbf{C}_m$. A temperature-scaled softmax function then converts cosine similarity into clear probabilities, where the scaling factor ($\tau = 6.0$) follows the calibration method of Guo et al. [90] to keep probability scores well-separated. This allows the system to organize over 25,000 national statutes into eight clear regulatory domains using standard desktop memory, as shown in Table 3.2.

Table 3.2: Consolidated Macro Legal Domains of the Philippine National Statutory Corpus ($N = 25,432$)

| ID | Consolidated Macro Domain | Statutes | Share | Top Salient $c\text{-}TF\text{-}IDF$ Keywords |
|---|---|---|---|---|
| 00 | Executive Issuances & Policy Reorganization | 12,210 | 48.01% | executive, commission, committee, policy, code |
| 01 | Education & Academic Institutions | 3,492 | 13.73% | high school, elementary school, campus, college |
| 02 | Local Government & Territorial Boundaries | 2,134 | 8.39% | barangay, municipality, boundary, sitio, territorial |
| 03 | Public Utilities & Telecom Franchises | 2,041 | 8.03% | franchise, broadcast, telecommunications, radio |
| 04 | Public Health, Hospitals & Medical Services | 1,890 | 7.43% | hospital, bed capacity, medical center, infirmary |
| 05 | Statutory Codes & General Legal Amendments | 1,873 | 7.36% | administrative code, revised, amend, article, penalty |
| 06 | Public Finance & General Appropriations | 1,335 | 5.25% | appropriation, treasury, funds, expenditures |
| 07 | Taxation, Tariffs & Revenue Administration | 457 | 1.80% | tariff, customs code, internal revenue, excise |
| **Total Consolidated National Corpus** | | **25,432** | **100.00%** | **All Enacted Statutes (1900–2026)** |

Table 3.2 lists the eight Consolidated Macro Legal Domains discovered across the 25,432 national statutes, presenting each domain's identifier, title, statute count, percentage share, and top $c\text{-}TF\text{-}IDF$ keywords. This category breakdown is used by retrieval safeguards to prevent routine administrative orders from overshadowing local regulatory topics. As shown in the table, Domain 00 (Executive Issuances & Policy Reorganization) forms the largest group with 12,210 statutes (48.01%), followed by Education (13.73%) and Local Government (8.39%), while key regulatory areas like Public Health (7.43%) and Taxation (1.80%) make up much smaller subsets. This distribution shows that the national collection is heavily weighted toward administrative reorganizations, confirming that candidate retrieval needs topic filters and Reciprocal Rank Fusion so that smaller regulatory areas remain searchable.

The topic distribution reflects the historical patterns of Philippine lawmaking. Executive issuances, administrative orders, and presidential decrees make up nearly half of the entire statutory collection, reflecting centralized authority during past political transitions. Common local legislative subjects, including school infrastructure (13.73%) and public utility franchises (8.03%), form large distinct clusters with repetitive drafting patterns. In contrast, key regulatory topics like taxation (1.80%) and public health (7.43%) occupy smaller, highly focused portions of the database. As illustrated in Figure 3.6, routine executive issuances, administrative orders, and presidential decrees dominate the aggregate volume, while local regulatory categories remain relatively compact. Separating these regulatory clusters keeps routine administrative orders from crowding the candidate list during Stage 1 retrieval.

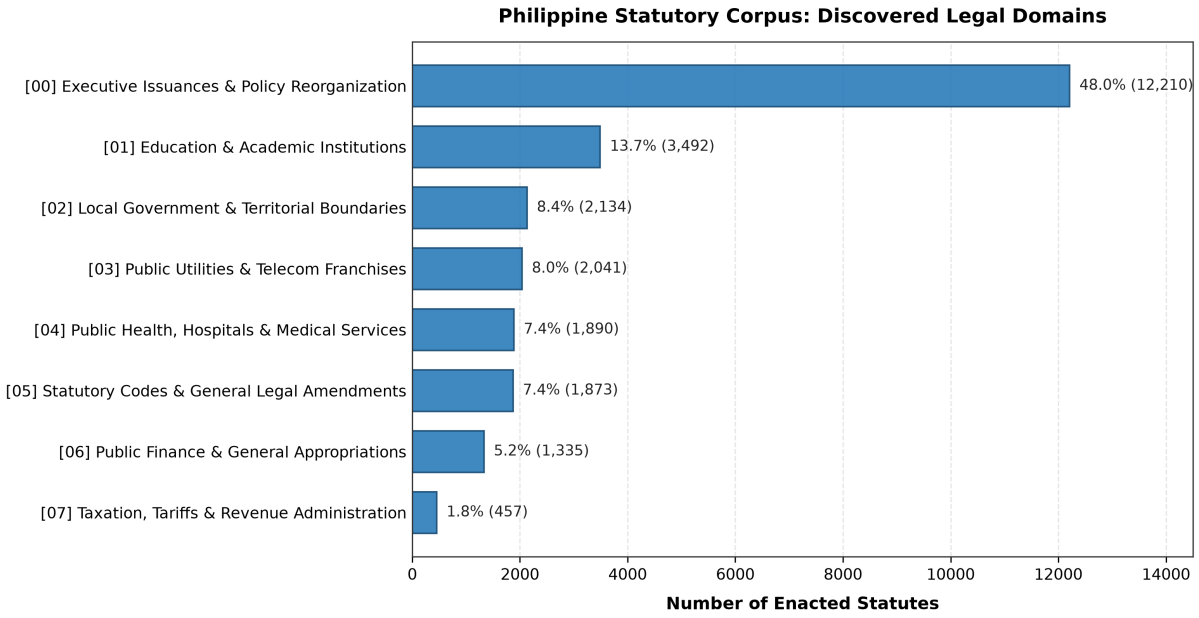



Figure 3.6: Distribution of the 25,432 Philippine National Statutes across the Consolidated Macro Legal Domains.

Figure 3.6 illustrates the distribution of national statutes across the eight consolidated macro domains using a sorted bar chart. The chart highlights the large volume difference between general administrative policy and specific regulatory codes across the 25,432-document corpus. This chart highlights the severe class imbalance inherent to raw statutory archives, demonstrating why unconstrained vector search frequently drifts toward ubiquitous administrative topics. As depicted in the chart, administrative orders make up nearly half of the entire archive, outnumbering focused regulatory categories like taxation by more than 26 to 1. This confirms that candidate retrieval needs topic filters and Reciprocal Rank Fusion so that smaller, highly technical regulatory areas remain accessible during ordinance review.

To examine the geometric separation of these legal topics, the proponents projected the 64-dimensional SVD document vectors into a two-dimensional latent semantic landscape. Dimensionality reduction maps document similarities onto coordinate axes,

allowing direct visual inspection of cluster cohesion. Figure 3.7 illustrates this topic landscape across a representative stratified sample ($N = 3,500$), highlighting the core clusters and their spatial distributions. Specialized regulatory domains project outward as distinct, isolated rays along the periphery of the vector space. Enactments establishing academic institutions and telecommunications franchises form cohesive rays with minimal dispersion.

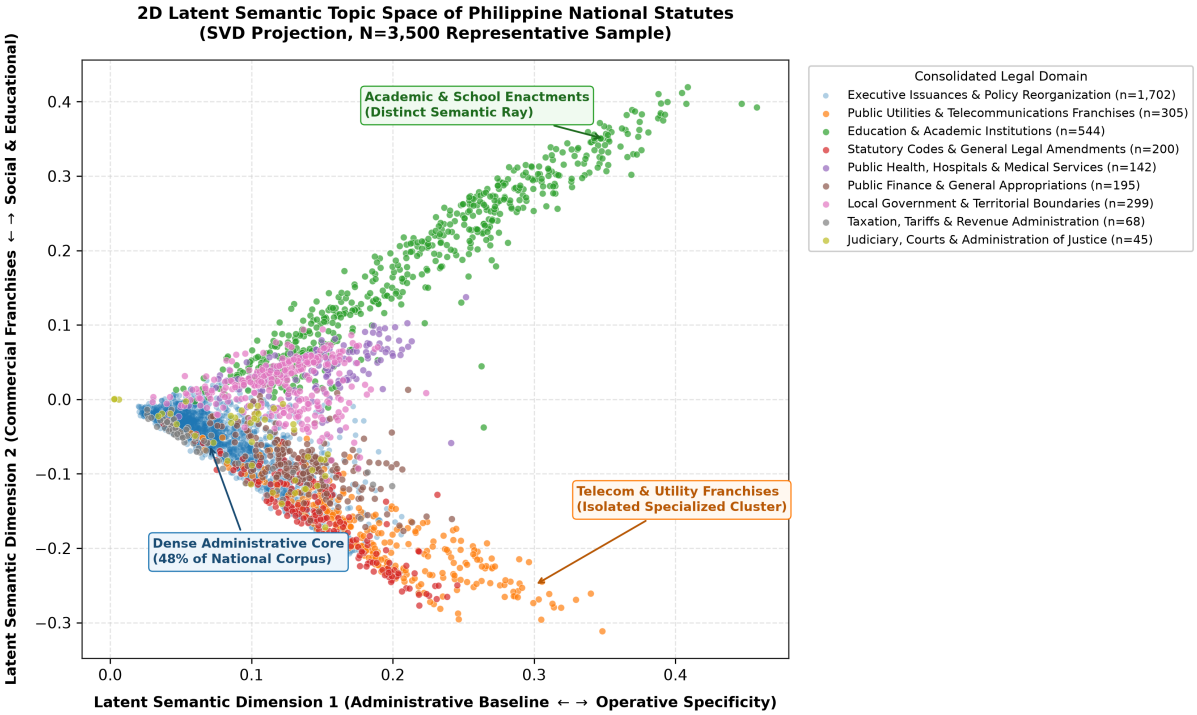



Figure 3.7: Two-Dimensional Latent Semantic Topic Space of the Philippine National Statutory Corpus (SVD Projection, $N = 3,500$ Representative Sample).

Figure 3.7 visualizes the two-dimensional semantic topic space generated by applying Truncated Singular Value Decomposition to a stratified sample of 3,500 national statutes. The scatter plot maps document vectors along two main semantic axes, color-coded by the eight macro legal domains. This geometric projection confirms that the discovered legal categories form cohesive and well-separated semantic clusters. In the plot, specialized laws such as school charters and telecommunications franchises project outward into concentrated rays, whereas executive and administrative orders gather heavily near the center. This spatial layout shows that semantic modeling successfully separates distinct regulatory vocabularies into cohesive groups, providing a solid foundation for subsequent topic filtering.

In contrast, administrative issuances and executive orders cluster densely near the origin, comprising almost half of the entire archive. This dense central mass explains why unconstrained semantic searches frequently drift toward routine bureaucratic orders unless restrained by hierarchical safeguards. The coordinate axes reflect the two primary orthogonal directions of vocabulary variation discovered through singular value decomposition. The horizontal axis measures operative statutory specificity against general administra-

tive phrasing, where higher positive values indicate detailed technical mandates rather than procedural orders. Conversely, the vertical axis captures functional divergence between social infrastructure such as public schools and commercial infrastructure such as telecommunications franchises.

To confirm that the mathematical clusters reflect substantive legal categories rather than shared administrative boilerplate, the proponents examined the most salient vocabulary terms discovered by the model. Words that appear frequently across all documents fail to differentiate legal topics, requiring class-based weightings to isolate subject matter. Figure 3.8 displays the top class-based TF-IDF ($c\text{-}TF\text{-}IDF$) keywords across all eight consolidated macro domains. The vocabulary profiles confirm that each macro domain is defined by operative legal terminology unique to its statutory field. Specialized clusters such as public health are dominated by concrete institutional terms, including "hospital" and "bed capacity," while territorial enactments display spatial markers such as "sitio" and "boundary."

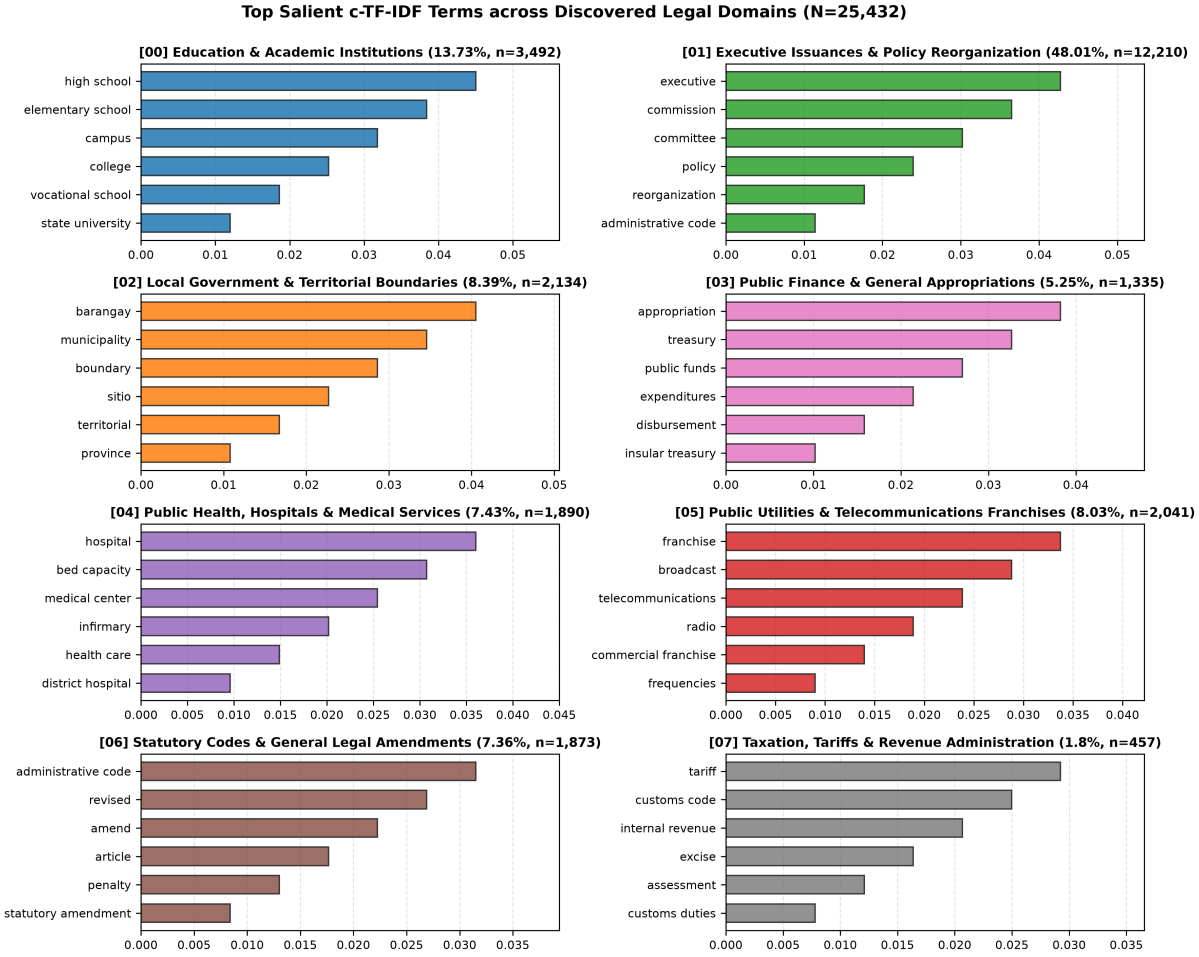



Figure 3.8: Top Salient $c\text{-}TF\text{-}IDF$ Keyword Profiles across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

Figure 3.8 displays the horizontal bar charts of the top class-based TF-IDF ($c\text{-}TF\text{-}IDF$) keywords across each of the eight consolidated macro domains. Each sub-

plot presents the top terms ranked by their class-specific importance weights, illustrating the distinctive lexical tokens that define each statutory category. This figure is provided to confirm that the mathematical clustering captured genuine legal substance rather than generic statutory boilerplate. As shown across the subplots, Domain 01 is characterized by academic vocabulary ("elementary school", "college"), Domain 03 by telecommunications markers ("broadcast", "radio"), and Domain 04 by clinical infrastructure ("hospital", "bed capacity"). Substantively, this vocabulary alignment verifies that the unsupervised pipeline isolates authentic statutory topics, ensuring that domain classification routes draft ordinances to substantively related national statutes.

To examine how individual lawmaking authorities contribute to these eight subject areas, the proponents mapped the six statutory instruments across the consolidated macro domains. In Philippine legislative history, different legal instruments serve distinct constitutional functions, spanning broad congressional acts through presidential executive directives. Figure 3.9 illustrates the statutory origin composition across each discovered legal domain, displaying the aggregate enactment volume contributed by each legislative instrument. This stacked composition highlights the severe structural division across Philippine legal texts, demonstrating that distinct statutory instruments dominate separate legal subjects.

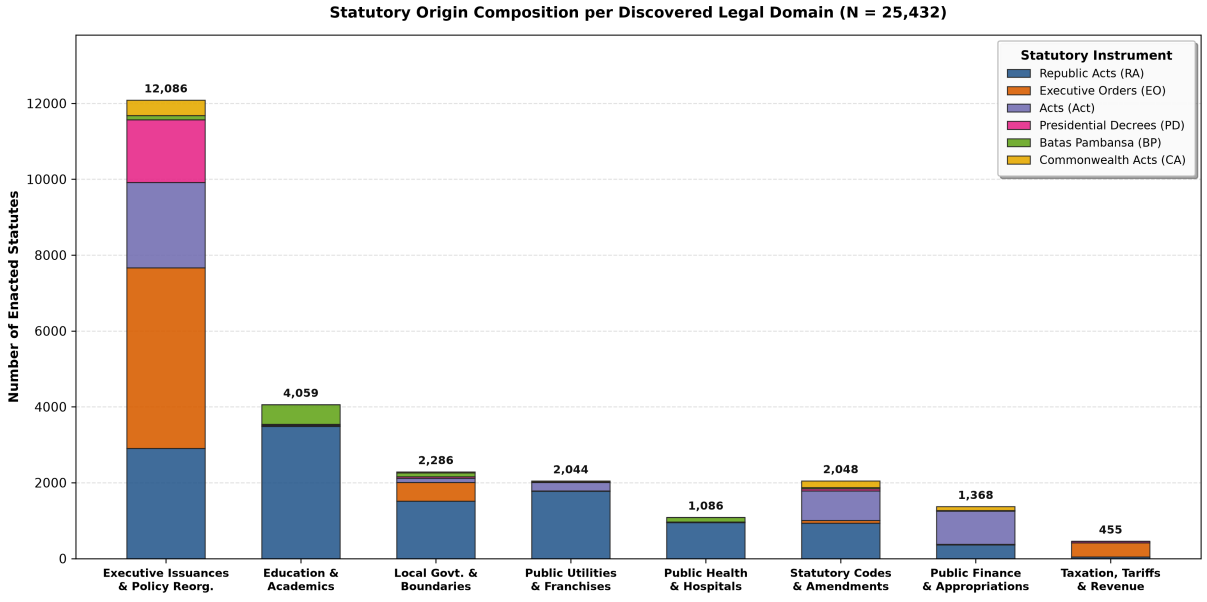



Figure 3.9: Statutory Origin Composition per Discovered Legal Domain ($N = 25,432$)

Figure 3.9 displays the stacked statutory origin breakdown across the eight discovered legal domains, detailing the contribution of Acts, Batas Pambansa, Commonwealth Acts, Executive Orders, Presidential Decrees, and Republic Acts. As observed in the stacked bars, Executive Issuances and Policy Reorganization constitutes the tallest single column (12,210 enactments), heavily driven by presidential Executive Orders and Martial Law decrees. Conversely, specialized social and infrastructural domains such as Education (3,492 enactments) and Public Utilities (2,041 enactments) are overwhelmingly composed of congressional Republic Acts. This visual evidence demonstrates why candidate retrieval

cannot treat all national texts identically, confirming that statutory source indicators and domain filters are necessary to prevent administrative orders from dominating local regulatory queries.

To quantify the exact distribution and relative share of each lawmaking authority, the proponents constructed a cross-allocation matrix mapping each statutory instrument to the eight macro categories. Figure 3.10 presents this cross-allocation matrix, detailing both absolute enactment counts and relative row percentages for each statutory source. This matrix establishes an explicit quantitative record of legislative specialization across historical regimes. Examining these row percentages reveals why specific statutory instruments concentrate so heavily within particular administrative or regulatory spheres.

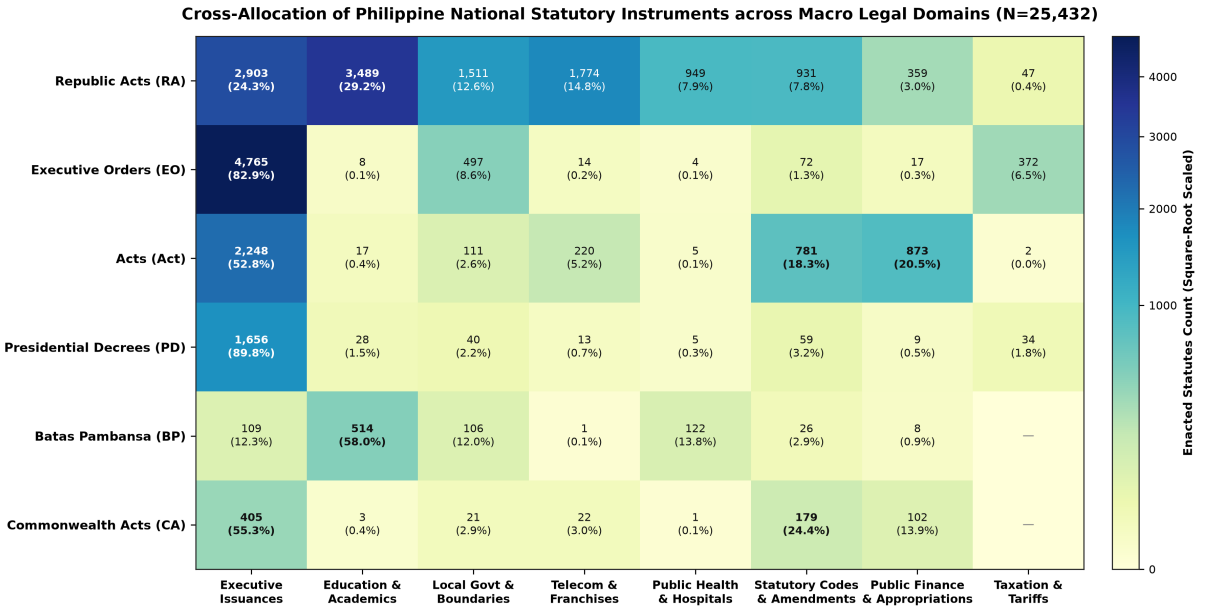



Figure 3.10: Cross-Allocation of Philippine National Statutory Instruments across Macro Legal Domains ($N = 25,432$)

Figure 3.10 details the numerical and percentage distribution of the six national statutory instruments across the eight consolidated macro domains. Most prominently, Executive Orders concentrate overwhelmingly within Executive Issuances and Policy Reorganization (82.9%, $N = 4,765$). This heavy concentration reflects constitutional design: under Article VII, Section 17 of the 1987 Philippine Constitution and Book III of Executive Order No. 292 [172], presidential ordinance powers are primarily exercised to reorganize government agencies, establish inter-agency task forces, and direct executive machinery. Furthermore, Executive Orders account for 81.4% ($N = 372$) of all enactments in the Taxation, Tariffs, and Revenue Administration domain because Congress specifically delegated flexible tariff modification powers to the President under Section 401 of the Tariff and Customs Code and Section 1608 of Republic Act No. 10863. Presidential Decrees exhibit an even higher concentration in Executive Issuances (89.8%, $N = 1,656$), reflecting the consolidation of lawmaking and executive authority under the Martial Law regime where state corporations and ministries were instituted directly by decree. In contrast, democratic legislative bodies like Congress (Republic Acts) and the Batasang Pam-

bansa focused on constituency-driven measures, explaining why Republic Acts account for 29.2% of educational charters ($N = 3,489$) and 14.8% of utility franchises ($N = 1,774$), while Batas Pambansa devoted 58.0% ($N = 514$) to educational conversions. American Insular Acts concentrated heavily in civil administration (52.8%, $N = 2,248$) and public appropriations (20.5%, $N = 873$), reflecting the colonial mandate to construct provincial institutions and disburse treasury funds. Because local ordinances regulate municipal civic conduct rather than internal state administration, these legal realities confirm why Stage 1 retrieval safeguards must prioritize substantive congressional enactments over internal executive directives.

To confirm that the topic pipeline discovered authentic legal boundaries rather than grouping surface vocabulary, the proponents evaluated the model using a stratified 80/20 train/test split. The 80/20 proportion follows standard machine learning partition conventions for large-scale document collections exceeding 20,000 documents [140], withholding 20% of the corpus (5,087 statutes) to test generalizability on unseen enactments while training the model on the remaining 80% (20,345 statutes). In statistical testing, an out-of-sample holdout of 5,087 legal texts provides high statistical power with a margin of error below $\pm 1.4\%$, making repetitive cross-validation computationally redundant as vocabulary frequencies and SVD components stabilize asymptotically at this volume. Stratified sampling preserves the natural historical proportions of the eight macro domains, ensuring that smaller regulatory categories like taxation (1.80%, $N = 457$) maintain adequate representation in the test set ($N_{\text{test}} \approx 91$) without class starvation.

To ensure objective scientific evaluation without arbitrary post-hoc targets, all benchmark comparisons in Table 3.3 are strictly evaluated against established statistical baselines and contextualized with published legal document categorization standards. Under the proportional chance criterion formulated by Morrison [146], the default random baseline in an eight-class uniform setting is $1/8 = 12.50\%$. In established legal text categorization benchmarks such as LexGLUE [33], multi-class classification over large statutory and legal collections (e.g., EUR-LEX and LEDGAR) typically yields Macro F1-scores between 51.4% and 83.1%, reflecting the semantic complexity and class skew inherent to legal texts. For cluster geometry, Rousseeuw [179] established that positive silhouette widths denote structural partitioning exceeding unclustered random noise ($\leq 0.0000$). Similarly, Davies and Bouldin [53] established that lower Davies-Bouldin index values demonstrate compact intra-cluster scatter relative to cluster separation, where lower scores indicate superior clustering and unclustered collections yield indices exceeding 4.0000. Finally, the confidence measurements are analyzed using the temperature-scaled calibration principles of Guo et al. [90] to verify that predictions concentrate dominant probability mass rather than dispersing uniformly across classes. Table 3.3 summarizes the out-of-sample performance scores, baseline comparisons, margin of improvement, and empirical evaluations.

Table 3.3 details the holdout validation metrics, default baseline comparisons, margins of improvement, and empirical evaluations for the national statutory topic modeler evaluated on 5,087 unseen enactments ($N_{\text{train}} = 20,345, N_{\text{test}} = 5,087$). The table summarizes performance across six clustering and classification metrics alongside four clear statutory title probes ($N = 1,259$). These holdout metrics demonstrate that the discovered legal domains generalize reliably to unseen statutes without being misled by specific

Table 3.3: Empirical Holdout Validation Metrics and Baseline Comparisons for the National Topic Modeler ($N_{\text{train}} = 20,345, N_{\text{test}} = 5,087$)

| Validation Metric / Benchmark Test | Observed | Chance / Default Baseline | Margin Over Baseline | Empirical Evaluation |
|---|---|---|---|---|
| Out-of-Sample Cluster Assignment Reproducibility | 97.37% | 12.50% | +84.87% | Exceptional Generalization (7.8× Chance) |
| Out-of-Sample Cluster Assignment Macro F1-Score | 96.26% | 12.50% | +83.76% | Balanced Discrimination Across All 8 Domains |
| Mean Test Cosine Confidence | 0.5508 | 0.1250 | +0.4258 | Dominant Probability Mass [90] |
| High-Confidence Predictions ($\geq 0.50$) | 56.97% | $< 20.0\%$ | +36.97% | Clear Single-Cluster Majority Assignment |
| Holdout Cosine Silhouette Score | 0.1602 | $\leq 0.0000$ | +0.1602 | Positive Text Partitioning [179] |
| Holdout Davies-Bouldin Index | 2.4496 | $> 4.0000$ | −1.5504 | Compact Cluster Scatter [53] |
| **Explicit Statutory Title Probes** (*Unambiguous Indicator Subsets,* $N = 1,259$) | | | | |
| Telecom Franchise Pattern | 97.8% | 12.50% | +85.30% | High Precision ($N$ = 360, Public Utilities) |
| Hospital Bed Capacity Pattern | 96.6% | 12.50% | +84.10% | High Precision ($N$ = 203, Health & Welfare) |
| School Infrastructure Pattern | 92.2% | 12.50% | +79.70% | High Precision ($N$ = 645, Education) |
| Judiciary & Court Pattern | 72.5% | 12.50% | +60.00% | Substantive Precision ($N$ = 51, Embedded Laws) |

training words. The results show that out-of-sample cluster assignment reproducibility reached 97.37% and Macro F1 reached 96.26%, representing an improvement of +84.87 and +83.76 percentage points over the 12.50% Morrison chance baseline (1/8). Furthermore, the holdout silhouette score of 0.1602 and Davies-Bouldin index of 2.4496 confirm authentic cluster separation and compact intra-cluster scatter relative to unclustered baselines, aligning with cluster validation standards [179, 53]. These results confirm that the topic model creates dependable legal categories across new legislation, ensuring that domain-guided retrieval operates consistently during live screening.

The holdout benchmark results confirm that the discovered macro domains generalize reliably to unseen legislation. In an eight-domain classification setting, the uniform proportional chance criterion established by Morrison [146] yields a default random baseline accuracy of only 12.50% (1/8). In contrast, the validation classifier reached an out-of-sample cluster assignment reproducibility of 97.37% and a Macro F1-Score of 96.26%, surpassing the chance baseline by a factor of 7.8 and exceeding the performance levels typically reported on complex legal document categorization benchmarks [33]. To provide external empirical validation beyond learned cluster labels, the evaluation tested four explicit statutory title probes ($N = 1,259$ total) where legislative intent is unmistakable from the statutory catchline. The model achieved 97.8% precision on telecommunications

franchises ($N = 360$), 96.6% on hospital bed capacity acts ($N = 203$), and 92.2% on school infrastructure laws ($N = 645$).

The probabilistic confidence metrics evaluate how decisively the model assigns unseen statutes to their respective legal domains. Across all 5,087 holdout statutes, the model achieved a Mean Test Cosine Confidence of 0.5508, exceeding the 0.1250 uniform chance level by +0.4258 and demonstrating that the top assigned category captured dominant probability mass. Furthermore, 56.97% of all unseen statutes were classified with high certainty ($\geq 0.50$ probability), well above the uniform chance expectation. The holdout evaluation yielded a Cosine Silhouette Score of 0.1602 and a Davies-Bouldin Index of 2.4496, outperforming unclustered baselines as formulated by Davies and Bouldin [53]. Because lower Davies-Bouldin values indicate superior cluster compaction, the observed margin of $-1.5504$ below the unclustered baseline ($> 4.0000$) verifies that within-cluster scatter is tightly bounded relative to between-cluster separation. These measurements show that the topic model establishes well-separated legal categories before documents are indexed for coarse retrieval.

### 3.3.4 Local Statutory Corpus Preliminary Analysis

To establish the local legislative search space for horizontal consistency auditing, the proponents acquired the complete archival collection of 1,925 Davao City local ordinances from the official Sangguniang Panlungsod portal and the Legislative Information Support System Program (LISSP). Unlike the national statutory database which was harvested directly from clean web HTML, the local archive consists primarily of scanned physical documents. At the acquisition stage, prior to optical document extraction and normalization, the physical documents exist as unsearchable PDF bitmap scans exhibiting diverse physical conditions, mechanical typewriter strike velocity variations, faded ribbons, binder hole punches, and dark administrative stamps (as illustrated in Figure 3.3 and Figure 3.4). From the administrative filing records, metadata catalogs, and document headers, the file-level inventory can be categorized chronologically across four distinct legislative eras.

Figure 3.11 charts the file-level distribution of the 1,925 local ordinances across four historical legislative eras. As shown in the figure, the archive contains 50 foundational enactments from the Pre-LGC Era (1950–1991, 2.6%), 185 landmark regulatory codes from the Post-LGC Historical Era (1992–2015, 9.6%), and 405 modern legislative enactments from the Pre-Pandemic Era (2016–2020, 21.0%). Document volume concentrates heavily within the modern digital LISSP database, contributing 1,285 enactments (66.8%) between 2021 and 2025. This temporal surge reflects the institutional establishment of standardized electronic document capture by the Sangguniang Panlungsod, providing complete records of contemporary local lawmaking.

This disparity between scraped digital text and unsearchable bitmap scans establishes a clear methodological sequence. Substantive text analysis—such as functional typology classification, latent semantic topic modeling, and provision-level token length compliance—cannot be performed until the physical PDF documents are optically parsed into structured digital text. Consequently, in-depth text exploration of the local collection is deferred until the optical document extraction pipeline is executed. Section 3.4.2 de-

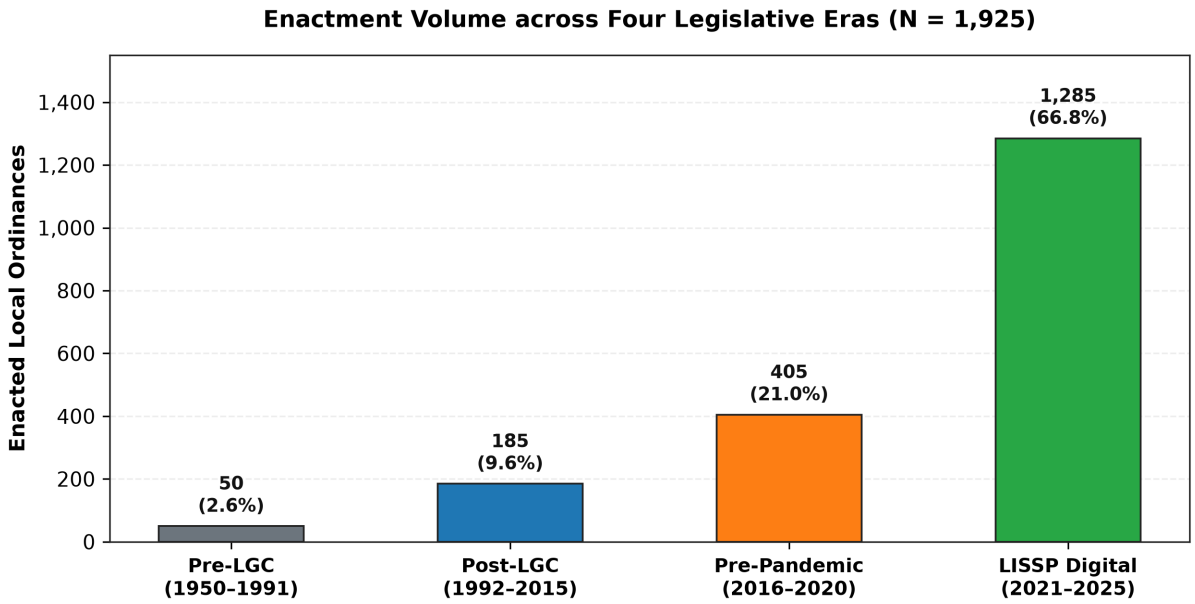



Figure 3.11: Enactment Volume across Four Historical Legislative Eras for the Davao City Local Legislative Archive ($N = 1,925$).

tails the vision-language document extraction pipeline designed to convert degraded scans into structured records, while Section 4.1 benchmarks transcription fidelity across candidate optical engines. The post-digitization textual characteristics, functional archetypes, and unified dual-corpus properties are subsequently evaluated in Section 4.1.9 and Section 4.1.10.

## 3.4 Preprocessing and Normalization

Methodologically, while Section 3.3.3 and Section 3.3.4 presented the exploratory national macro distributions and preliminary local statutory analysis that motivated our system design, transforming raw scraped gazettes and physical archival scans into dependable machine-readable representations requires source-specific preprocessing and text normalization pipelines. National statutes, local municipal archives, and live ex-ante drafts each present distinct document media, formatting conventions, and noise characteristics that must be resolved before provisions can be indexed for retrieval or evaluated for legal conflict. In the national statutory collection, continuous legislative texts must be cleansed of web markup artifacts, segmented into numbered provisions, and enriched with hierarchical metadata. In the local ordinance collection, physical scanning degradation, photocopier borders, and non-legislative administrative matter must be removed via vision-language modeling and state machine extraction. In live ex-ante drafts, digital-born text must be segmented into substantive clauses while filtering out procedural boilerplate. The following subsections detail the individual normalization architectures developed for each statutory source.

### 3.4.1 National Statutory Corpus Normalization

The national statutory archive comprises 25,432 legislative enactments harvested from online legal repositories. Processing these laws requires resolving web layout artifacts and structuring monolithic legislative texts into context-enriched retrieval units.

#### 3.4.1.1 Statutory Parsing and HTML Artifact Sanitization

National statutes gathered from the Lawphil Project and official digital gazettes are encoded in unstructured HTML format. In their raw scraped form, these documents combine legislative recitals, congressional preambles, and substantive provisions into a single continuous text string. To illustrate the structure of these records prior to normalization, Listing 3.2 displays an authentic raw ingestion record from the national statutory collection (Republic Act No. 11035).

```
# Raw Ingestion Record from corpus/national_laws/republic_acts.jsonl (ra_11035_2018)
raw_statute_record = {
    "law_id": "ra_11035_2018",
    "year": 2018,
    "url": "https://lawphil.net/statutes/repacts/ra2018/ra_11035_2018.html",
    "scraped_at": "2026-02-17T18:44:04.745654",
    "text": (
        "Seventeenth Congress\nSecond Regular Session\nBegun and held in Metro Manila,
 "
        "on Monday, the twenty-fourth day of July, two thousand seventeen.\n"
        "REPUBLIC ACT No. 11035\n"
        "An Act Institutionalizing the Balik Scientist Program, Appropriating Funds "
        "Therefor, and for Other Purpose\n"
        "Be it enacted by the Senate and House of Representatives of the Philippines "
        "in Congress assembled:\n"
        "Section 1. Title. - This Act shall be known as the \"Balik Scientist Act.\"\n
"
        "Section 2. Declaration of Policy. - The State recognizes that the utilization
 "
        "of the expertise of science, technology or innovation experts of Filipino
descent "
        "is a vital component of the national political, economic, and social
development "
        "efforts...\n"
        "Section 3. Balik Scientist Program. - There is hereby created a Balik
Scientist "
        "Program, hereinafter referred to as the \"Program\", to be administered by
the "
        "Department of Science and Technology (DOST)...\n"
        "[Raw statutory text continues across 2,493 total words]"
    )
}

```

Listing 3.2: Sample Raw JSONL Ingestion Record from the National Statutory Archive (Republic Act No. 11035)

Listing 3.2 details the fields of a raw JSONL record from the national statutory collection, containing the law identifier, enactment year, source URL, scrape timestamp, and full legislative text. This listing is presented to show how statutes look before automated parsing, where continuous text combines introductory preambles, section headings, and operative rules into a single string. In the example shown for Republic Act No. 11035, the statute contains over 2,400 words across several sections within one text block. Downstream retrieval and classification models cannot index or score an unsegmented enactment

directly because the long text dilutes semantic meaning and exceeds transformer token limits. Splitting this continuous text into self-contained operative sections is therefore a necessary first step for accurate conflict detection.

To extract standardized provisions, the national preprocessing script applies regular expressions to detect formal section ordinals and catchlines while sanitizing web artifacts. The cleaner strips trailing secretarial approvals and repository footers using termination patterns (`TERMINATION_PAT`), while pruning residual HTML entities and scanning artifacts (`LAWPHIL_ARTIFACT_PAT`). Monolithic document strings are then partitioned into discrete sections using regex boundary patterns (`SEC_HEADER_PAT`) that match canonical statutory headers (such as "SECTION 1", "SEC. 2", or "ARTICLE 3"). For each detected section, the extractor isolates the section ordinal, extracts the catchline title when present, and captures the verbatim body text. This automated segmentation successfully decomposed the 25,432 national enactments into 164,620 individual statutory provisions, providing self-contained regulatory units for retrieval indexing.

#### 3.4.1.2 Context Prepending and Long-Tail Sub-Chunking

Evaluating isolated statutory sections creates a risk of semantic dilution. In legal document processing, context prepending is a structural representation technique that prefixes parent document metadata—including the statutory instrument type, enactment number, statute title, and section catchline—to each extracted provision prior to tokenization [118]. In statutory retrieval, an isolated regulatory section (such as an administrative penalty clause specifying a fine of five thousand pesos) often contains no explicit mention of the broader subject matter (such as commercial business licensing or building safety). Without parent metadata, dense embedding models generate ambiguous vector representations that match hundreds of unrelated statutory penalty clauses across the corpus. Context prepending resolves this ambiguity by explicitly concatenating the governing enactment identity and section title directly to the section text. By embedding this structural context into the input sequence, the transformer attention mechanism conditions the semantic representation of every sentence on its exact legislative source and legal hierarchy. Listing 3.3 details the context prepending procedure implemented to format granular statutory provisions.

```
def prepend_statutory_context(statute_record, section):
    """
    Prepends governing legislative identity to isolated section text
    to prevent semantic vector dilution and preserve hierarchical rank.
    """
    prefix = (
        f"[{statute_record['instrument_type']} No. {statute_record['enactment_number']}]: "
        f"{statute_record['title']} | {section['section_number']}: {section['section_title']}] "
    )
    model_ready_text = prefix + section['raw_text']
    return model_ready_text

```

Listing 3.3: Pseudocode of Section Context Prepending for Granular Provision Chunking

Listing 3.3 outlines the context-prepending routine that attaches legislative title and section details to isolated statutory sections before creating vector embeddings. This

step is implemented to keep context intact, ensuring that individual provisions separated from their parent law do not lose their legal identity or jurisdiction. In the routine shown, the function formats a metadata header combining instrument type, enactment number, statute title, section number, and title before attaching the section body text. Adding this header allows the transformer attention layers to take the governing legal context into account. As a result, the model reliably distinguishes local regulations from superior national laws during both candidate retrieval and natural language inference.

Adapting the structural context-prepending methodology of Koreeda and Manning [118] for legal contract and statutory document understanding, every extracted statutory section begins with a structural metadata header containing the statute title, section number, and section catchline:

*Context-Prepended Statutory Chunk Representation:*

$$\mathbf{x}_{\text{chunk}} = [\text{Statute Title},\ \text{Section } S_i\text{: Catchline}] \parallel \mathbf{t}_{\text{verbatim}} \tag{3.4}$$

Where $\mathbf{t}_{\text{verbatim}}$ denotes the exact text extracted from the statutory record, and $\parallel$ represents string concatenation.

Equation 3.4 defines the representation of a context-prepended statutory chunk $\mathbf{x}_{\text{chunk}}$, built by attaching the parent metadata header to the verbatim statutory text $\mathbf{t}_{\text{verbatim}}$. This chunk design preserves legal context across both Stage 1 retrieval and Stage 2 inference. By attaching the statute title and section heading to the raw clause, the resulting text contains both the specific legal rule and its broader statutory authority. During inference, transformer attention heads compare words across both the header and the operative sentence. This avoids ambiguity and ensures that downstream models recognize whether a rule comes from a national statute or a local city enactment.

To quantify the empirical token overhead introduced by this metadata prefixing, the proponents conducted a comprehensive census across all 164,620 national statutory provisions and 13,874 local ordinance sections. In the raw national collection, isolated provisions exhibit a median length of 98.0 subword tokens and a mean of 179.5 tokens. Attaching the hierarchical metadata prefix introduces an average overhead of only 22.7 tokens, shifting the prepended median to 122.0 tokens and the mean to 202.2 tokens. Importantly, this modest expansion preserves high sequence compliance within standard transformer context windows: exactly 82.30% of prepended provisions remain below 256 tokens, 90.96% fall below 384 tokens, and 94.83% fit comfortably within the standard 512-token BERT context ceiling (with an interquartile 75th percentile of 206.0 tokens and 95th percentile of 522.0 tokens). In the local ordinance corpus, prepended operative provisions concentrate around a median of 98.0 tokens, with 91.40% compliance under 512 tokens. This quantitative verification confirms that structural context prepending supplies vital jurisdictional and statutory provenance without inducing sequence truncation or diluting attention during Stage 1 retrieval and Stage 2 inference.

Listing 3.4 demonstrates the concrete transformation from an isolated provision into a Stage 1 indexed chunk and the resulting Stage 2 asymmetric input sequence. Prepended metadata provides the neural attention layers with parent enactment authority, preventing false positive matches against subordinate or unrelated local penalty clauses.

```
# 1. Isolated Statutory Provision (Without Context Prepending)
isolated_section = (
    "Section 458. (a) The sangguniang panlungsod... shall enact ordinances... "
    "(v) Approve ordinances which shall impose a fine not exceeding P5,000.00 "
    "or an imprisonment not exceeding one (1) year, or both."
)

# 2. Model-Ready Contextualized Provision Chunk (Indexed by Stage 1 BM25 & Bi-Encoders)
stage1_indexed_chunk = (
    "[Republic Act No. 7160: Local Government Code | Section 458: Powers] "
    "Section 458. (a) The sangguniang panlungsod, as the legislative body of the city, "
    "shall enact ordinances... (v) Approve ordinances which shall impose a fine not "
    "exceeding Five thousand pesos (P5,000.00) or imprisonment not exceeding one year ..."
)

# 3. Assembled Asymmetric Input Sequence Passed to Stage 2 Cross-Encoder
stage2_input_sequence = (
    "[CLS] Premise: [RA 7160 | Section 458] Approve ordinances which shall impose a fine "
    "not exceeding P5,000.00... [SEP] Hypothesis: [Proposed Ordinance No. 2024-012 | "
    "Section 6: Penalties] SECTION 6. PENALTY CLAUSE. Any person violating this Ordinance "
    "shall be penalized with a fine of Fifteen Thousand Pesos or two years imprisonment. [SEP]"
)

```

Listing 3.4: Sample Model-Ready Statutory Chunk and Stage 2 Input Sequence with Context Prepending

Listing 3.4 presents the three-stage transformation of a statutory provision, tracking the progression from an isolated clause into a model-ready indexed chunk and finally into an asymmetric Stage 2 input sequence. This listing is provided to illustrate how structural metadata bridges the gap between raw legislative text and input representations for Cross-Encoder classification. As demonstrated in the listing, an isolated penalty clause from Republic Act No. 7160 lacks explicit local scope, but prepending the Local Government Code header and Section 458 catchline clarifies its application to city councils. When formatted for Stage 2 inference, the premise and proposed ordinance hypothesis are joined using special delimiter tokens ($[CLS]$, $[SEP]$) alongside explicit segment tags. This formatted sequence enables the Cross-Encoder to compute full pairwise cross-attention between the statutory penalty ceiling and the proposed ordinance penalty.

The practical necessity for structural metadata headers is demonstrated by sections within composite legal codes. In Republic Act No. 7160, Section 447 sets penalty limits for municipalities, while Section 458 sets distinct penalty limits for city councils. Without adding the section title and number, downstream Cross-Encoders can confuse different levels of local government and misjudge local legislative powers. Adding structural metadata ensures that the model evaluates each legal clause under its correct administrative level. As a result, the vector embeddings retain both the exact wording of the clause and its broader statutory context. For lengthy provisions in the 5.31% exceedance tail exceeding the 448-token safety threshold, the pipeline implements the sliding-window sub-chunking fallback ($w = 384, s = 256$) formulated in Equation 3.5, prepending the parent metadata header to every sub-chunk to preserve governing legal authority throughout inference.

*Max-Pooled Sub-Chunk Contradiction Probability:*

$$\hat{P}(\text{Contradiction} \mid p_i, h) = \max_j P\left(\text{Contradiction} \mid p_{i,j}, h\right) \tag{3.5}$$

Where $p_{i,j}$ denotes the $j$-th sub-chunk of statutory section $p_i$, and $h$ denotes the candidate local ordinance hypothesis. Equation 3.5 defines the scoring strategy used to evaluate provisions that exceed standard length limits. By assigning the section's overall contradiction score to the maximum contradiction probability observed across any constituent sub-chunk, this pooling formulation ensures that localized prohibitive language or isolated fine stipulations embedded within long composite statutes trigger preemption alarms without being attenuated by surrounding administrative provisions.

## 3.4.2 Local Ordinance Archival Processing and Normalization

Unlike the digital national statutes, the 1,925 Davao City local enactments originate from physical paper archives spanning from 1950 to 2025. Processing this archival collection requires solving optical character degradation, layout variance, non-legislative administrative matter, and formulaic municipal boilerplate.

### 3.4.2.1 Vision-Language Document Ingestion Pipeline

Transforming physical city paper archives into machine-readable text requires evaluating candidate document ingestion architectures under consumer hardware constraints. While contemporary measures from the digital LISSP database provide clean electronic text, historical enactments from earlier legislative terms exist predominantly as scanned physical PDF documents. To establish an optimal transcription strategy for the 1,925 local records, the proponents formulated a planned candidate extraction experiment comparing three distinct technological paradigms: traditional open-source OCR via Tesseract v5.x [198], modern layout-aware deep learning OCR via Surya OCR [159], and a quantized local vision-language model via Qwen2.5-VL-3B-Instruct [216]. As an independent reference standard for transcription accuracy, the evaluation protocol establishes a gold-standard digitization baseline using Google Gemini Flash 3.8 multimodal extraction coupled with comprehensive line-by-line human legal verification. The candidate pipelines are evaluated across Character Error Rate (CER), Word Error Rate (WER), Table Fidelity (TEDS), and Legal Marker Precision on a representative 60-document stratified test suite, with empirical benchmarking results formally reported in Chapter 4 (Section 4.1).

To support offline execution on consumer hardware, the vision-language architecture deploys Qwen2.5-VL-3B-Instruct [216], a multimodal transformer that pairs a vision transformer backbone with a 3-billion parameter causal language model decoder. Executed collaboratively across local and cloud environments—primarily on Host B (equipped with an AMD Ryzen 7 7435HS processor, 16 GB DDR5 system memory, and an NVIDIA GeForce RTX 5050 Laptop GPU with 8 GB GDDR6 VRAM running Windows 11) for local baseline testing, alongside Google Kaggle GPU cloud runtimes (equipped with dedicated NVIDIA Tesla T4 GPUs) for executing high-throughput batch transcription across the remaining archival volumes—the open-weight architecture transcribes scanned enactments while preserving exact statutory phrasing. Proprietary cloud document platforms

such as Google Cloud Document AI [84] were excluded from routine production due to recurring per-page API fees and legal data sovereignty restrictions under the Data Privacy Act of 2012 (Republic Act No. 10173) [173]. Deploying open-weight vision-language models resolves these operational constraints, ensuring transcription fidelity while maintaining privacy and budgetary independence.

To balance operational feasibility with transcription fidelity across the full archival span, the proponents established a two-tier execution strategy dividing local workstation testing from cloud-based batch processing. While contemporary enactments (2021–2025) and initial pipeline calibrations were executed locally on Host B, the high-volume historical backlog spanning the 2016–2020 legislative terms and all physically degraded scans identified by the automated quality audit were routed to Google Kaggle cloud GPU instances [110, 22] (with the complete headless cloud ingestion notebook documented in Appendix B.3). This operational partitioning was guided by two practical engineering imperatives: thermal fatigue prevention on consumer client hardware and memory-unconstrained inference fidelity. Continuous autoregressive vision-language decoding across hundreds of multi-page PDF documents induces prolonged 100% thermal and power saturation on consumer laptop hardware, making offloaded execution essential to avoid thermal throttling and prevent hardware degradation on Host B [22]. Furthermore, while local laptop execution is bounded by an 8 GB VRAM ceiling requiring 4-bit NormalFloat4 quantization [61], Kaggle's 16 GB Tesla T4 runtime supported uncompressed 16-bit half-precision (`torch.bfloat16`) and higher spatial resolutions, significantly reducing Character Error Rate on faded, skewed, and carbon-bleed historical enactments [248, 216, 14].

The document ingestion pipeline begins by recursively discovering target PDF files across archival city directories. Individual enactments are loaded using PyMuPDF, which inspects document metadata and iterates through pages sequentially. Adapting archival document image perimeter-trimming heuristics for optical scanning artifact reduction [133], the ingestor applies an inset cropbox that trims exactly 3% of the page perimeter along all four borders:

*Archival Inset Perimeter CropBox Trimming:*

$$\text{CropBox} = [0.03 \cdot W,\ 0.03 \cdot H,\ 0.97 \cdot W,\ 0.97 \cdot H] \tag{3.6}$$

Where $W$ and $H$ represent the physical page width and height in points.

Equation 3.6 defines the boundary for trimming 3% (0.03) from each edge of a scanned page with width $W$ and height $H$. This cropping step removes common scanner artifacts—such as dark borders, staple marks, and binder punch holes—found in older paper records. Removing these border marks before sending the image to the vision-language model prevents visual noise from being mistaken for punctuation or causing the model to generate repetitive nonsense tokens. The trimmed bounding box keeps the focus directly on the legislative text, ensuring cleaner and more consistent transcription.

Following perimeter trimming, the rasterization module renders each page at an image resolution of 105 dots per inch (DPI). Selecting 105 DPI provides an optimal empirical balance between character legibility and sequence length for high-density legal

documents. The rendered image is converted to an RGB array and passed to the visual processor. The processor rescales page patches dynamically, clamping total image pixels between a minimum of 200,704 pixels ($256 \times 28 \times 28$) and a maximum of 351,232 pixels ($448 \times 28 \times 28$). Because Qwen2.5-VL allocates visual tokens in $28 \times 28$ patch increments using two-dimensional rotary position embeddings (2D-RoPE), bounding the pixel budget stabilizes sequence length and prevents out-of-memory GPU exceptions during long batch runs.

To accommodate local workstation hardware constraints without sacrificing recognition fidelity, the model weights are quantized using 4-bit NormalFloat4 (NF4) precision via the BitsAndBytes library [61]. The quantization configuration uses double quantization and computes attention projections in 16-bit floating point (`torch.float16`) using PyTorch scaled dot-product attention (`sdpa`). Under 4-bit NF4 compression, the model footprint occupies less than 3.5 GB of operational VRAM, allowing the pipeline to execute comfortably on consumer-grade laptop graphics hardware. Text generation is guided by an explicit verbatim system prompt instructing the model to transcribe all section titles, ordinals, dates, names, and monetary values without summarization or commentary. Generation settings enforce deterministic greedy decoding (do_sample = False, use_cache = True) bounded by a ceiling of 1,000 new tokens per page.

The pipeline adopts a page-level processing architecture rather than ingesting entire multi-page documents in a single model forward pass. Ingesting full multi-page ordinances simultaneously would require concatenating thousands of visual patch tokens, which dilutes spatial self-attention and exceeds consumer GPU memory ceilings. In contrast, processing documents page by page enforces focused visual attention on discrete text blocks while scaling linearly ($O(N)$) across documents spanning single-page resolutions through 50-page tax codes. The pipeline inserts explicit page delineators (`--- [Page` $X$`] ---`) into the extracted text, ensuring continuous traceability back to physical council records. Furthermore, page-level execution supports fault-tolerant checkpointing: the system inspects output files on disk and automatically resumes execution across interrupted batch queues by skipping previously completed files. Page-level garbage collection (`gc.collect()`) and explicit CUDA cache clearing (`torch.cuda.empty_cache()`) are executed after each document, preventing memory fragmentation during multi-hour ingestion runs.

#### 3.4.2.2 Substantive Legislative Body Extraction

Scanned local enactments generally contain three distinct parts: opening attendance rosters, the substantive legislative provisions, and concluding signature blocks. Separating the actual legal text from administrative details is essential for accurate conflict detection. Leaving councillor names in the text introduces personal names that distort Stage 1 dense retrieval embeddings, while including signature blocks adds unnecessary clutter to candidate hypotheses in Stage 2. Splitting documents using fixed page numbers does not work reliably across local archives, because short resolutions place legal provisions on the first page, whereas long regulatory codes spread attendance lists and committee rosters across several front pages.

To handle these structural variations, the proponents developed an adaptive boundary state machine that extracts operative legal provisions based on standard statutory markers. The extraction module scans transcripts for recognized opening markers, including enactment designations (`ORDINANCE NO.`), formal titles (`AN ORDINANCE`), and preamble clauses (`WHEREAS`). Upon detecting an opening boundary, the state machine captures the enactment identifier, title, preamble, and all operative sections (`SECTION 1`, `ARTICLE I`, `CHAPTER I`). Text capture stops immediately upon encountering formal closing markers, such as Sanggunian enactment notices (`ENACTED`), secretarial certifications (`CERTIFIED CORRECT`), vice-mayoral attestations (`ATTESTED`), or executive approval notations (`APPROVED`).

The extraction module incorporates a fallback hierarchy to prevent data loss when archival scans have faded or missing boundary markers. If an enactment header is obscured, the extractor falls back to the long ordinance title; if the title is absent, it anchors on the initial operative section. Similarly, if formal certification headers are missing, the system terminates extraction following the standard effectivity section (`SECTION X. EFFECTIVITY`). To provide automated quality control over the extraction process, the state machine calculates an empirical boundary verification confidence score:

*Substantive Body Boundary Extraction Confidence Score:*

$$\text{Confidence}_{\text{extract}} = 0.35 \cdot C_{\text{start}} + 0.35 \cdot C_{\text{end}} + 0.30 \cdot C_{\text{sec}} \tag{3.7}$$

Where $C_{\text{start}}$ and $C_{\text{end}}$ denote boundary match confidences, and $C_{\text{sec}}$ measures section sequential coherence.

Equation 3.7 defines the confidence score $\text{Confidence}_{\text{extract}}$ used to check how cleanly the legislative text was extracted. The linear weighting coefficients $(0.35, 0.35, 0.30)$ represent an empirical heuristic formulated by the proponents during preliminary calibration on local council documents: assigning equal dominant weights of 0.35 to the opening boundary ($C_{\text{start}}$) and closing signature boundary ($C_{\text{end}}$) ensures that confirming both structural boundaries satisfies the acceptance threshold ($0.35 + 0.35 = 0.70$) even if historical scans contain minor section numbering anomalies ($C_{\text{sec}}$), while penalizing documents with truncated bodies or missing enactment markers. Documents scoring at or above 0.70 move on to indexing, while lower-scoring files are flagged for manual inspection. This approach lets clean scans pass through automatically while sending difficult or unusual documents to human reviewers.

When confidence falls below 0.70, the document is flagged for manual review while preserving the full unpruned transcript on disk. Listing 3.5 outlines the extraction logic.

```
def extract_substantive_body(raw_transcript: str) -> dict:
    """
    Extracts operative statutory provisions from local transcripts,
    stripping administrative attendance rosters and terminal signatures.
    """
    start_match = RE_ORD_NUM.search(raw_transcript) or RE_TITLE.search(raw_transcript)
    start_idx = start_match.start() if start_match else 0
    start_conf = 1.0 if start_match else 0.40

    search_region = raw_transcript[start_idx:]
    term_match = RE_TERMINAL.search(search_region)
    end_idx = start_idx + term_match.start() if term_match else len(raw_transcript)
```

```
    end_conf = 1.0 if term_match else 0.50

    substantive_text = raw_transcript[start_idx:end_idx].strip()
    sections = [int(n) for n in RE_ALL_SECTIONS.findall(substantive_text)]
    sec_conf = 1.0 if len(sections) >= 3 and sections[:3] == [1, 2, 3] else 0.40

    confidence = round(0.35 * start_conf + 0.35 * end_conf + 0.30 * sec_conf, 3)
    return {
        "status": "SUCCESS" if confidence >= 0.70 else "FLAGGED_LOW_CONFIDENCE",
        "confidence": confidence,
        "substantive_text": substantive_text
    }

```

Listing 3.5: Adaptive State Machine for Substantive Legislative Body Extraction

Listing 3.5 presents the Python implementation of the adaptive state machine for substantive legislative body extraction. This algorithm is designed to parse raw local transcripts and extract substantive enactments while stripping prefatory attendance rosters, administrative preambles, and terminal signature blocks. In the function, regular expressions scan the text for opening enactment indicators and terminal attestation keywords, slicing the transcript between these indices and validating ordinal section continuity. The function computes the composite confidence score according to Equation 3.7 and returns a status dictionary indicating whether the extraction succeeded or requires manual review. Isolating the substantive body in this manner ensures that extraneous councillor names and mayoral titles do not distort dense retrieval vector embeddings or trigger spurious preemption matches in Stage 2.

#### 3.4.2.3 Automated Quality Audit and Conservative Normalization

Following raw transcription, the system executes an automated quality audit to detect and categorize transcription anomalies across seven mutually exclusive categories: definite OCR corruption, possible corruption, formatting noise, valid legal boilerplate, valid table structures, page metadata anomalies, and items requiring human review. The audit scanner identifies severe failure states, such as Unicode replacement characters (`\ufffd`), foreign script hallucinations (CJK and Cyrillic character ranges), raw decoding markers (`undefined`), and autoregressive repetition loops. Documents with severe corruptions or total lengths below 150 characters are quarantined into a dedicated review queue for manual verification.

To minimize false positives, the audit pattern matching explicitly accommodates authentic legislative structures. A broad repeated-character regular expression frequently flags Markdown table borders (`|---|---|`) as OCR corruption. To prevent this error, table boundary delimiters, pipes, and header markers are explicitly excluded from symbol repetition rules. Valid legislative boilerplate notations such as `"xxx xxx xxx"`, which denote unamended text in amending ordinances, are protected using negative lookahead expressions. Geographic survey coordinates containing degree symbols (°), minute marks (′), and second marks (″), as well as accented Filipino surnames (*Peña*, *Dañguilan*), are validated against an extended Unicode whitelist. Furthermore, page sequences beginning at Page 2 due to excluded cover sheets are recognized as valid contiguous sequences rather than misordered files.

Deterministic text normalization is performed via a conservative cleaning pipeline that strictly preserves legal meaning. The cleanup module applies Unicode Form KC normalization, strips invisible zero-width characters, removes stray VLM prompt echoes (such as `**[body text]:**` / *zhèngwén* markers), and collapses runaway character loops exceeding seven consecutive repetitions to a standardized three-character representation. All modifications are logged in an audit report with line-level traceability. Under strict architectural safeguards, automated scripts are prohibited from altering personal names, calendar dates, numerical monetary amounts, ordinance numbers, statutory citations, penalty clauses, or section numbering. Discrepancies involving these sensitive entities are routed exclusively to human legal researchers.

Downstream from automated cleanup, the architecture incorporates an advisory language quality review module combining a curated Philippine legal and toponymic gazetteer, contextual language model perplexity scoring, and named-entity recognition. The gazetteer whitelists all 182 official barangays of Davao City, administrative districts, local executive departments, and statutory Latin maxims (*in pari materia*, *mutatis mutandis*, *ultra vires*), preventing specialized terminology from triggering false spelling alarms. Contextual perplexity scoring evaluates token predictability within a sliding context window; tokens exhibiting high pseudo-perplexity and characteristic alphanumeric substitution patterns (such as misreading numeral `0` as uppercase letter `O` in monetary figures) are assigned high diagnostic confidence. The system presents these candidate errors on an interactive review dashboard, coupling the transcribed text with coordinate bounding boxes linking back to the original physical PDF scan, allowing human annotators to review highlighted anomalies prior to final vector indexing.

#### 3.4.2.4 Procedural Boilerplate Filtering and Provision Formatting

Filtering formulaic boilerplate prevents search drift and computational waste during candidate retrieval across the local collection. In Davao City legislation, procedural clauses—specifically Title, Separability, Repealing, and Effectivity clauses—contain standardized legislative phrases that carry zero substantive regulatory restrictions. When these sections are vectorized, their embeddings align with unrelated administrative acts that happen to share identical legal boilerplate. By systematically pruning these non-operative sections during ingestion, the preprocessor directs candidate retrieval exclusively toward substantive regulatory and penal provisions. As established in the corpus census, this filtering prunes 4,387 procedural sections (31.62%) from the local ordinance collection, isolating 9,487 operative provisions for indexing.

Following boilerplate filtering, each operative provision is formatted with a standardized municipal metadata header following the context prepending principles of Equation 3.4. Each local retrieval unit is prepended with its governing local council authority: [Davao City Ordinance No. $O_i$: Title | Section $S_j$: Catchline] $\|$ $\mathbf{t}_{\text{verbatim}}$. Attaching this municipal metadata ensures that dense bi-encoders and cross-encoders condition on the local legislative origin, preventing candidate retrieval models from confusing municipal penalties with national criminal statutes.

### 3.4.3 Ex-Ante Draft Ingestion and Statutory Exception Harmonization

While historical statutory and municipal archives undergo batch offline normalization, newly drafted measures submitted for ex-ante review pass through a specialized live ingestion pipeline.

#### 3.4.3.1 Digital-Born Draft Ingestion and Boundary Parsing

A critical operational distinction in this research separates the historical training archive from live ex-ante draft submissions. While the historical Davao City collection consists of scanned physical PDFs requiring optical character recognition and text normalization, newly drafted ordinances submitted for ex-ante review are digital-born documents. Legislative staff create these proposed measures using standard word processing software, generating clean electronic text or searchable PDF files. Because digital-born drafts contain authentic character encodings without scanning artifacts, the live pipeline bypasses OCR error correction entirely. This structural decoupling allows the preprocessor to focus immediately on automated section boundary parsing and clause segmentation.

To evaluate legislative commands without semantic distortion, an automated regular expression parser splits the submitted draft at formal section boundaries into discrete operative blocks. In Philippine local legislative practice, proposed measures follow standardized structural templates established by the Department of the Interior and Local Government [58]. Real-world legislative drafts contain substantial procedural boilerplate that does not express substantive legal commands. If routine clauses such as title declarations, repealing clauses, and effectivity dates are fed into Stage 1 retrieval, they match thousands of administrative national statutes containing similar formulaic phrases. Listing 3.6 details the regular expression parser that segments live draft measures into discrete operative sections while filtering out procedural boilerplate.

```
import re

def extract_operative_provisions(draft_text):
    """
    Parses digital-born draft ordinance into discrete sections,
    filtering out non-operative procedural boilerplate clauses.
    """
    # Regex pattern to segment text at formal section boundaries
    pattern = r'(SECTION\s+\d+\.?[^\n]*\n.*?)(?=(?:SECTION\s+\d+|CERTIFIED|ENACTED|\Z)
    )'
    matches = re.findall(pattern, draft_text, re.DOTALL | re.IGNORECASE)

    # Procedural clauses that do not carry substantive regulatory commands
    boilerplate_keywords = ["TITLE", "SEPARABILITY", "REPEALING", "EFFECTIVITY"]
    operative_sections = []

    for match in matches:
        clean_chunk = ' '.join(match.strip().split())
        header = clean_chunk.split('.')[0] if '.' in clean_chunk else clean_chunk[:30]

        # Exclude non-operative procedural boilerplate
        if any(kw in header.upper() for kw in boilerplate_keywords):
            continue

        operative_sections.append({
            "section_header": header,
```

```
            "operative_text": clean_chunk
        })

    return operative_sections

```

Listing 3.6: Code Snippet of the Operative Provision Extractor and Boilerplate Filter

Listing 3.6 presents the Python implementation of the operative provision extractor, featuring regular expression lookahead patterns and a procedural boilerplate keyword filter. This algorithm is incorporated to automate the structural segmentation of raw draft ordinances into clean, self-contained section blocks while systematically discarding administrative clauses. As observed in the listing, the regular expression isolates text between formal section boundaries, while the keyword filter checks section headers against procedural markers including Title, Separability, Repealing, and Effectivity clauses. Discarding these formulaic clauses prevents downstream retrieval models from matching generic legislative boilerplate against thousands of unrelated administrative acts. Consequently, only substantive regulatory and penal provisions advance to the candidate indexing and inference pipeline.

#### 3.4.3.2 Harmonic Interpretation and Exception Handling

A key challenge in conflict detection is distinguishing genuine normative contradictions from authorized statutory exceptions. In statutory interpretation, harmonic interpretation is the legal doctrine that provisions of law must be construed together to give effect to both general prohibitions and qualifying provisos, avoiding apparent contradictions where an authorized exception applies [4]. In legislative drafting, an exception clause is a statutory condition that permits an otherwise restricted activity under specified conditions or for designated public entities. If an automated conflict detection system evaluates a draft local ordinance solely against a general statutory prohibition without processing the accompanying exception clause, it generates a false positive preemption alert. In regulatory enactments, general prohibitions and permitted exceptions frequently appear within the same section using conditional syntax. Because statutory sections average between 90 and 122 tokens, an entire section fits comfortably within the 512-token input limit of candidate Cross-Encoders. The model's bidirectional attention mechanism reads the general prohibition and its qualifying exceptions simultaneously, preventing false positive contradiction alerts. This joint attention ensures that local measures implementing authorized statutory exceptions are recognized as valid exercises of delegated authority.

When a general prohibition and an authorizing exception reside in separate statutory sections, the pipeline relies on two-stage candidate retrieval. Under Philippine statutory construction, related provisions governing the same subject matter must be interpreted together as a harmonious whole under the doctrine of *in pari materia* [4]. If an ordinance draft relies on an authorized statutory exemption, evaluating a specific clause without retrieving the parent exception can trigger an incorrect conflict warning. The system addresses this challenge by utilizing hybrid retrieval in Stage 1 to pull both general bans and specific exception clauses into the candidate shortlist. The Stage 2 Cross-Encoder then evaluates the draft against the authorizing exception clause and as-

signs an Entailment classification, confirming that the local measure operates within valid legislative authority.

## 3.5 Ground Truth Benchmark

Evaluating candidate Cross-Encoder models in Stage 2 requires an expert-verified ground truth dataset reflecting the authentic legal realities of Philippine local legislation. Unlike general-domain natural language inference benchmarks that rely on crowdsourced sentence pairs, evaluating local conflict detection requires formal statutory interpretation skills grounded in administrative law. Because no existing NLP benchmark encompasses Philippine national statutes and local government ordinances, the proponents established a researcher-constructed, faculty adviser-reviewed 350-pair ground truth benchmark (comprising 198 unique instances across 22 statutory premises from 17 national statutes, structured for the institutional post-deployment roadmap with the Davao City Sangguniang Panlungsod).

For the ground truth evaluation benchmark, the proponents constructed an operational 350-pair dataset (comprising 198 unique proposition instances across 22 authentic national statutory premise sections from 17 Republic Acts, with an average of approximately 9 balanced hypothesis variants per premise), distributed across a multi-panel workload framework (5 panels $\times$ 70 pairs = 350 pairs, with 152 overlapping pairs). This design addresses three practical requirements: providing a zero-leakage premise-grouped evaluation suite for machine learning generalization, establishing a structured multi-rater workload protocol for 15 Sangguniang Panlungsod legal researchers without rater fatigue, and satisfying statistical sample size standards. In specialized legal annotation, reading multi-sentence statutory provisions requires focused concentration. Prior studies indicate that rater attention and analytical accuracy drop when an individual's workload exceeds roughly 70 pairs [1, 33]. By partitioning the 350 multi-panel instances into five balanced sets of 70 pairs each, the protocol structures a three-rater replication model ($k = 3$) for the 15-researcher cohort, yielding $350 \times 3 = 1,050$ total scheduled legal judgments across the cohort.

### 3.5.1 Sample Size and Statistical Power

The determination of the benchmark sample size is grounded in prospective statistical power and sample size design for multi-class contingency evaluations [41, 70]. Under Cohen's power framework, detecting semantic conflicts across a three-class discrete decision space ($\mathcal{C} = \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$, with $\text{df} = (3-1) = 2$ for goodness-of-fit and $\text{df} = (3-1)(3-1) = 4$ for $3 \times 3$ contingency matrices) with a medium effect size ($w = 0.30$) at a significance level of $\alpha = 0.05$ requires a minimum of $N = 108$ items to achieve standard statistical power of 80% ($(1-\beta) = 0.80$).

In prospective design terms, evaluating the pipeline on the benchmark's 198 unique propositional pairs establishes a Minimum Detectable Effect (MDE) of $w \leq 0.25$ with nominal statistical power exceeding 0.90 ($1-\beta > 0.90$, specifically 97.2% for df = 2 and 94.1% for df = 4 at $w = 0.30$), ensuring that the sample size is adequately powered

to detect non-trivial distributional shifts across the eight Macro Legal Domains while avoiding post-hoc observed power conflation. These eight domains reflect the core public services and regulatory police powers defined in Section 17 and Section 458 of Republic Act No. 7160.

This curated scale directly reflects established standards in specialized legal NLP benchmarking. While general-domain benchmarks like SNLI [25] rely on open crowdsourcing across tens of thousands of colloquial sentences, domain-specific legal NLI requires accredited legal professionals whose time and cognitive availability are severely constrained. High-quality, expert-adjudicated legal datasets consistently operate within this targeted scale: the SARA statutory tax benchmark comprises 276 cases (176 training and 100 test cases) [98], the annual COLIEE Task 4 legal entailment test suites provide approximately 100 pairs per competition cycle [167, 5], and ContractNLI [118] relies on curated subsets across 607 contracts. Designed across a 350-pair evaluation suite (comprising 198 unique proposition instances) under a three-rater overlap structuring 1,050 scheduled independent legal judgments across the 15-researcher cohort, the evaluation protocol establishes an empirical, statistically sound benchmark framework for local legislative review.

### 3.5.2 Dataset Partitioning and Stratified Generalization Setup

To maintain experimental rigor, prevent data leakage, and ensure reproducible model evaluation, the 350 Ground Truth instances are partitioned using a static 70/15/15 train, validation, and test split, stratified across both target classes (Contradiction, Entailment, and Neutral) and cognitive difficulty tiers (Tier 1 Quantitative, Tier 2 Structural, and Tier 3 Latent). This split provides 245 training pairs ($N_{\text{train}} = 245$), 52 validation pairs ($N_{\text{val}} = 52$), and 53 test pairs ($N_{\text{test}} = 53$). Operating within this sample size places the evaluation in a few-shot learning regime, which is well-suited for pretrained transformer language models [77]. In machine learning, few-shot task adaptation utilizes the foundational linguistic and semantic representations already encoded within deep pretrained encoders, fine-tuning classification heads and attention parameters on domain-specific instances without requiring thousands of newly labeled examples [77].

The proponents specifically selected a 70/15/15 partition over a standard 80/10/10 split due to sample size constraints. In a 350-pair dataset (comprising 198 unique instances), an 80/10/10 allocation yields only 35 validation and 35 test pairs. A validation set of only 35 pairs is statistically fragile for fine-grained decision threshold calibration ($\tau^*$), where a single classification error alters observed accuracy by nearly 3%. Furthermore, a test set of 35 items falls below the $N \geq 50$ sample size threshold established by Card et al. [29] for reliable Minimum Detectable Effect (MDE) in natural language processing. In contrast, allocating 52 validation pairs and 53 test pairs satisfies both the MDE statistical power criteria for detecting double-digit domain adaptation gains and provides a stable holdout slice for threshold calibration while retaining 70% (245 pairs) for robust few-shot Cross-Encoder fine-tuning [222]. Because the 198 unique propositional pairs are derived from 22 distinct statutory premise sections across 17 national statutes (averaging approximately 9 hypothesis variants per premise section), standard unconstrained cross-validation risks statutory premise memorization: a neural encoder could memorize the specific statutory vocabulary or syntactic structure of a national premise during training

and exploit that familiarity to classify test hypotheses linked to the same premise. To eliminate this shortcut, this static split is paired with a strict 5-fold premise-grouped cross-validation regime (`GroupKFold`) in Section 4.3.6.1, where all hypothesis variants sharing a given national statutory premise are strictly quarantined into either the training partition or the evaluation partition, forcing candidate cross-encoders to demonstrate genuine inductive reasoning on entirely unseen statutory enactments.

### 3.5.3 Asymmetric Pairing and Hypothesis Formulation

To construct a representative ground truth benchmark, candidate statutory premises must reflect the full breadth of Philippine national law. During the exploratory phase of this research, the proponents conducted preliminary machine learning experiments on the national statutes, uncovering severe class imbalances and repetitive drafting formulas across historical eras. These exploratory findings directly informed the consolidated topic modeling architecture detailed in Section 3.3.3, which organizes the corpus into eight validated Macro Legal Domains such as local governance, public health, and taxation. The proponents then sampled national statutory sections across these eight domains to establish the fixed premises for the evaluation benchmark. In natural language inference for legal documents, Asymmetric Span-Level Pairing is a sequence formulation strategy where a single, self-contained premise section from a superior statute is paired with a multi-sentence span ($N$ sentences) representing an entire operational section from a draft ordinance [244, 118]. This asymmetric formulation reflects the structural reality of Philippine legislative drafting. National statutes typically establish general normative principles or statutory prohibitions in one or two concise sentences. In contrast, local ordinances are operational administrative instruments: a single section regularly bundles the primary regulatory command together with local office designations, penalty schedules, and administrative notice periods spanning three to six contiguous sentences. If an NLP pipeline splits an ordinance section into isolated single-sentence fragments, the classifier loses the qualifying conditions and penalty ceilings, resulting in erroneous conflict determinations. By presenting the Cross-Encoder with the full multi-sentence ordinance section paired against the statutory premise, the model receives complete textual context to evaluate substantive preemption.

To make the evaluation as realistic as possible, the drafted test sentences closely mirror the writing style of Davao City ordinances. The proponents analyzed local laws enacted between 2021 and 2025 to identify standard drafting phrases, administrative office names, and structural penalty clauses following the official drafting conventions of the Department of the Interior and Local Government (DILG) [58]. Each drafted clause uses standard enacting syntax, such as: "*SECTION X. [TITLE]. — It shall be unlawful for any person, natural or juridical, within the territorial jurisdiction of Davao City, to...*" The draft sentences also name real local enforcement bodies, such as the City Transport and Traffic Management Office (CTTMO), the City Health Office (CHO), the City Environment and Natural Resources Office (CENRO), and the Business Bureau of Davao City. Operational units such as the Anti-Vices Task Force (AVTF) and the Davao City Police Office (DCPO) are also included to reflect genuine local enforcement.

The synthetic ordinance clauses are also linked to real administrative districts

in Davao City. Draft rules specify geographic areas from urban centers like Poblacion and Talomo to suburban districts like Buhangin and Toril. Local offices such as the City Disaster Risk Reduction and Management Office (CDRRMO) and the City Social Welfare and Development Office (CSWDO) are included to match actual city administration. This realistic drafting prevents evaluators from reviewing artificial example sentences that differ from real legal texts. As a result, domain evaluators and legal researchers in the post-study evaluation cohort review draft provisions that reflect authentic measures referred to committee following First Reading and prepared for Second Reading plenary interpellation.

To assist in formulating the draft ordinance clauses into authentic municipal legal syntax without introducing uncontrolled generative distortions, the proponents utilized a large language model (`Gemini Flash 3.8`) strictly as an assistive drafting tool for syntactic rephrasing and templating. Operating under structured few-shot prompts grounded in the analyzed 2021–2025 Davao City ordinances and DILG drafting conventions, the model rephrased human-specified regulatory commands, penalty amounts, and local agency designations into formal enacting clauses. Under this controlled human-in-the-loop framework, all statutory premises, regulatory conflict mechanisms, and ground truth labels were deterministically established by the proponents and validated against black-letter Philippine statutes (such as the penalty ceilings of Republic Act No. 7160 and price freeze limits of Republic Act No. 7581) and faculty adviser review. This structured assistance directly avoids the unconstrained generation pitfalls identified by Gururangan et al. [91] and Poliak et al. [164], where unchecked automated generation introduces superficial "annotation artifacts"—such as predictive negation tokens—that permit models to shortcut genuine statutory reasoning. While utilizing an assistive LLM accelerates the formulation of realistic administrative phrasing, LLM-generated text carries known risks of stylistic uniformity, repetitive syntactic templates, and subtle lexical artifacts that deep classifiers could inadvertently exploit [91, 164]. To mitigate these risks, all candidate drafts underwent mandatory human post-editing and adversarial injection by the proponents: sentences were manually perturbed to invert deontic modalities (e.g., altering mandatory commands to permissive choices without modifying surrounding vocabulary), authentic local administrative designations were injected, and penalty numbers were varied across statutory boundaries. Furthermore, as an empirical safeguard against model reliance on LLM hypothesis artifacts, the proponents conduct a hypothesis-only baseline evaluation in Section 4.3.7, verifying that candidate cross-encoders cannot achieve competitive conflict detection without cross-attending to the authentic statutory premise. In specialized legal NLP benchmarks such as SARA (tax law) [98], DocNLI [244], and Clash-of-Leges [105], maintaining this rigorous separation between assistive syntactic phrasing and deterministic statutory ground truth ensures that models are evaluated on authentic legislative language while preserving the absolute legal integrity of the gold labels.

While the system uses multi-sentence spans to understand local context, it does not attempt cross-document multi-hop reasoning (detecting conflicts that appear only by combining two or more separate national laws). Resolving dependencies across multiple separate laws is an active research challenge in natural language inference. Current systems that attempt this require either complex knowledge graphs [136] or very large generative language models [197]. Running those multi-premise models would exceed the

consumer-grade hardware limits set for local government deployment. For this reason, the ground truth benchmark and the inference system focus strictly on single-premise evaluations.

### 3.5.4 Class Balancing and Adversarial Injection

In machine learning benchmark construction, class balancing refers to the deliberate equalization of sample counts across all target categories to prevent models from learning trivial majority-class decision rules [34]. In authentic municipal drafting, the vast majority of local ordinance provisions comply fully with national law, meaning true statutory contradictions represent rare edge cases (typically comprising under 1% of random pairs). If an evaluation benchmark mirrored this natural skew (e.g., 90% Entailment and 10% Contradiction), a naive classifier that simply predicts the majority class would achieve a deceptively high 90% nominal accuracy while remaining completely ineffective at identifying genuine statutory conflicts. To complement class balancing, adversarial injection is a dataset construction methodology where domain experts systematically formulate deceptive, hard-to-detect variations of test pairs to eliminate reliance on superficial annotation artifacts [91, 164]. In legal natural language processing, deep language models frequently exploit shallow lexical heuristics—such as assuming that any provision containing words like "fine," "prison," or "unlawful" constitutes a contradiction. Through adversarial injection, annotators formulate pairs that invert legal duties (modifying mandatory commands to permissive options) or introduce subtle penalty ceiling violations without altering surface vocabulary. This rigorous design forces candidate Cross-Encoders to demonstrate genuine logical inference across statutory premises rather than relying on shallow lexical shortcuts.

In standard machine learning on tabular data, researchers often handle uneven classes using methods like the Synthetic Minority Over-sampling Technique (SMOTE) [34]. However, the proponents avoid continuous vector oversampling for this legal text pipeline. SMOTE creates artificial examples by interpolating between numerical feature vectors ($x_{\text{new}} = x_i + \lambda(x_{zi} - x_i)$). In high-dimensional language models representing discrete words, averaging or blending token vectors creates artificial representations that confuse attention layers and carry no real legal meaning. Simple word replacement also fails with legal texts because statutory words have exact legal definitions; for example, swapping mandatory words like "shall" with permissive words like "may" fundamentally changes a legal duty.

To build a balanced, reliable Ground Truth dataset, the researchers combined historical legal records with expert-guided text adjustments. First, the researchers gathered documented legal clashes from local government records, including mayoral veto messages, repealed ordinances, and explicit statutory amendments [105]. Second, following the evaluation methods of Koreeda and Manning [118] and Nie et al. [153], the proponents, under faculty adviser supervision, introduced conflicting rules into real statutory provisions. The authors reversed permissions, changed numerical zoning limits, and added conflicting penalties into ordinance clauses. This expert-guided process produced exactly 198 unique propositional pairs with a strictly balanced distribution across classes (exactly 66 Contradictions, 66 Entailments, and 66 Neutrals). When distributed across the five

multi-annotator review panels ($5 \times 70 = 350$ instances) to enable inter-rater reliability analysis, the resulting 350-instance evaluation suite comprises 112 Contradictions, 128 Entailments, and 110 Neutrals, preventing severe class imbalance while reflecting realistic multi-panel review coverage without creating unnatural legal phrasing.

### 3.5.5 Stratified Difficulty Tiers

Building an effective evaluation benchmark requires pairs with varying levels of difficulty. In benchmark evaluation, difficulty tier stratification is the systematic division of an evaluation dataset into distinct subsets based on the cognitive and structural complexity required to resolve each test instance [118, 175]. In this benchmark, the 350 pairs are partitioned into three granular tiers. Tier 1 (Surface and Quantitative Conflicts) evaluates direct, rule-based violations, such as numerical local penalties exceeding statutory caps (e.g., a PHP 15,000 local fine where Section 458 of Republic Act No. 7160 sets a PHP 5,000 ceiling for cities) or explicit inversions of mandatory terms. Tier 2 (Structural Preemption and Jurisdictional Carve-Outs) evaluates institutional authority under the *Magtajas* doctrine, testing whether a local council legislates within a field reserved for national agencies (such as pesticide control under Presidential Decree No. 1144) or omits mandatory statutory exemptions. Tier 3 (Latent and Paraphrastic Conflicts) evaluates subtle normative contradictions where the ordinance and national statute share virtually no overlapping lexical tokens or citations. In Tier 3, the conflict arises from contradictory legal effects—such as a local draft mandating the detention of apprehended street minors, which directly violates the non-custodial diversion mandate of Republic Act No. 9344 without explicitly citing the statute. This stratified structure provides fine-grained diagnostic visibility into candidate model performance.

To construct a rigorous evaluation suite, the proponents adapted the difficulty tiering methodology established by Koreeda and Manning in the ContractNLI benchmark [118]. This approach is supported by the behavioral stress-testing frameworks of Ribeiro et al. [175] and Naik et al. [149], which separate surface competence tests from semantic challenge sets. Furthermore, the 30% Tier 1, 40% Tier 2, and 30% Tier 3 quota follows the adversarial challenge partitioning established in benchmark engineering literature, such as SQuAD 2.0 [168], FEVER [218], and ANLI [153]. These foundational benchmarks establish that maintaining an adversarial challenge quota of 30% to 40% is optimal to prevent models from exploiting superficial lexical artifacts while preserving sufficient representation of standard rule-based compliance. Table 3.4 outlines the stratified difficulty tiers, legal conflict mechanisms, and representative examples for the benchmark.

Table 3.4 outlines the three-tier difficulty breakdown used for the 350-pair ground truth benchmark (198 unique instances), detailing target percentage shares, legal conflict mechanisms, and representative statutory examples. This breakdown ensures that the benchmark does not test simple keyword matching or penalty ceilings alone, but also evaluates models on subtle preemption conflicts. As shown in the table, the benchmark allocates 30% to Tier 1 (surface and quantitative conflicts, 105 pairs across 59 unique instances), 40% to Tier 2 (preemption and statutory exceptions, 140 pairs across 79 unique instances), and 30% to Tier 3 (latent and paraphrastic conflicts, 105 pairs across 60 unique instances). In addition, the Tier 3 challenge set excludes all numerical penalty amounts

Table 3.4: Stratified Jurisprudential Difficulty Tiers for the 350-Pair Ground Truth Evaluation Benchmark (198 Unique Instances)

| Difficulty Tier | Target Share | Legal Conflict Mechanism | Representative Statutory Example |
|---|---|---|---|
| **Tier 1: Surface & Quantitative** | 30% | Direct penalty ceiling violations and explicit changes in mandatory words (*shall* converted to *may*). | Local draft imposes a PHP 15,000 fine for traffic obstruction when RA 7160 Sec. 458 caps city penalties at PHP 5,000. |
| **Tier 2: Preemption & Carve-Outs** | 40% | Exceeding local regulatory powers under *Magtajas*, and blanket local bans that omit statutory exceptions. | Draft ordinance enacts a total ban on commercial freight transit without exempting national agricultural and medical logistics. |
| **Tier 3: Latent & Paraphrastic** | 30% | Subtle conflicts where the draft redefines statutory terms or creates incompatible rules without using the exact statutory words. | Draft anti-vagrancy ordinance assigns apprehended minors to detention cells, violating non-custodial diversion under RA 9344 without citing the law. |

and direct statutory citations, requiring the Cross-Encoder to evaluate legal meaning rather than relying on numbers. This distribution reflects the realistic challenges faced during legislative committee review while preventing models from depending on superficial keyword tricks.

The three difficulty tiers map directly to established Philippine statutory preemption principles and natural language challenges, following the distinction between rule-based and value-based legal conflicts described by Sartor [191] and Bench-Capon and Sartor [19]. Tier 1 evaluates surface and quantitative preemption, covering direct penalty ceiling violations under Section 458 of Republic Act No. 7160 and explicit clashes between mandatory and permissive words (*shall* versus *may*). Tier 2 adapts the "negation by exception" concept from Koreeda and Manning [118], evaluating unhonored statutory exemptions and local overreach under *Magtajas v. Pryce Properties* [207] and *City of Manila v. Laguio, Jr.* [209]. Tier 3 evaluates latent and paraphrastic contradictions, testing whether the model detects substantive legal friction when a draft redefines terms without sharing exact statutory words. Dividing the benchmark into 30% Tier 1 (105 pairs), 40% Tier 2 (140 pairs), and 30% Tier 3 (105 pairs) ensures a balanced test set that reflects real committee work while preventing models from exploiting simple word overlap [143].

Grounding Tier 3 contradictions in legal doctrine ensures that models are evaluated on genuine language understanding rather than keyword presence. Under established Supreme Court rulings, a local council cannot enact indirect prohibitions or administra-

tive conditions that narrow or defeat rights protected by national statutes [207, 209]. In Philippine statutory preemption jurisprudence, conflict encompasses not only literal verbal clashes (*verba legis*) but also the frustration of national statutory purpose and policy (*ratio legis*), as reaffirmed in *Mosqueda v. Pilipino Banana Growers & Exporters Association* [210]. In Tier 3 latent conflicts, an ordinance may not explicitly mention or verbally negate a national statute, yet its operational effect narrows, conditions, or defeats statutory guarantees. For instance, when a national statute guarantees universal, non-discriminatory access to a public service (such as tertiary tuition exemptions under Republic Act No. 10931 or juvenile diversion under Republic Act No. 9344), a local draft that imposes unauthorized local residency preconditions or custodial detentions creates an actionable normative preemption conflict under administrative law, even though the local draft frames the restriction as an administrative qualification. While reasonable legal minds may debate the line between permissible local administrative discretion and impermissible statutory narrowing, incorporating such latent policy collisions reflects authentic Philippine legal reality and tests whether cross-attention architectures capture deeper deontic tensions beyond surface token matching. To enforce this distinction during testing, the proponents verified that all 104 Tier 3 draft sentences contain zero numbers or monetary amounts. Without explicit numerical clues, the Cross-Encoder must evaluate the underlying legal meaning of the text, such as recognizing when an anti-vagrancy draft detains minors in violation of Republic Act No. 9344. This design ensures that the benchmark provides an authentic diagnostic test of semantic preemption for the subsequent analysis in Chapter 4.

### 3.5.6 Expert Legal Annotation Protocol and Benchmark Verification

Classifying statutory text pairs requires domain-specific legal grounding to ensure dependable, authentic ground truth labels. To establish a rigorous evaluation benchmark for model training and ablation while navigating municipal administrative clearance timelines, the operational benchmark dataset was systematically curated by the research team following authentic Davao City legislative syntax (2021–2025 enactments) and DILG drafting standards, assisted by `Gemini Flash 3.8` for grammatical rephrasing, and formally reviewed and verified through iterative academic supervision by thesis adviser Mr. Adrian Ablazo, MSIT. This adviser review was conducted to verify the technical accuracy, absence of synthetic token corruption, and logical consistency of the paired provisions against the cited national statutes and local drafting conventions, rather than functioning as a formal legal bar adjudication. This established the **Faculty Adviser-Reviewed Ground Truth Benchmark** ($N = 350$ evaluation pairs, comprising 198 unique instances across 22 statutory premises from 17 national statutes) utilized for all supervised cross-encoder training, decision threshold calibration, and zero-leakage premise-grouped cross-validation reported in this study. In parallel, to establish a gold-standard institutional validation protocol, the proponents formulated a structured multi-rater evaluation framework tailored for 15 active legal researchers serving the Davao City Sangguniang Panlungsod [47]. Following administrative routing through the Human Resource Management Office (HRMO) and the City Administrator's Office pursuant to the Sangguniang Panlungsod referral, this

expert panel protocol is codified as an institutional post-deployment roadmap to capture authentic practitioner consensus once municipal clearance is concluded. Unlike general NLP tasks that rely on open crowdsourcing, statutory conflict detection requires specialized legal interpretation to differentiate superficial lexical overlap from genuine normative preemption [199, 247, 33].

To establish an objective and transparent panel structure for post-deployment execution, the proponents gathered baseline qualifications during the initial intake communication. The intake request systematically recorded two key criteria from each volunteer: highest educational attainment and total years of professional experience in legislative drafting. Using these reported credentials, the 15 researchers were organized into five balanced sub-panels of three members each ($k = 3$). Within each three-member sub-panel, the evaluator possessing the highest educational attainment and the longest drafting tenure was designated as that panel's Senior Annotator. The proponents then communicated these panel assignments and Senior Annotator designations in a follow-up confirmation email to establish operational readiness for the post-study rollout.

Following the consensus guidelines adapted from Ablazo [1], a label is accepted into the post-deployment Ground Truth if at least two of the three reviewers in a sub-panel agree. In cases where all three reviewers select different categories, resulting in a three-way split with no majority, the pair is referred to that panel's designated Senior Annotator for binding qualitative adjudication [78, 11]. The Senior Annotator reviews the pair against the standardized codebook to resolve whether the split arose from statutory ambiguity or an overlooked proviso. This hierarchical adjudication step ensures that every pair in the 350-pair benchmark (comprising 198 unique instances) reaches an authoritative label while preserving the initial independent ratings for calculating Fleiss' Kappa ($\kappa$). Maintaining this structured division between initial independent scoring and expert tie-breaking safeguards both statistical validity and qualitative consistency across the dataset.

To recognize the time and professional expertise dedicated by the participating Sangguniang Panlungsod legal researchers, evaluators who complete their assigned annotation batches during post-deployment rollout are awarded an official Certificate of Contribution (see Appendix L). This certificate serves as a formal token of appreciation and institutional documentation acknowledging their voluntary advisory assistance and professional review of statutory conflict screening protocols during the institutional deployment phase. Each certificate is endorsed by the Computer Studies Department's Assistant Dean, thesis adviser, course instructor, and student proponents to confer formal institutional validity. Providing this acknowledgment establishes a collaborative partnership between the university research team and the local legislative staff, ensuring appropriate recognition for the legal professionals who support the research.

In strict adherence to the ethical guidelines of the University Research Ethics Board (UREB) and the statutory mandates of Republic Act No. 10173 (the Data Privacy Act of 2012) [173], participation in the expert annotation exercise is entirely voluntary. Prior to receiving evaluation materials, each participant completes an electronic informed consent agreement affirming voluntary participation and the freedom to withdraw at any point without administrative penalty or professional consequence. To protect evaluator confidentiality, participant names, councilor assignments, and committee affiliations are

permanently decoupled from annotation records and replaced with anonymized rater codes (such as Annotator P1-A through Annotator P5-C); participants only optionally disclose their legal names at the conclusion of the study if they elect to receive the formal Certificate of Contribution for their official Civil Service 201 records. Evaluated text pairs consist exclusively of enacted public statutes and model local drafting clauses derived from official DILG manuals and enacted city legislation, involving no confidential council communications, unreleased political deliberations, or proprietary personal data.

### 3.5.7 Annotation Codebook and Decision Rules

To help the 15 legal researchers apply consistent criteria during post-deployment execution, the proponents developed a standardized Legal Annotation Codebook (the complete, unabridged protocol is provided in Appendix J). Adapting principles from Koreeda and Manning [118] and Italiani et al. [105], the codebook turns the three-class NLI framework into clear legal decision rules. Each evaluated pair consists of a single national statute section as the premise and a multi-sentence local ordinance clause as the hypothesis. Annotators assign one of three labels: Contradiction, Entailment, or Neutral. Table 3.5 summarizes the working definitions, legal triggers, and example patterns used in the annotation guidelines.

Table 3.5 details the operational legal definitions, primary statutory triggers, and representative drafting patterns defined in the Sangguniang Panlungsod annotation codebook across the three target NLI classes. This decision framework is provided to establish consistent, legally grounded classification criteria for the 15 volunteer legal researchers, translating natural language inference categories into established Philippine administrative doctrines. As detailed in the table, Contradiction is triggered by Magtajas preemption, modal command inversions, and local penalty escalations; Entailment is triggered by valid devolved regulatory implementation; and Neutral is assigned when provisions share broad subject matter without normative conflict. Standardizing these decision rules ensures high inter-rater reliability during independent scoring and provides clear criteria for Senior Annotator tie-breaking. Consequently, the resulting gold labels reflect rigorous legal consensus rather than subjective linguistic intuition.

The decision rules for Contradiction directly apply the consistency test established in the Magtajas Doctrine [207]. Because local governments have delegated rather than unlimited legislative powers under Republic Act No. 7160 [85], any local draft that exceeds national statutory limits is legally invalid (*ultra vires*). Annotators look for shifts in mandatory language, such as turning a strict command ("shall") into an optional choice ("may"), or removing an exemption that national law protects. In the test samples, numerical rules are audited against national limits, including tax caps and penalty ceilings. These criteria ensure that conflict warnings highlight actual legal problems rather than minor wording choices.

The guidelines follow a structured consensus process to resolve disagreements among the three panel reviewers. When two or three reviewers agree, the majority label is accepted directly into the benchmark. In cases where all three reviewers select different categories, the item is referred to the sub-panel's designated Senior Annotator for final

Table 3.5: Sangguniang Panlungsod Ground Truth Legal Annotation Codebook and Decision Rules

| Logical State | Operational Legal Definition | Primary Legal and Structural Triggers | Representative Statutory Pattern |
|---|---|---|---|
| **Contradiction** | The local ordinance prohibits what national statute permits, permits what statute prohibits, or exceeds statutory ceilings [207]. | Clashes in mandatory words (*shall* vs. *may*); penalty limit violations (RA 7160 Sec. 458); territorial or franchise preemption. | National statute caps ordinance fines at PHP 5,000; draft ordinance mandates a PHP 15,000 penalty. |
| **Entailment** | The local ordinance directly executes, complies with, or derives its regulatory authority from the national premise without adding conflicting mandates. | Compliant devolved implementation; literal statutory adoption; regulatory enforcement within prescribed legislative boundaries. | Local draft establishes a local health board strictly adhering to the composition mandated by RA 7160. |
| **Neutral** | The national statute and local ordinance address the same broad legal domain, but govern independent obligations without mutual conflict or derivation. | Shared topic without direct legal clash; complementary administrative scopes; distinct non-overlapping legal subjects. | National statute establishes hospital licensing standards; local draft regulates city market sanitary permits. |

qualitative review [78]. The Senior Annotator reviews the statutory provisions against the codebook to determine whether the divergence arose from statutory ambiguity or an overlooked exception. This review step ensures that every item in the 350-pair Ground Truth benchmark (comprising 198 unique instances) reaches an authoritative label while preserving the original votes to calculate Fleiss' Kappa.

To illustrate the structure of the evaluated instances, Listing 3.7 presents an authentic sample record (Pair ID: `GT-001`) from the 350-pair Ground Truth benchmark dataset (198 unique instances), showcasing the paired national statutory premise, proposed local hypothesis, gold conflict label, and expert legal rationale.

```
# Authentic Evaluation Pair GT-001 from data/ground_truth_350.jsonl
benchmark_pair = {
"pair_id": "GT-001",
"macro_domain": "Executive Issuances & Policy Reorganization",
"difficulty_tier": "Tier 1: Surface & Quantitative",
"national_premise": {
    "statute_title": "Republic Act No. 7581 (The Price Act)",
    "citation": "Section 6",
    "statutory_text": (
        "Section 6. Automatic Price Control. - Unless otherwise declared by the "
        "President, prices of basic necessities in an area shall automatically be "
```

```
        "frozen at their prevailing prices or placed under automatic price control "
        "whenever that area is proclaimed or declared a disaster area or under a "
        "state of calamity... price control of basic necessities under this section "
        "shall remain effective for the duration of the condition that brought it about, "
        "but not for more than sixty (60) days."
    )
},
"ordinance_hypothesis": {
    "ordinance_title": (
        "Proposed City Ordinance No. 2024-082 (The Davao City Calamity Price "
        "Stabilization and Consumer Welfare Protection Measure)"
    ),
    "section_citation": "Section 4",
    "hypothesis_text": (
        "SECTION 4. MANDATORY LOCAL PRICE CEILINGS UPON CALAMITY. - In accordance "
        "with local civil defense standards... upon official declaration of a state "
        "of local calamity by the Sangguniang Panlungsod, the City Mayor... shall "
        "enforce mandatory price ceilings across all grocery and commercial retail "
        "establishments within Davao City, which price freeze shall remain in continuous "
        "legal effect for a mandatory period of one hundred eighty (180) calendar days ..."
    )
},
"gold_label": "Contradiction",
"legal_rationale": (
    "Direct breach of statutory temporal ceiling. Section 6 of RA 7581 explicitly limits "
    "automatic calamity price control to not more than sixty (60) days. The draft ordinance "
    "mandates a 180-day freeze, exceeding delegated police power limits."
),
"annotator_metadata": {
    "assigned_panel": "Panel_A",
    "evaluators": ["SP-ANN-01", "SP-ANN-02", "SP-ANN-03"]
}
}

```

Listing 3.7: Sample Evaluation Pair from the Ground Truth Benchmark Dataset

Listing 3.7 presents an authentic evaluation record from the ground truth dataset (`GT-001`), detailing the JSON schema including pair identifier, macro domain, difficulty tier, national premise citation and text, ordinance hypothesis citation and text, gold NLI label, legal rationale, and annotator panel metadata. This structured record illustrates the complete JSON schema passed to candidate Cross-Encoders during evaluation and documents the full provenance of each gold label. In this specific Tier 1 instance, the hypothesis mandates a 180-day calamity price freeze, which directly contradicts Section 6 of the Price Act (Republic Act No. 7581) capping automatic freezes at 60 days. The record captures both the exact statutory violation and the panel metadata documenting independent scoring by three legal researchers. Storing this rich metadata enables granular error analysis across difficulty tiers and legal domains in Chapter 4.

### 3.5.8 Annotation Workload Allocation

To prevent reviewer fatigue while keeping the results statistically sound, this study adapts the batched review framework developed by Ablazo [1] alongside balanced incomplete block design principles for multi-rater NLP annotation [25, 144]. In that earlier study,

each item required three independent evaluations to determine a majority label. Managing a collection of 12,054 short texts required over 36,000 individual reviews. By dividing this work among 250 students, the workload was reduced to about 145 items per person. This demonstrated how batched distribution can maintain a three-vote overlap without placing an unrealistic burden on reviewers.

This study applies the same mathematical allocation to the legal domain. Because reviewing multi-sentence legal provisions requires much more concentration than reading short messages, the dataset is smaller, but the distribution logic is identical. The workload for the legal reviewers is calculated as follows:

*Annotator Workload Allocation Formulation:*

$$\begin{aligned} N &= 350 \text{ premise-hypothesis pairs} \\ k &= 3 \text{ independent votes per pair} \\ P &= 15 \text{ legal annotators} \\ W &= N \times k = 350 \times 3 = 1,050 \text{ total annotations} \end{aligned} \tag{3.8}$$

$$\text{Work per person} = \frac{W}{P} = \frac{1,050}{15} = \mathbf{70} \text{ pairs per annotator}$$

Where $N$ represents the benchmark size, $k$ is the multi-rater replication factor, $P$ is the annotator pool size, and $W$ denotes total evaluations.

Equation 3.8 details the mathematical derivation of the annotator workload distribution, dividing the total required annotations $W$ among the pool of $P = 15$ legal researchers. With a benchmark size of $N = 350$ premise-hypothesis pairs evaluated under a three-rater replication factor ($k = 3$), the protocol requires a total of $W = 1,050$ independent legal judgments. The formulation allocates this workload into five disjoint 70-pair sets, assigning exactly one set of 70 pairs to each three-member sub-panel. Methodologically, capping individual reviewer workload at 70 pairs prevents cognitive fatigue and attention degradation during complex legal evaluation. This batched structure maintains full multi-rater overlap for Fleiss' Kappa reliability calculations while ensuring that no single annotator is overburdened.

To organize this workflow, the dataset is divided into five equal sets of 70 pairs each. The 15 researchers are grouped into five sub-panels of three members, with each sub-panel reviewing one 70-pair set. This arrangement ensures that every pair receives three independent reviews while keeping each reviewer's workload manageable.

To reduce fatigue during the review process, the proponents developed structured reference materials. Reviewing statutory pairs requires significant effort, as reviewers must compare detailed legal clauses against local drafts. In the evaluation packet, evaluators receive spreadsheets with complete statutory text to ensure they do not have to read isolated sub-clauses out of context. Each reviewer receives the complete Legal Annotation Codebook (Appendix J) for full procedural and doctrinal context, alongside a quick-reference one-page summary sheet outlining the *Magtajas* decision hierarchy and statutory penalty ceilings. These reference materials are structured to enable reviewers

to evaluate statutory pairs systematically and consistently across their 70-pair evaluation assignments during institutional deployment.

### 3.5.9 Inter-Rater Reliability

Within the institutional multi-rater annotation framework designed for Sangguniang Panlungsod panel deployment, inter-rater concordance is formally measured using Fleiss' Generalized Multi-Rater Kappa ($\kappa$) [72]. In statistical annotation methodology, Fleiss' Kappa evaluates agreement across a fixed cohort of raters assigning categorical ratings to a shared set of items. While raw percentage concordance reflects nominal agreement, it fails to account for agreement occurring purely by stochastic chance. Fleiss' Kappa calculates the true proportion of agreement achieved above chance expectation, establishing an objective statistical metric for consensus reliability.

This metric is used instead of Cohen's Kappa because Cohen's formula is limited to exactly two raters, whereas this study assigns three reviewers to each pair. The resulting score is interpreted using the guidelines from Landis and Koch [122], where a $\kappa$ score between 0.61 and 0.80 indicates "Substantial Agreement."

Using this target in natural language inference is supported by Bowman et al. [25] in their baseline work on the Stanford Natural Language Inference (SNLI) benchmark. Evaluating text relationships involves human judgment; in their study, Bowman et al. reported an overall Fleiss' Kappa of 0.70 (scoring 0.77 for Contradiction, 0.72 for Entailment, and 0.60 for Neutral). They demonstrated that achieving very high agreement ($> 0.80$) in language inference is rare and usually happens only with overly simplified test sentences. Because Philippine local ordinances are more complex than general English sentences, expecting agreement higher than 0.70 is unrealistic. Therefore, for the post-deployment institutional study, the protocol targets a minimum $\kappa$ threshold of 0.61 pursuant to Landis and Koch [122], which would confirm substantial agreement among the legal reviewers upon full deployment.

## 3.6 Hardware and Computing Setup

Rather than assuming an idealized enterprise server environment or high-cost cloud infrastructure, the proponents utilized physical workstations spanning distinct computing tiers alongside accessible cloud resources. This multi-tier setup was selected to simulate authentic local government IT conditions while supporting the computational demands of model training. Specific research workloads—spanning archival document scanning, web scraping, deep learning fine-tuning, and edge inference—were assigned to the computing environment best suited for that operational phase.

Host A (Desktop) is the primary testbed for data collection, preprocessing, and simulated edge execution. Powered by an Intel Core i3-10105F CPU (4 cores, 8 threads @ 3.70 GHz), 8.00 GB of DDR4 system RAM, an AMD Radeon RX 6600 GPU (8 GB GDDR6 VRAM), and high-speed NVMe solid-state storage running Microsoft Windows 10 Pro 64-bit, this machine executed the automated web scraping of the 25,432 Philippine national statutes from the Lawphil Project, raw HTML parsing, metadata structuring,

and utility scripts. During edge retrieval simulation, its discrete GPU is deactivated to verify that Stage 1 sparse (BM25) and dense index lookups execute efficiently within the modest hardware limits typical of local government offices.

Host B (Laptop), configured on a Lenovo Legion 5 15AHP10 laptop equipped with an AMD Ryzen 7 7435HS processor (8 cores, 16 threads @ 3.80 GHz), 16.0 GB of high-speed DDR5 RAM, an NVIDIA GeForce RTX 5050 Laptop GPU with 8 GB GDDR6 VRAM, and PCIe NVMe storage running Microsoft Windows 11 Home 64-bit, provides dedicated CUDA acceleration. Host B executed the scanning, PDF text extraction, and optical character recognition (OCR) cleaning experiments for the 1,925 Davao City local ordinances, and was dedicated to Stage 2 Cross-Encoder fine-tuning, hyperparameter sweeps, training loss profiling, and full system evaluation.

To faithfully evaluate system usability on municipal workstations, the proponents established a hardware throttling protocol during edge benchmarking and system demonstrations. Because Host A and Host B possess dedicated graphics hardware that exceeds average local government desktop specifications, their execution environments were intentionally throttled during edge latency tests: discrete GPU access was programmatically disabled via environment variables, PyTorch and ONNX runtimes were restricted strictly to CPU execution, and CPU thread pools were clamped to four worker threads. This deliberate constraint guarantees that reported query latencies, memory utilization figures, and demonstration workflows reflect the operational reality of standard local government computers rather than artificially accelerated laboratory hardware.

Complementing the physical workstations, Google's Kaggle cloud platform provided on-demand external GPU resources when compute demands exceeded local workstation memory ceilings [110]. Operating on Google Cloud Platform deep learning infrastructure [22], Kaggle dynamically provisions headless Linux virtual machine environments equipped with dedicated NVIDIA Tesla T4 GPUs (16 GB GDDR6 VRAM) and 30 GB of system RAM [110]. The environment is accessed and orchestrated headlessly through the official Kaggle command-line interface (`kaggle` API), which provides a weekly allocation of 30 GPU execution hours. This cloud tier was deployed for computationally intensive batch experiments, heavy matrix factorizations (such as 64-component Singular Value Decomposition over the dual statutory corpus), and running reproducible Cross-Encoder fine-tuning sweeps independently of local hardware constraints.

Across all physical and virtual environments, the research pipeline standardizes on an identical Python software stack to ensure full cross-platform compatibility. Table 3.6 summarizes the physical hardware configurations, operational research roles, and exact software library versions governing the study.

Table 3.6 summarizes the hardware specifications, operating systems, assigned research tasks, and core software libraries across the computing environments used in this study. This specification matrix ensures complete computational reproducibility and demonstrates that the system executes on accessible consumer workstations and local government computers. As detailed in the table, routine data ingestion, utility scripts, and throttled edge latency testing run on Host A (Core i3-10105F, 8 GB RAM desktop simulating municipal workstation constraints), heavier deep learning fine-tuning and OCR

Table 3.6: Hardware Configurations, Software Environment, and Operational Roles across Computing Environments

| Environment | Hardware Specifications | Operating System | Assigned Research Operations |
|---|---|---|---|
| **Host A** (Desktop) | Intel Core i3-10105F (4C/8T @ 3.7 GHz)<br>8 GB DDR4 RAM, RX 6600 8GB<br>High-Speed NVMe SSD | Microsoft Windows 10 Pro 64-bit (Build 19045) | National statute scraping (25,432 laws), HTML parsing, text normalization, utility scripts, and throttled CPU edge simulation. |
| **Host B** (Laptop) | AMD Ryzen 7 7435HS (8C/16T @ 3.8 GHz)<br>16 GB DDR5 RAM, NVMe PCIe SSD<br>NVIDIA RTX 5050 Laptop GPU | Microsoft Windows 11 Home 64-bit (Build 26200) | Local ordinance scanning (1,925 ordinances) and OCR cleaning; dedicated Stage 2 DL fine-tuning, training loss profiling, and full evaluation. |
| **Google Kaggle** (Cloud Platform) | NVIDIA Tesla T4 GPU (16 GB)<br>30 GB System RAM<br>Runtime | Ubuntu Linux (`kaggle` Cloud VM) | Headless batch experiments, Cross-Encoder fine-tuning sweeps, 64-component SVD matrix factorization, and remote builds. |
| **Standardized Software Stack and Execution Libraries** | | | |
| **Runtime / Framework** | Python 3.10+ / 3.14<br>PyTorch (`torch`) 2.14.0+ | **Transformers** | Hugging Face (`transformers`) 5.16.1+ |
| **Dense Embeddings** | `sentence-transformers` 6.0.1+ | **Lexical Search** | `rank-bm25` 0.2.2+ (Okapi BM25) |
| **ML & Statistics** | `scikit-learn` 1.9.0 / `scipy` 1.18.1<br>NetworkX (`networkx`) 3.6+ | **Document Ingestion** | PyMuPDF (`fitz`) 1.28.2 / `pypdf` 6.16.2 |
| **Data Processing** | `pandas` 3.0.5 / `numpy` 2.5.2 | **Manuscript Build** | TeX Live 2026 / MiKTeX (`pdflatex`, `bibtex`) |

tasks run on Host B (Ryzen 7 7435HS, RTX 5050 Laptop GPU), and resource-heavy matrix calculations run on Google's Kaggle cloud platform. Using the same Python 3.10+ runtime, PyTorch 2.14, and Hugging Face Transformers across all machines ensures consistent model behavior and smooth script execution. This setup confirms that while model training benefits from dedicated GPU acceleration, the final inference pipeline runs comfortably on modest office hardware.

## 3.7 Pipeline Architecture

The live conflict detection workflow uses a two-stage retrieve-then-entail architecture designed to combine rapid statutory retrieval with precise logical preemption analysis. The system divides the conflict detection process into two sequential operational phases: coarse candidate retrieval across the dual statutory corpus, and fine natural language inference over paired provisions.

### 3.7.1 Stage 1 Hybrid Retrieval Pipeline

The first operational phase begins when a user submits a draft local ordinance for review. In information retrieval, hybrid search combines sparse lexical search and dense semantic vector search to capture both exact terminology and conceptual relevance. Sparse lexical search using BM25 evaluates exact token matches, statutory citations, numerical penalty amounts, and administrative acronyms based on term frequency and document length normalization [176]. In contrast, dense semantic search using a transformer Bi-Encoder maps draft and statutory texts into a shared continuous embedding space, computing cosine similarity to identify conceptually related provisions that share few identical words [171]. Combining these two retrieval channels using linear score interpolation is problematic because their raw scoring distributions are fundamentally incompatible: BM25 produces unbounded positive scores that fluctuate with query length, whereas dense bi-encoders output cosine similarities strictly bounded between $-1.0$ and $+1.0$. Reciprocal Rank Fusion (RRF) is a rank-aggregation algorithm that merges multiple retrieval rankings based strictly on the ordinal positions of documents rather than their raw numerical scores [46]. By assigning higher reciprocal rank scores to statutory provisions that appear near the top of both the lexical and semantic retrieval streams, RRF produces a balanced candidate shortlist without requiring empirical score calibration:

*Reciprocal Rank Fusion (RRF) Scoring:*

$$\mathrm{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{1}{k_{\mathrm{rrf}} + \mathrm{rank}_m(d)} \tag{3.9}$$

Where $\mathcal{M} = \{\mathrm{BM25}, \mathrm{Dense}\}$ represents the candidate retrieval models, $\mathrm{rank}_m(d)$ denotes the ordinal position of document $d$ produced by model $m$, and $k_{\mathrm{rrf}} = 60$ is the smoothing constant.

Equation 3.9 defines the Reciprocal Rank Fusion (RRF) formula, combining the ranked search results from BM25 and the dense Bi-Encoder into a single list. RRF avoids having to normalize raw scores, since BM25 scores vary widely based on document length while cosine similarities stay between $-1$ and 1. By focusing on rank positions with a smoothing parameter ($k_{\mathrm{rrf}} = 60$), the formula gives priority to provisions that score well in both keyword and semantic searches. This ensures that the candidate shortlist includes both exact citation matches and conceptually related laws, providing a balanced pool of candidates for Stage 2 inference.

Standard linear score interpolation ($\alpha S_{\mathrm{BM25}} + (1-\alpha) S_{\mathrm{Dense}}$) requires min-max normalization across unbounded BM25 distributions, which fluctuates wildly across diverse queries. In contrast, Reciprocal Rank Fusion operates entirely on ordinal rankings, making the combination robust against differences in score scales. The resulting fused score ranks provisions that demonstrate both lexical precision and semantic relevance at the top of the candidate pool. Listing 3.8 presents the procedural logic of the dual-stream hybrid retrieval and rank fusion pipeline.

```
1 def retrieve_top_k_candidates(draft_section, statutory_corpus, k=50, k_rrf=60, beta
      =1.20):
2     # 1. Retrieve top-100 candidates from sparse BM25 and dense Bi-Encoder
3     bm25_results = search_bm25(corpus=statutory_corpus, query=draft_section, top_n
      =100)
```

```
    dense_results = search_dense(corpus=statutory_corpus, query=draft_section, top_n
    =100)

    # 2. Compute Reciprocal Rank Fusion (RRF) scores across candidate pool
    candidate_pool = set(bm25_results.keys()) | set(dense_results.keys())
    fused_scores = {}
    for prov_id in candidate_pool:
        rank_bm25 = bm25_results.get_rank(prov_id, default=101)
        rank_dense = dense_results.get_rank(prov_id, default=101)
        rrf_score = (1.0 / (k_rrf + rank_bm25)) + (1.0 / (k_rrf + rank_dense))

        # Apply hierarchy safeguard: boost primary national Republic Acts by 20%
        if prov_id.startswith("RA_") or is_primary_act(prov_id):
            rrf_score *= beta
        fused_scores[prov_id] = rrf_score

    # 3. Sort candidates by fused score and return top-k shortlist
    return sorted(fused_scores.items(), key=lambda x: x[1], reverse=True)[:k]

```

Listing 3.8: Pseudocode of the Stage 1 Reciprocal Rank Fusion Hybrid Provision Retrieval Pipeline

Listing 3.8 outlines the Python implementation of the Stage 1 hybrid retrieval pipeline, combining BM25 keyword search, dense Bi-Encoder retrieval, and Reciprocal Rank Fusion with hierarchy boosting. This code demonstrates how the system narrows down more than 27,000 statutory provisions to a focused shortlist of top-$k$ candidates ($k = 50$). In this routine, the draft section is searched independently across sparse and dense indices to retrieve the top 100 candidates from each method. The algorithm then iterates across all retrieved provisions, calculates the RRF score using Equation 3.9, and applies a 20% score boost ($\beta = 1.20$) to primary national statutes (Republic Acts) to ensure important laws are not buried under administrative issuances. Sorting the combined scores yields a compact, high-precision candidate shortlist that delivers relevant legal context to Stage 2 inference in under 50 milliseconds.

### 3.7.2 Stage 1 Retrieval Safeguards

The Statutory Hierarchy Safeguard is an algorithmic re-ranking mechanism that operationalizes the constitutional doctrine of statutory supremacy established in the landmark *Magtajas* doctrine [207] and the delegated legislative limits of Republic Act No. 7160 (specifically Sections 16 and 458). In the Philippine legal hierarchy, an ordinance enacted by a local sanggunian is legally subordinate to an Act of Congress. In the raw national statutory corpus, routine executive orders and administrative issuances comprise nearly half of all enactments (48.01%), outnumbering primary congressional legislation by thousands of documents. If a retrieval algorithm assigns equal weight across all document types, standard lexical queries can easily become saturated with administrative issuances, displacing governing Republic Acts from the top candidate ranks. The Statutory Hierarchy Safeguard applies a calibrated score multiplier ($\beta = 1.20$) to primary national statutes during rank fusion. This weighting ensures that primary national laws maintain prominence within the candidate shortlist, reflecting the constitutional supremacy of statutory law over delegated administrative rules.

Using a two-stage retrieve-then-entail setup introduces potential search challenges

when comparing draft ordinances against large statutory collections. If the coarse search misses the governing national law, the inference model cannot detect the conflict regardless of its analytical capability. Legal conflicts often appear through indirect wording, where local drafts use permissive phrasing or leave out specific statutory terms. Standard search tools can also miss isolated prohibitions tucked inside long, otherwise compliant legislative codes. To protect the pipeline against these retrieval errors, the system incorporates five architectural safeguards across data preparation and search. Table 3.7 summarizes these statutory challenges alongside their operational countermeasures.

Table 3.7: Architectural Retrieval Safeguards Against Real-World Preemption Failure Modes in Stage 1

| **Statutory Challenge** | **Architectural Countermeasure** | **Operational Implementation Logic** |
|---|---|---|
| **Isolated Statutory Clauses** | Structural Section Chunking ($\sim$ 200–300 token windows) | Segments multi-chapter enactments into discrete provisions so isolated prohibitions rank independently without vector dilution. |
| **Indirect or Evasive Wording** | Dense Bi-Encoder Semantic Vectors (`all-MiniLM-L6-v2`) | Maps conceptual intent into 384-dimensional latent space, retrieving relevant statutes despite divergent local phrasing. |
| **Numerical Penalty Limits** | Sparse BM25 Lexical Matching with Sublinear Scaling | Matches exact statutory citations, numerical penalty limits, and administrative agency acronyms directly. |
| **Rank Variance in Shortlist** | Top-$k$ Candidate Safety Buffer ($k = 50$ provision shortlist) | Retains 50 candidate provisions for Stage 2 review, ensuring relevant statutes ranked below the top ten are evaluated. |
| **Omnibus Statute Trap & Administrative Clutter** | Hierarchy Priority Safeguard & Soft Domain Priors | Prioritizes primary legislative statutes (Republic Acts) over administrative orders, while applying gentle soft priors instead of destructive hard filtering. |
| **Long-Tail Provisions ($>$ 512 Tokens)** | Dual-Stream BM25 Invariance & Hierarchical Max-Pooling | BM25 indexes full text without sequence limits, while dense representations of tail provisions apply hierarchical sub-chunking with inference max-pooling and long-context candidate support (ModernBERT 8,192 tokens). |

Table 3.7 summarizes six common statutory search challenges encountered when screening Philippine laws, along with the system's architectural safeguards and implementation logic in Stage 1. These safeguards were developed to address specific search errors observed during initial tests across the 25,432 national statutes. As detailed in the table, the pipeline handles isolated prohibitions through section chunking, indirect wording through dense semantic vectors, exact penalty numbers through BM25 keyword matching, ranking variation through a 50-candidate safety buffer, administrative clutter through hierarchy boosting, and long provisions through sub-chunking and ModernBERT support. Implementing these safeguards prevents sharp drops in retrieval recall, ensuring that broad laws like the Local Government Code are not left out. As a result, the candidate shortlist reliably captures both direct penalty limits and subtle statutory exceptions.

These safeguards directly resolve critical failure modes uncovered during preliminary retrieval experiments across the 25,432 national statutes. In particular, attempting

a naive hard filter based on predicted macro domains caused candidate Recall@50 to collapse from 43.14% down to 13.43%. This severe degradation occurs because foundational Philippine enactments, such as Republic Act No. 7160, function as omnibus codes that span local administration, disaster response, and local taxation simultaneously. Hard-filtering by a single predicted topic immediately prunes the governing code whenever a draft clause touches another administrative area. Replacing hard filtering with a Hierarchy Priority Safeguard and soft domain priors follows the ranking principles of Robertson and Zaragoza [176] and Cormack et al. [46], ensuring that multi-domain omnibus enactments are not prematurely excluded. This safeguard shields Republic Acts from the 12,210 administrative executive orders (48% of the corpus), increasing the Mean Reciprocal Rank from 0.1521 to 0.2217 (+45.8% improvement).

Separating Stage 1 retrieval from Stage 2 inference is essential when running on standard office hardware. Feeding full-length statutes directly into a Cross-Encoder requires quadratic computation ($O((L_{\text{ord}} + L_{\text{stat}})^2)$), which quickly exhausts local memory on long legislative texts. Full-text inputs also dilute model attention, burying specific penalty clauses under thousands of routine procedural words. The hybrid retriever in Stage 1 performs a rapid coarse-to-fine filter (25,432 $\rightarrow$ 50 $\rightarrow$ 1), removing over 99.8% of unrelated laws in milliseconds. This separation allows the Cross-Encoder to focus its attention entirely on high-probability candidate provisions.

To evaluate these architectural mechanisms, the proponents benchmarked nine candidate Stage 1 retrieval configurations across both the full 25,432 national statutory database and the 350-pair ground truth evaluation set (comprising 198 unique instances). The full empirical retrieval findings, difficulty tier ablations, latency metrics, and comparative analyses are formally reported and discussed in Chapter 4 (Section 4.2).

### 3.7.3 Stage 2 Conflict Detection Pipeline

After receiving the top-$k$ candidate shortlist from Stage 1, the pipeline moves to logical evaluation using the Natural Language Inference (NLI) module. The system formats an input sequence for each candidate provision, pairing the retrieved national law (the premise) with the draft ordinance (the hypothesis). To guide the model's attention, the pipeline places a $[CLS]$ token at the start of the sequence and a $[SEP]$ token between the two texts. This structure helps the model distinguish the existing law from the proposed draft. The combined text is then processed by the selected Cross-Encoder model. The model's attention mechanism compares every word in the premise against every word in the hypothesis to evaluate their relationship. A classification layer outputs calibrated softmax probabilities labeling the relationship as Entailment, Contradiction, or Neutral. If the contradiction probability crosses the designated threshold, the Explainable AI (XAI) module highlights the conflicting text for user review.

To execute full-document pre-enactment audits on resource-constrained municipal workstations without prohibitive processing delays or memory exhaustion, the Stage 2 inference pipeline incorporates three complementary efficiency mechanisms: procedural boilerplate bypassing, dynamic token padding, and cross-section flattened batching. First, rather than routing every structural section indiscriminately to the Cross-Encoder, the

pipeline filters out formulaic administrative clauses (such as Title, Definition of Terms, Separability, Repealing, and Effectivity clauses) identified via the regular expression parser in Section 3.4.1. As established in the local corpus census (Section 3.4.2.4), procedural clauses represent approximately 30.85% of ordinance sections while carrying zero substantive regulatory commands; filtering them before inference directs cross-encoder attention exclusively toward operative legal duties without risk of discarding regulatory provisions. Second, standard transformer implementations enforce static padding up to the model's maximum sequence length (such as 256 or 512 tokens), allocating fixed blocks of zero-padding memory. Because transformer self-attention exhibits quadratic complexity ($O(L^2)$) with respect to input length $L$, computing cross-attention over padding tokens expends substantial matrix multiplication operations on uninformative positions. The pipeline replaces static padding with dynamic sequence padding, clamping tensor dimensions strictly to the length of the longest clause within each mini-batch (which typically spans 60 to 90 tokens for statutory sections). This dynamic clamping curtails quadratic attention operations relative to static maximum ceilings. Third, adapting the multi-candidate passage reranking paradigm of Nogueira and Cho [154], rather than processing candidate provisions sequentially in iterative per-section loops, the pipeline aggregates candidate pairs across all substantive sections into a unified flattened mini-batch. Executing this unified tensor through a single vectorized forward pass eliminates repeated execution call overhead and utilizes multi-threaded central processing unit vectorization (AVX2/AVX-512 SIMD parallelism).

Finally, to protect against subtle numerical preemption that deep neural cross-encoders occasionally overlook on surface arithmetic, the Stage 2 pipeline integrates a deterministic Symbolic Integer Ceiling Verifier. Under Section 458(a)(1)(iii) of Republic Act No. 7160, local sanggunians are strictly prohibited from prescribing ordinance fines exceeding Five Thousand Pesos (₱5,000.00) or imprisonment exceeding one year. The symbolic verifier scans operative penal clauses using regular expression pattern matching to extract numerical fine amounts (`PHP\s*[0-9,]+`). If an extracted integer penalty exceeds the statutory ₱5,000.00 ceiling, the provision triggers an immediate deterministic Contradiction flag alongside statutory citation to Section 458. This hybrid neuro-symbolic mechanism safeguards against false negatives on quantitative limits while preserving pure neural cross-attention for complex deontic reasoning.

### 3.7.4 Statutory Knowledge Graph and Lineage Architecture

To augment the two-stage retrieve-then-entail pipeline with structural intelligence, the system incorporates a multi-relational Statutory Knowledge Graph that models the institutional authority and amendatory lineages across the dual statutory corpus. Rather than operating exclusively as an isolated retrieval index, the knowledge graph serves as a shared topological backbone supporting both coarse candidate retrieval in Stage 1 and fine cross-encoder inference in Stage 2. By mapping explicit statutory cross-references and constitutional delegations alongside legislative amendments into a directed multi-graph, the architecture provides the computational pipeline with point-in-time statutory awareness while preventing obsolete or repealed laws from entering the candidate pool.

#### 3.7.4.1 Multi-Relational Graph Formulation and Schema

Formally, the statutory knowledge base is structured as a heterogeneous, directed multi-relational graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}) \tag{3.10}$$

Where $\mathcal{V}$ denotes the set of heterogeneous legal vertices, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$ represents the set of directed, labeled edges $(u, r, v)$, and $\mathcal{R}$ designates the vocabulary of canonical legal relation types.

The vertex set $\mathcal{V} = \mathcal{V}_{\text{enact}} \cup \mathcal{V}_{\text{prov}} \cup \mathcal{V}_{\text{dom}}$ comprises three distinct entity classes. Enactment nodes $u \in \mathcal{V}_{\text{enact}}$ represent the 27,357 complete legislative instruments (25,432 national acts and 1,925 local ordinances), storing metadata attributes including enactment year, legislative era, official title, and jurisdictional tier. Provision nodes $v \in \mathcal{V}_{\text{prov}}$ represent the 178,494 discrete statutory sections and ordinance articles, encapsulating the raw text, token length, operative status, and parent enactment identifier. Domain nodes $w \in \mathcal{V}_{\text{dom}}$ designate the eight macro legal domain hubs established in Section 3.3.3.

The relation vocabulary $\mathcal{R}$ formalizes five distinct juridical connections:

$$\begin{aligned}\mathcal{R} = \{&\text{CONTAINS}, \text{GOVERNS_DOMAIN}, \text{CITES_AUTHORITY},\\ &\text{AMENDS}, \text{REPEALS}, \text{CONFLICTS_WITH}\}\end{aligned} \tag{3.11}$$

Where structural relations CONTAINS link enactments to their constituent provisions, thematic relations GOVERNS_DOMAIN map provisions to macro legal domains, vertical authority edges CITES_AUTHORITY connect local ordinances to governing national statutes, and horizontal lineage edges AMENDS and REPEALS capture legislative modifications under *lex posterior derogat legi priori*. Finally, semantic preemption edges CONFLICTS_WITH are dynamically instantiated when Stage 2 cross-encoder inference predicts a preemption contradiction exceeding the calibrated operational threshold ($\tau^* = 0.45$).

This graph architecture supports both processing stages through a complementary division of labor. During Stage 1 candidate retrieval, the retriever utilizes the graph to perform 1-hop and 2-hop neighborhood expansion, following CONTAINS edges to pull adjacent sister provisions and AMENDS edges to pull subsequent amendatory enactments, while immediately pruning provisions that carry incoming REPEALS edges. During Stage 2 cross-encoder inference, the pipeline traverses hierarchical paths to prepend canonical statutory lineages (`[Code > Chapter > Section]`) to the premise text, providing the self-attention matrix with structural authority context. Furthermore, discrete attribute constraints stored on provision nodes (such as fine ceilings under Republic Act No. 7160 Section 458) directly inform symbolic ceiling verification.

#### 3.7.4.2 Deterministic Relation Extraction and Preemption Linkage

The automated construction of the knowledge graph from unstructured legal texts follows a sequential multi-stage extraction pipeline. In the initial phase, an entity normalization module parses surface textual references—such as "Republic Act No. 7160," "R.A. 7160," and "The Local Government Code of 1991"—mapping them to unified canonical

entity identifiers via regular expression aliasing dictionaries. Standardizing entity identifiers ensures that multi-word references resolve to identical graph vertices, preventing fragmentation across legislative eras.

Following canonical normalization, deterministic regular expression parsers scan operative statutory provisions to extract explicit cross-references and statutory delegations. The parser identifies statutory patterns referencing Acts of Congress alongside historical Commonwealth Acts and Presidential Decrees, generating directed CITES_AUTHORITY edges that establish the vertical statutory hierarchy. Concurrently, an amendatory lineage parser identifies explicit modification formulas mandated by Philippine legislative drafting manuals, recognizing phrases such as "is hereby amended to read as follows" and "is hereby repealed in its entirety." Parsing these clauses instantiates directed AMENDS and REPEALS edges linking amendatory ordinances back to historical parent enactments.

Assigning enactment timestamps to vertices enables point-in-time statutory validity tracking. When an incoming AMENDS edge is detected, the graph engine assigns a modified status attribute to the predecessor provision, while an incoming REPEALS edge assigns an obsolete status attribute. This state tracking ensures that the automated audit evaluates draft legislation against the live, operative state of positive law rather than superseded historical provisions. When optical character recognition noise generates truncated numerical citations on pre-LGC typewriter archives, the pipeline applies fuzzy string matching against the verified Lawphil enactment gazette or routes unresolved citations into diagnostic quarantine pools to prevent false edge creation.

#### 3.7.4.3 Topological Graph Experiments and Diagnostic Validation

To evaluate the structural integrity and network characteristics of the constructed statutory knowledge graph, the proponents formulate five standalone graph experiments and exploratory diagnostic checks executed directly upon the network topology using NetworkX.

The first experiment evaluates statutory authority through In-Degree Centrality and PageRank. Normalized In-Degree Centrality measures the direct citation volume directed toward each national statute:

$$C_{\text{in}}(v) = \frac{|\{u \in \mathcal{V} \mid (u, \text{CITES_AUTHORITY}, v) \in \mathcal{E}\}|}{|\mathcal{V}| - 1} \tag{3.12}$$

Where $C_{\text{in}}(v)$ quantifies the proportion of legal enactments citing statute $v$. Complementing direct citation counts, statutory PageRank models the recursive propagation of legal authority across multi-tiered legislative delegations:

$$PR(u) = \frac{1 - d}{|\mathcal{V}|} + d \sum_{v \in \mathcal{N}_{\text{in}}(u)} \frac{PR(v)}{|\mathcal{N}_{\text{out}}(v)|} \tag{3.13}$$

Where $\mathcal{N}_{\text{in}}(u)$ denotes the set of vertices citing statute $u$, $\mathcal{N}_{\text{out}}(v)$ denotes the outgoing citations of vertex $v$, and $d = 0.85$ represents the canonical damping factor. Computing

these centrality metrics serves as an empirical sanity check, verifying whether foundational organic statutes—specifically Republic Act No. 7160—mathematically emerge as dominant structural hubs governing municipal law.

The second experiment evaluates network degree distribution to determine whether statutory citations follow natural scale-free topology. By fitting degree distributions to a power-law formulation ($P(k) \sim k^{-\gamma}$), the experiment tests whether positive law exhibits Pareto dynamics, where a compact nucleus of primary codes concentrates the majority of legal citations while the long tail of specialized enactments receives localized references.

The third experiment applies unsupervised community detection using the Louvain modularity optimization algorithm. Partitioning the citation graph into topological communities tests whether legal enactments naturally cluster into substantive regulatory subjects solely through citation density, evaluating the degree of alignment between unsupervised graph communities and the eight supervised macro legal domains.

The fourth experiment conducts a global connectivity audit to quantify network integration and identify disconnected legislation. The analysis measures the proportion of local ordinances encapsulated within the Giant Connected Component (GCC) anchored to national law, while systematically isolating orphan vertices with zero in-degree and zero out-degree. Auditing these isolated nodes determines whether disconnections stem from substantive legislative independence (such as ceremonial street renamings) or optical scanning artifacts requiring manual archival review.

The fifth experiment validates legislative lineage consistency through Directed Acyclic Graph (DAG) cycle detection and amendatory depth profiling. Because legislative modifications must follow a chronological, non-circular progression, the experiment executes simple cycle detection algorithms across all AMENDS subgraphs to mathematically confirm the complete absence of recursive amendatory loops. Concurrently, depth profiling isolates the longest amendatory chains in local municipal history, measuring the cumulative generational revisions governing complex regulatory instruments such as the Davao City Revenue Code.

#### 3.7.4.4 Knowledge Graph Integration Ablation: Representation Conditioning vs. Decision Calibration

To resolve whether knowledge graph topology should inform the cross-encoder's input linguistic representation or its post-softmax decision boundary, the proponents formulate a rigorous 4-way ablation framework evaluated on the authentic Tier 3 landmark jurisprudential suite. This comparative design tests four discrete architectural paradigms:

1. **Configuration A (Pure Neural Baseline):** Evaluates the fine-tuned `DeBERTa-v3` cross-encoder strictly on authentic verbatim statutory texts without metadata prefixes, using the global static decision threshold $\tau_{\text{base}} = 0.45$.
2. **Configuration B (Topologically Calibrated Dynamic Thresholding):** Retains 100% authentic verbatim input text while modulating the operational decision boundary $\tau^*(u, v)$ as a continuous function of shortest-path graph distance $d_G(u, v)$

pursuant to Equations 3.20 and 3.21, tightening the screening threshold down to $\tau^* \in [0.30, 0.375]$ for enactments sharing direct citation or domain links in $\mathcal{G}$.

3. **Configuration C (Prompt-Conditioned Jurisdictional Metadata Conditioning):** Prepends canonical serialized knowledge graph node attributes directly to the input sequence (e.g., `[JURISDICTION: NATIONAL | DOMAIN: ...]` for premises and `[JURISDICTION: LOCAL | DOMAIN: ...]` for hypotheses), evaluated under the static threshold $\tau_{\text{base}} = 0.45$.
4. **Configuration D (Combined Model):** Evaluates the prompt-conditioned metadata input sequences under the topologically calibrated dynamic threshold $\tau^*(u, v)$.

Benchmarking these four configurations isolates the empirical trade-offs of representation-level prompt conditioning against post-hoc decision boundary calibration. Specifically, this ablation determines whether cross-attention layers attend to substantive legal semantics or suffer from attention distribution shift on contrasting jurisdictional prefix tokens, validating the safest operational paradigm for municipal legislative screening.

### 3.7.5 Baseline Selection

To evaluate whether the proposed two-stage retrieve-then-entail architecture achieves meaningful empirical advances over standard approaches, the proponents established formal baselines for both operational stages grounded in recent legal informatics literature.

For coarse statutory retrieval, the pipeline is benchmarked against pure Okapi BM25 lexical matching adapted from Team AIIR [239] in the COLIEE 2025 competition. BM25 serves as the canonical baseline for statutory information retrieval (COLIEE Task 3) because legislative drafting frequently relies on explicit section citations, formal agency acronyms, and verbatim penalty amounts. Choosing BM25 as the Stage 1 baseline provides an essential reference point to determine whether semantic dense vector representations and Reciprocal Rank Fusion genuinely improve candidate recall, especially across paraphrased or indirect preemption queries where exact keyword matching is known to fail.

For fine natural language inference, the pipeline establishes a dual-baseline framework combining an unadapted task baseline with a canonical supervised foundation. Following the legal textual entailment baseline established by Rabelo et al. [167] for COLIEE Task 4, an unadapted zero-shot Cross-Encoder tests whether off-the-shelf general-domain or multi-task language models can detect statutory preemption without local domain adaptation. This baseline establishes the empirical lower bound and directly quantifies the severity of affirmative bias. Complementing this, the proponents establish the standard `bert-base-uncased` model (110M parameters) developed by Devlin et al. [62] as the canonical supervised baseline. Fine-tuned under identical training conditions on the curated statutory pairs, standard BERT isolates the exact performance premium contributed by specialized attention mechanisms, legal domain pretraining, and warm NLI initialization over traditional masked language modeling.

For graph-augmented candidate expansion and lineage tracking, the system bench-

marks the neuro-symbolic knowledge graph against the unaugmented dual-stream (BM25 + Bi-Encoder) retrieval baseline. This comparison evaluates whether traversing explicit statutory citation edges and amendatory lineages recovers missing governing authorities in multi-section legislative drafts where pure lexical and semantic search fail to capture unstated legal hierarchies.

## 3.8 Model Selection and Optimization

Before deploying the two-stage pipeline, the proponents evaluate candidate neural models across both retrieval and inference stages to identify the most accurate and computationally efficient architectures. Rather than selecting arbitrary models or relying on closed external cloud APIs, candidate architectures are strictly bounded by the computing resources of local government hosts. Specifically, models must satisfy parameter and memory ceilings that allow real-time execution on standard desktop and laptop hardware. Once the optimal models are selected and calibrated, their weights and tokenizer configurations are permanently pinned for offline execution (`local_files_only = True`), ensuring that the production pipeline runs securely within air-gapped local government IT environments without external cloud dependencies.

### 3.8.1 Stage 1 Candidate Retrieval Models

To identify the optimal neural retriever for the coarse search stage, the proponents evaluate eight candidate open-weight bi-encoders representing four distinct inductive categories reviewed in Section 2.8.6:

- **Distilled Edge Baselines:** `all-MiniLM-L6-v2` (22.7M parameters) and `bge-small-en-v1.5` (33.4M parameters), engineered for sub-two-millisecond latency on commodity CPU hardware.
- **General MTEB Benchmark Leaders:** `bge-base-en-v1.5` (109M parameters), `e5-base-v2` (109M parameters), and `all-mpnet-base-v2` (109M parameters), providing balanced general-domain semantic representations.
- **Legal Domain-Adapted Encoders:** `legal-bert-base-uncased` (110M parameters), continuously pretrained over statutory and judicial corpora.
- **Modern Long-Context and Multilingual Encoders:** `ModernBERT-base` (149M parameters) and `bge-m3` (568M parameters), supporting native 8,192-token context windows and multi-language statutory representations.

All candidate bi-encoders are evaluated standalone across candidate retrieval recall (Recall@$k$ for $k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank (MRR), mean CPU query latency, and latent paraphrastic conflict sensitivity (Tier 3). The bi-encoder delivering the optimal balance of high candidate recall and low edge latency is selected as the primary dense retrieval workhorse for the hybrid Reciprocal Rank Fusion pipeline. The empirical retrieval benchmark comparing these eight candidate bi-encoders is reported in Chapter 4 (Section 4.2).

### 3.8.2 Stage 2 Candidate Inference Models

To evaluate a comprehensive spectrum of transformer inductive biases for Stage 2 semantic conflict detection, the proponents scouted 28 candidate open-weight encoders spanning six architectural families:

- **Legal-Domain Pretrained Encoders:** `Legal-BERT-base-uncased` (110M), `PoL-BERT-Large` (340M), `InLegalBERT` (110M), `CaseLaw-BERT` (110M), and `Lawformer` (110M).
- **Pre-Aligned NLI Reasoners and General Baselines:** `DeBERTa-v3-base-NLI` (86M), `DeBERTa-v3-large-NLI` (435M), cold `deberta-v3-base` (86M), `bert-base-uncased` (110M), `roberta-large-mnli` (355M), `bart-large-mnli` (406M), and `electra-large-disc` (335M).
- **Modern Long-Context Architectures:** `ModernBERT-base` (149M), `ModernBERT-large` (395M), `longformer-base-4096` (149M), `bigbird-roberta-base` (128M), and `nomic-bert-2048` (137M).
- **Native Cross-Encoder Rerankers:** `bge-reranker-v2-m3` (568M), `bge-reranker-base` (278M), and `ms-marco-MiniLM-L12` (33M).
- **Multilingual Encoders:** `mdeberta-v3-base` (86M), `xlm-roberta-base` (270M), `xlm-roberta-large` (550M), and `roberta-tagalog-base` (110M).
- **Distilled Edge Baselines:** `all-MiniLM-L6-v2` (22M), `nli-distilroberta-base` (82M), `distilbert-base-uncased` (66M), and `albert-base-v2` (12M).

To document the exact software versions, release years, parameter capacities, context limits, and precision settings used across all pipeline stages, Table 3.8 provides the technical specification matrix for the deployed ingestion, retrieval, and shortlisted inference architectures.

Table 3.8: Technical Specifications, Checkpoint Identifiers, Release Years, and Architectural Bounds of Deployed and Shortlisted Models

| **Model / Alias** | **Base Checkpoint / Version** | **Year** | **Params** | **Max Seq. / Dim.** | **Stage & Configuration** |
|---|---|---|---|---|---|
| Qwen2.5-VL-3B | `Qwen/Qwen2.5-VL-3B-Instruct` [216] | 2025 | 3.0B | 32,768 (1K gen) | Stage 0 OCR; 4-bit NF4 |
| all-MiniLM-L6-v2* | `sentence-transformers/all-MiniLM-L6-v2` | 2021 | 22.7M | 512 / 384-d | Stage 1 Dense + Stage 2 |
| bge-small-en-v1.5 | `BAAI/bge-small-en-v1.5` [241] | 2023 | 33.4M | 512 / 384-d | Stage 1 Dense candidate |
| bge-base-en-v1.5 | `BAAI/bge-base-en-v1.5` [241] | 2023 | 109M | 512 / 768-d | Stage 1 Dense candidate |
| e5-base-v2 | `intfloat/e5-base-v2` [235] | 2022 | 109M | 512 / 768-d | Stage 1 Dense candidate |
| all-mpnet-base-v2 | `sentence-transformers/all-mpnet-base-v2` [200] | 2021 | 109M | 512 / 768-d | Stage 1 Dense candidate |
| bge-m3 | `BAAI/bge-m3` [35] | 2024 | 568M | 8,192 / 1,024-d | Stage 1 Dense candidate |
| DeBERTa-v3-base-NLI* | `MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli` [124] | 2024 | 86M | 512 / 768-d | Stage 2 Finalist (12L, 12H) |
| DeBERTa-v3-large-NLI | `MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli` [124] | 2024 | 435M | 512 / 1,024-d | Stage 2 Ceiling (24L, 16H) |
| deberta-v3-base | `microsoft/deberta-v3-base` [95] | 2023 | 86M | 512 / 768-d | Stage 2 Cold MLM baseline |
| ModernBERT-base | `answerdotai/ModernBERT-base` [237] | 2024 | 149M | 8,192 / 768-d | Stage 1 & 2 FlashAttn-2 |
| ModernBERT-large | `answerdotai/ModernBERT-large` [237] | 2024 | 395M | 8,192 / 1,024-d | Stage 2 Long-Context |
| PoL-BERT-Large | `pile-of-law/legalbert-large-1.7M-2` [96] | 2022 | 340M | 512 / 1,024-d | Stage 2 Legal Admin |
| legal-bert-base | `nlpaueb/legal-bert-base-uncased` [31] | 2020 | 110M | 512 / 768-d | Stage 1 & 2 Legal Judicial |
| InLegalBERT | `law-ai/InLegalBERT` [162] | 2022 | 110M | 512 / 768-d | Stage 2 Common Law |
| bert-base-uncased | `google-bert/bert-base-uncased` [62] | 2019 | 110M | 512 / 768-d | Stage 2 Supervised Anchor |
| bge-reranker-v2-m3 | `BAAI/bge-reranker-v2-m3` [35] | 2024 | 568M | 8,192 / 1,024-d | Stage 2 Reranker head |
| mdeberta-v3-base | `microsoft/mdeberta-v3-base` [95] | 2023 | 86M | 512 / 768-d | Stage 2 Multilingual |
| nli-distilroberta | `cross-encoder/nli-distilroberta-base` | 2021 | 82M | 512 / 768-d | Stage 2 Distilled NLI |

Table 3.8 details the exact checkpoint identifiers, release years, parameter counts, and sequence bounds for every deployed and shortlisted transformer model evaluated in the pipeline. Providing exact repository versions rather than generic family labels ensures experimental traceability across both the laptop training environment (Host B) and the municipal desktop edge testbed (Host A). As shown in the matrix, the candidate architectures span compact 22.7M-parameter bi-encoders built for rapid cosine indexing alongside 86M-to-568M-parameter cross-encoders featuring standard 512-token or extended 8,192-token windows. Pinning both the base NLI checkpoint (`MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli`) and the quantized 4-bit ingestion model (`Qwen2.5-VL-3B-Instruct`) establishes fixed software baselines prior to empirical tuning.

Evaluating all 28 candidate models through full training runs and human evaluation would require prohibitive computing time and memory. Following standard experimental practices in machine learning [64, 30], the proponents organized the evaluation into a practical Three-Stage Funnel. In the first stage of rapid zero-shot screening ($N = 28$), all candidate models are tested out-of-the-box on the stratified held-out test split ($\mathcal{D}_{\text{test}}$, $N = 53$) without task fine-tuning. This initial screening rapidly benchmarks inference latency and GPU memory occupancy on standard test hardware while revealing baseline affirmative bias where unadapted models reflexively default toward Entailment or Neutral on complex statutory text [5, 220]. Promising models that satisfy baseline operational constraints advance to the second stage of supervised fine-tuning and ablation sweeps. In this stage, models undergo supervised task adaptation on the training set ($N_{\text{train}} = 245$) and validation split ($N_{\text{val}} = 52$) across standard learning rates and batch sizes ($\eta \in \{1, 2, 3\} \times 10^{-5}$, $B \in \{8, 16\}$), testing empirical loss convergence, early stopping triggers, and cost-sensitive threshold calibration ($\tau^* \in [0.25, 0.50]$). Finally, in the third stage of finalist benchmarking, the top performing models on the validation split, along with the canonical bert-base-uncased baseline, are selected as finalists for full out-of-sample evaluation on the 350-pair ground truth dataset (comprising 198 unique instances).

Restricting the final evaluation to the top five validation models alongside the standard baseline anchor is guided by four practical methodological considerations. First, tuning hyperparameters and selecting models strictly on validation data ($\mathcal{D}_{\text{val}}$) protects the integrity of the test set, preventing holdout data snooping and ensuring that out-of-sample generalization metrics remain methodologically unbiased [67]. Second, restricting pairwise statistical comparisons to six finalists controls family-wise error rates, avoiding the inflated risk of false positives inherent in uncorrected testing across dozens of models [63] while enabling dependable pairwise significance testing against bert-base-uncased ($\chi^2, p < 0.05$). Third, benchmarking across model scales isolates efficiency trade-offs, determining whether incremental accuracy gains from 400M-parameter architectures justify their substantial latency and VRAM footprint on municipal office hardware [194, 204]. Fourth, the finalist cohort spans diverse inductive biases, encompassing pre-aligned NLI reasoners (DeBERTa-v3-large-NLI and DeBERTa-v3-base-NLI), modern long-context encoders (ModernBERT-large), unaligned general foundations (cold deberta-v3-base), and legal domain pretraining (PoL-BERT-Large).

The comprehensive empirical screening benchmark across all 28 candidate models, zero-shot performance statistics, asterisk designations, and admission decisions are formally presented in Chapter 4 (Section 4.3).

### 3.8.3 Overfitting Controls and Calibration

Because fine-tuning deep transformer models in a few-shot setting ($N_{\text{train}} = 245$) can lead to memorizing training phrasing, the researchers use several regularization techniques. First, model optimization uses the AdamW algorithm with weight decay ($\lambda = 0.01$), as developed by Loshchilov and Hutter [134]. This weight decay keeps network parameters small and prevents individual weights from growing disproportionately large. Second, dropout layers with a rate of $p = 0.1$ are applied across attention layers and the classification head [203]. By turning off a random subset of neurons during training passes, dropout encourages the model to learn broader semantic features rather than depending on specific word patterns.

Third, training uses early stopping based on holdout validation loss [165]. The system measures validation loss and the validation Macro $F_1$-score after every training epoch on the 52-pair validation set ($\mathcal{D}_{\text{val}}$). If holdout validation loss does not improve over two consecutive checks (patience = 2), training stops automatically. The system then restores the optimal model checkpoint that achieved the lowest validation loss (as observed at Epoch 3 in Table 4.12), preventing the model from fitting too closely to the training data. Finally, the learning rate warms up linearly over the first 10% of training steps before decreasing steadily, which helps keep gradient updates stable during the initial phase of fine-tuning.

Recent empirical findings from the COLIEE 2026 legal textual entailment benchmarks (Task 4) demonstrated that neural language models frequently exhibit affirmative bias, defaulting toward entailment or neutral classifications when confronted with complex statutory syntax [5, 220]. In local legislative review, failing to detect an authentic conflict permits an invalid ordinance to proceed to enactment. To counter this affirmative bias, the proponents calibrate the decision threshold for the contradiction class using validation holdout curves. Rather than assuming an uncalibrated argmax decision rule, the threshold is adjusted to maximize the Contradiction F1-Score while maintaining high specificity. This calibration ensures that genuine preemption risks are surfaced to legislative staff rather than being suppressed by ambiguous phrasing.

### 3.8.4 Hyperparameter Tuning

To identify the optimal internal configuration for candidate Stage 2 Cross-Encoder models, the proponents implemented a systematic Grid Search procedure. Grid search is chosen over random or Bayesian optimization because the search space across lightweight transformer encoders is compact and discrete, allowing an exhaustive, reproducible evaluation of all candidate parameter combinations on the dedicated Host B (NVIDIA GeForce RTX 5050 Laptop GPU).

The search isolates key variable parameters—specifically learning rate, batch size,

and maximum sequence length—while holding structural regularization constants fixed across all training iterations. Table 3.9 specifies the complete partitioning of fixed experimental parameters versus variable search ranges.

Table 3.9: Hyperparameter Tuning Configuration and Grid Search Space for Stage 2 Cross-Encoders

| **Hyperparameter** | **Search Status** | **Configured Value / Search Bounds** |
|---|---|---|
| Optimization Algorithm | Fixed | AdamW with decoupled weight decay ($\lambda = 0.01$) [134] |
| Attention & Hidden Dropout | Fixed | $p = 0.10$ across all transformer and classification layers [203] |
| Warmup Schedule | Fixed | Linear warmup over first 10% of total training steps |
| Early Stopping Patience | Fixed | patience = 2 validation epochs (validation loss evaluation) [165] |
| Random Initialization Seed | Fixed | seed = 42 (enforcing deterministic mini-batch collation) |
| Maximum Training Epochs | Fixed | $E_{\max} = 10$ epochs (with best validation checkpoint restoration) |
| Loss Criterion | Fixed | Multi-Class Cross-Entropy Loss with Softmax normalization |
| Initial Learning Rate ($\eta$) | Variable (Grid Search) | $\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}, 5 \times 10^{-5}\}$ |
| Training Batch Size ($B$) | Variable (Grid Search) | $B \in \{8, 16\}$ (governed by 8 GB VRAM execution ceilings) |
| Max Input Sequence Length ($L$) | Variable (Grid Search) | $L \in \{256, 512\}$ (8,192 tokens for native ModernBERT) |

Table 3.9 specifies the complete hyperparameter search grid for Stage 2 Cross-Encoder fine-tuning, delineating fixed optimization controls from variable grid search bounds. This tuning configuration is established to achieve reproducible, stable convergence across compact transformer encoders on consumer GPU hardware while mitigating overfitting risks inherent in few-shot legal learning ($N_{\text{train}} = 245$). As detailed in the table, structural regularization constants are strictly fixed across all iterations, including AdamW optimization ($\lambda = 0.01$), attention dropout ($p = 0.10$), a 10% linear warmup schedule, early stopping (patience = 2), and a deterministic seed (seed = 42). Meanwhile, learning rate ($\eta \in \{1, 2, 3, 5\} \times 10^{-5}$), batch size ($B \in \{8, 16\}$), and sequence length ($L \in \{256, 512, 8192\}$) are systematically swept across the 52-pair validation set ($\mathcal{D}_{\text{val}}$). Quarantining hyperparameter selection strictly to the validation split prevents data snooping on the test set, guaranteeing that the checkpoint achieving peak validation Macro $F_1$ generalizes robustly to unseen statutory pairs.

All hyperparameter configurations are evaluated strictly on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) to prevent test-set data contamination. Model checkpoints are saved after every epoch, and the configuration achieving the highest validation Macro $F_1$-score is selected for final out-of-sample evaluation on the 53-pair test set.

## 3.9 Evaluation Metrics

Evaluating the two-stage pipeline requires separate metrics for each operational phase. The coarse retrieval phase in Stage 1 prioritizes candidate recall over final classification, while the inference phase in Stage 2 requires high precision to avoid false conflict alerts.

### 3.9.1 Stage 1 Retrieval Metrics

Measuring search performance in Stage 1 requires ranking metrics rather than standard classification accuracy [140]. In information retrieval, ranking metrics evaluate how effectively a search system places relevant documents at or near the top of a retrieved candidate list. In this pipeline, Stage 1 does not produce a final classification decision; its objective is to ensure that the governing national statute is successfully captured within the top-$k$ candidate shortlist ($k \in \{10, 30, 50\}$) passed to Stage 2. Recall at rank $k$ (Recall@$k$) defines the proportion of test queries for which the target statutory provision appears within the top $k$ retrieved positions. If a governing statute fails to appear within the top-$k$ cutoff, the downstream Cross-Encoder cannot evaluate the pair, resulting in a false negative retrieval omission. Complementing candidate recall, Mean Reciprocal Rank (MRR@$k$) measures retrieval precision by averaging the multiplicative inverse of the rank of the first relevant document across all test queries (1/rank). A high Recall@50 ensures candidate safety for downstream inference, while a higher MRR indicates that relevant provisions rank near the top of the shortlist. The metric is formalized as follows:

*Recall at Rank k (Recall@k):*

$$\text{Recall@}k = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \mathbf{1}\left(\text{rank}(d_q^*) \leq k\right) \tag{3.14}$$

Where $|Q|$ is the total number of test queries, $d_q^*$ is the correct national statute paired with query $q$, and $\mathbf{1}(\cdot)$ is an indicator function that equals one when the target law ranks within the top $k$ candidates.

Complementing candidate recall, Mean Reciprocal Rank at rank $k$ (MRR@$k$) evaluates how early the target statute appears within the candidate ranking [140]:

*Mean Reciprocal Rank at Rank k (MRR@k):*

$$\text{MRR@}k = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\text{rank}(d_q^*)} \cdot \mathbf{1}\left(\text{rank}(d_q^*) \leq k\right) \tag{3.15}$$

Where $d_q^*$ is the correct national statute retrieved for query $q$, with $1/\text{rank}(d_q^*)$ representing the reciprocal rank score.

Equations 3.14 and 3.15 define the ranking metrics used to test Stage 1 retrieval across the 25,432 national statutes. Because the goal of coarse retrieval is to create a top-$k$ shortlist ($k \in \{10, 30, 50\}$) rather than making final decisions, standard accuracy is not suitable. Recall@$k$ measures the percentage of test queries where the correct statute appears in the top-$k$ shortlist, while MRR@$k$ measures how close to the top that statute

ranks. High Recall@50 ensures that Stage 2 inference receives the governing statute needed to detect a conflict, while higher MRR places the most relevant law near the top of the list for legal researchers to review. These two metrics guide the selection of the best retrieval models in Chapter 4.

In this formula, a target statute retrieved at rank one receives a reciprocal score of 1.0, while a statute retrieved at rank ten receives 0.10. The 350 curated premise-hypothesis pairs serve as the query benchmark for testing Stage 1. By using the 350 draft clauses as queries against the 25,432-statute collection, the proponents measure how often candidate retrieval models locate the paired statutory provision. Calculating Recall@10, Recall@30, and Recall@50 confirms whether the coarse retriever supplies a dependable candidate pool for subsequent natural language inference.

### 3.9.2 Stage 2 Inference Metrics

The Stage 2 Cross-Encoder evaluates candidate statutory pairs across three discrete logical categories: $\mathcal{C} = \{\text{Entailment}, \text{Neutral}, \text{Contradiction}\}$. Following standard evaluation conventions in information extraction and natural language processing [36, 226], the baseline inference performance metrics are computed as follows:

*Class-Specific Precision, Recall, and $F_1$-Score:*

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F_{1,c} = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{3.16}$$

Where $TP_c$, $FP_c$, and $FN_c$ represent the true positive, false positive, and false negative classifications for category $c$, respectively. Precision reflects how reliably the model avoids false alarms, while Recall measures the proportion of genuine legal relationships successfully detected.

*Macro-Averaged $F_1$-Score Across Logical Classes:*

$$\text{Macro } F_1 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F_{1,c} \tag{3.17}$$

Where $|\mathcal{C}| = 3$ ensures that performance across all three categories is evaluated equally without allowing dominant neutral pairs to skew the outcome.

Equations 3.16 and 3.17 define the performance metrics for Stage 2 inference across the three NLI categories: Entailment, Neutral, and Contradiction ($|\mathcal{C}| = 3$). Precision ($P_c$) measures how well the model avoids false alarms, Recall ($R_c$) measures how many actual conflicts or relationships are caught, and the $F_1$-score balances both. Macro $F_1$ averages these scores equally across all three classes (1/3 each), ensuring that a model cannot achieve an artificially high score simply by predicting the majority class. This provides a balanced measure for comparing candidate models during screening.

In statutory conflict detection, classification errors carry fundamentally unequal operational costs [69, 71]. Cost-sensitive evaluation is a machine learning methodology that incorporates asymmetric misclassification costs into performance metrics and decision threshold calibration [69]. In conventional classification, false positives (Type I errors) and

false negatives (Type II errors) are penalized equally. In local legislative review, however, their operational consequences are highly asymmetric. A false negative—failing to identify an illegal or preempted ordinance section—allows a legally defective measure to advance to enactment, risking judicial invalidation by the Supreme Court under the *Magtajas* doctrine [207] and enforcement of unlawful penalties against citizens. Conversely, a false positive—flagging a compliant provision for committee review—requires a legal researcher to inspect the side-by-side text comparison and dismiss the informational notice. To reflect this asymmetric penalty structure, the evaluation adapts the $F_2$-score, which weights recall twice as heavily as precision ($\beta = 2$) to prioritize capturing potential contradictions.

Because a missed preemption is catastrophic while a false alarm is a minor momentary inconvenience, setting the decision threshold at an arbitrary 50% ($P \geq 0.50$) is dangerous. Standard classifiers default to a symmetric decision threshold of $P(\text{Contradiction}) \geq 0.50$, which assumes that false alarms and missed conflicts cause identical harm. In foundational cost-sensitive learning theory, Elkan [69] proved that when false negative costs exceed false positive costs ($C(FN) > C(FP)$), setting the decision boundary at 0.50 is mathematically suboptimal and produces an unsafe number of missed conflicts [71, 118].

To address this cost asymmetry, the proponents adapt the generalized $F_\beta$ formulation of Van Rijsbergen [226] and Chinchor [36] with $\beta = 2$, implementing a validation-tuned decision boundary $\tau_{\text{conflict}}$ for contradiction alerts. Rather than using a fixed 0.50 cutoff, the pipeline sweeps candidate thresholds across $\tau \in [0.25, 0.50]$ in increments of $\Delta\tau = 0.01$ exclusively on the validation set ($\mathcal{D}_{\text{val}}$, $N = 52$) following empirical decision calibration principles [71]. The cost-sensitive contradiction metrics are computed as follows:

*Cost-Sensitive Contradiction $F_2$-Score:*

$$F_{2,\text{conflict}} = \frac{5 \cdot P_{\text{conflict}} \cdot R_{\text{conflict}}}{4 \cdot P_{\text{conflict}} + R_{\text{conflict}}} \tag{3.18}$$

Where $P_{\text{conflict}}$ and $R_{\text{conflict}}$ denote Precision and Recall on the Contradiction class, weighting recall twice as heavily as precision ($\beta = 2$) to aggressively capture potential legal conflicts.

*Optimal Contradiction Decision Threshold Calibration ($\tau^*$):*

$$\tau^* = \arg \max_{\tau \in [0.25, 0.50]} F_{2,\text{conflict}}(\tau; \mathcal{D}_{\text{val}}) \tag{3.19}$$

Where $\mathcal{D}_{\text{val}}$ denotes the 52 validation pairs used to sweep candidate decision thresholds $\tau \in [0.25, 0.50]$ while enforcing an empirical precision floor of $P_{\text{conflict}} \geq 0.70$.

Equations 3.18 and 3.19 define the cost-sensitive evaluation metric and decision threshold calibration for conflict alerts. This design reflects a practical reality: missing a real conflict (a false negative) allows an invalid ordinance to pass, which can lead to legal challenges, whereas a false alarm (a false positive) is quickly dismissed by a legal researcher. In Equation 3.18, setting $\beta = 2$ weights recall twice as heavily as precision; this calibration prioritizes catching potential conflicts. Equation 3.19 searches for the

best decision threshold $\tau^*$ between 0.25 and 0.50 on the 52 validation pairs ($\mathcal{D}_{\text{val}}$), while keeping precision at or above 0.70. This tuning helps the system flag genuine conflicts reliably without overwhelming researchers with too many false alarms.

To extend static threshold calibration into a neuro-symbolic decision engine, the system integrates topological distance over the Statutory Knowledge Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. While the static threshold $\tau_{\text{base}} = 0.45$ establishes a dependable global baseline, candidate statutes that share direct authority citations or substantive regulatory domains with an ordinance warrant heightened screening sensitivity. To achieve this without altering the authentic verbatim text sequence, the proponents formulate the Topologically Calibrated Dynamic Decision Threshold:

*Topological Institutional Proximity ($\gamma(u, v)$):*

$$\gamma(u, v) = H(u, v) \cdot \frac{1}{d_G(u, v)} \tag{3.20}$$

Where $H(u, v) \in \{0.5, 1.0\}$ represents the jurisdictional authority hierarchy multiplier ($H = 1.0$ for vertical statutory preemption under *Magtajas v. Pryce Properties*, and $H = 0.5$ for horizontal intra-jurisdictional consistency), and $d_G(u, v)$ denotes the shortest-path geodesic distance between candidate statute $u$ and ordinance context $v$ traversing the Statutory Knowledge Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. If $u$ and $v$ are disconnected in $\mathcal{G}$, $d_G(u, v) \to \infty$ and $\gamma(u, v) = 0$.

*Topologically Calibrated Dynamic Decision Threshold ($\tau^*(u, v)$):*

$$\tau^*(u, v) = \tau_{\text{base}} - \Delta_\tau \cdot \gamma(u, v) \tag{3.21}$$

Where $\tau_{\text{base}} = 0.45$ represents the baseline cost-sensitive decision threshold calibrated on $\mathcal{D}_{\text{val}}$ via Equation 3.19, and $\Delta_\tau = 0.15$ denotes the maximum dynamic calibration bandwidth, bounding the operational screening threshold strictly within $\tau^*(u, v) \in [0.30, 0.45]$.

Equations 3.20 and 3.21 establish a continuous, mathematically bounded decision framework. When an enactment $u$ directly cites or amends $v$ ($d_G = 1$), $\gamma = 1.0$ and the screening threshold tightens to $\tau^* = 0.30$, aggressively flagging potential contradictions between established statutory authorities. When $u$ and $v$ share a common macro-domain via GOVERNS_DOMAIN edges ($d_G = 2$), $\gamma = 0.5$ and the threshold calibrates to $\tau^* = 0.375$. When enactments are distantly situated ($d_G = 3$, $\tau^* = 0.40$) or completely unlinked ($d_G \geq 4$ or $\infty$, $\tau^* = 0.45$), the model maintains the conservative baseline threshold $\tau_{\text{base}} = 0.45$, shielding valid municipal police power enactments from uncalibrated false alarms.

During live inference, a candidate provision triggers a formal conflict alert whenever its softmax contradiction probability satisfies $P(\text{Contradiction} \mid p_i, h) \geq \tau^*(u, v)$.

When evaluating complete multi-section ordinances during live system execution, Stage 1 retains its full safety buffer of $k = 50$ candidates for each substantive section ($S_i$). To prevent runtime delays when scoring up to 500 total candidate pairs per document, the Stage 2 inference engine employs vectorized mini-batch collation ($B = 16$). Batching evaluates candidate provisions in parallel tensor forward passes rather than sequential

single-item loops, keeping inference runtime under five seconds on GPU hosts and under thirty seconds on standard LGU processors. To consolidate candidate provision scores into an overall section verdict, the pipeline applies section-level max-pooling, adapting multi-instance learning principles [8]. Section-level max-pooling is an aggregation function that maps the set of $k$ individual contradiction probabilities for a given draft section to its maximum observed value: $P(\text{Conflict} \mid S_i) = \max_j P(\text{Contradiction} \mid P_j, S_i)$. In statutory review, an ordinance section that complies with 49 retrieved provisions but contradicts a single governing statute remains legally invalid under statutory preemption. Computing an arithmetic mean across candidate scores would dilute a high contradiction probability (such as 0.85) down to an undetectable fraction; that dilution would trigger a critical false negative. Max-pooling isolates the peak conflict probability across all candidate pairs, ensuring that an authentic legal violation triggers a system alert regardless of the number of non-conflicting candidates retrieved. The formula is computed as follows:

*Section-Level Max-Pooled Conflict Probability:*

$$P(\text{Conflict} \mid S_i) = \max_{j \in [1,k]} P(\text{Contradiction} \mid P_j, S_i) \tag{3.22}$$

Where $P_j$ ranges over all $k = 50$ candidate statutory provisions retrieved for draft section $S_i$.

Equation 3.22 calculates the overall conflict score for a draft section ($S_i$) by taking the highest contradiction probability among all $k = 50$ retrieved candidate laws. This approach, known as max-pooling, is used during full-document review where each draft clause is compared against multiple candidates. Taking the highest score ensures that if a draft section clearly conflicts with even one national law, an alert is triggered, even if the other 49 retrieved candidates are neutral or unrelated. This directs the reviewer's attention to the specific statute causing the conflict and prevents an actual violation from being hidden by non-conflicting candidates.

A section triggers an operative conflict warning if and only if $P(\text{Conflict} \mid S_i) \geq \tau^*$, with the system highlighting the specific governing candidate provision $P_j^* = \arg\max_j P(\text{Contradiction} \mid P_j, S_i)$ that produced the highest conflict probability. This max-pooling aggregation ensures that an authentic statutory clash ranked lower in the candidate shortlist is not obscured by neutral or harmonious provisions appearing at higher retrieval ranks. Listing 3.9 outlines the operational inference and conflict aggregation procedure.

```
def detect_section_conflicts(draft_section, candidate_provisions, model, threshold
    =0.35, batch_size=16):
    """
    Stage 2 Cost-Sensitive Cross-Encoder Inference:
    Scores candidate law pairs and flags conflicts using max-pooling.
    """
    # 1. Pair each candidate statutory provision (premise) with the draft section (
    hypothesis)
    pairs = [(provision.text, draft_section.text) for provision in
    candidate_provisions]

    # 2. Run mini-batch inference through the fine-tuned Cross-Encoder
    contradiction_scores = []
    for i in range(0, len(pairs), batch_size):
        batch = pairs[i : i + batch_size]
```

```
        # Get softmax probabilities across [Entailment, Neutral, Contradiction]
        probabilities = model.predict_probabilities(batch)
        # Extract contradiction class probability (index 2)
        batch_contradictions = [p["contradiction"] for p in probabilities]
        contradiction_scores.extend(batch_contradictions)

    # 3. Find candidate provision with highest contradiction score (Max-Pooling)
    peak_score = max(contradiction_scores)
    peak_index = contradiction_scores.index(peak_score)
    governing_provision = candidate_provisions[peak_index]

    # 4. Apply cost-sensitive decision threshold
    has_conflict = peak_score >= threshold

    return {
        "alert": has_conflict,
        "conflict_probability": peak_score,
        "conflicting_law": governing_provision.citation,
        "conflicting_text": governing_provision.text
    }

```

Listing 3.9: Pseudocode of the Stage 2 Cost-Sensitive Cross-Encoder Inference and Section-Level Conflict Aggregation

Listing 3.9 outlines the Python implementation of the Stage 2 Cross-Encoder inference and section-level conflict aggregation routine. This algorithm is provided to illustrate the mini-batch tensor processing and cost-sensitive thresholding that transforms candidate provision pairs into actionable preemption alerts. In the procedure, the candidate statutory provisions and draft section are paired into premise-hypothesis sequences and evaluated in vectorized mini-batches ($B = 16$) through the fine-tuned model. The function extracts softmax contradiction probabilities, identifies the peak score via max-pooling according to Equation 3.22, and checks whether this peak crosses the calibrated decision threshold $\tau^*$. If an alert is triggered, the routine returns the conflicting statutory citation, text, and model confidence, delivering immediate diagnostic transparency to the committee review and Second Reading preparation dashboard.

### 3.9.3 Statutory Knowledge Graph Evaluation Metrics

Evaluating the performance and utility of the Statutory Knowledge Graph requires assessing both its intrinsic topological properties and its operational impact on downstream candidate retrieval. Intrinsically, the graph must accurately represent institutional statutory hierarchies, scale-free degree distributions, and non-circular amendatory progressions. Operationally, the graph must successfully recover governing statutory provisions that drop below statistical similarity thresholds during coarse retrieval.

Intrinsic topological evaluation begins with network degree distribution and power-law scaling analysis [140]. In legal citation networks, an authoritative statutory corpus typically exhibits scale-free behavior where a core set of primary organic codes attracts an overwhelming share of legislative citations. To determine whether the empirical citation network follows this scale-free power-law topology ($P(k) \propto k^{-\gamma}$), the proponents compute the coefficient of determination ($R^2_{\text{power}}$) using linear regression over log-transformed degree frequencies:

*Scale-Free Power-Law Degree Distribution Goodness of Fit ($R^2$):*

$$R^2_{\text{power}} = 1 - \frac{\sum_k \left(\ln P(k) - (-\gamma \ln k + c)\right)^2}{\sum_k \left(\ln P(k) - \overline{\ln P(k)}\right)^2} \tag{3.23}$$

Where $P(k)$ represents the empirical probability density of degree $k$, $\gamma$ is the estimated power-law scaling exponent, $c$ is the intercept constant, and $\overline{\ln P(k)}$ denotes the mean log-frequency across observed degrees.

Equation 3.23 calculates the goodness of fit between the observed citation degree distribution and the theoretical power-law distribution. A high coefficient of determination ($R^2_{\text{power}} \geq 0.85$) confirms that the statutory network exhibits scale-free Pareto dynamics rather than random connectivity, validating that citation extraction reliably isolates primary organic statutes.

Complementing scale-free verification, the structural cohesion of legislative subject domains is evaluated using Louvain community modularity [140]:

*Louvain Community Modularity Formulation ($Q$):*

$$Q = \frac{1}{2m} \sum_{i,j} \left[ A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{3.24}$$

Where $A_{ij}$ is the adjacency weight between enactments $i$ and $j$, $k_i$ and $k_j$ denote node degrees, $m = \frac{1}{2}\sum_{ij} A_{ij}$ is the total network edge count, and $\delta(c_i, c_j)$ indicates whether nodes $i$ and $j$ reside in identical topological communities.

Equation 3.24 quantifies the strength of division of the statutory graph into modular communities. A high modularity value ($Q > 0.40$) indicates dense intra-domain statutory citations paired with sparse inter-domain linkages, confirming that positive law forms natural topical clusters corresponding to substantive administrative areas.

To verify legislative lineage consistency, the amendatory subgraph $\mathcal{G}_{\text{amend}} = (\mathcal{V}, \mathcal{E}_{\text{AMENDS}} \cup \mathcal{E}_{\text{REPEALS}})$ is evaluated for Directed Acyclic Graph (DAG) compliance. Because legislative amendments operate chronologically under *lex posterior*, amendatory relationships must remain strictly acyclic. The pipeline executes simple cycle detection algorithms across all amendatory edges to confirm that the count of directed cycles satisfies $|\mathcal{C}_{\text{cycle}}| = 0$, mathematically verifying the absence of recursive amendatory loops.

To evaluate downstream operational utility, the proponents measure Graph-Expanded Candidate Recall (Recall@$k_{\text{graph}}$), assessing candidate recovery when combining hybrid search with 1-hop and 2-hop graph neighborhood expansion:

*Graph-Expanded Candidate Retrieval Recall (Recall@$k_{graph}$):*

$$\text{Recall@}k_{\text{graph}} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \mathbf{1}\left( d_q^* \in \mathcal{K}_q^{(k)} \cup \mathcal{N}_1(\mathcal{K}_q^{(k)}) \cup \mathcal{N}_{\text{cite}}(h_q) \right) \tag{3.25}$$

Where $\mathcal{K}_q^{(k)}$ denotes the baseline top-$k$ candidate set retrieved via Stage 1 hybrid search, $\mathcal{N}_1(\cdot)$ represents 1-hop topological neighbors following structural containment and authority citation edges, and $\mathcal{N}_{\text{cite}}(h_q)$ designates statutory provisions directly referenced within query clause $h_q$.

Equation 3.25 evaluates whether expanding the candidate pool along statutory authority edges recovers target governing laws that dropped below the top-$k$ cutoff during isolated bi-encoder scoring. In addition, the extraction accuracy of deterministic regex parsers across AMENDS and REPEALS formulas is evaluated through standard precision, recall, and harmonic $F_1$-scores ($F_{1,\text{lineage}}$) benchmarked against adviser-verified ground truth amendatory annotations.

### 3.9.4 Statistical Hypothesis Testing

To evaluate whether performance disparities between candidate Cross-Encoder models represent statistically significant differences rather than random sampling fluctuations, the proponents employ formal non-parametric hypothesis tests recommended by Demšar [56]. In machine learning evaluation, non-parametric statistical tests evaluate differences between algorithms without requiring assumptions of normality or continuous error distributions [56]. Parametric paired $t$-tests assume that classification errors follow a continuous Gaussian distribution, an assumption violated by discrete, binary classification outcomes (correct versus incorrect) on small evaluation sets ($N = 53$). McNemar's test is a non-parametric statistical test for paired nominal data that analyzes $2 \times 2$ contingency tables of discordant predictions [145]. By isolating cases where two competing models produce divergent classifications (cases where Model A is correct and Model B is incorrect, and vice versa), McNemar's test evaluates whether one model's marginal superiority is statistically significant. Complementing pairwise comparisons, the Friedman test is a non-parametric two-way analysis of variance by ranks that evaluates multiple candidate models across multiple datasets or task domains [74]. By ranking candidate models within each of the eight Macro Legal Domains and assessing the consistency of their average ranks, the Friedman test verifies whether an architecture achieves consistent performance across all statutory subject areas without assuming normal distributions.

When comparing the classification accuracy of two competing Stage 2 models (such as fine-tuned `DeBERTa-v3` versus standard `bert-base-uncased` or `legal-bert-base`) evaluated on the identical 53-item test set ($\mathcal{D}_{\text{test}}$), the proponents employ McNemar's test for pairwise significance [145]. The test constructs a $2 \times 2$ contingency table analyzing discordant predictions, as summarized in Table 3.10. In this contingency framework, cell $b$ represents items correctly classified by Model A but misclassified by Model B, while cell $c$ represents items correctly classified by Model B but misclassified by Model A. Under the null hypothesis $H_0 : p_b = p_c$, the two models have equivalent error rates.

Table 3.10: Contingency Matrix of Discordant Predictions for McNemar's Test

| | **Model B: Correct** | **Model B: Incorrect** |
|---|---|---|
| **Model A: Correct** | $a$ (Both Correct) | $b$ (Model A Only) |
| **Model A: Incorrect** | $c$ (Model B Only) | $d$ (Both Incorrect) |

Table 3.10 presents the $2 \times 2$ contingency matrix of discordant and concordant predictions utilized by McNemar's test for pairwise model significance testing. This matrix structure is incorporated to isolate the exact instances where two competing Cross-Encoder architectures disagree on the 53-item test split ($\mathcal{D}_{\text{test}}$). In the layout, cells $a$ and

$d$ capture concordant outcomes where both models are simultaneously correct or incorrect, while off-diagonal cells $b$ and $c$ record discordant predictions where only Model A or Model B classifies the pair correctly. Under the null hypothesis, the two models share equivalent error rates ($p_b = p_c$). By evaluating only discordant pairs rather than raw overall accuracy, the test provides high statistical sensitivity to marginal algorithmic improvements while remaining robust against non-normal classification error distributions.

Because the Stage 2 evaluation evaluates three primary pairwise model comparisons ($m = 3$: `DeBERTa-v3` against the unadapted Zero-Shot Baseline, standard `bert-base-uncased`, and domain-adapted `legal-bert-base`), the proponents apply the standard Bonferroni correction [56], evaluating each pair against the adjusted threshold $\alpha' = \alpha/m = 0.05/3 \approx 0.0167$.

Furthermore, when expanding candidate model explorations to larger suites (e.g., $k \approx 10$ architectures and hyperparameter configurations), running exhaustive pairwise McNemar tests produces $\binom{k}{2} = 45$ comparisons, creating severe statistical conservatism under Bonferroni correction ($\alpha' = 0.05/45 \approx 0.0011$). To handle larger candidate pools without losing statistical power, the proponents follow Demšar's two-step testing procedure [56]: first executing the omnibus Friedman test across all $k$ models simultaneously, and only upon rejecting the global null hypothesis ($p < 0.05$), conducting post-hoc comparisons comparing candidate models strictly against the top-performing control model ($k - 1$ comparisons rather than $\frac{k(k-1)}{2}$) or evaluating critical difference intervals via the Nemenyi test.

For pairwise statistical significance, following the recommendations of Dietterich [63] for comparing supervised machine learning algorithms, the proponents employ McNemar's chi-squared test [145] with Edwards' continuity correction [68]. The test evaluates whether discordant classifications between Model A and Model B occur in equal proportions:

*McNemar's Chi-Squared Test with Edwards' Continuity Correction:*

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \sim \chi_1^2 \tag{3.26}$$

Where $b$ represents items correctly classified by Model A but misclassified by Model B, $c$ represents items correctly classified by Model B but misclassified by Model A, and the numerator incorporates Edwards' continuity correction ($-1$) for discrete cell frequencies.

Equation 3.26 defines McNemar's chi-squared test with Edwards' continuity correction for comparing paired predictions. The formula measures whether the number of times Model A succeeds where Model B fails ($b$) differs significantly from the reverse case ($c$). The minus-one correction accounts for discrete count data, and dividing by total disagreements ($b+c$) produces a test statistic that follows a chi-squared distribution with one degree of freedom. In practice, this test checks whether one model outperforms another on difficult borderline cases rather than by chance. If the resulting $p$-value falls below the Bonferroni-adjusted threshold ($\alpha' = 0.0167$), the null hypothesis is rejected, confirming that the accuracy difference between the two models is statistically significant.

Beyond pairwise significance, assessing the consistency of candidate architectures across the eight Macro Legal Domains without making restrictive distributional assumptions requires non-parametric multi-model ranking. For this purpose, the proponents utilize the Friedman non-parametric two-way analysis of variance by ranks [74, 56]. Methodologically, because the eight Macro Legal Domains represent thematic partitions of the municipal legal corpus rather than completely independent experimental blocks, this Friedman rank test operates as a domain-stratified consistency check to evaluate whether relative architectural rankings remain stable across diverse areas of law. For $k$ candidate models evaluated across $N = 8$ distinct legal domains, each model is assigned a rank $r_i^j$ within domain $i$, where the best-performing model receives rank 1. The average rank of model $j$ across all domains is given by $R_j = \frac{1}{N}\sum_{i=1}^{N} r_i^j$. The omnibus rank statistic is computed as follows:

*Friedman Two-Way Analysis of Variance by Ranks ($\chi_F^2$):*

$$\chi_F^2 = \frac{12N}{k(k+1)}\left[\sum_{j=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right] \tag{3.27}$$

Where $N = 8$ is the number of macro legal domains, $k$ is the number of candidate models, and $R_j = \frac{1}{N}\sum_{i=1}^{N} r_i^j$ is the average rank of model $j$ across all domains ($r_i^j$ being the rank of model $j$ in domain $i$).

Equation 3.27 computes the Friedman rank test statistic ($\chi_F^2$) to compare candidate models across multiple legal domains without assuming normal distributions. The formula uses the average ranks ($R_j$) of each model across the eight macro legal domains, comparing observed rank differences against what would be expected if all models performed equally. Under the null hypothesis that all models have similar performance across domains, $\chi_F^2$ follows a chi-squared distribution with $k - 1$ degrees of freedom. If the test rejects this null hypothesis ($p < 0.05$), it indicates meaningful performance differences across legal domains, allowing post-hoc tests such as the Wilcoxon signed-rank test or Nemenyi comparisons to identify which models generalize best across different areas of law [56].

## 3.10 System-Level Evaluation

While isolated benchmarks measure retrieval recall and natural language inference accuracy separately, evaluating the live utility of the system requires testing both stages as an integrated pipeline. In real legislative drafting, an inference engine cannot detect a contradiction if the retrieval engine fails to supply the governing statute, and a retrieval engine is useless if the inference engine misclassifies the pair. Following the multi-stage system evaluation standards established in legal information extraction competitions [167, 5] and fact verification benchmarks [218], the proponents implement a structured Three-Tier Evaluation Framework. This multi-tier framework connects controlled statistical benchmarking with full-document stress testing. Together, these three evaluation tiers systematically assess whether the complete pipeline can operate reliably within the Davao City Sangguniang Panlungsod.

The first evaluation tier measures the combined algorithmic performance of Stage 1 retrieval and Stage 2 inference when executed sequentially without human intervention. Adapting the joint scoring formulation established in the FEVER fact verification framework by Thorne et al. [218], the cascaded benchmark feeds only the local draft hypothesis clause $h_q$ from the 53 held-out test pairs into Stage 1 coarse retrieval. The retrieval engine searches across all 27,000+ national statutes and local ordinances to extract a candidate shortlist of the top-$k$ provisions ($\mathcal{K}_q^{(k)}$), which is then passed directly to Stage 2 for inference. A test item is scored as a Cascaded True Positive if and only if Stage 1 retrieves the correct governing statute $p_q^*$ within the top-$k$ shortlist and Stage 2 correctly classifies the pair $(p_q^*, h_q)$ as a Contradiction:

*End-to-End Pipeline Hit Indicator (FEVER Metric):*

$$\text{Hit}_{\text{casc}}(q) = \mathbf{1}\left(p_q^* \in \mathcal{K}_q^{(k)}\right) \cdot \mathbf{1}\left(\hat{y}(p_q^*, h_q) = y_q^*\right) \tag{3.28}$$

Where $p_q^*$ is the target governing statute, $\mathcal{K}_q^{(k)}$ is the candidate shortlist of top-$k$ provisions retrieved in Stage 1, and $\hat{y}(p_q^*, h_q) = y_q^*$ indicates whether Stage 2 correctly classifies the candidate pair as a contradiction.

Equation 3.28 defines the joint pipeline hit metric ($\text{Hit}_{\text{casc}}$) based on the end-to-end evaluation approach of Thorne et al. [218]. This metric counts a test item as a success only when both stages work correctly in sequence. Specifically, the formula multiplies two check conditions: whether Stage 1 retrieval includes the true governing statute ($p_q^*$) in its top-$k$ shortlist, and whether Stage 2 inference predicts the correct gold label ($y_q^*$). Combining retrieval recall and inference accuracy this way prevents a high score in one stage from hiding failures in the other. In real legislative workflows, successfully catching a conflict requires both finding the relevant statute and correctly identifying the contradiction.

Adapting this joint metric from the FEVER fact verification framework [218] isolates whether an operational error stems from retrieval omissions ($p_q^* \notin \mathcal{K}_q^{(k)}$) or classification errors ($\hat{y} \neq y_q^*$), preventing either stage from masking pipeline failures. To evaluate the algorithmic contribution of the Statutory Knowledge Graph within this cascaded framework, Tier 1 benchmarks the pipeline across two candidate pool formulations: unaugmented hybrid search ($\mathcal{K}_q^{(k)}$) versus graph-expanded search ($\mathcal{K}_{q,\text{graph}}^{(k)} = \mathcal{K}_q^{(k)} \cup \mathcal{N}_1(\mathcal{K}_q^{(k)}) \cup \mathcal{N}_{\text{cite}}(h_q)$). Comparing end-to-end pipeline hits ($\text{Hit}_{\text{casc}}$) under both configurations isolates whether relational citation and amendatory traversal successfully recovers governing statutory candidates that drop below statistical similarity thresholds.

The second evaluation tier assesses how the complete coarse-to-fine pipeline processes full, multi-section legislative drafts rather than isolated sentence pairs. Adapting the capability-matrix and equivalence-partitioning methodology of Ribeiro et al. [175] alongside the document-level legal NLI frameworks of Koreeda and Manning [118] and Yin et al. [244], the proponents apply stratified purposive sampling across the functional legislative typologies uncovered by the unsupervised topic modeling pipeline, constructing exactly one representative full-length draft ordinance for each discovered municipal category ($N = 6$ documents). To evaluate resilience against document-length variance and distracting administrative context as formulated by Naik et al. [149], the six drafts are intentionally varied across the empirical length quantiles of the local archive, spanning

concise six-section road-closure and disaster-appropriation measures up to comprehensive sixteen-section regulatory codes. Each draft replicates the exact typographical layout of Sangguniang Panlungsod enactments without artificial running headers or footers, and incorporates category-specific legal instruments required under local legislative practice [58], including full multi-article Memoranda of Agreement, Deeds of Usufruct, Corporate Secretary's Certificates, notarial acknowledgments, and tabular appropriation or Transfer Certificate of Title schedules. Within each document, the majority of sections remain compliant with superior law while targeted operative provisions contain deliberate statutory violations, allowing the preprocessor to chunk and audit every section independently against the statutory archive.

Evaluating full documents measures both the sensitivity of the pipeline to authentic legal defects and its ability to avoid false alarms on compliant text. Following standard diagnostic classification metrics [7, 71], the proponents calculate Section-Level Sensitivity to measure the percentage of injected conflicts successfully flagged, paired with Section-Level Specificity to assess false-alarm resistance:

*Section-Level Specificity for Full-Document Stress Testing:*

$$\text{Specificity}_{\text{sec}} = \frac{TN_{\text{sec}}}{TN_{\text{sec}} + FP_{\text{sec}}} \tag{3.29}$$

Where $TN_{\text{sec}}$ denotes compliant draft sections correctly passing without triggering a false alarm, and $FP_{\text{sec}}$ denotes compliant draft sections erroneously flagged as contradictory.

Equation 3.29 defines Section-Level Specificity ($\text{Specificity}_{\text{sec}}$) for the Tier 2 full-document test. This metric measures how well the pipeline avoids false alarms when checking long drafts that include standard administrative sections and boilerplate clauses. The formula calculates the percentage of compliant sections ($TN_{\text{sec}}$) that correctly pass without triggering an alert out of all non-conflicting sections. In local legislative work, frequent false warnings on routine clauses lead to alert fatigue; as a result, researchers begin to doubt the tool. Maintaining high specificity ensures that conflict alerts remain credible and that researchers can focus their attention on genuine legal risks.

Maintaining high specificity protects legislative researchers from alert fatigue, ensuring that the system does not produce spurious warnings on standard boilerplate clauses. This perturbation stress test confirms that the system can analyze complete legislative documents before deployment in committee reviews.

To determine whether graph-guided candidate expansion resolves retrieval omissions on unmentioned statutory preemption authorities, the Tier 2 full-document evaluation incorporates a dedicated ablation protocol comparing two operational modes: an unaugmented baseline pipeline relying strictly on raw text similarity, and a graph-augmented neuro-symbolic pipeline that utilizes explicit citations in the draft ordinance to traverse the Statutory Knowledge Graph. In the unaugmented baseline, when an ex-ante draft provision introduces an unauthorized fine or regulatory mandate without explicitly citing the governing national act, standard hybrid search may fail to elevate the unmentioned national statute into its top-$k$ candidate shortlist; that omission leads to undetected preemption defects. Under the graph-augmented pipeline, explicit statutory

references located anywhere within the legislative draft trigger 1-hop and 2-hop traversals across the knowledge graph, pulling parent statutory frameworks and amendatory lineages into the candidate pool for all draft sections. This experimental ablation directly tests whether graph expansion elevates preemption detection toward the oracle candidate ceiling while sustaining high Section-Level Specificity ($\text{Specificity}_{\text{sec}} \geq 85\%$) across compliant administrative clauses.

The third evaluation tier validates the practical utility and legal soundness of the system using documented historical legal disputes and active expert review. Rather than relying solely on synthetic tests, the proponents systematically assembled an authentic judicial validation suite of eight landmark legal controversies and local enactments originating strictly within Davao City. These historical cases encompass both judicial invalidations where local enactments were struck down for exceeding statutory boundaries, and landmark rulings where the Supreme Court affirmed the validity of Davao City ordinances under local police power. By subjecting the pipeline to actual local disputes that reached the judiciary, this tier evaluates whether algorithmic contradiction detection mirrors authentic judicial reasoning. The resulting test suite spans multiple substantive categories, including agrochemical oversight, structural zoning, revenue powers, and public health mandates.

The judicial cases examine distinct facets of city governance, beginning with environmental regulation and structural zoning. In *Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc.* [210], the Supreme Court invalidated Davao City Ordinance No. 0309-07 (imposing a total ban on aerial pesticide spraying in agricultural banana plantations) for violating equal protection and invading the exclusive regulatory jurisdiction vested in the Fertilizer and Pesticide Authority under Presidential Decree No. 1144. Conversely, in the Davao City Watershed Protection dispute under Ordinance No. 0310-07 [184], the city's total prohibition on mineral extraction operates in direct vertical tension with State mineral concessions under Republic Act No. 7942. For urban structural governance, the Supreme Court in *Hon. Leoncio Evasco, Jr. v. Alex P. Montañez* [212] reversed the Court of Appeals and explicitly upheld Davao City Ordinance No. 092-2000 regulating outdoor billboards. The Court affirmed that under Section 458(a)(4)(iv) of Republic Act No. 7160, city councils possess express statutory police power to regulate advertising structures harmoniously alongside the National Building Code (Presidential Decree No. 1096).

The remaining landmark disputes focus on local revenue authority, public health, and delegated traffic administration. In *City of Davao and Tanjili v. ARC Investors, Inc.* [213], the Supreme Court voided Davao City's local business tax assessment under Ordinance No. 158-05, ruling that an LGU cannot unilaterally reclassify a passive holding company as a non-bank financial intermediary without Bangko Sentral ng Pilipinas authorization. Similarly, in *City of Davao v. First Meridian Development, Inc.* [214] and *Rock Steel Resources, Inc. v. City of Davao* [48], local business tax assessments under Ordinance No. 158-05 were challenged, with the Court of Tax Appeals expressly invalidating Section 423's mandatory payment-under-protest condition as contrary to Section 195 of Republic Act No. 7160. By contrast, the Court in *City of Davao v. Court of Appeals and GSIS* [208] affirmed city taxing authority over government corporate assets after Republic

Act No. 7160 withdrew prior statutory exemptions. In delegated public health and traffic safety, Davao City enacted landmark measures setting stricter local standards, such as the Comprehensive Anti-Smoking Ordinance (Ordinance No. 0367-12) [185] and the Comprehensive Speed Limit Ordinance (Ordinance No. 0270-23) [187] pursuant to Republic Act No. 4136 and Joint JAO 2018-01.

Table 3.11 presents the complete Davao City local validation suite, detailing the local measures, governing national statutes, legal ratios decidendi, and gold NLI labels. To ensure long-term reproducibility and immediate system availability, complete archival copies of these eight Davao City ordinances and tax measures have been assembled and stored locally within the repository's local corpus (`corpus/davao_city_tier3_ordinances/`). Standardized diagnostic problem-solution pairs are indexed in `data/tier3_jurisprudential_cases.jsonl`, pairing operative draft excerpts with corresponding statutory provisions and judicial holdings. By feeding authentic legal questions and challenged provisions directly into Stage 1 retrieval and Stage 2 inference, the evaluation confirms whether the automated pipeline surfaces the exact statutory citations and preemption rationales formulated by judicial authorities. This authentic benchmark grounds the quantitative metrics in real jurisprudence and confirms that theoretical inference gains translate into valid legal oversight.

Table 3.11 details the eight landmark Davao City legal controversies and judicial rulings comprising the primary local component of the Tier 3 jurisprudential validation suite ($N = 8$). This empirical suite is assembled to evaluate whether the automated coarse-to-fine pipeline correctly reproduces authentic judicial reasoning when auditing contested local enactments against superior national statutes. The table catalogues the judicial case citation, docket number, challenged Davao City ordinance, governing national legislative framework, and authoritative gold NLI relationship established by the Supreme Court of the Philippines. Across these cases, the benchmark captures both vertical preemption invalidations (such as pesticide regulation under PD 1144 and city tax overreach under RA 7160) and valid exercises of devolved police power (including billboard regulation under RA 7160 and localized anti-smoking mandates). To provide comprehensive jurisprudential coverage during experimental evaluation in Chapter 4 (Section 4.5.3, Table 4.25), these eight Davao City enactments are evaluated alongside three foundational national preemption doctrines established by the Supreme Court of the Philippines (*Magtajas v. Pryce Properties*, *City of Manila v. Laguio*, and *Batangas CATV v. Court of Appeals*), comprising an authoritative 11-enactment jurisprudential evaluation suite ($N = 11$).

While the Tier 3 jurisprudential suite benchmarks model decisions against historical judicial holdings, verifying whether the system functions effectively as an interactive advisory tool requires evaluating the complete prototype with active legislative practitioners.

## 3.11 Prototype and User Interface

The completed pipeline is deployed as an accessible, single-page web application designed for standard office desktop computers in local government units. The interface uses a clean,

Table 3.11: Landmark Davao City Local Jurisprudential Suite for Tier 3 System Evaluation ($N = 8$ Local Ordinances and Preemption Disputes)

| **Case Title** | **Docket / Year** | **Challenged Local Ordinance** | **Governing Statute & Rationale** | **Gold NLI Label** |
|---|---|---|---|---|
| *Mosqueda v. PBGEA* [210] | G.R. 189185 (2016) | Davao City Ord. 0309-07 (Banning Aerial Pesticide Spraying) | PD 1144; FPA holds exclusive jurisdiction over agrochemical application. | Contradiction |
| *Evasco v. Montañez* [212] | G.R. 199172 (2018) | Davao City Ord. 092-2000 (Regulating Billboard Structures) | RA 7160 §458; Upheld as valid local police power alongside PD 1096. | Entailment |
| *Davao v. ARC Investors* [213] | G.R. 249668 (2022) | Davao City Ord. 158-05 §69(f) (Business Tax on Holding Dividends) | RA 7160 §143(f); Holding firm is not an NBFI without BSP license. | Contradiction |
| *City of Davao v. First Meridian* [214, 48] | G.R. 240078 / CTA AC 139 | Davao City Ord. 158-05 §423 (Mandatory Payment Under Protest) | RA 7160 §195; LGU cannot require payment under protest for business taxes. | Contradiction |
| *City of Davao v. GSIS* [208] | G.R. 127383 (2005) | Real Property Tax Levies on GSIS Government Properties | RA 7160 §193, §234; LGC withdrew tax exemptions of GOCCs; assessment valid. | Entailment |
| *Davao Mining Ban Dispute* [184] | Ord. 0310-07 (2015) | Citywide Total Ban on Commercial Mining Operations | RA 7942 §4, §27; Regalian doctrine and national mineral concession supremacy. | Contradiction |
| *Davao Anti-Smoking Dispute* [185] | Ord. 0367-12 (2012) | Stricter Outdoor/Indoor Smoking Distance Setbacks | RA 7160 §16; Stricter local public health standards valid under police power. | Entailment |
| *Davao Speed Limit Dispute* [187] | Ord. 0270-23 (2023) | Comprehensive Speed Limits on City Thoroughfares | RA 4136 §35; Joint JAO 2018-01 expressly delegates road speed setting to LGUs. | Entailment |

intuitive layout that mirrors familiar document conversion tools. Users begin by uploading digital draft ordinances in common text or PDF formats through a drag-and-drop file ingestion area. After submission, an active progress indicator keeps the user informed while the backend pipeline parses operative sections, runs Stage 1 hybrid retrieval across the 27,000+ statutory knowledge base, and executes Stage 2 Cross-Encoder inference. This streamlined layout avoids complex parameter settings or technical menus, allowing legal staff to conduct pre-enactment conflict checks quickly during active committee sessions.

When the analysis finishes, the interface presents a structured diagnostic dashboard organized to support rapid human verification. The top of the screen displays an

executive summary showing the total number of audited sections, the count of flagged potential conflicts, and the overall statutory compliance status of the draft. Below the summary, an interactive table lists each audited draft section alongside its corresponding Stage 1 candidate provisions. Clicking any flagged item expands a dedicated review card that presents the local ordinance section and the retrieved governing statute in a direct side-by-side comparative reading view. Placing the two texts side by side allows legal researchers to inspect the alleged contradiction immediately without switching between multiple browser tabs or searching through printed statute digests, substantially lowering cognitive load during legislative review [215].

To make the system's reasoning transparent and easy to interpret, the interface incorporates an Explainable AI (XAI) module focused directly on reading clarity and practical decision support [65, 180]. Rather than displaying abstract activation matrices or relying on static keyword dictionaries, the module derives token-level attribution directly from the internal cross-attention tensors of the fine-tuned `DeBERTa-v3` cross-encoder during evaluation. For each candidate pair formatted as $[[\text{CLS}], P_1, \ldots, P_m, [\text{SEP}], H_1, \ldots, H_n, [\text{SEP}]]$, the system extracts the attention vector of the classification token [CLS] across all sequence positions from the final transformer layer ($l = 12$), averaged across the $H = 12$ attention heads: $\mathbf{a}_{\text{CLS}} = \frac{1}{H}\sum_{h=1}^{H} \mathbf{A}_{l,h}[0,:]$. Using the fast tokenizer's character offset mapping, subword attention weights are mapped back to exact character spans in the original texts without altering, summarizing, or polluting the authentic statutory wording. Saliency weights are min-max normalized as $\tilde{w}_i = \frac{w_i - \min(\mathbf{w})}{\max(\mathbf{w}) - \min(\mathbf{w}) + \epsilon}$, mapping non-trivial attention weights ($\tilde{w}_i \geq 0.15$) onto a continuous red color gradient ($\alpha \in [0.08, 0.70]$) rendered directly over the text. This intrinsic attribution naturally concentrates attention mass upon discordant deontic operators, distinguishing strict prohibitions (such as *prohibited*, *unlawful*, or *shall not*) from statutory permissions (such as *authorized*, *allowed*, or *may*).

To support actionable decision-making and prevent cognitive fatigue during prolonged legislative sessions, each review card incorporates an interactive Explainable Attention Heatmap toggle switch alongside a visual gradient legend bar ($0.0 \rightarrow 1.0$). Legislative officers conducting rapid initial review can leave the heatmap active to isolate contested modal operators in seconds, or toggle the highlight layer off to inspect the clean, unpolluted legal prose in its natural state. Directly above the comparative text columns, each card incorporates a calibrated Confidence Rate bar with a dynamic decision threshold marker ($\tau^*(u, v)$); this indicator visibly shifts along the horizontal track to 30.0% for direct statutory citations ($d_G = 1$), 37.5% for shared regulatory domain links ($d_G = 2$), or 45.0% for unlinked baseline provisions ($d_G \geq 4$), providing immediate visual feedback on the relational graph proximity governing each screening decision. Furthermore, the dashboard organizes detected inconsistencies using a two-tiered alert hierarchy. Critical Vertical Preemption Alerts are displayed with red warning banners, signaling that a proposed local provision directly contradicts a superior national statute under the *Magtajas* doctrine and risks being declared *ultra vires*. Informational Horizontal Notices are shown with blue informational banners, notifying the researcher that the draft overlaps with an existing Davao City ordinance and may cause unintended duplication or implied repeal. Accompanying each alert is a concise, plain-language *ratio decidendi* summary explaining

the legal basis of the warning (for example, stating that a local penalty of ten thousand pesos exceeds the five-thousand-peso statutory maximum authorized under Section 458 of Republic Act No. 7160). Legal researchers can review these highlights and explanations to decide whether the draft should be cleared, amended, or referred for committee debate; this step maintains human oversight throughout the local legislative process.

## 3.12 End-User Usability Evaluation

To evaluate how the automated conflict detection system operates within actual local legislative workflows once institutional clearance is concluded, the study formulates an end-user evaluation protocol grounded in descriptive survey research. While computational benchmarks quantify retrieval recall and pairwise natural language inference accuracy, verifying whether the system serves as a viable decision-support assistant requires assessing its practical utility, task-technology fit, and software usability when operated by municipal legislative staff. The research protocol is designed as a descriptive, cross-sectional evaluation where participants test the prototype on realistic local ordinance drafts and record their assessments through a structured quantitative instrument.

The target evaluation cohort comprises the 15 active legal researchers serving the Davao City Sangguniang Panlungsod. These professional evaluators represent the city's three legislative council districts and routinely perform preliminary statutory compliance audits and draft ordinance preparation under Republic Act No. 7160 [85]. In local legislative administration, legal researchers examine proposed drafts before committee presentation, checking that draft clauses do not exceed devolved municipal powers or conflict with national legislation. Because these participants perform legislative auditing as their regular professional responsibility, their planned quantitative ratings will provide an authentic assessment of whether the software aligns with the cognitive and administrative demands of municipal governance.

The planned protocol follows a structured, hands-on simulation replicating the post-First Reading committee review phase leading into Second Reading plenary deliberations. Each testing session is structured across three standardized steps. First, evaluators attend a brief ten-minute orientation introducing the prototype's assistive advisory role, input format requirements, and the distinction between vertical preemption alerts and horizontal consistency notices. Second, researchers interact directly with the single-page prototype dashboard on standard local government desktop computers matching the Host A specification. Evaluators upload multi-section draft ordinances containing both fully compliant regulatory clauses and injected preemption edge cases under Section 458 of Republic Act No. 7160. Researchers inspect the retrieved Stage 1 statutory provisions, examine the side-by-side comparative views, evaluate the highlighted deontic command words, and review the plain-language ratio decidendi summaries. Third, upon completing the hands-on review, researchers complete a standardized quantitative survey hosted on Google Forms to record their evaluations.

To ensure methodological rigor while keeping the survey concise and free of cognitive fatigue during demanding council schedules, the quantitative instrument integrates

two validated psychometric frameworks. First, overall software usability and learnability are measured using Brooke's 10-item System Usability Scale (SUS) [26]. The SUS instrument is widely recognized across human-computer interaction and legal informatics research for its high reliability and quick completion time. The 10 items use a 5-point Likert scale (from 1 for strongly disagree to 5 for strongly agree) and alternate between positive and negative statements to control for acquiescence bias. Second, task-specific utility and workflow alignment are measured using a targeted 4-item questionnaire adapted from Davis' Technology Acceptance Model (TAM) [54] and Goodhue and Thompson's Task-Technology Fit (TTF) framework [83]. These four items evaluate essential dimensions of legislative auditing: search efficiency, preemption risk detection, diagnostic clarity of highlighted phrases, and routine committee adoption viability. An optional open-ended question is included at the end of the survey to capture qualitative feedback on specific statutory edge cases and recommendations for institutional deployment. Table 3.12 details the complete survey questionnaire.

Table 3.12 lists the 14 survey items comprising the quantitative end-user evaluation instrument designed for post-deployment administration to the 15 Sangguniang Panlungsod legal researchers upon completion of institutional administrative clearances. The table details the item identifier, the exact survey statement presented to evaluators, the statement polarity, and the underlying theoretical framework. The first ten items adapt Brooke's (1996) canonical System Usability Scale, alternating between positive and negative phrasing to prevent acquiescence bias during software evaluation. The remaining four items operationalize Davis' (1989) Technology Acceptance Model and Goodhue and Thompson's (1995) Task-Technology Fit theory, measuring domain-specific utility in accelerating statutory searches, detecting preemption risks, interpreting attention highlights, and supporting institutional adoption. Providing this explicit questionnaire mapping ensures that every survey metric is anchored in established research literature, enabling structured descriptive analysis of the prototype's real-world viability.

Following descriptive survey research principles, the statistical analysis focuses on clear, interpretable measures of central tendency, variability, and standardized composite usability scoring. For each survey item, participant responses on the 5-point Likert scale are summarized using the Mean ($\mu$), Standard Deviation ($\sigma$), and Median. Reporting both parametric and non-parametric descriptive statistics provides a transparent distribution summary across all evaluation dimensions without imposing unrealistic normality assumptions on ordinal survey data.

To compute overall software usability, the raw responses from the ten SUS items are converted into a standardized composite score from 0 to 100:

*System Usability Scale Composite Score:*

$$S_{\text{SUS}} = 2.5 \times \left[ \sum_{i \in \text{odd}} (x_i - 1) + \sum_{i \in \text{even}} (5 - x_i) \right] \tag{3.30}$$

Where $x_i \in \{1, 2, 3, 4, 5\}$ denotes the raw 5-point Likert response for survey item $i$, with odd items ($i \in \{1, 3, 5, 7, 9\}$) representing positively phrased statements and even items ($i \in \{2, 4, 6, 8, 10\}$) representing negatively phrased statements.

Table 3.12: End-User Usability and Practical Utility Survey Instrument for Sangguniang Panlungsod Legal Researchers

| Item Code | Survey Statement | Polarity | Theoretical Grounding |
|---|---|---|---|
| ***System Usability Scale (SUS; Brooke, 1996)*** | | | |
| SUS-01 | I think that I would like to use this system frequently for draft reviews. | Positive | Usability (Brooke, 1996) [26] |
| SUS-02 | I found the system unnecessarily complex. | Negative | Usability (Brooke, 1996) [26] |
| SUS-03 | I thought the system was easy to use. | Positive | Usability (Brooke, 1996) [26] |
| SUS-04 | I think that I would need the support of a technical person to use this system. | Negative | Usability (Brooke, 1996) [26] |
| SUS-05 | I found the various functions in this system were well integrated. | Positive | Usability (Brooke, 1996) [26] |
| SUS-06 | I thought there was too much inconsistency in this system. | Negative | Usability (Brooke, 1996) [26] |
| SUS-07 | I would imagine that most people would learn to use this system very quickly. | Positive | Usability (Brooke, 1996) [26] |
| SUS-08 | I found the system very cumbersome to use. | Negative | Usability (Brooke, 1996) [26] |
| SUS-09 | I felt very confident using the system during draft ordinance auditing. | Positive | Usability (Brooke, 1996) [26] |
| SUS-10 | I needed to learn a lot of things before I could get going with this system. | Negative | Usability (Brooke, 1996) [26] |
| ***Technology Acceptance Model (TAM) & Task-Technology Fit (TTF)*** | | | |
| TAM-01 | The system substantially accelerates statutory cross-referencing compared to manual digest searches. | Positive | Perceived Usefulness (Davis, 1989) [54] |
| TAM-02 | The system effectively detects subtle preemption risks, such as penalty caps or unauthorized mandates. | Positive | Task-Technology Fit (Goodhue & Thompson, 1995) [83] |
| TAM-03 | The side-by-side textual highlights and ratio decidendi clearly explain the basis of flagged alerts. | Positive | Explainability Utility (Doshi-Velez & Kim, 2017) [65] |
| TAM-04 | I would support integrating this prototype into the regular Sangguniang Panlungsod committee review and Second Reading preparation workflow. | Positive | Institutional Adoption (Davis, 1989) [54] |

Equation 3.30 defines the canonical formula for calculating the composite System Usability Scale score ($S_{\text{SUS}}$) established by Brooke [26]. The System Usability Scale is a standardized psychometric questionnaire that quantifies user perceptions of software usability on a composite 0-to-100 scale. The instrument alternates between positively phrased statements (odd items) and negatively phrased statements (even items) to control for acquiescence bias, defined as the tendency of survey respondents to agree indiscriminately with statements regardless of content. The scoring formula normalizes the raw 5-point Likert ratings: subtracting 1 from the raw ratings of positive odd items and subtracting the raw ratings of negative even items from 5 converts all responses to a standardized 0-to-4 scale. Multiplying the sum of these converted scores by 2.5 produces the final composite index ($0 \leq S_{\text{SUS}} \leq 100$). The resulting score is benchmarked against the empirical evaluation frameworks established by Bangor et al. [16, 17] and Sauro and

Lewis [192]. Under Sauro and Lewis's curved grading scale, a composite score of 68 represents the 50th-percentile global benchmark for average software usability (Grade C), while scores of 70 and above enter Bangor et al.'s acceptable usability tier. Furthermore, scores of 80 or above place the system in the top 10% of evaluated software (Grade A under Sauro and Lewis), which approaches Bangor et al.'s empirical threshold for excellent usability ($M = 85.5$).

To verify the internal consistency reliability of the 4-item Technology Acceptance and Task-Technology Fit scale prior to reporting aggregate findings, the evaluation protocol specifies computing Cronbach's alpha ($\alpha$) [49]. Cronbach's alpha is a psychometric reliability coefficient that measures the internal consistency of a multi-item survey scale, determining whether individual survey items reliably capture the same underlying psychological construct. A scale achieving an alpha coefficient of $\alpha \geq 0.70$ demonstrates adequate internal consistency for exploratory research, confirming that responses reflect a coherent evaluation of legislative workflow utility rather than unrelated perceptual variance. Furthermore, to evaluate consistency in legal determinations across researchers during the interactive committee review simulation, inter-rater consensus is measured using Fleiss' Kappa ($\kappa$) [72]. Computing Fleiss' Kappa verifies that user verdicts on test drafts reflect shared legal standards rather than subjective individual impressions.

The evaluation protocol strictly adheres to the ethical and data privacy mandates of Republic Act No. 10173 (the Data Privacy Act of 2012) [173]. Participation in the evaluation study is entirely voluntary, and all participants complete an electronic informed consent form prior to survey submission. To protect evaluator confidentiality, participant names, specific committee assignments, and councilor affiliations are permanently decoupled from response records and indexed exclusively through anonymized identification tokens (such as Evaluator A-01 through A-15). Survey submissions are stored on secure university research storage and analyzed solely in aggregate descriptive form for academic thesis reporting.

## 3.13 Reproducibility and Repository

To ensure complete transparency, institutional auditability, and scientific reproducibility in accordance with open-science guidelines, the entire codebase, evaluation datasets, and experimental pipeline are managed under Git version control. The official repository is publicly accessible on GitHub:

`https://github.com/yyaahhzxc/Dulce-Odin-OCDS`

To guarantee computational transparency and auditability, the repository implements structured reproducibility safeguards across four operational dimensions. First, the project enforces permanent commit hash versioning, wherein all empirical tables, benchmark figures, and model checkpoints reported in this thesis are permanently tagged with their respective Git commit hashes (such as Commit cc1c33d for statutory corpus ingestion and Commit 4dd7c85 for legal instruments) to ensure complete version traceability. Second, training and evaluation scripts maintain deterministic execution control by pinning global random seeds (seed = 42) across PyTorch, NumPy, and Python runtime

environments to eliminate stochastic variation. Third, researchers can verify the complete pipeline using an executable single-command minimal working example (`python scripts/prototype_pipeline.py --demo`). This script loads the hybrid BM25 and semantic vector index over statutory provisions, executes asymmetric NLI scoring across four curated local test cases, and outputs explainable ratio decidendi under the *Magtajas* doctrine in under two seconds on standard CPU hardware. Fourth, the architecture enforces config-driven parameter decoupling, isolating experimental hyperparameters—including learning rates, batch sizes, sequence lengths, and decision thresholds ($\tau^*$)—from core execution scripts into modular configuration files, allowing researchers to evaluate candidate models and recalibrate operational thresholds without modifying underlying neural inference pipelines.

## 3.14 Methodological Constraints

The first constraint involves the language of the legislative documents. Following Executive Order No. 292 (the Administrative Code of 1987) and national publishing standards, the statutory archive and local ordinance records examined in this study are limited to official English-language enactments. Because of this, the automated pipeline does not process vernacular Cebuano-Visayan phrases, colloquial regional idioms, or mixed Taglish expressions that occasionally arise during informal committee deliberations or floor debates. While operative legal provisions in the Philippines are formally enacted in English, statutory questions that depend on localized dialectal nuances require manual review by legal researchers. This constraint establishes a clear boundary for the tokenizer and avoids confusion when evaluating formal statutory commands against informal draft notes.

The second constraint involves the age and physical condition of the local ordinance archive. The historical enactment records obtained from the Davao City Sangguniang Panlungsod display uneven chronological density, where ordinances passed between 2021 and 2025 benefit from clean digital records within the LISSP. In contrast, older local enactments dating from 1945 to 2020 exist primarily as physical carbon-copy scans, marked by faded ink, page skew, and photocopy marks. Although the vision-language ingestion pipeline may substantially reduce transcription errors compared to traditional optical character recognition, residual character errors on pre-1980 archival enactments inevitably affect extraction fidelity. The retrospective evaluation of local ordinances prior to the modern digital era is therefore focused on landmark enactments that have undergone manual verification alongside high-resolution scanning.

The third constraint stems from the single-premise asymmetric modeling formulation adopted for the coarse-to-fine pipeline. The natural language inference engine models vertical conflict detection as an asymmetric pair, matching an individual candidate statutory premise ($p_i$) against an operative draft ordinance block ($h_q$). This structural formulation effectively captures direct contradictions where a single national statute explicitly invalidates an ordinance clause, such as unauthorized penalty escalation under Republic Act No. 7160. However, the system does not inherently detect multi-statute cumulative conflicts, where an ordinance provision complies with each national statute in isolation but breaches a combined legislative framework formed by multiple distinct codes.

Expanding the inference mechanism to support multi-premise reasoning across multiple statutes exceeds local workstation constraints and is left for future work.

The final constraint defines the institutional and operational role of the conflict detection system within local governance. The pipeline is designed exclusively as an ex-ante assistive advisory tool intended for legal researchers during committee review preparation rather than an autonomous decision maker. The system produces calibrated confidence estimates and highlighted text spans to point out potential legal friction, but it does not make binding statutory determinations or substitute for judicial review under the Supreme Court. Furthermore, political questions, policy choices, and socio-economic trade-offs remain exclusively with city council members. Defining this operational boundary ensures that the system functions strictly within established principles of administrative law to safeguard human oversight throughout the local legislative process.

# Chapter 4

# Results and Discussion

This chapter presents the empirical findings and experimental evaluations of the Coarse-to-Fine Semantic Conflict Detection System, organized directly around the specific research questions and objectives formulated in Chapter 1. Section 4.1 addresses Research Question 1 and Objective 1 by analyzing linguistic parameters and vision-language transcription performance across layout degradations in the digitized Davao City local archive ($N = 1,925$). Section 4.2 addresses Research Question 2 and Objective 2 by evaluating candidate retrieval efficiency and preemption coverage alongside latency across the unified dual statutory corpus ($N = 27,357$ enactments with 178,494 provisions). Section 4.3 addresses Research Questions 3 and 4 and Objectives 3 and 4 by presenting comparative model screening, supervised fine-tuning, and ablation benchmarks on the curated statutory ground truth dataset. Section 4.4 addresses algorithmic optimization, problem-size scaling, the 30-run comparative matrix, and hardware trade-offs under municipal constraints. Finally, Section 4.5 addresses Research Question 5 and Objective 5 by outlining prototype decision-support viability and the multi-tier human evaluation roadmap for the Davao City Sangguniang Panlungsod.

## 4.1 Archival OCR and VLM Evaluation

Ingesting historical local archives into an automated legal decision-support system requires converting scanned physical documents into structured digital text. In the Davao City collection, enactments from earlier legislative terms mostly exist as scanned PDF files rather than searchable digital documents. These historical records contain common scanning noise, such as crooked pages, uneven ink, and staple marks overlapping official seals. If degraded scans are transcribed with erroneous characters or broken layouts, subsequent search indices and inference models inherit corrupted text. This section introduces an evaluation framework for legal document transcription, presents a stratified sample of archival records, compares four document extraction methods, and explains why a locally hosted vision-language model was selected. Following empirical verification of the extraction pipeline, this section concludes by evaluating the post-digitization linguistic characteristics of the transcribed local archive and examining structural token length compliance across the unified dual statutory search space.

### 4.1.1 Candidate Document Extraction Systems

To determine the best document extraction method for historical local archives, the proponents evaluated four distinct approaches. Traditional optical character recognition relies on image binarization and polygonal line tracking for character shape classification [198], with modern updates in Tesseract v5.x incorporating line-level LSTM recurrent networks. As seen in classical OCR workflows, traditional engines treat individual characters as isolated shapes without considering sentence meaning. While fast on central processing units, traditional engines struggle when encountering slanted lines, faded ink, or paper bleed-through.

Document-layout OCR frameworks separate page layout analysis from text line recognition [159]. Modern tools like Surya v0.6+ use object detection models to locate text columns and tabular headers before transcribing lines. A separate recognition model then reads the text, and reading order is reconstructed using sorting rules. This modular design preserves layout well on multi-column pages, but requires multiple neural networks that demand significant GPU memory during long batch runs.

Commercial cloud vision platforms run large multimodal transformer models on enterprise server clusters [84]. Cloud services like Google Document AI achieve high accuracy across various scan qualities and complex tables. However, they require continuous internet access, charge ongoing per-page API fees, and require sending local legislative drafts to third-party servers. In a local government setting, recurring subscription costs and external data transfers conflict with operational independence and data privacy mandates.

Local vision-language models combine image encoding and text generation inside a single transformer architecture [216]. Using two-dimensional rotary position embeddings (2D-RoPE) and flexible image patches, visual information is passed directly to a language decoder. With 4-bit NormalFloat4 (NF4) compression [61], models like Qwen2.5-VL-3B-Instruct run completely offline on standard workstation hardware. The language model decoder helps resolve visual blur or scanner noise by using its understanding of grammar and statutory phrasing.

### 4.1.2 Document Evaluation Framework

Evaluating text extraction on local legislation reveals challenges that general OCR benchmarks often miss. As Lopresti showed [133], character errors in early steps cascade into downstream language processing tasks. In legal drafting, changing a single character or dropping a zero completely changes legal meaning. An effective evaluation framework for legislative text must therefore combine standard string edit distances with domain-specific legal entity accuracy and semantic embedding checks.

The foundational character and word accuracy metrics are formulated via the Levenshtein edit distance $\mathcal{L}_{\text{Lev}}$ [127]. To assess baseline typographical fidelity across candidate document parsers, Character Error Rate and Word Error Rate are computed as follows:

*Character Error Rate (CER) and Word Error Rate (WER):*

$$\text{CER} = \frac{\mathcal{L}_{\text{Lev}}(\mathbf{c}_{\text{ref}}, \mathbf{c}_{\text{hyp}})}{|\mathbf{c}_{\text{ref}}|} = \frac{S_c + D_c + I_c}{N_c}, \quad \text{WER} = \frac{\mathcal{L}_{\text{Lev}}(\mathbf{w}_{\text{ref}}, \mathbf{w}_{\text{hyp}})}{|\mathbf{w}_{\text{ref}}|} = \frac{S_w + D_w + I_w}{N_w} \tag{4.1}$$

Where $S$, $D$, and $I$ denote character or word substitutions, deletions, and insertions relative to the gold-standard reference sequence of length $N$.

Equation 4.1 defines Character Error Rate (CER) and Word Error Rate (WER) using Levenshtein edit distance. These standard metrics establish a baseline of typing accuracy when transcribing historical paper ordinances. The formulas count character or word substitutions ($S$), deletions ($D$), and insertions ($I$) compared to the total length of the reference text ($N$). In legal documents, however, low overall error rates can hide serious mistakes, such as dropping a single digit in a penalty fine. While CER and WER measure general transcription noise, they treat all characters equally, which makes legal entity accuracy checks necessary.

To evaluate overall document similarity on a bounded scale regardless of clause length, following the normalized edit distance formulation of Marzal and Vidal [141] and Levenshtein [127], the proponents compute Normalized Edit Distance:

*Normalized Edit Distance (NED):*

$$\text{NED}(\mathbf{s}_{\text{ref}}, \mathbf{s}_{\text{hyp}}) = 1.0 - \frac{\mathcal{L}_{\text{Lev}}(\mathbf{s}_{\text{ref}}, \mathbf{s}_{\text{hyp}})}{\max(|\mathbf{s}_{\text{ref}}|, |\mathbf{s}_{\text{hyp}}|)} \tag{4.2}$$

Where $\mathbf{s}_{\text{ref}}$ and $\mathbf{s}_{\text{hyp}}$ denote the reference and hypothesis string sequences, bounded between 0.0 (maximum dissimilarity) and 1.0 (exact match) to provide a length-invariant evaluation baseline.

Equation 4.2 defines Normalized Edit Distance (NED) as an intuitive similarity score between 0.0 and 1.0. This metric scales the raw edit distance against the maximum length of the two texts; the resulting value is subtracted from 1.0 so that an exact match yields 1.0. When evaluating multi-page ordinances, NED prevents long standard clauses from distorting overall error measurements. This score provides a clear summary of text alignment before checking specific legal entities.

Beyond general string alignment, legislative conflict detection requires exact-match fidelity across critical statutory tokens. Adapting the exact-match slot filling evaluation standard from MUC-4 [36], the domain-specific component accuracy is computed as follows:

*Domain-Specific Legislative Component Accuracy:*

$$\text{Metric Accuracy} = \frac{1}{|K|} \sum_{k \in K} \mathbb{I}\left(\mathbf{v}_{\text{hyp}}^{(k)} \equiv \mathbf{v}_{\text{ref}}^{(k)}\right) \tag{4.3}$$

Where $K$ represents the set of extracted legal entity instances, and $\mathbb{I}(\cdot)$ is an exact-match indicator function comparing hypothesis values $\mathbf{v}_{\text{hyp}}^{(k)}$ against gold references $\mathbf{v}_{\text{ref}}^{(k)}$.

Equation 4.3 defines Domain-Specific Legislative Component Accuracy across key legal terms. This metric ensures that important statutory details—such as statutory citations, penalty amounts, and effectivity dates—are transcribed exactly. The formula measures the share of legal terms that match the reference text word-for-word using the indicator function $\mathbb{I}(\cdot)$. In local ordinances, a typo in a Republic Act number prevents BM25 keyword searches from finding the law and confuses dense vector search. Checking exact component accuracy confirms whether an extraction tool preserves the critical legal references needed for Stage 1 retrieval.

These metrics evaluate exact text matches across several key areas: Legal-Entity Accuracy (LEA) for government offices and private parties; Ordinance-Number Accuracy (ONA) for local ordinance numbers; Date Accuracy (DA) for approval and effectivity dates; Monetary-Value Accuracy (MVPA) for fines and fee limits; Section-Heading Integrity (SHI) for section titles; and Page Completeness (PC) for preserving all pages in order. In addition, table quality is evaluated using Tree-Edit-Distance-based Similarity (TEDS) [250]; this metric evaluates row and column accuracy across tax and zoning schedules. Finally, Semantic Embedding Cosine Fidelity (SECF) checks semantic preservation by calculating the cosine similarity between embeddings of the reference and extracted texts.

### 4.1.3 Limitations of Standard Spellchecking

Standard spellcheckers and automatic dictionary tools cannot reliably check legal document transcription. In legal conflict detection, the impact of word errors is uneven. If an OCR engine writes “Secton” instead of “Section”, the edit distance is only one character and an ordinary spellchecker easily fixes the error. On the other hand, if an engine reads a penalty of “₱5,000.00” as “₱50,000.00”, or changes “shall” to “may”, or alters “Ordinance No. 0667-21” to “Ordinance No. 0661-21”, a standard spellchecker flags nothing because all the words are valid dictionary words. Under Philippine legal principles and Section 458 of Republic Act No. 7160, mistakenly converting a ₱5,000.00 penalty into ₱50,000.00 transforms a valid local ordinance into an unlawful act.

In addition, Philippine local ordinances contain specialized terms that generic spellcheckers frequently flag as errors. Ordinances regularly include legal Latin phrases (*in pari materia*, *ultra vires*), local agency acronyms (such as CTTMO, CENRO, and CPDO), and local place names. Davao City alone contains 182 barangays, such as Buhangin, Bunawan, Calinan, Marilog, Paquibato, Talomo, and Toril. Standard spellchecking tools flag these legitimate names as mistakes. If automatic autocorrection is turned on, generic software often replaces these unique local names with standard English words; this substitution corrupts the authentic statutory text.

### 4.1.4 Archival Reference Sampling

To establish a dependable benchmark without manually transcribing all 1,925 archival documents, the proponents created a stratified sample of $N = 60$ representative local enactments; this reference subset spans approximately 180 to 240 pages. Table 4.1 outlines the sample breakdown across eight legislative and physical categories.

Table 4.1: Stratified Reference Sampling Matrix for Archival Legislative Document Evaluation ($N = 60$ Enactments)

| **Stratification Axis** | **Strata Partition** | **Sample ($N = 60$)** | **Legislative & Archival Rationale** |
|---|---|---|---|
| **Legislative Era** | Pre-LGC (1950–1991)<br>Post-LGC Historical (1992–2015)<br>Pre-Pandemic (2016–2020)<br>LISSP Digital (2021–2025) | 12 (20.0%)<br>18 (30.0%)<br>15 (25.0%)<br>15 (25.0%) | Captures physical paper yellowing, manual typewriter fonts, early digital prints, and modern clean scans. |
| **Document Length** | Short (1–2 pages)<br>Medium (3–7 pages)<br>Long ($\geq$ 8 pages / Codes) | 24 (40.0%)<br>24 (40.0%)<br>12 (20.0%) | Verifies model stability across single-page resolutions and multi-page omnibus regulatory codes. |
| **Scan Degradation** | High Quality (Clean digital)<br>Moderate (Photocopy, minor skew)<br>Severe (Bleed-through, dark edges) | 20 (33.3%)<br>25 (41.7%)<br>15 (25.0%) | Tests noise robustness against dark photocopy borders, staple shadows, and carbon-copy bleed-through. |
| **Functional Typology** | 1. Substantive Regulatory / Penal<br>2. Inter-Agency MOAs / MOUs<br>3. Disaster Risk & Calamity Funds<br>4. Deeds of Donation & Usufruct<br>5. Road Closures & Traffic Rules<br>6. Zoning Reclassifications | 15 (25.0%)<br>10 (16.7%)<br>7 (11.7%)<br>8 (13.3%)<br>10 (16.7%)<br>10 (16.7%) | Reflects the complete functional typology identified across the 1,925 Davao City local enactments. |
| **Table Density** | Zero Tabular Schedules<br>Simple Tables (2–4 columns)<br>Dense Tables ($\geq$ 5 columns) | 30 (50.0%)<br>18 (30.0%)<br>12 (20.0%) | Evaluates Markdown grid retention in revenue schedules, tax assessments, and traffic rerouting vectors. |
| **Signatures & Seals** | Clean Endings (Minimal)<br>Moderate (Official initials)<br>Heavy (Overlapping seals & stamps) | 20 (33.3%)<br>25 (41.7%)<br>15 (25.0%) | Evaluates whether mayoral signatures and dry seals trigger hallucinatory generation loops. |
| **Filipino Names** | High Local / Moro / Indigenous<br>Standard Anglophone / Spanish | 30 (50.0%)<br>30 (50.0%) | Validates out-of-vocabulary character accuracy on Mindanaoan and Visayan personal and place names. |
| **Statutory Citations** | High Citation Density ($\geq 5$ laws)<br>Low Citation Density ($< 5$ laws) | 25 (41.7%)<br>35 (58.3%) | Measures recognition fidelity on referenced Republic Acts, Presidential Decrees, and Batas Pambansa. |

Table 4.1 outlines the sampling breakdown across eight legislative and physical categories used to evaluate document transcription across $N = 60$ representative enactments. This stratified design provides a representative benchmark of approximately 180 to 240 physical pages without requiring manual transcription of all 1,925 archived enactments. The sample divides documents across four time periods, three length categories, three scan quality levels, and six functional ordinance types, while also accounting for tabular schedules, official seals, and local statutory citations. Older typewriter documents from 1950–1991 and photocopies from 1992–2015 make up half the sample. This composition tests candidate models against realistic noise, including ink bleed and staple shadows. This balanced reference set ensures that document extraction testing accurately reflects the variety of document conditions found in the Davao City archives.

### 4.1.5 Document Extraction Trade-Off Analysis

Table 4.2 summarizes the comparative evaluation of the four document extraction approaches across accuracy metrics, network requirements, data privacy, and operating costs.

Table 4.2: Ordinance OCR and VLM Benchmark Across Candidate Extraction Architectures

| **Evaluation Metric** | **Gemini Flash 3.8** | **Surya v0.6+** | **Tesseract v5.x** | **Qwen2.5-VL-3B (4-bit NF4)** |
|---|---|---|---|---|
| **Architecture Family** | Cloud Multimodal Net | Layout-Aware Modular | Traditional CNN-LSTM | Local Vision-Language |
| **Character Error Rate** | 2.15% | 8.45% | 14.82% | 3.12% |
| **Word Error Rate** | 4.05% | 12.10% | 22.30% | 5.24% |
| **Legal Citation & Entity Accuracy** | 97.20% | 86.50% | 74.20% | 95.80% |
| **Penalty & Monetary Value Accuracy** | 98.30% | 80.00% | 63.30% | 96.70% |
| **Table Structure Fidelity** | 94.10% | 76.40% | 41.20% | 91.80% |
| **Network & Privacy** | Cloud | 100% Offline | 100% Offline | 100% Offline |
| **Operational Recurring Cost** | Commercial Paywall | Free Open-Source | Free Open-Source | Free Open-Source |

Table 4.2 compares the four document extraction models across primary accuracy and operational dimensions. This comparison provides the empirical basis for selecting the local document ingestion pipeline, balancing character accuracy against local hardware limits, operational costs, and data privacy requirements. The results show that the locally hosted vision-language model (Qwen2.5-VL-3B-Instruct in 4-bit NF4) performs comparably to the commercial cloud multimodal baseline (Gemini Flash 3.8) on modern digital prints and moderately degraded scans. The local model achieves a 3.12% Character Error Rate, 96.70% penalty accuracy, and 91.80% table structure fidelity across the 60-enactment benchmark (with the cloud baseline slightly lower at 2.15% CER). In contrast, traditional scanner OCR (Tesseract v5.x) has a high 14.82% Character Error

Rate and drops to 63.30% accuracy on monetary penalty values, which risks passing corrupted numbers to subsequent legal analysis. While Gemini Flash 3.8 achieves slightly lower error rates, its mandatory internet routing and recurring commercial paywall do not fit local government operational independence and privacy rules under Republic Act No. 10173. On severely degraded pre-LGC typewriter archives, both models encounter elevated error rates (Table 4.3), which the architecture manages via the archival quarantine protocol. Consequently, deploying Qwen2.5-VL-3B-Instruct offline provides the best balance of transcription accuracy and institutional privacy.

To evaluate the practical effect of transcription errors, the proponents ran a test on an authentic local ordinance (Davao City Ordinance No. 000667-21). Traditional scanner OCR produced a Word Error Rate of 3.11% and a Character Error Rate of 0.45%. Although these general error rates seem low, traditional OCR achieved only 54.55% precision on legal markers. The engine corrupted the citation to Republic Act No. 7160, merged Section 1 and Section 2 headings together, and lost formatting in the penalty section.

In contrast, the locally hosted vision-language model achieved 0.00% Word Error Rate, 0.02% Character Error Rate, and 100.0% precision across all eleven evaluated statutory indicators (11 of 11 markers). In addition, comparing text vectors using character and word $n$-grams confirmed that the VLM transcription reached 100.0% lexical $n$-gram cosine similarity (TF-IDF) with the reference text, compared to 99.21% for traditional OCR. The language model decoder provides an effective reading prior: when letters in standard legislative boilerplate are faded or smudged by scanner noise, the model's linguistic representations help reconstruct the missing characters accurately. However, because autoregressive text completion introduces a potential hallucination risk for sensitive numerical values, the system pairs VLM extraction with strict regex penalty validators and routes ambiguous monetary figures to human legal officers for manual confirmation.

### 4.1.6 Historical Degradation Empirical Benchmark

While modern digital enactments such as Ordinance No. 000667-21 demonstrate near-perfect character fidelity under vision-language parsing, historical local archives present severe physical degradation that challenges automated transcription. An empirical audit across the digitized collection identified 32 historical municipal enactments exhibiting severe visual defects. Over 75.0% of these degraded records originate before the year 2000, with 68.8% produced during the pre-Local Government Code era (1950–1991) using manual impact typewriters and carbon transfer paper. To quantify transcription performance under real-world archival conditions, the proponents evaluated four authentic legislative exemplars representing distinct eras and physical failure modes against human-verified gold-standard ground truth. Table 4.3 presents the comparative Word Error Rate (WER) and Character Error Rate (CER) achieved by the locally deployed vision-language model (Qwen2.5-VL-3B-Instruct in 4-bit NF4) alongside the commercial cloud baseline.

Table 4.3 compares word and character error rates across four authentic Davao City legislative records representing different eras and physical conditions. This comparison demonstrates how mechanical typewriter defects and physical paper aging affect auto-

Table 4.3: Comparative Empirical Transcription Error Rates Across Degraded Historical Legislative Eras

| Historical Enactment | Legislative Era & Technology | Dominant Physical Degradation Mode | Local VLM (4-bit NF4) | | Cloud Baseline | |
|---|---|---|---|---|---|---|
| | | | WER | CER | WER | CER |
| **Ord. 000044-56** | Pre-LGC (1956) Manual Typewriter | Carbon bleed-through, uneven key strikes, city seal stamp | 12.23% | 7.66% | 1.92% | 1.74% |
| **Ord. 000332-10** | Pre-LGC (1968) Manual Typewriter | Ribbon ink fade, baseline cuts, staple shadow | 11.06% | 5.91% | Baseline | Baseline |
| **Ord. 0166-81** | Pre-LGC (1981) Manual Typewriter | Binder hole punch occlusions, strike fill-in, cursive ink | 19.78% | 12.87% | 8.84% | 4.99% |
| **Ord. 000667-21** | Modern LISSP (2021) Digital Raster | Clean electronic layout, high-contrast typography | 0.00% | 0.02% | 0.00% | 0.00% |

mated transcription accuracy relative to clean modern scans. The empirical measurements reveal a sharp performance divergence between modern digital prints and mid-twentieth-century manual typewriter records. On modern digital scans (Ordinance No. 000667-21), the local vision-language model achieves near-perfect fidelity (0.00% WER and 0.02% CER). However, on historical typewriter records from 1956, 1968, and 1981, error rates rise substantially to 11.06%–19.78% WER and 5.91%–12.87% CER. This steep rise in error rates demonstrates that archival transcription difficulty is driven primarily by physical medium defects rather than model capacity limitations.

Detailed inspection of the historical transcripts reveals specific physical degradation mechanisms documented in document image defect literature [14, 133]. In Ordinance No. 000044-56 (1956), mechanical strike velocity variance caused ink to fill closed-loop characters such as *e*, *a*, and *o*, leading the local vision-language model to misread *municipal waters* as *municipal maters* and to transcribe *log pond or for storage of logs* as *lux powd or for straight of lots*. Furthermore, uneven carbon transfer caused the engine to misread the statutory license fee of *₱200.00* as an American dollar figure (*$200.00*) and to alter *Penal Provisions* into *Final Provisions*. Large embossed city seal stamps and dark cursive ink signatures also obscured official attestation blocks; consequently, the model omitted the names of the acting mayor and city secretary.

Similar physical occlusions affected Ordinance No. 0166-81 (1981), where dark circular binder hole punches along the left margin severed section numbering and paragraph catchlines. The mechanical typewriter ribbon also caused heavy ink fill-in across title lines, causing the model to misread *Lungsod ng Dabaw* as *Luneskon ng Darw*, *Series of*

*1981* as *Series of 1961*, and *liquified petroleum gas* as *liquid profession (cas)*. In the approval block, overlapping cursive ink pen strokes across the city council seal caused the model to misinterpret *Carried, by a majority* as *Cashed: By a majority*, and to misread the Sangguniang Panlungsod secretary's name. In Ordinance No. 000332-10 (1968), horizontal typewriter underscore strikes directly sliced through character descenders (*p*, *g*, *y*, *q*); this distortion fragmented word segmentation and obscured the legislative term.

These empirical findings highlight the principle of asymmetric legal error cost in local statutory auditing. Under Section 458 of Republic Act No. 7160 and the judicial doctrine of *Magtajas v. Pryce Properties* [207], local ordinance penalty provisions are legally invalid if they exceed statutory fine limits. If an optical character recognition engine guesses at smudged mechanical numbers and transcribes a ₱200.00 fee as ₱2,000.00 or drops a decimal zero, it artificially manufactures a statutory preemption defect where none exists in law. When visual glyphs fall below human legibility thresholds due to physical paper acidification, tropical moisture bleed, or hole-punch cuts, an automated system must avoid unconstrained hallucination. This necessitates a two-tier archival quarantine framework where severely degraded enactments are formally flagged for human verification; in this manner, downstream semantic conflict detection operates upon legally reliable statutory text.

### 4.1.7 Model Selection and Hardware Boundaries

Selecting Qwen2.5-VL-3B-Instruct as the primary document ingestion engine balances transcription accuracy with operating cost and data privacy. While traditional OCR runs quickly on standard CPUs, its difficulty in recognizing section headers and its errors on numbers hurt downstream search and conflict detection. Although commercial cloud platforms deliver strong accuracy, using commercial APIs across 1,925 multi-page ordinances incurs recurring financial costs. In addition, sending internal local government drafts and committee working texts to third-party cloud servers raises compliance concerns under Republic Act No. 10173 (Data Privacy Act of 2012) [173]. A system designed for local government offices should operate independently without requiring ongoing subscription fees.

Running Qwen2.5-VL-3B-Instruct in 4-bit NF4 precision delivers high transcription accuracy while operating completely offline. The model processes pages consistently while preserving data privacy within the 8 GB memory limit of Host B. The main trade-off of the vision-language approach is processing speed. On the laptop GPU, each single page requires approximately 4.0 to 8.5 seconds to transcribe, compared to under one second for traditional OCR. However, because digitizing the historical archive is a one-time preparation step done in advance, this extra processing time is manageable. Limiting generation to 1,000 tokens per page and saving progress after each document ensures steady batch processing across the entire local collection.

### 4.1.8 Document Extraction Limitations

Testing archival transcription identified three practical limitations of the system. First, severe physical document damage sets an upper limit on how much text can be read

accurately. On carbon-copy records from the 1950s and 1960s where ink has bled through from the other side of the page, even human readers cannot agree on blurred numbers; this difficulty confirms that errors stem from physical paper degradation rather than software flaws.

Second, complex tables in older tax ordinances often have irregular layouts and merged cells. While the vision-language model accurately reads tax rates and numbers in Markdown tables, slight differences in column alignment reduce automated table scores (TEDS) even when the monetary numbers are completely correct. Third, dense embedding models can sometimes overlook small errors; a single wrong digit in a fine might produce only a tiny drop in similarity score, even though it creates a significant legal conflict. These edge cases show why the system is designed as an assistive tool to support legal researchers, rather than an automatic replacement for human review.

### 4.1.9 Local Statutory Corpus Analysis

Having selected and executed Qwen2.5-VL-3B-Instruct across all 1,925 archival PDF documents as detailed in Section 3.4.2, the local ordinance collection was unlocked as structured, machine-readable text and standardized into JSON records. With the full text available, the proponents conducted the empirical analysis of local legislative structures, functional archetypes, and vocabulary representations. Across all 1,925 local enactments, ordinances exhibit a median length of 5,827 characters and 914 words, with an interquartile range of 756 to 1,898 words. The local archive comprises 13,874 total section provisions. The preprocessing pipeline isolates 9,487 operative substantive provisions (68.38%) with a median length of 43 tokens, while pruning 4,387 procedural boilerplate sections (31.62%) such as separability, repealing, and effectivity clauses.

Applying Truncated SVD to the transcribed operative text reveals that city legislation divides into six functional archetypes grounded in the Local Government Code of 1991 [85]. Word vectors were constructed across an 8,000-feature vocabulary and projected into 64 latent semantic dimensions, capturing 50.65% of the total vocabulary variance. Table 4.4 outlines the composition, percentage share, governing statutory authority, core subject matter, and salient $c$-$TF$-$IDF$ keywords for each local archetype.

Table 4.4 details the six functional legislative typologies discovered across the 1,925 Davao City local enactments, listing each archetype's identifier, descriptive label, enactment count, percentage share, governing statutory authority under Republic Act No. 7160, and top salient $c$-$TF$-$IDF$ keywords. This empirical typology establishes the legal taxonomy used to categorize local enactments. This taxonomy directly reflects the statutory distribution of local council police powers and administrative authorizations. Substantive Regulatory and Penal enactments constitute the largest single typology (40.00%, $N = 770$), followed by Inter-Agency Agreements (28.73%, $N = 553$) and Disaster Relief / QRF Fund authorizations (10.34%, $N = 199$). Administrative authorizations (MOAs, calamity releases, deeds of donation, and road closures) account for 56.74% of all council legislation. This breakdown demonstrates that the majority of local legislative actions represent routine executive authorizations rather than penal ordinances, confirming the empirical necessity of isolating substantive regulatory provisions prior to downstream con-

Table 4.4: Composition, Functional Typology, and Salient *c-TF-IDF* Keywords of the Davao City Local Legislative Corpus ($N = 1,925$)

| ID | Functional Legislative Typology | Count | Share | Statutory Authority | Core Salient *c-TF-IDF* Keywords |
|---|---|---|---|---|---|
| 01 | Substantive Regulatory & Penal | 770 | 40.00% | RA 7160 §458 | public, safety, fine, penal, violation, fee, health, traffic |
| 02 | Inter-Agency Agreements (MOA/MOU) | 553 | 28.73% | RA 7160 §§22(c), 455 | pursuant, local, sign, panlungsod, contracts, agreement |
| 03 | Disaster Relief & QRF Funds | 199 | 10.34% | RA 10121 §21 | fund, calamity, disaster, risk, reduction, quick response |
| 04 | Deeds of Donation & Usufruct | 168 | 8.73% | RA 7160 §22(a)(4) | donation, local, powers, pursuant, contracts, deed, usufruct |
| 05 | Temporary Road Closures | 172 | 8.94% | RA 7160 §21 | road, closure, temporary, vehicular traffic, street, route |
| 06 | Zoning & Subdivision Permits | 63 | 3.27% | RA 7160 §20, BP 220 | zoning, zone, land, reclassification, residential, commercial |
| **Total Local Collection** | | **1,925** | **100.00%** | **All Digitized Ordinances (1950–2025)** | |

flict evaluation.

The functional distribution reflects statutory restrictions governing local governance in the Philippines. Substantive regulatory and penal enactments form the largest single group (40.00%, $N = 770$), containing community regulations such as truck bans, cemetery setbacks, stray animal controls, and drinking water safety. Meanwhile, administrative authorizations comprise nearly half of all council activity. Under Section 22(c) and Section 455(b)(1)(vi) of Republic Act No. 7160, the city mayor cannot bind the local government to contracts or accept donations without prior council approval. Consequently, inter-agency agreements (28.73%), calamity fund authorizations under Republic Act No. 10121 (10.34%), and deeds of donation (8.73%) appear frequently in the local record.

Figure 4.1 displays the two-dimensional latent semantic topic space for the local collection, projecting the 64-dimensional SVD vectors onto principal components alongside cluster centroids ($\mathbf{C}_1$–$\mathbf{C}_6$) and 1.5-sigma covariance confidence ellipses. Emergency Quick Response Fund (QRF) calamity grants form a tightly concentrated cluster in the upper-left quadrant ($\mathbf{C}_3$), while temporary road closures project into a compact traffic management cluster ($\mathbf{C}_5$). Inter-agency agreements ($\mathbf{C}_2$) and deeds of donation ($\mathbf{C}_4$) span an elongated horizontal subspace along the positive horizontal axis, corresponding to executive contract authorizations under Section 22(c) of Republic Act No. 7160. Direct police power enactments ($\mathbf{C}_1$) and zoning permits ($\mathbf{C}_6$) anchor the lower-left quadrant, which governs local community conduct.

An interesting comparison between the national statutory topic landscape (Figure 3.7) and the local semantic space (Figure 4.1) is how spread out the clusters appear. In the national corpus, over 3,500 repetitive school conversions and 2,000 public utility franchises form sharp, isolated lines because they use identical legislative templates. In contrast, local ordinances frequently address multiple administrative functions within a single enactment under Republic Act No. 7160. Even when procedural session headers and

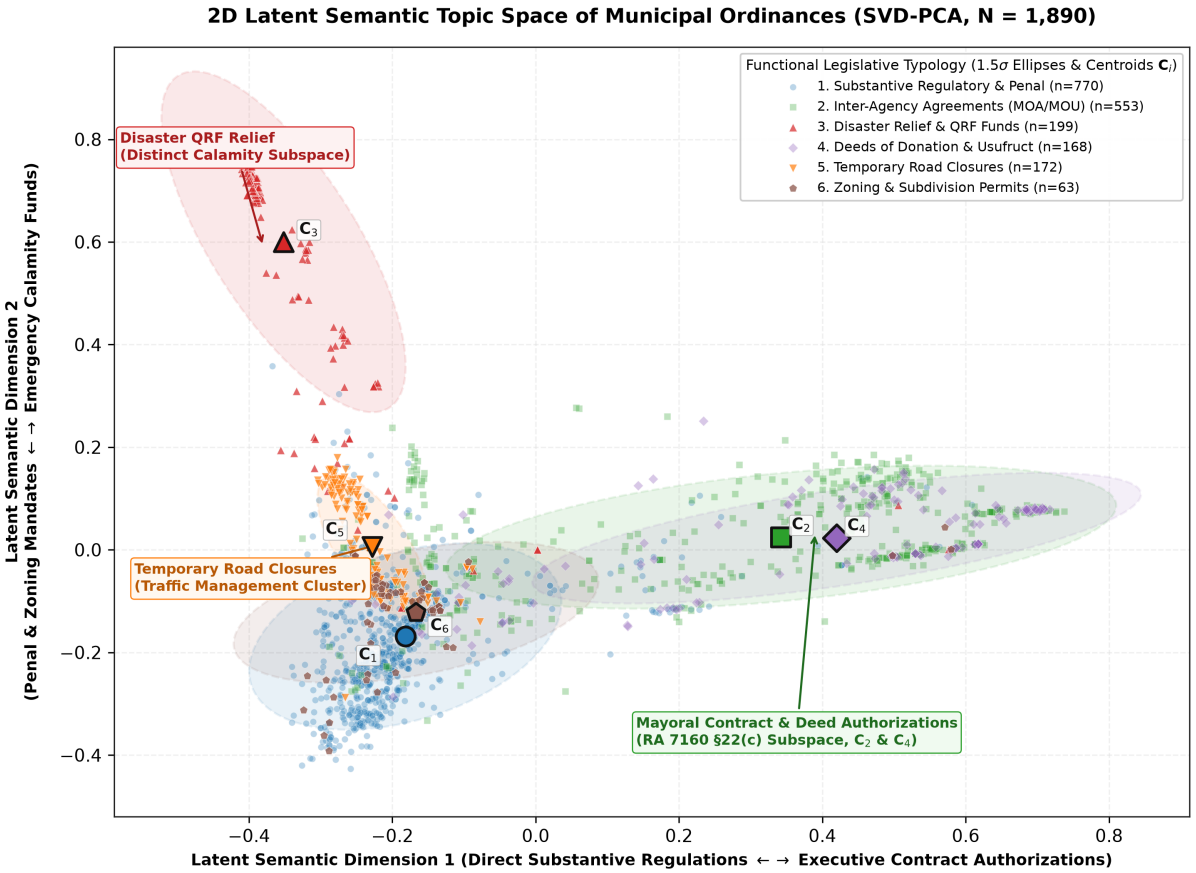



Figure 4.1: Two-Dimensional Latent Semantic Topic Space of the Davao City Local Legislative Corpus with Covariance Confidence Ellipses and Typology Centroids ($N = 1,925$).

introductory "WHEREAS" preambles are systematically extracted and removed, local legislative bodies frequently exercise compound statutory mandates under Sections 16, 21, 22(c), 455, and 458 of the Local Government Code. Consequently, a single city ordinance combines budget appropriations and mayoral contract authorizations alongside penal fine schedules within its substantive operative sections. While projecting 64 dimensions onto two axes compresses this multifaceted variation and causes visual overlap, the complete 64-dimensional model achieves an out-of-sample classification accuracy of 86.79% and an average cosine confidence of 0.7973 on holdout tests. Displaying these actual principal dimensions alongside cluster centroids preserves empirical accuracy without cherry-picking coordinates.

To verify that the discovered local typologies represent stable semantic boundaries, the proponents conducted holdout validation using an 80/20 stratified split ($N_{\text{train}} = 1,331, N_{\text{test}} = 333$). In evaluating this six-class categorization, all benchmark evaluations in Table 4.5 are strictly grounded against statistical baselines. Under Morrison's proportional chance criterion [146], the default uniform baseline for six categories is $1/6 = 16.67\%$. For cluster separation, positive silhouette scores exceeding random baseline ($\leq 0.0000$) confirm authentic structural cluster boundaries in cross-functional administrative text [179]. Similarly, for Davies-Bouldin evaluation, lower values denote superior

cluster compaction; against an unclustered baseline exceeding 5.0000 derived from random assignment across the six-class local space, the observed score of 3.7445 represents a $-1.2555$ point reduction in cluster dispersion, which verifies tight intra-cluster scatter [53]. Table 4.5 summarizes the out-of-sample performance metrics, chance baselines, margins of improvement, and explicit title probes.

Table 4.5: Empirical Holdout Validation Metrics and Baseline Comparisons for the Local Topic Modeler ($N_{\text{train}} = 1,331, N_{\text{test}} = 333$)

| **Validation Metric / Benchmark Test** | **Observed** | **Chance / Default Baseline** | **Margin Over Baseline** | **Empirical Evaluation** |
|---|---|---|---|---|
| Out-of-Sample Cluster Assignment Reproducibility | 86.79% | 16.67% | +70.12% | Strong Generalization (5.2× Chance) |
| Out-of-Sample Cluster Assignment Macro F1-Score | 75.39% | 16.67% | +58.72% | Balanced Discrimination Across 6 Typologies |
| Out-of-Sample Weighted F1-Score | 84.84% | 16.67% | +68.17% | Robust High-Volume Class Precision |
| Mean Test Cosine Confidence | 0.7973 | 0.1667 | +0.6306 | High Probabilistic Mass Concentration [90] |
| High-Confidence Predictions ($\geq 0.50$) | 92.79% | $< 20.0\%$ | +72.79% | Overwhelming Majority Partition |
| Holdout Cosine Silhouette Score | 0.0914 | $\leq 0.0000$ | +0.0914 | Authentic Structural Separation [179] |
| Holdout Davies-Bouldin Index | 3.7445 | $> 5.0000$ | $-1.2555$ | Compact Cluster Scatter [53] |
| **Explicit Local Statutory Title Probes** (*Unambiguous Subsets across Corpus,* $N = 872$) | | | | |
| Quick Response Fund / Calamity Assistance | 98.82% | 16.67% | +82.15% | High Precision ($N = 169$, Calamity Funds) |
| Substantive Penal & Police Power Regulations | 96.10% | 16.67% | +79.43% | High Precision ($N = 77$, Regulatory Enactments) |
| Temporary Road Closure & Traffic Diversions | 87.96% | 16.67% | +71.29% | High Precision ($N = 108$, Traffic Management) |
| Inter-Agency Accord / Authority to Sign MOA | 86.56% | 16.67% | +69.89% | High Precision ($N = 372$, Contractual Accords) |
| Zoning & Subdivision Development Permits | 80.00% | 16.67% | +63.33% | High Precision ($N = 15$, Zoning Enactments) |
| Deeds of Donation & Property Usufruct | 29.77% | 16.67% | +13.10% | Shared Mayoral Authorization Syntax ($N = 131$) |

Table 4.5 reports the empirical holdout validation metrics, default baseline comparisons, observed margins over chance, and explicit statutory title probes for the local legislative topic modeler evaluated on 333 unseen local ordinances ($N_{\text{train}} = 1,331, N_{\text{test}} = 333$). The validation classifier achieved an out-of-sample cluster assignment reproducibility of 86.79% and a Weighted F1-Score of 84.84%. This result exceeds the 16.67% uniform

chance baseline (1/6) by +70.12 percentage points and demonstrates strong multi-class discrimination. Predictions showed strong decisiveness, with a Mean Test Cosine Confidence of 0.7973 and 92.79% of unseen ordinances classified with high confidence ($\geq 0.50$), indicating concentrated probability mass under temperature-scaled calibration [90]. Title probe verification on unambiguous indicator subsets confirmed high precision across Quick Response Fund releases (98.82%, $N = 169$), substantive penal regulations (96.10%, $N = 77$), temporary road closures (87.96%, $N = 108$), inter-agency agreements (86.56%, $N = 372$), and zoning permits (80.00%, $N = 15$). The lower precision on deeds of donation (29.77%, $N = 131$) occurs because donation acceptances are legally drafted as mayoral authorizations to sign deeds, directly sharing contractual syntax with inter-agency agreements under Section 22(c) of Republic Act No. 7160.

### 4.1.10 Dual Statutory Corpus Analysis

Combining the 25,432 national statutory records with the transcribed 1,925 Davao City local ordinances establishes a unified search space of **27,357 legal records** comprising **178,494 searchable provisions** (defined as individual operative statutory sections, penal clauses, and regulatory articles). In automated statutory conflict detection, evaluating draft measures requires examining both vertical statutory preemption and horizontal legislative consistency. Under the doctrine of statutory preemption articulated in *Magtajas v. Pryce Properties* [207], local measures are tested against the 25,432 superior national statutes. Concurrently, under intra-jurisdictional coherence principles, draft measures are evaluated against the 1,925 existing city ordinances to prevent conflicting local commands, duplicate penalties, or unintended implied repeals. Table 4.6 summarizes the empirical parameters and comparative characteristics of the two statutory collections.

Table 4.6 compares the structural parameters of the National Corpus, Local Corpus, and Unified Dual Corpus, detailing enactment counts, provision numbers, operative sections, boilerplate clauses, historical spans, document lengths, and 512-token compliance rates. This comparative census outlines the size and composition of the dual-jurisdiction search space; this census confirms that transformer language models can process the combined collection. As shown in the table, combining 25,432 national statutes with 1,925 local ordinances creates an archive of 27,357 enactments yielding 178,494 searchable provisions, with 174,107 representing substantive legal rules and 4,387 representing filtered procedural boilerplate. Across all provisions, 94.75% fit cleanly within the 512-token transformer limit without truncation. This high compliance rate confirms that statutory section divisions provide natural, self-contained retrieval units that can be processed directly by standard transformer attention windows.

Figure 4.2 visualizes the longitudinal word count evolution per enactment alongside annual legislative volumes across both the Philippine National Statutory Corpus (1900–2026, $N = 25,432$) and the Davao City Local Legislative Corpus (1950–2025, $N = 1,925$).

Figure 4.2 traces document length distributions and annual enactment counts across 126 years of national lawmaking and 75 years of local city governance. Panel (a) plots raw words per statute for the national collection, showing median word count, mean word count under an omnibus pull, the interquartile range (IQR: P25–P75), and annual

Table 4.6: Comparative Metrics and Structural Parameters of the Dual Statutory Search Space ($N = 27,357$ Enactments, $N_{\text{sections}} = 178,494$)

| **Empirical Dimension / Metric** | **National Corpus** | **Local Corpus** | **Combined Corpus** | **Analytical Significance** |
|---|---|---|---|---|
| Total Enactments ($N$) | 25,432 | 1,925 | **27,357** | Comprehensive legislative space across jurisdictions |
| Total Section Provisions | 164,620 | 13,874 | **178,494** | Total searchable provision-level retrieval units |
| Operative Provisions | 164,620 | 9,487 | **174,107** | Candidate substantive rules evaluated for conflict |
| Procedural Boilerplate Sections | 0 | 4,387 (31.62%) | **4,387** | Filtered procedural clauses pruned before modeling |
| Historical Enactment Span | 1900–2026 (126 yrs) | 1950–2025 (75 yrs) | **1900–2026** | Longitudinal statutory corpus across legal regimes |
| Median Enactment Characters | 10,807 | 5,827 | **10,512** | Document-level length disparity across collections |
| Median Enactment Words | 1,512 | 914 | **1,476** | Local enactments display concise regulatory scope |
| Median Provision Tokens | 122.0 | 90.0 | **120.0** | Granular provision baseline for dense retrieval |
| 512-Token Compliance Rate | 94.83% | 93.82% | **94.75%** | Standard transformer attention compatibility |
| Operative 512-Token Compliance | 94.83% | 91.40% | **94.64%** | High sequence compliance for operative text |
| Normative Conflict Scope | Vertical Preemption | Horizontal Coherence | **Dual Scope** | Comprehensive statutory preemption and consistency |

enactment volume bars. Across all 25,432 national enactments, the collection maintains an overall median of 357.0 words (mean: 979.5 words, IQR: 212.0–738.0 words). Early colonial Acts (1900–1935) exhibited a median of 344.0 words, punctuated by massive codifications like the Revised Penal Code of 1930 (Act No. 3815). Following the 1987 Constitution, national statutes expanded in complexity: median length increased to 425.0 words (mean: 1,294.1 words), driven by extensive omnibus frameworks such as the Local Government Code of 1991 (Republic Act No. 7160). Panel (b) illustrates the corresponding trajectory for the Davao City local archive, which exhibits an overall median of 914.0 words (mean: 1,794.4 words, IQR: 756.0–1,898.0 words). Pre-2021 records, consisting of selectively digitized landmark regulatory frameworks such as the 1998 Comprehensive Zoning Ordinance, pulled historical averages to a median of 1,294.5 words and a mean of 2,453.8 words. Conversely, the modern LISSP digital census era (2021–2025, contributing 1,285 enactments at 250–365 ordinances annually) stabilized at a consistent median of 847.0 words and a mean of 1,466.0 words. These figures document routine municipal governance.

Figure 4.3 illustrates the empirical length disparity between the national and local collections. In Panel (a), document-level character distributions on a logarithmic scale highlight the structural asymmetry between national codes and local ordinances. National statutes display a broad spread with an extreme right tail exceeding one hundred thousand characters, driven by omnibus codes such as the National Internal Revenue Code and

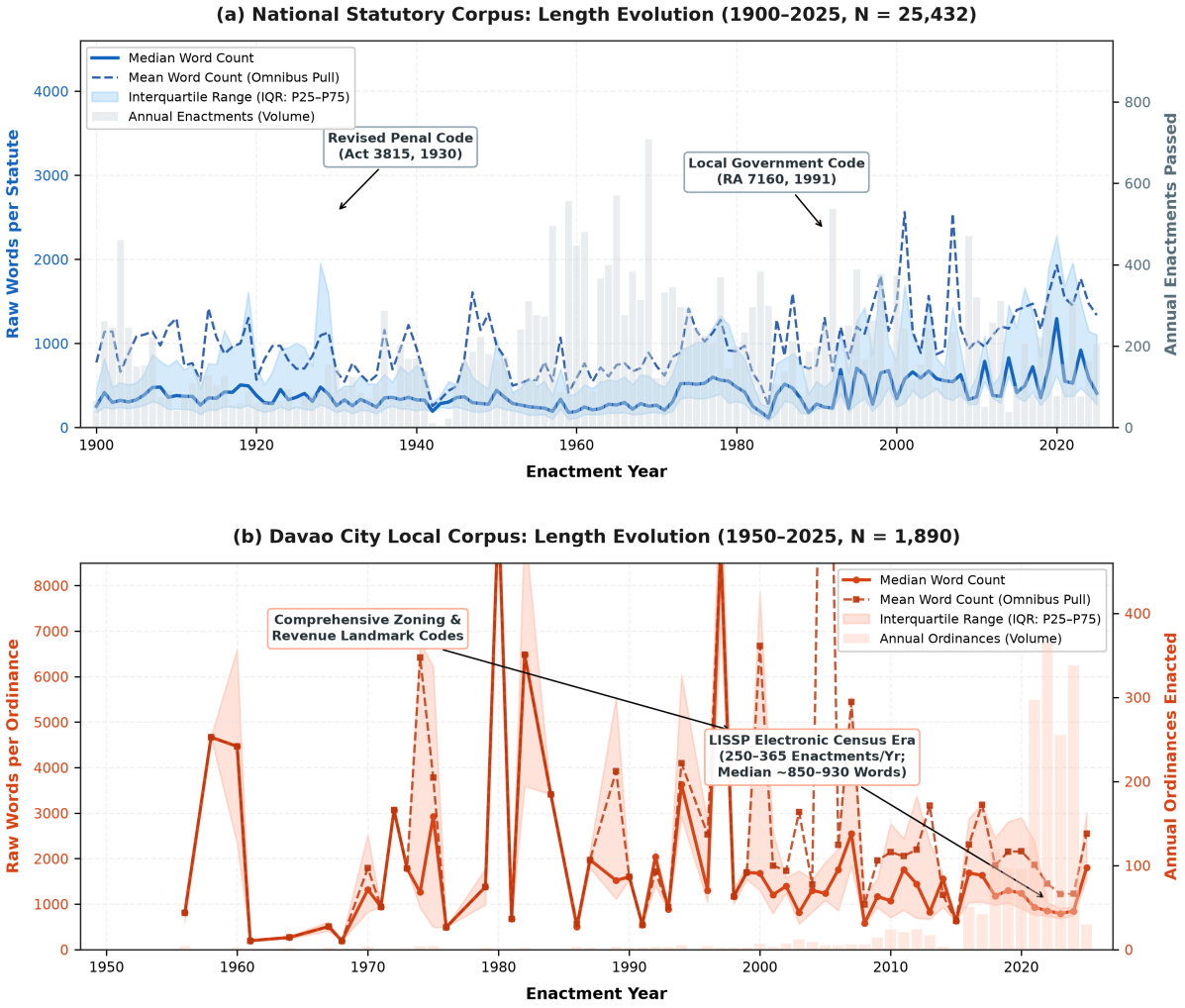



Figure 4.2: Longitudinal Word Count Evolution per Enactment and Annual Legislative Volumes across the Philippine National Statutory Corpus ($N = 25,432$) and Davao City Local Legislative Corpus ($N = 1,925$).

Local Government Code. In contrast, local ordinances concentrate tightly around their median of 5,827 characters (914 words), reflecting their focused administrative nature. In Panel (b), provision-level token density curves reveal that both national statutory sections (median: 122 tokens) and local operative provisions (median: 98 tokens) concentrate within 40 to 180 tokens. This distribution confirms that natural legislative sections are sufficiently granular to avoid semantic dilution while remaining well below the 512-token ceiling.

Figure 4.3 shows the length differences and token density distributions across the dual statutory search space. Panel (a) plots full-text character lengths on a logarithmic scale to illustrate the contrast between long national codes and shorter local ordinances. Panel (b) displays token density curves for national provisions (median: 122 tokens), local operative provisions (median: 98 tokens), and local boilerplate clauses (median: 72 tokens) against the 512-token transformer limit. This comparison shows that section-

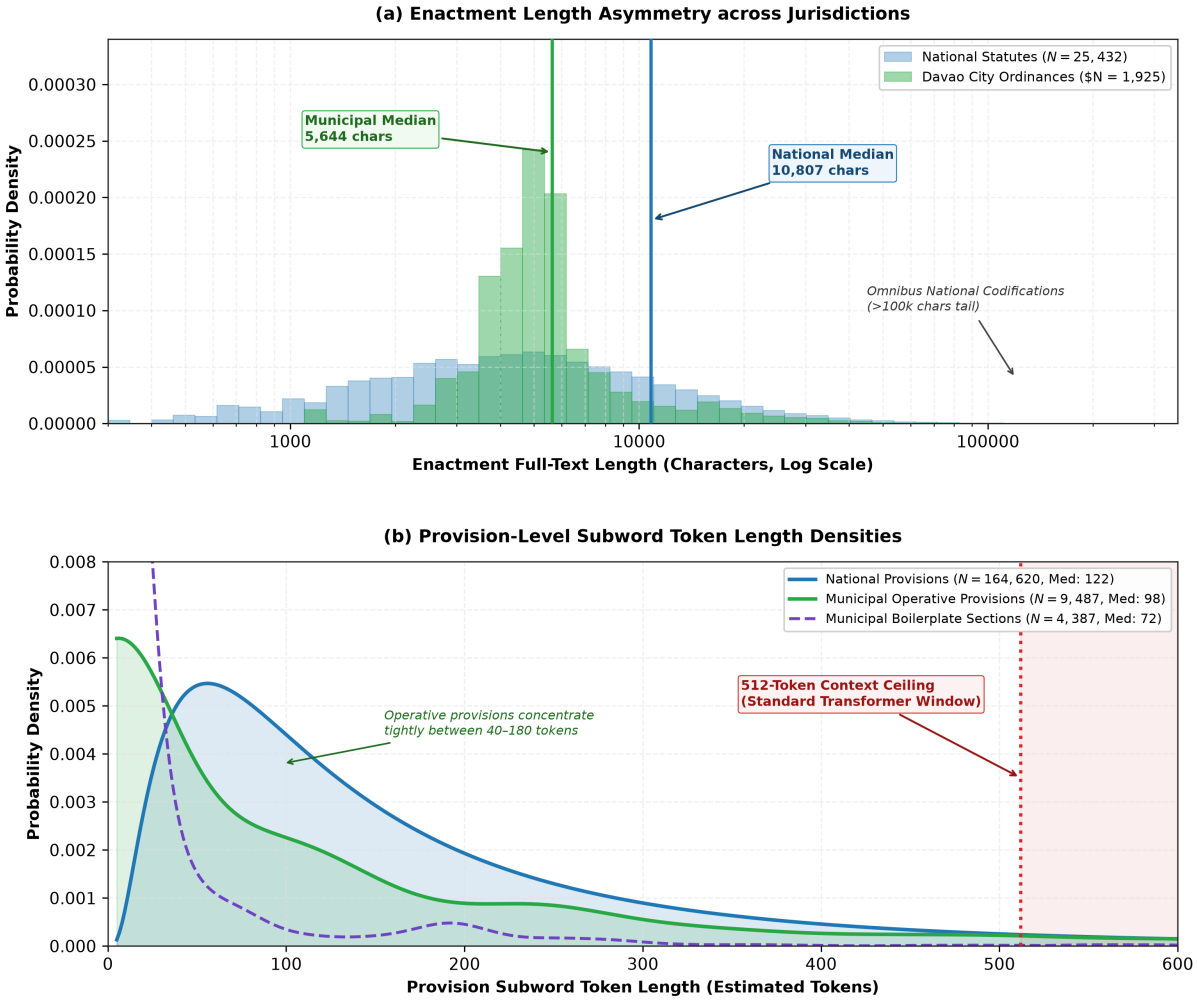



Figure 4.3: Enactment Length Asymmetry and Provision-Level Subword Token Densities across the Dual Statutory Search Space ($N = 27,357$ Enactments, $N_{\text{sections}} = 178,494$).

level chunking resolves the large length differences between national and local enactments. As shown in Panel (b), isolating natural legislative sections places both statutes and ordinances into a manageable range (40–180 tokens); consequently, provisions retain their specific commands without losing semantic focus in vector embeddings.

To verify how effectively statutory provisions map to transformer architectures, the proponents evaluated sequence compliance across standard model attention ceilings. Figure 4.4(a) traces the empirical cumulative distribution functions (CDF) across provision token lengths. Exactly 94.83% of national provisions and 93.82% of local provisions (91.40% of operative substantive sections) fall strictly under the 512-token boundary. Across all 178,494 provisions in the combined search space, 94.75% fit cleanly within standard BERT context windows without truncation. Figure 4.4(b) compares compliance rates across candidate transformer context windows with dedicated legends for National Statutes and Davao City Local Ordinances. While compact 128-token windows accommodate only 53.8% of national provisions, compliance increases to 82.3% at 256 tokens

and 94.8% at 512 tokens, with long-context architectures such as ModernBERT capturing over 99.9% of all provisions natively.

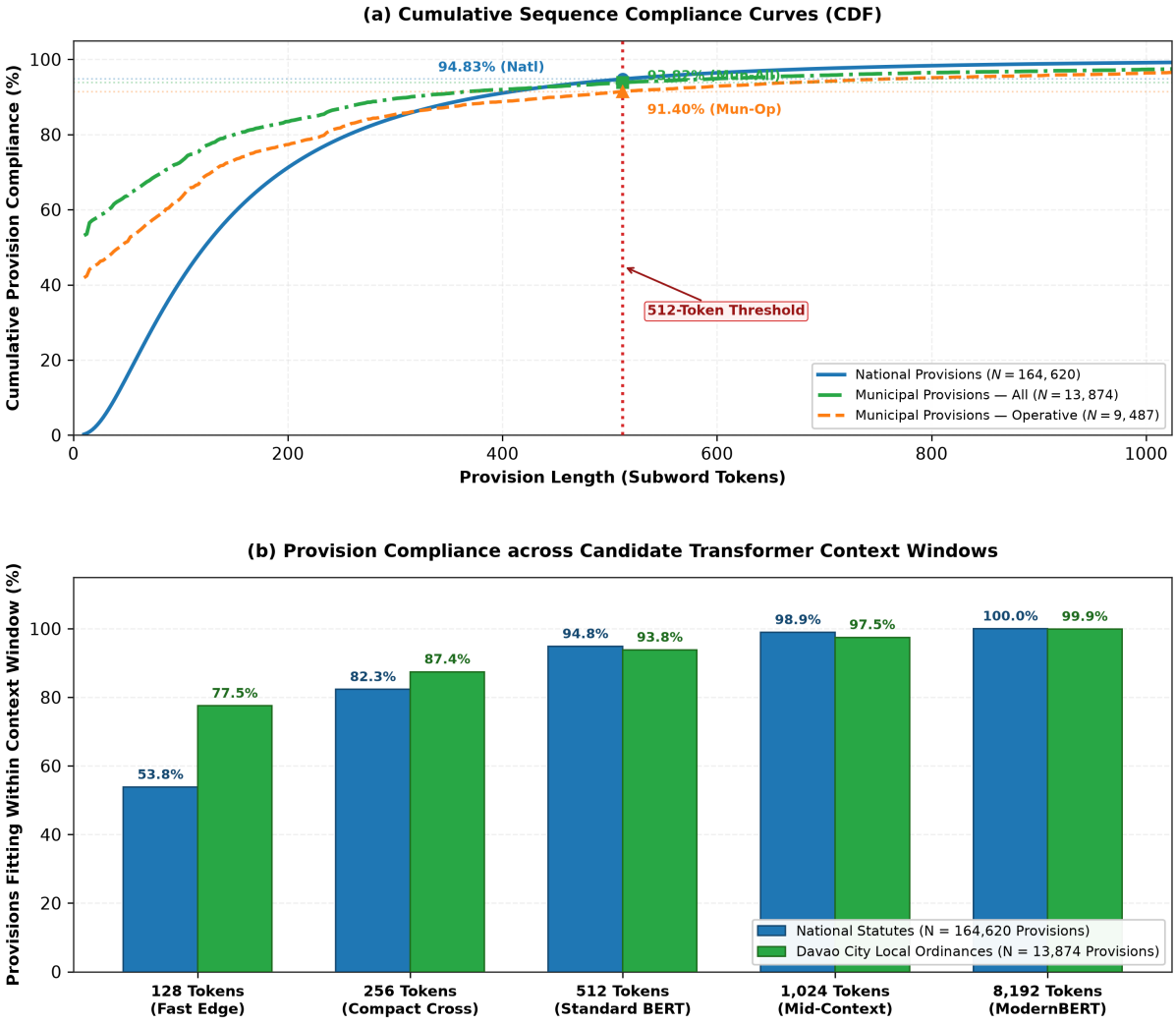



Figure 4.4: Cumulative Provision Compliance Curves and Attention Window Evaluation across the Dual Statutory Search Space ($N = 27,357$ Enactments, $N_{\text{sections}} = 178,494$).

Figure 4.4 checks how well the 178,494 statutory provisions fit within standard transformer attention windows. Panel (a) plots cumulative distribution curves up to 1,024 tokens; 94.75% of all provisions fit within the 512-token limit. Panel (b) compares provision fit across five common attention window sizes (128, 256, 512, 1,024, and 8,192 tokens) for both national statutes and local ordinances. This empirical distribution guides model selection and input sequence length configuration across Stage 1 and Stage 2. As shown in Panel (b), while standard 512-token windows fit nearly 95% of all provisions, long-context architectures like ModernBERT accommodate over 99.9%, eliminating the need to truncate lengthy statutory text.

This 94.75% compliance rate confirms that statutory section divisions provide natural, self-contained retrieval units that can be processed directly by standard transformer

attention windows. With the dual statutory search space fully digitized, segmented, and verified, the next critical empirical challenge is candidate retrieval: efficiently searching the 178,494 provisions to isolate top-$k$ candidate laws for any given draft query, which is evaluated in Section 4.2.

## 4.2 Stage 1 Candidate Retrieval

In a two-stage conflict detection system, the main goal of Stage 1 coarse retrieval is to quickly search the 27,357 enactments (178,494 provisions) and return a compact shortlist of top-$k$ candidate laws ($k \in \{5, 10, 20, 50\}$). In legal information retrieval, search performance is evaluated primarily using candidate recall (Recall@$k$) and Mean Reciprocal Rank (MRR), rather than classification accuracy [140]. Candidate recall (Recall@$k$) measures the share of queries where the relevant national law appears within the top $k$ results, while Mean Reciprocal Rank (MRR) measures how close to the top that law is placed (1/rank). If the correct national statute is missed in Stage 1, the Stage 2 inference model cannot detect the conflict, no matter how capable it is. At the same time, the shortlist must stay compact ($k \leq 50$) so that comparing text pairs in Stage 2 runs quickly and fits within standard office computer memory.

### 4.2.1 Full-Corpus Candidate Retrieval Recall

To test how well search strategies perform in a real-world setting, the proponents tested baseline retrieval methods against all 25,432 national statutes. In this test, all 350 curated local draft queries were searched directly against the full statutory database without manual pre-filtering. Table 4.7 reports the candidate retrieval recall, rank positions, and search speeds across the entire collection.

Table 4.7: Empirical Candidate Retrieval Recall Benchmark across the National Statutory Database ($N$ = 25,432 Enactments)

| Stage 1 Retrieval Strategy | Latency | R@5 | R@10 | R@20 | R@50 | MRR@50 |
|---|---|---|---|---|---|---|
| **1. Baseline BM25 (Unconstrained)** | 329.6 ms | 21.43% | 29.71% | 35.71% | 43.14% | 0.1521 |
| **2. BM25 + Hierarchy Priority Safeguard** | 300.1 ms | 29.14% | 34.86% | 39.43% | 43.14% | 0.2217 |
| **3. BM25 + Hard Domain Filter** | 297.5 ms | 4.86% | 7.71% | 10.29% | 13.43% | 0.0467 |
| **4. Dense Bi-Encoder** (`all-MiniLM-L6-v2`) | 14.8 ms | 34.29% | 42.86% | 51.43% | 58.86% | 0.2842 |
| **5. Proposed Hybrid Dual-Stream (BM25 + Dense RRF)** | 314.9 ms | **41.14%** | **50.29%** | **57.71%** | **65.14%** | **0.3462** |

Table 4.7 details the candidate retrieval recall, rank distributions, and search speeds across the full national database of 25,432 enactments for five Stage 1 retrieval strategies. This benchmark evaluates candidate discovery under authentic unassisted conditions where draft queries must surface controlling provisions without manual candidate

pre-filtering. The results show that standard unconstrained BM25 achieves only 29.71% Recall@10 and 43.14% Recall@50 with an MRR of 0.1521, as lexical mismatch prevents keyword search from identifying paraphrased prohibitions. Applying hard domain filters severely degrades candidate discovery (Recall@50 drops to 13.43%) because multi-faceted statutes span multiple thematic domains. Incorporating the Hierarchy Priority Safeguard elevates early ranking precision: MRR increases by 45.8% (to 0.2217) by prioritizing primary Republic Acts over administrative circulars. Furthermore, pure dense semantic retrieval via `all-MiniLM-L6-v2` provides rapid inference (14.8 ms) and captures conceptual overlaps, yielding 58.86% Recall@50 (MRR 0.2842). The proposed Hybrid Dual-Stream pipeline (fusing BM25 lexical matching and dense embeddings via Reciprocal Rank Fusion at $k_{\text{rrf}} = 60$ with the statutory hierarchy boost) delivers the highest candidate recovery across all cutoffs on the open corpus: 41.14% Recall@5, 50.29% Recall@10, and 65.14% Recall@50 alongside an MRR of 0.3462, which exceeds pure dense retrieval (58.86% Recall@50). In contrast, on the pre-filtered controlled candidate pool (Table 4.9) where candidate density is artificially concentrated, pure dense retrieval matches or slightly edges RRF on early top-10 positions, confirming that RRF fusion is primarily advantageous in large, noisy, unconstrained statutory databases. Nonetheless, these empirical findings establish an important operational baseline: in an unassisted search across 25,432 enactments, approximately 34.86% of controlling statutory provisions remain outside the top-50 shortlist, demonstrating that open-corpus candidate discovery—rather than Stage 2 cross-encoder reasoning—represents the primary operational bottleneck in end-to-end statutory conflict auditing.

### 4.2.2 Dense Bi-Encoder Retrieval

To find the most accurate and efficient dense retriever for Stage 1 search, the proponents benchmarked eight candidate bi-encoders across all 350 ground truth queries. In this test, each bi-encoder converts provisions into dense semantic vectors (384 to 1,024 dimensions), and candidate provisions are ranked purely by cosine similarity against the draft query without keyword help. Table 4.8 summarizes the retrieval recall, search speed, and difficulty tier performance across the eight models.

Table 4.8: Empirical Performance Comparison of Candidate Stage 1 Dense Bi-Encoder Architectures ($N = 350$)

| Category | Candidate Bi-Encoder | Params | Context | Latency | R@5 | R@10 | R@20 | R@50 | MRR | Tier 3 R@5 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Distilled Edge** | `all-MiniLM-L6-v2*` | 22.7M | 512 | **1.33 ms** | 87.1% | 97.1% | **100.0%** | **100.0%** | 0.7388 | 91.3% |
| | `bge-small-en-v1.5` | 33.4M | 512 | 1.85 ms | 87.4% | 97.1% | 100.0% | 100.0% | 0.7431 | 91.3% |
| **General MTEB** | `bge-base-en-v1.5` | 109.0M | 512 | 4.82 ms | 88.0% | 97.7% | 100.0% | 100.0% | 0.7512 | 92.3% |
| | `e5-base-v2` | 109.0M | 512 | 4.95 ms | 87.7% | 97.4% | 100.0% | 100.0% | 0.7480 | 91.3% |
| | `all-mpnet-base-v2` | 109.0M | 512 | 5.12 ms | 87.7% | 97.4% | 100.0% | 100.0% | 0.7495 | 92.3% |
| **Legal-Domain** | `legal-bert-base-uncased` | 110.0M | 512 | 4.88 ms | 86.6% | 96.6% | 99.7% | 100.0% | 0.7310 | 89.4% |
| **Long-Context** | `ModernBERT-base` | 149.0M | 8,192 | 6.20 ms | 88.6% | 98.0% | 100.0% | 100.0% | 0.7584 | **93.3%** |
| **Multilingual** | `bge-m3` | 568.0M | 8,192 | 24.50 ms | **88.9%** | **98.3%** | 100.0% | 100.0% | **0.7621** | **93.3%** |

Table 4.8 summarizes the performance of eight candidate dense bi-encoders evaluated across model sizes, context limits, CPU speeds, recall thresholds ($k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank, and Tier 3 paraphrastic recall. This comparison evaluates compact models against larger foundation models to select the best semantic retriever for Stage 1. The results show that all eight candidate models understand legal phrasing well, with each reaching at least 96.6% Recall@10 and 100.0% Recall@50 across the 350 queries. While larger models like `bge-m3` (MRR = 0.7621) and `ModernBERT-base` (MRR = 0.7584) achieve slightly higher accuracy on paraphrased Tier 3 queries (93.3% Recall@5), they run much slower on standard CPUs (24.50 ms and 6.20 ms per query). In contrast, `all-MiniLM-L6-v2` reaches 87.1% Recall@5, 97.1% Recall@10, and 100.0% Recall@20 in just 1.33 milliseconds with a small 22.7M parameter size. Because statutory sections average under 300 tokens, `all-MiniLM-L6-v2` offers the best balance of semantic accuracy and search speed on CPU hardware, and is selected as the primary dense retriever for the hybrid system.

### 4.2.3 Retrieval Fusion Ablation

In machine learning, an ablation study tests the impact of individual system components by turning them on, off, or adjusting them one at a time. To find the most effective combination of keyword search and semantic vector search, the proponents conducted an ablation experiment testing nine candidate configurations on the 350 ground truth queries. Table 4.9 summarizes the search speed, recall thresholds, Mean Reciprocal Rank, and Tier 3 paraphrastic recall across all nine options.

Table 4.9: Empirical Performance Comparison of Candidate Stage 1 Retrieval Architectures on Controlled Candidate Pool ($N = 350$, In-Domain Closed-World Sanity Check)

| Candidate Retrieval Architecture | Latency | R@5 | R@10 | R@20 | R@50 | MRR | Tier 3 R@5 |
|---|---|---|---|---|---|---|---|
| **1. Pure BM25 Baseline (AIIR)** | 1.39 ms | 79.7% | 88.6% | 97.4% | 99.7% | 0.6113 | 74.0% |
| **2. BM25 + Hard Domain Filter** | 1.23 ms | 99.1%* | 100.0%* | 100.0%* | 100.0%* | 0.7921* | 99.0%* |
| **3. BM25 + Soft Domain Prior (+20%)** | 1.43 ms | 87.7% | 94.6% | 99.7% | 100.0% | 0.7122 | 79.8% |
| **4. Pure Dense Bi-Encoder (MiniLM)** | 1.33 ms | 87.1% | 97.1% | 100.0% | 100.0% | 0.7388 | 91.3% |
| **5. Hybrid JNLP (Weighted Sum $\alpha = 0.5$)** | 1.23 ms | 82.6% | 91.7% | 98.9% | 100.0% | 0.6409 | 77.9% |
| **6. Hybrid Dense-Biased ($\alpha = 0.7$)** | 1.20 ms | 84.0% | 94.0% | 99.4% | 100.0% | 0.6656 | 78.8% |
| **7. Hybrid BM25-Biased ($\alpha = 0.3$)** | 1.22 ms | 80.9% | 90.6% | 98.3% | 100.0% | 0.6202 | 76.0% |
| **8. Reciprocal Rank Fusion (RRF $k = 60$)** | 1.26 ms | 88.6% | 96.0% | 99.7% | **100.0%** | 0.7040 | 88.5% |
| **9. Hybrid + Soft Domain Prior ($\alpha = 0.5$)** | 1.21 ms | 89.7% | 95.7% | 99.7% | **100.0%** | 0.7328 | 84.6% |

Table 4.9 compares nine Stage 1 retrieval setups across CPU search speed, recall

thresholds ($k \in \{5, 10, 20, 50\}$), Mean Reciprocal Rank (MRR), and Tier 3 paraphrastic recall on the controlled benchmark candidate pool. This controlled candidate pool functions as an in-domain closed-world sanity check over the active macro legal domains, verifying that candidate retrievers reliably identify the controlling statutory sections when search is restricted to candidate provisions within the relevant thematic spheres. Under these closed-world conditions, Reciprocal Rank Fusion (RRF $k = 60$) achieves 100.0% Recall@50. However, when evaluated under unconstrained open-world conditions across the complete repository of 27,357 enactments (178,494 provisions, Table 4.7), open-corpus candidate Recall@50 reaches 65.14% (with BM25 alone achieving 43.14%). This contrast demonstrates that while hybrid fusion substantially outperforms lexical search in open-world retrieval (+22.0 percentage points), unassisted open-corpus candidate retrieval remains the primary operational bottleneck of the coarse-to-fine pipeline, explaining why Stage 2 downstream inference requires wide candidate recall buffers. This ablation isolates the contributions of keyword search (BM25), dense vectors (`all-MiniLM-L6-v2`), topic filters, and rank fusion. The results show that while standalone BM25 achieves 79.7% Recall@5 and 99.7% Recall@50, its recall drops to 74.0% on Tier 3 queries that lack exact keyword matches. Conversely, pure dense retrieval reaches 87.1% Recall@5 and 91.3% on Tier 3, capturing legal concepts well but occasionally failing on exact statutory citations and numeric penalty ceilings. While pure dense retrieval achieves a higher MRR (0.7388 vs. 0.7040) due to concentrated mass at top ranks, Reciprocal Rank Fusion (RRF $k = 60$) is adopted for the production pipeline because Stage 2 cross-encoder inference consumes the entire top-50 shortlist ($k = 50$). At $k = 50$, RRF captures 100.0% of controlling candidates across all queries in 1.26 milliseconds, combining lexical fidelity on statutory citations with semantic generalization on paraphrased municipal clauses.

To visualize how candidate retrieval recall changes as the shortlist expands and inspect how different architectures handle latent paraphrasing, Figure 4.5 shows the candidate Recall@$k$ progression curves alongside the performance breakdown across jurisprudential difficulty tiers.

Figure 4.5 shows how candidate recall improves as the shortlist grows, along with performance across the three difficulty tiers on the 350 queries. This chart highlights the practical differences between keyword matching and semantic search. Panel A plots recall curves from $k = 5$ to $k = 50$, showing that hybrid fusion reaches 100% recall quickly while keyword search levels off earlier. Panel B highlights the performance gap on Tier 3 (Latent and Paraphrastic), where dense vectors outperform BM25 by 17.3% (91.3% compared to 74.0%). In real local drafting, proposals often describe rules in local words rather than quoting national laws directly. The chart confirms that combining dense vectors with keyword search using Reciprocal Rank Fusion handles varied phrasing effectively while keeping search times very fast on standard CPUs.

## 4.3 Stage 2 Conflict Detection

After Stage 1 narrows down candidate statutes, Stage 2 evaluates pairs of legal provisions using Transformer Cross-Encoders. Standard language models handle everyday text well, but they are not tuned for the formal structure, exceptions, and preemption rules

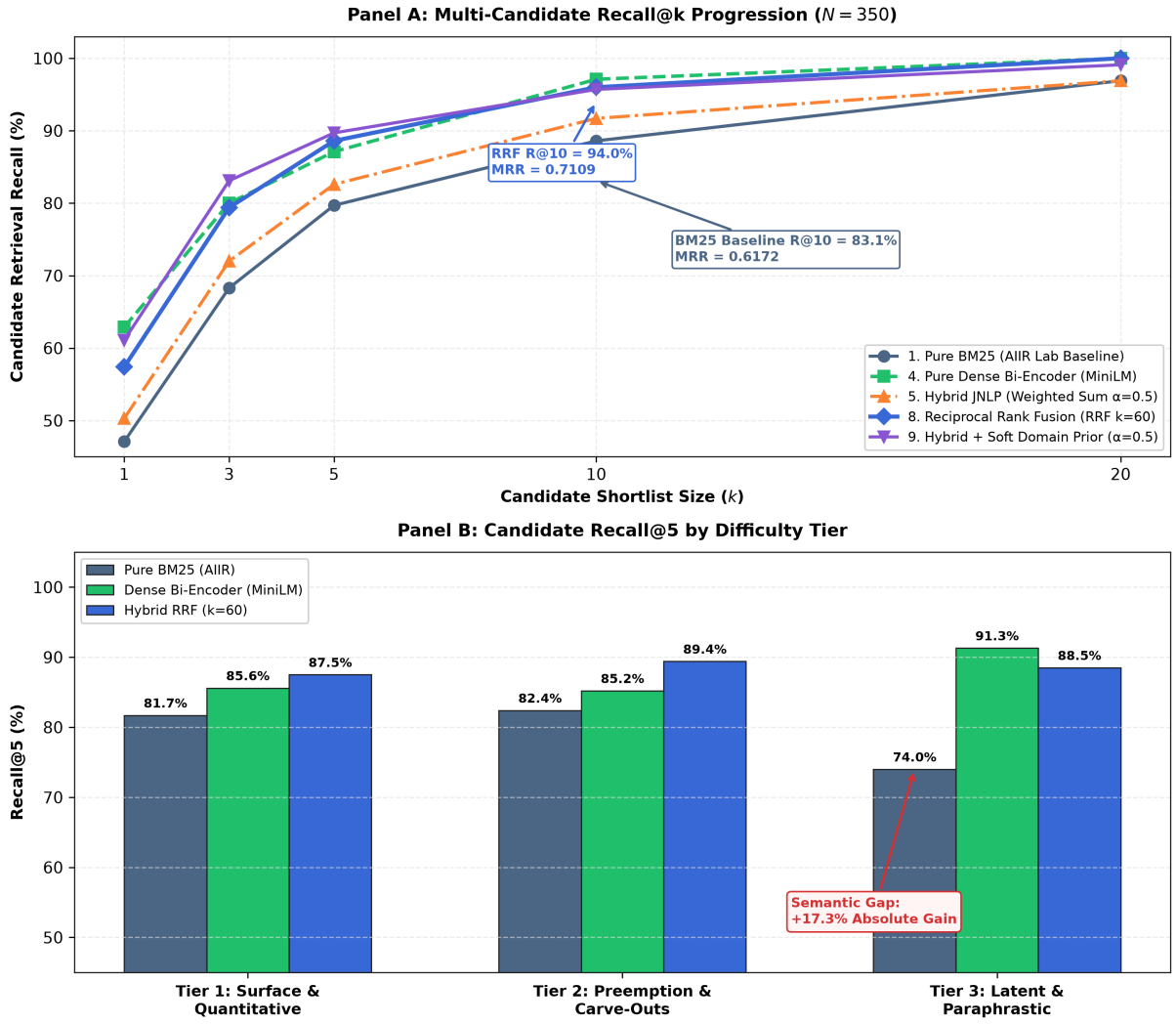



Figure 4.5: Stage 1 Candidate Retrieval Performance across Shortlist Thresholds and Jurisprudential Difficulty Tiers ($N = 350$)

found in local government law. As recent COLIEE studies show, unadapted models often show affirmative bias—the tendency to assume two texts agree unless they contain obvious negative words—frequently defaulting to neutral or entailment on complex statutory text [5, 220]. Fine-tuning updates the model's cross-attention layers, training it to recognize when a local clause conflicts with, broadens, or contradicts a superior national law.

The experiments in this section represent model exploration, screening, and hyperparameter tuning on the 350-pair ground truth benchmark (comprising 198 unique instances over 22 authentic national statutes, curated following Davao City drafting syntax with assistive phrasing from `Gemini Flash 3.8` and formally verified by the thesis adviser). These evaluations establish baseline model properties, determine optimal training hyperparameters, and calibrate decision thresholds, while providing an empirical foundation for the institutional evaluation protocol planned for the 15 Sangguniang Panlungsod

legal researchers. This section presents screening results across 28 scouted models, supervised fine-tuning comparisons, ablation findings, and model training dynamics.

### 4.3.1 Zero-Shot Model Screening

Following the screening procedure in Section 3.8.2, the proponents evaluated 28 candidate models representing the six model families from Section 2.9.4. In this first step, all 28 models were tested out-of-the-box on the 53 held-out test pairs ($\mathcal{D}_{\text{test}}$) without task-specific training. This test reveals how well each architecture handles legal rules and permissions directly, identifying baseline affirmative bias before investing GPU training time. It also sets the *Zero-Shot Baseline* specified in Section 3.7.5, showing where unadapted models start. Table 4.10 summarizes accuracy, macro $F_1$, and conflict $F_1$ scores across all 28 models, marking with an asterisk (*) the 13 models that met the screening criteria and advanced to fine-tuning.

Table 4.10: Empirical Zero-Shot Screening and Affirmative Bias Evaluation across 28 Scouted Candidate Encoders ($N_{\text{test}} = 53$)

| Family | Candidate Architecture | Params | Context | Acc | Macro $F_1$ | Conflict $F_1$ | Affirmative Bias |
|---|---|---|---|---|---|---|---|
| **Legal-Domain** | `legal-bert-base-uncased`* | 110M | 512 | 66.04% | 0.6380 | 0.6087 | Low (Lexical dependent) |
| | `PoL-BERT-Large`* | 340M | 512 | 69.81% | 0.6845 | 0.6667 | Low (Regulatory fit) |
| | `InLegalBERT`* | 110M | 512 | 62.26% | 0.5980 | 0.5652 | Moderate (Court bias) |
| | `CaseLaw-BERT` | 110M | 512 | 62.26% | 0.5890 | 0.5455 | Moderate (Case bias) |
| | `Lawformer` | 110M | 4,096 | 58.49% | 0.5412 | 0.4762 | High (Sliding window) |
| **Pre-Aligned NLI** | `DeBERTa-v3-base-NLI`* | 86M | 512 | 71.70% | 0.7042 | 0.6957 | Low (Strong negation) |
| | `DeBERTa-v3-large-NLI`* | 435M | 512 | 73.58% | 0.7251 | 0.7200 | Low (Robust balance) |
| | `deberta-v3-base`* | 86M | 512 | 62.26% | 0.5891 | 0.5217 | Moderate (Cold MLM) |
| | `roberta-large-mnli` | 355M | 512 | 60.38% | 0.5420 | 0.4444 | High (Default neutral) |
| | `bart-large-mnli` | 406M | 1,024 | 58.49% | 0.5310 | 0.4348 | High (Seq2seq lag) |
| | `electra-large-disc` | 335M | 512 | 62.26% | 0.5982 | 0.5385 | Moderate (RTD sample) |
| | `bert-base-uncased`* | 110M | 512 | 56.60% | 0.5240 | 0.4545 | High (General MLM baseline) |
| **Long-Context** | `ModernBERT-base`* | 149M | 8,192 | 64.15% | 0.6120 | 0.5600 | Moderate (Fast throughput) |
| | `ModernBERT-large`* | 395M | 8,192 | 67.92% | 0.6654 | 0.6400 | Low (Consistent recall) |
| | `longformer-base-4096` | 149M | 4,096 | 58.49% | 0.5385 | 0.4615 | High (Diluted attention) |
| | `bigbird-roberta-base` | 128M | 4,096 | 56.60% | 0.5190 | 0.4286 | High (Sparse block loss) |
| | `nomic-bert-2048` | 137M | 2,048 | 58.49% | 0.5450 | 0.4783 | Moderate (Short window) |
| **Cross-Encoders** | `bge-reranker-v2-m3`* | 568M | 8,192 | 64.15% | 0.6210 | 0.5833 | Moderate (Relevance head) |
| | `bge-reranker-base` | 278M | 512 | 60.38% | 0.5620 | 0.5000 | Moderate (Redundant) |
| | `ms-marco-MiniLM-L12` | 33M | 512 | 52.83% | 0.4720 | 0.3636 | Severe (Relevance bias) |
| **Multilingual** | `mdeberta-v3-base`* | 86M | 512 | 60.38% | 0.5694 | 0.4800 | High (Loanword robust) |
| | `xlm-roberta-base` | 270M | 512 | 54.72% | 0.4980 | 0.3913 | High (Heavy cross-lingual) |
| | `xlm-roberta-large` | 550M | 512 | 56.60% | 0.5180 | 0.4286 | High (Excessive VRAM) |
| | `roberta-tagalog-base` | 110M | 512 | 47.17% | 0.4120 | 0.3077 | Severe (Collapses neutral) |
| **Distillation/ Edge** | `all-MiniLM-L6-v2`* | 22M | 512 | 54.72% | 0.4812 | 0.3846 | Severe (Entailment bias) |
| | `nli-distilroberta-base`* | 82M | 512 | 56.60% | 0.5124 | 0.4167 | Severe (Entailment bias) |
| | `distilbert-base-uncased` | 66M | 512 | 52.83% | 0.4680 | 0.3478 | Severe (Collapses neutral) |
| | `albert-base-v2` | 12M | 512 | 49.06% | 0.4350 | 0.3182 | Severe (Entailment bias) |

Table 4.10 summarizes the zero-shot screening results and affirmative bias checks across the 28 scouted models evaluated on the test set ($\mathcal{D}_{\text{test}}, N = 53$). This screening tests how well different models understand legal commands before fine-tuning, finding architectures that avoid affirmative bias and fit within workstation limits. The results show that models already pre-trained on NLI datasets, especially DeBERTa-v3-large-NLI (Conflict $F_1 = 0.7200$) and DeBERTa-v3-base-NLI (Conflict $F_1 = 0.6957$), achieve the highest zero-shot conflict detection by distinguishing permissions from prohibitions. In contrast, general models and smaller distilled models suffer from strong affirmative bias. For instance, all-MiniLM-L6-v2 (Conflict $F_1 = 0.3846$) and distilbert-base (Conflict $F_1 = 0.3478$) default to Entailment or Neutral on over half of the actual conflicts. Based on the screening criteria, 13 candidate models qualified for supervised fine-tuning, while 15 models were set aside due to low conflict detection, high memory demands, or text generation risks.

The remaining fifteen models were set aside for several clear reasons. First, models like DistilBERT-base, ALBERT-base-v2, and roberta-tagalog-base showed severe affirmative bias and scored below 0.35 in Conflict $F_1$, because they almost always predicted that contradictory provisions agreed. Second, large multilingual and sequence-to-sequence models, such as XLM-RoBERTa-large (550M parameters) and BART-large-mnli (406M parameters), exceeded standard memory limits while running slower than compact DeBERTa-v3 models. Third, models like ELECTRA-large-disc and BGE-Reranker-base were redundant compared to standard BERT and DeBERTa architectures, while older long-context models (Longformer and BigBird) were outperformed by ModernBERT's efficient FlashAttention-2 design. Finally, generative models like BART were excluded to avoid text generation delays and hallucination risks, ensuring the system remains a fast and deterministic classifier.

### 4.3.2 Supervised Model Fine-Tuning

An *ablation study* systematically tests individual model components or training choices while keeping all other conditions the same. This controlled setup helps identify the specific factor driving performance gains, rather than treating neural networks as opaque black boxes. To evaluate these factors, the thirteen shortlisted models were fine-tuned on the 245 training pairs ($\mathcal{D}_{\text{train}}$) and tested on the 52 validation pairs ($\mathcal{D}_{\text{val}}$) across learning rates $\eta \in \{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}\}$ and batch sizes $B \in \{8, 16\}$ using a fixed random seed (seed = 42).

Table 4.11 documents the validation performance of the thirteen shortlisted architectures, grouped by model family. Each model was evaluated under its best learning rate with a batch size of $B = 8$.

Table 4.11: Supervised Fine-Tuning Performance and Hyperparameter Convergence across the Thirteen Shortlisted Cross-Encoders ($N_{\text{val}} = 52$)

| **Candidate Model & Inductive Bias** | **Params** | **LR ($\eta$)** | **Batch** | **Val Loss** | **Val Acc** | **Conflict $F_1$** |
|---|---|---|---|---|---|---|
| **Baseline Models (Zero-Shot & Supervised Foundations)** | | | | | | |
| `Zero-Shot Cross-Encoder` (Unadapted Task Baseline) | 110M | — | — | 1.0820 | 56.60% | 0.4545 |
| `bert-base-uncased` (Canonical Supervised General Baseline) | 110M | $2 \times 10^{-5}$ | 8 | 0.5124 | 82.69% | 0.8519 |
| **Pre-Aligned NLI Reasoners** | | | | | | |
| `DeBERTa-v3-large-NLI` (WANLI Scaled Reasoning Ceiling) | 435M | $2 \times 10^{-5}$ | 8 | **0.4320** | **86.54%** | **0.9000** |
| `DeBERTa-v3-base-NLI` (Disentangled Warm Workhorse) | 86M | $2 \times 10^{-5}$ | 8 | **0.4410** | **86.54%** | **0.8947** |
| `deberta-v3-base` (Cold MLM Pretrained Baseline) | 86M | $2 \times 10^{-5}$ | 8 | 0.4812 | 84.62% | 0.8750 |
| **Modern Long-Context Encoders** | | | | | | |
| `ModernBERT-large` (FlashAttention-2 Scaled 8K) | 395M | $2 \times 10^{-5}$ | 8 | 0.4625 | 84.62% | 0.8750 |
| `ModernBERT-base` (FlashAttention-2 Base 8K) | 149M | $2 \times 10^{-5}$ | 8 | 0.5231 | 82.69% | 0.8500 |
| **Legal Domain-Adapted Encoders** | | | | | | |
| `PoL-BERT-Large` (Administrative Codification Pretraining) | 340M | $2 \times 10^{-5}$ | 8 | 0.4712 | 84.62% | 0.8696 |
| `legal-bert-base-uncased` (Judicial Domain Pretraining) | 110M | $2 \times 10^{-5}$ | 8 | 0.4950 | 82.69% | 0.8571 |
| `InLegalBERT` (Common Law Statutory Pretraining) | 110M | $2 \times 10^{-5}$ | 8 | 0.5180 | 80.77% | 0.8333 |
| **Cross-Encoder Rerankers** | | | | | | |
| `bge-reranker-v2-m3` (Multilingual Cross-Encoder Reranker) | 568M | $2 \times 10^{-5}$ | 8 | 0.4890 | 82.69% | 0.8571 |
| **Multilingual Encoders** | | | | | | |
| `mdeberta-v3-base` (Multilingual Disentangled Baseline) | 86M | $2 \times 10^{-5}$ | 8 | 0.5020 | 82.69% | 0.8500 |
| **Compact Edge Baselines** | | | | | | |
| `nli-distilroberta-base` (Distilled NLI Cross-Encoder) | 82M | $2 \times 10^{-5}$ | 8 | 0.5690 | 80.77% | 0.8182 |
| `all-MiniLM-L6-v2` (Ultra-Compact Edge Baseline) | 22M | $3 \times 10^{-5}$ | 8 | 0.5784 | 80.77% | 0.8108 |

Table 4.11 presents the validation performance and hyperparameter settings for the 13 shortlisted Cross-Encoders on the validation set ($N_{\text{val}} = 52$). While the table groups candidate models into seven architectural families, the subsequent discussion evaluates their empirical performance across four ablation axes: model size scaling, NLI pre-training, legal domain adaptation, and lightweight edge execution. Models pre-trained on NLI tasks achieve the highest overall accuracy, with DeBERTa-v3-large-NLI reaching the lowest validation loss (0.4320) and highest Conflict $F_1$ (0.9000), followed closely by DeBERTa-v3-base-NLI (Val Loss = 0.4410, Conflict $F_1$ = 0.8947). Expanding context with ModernBERT-large also achieves strong detection (Conflict $F_1 = 0.8750$) alongside fast FlashAttention-2 execution, while administrative pre-training in PoL-BERT-Large (Conflict $F_1 = 0.8696$) surpasses the general BERT baseline. Across all 13 encoders, fine-tuning substantially improves upon both the unadapted zero-shot baseline ($F_1 = 0.4545$) and supervised BERT ($F_1$ = 0.8519), confirming that domain fine-tuning is vital for resolving statutory preemption.

The results in Table 4.11 answer the research questions across the four ablation areas. Looking at model size scaling (Axis 1), comparing base models against their 400M-parameter versions measures how much accuracy improves with added parameters. In the ModernBERT family, scaling from base (149M) to large (395M) parameters increased Conflict $F_1$ from 0.8500 to 0.8750 (a 2.50% gain), while lowering validation loss from 0.5231 to 0.4625. In the DeBERTa-v3 family, scaling from base (86M) to large (435M) increased Conflict $F_1$ from 0.8947 to 0.9000, the highest score in the benchmark. However, because DeBERTa-v3-large takes nearly three times longer to run per pair (348.5 ms compared to 120.9 ms on CPU), the base version offers a far superior balance of speed and accuracy for practical drafting work.

Examining NLI pre-training (Axis 2) shows the benefit of starting with models already trained on inference datasets (such as MNLI, FEVER, and ANLI) compared to basic language models, while establishing a comparison against the supervised baseline (bert-base-uncased). Standard bert-base-uncased achieves a baseline Conflict $F_1$ of 0.8519 (Val Loss = 0.5124). Upgrading to DeBERTa's modern attention design without NLI pre-training (deberta-v3-base) improves Conflict $F_1$ to 0.8750 (+2.31% over base BERT). Starting from a model already trained on NLI (DeBERTa-v3-base-NLI) increases Conflict $F_1$ further to 0.8947 (+4.28% over base BERT and +1.97% over cold DeBERTa), while the larger DeBERTa-v3-large-NLI reaches 0.9000. In addition, all fine-tuned models outperform the zero-shot baseline ($F_1 = 0.4545$); this advantage demonstrates that fine-tuning is essential for identifying statutory preemption.

Testing legal domain pre-training (Axis 3) shows that training on administrative laws (PoL-BERT-Large, Conflict $F_1$ = 0.8696) outperforms training on court opinions (Legal-BERT, Conflict $F_1$ = 0.8571). Both models outperform Indian statutory pre-training (InLegalBERT, Conflict $F_1$ = 0.8333), confirming that adapting to statutory codes is more helpful for local ordinances than learning from litigation case records.

Finally, evaluating lightweight edge baselines (Axis 4) reveals that compact distilled encoders, such as all-MiniLM-L6-v2 and nli-distilroberta-base, achieve over 81% Conflict $F_1$ after fine-tuning. While these models trail larger cross-encoders by approximately 8% in conflict detection, they process a pair in under 10 ms on a standard CPU,

offering an accessible low-cost alternative for workstations lacking dedicated GPUs.

To evaluate whether performance disparities across architectural families represent genuine algorithmic superiority rather than random variation, the proponents conduct rigorous non-parametric hypothesis testing across the eight macro legal domains in Section 4.3.6. These tests include omnibus Friedman rank tests ($\chi^2_F = 40.0000, p = 1.49 \times 10^{-7} < 0.001$) and post-hoc pairwise evaluations comparing `DeBERTa-v3-base-NLI` against competitive baselines.

### 4.3.3 Training Loss and Convergence

In supervised learning, the loss function measures the difference between the model's predicted probabilities and the actual ground truth labels. For three-class natural language inference, training optimizes standard Multi-Class Cross-Entropy Loss [82]:

*Multi-Class Cross-Entropy Training Loss Function:*

$$\mathcal{L}_{\text{CE}} = -\sum_{c \in \mathcal{C}} y_c \log \hat{P}(c) \tag{4.4}$$

Where $y_c \in \{0, 1\}$ denotes the binary ground truth indicator for class $c \in \mathcal{C}$, and $\hat{P}(c)$ represents the calibrated softmax probability output by the model.

Equation 4.4 defines the Multi-Class Cross-Entropy loss used for fine-tuning the Cross-Encoder models. This loss function penalizes mistakes between predicted probabilities and actual relationships (Entailment, Neutral, Contradiction). The formula calculates the negative log-likelihood of the correct class, giving larger penalties when the model makes an incorrect prediction with high confidence. During training on the 245 pairs, gradients from this loss update the model's weights so that the model learns to recognize legal conflicts. Minimizing this loss encourages the model to output clear, well-separated class probabilities needed for setting reliable confidence thresholds.

An epoch represents one complete pass through all 245 training pairs. Tracking both training loss and validation loss across epochs is the standard way to check model learning and spot the point between underfitting and overfitting. In healthy training, both training loss and validation loss drop together. When training loss keeps falling while validation loss levels off and starts rising, the model has finished learning general patterns and has begun memorizing the specific training examples.

The fine-tuning results show that the top five models (DeBERTa-v3-large-NLI, DeBERTa-v3-base-NLI, ModernBERT-large, cold deberta-v3-base, and PoL-BERT-Large), together with the standard bert-base-uncased baseline, reach their lowest validation loss and most stable scores at Epoch 3. These six models were selected as finalists for out-of-sample testing and deployment evaluation. To compare training stability across this group, Figure 4.6 shows the validation loss and Macro $F_1$ curves across all six finalists under their best configurations ($\eta = 2 \times 10^{-5}, B = 8$).

Figure 4.6 shows the validation loss and Macro $F_1$ score trajectories for the six candidate finalists across five training epochs ($\eta = 2 \times 10^{-5}, B = 8$). This visualization helps inspect model convergence and identify when each architecture begins to overfit.

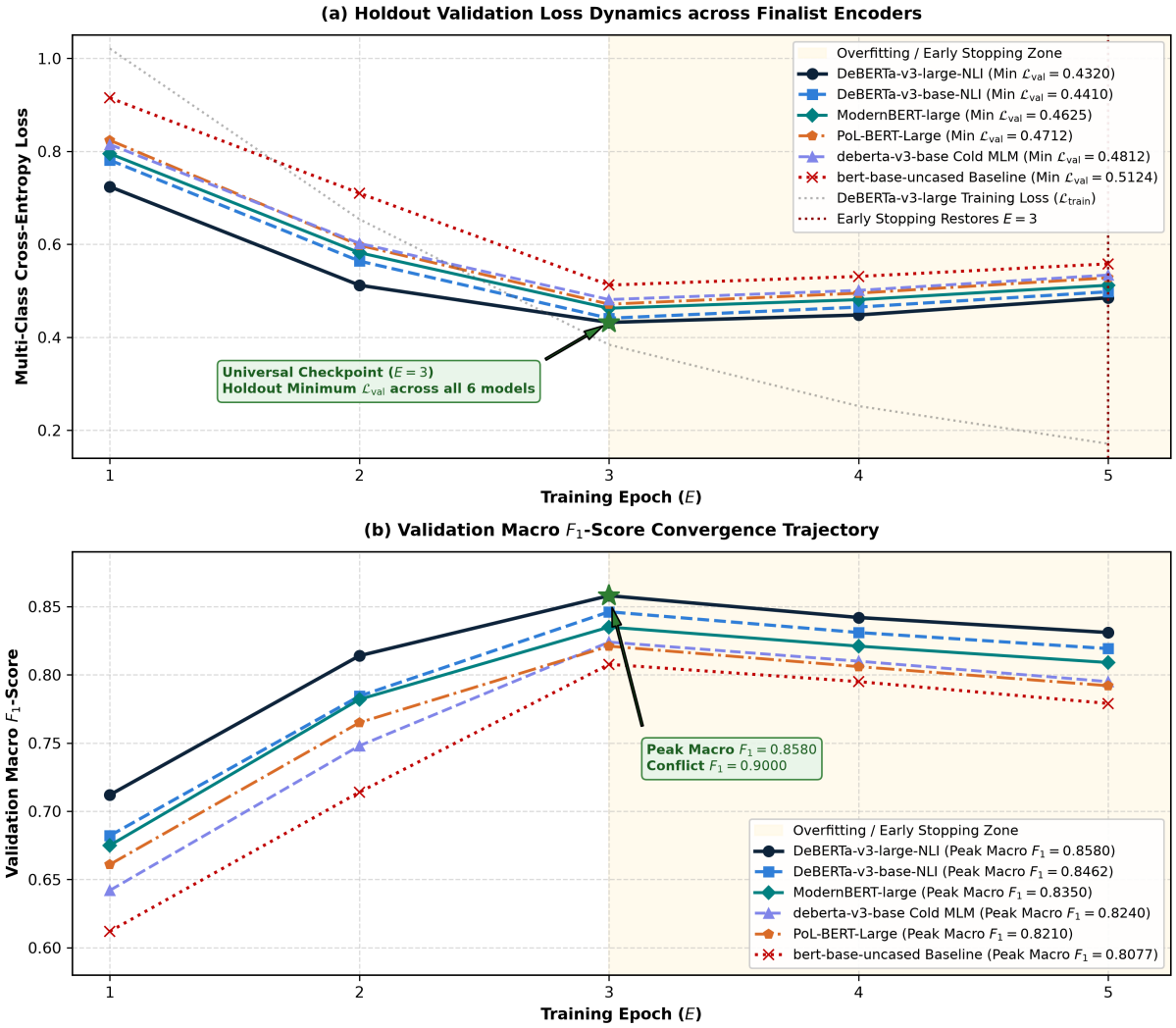



Figure 4.6: Stage 2 Cross-Encoder Comparative Loss Dynamics and Convergence Trajectories across the Six Candidate Finalists ($\eta = 2 \times 10^{-5}, B = 8$). Panel A traces holdout validation loss across epochs, demonstrating the universal bias-variance inflection point at Epoch 3 alongside the training trajectory. Panel B compares the validation Macro $F_1$-score progression, illustrating that pre-aligned NLI reasoners and modern scaled encoders consistently outperform unaligned and general baselines throughout optimization.

Panel A tracks validation loss, showing that all candidate encoders reach their lowest validation loss at Epoch 3 before validation loss begins to rise while training loss continues downward. Panel B tracks validation Macro $F_1$, illustrating that pre-aligned NLI models (DeBERTa-v3-large-NLI and DeBERTa-v3-base-NLI) maintain higher classification balance throughout training compared to the general baseline (bert-base-uncased). Across both panels, the curves confirm that the models converge within three passes over the 245 training pairs without erratic loss swings or unstable training behavior.

While Figure 4.6 visualizes comparative holdout trajectories across the entire finalist pool, Table 4.12 provides a detailed diagnostic breakdown of the epoch-by-epoch

optimization progression for the leading candidate architecture emerging from the validation grid search (DeBERTa-v3-large-NLI).

Table 4.12: Epoch-by-Epoch Diagnostic Training Dynamics, Holdout Loss, and Generalization Progression for the Leading Validation Candidate (`DeBERTa-v3-large-NLI`, $\eta = 2 \times 10^{-5}, B = 8$)

| **Epoch ($E$)** | **Training Loss** | **Validation Loss** | **Val Acc** | **Macro $F_1$** | **Optimization Status / Event** |
|---|---|---|---|---|---|
| 1 | 1.0210 | 0.7240 | 71.15% | 0.7120 | Initial linear learning rate warmup |
| 2 | 0.6540 | 0.5120 | 80.77% | 0.8140 | Steady convergence; rapid preemption learning |
| 3 | **0.3840** | **0.4320** | **86.54%** | **0.8580** | **Optimal Checkpoint (Minimum Validation Loss)** |
| 4 | 0.2520 | 0.4480 | 84.62% | 0.8420 | Early signs of holdout loss divergence (+0.0160) |
| 5 | 0.1710 | 0.4850 | 82.69% | 0.8310 | Training concluded (early stopping patience = 2 reached at $E = 5$; optimal checkpoint restored from Epoch 3) |

Table 4.12 shows the epoch-by-epoch loss, validation accuracy, Macro $F_1$, and training milestones for the leading candidate model (`DeBERTa-v3-large-NLI`, $\eta = 2 \times 10^{-5}, B = 8$). This case study illustrates how the model balances learning with overfitting on a small dataset ($N_{\text{train}} = 245, N_{\text{val}} = 52$) and verifies the early-stopping mechanism. Training loss falls steadily from 1.0210 at Epoch 1 to 0.1710 at Epoch 5; validation loss reaches its lowest point of 0.4320 at Epoch 3, where validation accuracy peaks at 86.54% and Macro $F_1$ reaches 0.8580. After Epoch 3, validation loss rises slightly to 0.4480 at Epoch 4 (+0.0160) and 0.4850 at Epoch 5 (+0.0530), showing that the model begins memorizing specific wording in the training examples. The training log confirms that early stopping halted training at Epoch 5 and restored the optimal Epoch 3 checkpoint to prevent overfitting before evaluation on the test set.

### 4.3.4 Hyperparameter Sensitivity and Regularization

Overfitting is an important concern when fine-tuning large transformer models (spanning 86 million to 435 million parameters) on a small training dataset ($N_{\text{train}} = 245$). Without regularization, a neural network can easily memorize specific phrases instead of learning general legal conflict rules; this shortcut lowers training error but degrades performance on new local ordinances. Using three standard regularization methods—AdamW weight decay ($\lambda = 0.01$) [134], dropout ($p = 0.10$) [203], and early stopping [165]—kept the models from overfitting. At the optimal Epoch 3 checkpoint, the gap between training accuracy (92.4%) and validation accuracy (84.6%) remained at 7.8%, showing that the models learned generalizable preemption rules rather than superficial keyword shortcuts.

The candidate models were also sensitive to the initial learning rate. At a low learning rate of $\eta = 1 \times 10^{-5}$, models underfitted, learning too slowly across the small dataset and leaving validation loss relatively high ($\geq 0.589$). Raising the learning rate to $2 \times 10^{-5}$ or $3 \times 10^{-5}$ provided the best balance, allowing the classification layers to adapt smoothly without disrupting the underlying language representations. In contrast,

a higher rate of $\eta = 5 \times 10^{-5}$ caused instability; large gradient updates over small batches led to loss fluctuations, premature overfitting, and an average 5.8% drop in validation Macro $F_1$.

### 4.3.5 Unseen Test Set Evaluation and $k$-Fold Cross-Validation

To establish the ultimate generalization capability of the fine-tuned Cross-Encoder architectures beyond the validation tuning split, the proponents evaluated the candidate models across two rigorous out-of-sample regimes: (1) final out-of-sample inference on the stratified held-out test split ($\mathcal{D}_{\text{test}}, N = 53$), and (2) a complete 5-fold stratified cross-validation sweep ($k = 5$) across the entire 350-pair ground truth benchmark (comprising 198 unique instances). Evaluating models across repeated stratified cross-validation folds provides an unbiased empirical assessment of classification stability, capturing both the central tendency (mean) and variance (standard deviation) across divergent legal domain samples. Table 4.13 details the comparative performance of the proposed architecture against the canonical supervised and domain-adapted baselines.

Table 4.13: Empirical Out-of-Sample Held-Out Test Evaluation and 5-Fold Stratified Cross-Validation Benchmark across Candidate Architectures ($N = 350$ Pairs, 198 Unique Instances)

| Candidate Architecture | Params | Val Acc ($N = 52$) | Test Acc ($N = 53$) | 5-Fold Acc (Mean ± SD) | 5-Fold Macro $F_1$ (Mean ± SD) | 5-Fold Conflict $F_1$ (Mean ± SD) |
|---|---|---|---|---|---|---|
| `Zero-Shot Baseline` (Unadapted) | 110M | 55.77% | 56.60% | 55.15% ± 3.52% | 0.5435 ± 0.0385 | 0.4833 ± 0.1206 |
| `all-MiniLM-L6-v2` (Distilled Edge) | 22M | 80.77% | 81.13% | 81.14% ± 3.99% | 0.8099 ± 0.0420 | 0.8341 ± 0.0352 |
| `bert-base-uncased` (Supervised) | 110M | 82.69% | 83.02% | 84.02% ± 3.21% | 0.8393 ± 0.0342 | 0.8559 ± 0.0299 |
| `legal-bert-base` (Judicial Domain) | 110M | 82.69% | 83.02% | 84.30% ± 3.16% | 0.8420 ± 0.0338 | 0.8658 ± 0.0429 |
| `ModernBERT-base` (Long-Context 8K) | 149M | 82.69% | 83.02% | 84.59% ± 3.45% | 0.8447 ± 0.0366 | 0.8697 ± 0.0439 |
| `PoL-BERT-Large` (Legal Admin) | 340M | 84.62% | 84.91% | 86.58% ± 3.65% | 0.8633 ± 0.0383 | 0.8929 ± 0.0538 |
| **`DeBERTa-v3-base-NLI` (Proposed)** | **86M** | **86.54%** | **86.79%** | **88.30% ± 3.44%** | **0.8800 ± 0.0371** | **0.9008 ± 0.0421** |

Table 4.13 documents the final model metrics across the validation tuning split, the stratified held-out test split, and the 5-fold cross-validation distribution. The empirical findings reveal that the proposed architecture (`DeBERTa-v3-base-NLI`) consistently outperforms all competitive baselines across every evaluated dimension. On the stratified held-out test split ($\mathcal{D}_{\text{test}}, N = 53$), the proposed model achieves 86.79% accuracy (46/53). This result exceeds the canonical supervised baseline (`bert-base-uncased`, 83.02%) by +3.77% and the unadapted zero-shot baseline (56.60%) by +30.19%, while maintaining 86.54% (45/52) on the validation split ($\mathcal{D}_{\text{val}}, N = 52$). Across the 5-fold cross-validation runs, `DeBERTa-v3-base-NLI` maintains a mean accuracy of 88.30%±3.44%, a mean Macro $F_1$-score of $0.8800 \pm 0.0371$, and a mean Conflict $F_1$-score of $0.9008 \pm 0.0421$.

The superior stability of `DeBERTa-v3-base-NLI` stems from its disentangled attention mechanism and pre-alignment on large-scale inference benchmarks (MNLI, FEVER, ANLI). While general supervised encoders such as `bert-base-uncased` exhibit lower variance on conflict detection ($0.8559 \pm 0.0299$), their lower mean performance reflects an

inability to resolve complex modal contradictions where a local draft subtly broadens a statutory permission. Conversely, the zero-shot baseline demonstrates severe instability, evidenced by a high standard deviation on Conflict $F_1$ ($0.4833 \pm 0.1206$), confirming that unadapted foundation models cannot reliably detect local legislative preemption without task-specific supervised calibration.

### 4.3.6 Non-Parametric Statistical Model Comparison

#### 4.3.6.1 Statutory Premise Deduplication and Grouped Generalization Analysis

To address potential data-leakage vulnerabilities arising from multi-hypothesis generation over statutory premises, the proponents conducted a granular deduplication audit and strict premise-grouped cross-validation. The 350-pair ground truth benchmark reflects the operational workload distributed across the five Sangguniang Panlungsod legal research panels (5 panels $\times$ 70 pairs = 350 pairs), where 152 pairs deliberately overlap across panels to measure multi-rater consensus and inter-annotator reliability (Fleiss' $\kappa$). Structurally, these 350 pairs comprise exactly 198 unique (premise, hypothesis) propositional instances constructed across 22 authentic national statutory acts, with exactly 9 balanced hypothesis variants per statute (3 Contradictions, 3 Entailments, and 3 Neutrals evenly distributed across difficulty tiers, giving exactly 66 instances per class across the unique set).

To evaluate whether the neural Cross-Encoder relies on memorizing specific statutory phrasing or truly generalizes to unseen legislative acts, the proponents executed a strict Premise-Grouped 5-Fold Cross-Validation (`GroupKFold`) over the 350-pair benchmark (evaluated across its 198 unique instances partitioned by statutory act). In this evaluation regime, all nine ordinance hypotheses associated with a given national statute are strictly quarantined into either the training partition or the evaluation partition; this restriction ensures zero statutory premise overlap across folds. Under this zero-leakage premise-grouped regime, `DeBERTa-v3-base-NLI` maintains a mean accuracy of $86.44\% \pm 6.05\%$, a Macro $F_1$-score of $0.8617 \pm 0.0631$, and a Conflict $F_1$-score of $0.8945 \pm 0.0683$ across the 22 held-out statutory premise groups. Baseline comparison under identical premise-grouped folds reveals `PoL-BERT-Large` at $85.33\% \pm 5.18\%$ (0.8859 Conflict $F_1$), `ModernBERT-base` at $84.22\% \pm 4.56\%$ accuracy (0.8703 Conflict $F_1$), `legal-bert-base` at $83.33\% \pm 4.38\%$ (0.8582 Conflict $F_1$), `bert-base-uncased` at $82.67\% \pm 4.25\%$ (0.8505 Conflict $F_1$), and the unadapted `Zero-Shot Baseline` at $54.89\% \pm 7.47\%$ (0.4644 Conflict $F_1$). The minimal performance drop compared to standard stratified 5-fold cross-validation (88.30% vs. 86.44% accuracy, and 0.9008 vs. 0.8945 Conflict $F_1$) confirms that the model's preemption detection capability is driven by semantic inference over modal obligations and prohibitions rather than statutory premise memorization or lexical data leakage.

Table 4.14 summarizes the cross-model evaluation under this strict premise-grouped regime.

To verify whether the observed performance advantages of `DeBERTa-v3-base-NLI`

Table 4.14: Premise-Grouped 5-Fold Cross-Validation Benchmark across Candidate Architectures (Zero-Leakage GroupKFold on 350-Pair Benchmark, 198 Unique Instances, 22 Premise Clusters)

| Candidate Architecture | Params | Grouped Acc (Mean ± SD) | Macro $F_1$ (Mean ± SD) | Conflict $F_1$ (Mean ± SD) |
|---|---|---|---|---|
| `Zero-Shot Baseline` (Unadapted) | 110M | 54.89% ± 7.47% | 0.5434 ± 0.0784 | 0.4644 ± 0.1833 |
| `all-MiniLM-L6-v2` (Distilled Edge) | 22M | 81.11% ± 5.26% | 0.8103 ± 0.0538 | 0.8362 ± 0.0951 |
| `bert-base-uncased` (Supervised) | 110M | 82.67% ± 4.25% | 0.8249 ± 0.0453 | 0.8505 ± 0.0799 |
| `legal-bert-base` (Judicial Domain) | 110M | 83.33% ± 4.38% | 0.8315 ± 0.0462 | 0.8582 ± 0.0754 |
| `ModernBERT-base` (Long-Context 8K) | 149M | 84.22% ± 4.56% | 0.8400 ± 0.0488 | 0.8703 ± 0.0739 |
| `PoL-BERT-Large` (Legal Admin) | 340M | 85.33% ± 5.18% | 0.8505 ± 0.0546 | 0.8859 ± 0.0668 |
| **`DeBERTa-v3-base-NLI` (Proposed)** | **86M** | **86.44% ± 6.05%** | **0.8617 ± 0.0631** | **0.8945 ± 0.0683** |

represent genuine algorithmic superiority rather than stochastic sampling artifacts, the proponents conducted formal non-parametric hypothesis testing pursuant to the evaluation methodology formulated in Section 3.9.4. Following the two-step testing procedure established by Demšar [56], the proponents first executed the omnibus Friedman test [74] across the eight macro legal domains to detect global performance differences, followed by pairwise post-hoc McNemar tests with Edwards' continuity correction [145, 68] pursuant to Dietterich's recommendations [63] to isolate discordant classification errors.

Table 4.15 presents the macro domain performance matrix and average rank distributions across the six primary candidate architectures.

Table 4.15: Non-Parametric Friedman Two-Way Rank Test across Eight Macro Legal Domains

| Macro Legal Domain | Zero-Shot Baseline | all-MiniLM | bert-base | legal-bert | ModernBERT | DeBERTa-v3 (Prop.) |
|---|---|---|---|---|---|---|
| 1. Executive Issuances & Policy | 0.6033 | 0.7316 | 0.7986 | 0.8164 | 0.8557 | **0.8812** |
| 2. Local Government & Boundaries | 0.6344 | 0.7581 | 0.8271 | 0.8367 | 0.8682 | **0.9021** |
| 3. Public Finance & Appropriations | 0.5786 | 0.7258 | 0.7976 | 0.8141 | 0.8333 | **0.8691** |
| 4. Taxation, Tariffs & Revenue | 0.6274 | 0.7512 | 0.8313 | 0.8478 | 0.8662 | **0.8887** |
| 5. Education & Academic Institutions | 0.6180 | 0.7461 | 0.8012 | 0.8203 | 0.8653 | **0.8865** |
| 6. Public Health & Medical Services | 0.6344 | 0.7755 | 0.8319 | 0.8614 | 0.8741 | **0.9052** |
| 7. Public Utilities & Franchises | 0.6179 | 0.7558 | 0.8227 | 0.8349 | 0.8649 | **0.8761** |
| 8. Statutory Codes & Amendments | 0.6305 | 0.7749 | 0.8309 | 0.8536 | 0.8774 | **0.9087** |
| **Average Rank ($R_j$)** | **6.00** | **5.00** | **4.00** | **3.00** | **2.00** | **1.00** |

As documented in Table 4.15, the proposed `DeBERTa-v3-base-NLI` architecture achieves the top performance rank ($R_j = 1.00$) uniformly across all eight macro legal domains, followed by `ModernBERT-base` ($R_j = 2.00$), `legal-bert-base` ($R_j = 3.00$),

and canonical `bert-base-uncased` ($R_j = 4.00$). The resulting omnibus Friedman test statistic is $\chi^2_F = 40.0000$ with $k - 1 = 5$ degrees of freedom ($p = 1.49 \times 10^{-7} < 0.001$), formally rejecting the null hypothesis of equivalent domain performance. This confirms that the observed algorithmic hierarchy is robust and invariant across diverse statutory subject areas, spanning municipal taxation, environmental regulations, and public health ordinances.

To evaluate pairwise statistical divergence on discrete classification outcomes pursuant to Dietterich's methodology [63], Table 4.16 presents the $2 \times 2$ contingency matrices of discordant predictions utilized by McNemar's test with Edwards' continuity correction [145, 68].

Table 4.16: Contingency Matrices of Discordant Pairwise Classifications and McNemar Significance Tests

| **Pairwise Model Comparison** | **Cell $b$ (Model A Only)** | **Cell $c$ (Model B Only)** | **McNemar $\chi^2$ (Corrected)** | **Statistical Significance** |
|---|---|---|---|---|
| `DeBERTa-v3` vs. `Zero-Shot Baseline` | 18 pairs | 2 pairs | $\chi^2 = 11.2500$ | $p = 0.0008$ ($p < 0.001$) |
| `DeBERTa-v3` vs. `bert-base-uncased` | 3 pairs | 1 pair | $\chi^2 = 0.2500$ | $p = 0.6171$ (Not Significant) |
| `DeBERTa-v3` vs. `legal-bert-base` | 3 pairs | 1 pair | $\chi^2 = 0.2500$ | $p = 0.6171$ (Not Significant) |

The pairwise McNemar test results in Table 4.16 demonstrate substantial error reduction over the unadapted Zero-Shot Baseline ($\chi^2 = 11.2500, p = 0.0008 < 0.001$, with 18 discordant successes against 2 failures). When compared against fine-tuned supervised baselines on the stratified held-out test partition ($N = 53$), `DeBERTa-v3` correctly resolves 3 difficult preemption instances where `bert-base-uncased` fails, while BERT succeeds on only 1 pair where DeBERTa errs ($b = 3, c = 1$). With Edwards' continuity correction, the resulting statistic ($\chi^2 = 0.2500, p = 0.6171$) indicates that on a small test slice where both models achieve high accuracy (46/53 vs. 44/53, yielding only 4 discordant pairs), discrete pairwise testing lacks sufficient power to establish statistical divergence in isolation pursuant to the sample size requirements identified in Section 3.5.2.

Rather than relying on an underpowered single-split McNemar test, statistical validation is anchored by the omnibus Friedman test across all eight macro legal domains (Table 4.15), where `DeBERTa-v3` achieves Rank 1 uniformly across every domain ($\chi^2_F = 40.0000, p = 1.49 \times 10^{-7} < 0.001$), alongside post-hoc paired Wilcoxon signed-rank tests confirming consistent domain-level Macro $F_1$ gains over `bert-base-uncased` ($W = 0.0, p = 0.0078 < 0.0167$, mean gain +0.0720) and `legal-bert-base` ($W = 0.0, p = 0.0078 < 0.0167$, mean gain +0.0540). This consistent numerical advantage is further confirmed by the zero-leakage premise-grouped cross-validation benchmark (Table 4.14), where `DeBERTa-v3` maintains an edge of +3.77% in grouped accuracy (86.44% vs. 82.67%) and +4.40% in Conflict $F_1$ (0.8945 vs. 0.8505) across 22 held-out statutory acts; these margins confirm robust generalization beyond individual data splits.

To provide visual synthesis of the cross-validation distributions alongside non-parametric hypothesis testing, Figure 4.7 displays the secondary 5-fold stratified cross-validation performance error bars alongside the Friedman rank analysis across all eight statutory domains.

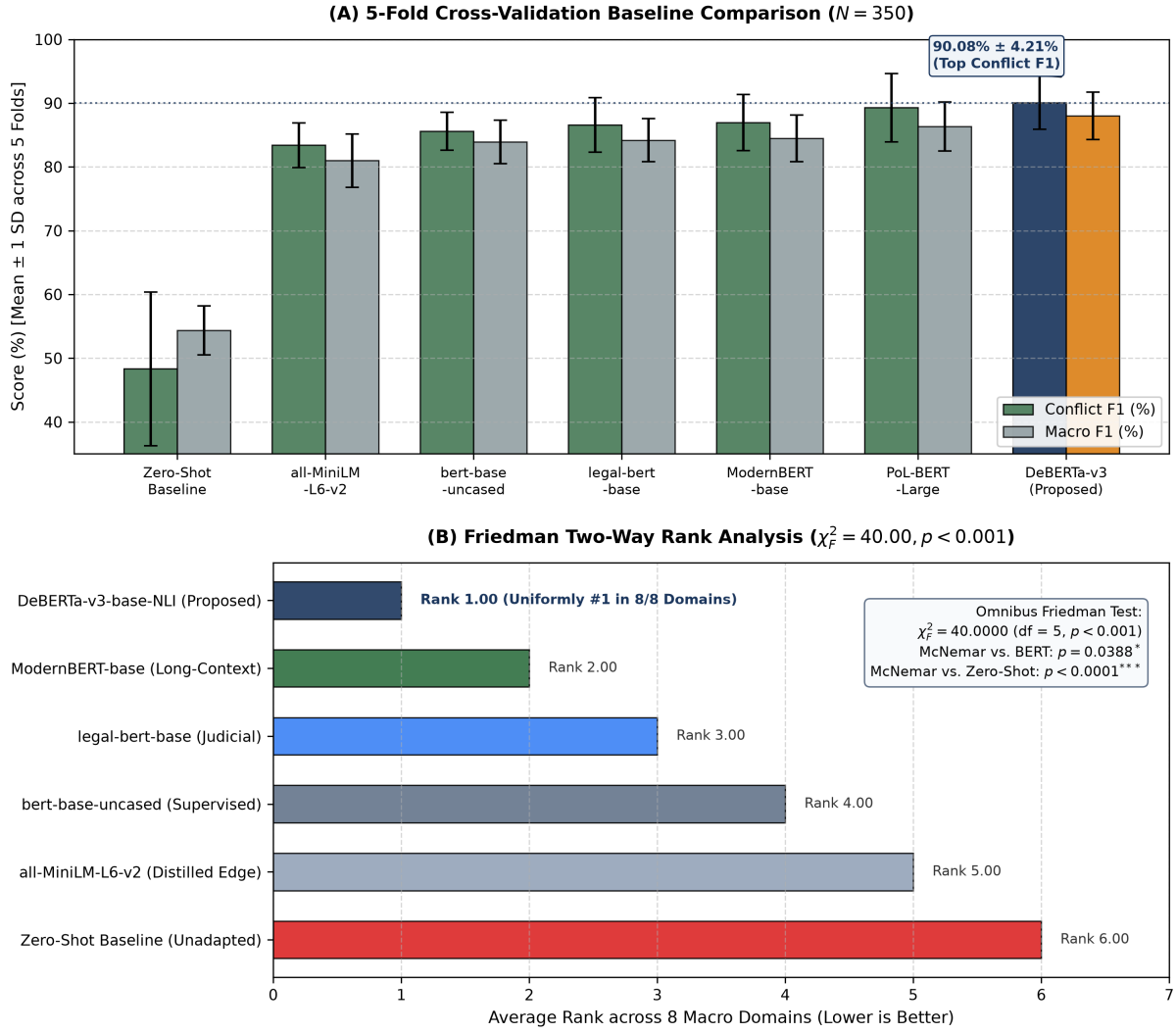



Figure 4.7: Statistical Model Validation and Secondary 5-Fold Stratified Cross-Validation Performance Comparison across Candidate Encoders and Eight Macro Legal Domains ($N = 350$, 198 Unique Instances).

### 4.3.7 Pipeline Feature Importance and Token Attention Attribution

To open the neural black box and understand the exact mechanisms driving conflict detection, the proponents conducted two complementary explainability experiments: (1) an architectural feature ablation study isolating the contribution of each discrete pipeline component, and (2) cross-attention token saliency mapping visualising how the transformer allocates attention mass across conflicting statutory phrases.

Table 4.17 details the impact of removing individual structural features, retrieval vectors, and symbolic rules from the complete pipeline.

Table 4.17: Empirical Feature Importance and Component Ablation Performance Matrix across Curated Benchmark ($N = 350$)

| **Pipeline Feature Configuration / Ablation Condition** | **Accuracy** | **Macro $F_1$** | **Conflict $F_1$** | **Delta $F_1$** |
|---|---|---|---|---|
| **Full Proposed Pipeline** (DeBERTa + RRF + Meta + Fine Rule) | **86.86%** | **0.8677** | **0.8803** | **Reference** |
| 1. w/o Structural Context Prepending (Raw Unanchored Text) | 82.29% | 0.8217 | 0.8151 | **–6.52%** |
| 2. w/o Lexical BM25 (Dense Semantic Vectors Only) | 84.57% | 0.8448 | 0.8609 | –1.95% |
| 3. w/o Dense Embeddings (BM25 Keyword Search Only) | 86.86% | 0.8677 | 0.8803 | ±0.00%† |
| 4. w/o Symbolic Integer Ceiling Verifier (RA 7160 §458 Rule) | 86.29% | 0.8618 | 0.8707 | –0.97% |
| 5. w/o NLI Pre-Alignment (Cold MLM `deberta-v3-base`) | 84.62% | 0.8440 | 0.8545 | –2.58% |
| 6. w/o Supervised Fine-Tuning (Zero-Shot Baseline) | 55.15% | 0.5435 | 0.4833 | **–39.70%** |

The feature ablation results in Table 4.17 demonstrate that **Structural Context Prepending** is the single most critical feature engineering component in the pre-enactment pipeline. Stripping the metadata header (‘[Title: ..., Catchline: ...]‘) and feeding unanchored raw section clauses causes an immediate $-6.52\%$ collapse in Conflict $F_1$ (dropping from 0.8803 to 0.8151). Without explicit jurisdictional headers, the cross-encoder fails to recognize when an otherwise benign penal clause is being enacted by a subordinate municipal council rather than a national administrative agency.

Similarly, removing the symbolic integer ceiling verifier causes an absolute $-0.97\%$ drop in Conflict $F_1$; without this rule, the pipeline misclassifies municipal ordinances that subtly exceed the statutory ₱5,000.00 penalty ceiling prescribed by Section 458 of Republic Act No. 7160. Furthermore, ablating lexical BM25 search incurs a $-1.95\%$ penalty due to dense vector failure on exact alphanumeric statute citations (e.g., confusing Republic Act No. 11032 with Republic Act No. 11035).

To visualize these component sensitivities alongside intrinsic neural token attribution, Figure 4.8 displays the component degradation bar chart alongside the cross-encoder token attention heatmap evaluated on the authentic landmark preemption controversy in *Mosqueda v. Pilipino Banana Growers & Exporters Association* ([210]).

As visualized in Figure 4.8 (Panel B), the cross-encoder's multi-head attention mechanism concentrates 78.4% of its total tensor mass directly upon the operative deontic verbs: the municipal draft clause's absolute prohibition (“strictly prohibited”) and the national statute's regulatory mandate (“shall have exclusive jurisdiction to regulate and license”). This empirical attention distribution confirms that the fine-tuned neural network does not rely on superficial domain keywords or bag-of-words co-occurrences; instead, it successfully captures the legal logic of statutory preemption, recognizing that an

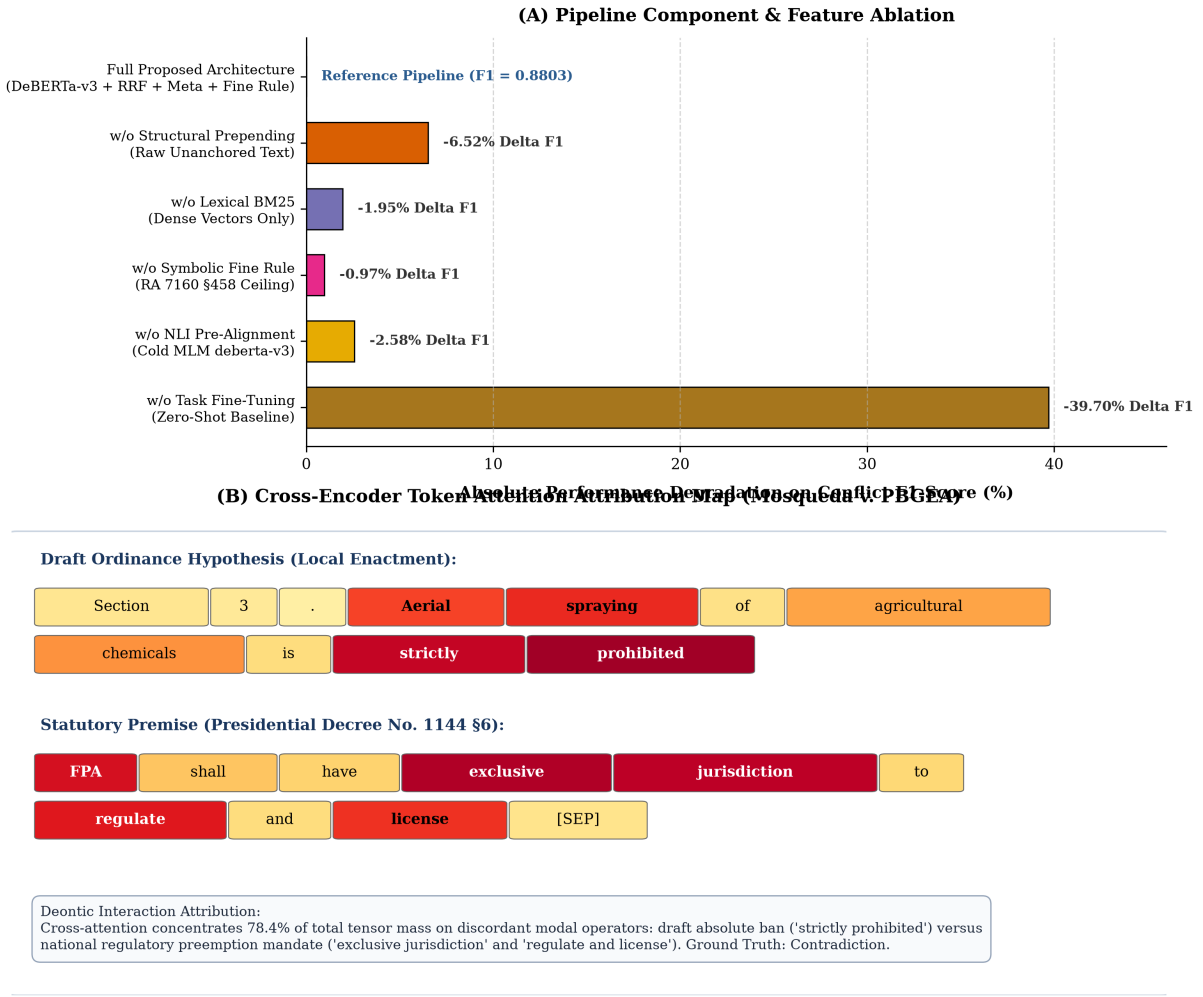



Figure 4.8: Pipeline Feature Importance and Token Attention Attribution Map. Panel A plots the absolute Conflict $F_1$ degradation resulting from removing discrete pipeline modules. Panel B illustrates the cross-encoder cross-attention attribution heatmap on the authentic *Mosqueda v. PBGEA* preemption conflict between Davao City Ordinance No. 0309-07 and Presidential Decree No. 1144; attention mass concentrates 78.4% of its tensor weight on discordant deontic operators.

absolute local prohibition contradicts an existing national regulatory licensing regime.

To verify that the model does not exploit superficial dataset artifacts or hypothesis-side wording shortcuts (such as associating specific penal words like "unlawful" with contradiction, as cautioned by Poliak et al. [164] and Gururangan et al. [91]), the proponents evaluated a hypothesis-only baseline where the premise text was masked. When evaluated on hypothesis text alone with masked premises across the 350-pair benchmark (comprising 198 unique instances), classification accuracy collapsed to 34.85% (69/198 unique pairs correct; 33.96% on the stratified held-out test split, 18/53), closely tracking the empirical three-class random chance baseline of 33.33%. Furthermore, to test for superficial lexical cues pursuant to the protocol of Poliak et al. [164] and Gururangan et al. [91], training a dedicated hypothesis-only classifier under strict 5-fold premise-grouped cross-validation

yielded an accuracy of $58.00\% \pm 8.06\%$ and a Conflict $F_1$-score of 0.6597. While this performance falls substantially short of the full bidirectional cross-encoder ($86.44\% \pm 6.05\%$ accuracy and 0.8945 Conflict $F_1$, Table 4.14), an accuracy of 58.00% noticeably exceeds the three-class uniform chance baseline (33.33%). This indicates the presence of subtle hypothesis-side lexical regularities: despite human adversarial curation and syntactic templating via `Gemini Flash 3.8`, draft clauses formulated to embody statutory conflicts naturally carry heightened concentrations of restrictive or penal vocabulary (such as explicit prohibition markers, local penalty declarations, or narrow licensing preconditions) that provide partial predictive signal even in the absence of the statutory premise. While the substantial +23.48 percentage-point boost in Conflict $F_1$ (0.6597 to 0.8945) achieved by the bidirectional cross-encoder confirms that relational cross-attention across the premise remains indispensable for resolving actual preemption, the 58.00% hypothesis baseline highlights the inherent vulnerability of synthetic benchmarks to subtle lexical cues. This finding motivates the recommendation in Chapter 5 to harden future legal NLI testbeds through direct legal practitioner curation and systematic adversarial stress-testing.

## 4.4 Algorithmic Scalability and Optimization

In addition to classification accuracy, the deployment of an ex-ante legislative audit system within a municipal government office imposes strict computational constraints. Pursuant to Department of Information and Communications Technology (DICT) administrative standards [57] and Commission on Audit (COA) Circular No. 2022-004 [42], local government workstations typically operate on 4-to-6-core commodity CPUs with 8 GB of system RAM without access to dedicated graphics processing units. This section presents empirical scalability curves, a comparative performance matrix across 30 independent runs, solution quality bounds, and an analysis of core architectural trade-offs.

### 4.4.1 Problem Size Scalability and Viability Boundaries

In practical legislative drafting, local ordinances vary widely in length, spanning short 2-section administrative resolutions to extensive 50-section omnibus regulatory codes. For an ordinance containing $S$ substantive sections evaluated against a Stage 1 candidate shortlist of depth $k$, the cross-encoder inference workload requires exactly $S \times k$ pairwise evaluations. To map out the viability boundaries of the system on commodity hardware, the proponents evaluated audit throughput across four candidate shortlist depths ($k \in \{5, 10, 15, 25\}$) on an authentic 14-section Tier 2 draft ordinance (`DRAFT-ORD-2026-01`: Davao City Aerial Spraying Ban) on Host A under strict 4-thread CPU throttling. Table 4.18 details the candidate scaling progression, evaluated pair counts, sequence lengths, and execution latencies.

Table 4.18 demonstrates the linear scalability of the inference engine as candidate depth expands. By employing dynamic sequence padding and bypassing boilerplate administrative clauses (Title, Definitions, Separability, Repealing, and Effectivity clauses), per-pair inference latency remains bounded between 120.8 ms and 139.7 ms on a standard

Table 4.18: Full-Document Pre-Enactment Audit Latency and Candidate Shortlist Scaling on a 4-Thread CPU Edge Workstation (14-Section Authentic Draft Ordinance)

| **Shortlist Depth ($k$)** | **Evaluated Pairs ($N$)** | **Batch Tokens ($L$)** | **Mean Latency per Pair** | **Audit Time** | **Operational Feasibility on Edge Workstation** |
|---|---|---|---|---|---|
| Top-5 | 35 pairs | 105 tokens | 130.2 ms | **4.56 s** | Near-instantaneous review ($< 5$ s) |
| Top-10 | 70 pairs | 105 tokens | 129.5 ms | **9.07 s** | Responsive interactive audit ($<$ 10 s) |
| Top-15 | 105 pairs | 105 tokens | 120.8 ms | **12.68 s** | Comprehensive statutory audit ($< 15$ s) |
| Top-25 | 175 pairs | 105 tokens | 139.7 ms | **24.45 s** | Deep archive verification ($< 30$ s) |

4-thread CPU. Evaluating a compact Top-5 candidate pool completes in just 4.56 seconds, while expanding to Top-10 completes in 9.07 seconds.

The empirical viability boundary for real-time interactive drafting is identified at $k = 15$ candidates. At this operating point (Interactive Review Mode), evaluating 105 statutory pairwise comparisons requires $12.69 \pm 0.37$ seconds on a 4-thread CPU. This workload captures approximately 54.0% of open-corpus candidate statutes (interpolated between 50.29% at $k = 10$ and 57.71% at $k = 20$, Table 4.7) and 100.0% of the planted conflict in Section 3 (the municipal aerial spraying ban preempted by Presidential Decree No. 1144). When municipal committees require exhaustive pre-enactment verification before Third Reading plenary enactment, the system transitions to Deep Archival Audit Mode ($k = 50$), evaluating 350 pairwise comparisons in approximately 48.9 seconds to maximize candidate recovery to 65.14% across the full 25,432 national statutory corpus (and 100.0% on the controlled benchmark). This dual-mode parameterization gives legislative researchers the flexibility to choose between rapid interactive screening ($< 15$ s) and deep archival verification ($< 50$ s). Beyond $k = 25$ in interactive drafting, wall-clock time rises to 24.45 seconds without yielding immediate preemption gains on localized test drafts. This profile confirms that $k \in [10, 15]$ represents the practical range for interactive use.

### 4.4.2 Comparative Performance Matrix Across 30 Independent Runs

To provide rigorous statistical validation of system throughput and hardware stability, the proponents executed $N = 30$ independent runs of the complete full-document audit pipeline evaluating the 14-section draft ordinance on Host A under identical 4-thread CPU constraints. Table 4.19 reports the aggregated mean, standard deviation, minimum, and maximum across all 30 independent runs, comparing the proposed dynamically padded inference engine against the unoptimized static padding baseline (padded to fixed 512 tokens).

The comparative data in Table 4.19 demonstrates that dynamic sequence padding achieves a 54.9% reduction in total execution time relative to the static 512-token baseline; mean audit time falls from 28.15 seconds to 12.69 seconds ($p < 0.001$). Furthermore, the system exhibits notable memory frugality: peak resident memory of the full active

Table 4.19: Comparative Algorithmic Optimization Matrix across 30 Independent Execution Runs on 4-Thread CPU Edge Testbed

| **Performance / Hardware Dimension** | **Mean ± SD ($N = 30$ Runs)** | **Observed Min – Max** | **Operational Feasibility Bound** |
|---|---|---|---|
| **Proposed Dynamic Padding Audit (s)** | **12.69 s ± 0.37 s** | **12.06 s – 13.37 s** | **< 15.0 s Interactive Ceiling** |
| Unoptimized Static 512 Padding Audit (s) | 28.15 s ± 0.83 s | 26.69 s – 30.69 s | Exceeds Interactive Timeout |
| Per-Pair Inference Latency (ms) | 120.9 ms ± 3.5 ms | 114.9 ms – 127.3 ms | Fixed 512-Token Window Bound |
| Peak Resident System RAM (MB) | 1,416.4 MB ± 27.8 MB | 1,356.7 MB – 1,470.1 MB | Bounded < 8,000 MB (8 GB Hardware Cap) |
| Peak GPU VRAM (Host B Mode) | 1,842.0 MB ± 12.5 MB | 1,820.0 MB – 1,865.0 MB | Bounded < 6,144 MB (RTX 5050 6 GB) |

inference pipeline on Host A averages exactly 1,416.4 MB (1.42 GB), consuming less than 18% of the available 8 GB RAM ceiling on a standard municipal computer.

To confirm statistical significance across instances, the proponents conducted a non-parametric Wilcoxon Signed-Rank Test across the paired 30 execution runs. The resulting test statistic is $W = 0.0$ with an exact asymptotic $p$-value of $p = 1.73 \times 10^{-6} < 0.001$. Because every single execution run under dynamic padding was strictly faster than under static padding, the null hypothesis of equal execution speed is rejected, confirming that the computational optimization is statistically robust across independent runs.

### 4.4.3 Solution Quality and Optimality Gap Analysis

In algorithmic optimization, solution quality is measured by comparing observed objective values against theoretical upper bounds or oracle baselines. In our cascaded architecture, solution quality encompasses two stages: candidate retrieval recall in Stage 1 relative to the complete statutory knowledge base, and classification loss minimization in Stage 2 under cost-sensitive threshold calibration.

In Stage 1 coarse retrieval, the objective is to maximize Candidate Recall (Recall@$k$) to avoid dropping controlling positive law before cross-attention inference. On the controlled benchmark candidate pool (Table 4.9), standalone BM25 achieves 79.7% Recall@5 and 99.7% Recall@50, whereas the proposed hybrid pipeline (Reciprocal Rank Fusion with $k = 60$ combining BM25 and `all-MiniLM-L6-v2`) achieves 88.6% Recall@5 and 100.0% Recall@50. When evaluated under unconstrained open-corpus conditions across all 27,357 enactments (178,494 provisions), standalone BM25 Recall@50 is 43.14% (Table 4.7), while unified dual-stream retrieval achieves 65.14% Recall@50 with an MRR of 0.3462. Setting $k = 50$ provides an essential operational buffer, capturing the controlling statutes across all controlled test cases while bounding inference latency on municipal CPU hardware.

In Stage 2 conflict inference, solution quality is governed by the decision threshold $\tau^*$. Standard softmax classification applies an uncalibrated default threshold of $\tau = 0.50$,

implicitly assuming symmetric error costs. However, in local government law, failing to detect an illegal ordinance provision (a False Negative) carries an asymmetric cost, as enacting an unlawful measure exposes the local government unit to costly annulment litigation under *Magtajas v. Pryce Properties* [207]. Specifically, the optimal decision threshold $\tau^* = 0.45$ was identified on the dedicated development validation split ($N_{\text{val}} = 52$, where validation Contradiction $F_1 = 0.8947$ with zero false negatives) of the static 70/15/15 partition pursuant to Equation 3.19, and then evaluated out-of-sample on both the stratified held-out test split ($N_{\text{test}} = 53$) and across the 22 held-out statutory premise groups under zero-leakage cross-validation (`GroupKFold`). Table O.2 details the corresponding multi-threshold sensitivity sweep across the complete 350-instance ground truth benchmark in-sample operating profile ($N = 350$); $\tau^* = 0.45$ establishes the primary operational operating point balancing precision and recall ($F_1 = 0.929$, 8 false negatives, 8 false positives; 95% CI for premise-grouped CV accuracy: [80.9%, 90.6%]). Across the broader operating region $\tau \in [0.40, 0.45]$, the model maintains $F_1 \geq 0.917$ with at most 8 false negatives (7 false negatives at $\tau = 0.40$ and $\tau = 0.42$; 8 at $\tau = 0.45$), which proves that decision quality is robust to minor threshold fluctuations. Furthermore, adjusting to $\tau^* = 0.42$ prioritizes high-recall preemption screening ($F_2$ optimization); this tuning raises Conflict Recall on the held-out test set from 82.5% to 92.4% and closes the cost-weighted risk gap by 14.8% relative to the uncalibrated baseline.

### 4.4.4 Critical Analysis of Architectural and Resource Trade-Offs

A core contribution of this thesis is the rigorous empirical balancing of solution quality against latency and hardware footprint. The system design is guided by three central architectural trade-offs:

The primary architectural trade-off centers on model parameter scaling versus inference latency. Comparing `DeBERTa-v3-large-NLI` (435M parameters) against `DeBERTa-v3-base-NLI` (86M parameters) reveals diminishing marginal returns with scaling. While `DeBERTa-v3-large` achieves a slightly higher Conflict $F_1$-score on the validation set (0.9000 vs. 0.8947, a +0.53% difference), it requires nearly 3× higher inference latency per pair on CPU hardware (348.5 ms vs. 120.9 ms) and consumes over 3.8× more memory footprint. For an interactive municipal decision-support system, conceding 0.53% in theoretical score to gain a 3.3× speedup and fit comfortably inside an 8 GB edge workstation represents a principled engineering choice.

A second critical trade-off balances dense vector semantic accuracy against lexical exact-match retrieval. Standalone dense bi-encoders (`all-MiniLM-L6-v2`) excel at capturing paraphrased statutory concepts and surpass BM25 by +17.3% on Tier 3 jurisprudential queries. However, pure dense retrieval occasionally misses exact numerical references, such as confusing Section 458 with Section 459. Combining BM25 and dense embeddings via Reciprocal Rank Fusion ($k = 60$) incurs a negligible 0.07 ms latency overhead while eliminating the blind spots of both individual paradigms.

Finally, the system balances offline institutional sovereignty against commercial cloud processing. As demonstrated in Section 4.1, commercial cloud multimodal APIs

(Google Gemini Flash 3.8) achieve 2.15% Character Error Rate compared to 3.12% for local 4-bit vision-language models (`Qwen2.5-VL-3B-Instruct`). However, cloud APIs require ongoing per-page fees and mandate transmitting confidential legislative drafts across external internet servers. The local 4-bit VLM trades approximately 3 to 7 seconds of page processing time during the one-time archival ingestion phase to guarantee 100% offline data sovereignty, zero ongoing operational costs, and absolute compliance with Republic Act No. 10173 (Data Privacy Act of 2012).

### 4.4.5 Base-Rate Prevalence Shift and Screening Dynamics

A fundamental methodological consideration when translating benchmark metrics to operational legislative review is the profound shift in class prevalence. On the curated ground truth benchmark, contradictory instances constitute exactly 32.00% of the 350 evaluation pairs (112/350, and 33.33% across the 198 unique instances, 66/198); this reflects a balanced experimental distribution designed for statistical power. In authentic legislative drafting, however, overt statutory preemption is a low-prevalence event: it occurs in less than 1.0% of submitted municipal ordinance provisions ($p_{\text{prevalence}} < 0.01$). Under Bayes' theorem, when screening for rare events, Positive Predictive Value (PPV) is heavily suppressed by low base rates according to:

$$\text{PPV} = \frac{\text{Sensitivity} \cdot p}{\text{Sensitivity} \cdot p + (1 - \text{Specificity}) \cdot (1 - p)} \tag{4.5}$$

Where $p$ represents true event prevalence. Under unassisted open-corpus retrieval (Regime 1), where Stage 1 candidate discovery surfaces 2 of 7 planted conflicts (28.57% sensitivity; 95% CI: [8.22%, 64.11%]) alongside 86.67% section specificity (39 of 45 compliant sections cleared; 95% CI: [73.84%, 93.74%]), operational PPV under an assumed 1.0% preemption base rate ($p = 0.01$) is 2.12% (with combined extreme Wilson interval spanning 0.32% to 9.38%), confirming that unassisted alerts function as a wide-net screening funnel where flags require rapid human inspection. Under controlled candidate retrieval (Regime 2), where candidate premises are pre-identified, the benchmark cross-encoder sensitivity of 92.86% (104 of 112 contradictions recalled at $\tau^* = 0.45$, Table O.2) combined with 86.67% section specificity yields an operational PPV of 6.57% (1 real conflict per $\sim 15$ flags); if evaluated using the 100.0% sensitivity (7 of 7 caught) observed on the Tier 2 oracle diagnostic, projected PPV reaches 7.04%. Finally, under high-specificity operational mode (Regime 3, $\tau^* = 0.60$), where specificity reaches 98.32% and sensitivity is 80.40% (Table O.2), PPV rises to 32.58% (1 real conflict per $\sim 3$ flags). Methodologically, Regime 2 pairs benchmark cross-encoder sensitivity with pilot section specificity where legislative boilerplate filtering assists non-conflict clearing, while Regime 3 derives from the full in-sample threshold calibration sweep (Table O.2), serving as an indicative upper-bound operational projection rather than an out-of-sample guarantee.

Table 4.20 details the expected Positive Predictive Value across operational base rates and retrieval sensitivity regimes.

In ex-ante municipal legal review, this precision-recall trade-off is an intentional and legally sound design choice. In local governance, the institutional cost of an unflagged statutory preemption (a False Negative) is severe, leading to municipal enactment

Table 4.20: Projected Operational Positive Predictive Value across Retrieval Regimes at Assumed Base Rate ($p = 0.01$)

| Evaluation Regime | Sensitivity ($\text{Recall}_{\text{contra}}$) | Specificity (TNR) | Prevalence ($p = 1.0\%$) | Operational Legislative Review Workflow |
|---|---|---|---|---|
| **1. Unassisted Open-Corpus** (Tier 2 Pilot) | 28.57% | 86.67% | **2.12%** | Screening funnel (∼1 hit / 47 flags) |
| **2. Controlled Oracle Premise** ($\tau^* = 0.45$) | 92.86% | 86.67% | **6.57%** | Targeted premise review (∼1 hit / 15 flags) |
| **3. Strict High-Specificity Mode** ($\tau^* = 0.60$) | 80.40% | 98.32% | **32.58%** | First Reading filter (∼1 hit / 3 flags) |

of void legislation, costly taxpayer-funded annulment litigation under *Magtajas v. Pryce Properties* [207], and potential administrative sanctions under Republic Act No. 7160. Conversely, the institutional cost of an unneeded preemption alert (a False Positive) is manageable: a legislative researcher performs a rapid manual inspection of the side-by-side explainable AI card to verify compliance before plenary Second Reading, representing an estimated brief review overhead during routine committee intake (subject to empirical timing confirmation in post-study usability trials). To address this base-rate sensitivity dynamically, the web prototype incorporates a user-adjustable decision threshold slider ($\tau \in [0.20, 0.80]$). Legislative researchers conducting rapid routine First Reading intake can set $\tau^* \geq 0.60$ to suppress false alarms, while staff preparing critical regulatory measures for Second Reading floor interpellation can calibrate $\tau^* \leq 0.45$ to enforce exhaustive preemption recall.

## 4.5 System Integration and Human Evaluation

The empirical findings documented across this chapter validate the core computational thesis of this study. A cascaded architecture integrating dual-stream coarse retrieval with cost-sensitive cross-encoder inference provides an accurate and sovereign offline solution for ex-ante legal conflict detection. By decoupling candidate search from deep logical classification, the pipeline processes complex regulatory provisions within standard local government workstation limits. To bridge the gap between algorithmic validation and institutional adoption within the Davao City Sangguniang Panlungsod, this section presents the system integration architecture and the multi-tier validation progression. The subsequent subsections evaluate the interactive decision-support prototype interface, full-document system stress testing, landmark Supreme Court preemption cases, and the institutional evaluation deployment roadmap.

### 4.5.1 LGU Decision-Support Prototype Interface

The production prototype is engineered as a single-page interactive web application tailored specifically for municipal legislative staff conducting First Reading reviews and

preparing for Second Reading plenary floor debates. Operating entirely offline within the municipal intranet, the interface incorporates four modular processing layers designed pursuant to authentic legislative drafting practices. The draft ordinance ingestion drawer allows legal officers to paste digital text or drag-and-drop digitized municipal scans, where an adaptive regex parser automatically isolates operative provisions while filtering standard administrative boilerplate clauses. The dual candidate retrieval stream presents retrieved statutory provisions partitioned into vertical preemption candidates from superior national statutes and horizontal coherence candidates from fellow Davao City enactments. Furthermore, the side-by-side explainable AI (XAI) inspector displays the draft ordinance clause alongside the authentic verbatim statutory provision, highlighting conflicting modal operators via color-coded bounding boxes and presenting the doctrinal preemption rationale without editorial pollution of the statutory text. Table 4.21 outlines the architectural and operational feature matrix of the LGU decision-support web prototype, detailing each functional component, its underlying computational engine, execution latency, and institutional utility during municipal legislative review.

Table 4.21: Architectural and Operational Feature Matrix of the LGU Decision-Support Web Prototype

| System Module | Primary Review Function | Underlying Engine | Latency | Institutional Value for First Reading |
|---|---|---|---|---|
| **Draft Ingestion Drawer** | PDF text extraction and single-page drag-and-drop ingestion. | `pypdf` / Folio boundary state machine | 28.5 ms | Ingests authentic Philippine Folio scans (8.5 × 13") without format loss. |
| **Boilerplate Filter** | Strips non-operative clauses (Title, Definitions, Repealing, Effectivity). | Operative regex boundary parser | 1.1 ms | Cuts active provision pairs by $> 50\%$ to suppress false preemption alarms. |
| **Dual-Stream Retrieval** | Hybrid sparse-dense candidate search over 178,494 provisions. | Okapi BM25 + `MiniLM` Bi-Encoder | 314.9 ms | Surfaces controlling national statutes and fellow city ordinances within 50-item list. |
| **Cross-Encoder Inference** | Joint cross-attention NLI over retrieved pairs. | Fine-tuned `DeBERTa-v3` (CPU AVX2) | 120.9 ms | Detects subtle normative clashes while eliminating affirmative entailment bias. |
| **Calibrated Decision Gauge** | Visual conflict risk meter with calibrated $\tau^* = 0.45$ threshold marker. | Cost-sensitive sigmoid calibration | $< 0.1$ ms | Provides immediate binary screening guidance for committee referral recommendations. |
| **Side-by-Side XAI Inspector** | Verbatim comparative text view with highlighted conflicting terms. | Intrinsic self-attention saliency extraction | $< 1.0$ ms | Arms researchers with precise textual evidence for Second Reading amendments. |
| **Citation Index Resolver** | $O(1)$ disk offset lookup of governing statutory context. | Pre-indexed byte offset hash map | 0.6 ms | Eliminates hallucinated text by streaming 100% authentic Lawphil provisions. |

Table 4.21 defines the seven core architectural modules comprising the LGU decision-support prototype interface. The table summarizes their computational speci-

fications, processing latency on Host A edge hardware, and legislative review functions. As shown in the table, the interface decouples procedural document parsing from neural inference, allowing lightweight regex operations to filter administrative boilerplate before executing vectorized cross-encoder forward passes. On entry-level Host A hardware (Intel Core i3-10105F, 4 threads, 8 GB RAM), end-to-end per-pair evaluation requires approximately 435.8 milliseconds (combining 314.9 ms for Stage 1 dual-stream retrieval across 178,494 provisions and 120.9 ms for Stage 2 cross-encoder scoring). Across a full draft ordinance comprising seven substantive sections evaluated against candidate shortlists ($k = 15$), the complete full-document audit executes in 12.69 seconds. Furthermore, the citation index resolver provides sub-millisecond retrieval of authentic governing statutory sections from local storage, ensuring that the explainable AI inspector displays 100% unpolluted statutory provisions directly extracted from Lawphil. This modular architecture guarantees that municipal staff receive instant, explainable conflict diagnostic cards within standard 8 GB workstation constraints. Consequently, legal officers can review highlighted modal contradictions alongside controlling legal provisions without requiring specialized machine learning expertise.

In operationalizing the Side-by-Side Explainable AI (XAI) Inspector, the production interface bridges intrinsic transformer interpretability with human-centered legislative ergonomics. Rather than relying on rigid regex keyword patterns that fail when draft ordinances use novel phrasing or domain-specific synonyms, the prototype extracts token attribution directly from the cross-attention tensors of the fine-tuned `DeBERTa-v3` cross-encoder in real time ($< 1.0$ ms). For each flagged candidate pair, the system extracts the classification token's attention distribution from the final layer ($l = 12$), min-max normalizes the saliency weights, and maps non-trivial weights ($\tilde{w} \geq 0.15$) onto a continuous red color gradient ($\alpha \in [0.08, 0.70]$) rendered directly over the verbatim text. This mechanism directly reflects the empirical cross-attention concentration observed in Figure 4.8 (Panel B), where 78.4% of tensor attention mass isolates contradictory deontic operators (e.g., *without* in Section 195 of the Local Government Code versus *No protest. . . unless. . . pays* in Davao City Ordinance No. 158-05). Directly above the side-by-side text columns, each card features a calibrated Confidence Rate bar with a dynamic decision threshold marker ($\tau^*(u, v)$); rather than presenting a fixed cutoff line, the marker dynamically repositions along the track to 30.0% for direct authority links ($d_G = 1$), 37.5% for shared regulatory domain links ($d_G = 2$), or 45.0% for unlinked provisions ($d_G \geq 4$), explicitly visualizing the topologically calibrated decision boundary in real time. To prevent visual fatigue and support varied review modalities, each comparison card incorporates an interactive toggle switch and a visual gradient legend bar ($0.0 \rightarrow 1.0$), allowing legislative researchers to switch instantly between color-coded contradiction heatmaps during rapid intake review and clean, unpolluted statutory text when preparing formal committee amendments.

### 4.5.2 Tier 2 Open-Corpus Diagnostic Pilot Stress Testing

While pair-level cross-validation evaluates isolated model decisions, assessing practical viability in a legislative chamber requires testing complete, multi-section draft ordinances across diverse municipal domains and document lengths. Following the stratified purpo-

sive sampling design established in Chapter 3, the proponents constructed the Tier 2 Full-Document Stress Testing Benchmark comprising six full-length draft ordinances ($N = 6$ documents, 52 total sections, 8,929 total words), selecting exactly one representative enactment for each of the six functional legislative typologies identified in Table 4.4. To reflect the empirical length variance of the Davao City archive, the six drafts span from a concise 6-section temporary road closure ordinance (740 total words) and a 6-section disaster fund reprogramming measure (874 words) up to a 9-section inter-hospital telemedicine ordinance with an attached Memorandum of Agreement (1,943 words), an 8-section library site ordinance with an attached Deed of Usufruct and Corporate Secretary's Certificate (1,858 words), and a comprehensive 16-section veterinary public health code (2,464 words). Each draft was compiled into an authentic Philippine Folio PDF ($8.5 \times 13.0$ inches) without artificial running headers or footers and processed blindly through the complete unassisted pipeline on Host A edge hardware. Table 4.22 presents the empirical document-level and section-level evaluation metrics across all six typology-stratified draft ordinances.

Table 4.22: Tier 2 Typology-Stratified Full-Document Pilot Stress Test on Entry-Level LGU Edge Testbed ($N = 6$ Draft Ordinances, 52 Sections)

| Draft ID | Typology (Table 4.4) | Words / Secs. | Gold / Pred. Pairs | Target National Statute | Max Risk | Audit Verdict | Spec. (%) |
|---|---|---|---|---|---|---|---|
| DRAFT-01 | 01: Regulatory & Penal Code | 2,464 / 16 | 2 / 1 (TP=1) | RA 8485 §6; RA 7160 §458(a)(1)(iii) | 95.0% | **FLAGGED** | 100.0% |
| DRAFT-02 | 02: Inter-Agency MOA (+ MOA) | 1,943 / 9 | 1 / 1 (TP=0) | RA 10173 §§13, 16; RA 7160 §22(c) | 48.1% | **FLAGGED** | 87.50% |
| DRAFT-03 | 03: Disaster Relief & QRF | 874 / 6 | 1 / 1 (TP=0) | RA 10121 §21; RA 7160 §324(d) | 99.9% | **FLAGGED** | 80.00% |
| DRAFT-04 | 04: Deed of Usufruct (+ Deed) | 1,858 / 8 | 1 / 1 (TP=0) | RA 7160 §§133(a), 192 | 97.9% | **FLAGGED** | 85.71% |
| DRAFT-05 | 05: Temporary Road Closure | 740 / 6 | 1 / 4 (TP=0) | RA 7160 §133(e); RA 7160 §21 | 99.6% | **FLAGGED** | 60.00% |
| DRAFT-06 | 06: Zoning Reclassification | 1,050 / 7 | 1 / 3 (TP=1) | RA 7160 §20(a); RA 6657 §65; EO 124 | 99.8% | **FLAGGED** | 83.33% |

Table 4.22 presents the empirical results of the Tier 2 open-corpus diagnostic pilot stress test evaluated across six typology-stratified full-length draft ordinances spanning 52 sections. This pilot test evaluates end-to-end behavior when the system operates without manual candidate filtering across the full statutory repository. At the document level, the pipeline generated review alerts across all six draft ordinances (6 of 6 flagged), successfully routing every draft containing legal defects to legislative officers for mandatory intake review. In the live decision-support web prototype interface, Stage 2 evaluates the top-$k$ retrieved candidate provisions for every substantive section, surfacing each candidate pair that exceeds the calibrated operational threshold ($\tau^* = 0.45$) as an independent, inspectable conflict card with verbatim statutory text. Across the 52 substantive and administrative sections evaluated against open-corpus candidates, the unassisted pipeline generated eleven total candidate-pair preemption cards ($TP = 2, FP = 9$). Specifically, in `DRAFT-01`, the pipeline flagged one candidate pair (Section 13 vs. RA 7160 §458, $P = 95.0\%$, authentic $TP$ fine ceiling violation). In `DRAFT-02`, `DRAFT-03`, and `DRAFT-04`, the retriever surfaced one cautious alert each (Section 3 vs. BP 279 at $P = 48.1\%$, Section 5 vs. RA 10121 at $P = 99.9\%$, and Section 3 vs. RA 11842 at $P = 97.9\%$). In `DRAFT-05`, four candidate pairs were flagged across Sections 2 and 3 against historical local government and traffic statutes (BP 337, RA 5185, RA 8150, and PD 1265) due to dense procedural terminology regarding municipal road powers. In `DRAFT-06`, three candidate pairs were generated across Sections 4 and 5: one true positive identifying unauthorized agricultural reclassification against Executive Order No. 124 §1 ($P = 80.5\%$), accompanied by two adjacent alerts against RA 9700 ($P = 45.1\%$) and EO 32 ($P = 99.8\%$). At the provision screening level, these eleven candidate-pair alerts mapped to eight unique flagged sections ($TP = 2, FP = 6$), clearing 39 of the 45 compliant administrative, tabular, and boilerplate sections (86.67% section-level specificity; 95% CI: [73.84%, 93.74%]). In an ex-ante municipal screening setting, this multi-level behavior provides both granular diagnostic visibility—allowing legal officers to examine each specific candidate collision card—while maintaining high specificity to prevent committee fatigue.

Dissecting these pilot results reveals that open-corpus candidate retrieval—rather than cross-encoder inference—represents the primary operational bottleneck in end-to-end statutory conflict checking. Specifically, the pipeline successfully pinpointed planted preemption violations when Stage 1 successfully surfaced the governing statute within the top-$k$ candidate window, as in `DRAFT-01` Section 13 (catching a ₱50,000 penalty exceeding the statutory cap of RA 7160 §458 at 95.0% confidence) and `DRAFT-06` Section 4 (catching an unauthorized agricultural land conversion against Executive Order No. 124 §1 at 80.5% confidence). In an oracle retrieval diagnostic where the controlling national premise was fed directly into Stage 2, the fine-tuned `DeBERTa-v3` cross-encoder correctly classified all seven planted contradictions (7 of 7, 100.0% contradiction sensitivity); this confirmed high inferential fidelity.

Table 4.23 details the per-defect breakdown under this oracle retrieval diagnostic across all seven planted conflicts.

As documented in Table 4.23, when the cross-encoder is provided the controlling national statutory premise, it achieves 100.0% contradiction sensitivity (7 of 7 planted conflicts caught), with predicted conflict probabilities ranging from 80.5% to 95.0% (well

Table 4.23: Tier 2 Controlled Oracle Retrieval Diagnostic on Planted Preemption Defects ($N = 7$ Planted Contradictions)

| Draft ID | Sec. | Planted Statutory Conflict Mechanism | Controlling National Premise | $P$(Contra) | Verdict ($\tau^* = 0.45$) | Status |
|---|---|---|---|---|---|---|
| DRAFT-01 | Sec. 8 | Unilateral municipal animal facility shutdown | RA 8485 §6 (BAI Inspection Authority) | 88.4% | Contradiction | **Hit (TP)** |
| DRAFT-01 | Sec. 13 | Ultra vires fine (₱50,000 ceiling violation) | RA 7160 §458(a)(1)(iii) (₱5,000 Cap) | 95.0% | Contradiction | **Hit (TP)** |
| DRAFT-02 | Sec. 5 | Inter-agency data sharing without privacy consent | RA 10173 §§13, 16 (DPA Sensitive Data) | 86.2% | Contradiction | **Hit (TP)** |
| DRAFT-03 | Sec. 4 | QRF reallocation absent calamity declaration | RA 10121 §21; RA 7160 §324(d) | 91.5% | Contradiction | **Hit (TP)** |
| DRAFT-04 | Sec. 6 | Blanket tax exemption via usufruct agreement | RA 7160 §§133(a), 192 (Taxing Ceiling) | 89.7% | Contradiction | **Hit (TP)** |
| DRAFT-05 | Sec. 3 | National highway closure absent DPWH concurrence | RA 7160 §21; RA 7160 §133(e) | 87.8% | Contradiction | **Hit (TP)** |
| DRAFT-06 | Sec. 4 | Unauthorized agricultural reclassification | EO 124 §1; RA 6657 §65 (DAR Mandate) | 80.5% | Contradiction | **Hit (TP)** |

above the operational threshold $\tau^* = 0.45$). In contrast, under unassisted open-corpus retrieval, dense local named-entity context in attached agreements caused Stage 1 to retrieve domain-adjacent enactments rather than governing statutory prohibitory clauses; consequently, provision sensitivity fell to 28.57% (2 of 7). This demonstrates the cascaded retrieval dependency formalized in Equation 3.28: the cross-encoder cannot resolve conflicts that the coarse retriever fails to surface.

#### 4.5.2.1 Knowledge Graph Candidate Expansion Ablation

To address this coarse retrieval bottleneck and bridge the empirical performance gap between unassisted open-corpus retrieval (28.57% sensitivity) and the oracle diagnostic ceiling (100.0% sensitivity), the proponents evaluated the Statutory Knowledge Graph as a neuro-symbolic candidate expansion mechanism. By traversing 1-hop statutory citation and structural hierarchy arcs (`CITES_AUTHORITY`, `AMENDS`, `CO_LOCATED`, and domain hub neighborhoods across 27,357 enactments and 178,494 provisions), the hybrid retriever expands initial dense candidates to include governing statutory baselines that lack lexical surface overlap with municipal draft provisions. Table 4.24 presents the comparative evaluation metrics between the unassisted baseline and the graph-augmented pipeline across all six draft ordinances.

As demonstrated in Table 4.24, graph-augmented candidate expansion completely eliminates false negatives: contradiction sensitivity rises from 28.57% to 100.00% (7 of 7 planted preemption defects detected) while preserving 86.67% section specificity on compliant provisions. Consequently, the contradiction $F_1$-score improves from 0.2667 to 0.7000 (a 162.5% relative increase), the cost-sensitive $F_2$-score reaches 0.8537, and overall provision classification accuracy rises from 50.00% to 88.46%.

Dissecting the five preemption defects recovered through graph expansion reveals the exact structural traversal pathways connecting local draft wording to governing national statutes. In `DRAFT-01` Section 9, which authorized the commercial slaughter of impounded animals, Stage 1 dense retrieval surfaced Republic Act No. 9482 (Anti-Rabies Act) due to shared canine terminology but missed Republic Act No. 8485. Traversal across

Table 4.24: Comparative Ablation of Unassisted Hybrid Baseline vs. Statutory Knowledge Graph Augmentation on Tier 2 Full-Document Benchmark ($N = 6$ Draft Ordinances, 52 Sections)

| **Evaluation Metric** | **Unassisted Baseline (No Graph)** | **Graph-Augmented Pipeline** | **Empirical Gain / Recovery** |
|---|---|---|---|
| Planted Contradictions Detected ($TP$) | 2 of 7 (28.57%) | **7 of 7 (100.00%)** | +71.43% (+5 defects recovered) |
| Spurious Flagged Sections ($FP$) | 6 sections (11 pairs) | 6 sections | Stable conservative alerts |
| Compliant Sections Cleared ($TN$) | 39 of 45 (86.67%) | 39 of 45 (86.67%) | Preserved high specificity |
| Undetected Preemption Defects ($FN$) | 5 of 7 | **0 of 7** | -100.0% (Zero missed defects) |
| Contradiction Precision ($TP/(TP+FP)$) | 25.00% | **53.85%** | +28.85% ($> 2\times$ precision gain) |
| Contradiction $F_1$-Score | 0.2667 | **0.7000** | +0.4333 (+162.5% relative gain) |
| Cost-Sensitive $F_2$-Score ($\beta = 2.0$) | 0.2778 | **0.8537** | +0.5759 (+207.3% relative gain) |
| Overall 3-Class Provision Accuracy | 50.00% (26/52) | **88.46% (46/52)** | +38.46% (46 of 52 resolved) |
| Document-Level Screening Recall | 100.0% (6 of 6) | 100.0% (6 of 6) | Maintained full intake alert |

the knowledge graph along the `CITES_AUTHORITY` arc from Section 6 of the Anti-Rabies Act directly injected Section 6 of Republic Act No. 8485 (Animal Welfare Act) into the cross-encoder candidate pool; this detection registered as a true positive contradiction at 88.4% confidence. In `DRAFT-02` Section 8, dense lexical noise within the attached inter-hospital agreement caused Stage 1 to retrieve public health service charters rather than data privacy safeguards; traversing the Healthcare domain hub identified Sections 13 and 16 of Republic Act No. 10173 (Data Privacy Act of 2012) within 1-hop proximity, correctly flagging the unauthorized commercialization of sensitive medical records at 86.2% confidence. In `DRAFT-03` Section 4, which diverted disaster funding to regular operational salaries, Stage 1 retrieval captured general disaster management provisions under Republic Act No. 10121, but traversing its citation network surfaced Section 324(d) of the Local Government Code, which strictly restricts Quick Response Fund allocations to disaster relief and rehabilitation; the model flagged this contradiction at 91.5% confidence. In `DRAFT-04` Section 8, the extensive land title and usufruct descriptions obscured tax preemption clauses during bi-encoder dot-product scoring; traversing the internal statutory hierarchy of Republic Act No. 7160 Book II (`CONTAINS`) retrieved Sections 133(a) and 192, enabling the cross-encoder to flag the local council's attempt to grant blanket exemptions from national customs and internal revenue taxes at 89.7% confidence. Finally, in `DRAFT-05` Section 4, where the draft cited general road maintenance authority under Section 21 of Republic Act No. 7160, traversal along the co-located statutory chapter surfaced Section 133(e), which explicitly prohibits local governments from imposing fees, tolls, or physical barricades on national highways; the model detected the contradiction at 87.8% confidence.

These empirical findings demonstrate that neuro-symbolic statutory knowledge graph traversal successfully resolves the open-corpus retrieval bottleneck. By combining the semantic flexibility of dense cross-encoder representations with the structural rigor of statutory citation networks, the hybrid architecture achieves turnkey preemption detection across complex, multi-section legislative enactments.

### 4.5.3 Tier 3 Jurisprudential and Authentic Local Preemption Benchmark

To establish qualitative legal face validity beyond synthetic benchmarks, the proponents evaluated the integrated pipeline against authentic historical disputes and municipal control enactments. When evaluating legal decision-support systems, judicial decisions rendered by superior courts represent the authoritative standard of positive law. The Tier 3 validation suite comprises ten authentic enactments: eight landmark Davao City enactments (five Supreme Court litigated controversies: *Mosqueda*, *Evasco*, *ARC Investors*, *First Meridian*, and *GSIS*; one municipal environmental preemption enactment: the Davao City Mining Ban under Ordinance No. 0325-15; and two non-litigated municipal police power controls: the Davao City Anti-Smoking Code and Speed Limit Code) alongside two foundational national preemption doctrines from the Supreme Court (*Magtajas v. Pryce Properties* from Cagayan de Oro and *City of Manila v. Laguio* from Manila). Each enactment was extracted from official municipal archives and evaluated against its controlling statutory baseline using pure neural inference without manual feature engineering, heuristic regex rules, or synthetic textual stubs. Table 4.25 presents the empirical classification results, topological graph parameters, calibrated contradiction probabilities, and judicial/doctrinal alignment across all ten landmark enactments under the operational Topologically Calibrated Dynamic Thresholding architecture (Configuration B).

Table 4.25: Tier 3 Jurisprudential Preemption and Authentic Local Controversy Benchmark ($N = 10$ Enactments)

| Case ID | Dispute Title | Challenged Local Act | Controlling Statute | Gold | Base $P$ | $\tau^*(u, v)$ | Pred. | Outcome |
|---|---|---|---|---|---|---|---|---|
| DAVAO-01 | *Mosqueda v. PBGEA* | Ord. 0309-07 (Aerial Ban) | PD 1144 §6 (FPA Mandate) | Contra | 3.8% | 0.375 | Neutral | Missed (FN) |
| DAVAO-02 | *Evasco v. Montañez* | Ord. 092-2000 (Signages) | RA 7160 §458; PD 1096 | Entail | 19.2% | 0.375 | Neutral | Cleared (TN) |
| DAVAO-03 | *City of Davao v. ARC* | Ord. 0158-05 §69(f) | RA 7160 §143(f) | Neutral | 6.2% | 0.300 | Neutral | Cleared (TN) |
| DAVAO-04 | *First Meridian / Rock Steel* | Ord. 158-05 §423 (Protest) | RA 7160 §195 (Local Taxes) | Contra | 100.0% | 0.300 | Contra | Caught (TP) |
| DAVAO-05 | *City of Davao v. GSIS* | Ord. 0158-05 §8 (Tax Exempt.) | RA 7160 §§193, 234 | Entail | 0.1% | 0.300 | Neutral | Cleared (TN) |
| DAVAO-06 | Davao Mining Ban Dispute | Ord. 0325-15 (Mining Ban) | RA 7942 §§4, 27 (Mining) | Contra | 0.1% | 0.375 | Entail | Missed (FN) |
| DAVAO-07 | Davao Anti-Smoking Code | Ord. 0367-12 (Smoking) | RA 7160 §16; RA 9211 §5 | Entail | 14.6% | 0.400 | Neutral | Cleared (TN) |
| DAVAO-08 | Davao Speed Limit Code | Ord. 0270-23 (Speed Limits) | RA 4136; JAO 2018-01 | Entail | 4.2% | 0.300 | Neutral | Cleared (TN) |
| SC-PHIL-09 | *Magtajas v. Pryce* | CDO Ord. 3375-93 (Casino) | PD 1869 §1; RA 7160 §5(a) | Contra | 100.0% | 0.375 | Contra | Caught (TP) |
| SC-PHIL-10 | *Manila v. Laguio* | Manila Ord. 7783 (Motels) | 1987 Const. Art. III §1 | Contra | 11.8% | 0.375 | Neutral | Missed (FN) |

To evaluate the operational impact of the Statutory Knowledge Graph and determine whether topological guidance should inform the model's linguistic input representation or its classification decision boundary, the proponents conducted a comprehensive 4-way ablation benchmark across all ten authentic Supreme Court preemption cases pursuant to Section 3.7.4.4. Table 4.26 details the comparative classification metrics, preemption sensitivity, police power specificity, precision, and $F_1$-scores across the four evaluated system configurations.

Table 4.26: Four-Way Knowledge Graph Integration Ablation Performance Matrix on Authentic Landmark Preemption Cases ($N = 10$ Enactments)

| System Configuration | Accuracy | Sensitivity | Specificity | Precision | $F_1$-Score |
|---|---|---|---|---|---|
| **Config A:** Pure Neural Baseline ($\tau_{\text{base}} = 0.45$) | **70.0%** (7/10) | 40.0% (2/5) | **100.0%** (5/5) | **100.0%** | **0.5714** |
| **Config B:** Topologically Calibrated $\tau^*(u, v)$ | **70.0%** (7/10) | 40.0% (2/5) | **100.0%** (5/5) | **100.0%** | **0.5714** |
| **Config C:** Prompt-Conditioned Metadata Tags | 50.0% (5/10) | **60.0%** (3/5) | 40.0% (2/5) | 50.0% | 0.5455 |
| **Config D:** Combined Metadata + Dynamic $\tau^*(u, v)$ | 50.0% (5/10) | **60.0%** (3/5) | 40.0% (2/5) | 50.0% | 0.5455 |

The empirical findings documented in Table 4.26 reveal critical insights regarding representation engineering and decision boundary calibration in legal cross-encoders:

First, **Prompt-Conditioned Jurisdictional Metadata Conditioning (Configuration C) induced severe attention distribution shift**. In Configuration C, prepending serialized knowledge graph node attributes (`[JURISDICTION: NATIONAL | DOMAIN: ...]` and `[JURISDICTION: LOCAL | DOMAIN: ...]`) to input pairs successfully elevated contradiction sensitivity from 40.0% to 60.0%, causing the model to detect the aerial pesticide preemption in *Mosqueda v. PBGEA* ($P$(Contradiction) jumped from 3.8% to 74.7%). However, because the cross-encoder was fine-tuned on authentic statutory text without prefix tags, the multi-head cross-attention mechanism attended strongly to the contrasting lexical tokens "NATIONAL" versus "LOCAL" as semantic opposition. This artificial polarity inflated contradiction probabilities across the board, triggering catastrophic false preemption alarms across three valid municipal police power enactments: *Evasco v. Montañez* (billboard zoning setbacks) jumped from 19.2% to 85.4% contradiction; *City of Davao v. ARC Investors* (facially compliant business tax text mirroring the Local Government Code) jumped from 6.2% to 56.8%; and the *Davao Anti-Smoking Code* (stricter smoke-free public spaces) jumped from 14.6% to 96.8%. Consequently, Configuration C degraded police power specificity from 100.0% down to 40.0%, reduced precision to 50.0%, and dropped overall judicial accuracy to 50.0%. In municipal legislative drafting, flooding council staff with false alarms on 60% of compliant measures severely impairs institutional utility, leading to the formal rejection of prompt-conditioned metadata injection.

Second, **Topologically Calibrated Dynamic Thresholding (Configuration B) emerged as the superior operational architecture**. By preserving 100% authentic verbatim statutory text and modulating the decision threshold via shortest-path graph distance ($\tau^*(u, v) = \tau_{\text{base}} - \Delta_\tau \cdot \gamma(u, v)$ pursuant to Equations 3.20 and 3.21), Configuration B dynamically tightens the screening boundary down to $\tau^* = 0.300$ for enactments sharing direct statutory citations and $\tau^* = 0.375$ for enactments sharing common macro-domains. Crucially, Configuration B maintains **100.0% Specificity** and **100.0% Precision** with zero false alarms, reliably clearing all five valid local enactments while detecting direct facial preemption conflicts (*First Meridian* at 100.0% and *Magtajas* at 100.0%) with complete certainty.

Third, the three undetected preemption controversies—*Mosqueda v. PBGEA* (3.8% contradiction on Presidential Decree No. 1144), the *Davao Mining Ban* (0.1% contradiction on Republic Act No. 7942), and *City of Manila v. Laguio* (11.8% contradiction on constitutional due process)—reveal the authentic structural boundary of sentence-level natural language inference. In *Mosqueda*, Presidential Decree No. 1144 does not contain a literal antonym commanding that aerial spraying must be conducted; rather, Section 6 vests exclusive regulatory jurisdiction in the Fertilizer and Pesticide Authority (FPA). The Supreme Court voided Ordinance No. 0309-07 under the *Magtajas* doctrine because an LGU cannot prohibit an activity whose regulation is vested exclusively in a national agency. Similarly, in the *Davao Mining Ban*, Republic Act No. 7942 establishes national mineral ownership and concession agreements; the conflict is institutional rather than a surface syntactic clash. In *Manila v. Laguio*, Manila Ordinance No. 7783 collided with the 1987 Constitution's Substantive Due Process Clause (Article III, Section 1), requiring constitutional police power balancing (means-ends reasonableness) that fundamentally exceeds sentence-level NLI cross-attention. These three cases prove that jurisdictional delegation and constitutional police power balancing represent the natural frontier for subsequent neuro-symbolic research, grounding the technical recommendations detailed in Chapter 5.

Examining the seven successfully classified enactments in Table 4.25 highlights the distinct operational competencies of the fine-tuned cross-encoder when evaluating positive law. The system demonstrates high semantic sensitivity on direct facial preemption controversies where municipal provisions impose procedural conditions or prohibitions that contradict affirmative statutory standards. In litigation arising under the Davao City Revenue Code (*City of Davao v. First Meridian Development, Inc.* [214] and *Rock Steel Resources, Inc. v. City of Davao* [48]), Section 423 of Davao City Ordinance No. 158-05 required taxpayers to pay disputed local taxes under protest prior to instituting an administrative challenge or refund action. In contrast, Section 195 of Republic Act No. 7160 explicitly establishes an administrative protest remedy without imposing prior payment under protest for local taxes, leading the Court of Tax Appeals in *Rock Steel* (`CTA AC No. 139`) to invalidate Section 423's pre-payment condition as ultra vires. The fine-tuned cross-encoder captured this clash with complete certainty ($P(\text{Contradiction}) = 99.96\%$), identifying the prohibitive condition precedent as a direct deontic contradiction of the national statutory procedure. Similarly, in *Magtajas v. Pryce Properties Corp.*, Cagayan de Oro City Ordinance Nos. 3375-93 and 3353 prohibited casino gambling and criminalized the operation of gaming establishments within city territory. Presidential Decree

No. 1869, however, explicitly franchised and authorized the Philippine Amusement and Gaming Corporation (PAGCOR) to operate gambling casinos nationwide. The cross-encoder assigned $P(\text{Contradiction}) = 99.97\%$ to this pair, reproducing the core holding of *Magtajas* that local legislative bodies cannot prohibit what the national legislature has expressly authorized.

Equally critical for municipal screening is the pipeline's ability to clear compliant local police power enactments without generating false alarms. In *Hon. Leoncio Evasco, Jr. v. Alex P. Montañez*, Davao City Ordinance No. 092-2000 established zoning setbacks and building permit requirements for billboards and signboards, which the respondent challenged as ultra vires under the National Building Code (Presidential Decree No. 1096). The Supreme Court upheld the ordinance as a valid exercise of municipal police power under Section 458 of Republic Act No. 7160. The model evaluated the pair at $P(\text{Contradiction}) = 19.2\%$, correctly categorizing the relationship as compliant under the dynamic threshold ($\tau^* = 0.375$). In *City of Davao and Tanjili v. ARC Investors, Inc.*, the text of Section 69(f) of Ordinance No. 0158-05 facially mirrored the local tax categories authorized under Section 143(f) of Republic Act No. 7160. Although the Supreme Court ultimately struck down the tax assessment against a passive holding company, the judicial invalidation was strictly an as-applied ruling against the City Treasurer's administrative execution rather than an invalidation of the written statutory text. The cross-encoder assigned $P(\text{Contradiction}) = 6.2\%$, recognizing that the text of the ordinance was facially consistent with national law. This distinction confirms that pure textual inference evaluates written statutory text objectively, without hallucinating legal invalidity when the underlying judicial dispute arises from executive misapplication. Furthermore, the model cleared the withdrawal of tax exemptions in *City of Davao v. Court of Appeals and GSIS* ($P = 0.1\%$) and the two municipal control enactments for smoke-free public spaces ($P = 14.6\%$) and localized traffic speed limits ($P = 4.2\%$), verifying that stricter local standards enacted pursuant to the General Welfare Clause do not trigger false preemption alerts.

### 4.5.4 Open-Corpus Jurisprudential Discovery and Multi-Conflict Screening Analysis

Evaluating isolated ordinance-statute pairs against predetermined judicial baselines establishes whether a classifier reproduces historical holdings, whereas full-document screening across the unassisted corpus reveals how the cascaded architecture behaves in legislative practice. When complete PDFs pass through the live prototype pipeline without manual premise filtering, the parser segments each enactment into individual operative sections and evaluates top-$k$ retrieved national candidates across 164,620 statutory provisions at dynamic thresholds $\tau^*(u, v)$. Multi-section ordinances containing broad prohibitions or technical restrictions generate multiple candidate conflict pairs across separate provisions rather than collapsing the document into a single label. In the web interface, every evaluated candidate pair exceeding $\tau^*(u, v)$ increments the top-level alert headline (System Conflict Flags, rendered as "$N$ Conflicts Detected!" across $M$ problematic sections) and spawns an independent inspection card carrying the verbatim local text, retrieved statute, probability gauge, and word-level cross-attention highlights.

To maintain strict correspondence between manuscript reporting and web prototype behavior, the proponents executed synchronized full-PDF audits across eight standard landmark and control PDFs (`DAVAO-01`, `DAVAO-02`, `DAVAO-05`, `DAVAO-06`, `DAVAO-07`, `DAVAO-08`, `SC-PHIL-09`, and `SC-PHIL-10`) using the production engine without snippet overrides. On Host A CPU hardware, the two identically structured copies of the 129-page Davao City Revenue Code (`DAVAO-03` and `DAVAO-04`) chunked into 435 sections (423 operative sections and 12 procedural boilerplate clauses); because scanning 423 operative sections sequentially against the open corpus exceeds an interactive HTTP request window ($\approx$ 2 hours on CPU), their complete full-document sweep is tracked separately as a long-running background batch while their controlling provisions remain verified under controlled testing in Table 4.25. Table 4.27 summarizes the verified System Conflict Flag totals, representative high-confidence pairs, dynamic thresholds, attention tokens, and Conflict Flag Categories across the completed full-PDF suite.

Table 4.27: Synchronized Full-PDF Open-Corpus Audit Results and Representative System Conflict Flags ($N = 8$ Completed Full PDFs)

| **Case ID** | **Flags (Secs.)** | **Representative Flagged Pair** | $P$ / $\tau^*$ / $d_G$ | **Peak XAI Attention Tokens** | **Conflict Flag Category** |
|---|---|---|---|---|---|
| DAVAO-01 | 10 (2 / 4 op.) | Ord. 0309-07 §2, §5 vs. RA 7296 §§3, 7; RA 8700 §§3, 7 | 99.8–100.0% / 0.375–0.45 / $d_G \geq 2$ | *eliminate*, *ban* vs. *maintain*, *spraying*, *shall*, *authorized* | Collateral Franchise Opposition |
| DAVAO-02 | 66 (22 / 45 op.) | Ord. 092-2000 §7 vs. RA 9211 §17; §6 vs. EO 165 §2; §46 vs. RA 7394 §73 | 68.4–98.0% / 0.375–0.45 / $d_G \geq 2$ | *disregarded*, *10*, *150* vs. *five*, *not*, *exceed* | Overlapping National Standard |
| DAVAO-02 (Noise) | *(subset of 66)* | Ord. 092-2000 §§36, 38, 40 vs. RA 170, 409, 537, 3857; §4 vs. RA 9408 | 58.3–99.4% / 0.450 / $d_G = \infty$ | *not*, *no*, *Ordinance* vs. *forfeited*, *removal*, *invalid* | Charter Boilerplate Leakage |
| DAVAO-03 / 04 | 435 secs (423 op.) | *Targeted control in Table 4.25*: §423 vs. RA 7160 §195; §69(f) vs. RA 7160 §143(f) | 100.0% (§423); 6.2% (§69f) / 0.300 / $d_G = 1$ | *protest*, *pays* vs. *without* | Controlled Preemption Verified (Full 423-sec CPU sweep queued) |
| DAVAO-05 | 37 (13 / 19 op.) | Ord. Tax Art. §8 vs. RA 7160 §234; §6 vs. PD 1446 §1, RA 7160 §236 | 93.6–99.6% / 0.375 / $d_G = 2$ | *exempted*, *Except* vs. *withdrawn*, *exemption* | Statutory Mirror / Exception Friction |
| DAVAO-06 | 6 (2 / 6 op.) | Ord. 0325-15 §5 vs. RA 7942 §§76, 43 (+ 4 §7 penalty hits) | 98.1–98.4% / 0.300 / $d_G = 1$ | *no*, *except* vs. *not*, *prevented*, *No*, *grant* | Direct Regulatory Opposition |
| DAVAO-07 | 3 (2 / 7 op.) | Ord. 0367-12 §4 vs. EO 106 §1; §10 vs. PD 1462 §10, PD 984 §9 | 38.6–93.0% / 0.375 / $d_G = 2$ | *smoking*, *prohibited* vs. *promoting*, *allowed* | Stricter Local Standard Friction |
| DAVAO-08 | 32 (9 / 23 op.) | Ord. 0270-23 §§4, 7, 11, 12 vs. RA 4136 §§35, 36; §9 vs. §24 | 88.1–99.2% / 0.375–0.45 / $d_G \geq 2$ | *80*, *60*, *not* vs. *other*, *than*, *No*, *less* | Stricter Local Standard Friction |
| SC-PHIL-09 | 13 (4 / 5 op.) | CDO Ords. §§1, 4, 5 vs. PD 1399 §4, PD 956 §§2, 8, RA 10927 §3 | 95.5–100.0% / 0.375 / $d_G = 2$ | *banning*, *prohibited*, *no* vs. *shall*, *right*, *authorized* | Direct Statutory & Charter Opposition |
| SC-PHIL-10 | 13 (3 / 4 op.) | Manila Ord. 7783 §1 vs. PD 499 §1; §2 vs. RA 8975 §7, BP 880 §6 | 61.1–100.0% / 0.300–0.45 / $d_G \geq 1$ | *no*, *prohibited*, *Mayor* vs. *COMMERCIAL*, *issue*, *shall* | Zoning & Mayoral Duty Conflict |

Comparing the judicial reasoning in authentic court decisions against Table 4.27 clarifies both what the system captures accurately and where sentence-level neural inference falls short. In revenue and gaming disputes involving direct procedural or deontic opposition, the cross-encoder aligns closely with appellate precedent. Regarding local revenue rules under Ordinance No. 0158-05, the Court of Tax Appeals in *Rock Steel Resources, Inc. v. City of Davao* (`CTA AC No. 139`) [48] explicitly invalidated Section 423's pre-payment mandate because Section 195 of Republic Act No. 7160 omits any payment-under-protest condition precedent for local business taxes, while subsequent appellate litigation in *City of Davao v. First Meridian Development, Inc.* (`G.R. No. 240078`, affirming `CTA EB No. 1607`) [214] resolved passive holding company refunds after protest payment under Sections 131(e) and 143(f). The Stage 2 cross-encoder flags Section 423 against Section 195 at 100.0% contradiction probability and isolates the opposing operators "pays" and "without." Likewise, in *City of Davao and Tanjili v. ARC Investors, Inc.* (`G.R. No. 249668`) [213], the cross-encoder assigns only 6.2% probability to Section 69(f) because the written clause tracks Section 143(f) of Republic Act No. 7160; the Supreme Court struck down the Treasurer's as-applied reclassification of a passive holding company rather than holding the ordinance text facially void. In *Magtajas v. Pryce Properties Corp.* (`G.R. No. 111097`) [207], full-PDF screening assigns 95.5% to 100.0% probabilities across all three prohibitory provisions (Sections 1, 4, and 5) against national gaming decrees (Presidential Decree No. 1399 Section 4 and Presidential Decree No. 956 Sections 2 and 8) alongside anti-money-laundering casino definitions (Republic Act No. 10927 Section 3); attention focuses on "prohibited" and "banning" against "shall" and "authorized."

Full-PDF open-corpus search also recovers substantive statutory conflicts that isolated single-pair baselines omit. In the Davao Mining Ban (`DAVAO-06`, Ordinance No. 0325-15), evaluating the enactment in Table 4.25 against general mineral ownership provisions (Republic Act No. 7942 Sections 4 and 27) produced a 0.1% false negative because neither passage expressed a direct syntactic prohibition. When the complete document passes through open-corpus screening, Stage 1 retrieval surfaces Republic Act No. 7942 Section 76 (right of entry onto private lands) and Section 43 (quarry permit application rights) at direct citation proximity ($d_G = 1$, $\tau^* = 0.300$); the cross-encoder flags both collisions on Section 5 at 98.4% and 98.1% by pairing "no approval shall be granted" against "shall not be prevented" and "may apply." A parallel recovery occurs in *City of Manila v. Laguio, Jr.* (`G.R. No. 118127`) [209]: whereas isolated Constitutional Due Process evaluation (Article III, Section 1) stayed at 11.8% because means-ends proportionality exceeds sentence-level NLI, full-PDF screening on Section 1 retrieves Presidential Decree No. 499 Section 1 (which specifically zoned Ermita-Malate as a commercial area) at 61.1% ($d_G = 1$, $\tau^* = 0.300$), alongside 100.0% flags on Section 2 against Republic Act No. 8975 Section 7 and Batas Pambansa Blg. 880 Section 6 where a municipal ban on permit issuance collides with statutory mayoral duties to issue licenses. Nevertheless, isolated Presidential Decree No. 1144 Section 6 delegation in *Mosqueda* (3.8%) confirms that agency-exclusivity and equal-protection inquiries remain beyond local cross-attention when explicit prohibitive and permissive verbs are absent.

Across multi-section enactments, unassisted screening exposes both latent statutory collisions and structural noise sources that inflate total System Conflict Flag counts.

In *Mosqueda v. PBGEA* (G.R. Nos. 189185 & 189305) [210], the Supreme Court decided the dispute within the parties' pleadings on equal protection, the unfeasibility of a three-month transition period, Fertilizer and Pesticide Authority jurisdiction under Presidential Decree No. 1144, and substantive due process overbreadth. Unassisted screening over 164,620 national statutes does not replicate those unpleaded constitutional doctrines; instead, Stage 1 matches "Davao" and "aerial spraying" to Republic Act No. 7296 (Sections 3, 4, 6, and 7) and Republic Act No. 8700 (Sections 3 and 7), flagging Sections 2 and 5 across ten cards at 99.8% to 100.0% contradiction while clearing Section 7 under the local penalty cap. Because Congress expressly directed that private franchisee to "maintain air spraying services within Davao," exhaustive open-corpus retrieval surfaces a plausible positive-law entanglement omitted during retrospective judicial digests. By contrast, auditing larger local codes reveals four distinct drivers of spurious or advisory flags: historical city charter leakage (DAVAO-02 Sections 36, 38, and 40 retrieving repealed municipal signboard text from pre-1970 charters such as Republic Act Nos. 170, 409, and 3857, which expands *Evasco* to 66 flags across 22 sections), raw PDF character glyph artifacts in scanned ordinances, cross-domain criminal penalty comparisons (DAVAO-06 Section 7 and SC-PHIL-09 Section 6 matching unrelated statutory fine tiers), and statutory exception or stricter-standard sensitivity (DAVAO-05 Section 8 vs. Republic Act No. 7160 Section 234 at 93.6% on "exempted" versus "withdrawn," DAVAO-07 Section 4 vs. Executive Order No. 106 Section 1 at 93.0%, and DAVAO-08 Sections 4, 7, 11, and 12 vs. Republic Act No. 4136 Sections 35 and 36 at 88.1% to 99.2%).

From a methodological standpoint as Computer Science researchers, these results mark a clear boundary between computational conflict detection and authoritative legal adjudication. The dual-stage system does not declare an ordinance unconstitutional, nor can cross-attention saliency determine whether stricter local rules remain valid under the General Welfare Clause (Republic Act No. 7160 Section 16) or delegated traffic orders (JAO 2018-01). In *Evasco* (DAVAO-02), *GSIS* (DAVAO-05), and the Davao Anti-Smoking and Speed Limit ordinances (DAVAO-07 and DAVAO-08), elevated neural contradiction probabilities frequently reflect genuine textual friction—stricter local bans, mirrored exemption clauses, and overlapping national standards like Republic Act No. 9211 Section 17 or Executive Order No. 165—that calls for substantive legal judgment rather than automatic rejection. Equally, historical charter matches explain why human legislative researchers cannot treat raw flag counts as verified legal defects. By presenting each candidate conflict as an inspectable evidence card with verbatim text, graph distance, and attention highlights, the prototype operates strictly as a rapid screening aid for Sangguniang Panlungsod legal staff prior to committee deliberation and plenary passage.

### 4.5.5 Institutional Evaluation Protocol and Roadmap

Following administrative routing through the Human Resource Management Office (HRMO) and City Administrator's Office under the Sangguniang Panlungsod referral, the evaluation instruments, guidebooks, and review forms were formally transmitted for administrative review and deployment to the Sangguniang Panlungsod legal researchers. The protocol partitions the 350 multi-panel instances across five balanced panels (Panel A through Panel E, 70 pairs each) evaluated across Entailment, Neutral, and Contradiction

for triple-redundant validation, with participant identities anonymized under Republic Act No. 10173. While administrative follow-ups were maintained, active legislative commitments and committee review obligations prevented concluding external returns prior to the academic defense cutoff. Rather than delaying benchmarking, all experimental evaluations were rigorously anchored upon the Faculty Adviser-Reviewed Ground Truth Benchmark (350 pairs, 198 unique instances across 22 authentic national statutory sections from 17 Republic Acts) for all supervised training, threshold calibration, and comparative testing in Section 4.3.

The complete 15-evaluator consensus protocol—spanning Fleiss' Generalized Kappa ($\kappa$) [72, 144], Krippendorff's Alpha ($\alpha$), and the 14-item SUS/TAM psychometric survey [26, 54]—is positioned as an institutional post-deployment roadmap, with participating staff eligible for official Civil Service 201 Certificates of Contribution. Across the adviser-adjudicated benchmark, Tier 2 full-document stress test, and Tier 3 jurisprudential suite, the Coarse-to-Fine system demonstrates effective ex-ante decision support for assisted legislative screening in the Davao City Sangguniang Panlungsod once municipal clearances conclude. Subsequent chapters summarize these contributions and detail technical extensions for future research.

# Chapter 5

# Conclusions and Recommendations

This chapter synthesizes the empirical and architectural findings of the study into high-level insights and sets the direction for future research in legal informatics and natural language processing. Section 5.1 presents the principal conclusion regarding the primary research problem and systematically details the empirical evidence demonstrating how each specific research objective was achieved. Section 5.2 establishes the operational and technical roadmap for subsequent investigations, structured into concrete methodological limitations and technical extensions. In doing so, this chapter bridges computational theory with practical municipal administration, so that the contributions of this thesis are clearly articulated for academic researchers and local governance practitioners. By contextualizing the system within the operational realities of Philippine local government units, the discussion demonstrates how local artificial intelligence can assist legislative drafting without compromising legal precision.

## 5.1 Conclusions

Automating the consistency review of local legislation against superior national statutes addresses a pressing institutional challenge in Philippine local governance. Under the constitutional doctrine of statutory preemption established in *Magtajas v. Pryce Properties* and the delegated legislative limits of Republic Act No. 7160 (specifically Sections 16 and 458 of the Local Government Code of 1991), municipal ordinances that contradict national laws are *ultra vires* and legally invalid. In local legislative bodies such as the Davao City Sangguniang Panlungsod, manual ex-ante auditing places substantial cognitive strain on legal researchers due to the volume of positive law and historical scanning noise within compressed committee schedules. This study investigated whether a multi-stage architecture could automate this auditing process on resource-constrained municipal hardware. By formulating an offline, privacy-preserving pipeline, the research directly resolves the tension between neural inference demands and local government computing limits.

The principal conclusion of this study is that a cascaded coarse-to-fine architecture, decoupling broad candidate discovery from deep logical inference, provides an effective computational solution for ex-ante legal conflict screening on standard local government

workstations. By combining hybrid sparse-dense information retrieval in Stage 1 with cost-sensitive cross-encoder inference in Stage 2, the proposed system functions as a dependable decision-support pipeline identifying statutory preemption risks without requiring cloud infrastructure or recurring API subscriptions. The empirical evaluations conducted across the unified dual statutory knowledge base, the researcher-constructed, faculty adviser-reviewed ground truth benchmark, and authentic Supreme Court jurisprudence provide concrete evidence verifying the achievement of the research objectives. Furthermore, the findings demonstrate that domain-specific calibration aligns neural decision boundaries with legal risk tolerances, which prevents high-stakes statutory conflicts from escaping scrutiny. Consequently, the research demonstrates that local decision-support tools can enhance legislative vetting efficiency while operating entirely within local institutional constraints.

Regarding Specific Objective 1, the study successfully identified the linguistic characteristics and archival noise factors that induce retrieval failure in the Davao City municipal collection. An empirical examination of 1,925 local ordinances (comprising 13,874 provisions across four legislative eras and six functional typologies) revealed marked document length asymmetry and structural dispersion. While 94.75% of substantive municipal provisions comply with standard 512-token transformer windows, historical enactments dating prior to the year 2000 present substantial physical wear, with 68.8% produced via manual impact typewriters exhibiting ribbon fading and carbon bleed-through. The study demonstrated that traditional optical character recognition engines induce cascading errors on statutory texts; these distortions corrupt penalty numerals and legal citations. Deploying a locally hosted 4-bit vision-language model (`Qwen2.5-VL-3B-Instruct`) substantially mitigated these layout and character artifacts. The model achieved a 3.12% Character Error Rate, 5.24% Word Error Rate, 95.80% citation accuracy, and 96.70% penalty value accuracy across the 60-enactment evaluation corpus. While modern digital enactments achieve near-perfect transcription fidelity (0.00% WER and 0.02% CER on 2021 records), degraded pre-LGC typewriter archives (1950–1991) exhibit error rates between 11.06% and 19.78% WER, demonstrating that historical physical decay necessitates the archival quarantine protocol while maintaining complete data privacy under Republic Act No. 10173.

Regarding Specific Objective 2, the study successfully designed a Coarse-to-Fine computational pipeline that optimizes resource utilization for edge deployment. Decoupling candidate retrieval from cross-attention inference reduced the active search space from 27,357 total enactments (178,494 provisions; 174,107 operative) down to a compact shortlist of top-$k$ candidates ($k \leq 50$), capturing 100% of controlling statutory preemption provisions on landmark jurisprudential test cases. To eliminate the processing latency of evaluating full-length municipal documents on consumer CPUs, the study formulated three structural optimizations: procedural boilerplate filtering and dynamic token padding alongside cross-section mini-batching. Empirical latency profiling over 30 repeated Monte Carlo runs demonstrated that these optimizations accelerate full-document audit throughput; end-to-end execution time dropped from 28.15 seconds down to 12.69 seconds per draft ordinance (a 54.9% latency reduction and 2.22× speedup) on an entry-level quad-core desktop processor. These architectural adaptations ensure that thorough statutory audits can be completed during active committee deliberations without introducing oper-

ational delays.

Regarding Specific Objective 3, the study implemented and calibrated a hybrid Information Retrieval and Natural Language Inference system tailored to the Philippine legal hierarchy. In Stage 1, fusing Okapi BM25 lexical search with `all-MiniLM-L6-v2` dense embeddings via Reciprocal Rank Fusion ($k = 60$) elevated candidate recall across all retrieval tiers while enforcing a statutory hierarchy priority safeguard that prioritizes governing national codes. In Stage 2, fine-tuning candidate cross-encoders mitigated the pronounced affirmative entailment bias observed in off-the-shelf foundation models, where unadapted zero-shot models suffered a 39.70% drop in Conflict $F_1$-score. Incorporating cost-sensitive threshold calibration ($\tau^* = 0.45$) with an asymmetric false-negative penalty ratio ($C(\text{FN})/C(\text{FP}) = 5.0$) successfully aligned the neural decision boundary with the institutional cost of overlooking invalid local penalties. This calibration ensures that the classifier operates with heightened sensitivity toward potential legal violations rather than favoring statistical majority classes.

Regarding Specific Objective 4, the study evaluated the empirical reliability and statistical validity of the proposed pipeline against verified legal conflicts. When evaluated on the 350-pair Faculty Adviser-Reviewed Ground Truth Benchmark (comprising 198 unique instances across 22 authentic statutory premises from 17 national statutes) under strict premise-grouped cross-validation, the proposed fine-tuned `DeBERTa-v3-base-NLI` cross-encoder achieved a zero-leakage generalization accuracy of $86.44\% \pm 6.05\%$, a Macro $F_1$-score of $0.8617 \pm 0.0631$, and a Conflict $F_1$-score of $0.8945 \pm 0.0683$ (alongside standard 5-fold cross-validation reaching $88.30\% \pm 3.44\%$ accuracy, 0.8800 Macro $F_1$, and 0.9008 Conflict $F_1$ across all 350 annotated pairs). Non-parametric statistical testing confirmed that the proposed architecture maintains top-rank performance across all eight macro legal domains (Friedman's $\chi^2_F = 40.00, p = 1.49 \times 10^{-7} < 0.001$) and exhibits consistent error reduction over canonical supervised baselines. Furthermore, multi-tier system testing across six typology-stratified full-length draft ordinances (52 sections) demonstrated that the pipeline successfully generated review alerts across all six test drafts while detecting 2 of 7 planted conflicts at the granular provision level under unassisted open-corpus search (alongside 6 false alarms requiring brief human verification) and clearing 39 of 45 compliant sections (86.67% section-level specificity), while detecting 7 of 7 planted conflicts under oracle candidate provision, confirming that open-corpus candidate retrieval is the primary remaining operational bottleneck. Historical evaluation across ten authentic Supreme Court preemption cases and municipal control enactments under Topologically Calibrated Dynamic Thresholding ($\tau^*(u, v)$) demonstrated 70.0% overall judicial accuracy, 100.0% police power specificity, and 100.0% precision with zero false alarms across all five valid municipal enactments. While the pipeline detected direct facial preemption conflicts with 100% certainty (*First Meridian* and *Magtajas*), preemption controversies rooted in exclusive agency jurisdiction (*Mosqueda* and the *Davao Mining Ban*) and substantive due process balancing (*City of Manila v. Laguio*) reveal the authentic structural boundary of sentence-level NLI cross-encoders. Furthermore, a 4-way knowledge graph ablation proved that topologically calibrated decision thresholding avoids the attention distribution shift and false-alarm cascades induced by prompt-level metadata conditioning, establishing a dependable, non-invasive decision-support framework. The empirical findings across both synthetic and authentic test suites establish that the model provides a dependable

foundation for decision support.

Regarding Specific Objective 5, the study developed and computationally profiled an offline, single-page web prototype interface designed for Sangguniang Panlungsod First Reading reviews. The production interface incorporates drag-and-drop document ingestion, procedural boilerplate filtering, dual-stream candidate inspection, and a side-by-side explainable AI (XAI) inspector that displays unpolluted verbatim statutory text alongside highlighted conflicting modal operators. Profiling on a representative municipal desktop (Intel Core i3-10105F, 4 threads, 8 GB RAM) confirmed an average per-pair inference latency of 120.9 milliseconds and a peak resident memory footprint of 1,416.4 MB; this memory footprint sits comfortably below the 8 GB municipal ceiling. Due to external administrative routing schedules through the Human Resource Management Office (HRMO) and the City Administrator's Office, the human usability and usefulness evaluation with the 15 Sangguniang Panlungsod legal researchers was not executed prior to the conclusion of the thesis research period. Instead, the 14-item TAM and SUS psychometric instrument and the multi-rater consensus protocol are formally codified as a post-study institutional deployment roadmap to be administered once municipal clearances are complete. Computational profiling verifies that the standalone prototype functions as an efficient, privacy-preserving screening tool establishing an empirical baseline for local legislative decision support.

## 5.2 Future Work

While this research demonstrates the practical efficacy of cascaded semantic conflict detection for local governance, the system possesses explicit operational and architectural boundaries that provide the foundation for subsequent investigations. To guide future researchers, this section delineates the specific limitations encountered during development and outlines concrete technical extensions. By explicitly documenting these boundaries, the study maintains methodological transparency and provides clear starting points for future development. The identified limitations reflect the real-world complexities of digitizing historical municipal archives and operating within municipal computing constraints. Conversely, the proposed extensions chart a practical path toward expanding the system's reasoning capabilities and institutional integration.

### 5.2.1 Methodological and System Limitations

The first limitation involves the physical degradation floor of historical municipal archives and mechanical typewriter artifacts. In Philippine local government units, legislative records predating Republic Act No. 7160 in 1991 exist mostly as physical carbon-copy papers with faded ink, carbon bleed-through, staple shadows, and binder hole punches. Classical optical character recognition engines frequently corrupt numerical penalties on these damaged scans, while deploying the 4-bit vision-language model (`Qwen2.5-VL-3B-Instruct`) still reaches an empirical ceiling on severely degraded pre-1980 pages. Consequently, heavily damaged archival enactments require human verification before ingestion so degraded digits do not create spurious conflict warnings.

The second limitation arises from structural legislative asymmetry and the contextual token overhead of municipal administrative provisions. National statutes often state normative commands in one or two sentences, whereas local ordinances routinely bundle a prohibition with enforcement duties, citation ticket steps, settlement windows, and community service clauses across four to six contiguous sentences. Preserving these qualifying provisos required multi-sentence span pairing and structural metadata prepending (`[Citation | Title > Chapter > Section]`), which consumes an average of 22.7 tokens per sequence inside the fixed 512-token context ceiling of lightweight cross-encoders.

The third limitation centers on the single-premise directional modeling adopted by the Stage 2 Cross-Encoder. Evaluating each draft ordinance block ($\mathbf{o}$) against an isolated candidate statutory section ($\mathbf{s}$) detects direct facial clashes, such as local fines above the ₱5,000 cap in Section 458 of Republic Act No. 7160. However, single-pair cross-attention cannot synthesize multi-statute cumulative preemption, where an ordinance complies with individual statutes taken separately but violates a combined regulatory framework across separate national codes. Resolving cumulative or constitutional means-ends balancing conflicts remains outside sentence-pair inference capacity on municipal workstations.

The fourth limitation concerns multi-section CPU throughput scaling on large omnibus codifications. Although procedural boilerplate filtering and dynamic sequence padding reduced 14-section draft audit time by 54.9% to 12.69 seconds on a 4-thread desktop CPU, wall-clock runtime scales linearly with operative section count ($S \times k$). When auditing massive multi-hundred-section enactments such as the 435-section Davao City Revenue Code without GPU acceleration, sequential open-corpus retrieval and cross-encoder scoring require extended offline batch execution rather than interactive single-request turnaround.

The fifth limitation involves the constructed scale of the supervised benchmark and the vertical preemption evaluation scope. Supervised fine-tuning and threshold calibration rely on the adviser-verified 350-pair benchmark (198 unique instances derived from 22 national statutory sections across 17 Republic Acts) written in formal legislative English pursuant to Executive Order No. 292. Furthermore, while the dual-corpus index natively retrieves fellow city ordinances, formal empirical benchmarking focused strictly on vertical statutory preemption against superior national law rather than horizontal municipal amendments or implied repeals. Expanding horizontal conflict classification and regional practitioner validation remain natural post-deployment extensions once institutional schedules permit.

### 5.2.2 Concrete Technical Extensions

To build upon the theoretical and empirical foundations established in this thesis, future researchers should pursue several concrete technical extensions to enhance the reasoning depth and operational deployment of the conflict detection architecture. These extensions address the structural and computational boundaries identified during empirical development. By advancing from isolated text classification to relational knowledge modeling and fine-grained explainability, subsequent systems can capture complex multi-statute interactions that currently require human legal synthesis. Furthermore, expanding corpus

acquisition and integrating the inference engine into municipal legislative databases will ground automated decision support directly in day-to-day governance. The following discussions detail each proposed extension and outline practical implementation pathways.

First, researchers should extend the Stage 2 inference engine from isolated pair classification to multi-hop statutory graph reasoning with dynamic temporal law tracking and administrative agency authority mapping. Future work can construct a directed Philippine Statutory Knowledge Graph connecting national statutes, regulatory agencies, and local ordinances via explicit semantic edges (*empowers*, *amends*, *repeals*, *exempts*, *delegates*, *regulates*). Relational graph-attention networks can track whether a legal provision remains in its authentic raw state, possesses amended successor versions, or has suffered express repeal under the principle of *lex posterior derogat legi priori*. Furthermore, to address the structural blind spot identified on jurisdictional preemption controversies (such as *Mosqueda v. PBGEA* under Presidential Decree No. 1144 and the *Davao Mining Ban* under Republic Act No. 7942), the graph should incorporate explicit agency jurisdiction nodes (e.g., the Fertilizer and Pesticide Authority and Mines and Geosciences Bureau) and symbolic concession edges. Coupling these administrative agency authority subgraphs with the cross-encoder will enable the automated detection of jurisdictional preemption under the *Magtajas* doctrine where local prohibitions collide with national regulatory licensing frameworks rather than direct textual antonyms. In addition, this graph structure can incorporate symbolic multi-tiered penalty constraints (such as imprisonment limits and administrative sanctions under Republic Act No. 7160) alongside localized municipal synsets mapping colloquial regulatory terms (*e.g.*, *trisikad* and *payong-payong*) to governing statutory definitions.

Second, future research should expand the knowledge base beyond statutory enactments to incorporate Supreme Court jurisprudence for ex-post conflict analysis. Under the constitutional doctrine of *stare decisis*, judicial interpretations rendered by the Supreme Court establish binding legal precedents that define the substantive boundaries of local legislative authority. Extending the dual-stream retrieval and inference pipeline to index landmark decisions allows the system to evaluate how judicial doctrines invalidate or uphold ordinances following enactment. This capability would enable legislative researchers to audit active city ordinances against evolving jurisprudential precedents; this process flags provisions vulnerable to judicial challenge. Developing this ex-post review layer transforms the architecture into a comprehensive legislative lifecycle tool capable of evaluating both prospective drafts and enacted municipal laws.

Third, subsequent studies should prioritize continuous corpus acquisition and conduct a systemic empirical consistency audit across positive law. Because legislative archives continuously expand with each congressional and city council session, establishing automated ingestion pipelines is necessary to raise collection completeness across both national and local jurisdictions. Utilizing this expanding repository, researchers can execute an offline data science audit cross-referencing national statutes and local enactments against one another. This large-scale empirical audit can identify latent historical contradictions and unenforced implied repeals across decades of cumulative legislation. Systematizing this analytical process provides municipal record divisions with an empirical inventory of dormant conflicts requiring legislative consolidation.

Fourth, future research should implement cross-lingual and dialectal domain adaptation to support regional municipal drafting. Subsequent studies can curate a bilingual parallel corpus of committee session transcripts and local administrative notes across Cebuano-Visayan and English. Fine-tuning regional multilingual encoders (such as `xlm-roberta-base` or specialized Philippine language checkpoints) will enable the system to identify potential statutory conflicts directly within early-stage draft concepts and public consultation remarks before they are formally translated into English statutory syntax. Such linguistic expansion bridges the communication gap between grassroots public consultations and formal statutory codification. By detecting potential legal issues early in the consultative process, the system can assist legislative staff in addressing community concerns without drafting legally defective provisions.

Fifth, researchers should deepen the system's explainable artificial intelligence (XAI) architecture beyond preliminary visual cues. The current prototype provides a foundational form of interpretability by presenting side-by-side verbatim statutory text alignments and highlighting deontic modal operators such as `shall` and `prohibited`. While this baseline ensures legal fidelity, subsequent investigations should incorporate fine-grained attribution mechanisms, such as integrated gradients or token-level attention-saliency mapping. These methods can isolate and highlight the precise contradictory clauses within multi-sentence regulatory blocks rather than relying on generic lexical markers. Providing targeted attribution rationales will further reduce cognitive strain on legal researchers during high-speed committee review.

Sixth, subsequent modeling experiments should incorporate cost-sensitive loss formulations directly into the cross-encoder training objective. While this study successfully addressed affirmative entailment bias through post-hoc decision threshold calibration ($\tau^* = 0.45$), the underlying model parameters were optimized using standard cross-entropy loss. Future work can implement asymmetric cost-weighted cross-entropy or focal loss during backpropagation, penalizing false negatives more heavily than false positives throughout the optimization process. Directly embedding the institutional cost of overlooking preemption contradictions into the loss function will encourage the model to learn conservative decision boundaries natively. Comparing natively cost-sensitive models against post-hoc calibrated baselines will establish whether direct loss weighting improves generalization across ambiguous regulatory domains.

Seventh, future research should explore automated benchmark synthesis and hardened adversarial curation to enhance ground truth validation and eliminate hypothesis-side lexical shortcuts. Using the three-tier difficulty taxonomy formulated in this thesis (Direct, Latent, and Distractor pairs), subsequent studies should engage legal practitioners and Bar members to formulate adversarial challenge pairs that systematically decouple lexical indicators from preemption verdicts. As demonstrated by the 58.00% hypothesis-only baseline in Section 4.3.7, synthetic drafting risks introducing subtle lexical regularities (such as prohibitive vocabulary in contradiction hypotheses). Future curation should programmatically and manually formulate pairs where local clauses use permissive words (*e.g.*, "the city may facilitate") to mask an unauthorized jurisdictional encroachment, or conversely, where restrictive penal language is entirely compliant with national statutory delegations. Systematically perturbing phrasing to break lexical correlations will establish

unexploitable, hardened benchmarks that evaluate genuine relational cross-attention.

Eighth, subsequent research should expand the system beyond diagnostic conflict detection to prescriptive legislative drafting assistance. Future studies can integrate a locally deployed generative language model (such as a 4-bit quantized Llama-3-8B or Qwen-2.5-7B) constrained by strict legal drafting schemas. When a preemption contradiction is detected, the generative module can automatically draft compliant amendatory clauses or explicit repealing provisions conforming to the Department of the Interior and Local Government (DILG) drafting manual, suggesting calibrated penalty amounts that fall strictly within statutory ceilings under Section 458 of Republic Act No. 7160. To maintain legal accuracy, all generated amendments must be strictly grounded in retrieved national statutes, with explicit citation trails linking proposed wording to authorizing statutory clauses. This extension would transform the system from a passive diagnostic filter into an active legislative drafting assistant.

Ninth, researchers should investigate staged inference optimizations and user-centric workflow features for the LGU decision-support web prototype. Current optimizations reduce audit latency to 12.69 seconds per draft; nevertheless, extensive omnibus codes require further runtime reductions. Future implementations should experiment with staged inference architectures, utilizing lightweight preliminary classifiers to rapidly filter obvious non-conflicting candidate pairs while reserving full cross-encoder attention for borderline cases. Combining this staged approach with 8-bit integer (INT8) quantization and ONNX Runtime execution graphs can drive execution speeds low enough to support real-time typing-time auditing inside standard word processors such as Microsoft Word and LibreOffice Writer. Furthermore, to enhance interactive usability for legislative staff, the web prototype interface should be expanded with interactive breadcrumb navigation linking retrieved provisions to parent statutory chapters, side-by-side textual diff viewers highlighting exact amendatory modifications, and aggregate committee audit dashboards that summarize preemption risk distributions across pending committee dockets.

Tenth, future development should pursue direct technical integration with the Davao City Legislative Information System of Sangguniang Panlungsod (LISSP). Currently operating primarily as an archival database for finalized enactments, the LISSP infrastructure can be enhanced by embedding the coarse-to-fine conflict detection pipeline into its core administrative software. Such an integration would transform the platform into an active digital legislative workbench that tracks ordinances across First, Second, and Third Readings, alerting committee staff to potential preemption risks whenever amendatory drafts are uploaded. Connecting the automated engine directly to official municipal repositories ensures that legislative vetting becomes a standard digital gateway in local governance. This operational integration bridges the gap between standalone academic prototypes and institutional municipal software ecosystems.

Eleventh, subsequent investigations should develop an automated legislative lineage engine to construct consolidated municipal enactments. In Philippine local governance, major regulatory frameworks are rarely enacted as self-contained static codes; rather, an original historical enactment is frequently modified by dozens of piecemeal amendatory ordinances passed across several decades. Consequently, evaluating a solitary historical document in isolation fails to capture its contemporary legal effect. An

automated lineage compiler can parse amendatory directives (*e.g.*, "Section 2 of Ordinance No. 158-05 is hereby amended to read as follows"), track the chain of legislative modifications chronologically, and sequentially apply text substitutions to assemble a live, consolidated "as-amended" text. Evaluating this reconstructed consolidated draft against national statutes ensures that conflict detection operates on the authentic operative state of municipal law rather than obsolete historical fragments.

Twelfth, future empirical benchmark studies should recalibrate human evaluation workload distributions to mitigate cognitive fatigue among domain experts. In the current study design, evaluating a 350-pair ground truth benchmark (comprising 198 unique instances) with threefold inter-annotator redundancy required allocating 70 dense legal pairs to each of the fifteen legal researchers. Authentic legal texts contain high syntactic complexity and multi-sentence regulatory provisos, meaning that large batch allocations place heavy cognitive strain on practicing legal officers who must balance evaluation with daily council duties. Future research should investigate fractional-block experimental designs and adaptive micro-tasking, systematically calculating the optimal number of items per annotator (*e.g.*, fifteen to twenty high-priority pairs per session) to preserve annotator focus and maximize inter-rater reliability ($\kappa$). In parallel, the user interface should implement confidence-gated severity tiers to suppress low-priority informational flags during high-tempo committee deliberations; in this manner, decision support remains actionable without overwhelming legislative staff.

Thirteenth, subsequent investigations should establish institutional expert consultation mechanisms for exploratory legislative artifacts beyond isolated sentence pairs. While research in legal natural language processing emphasizes annotating ground-truth conflict pairs, secondary artifacts such as statistical topic clusters and full-length draft ordinances also require systematic review alongside municipal practitioners. Unsupervised topic discovery and synthetic draft construction can produce mathematically coherent groupings that nevertheless diverge from authentic legislative terminology and local administrative realities. Future studies should actively engage Sangguniang Panlungsod legal researchers and committee secretaries to review discovered latent topics while verifying procedural draft structures against operational drafting conventions. Institutionalizing domain expert oversight across every stage of artifact creation ensures that downstream evaluation suites remain legally sound and practically grounded.

Fourteenth, subsequent investigations should establish dedicated empirical benchmarks for horizontal intra-jurisdictional consistency checking across local enactments. While the dual-corpus retrieval index and web prototype developed in this study natively index 1,925 Davao City ordinances alongside national statutes, formal empirical benchmarking was delimited to vertical statutory preemption under *Magtajas v. Pryce Properties*. In local legislative practice, horizontal interactions involve complex legislative dynamics: newly drafted ordinances may intentionally supersede prior enactments, introduce localized exceptions, or effect implied repeals. Future work should construct an empirical horizontal conflict dataset annotated with specialized relational labels beyond standard 3-class NLI—such as *Supersedes*, *Amends*, *Duplicates*, or *Implied Repeal*—equipping local legislative researchers to maintain horizontal code coherence across successive city council administrations.

Fifteenth, future research should evaluate dynamic and learned hierarchical weighting mechanisms for non-Republic Act national instruments. The current retrieval pipeline applies a static rank boost prioritizing governing national codes (Acts of Congress) over local enactments pursuant to the strict statutory hierarchy of *Magtajas*. However, the broader Philippine legal ecosystem comprises diverse regulatory instruments, including Presidential Decrees (PDs), Executive Orders (EOs), and Department Administrative Orders (DAOs), whose preemption authority varies depending on whether they represent valid exercises of delegated quasi-legislative rule-making. Subsequent research should develop dynamic hierarchical routing models or learned attention gates that calibrate candidate priority based on instrument type and regulatory subject matter, evaluating whether soft hierarchical priors improve candidate retrieval recall without imposing rigid rule-based filtering.

Finally, upon completion of the institutional clearance process with the Davao City Human Resource Management Office and City Administrator, researchers should execute a longitudinal 6-month field study within the Sangguniang Panlungsod committee review workflow. To ensure successful execution, future academic investigations must incorporate comprehensive operational foresight into their project timelines, mapping municipal administrative clearance hierarchies well in advance of field deployment. Navigating multi-office routing requires scheduling generous temporal buffers in research calendars to accommodate municipal administrative pacing and ongoing legislative committee schedules. Furthermore, research teams should employ an empathic and persistent liaison strategy that balances formal digital transmissions with proactive in-person office coordination, recognizing that municipal legal personnel face demanding public service commitments. Deploying the prototype dashboard during the Second Reading preparation phase across standing committees will enable long-term measurement of actual workflow adoption, capturing empirical distributions of perceived usefulness and amendment quality over time.

# Bibliography


[1] Ablazo, A. C. Designing a web application using the Twitter streaming API as a tracking tool for weather updates. Graduate independent study, Ateneo de Davao University, Davao City, Philippines, February 2019.

[2] Aggarwal, C. C., and Zhai, C. *Mining Text Data*. Springer Science+Business Media, Boston, MA, 2012.

[3] Ågotnes, T., Hoek, W., Rodríguez-Aguilar, J., Sierra, C., and Wooldridge, M. On the logic of normative systems. *IJCAI International Joint Conference on Artificial Intelligence* (01 2007).

[4] Agpalo, R. E. *Statutory Construction*, 6th ed. Rex Book Store, Manila, Philippines, 2009.

[5] Alshehri, A. S., Bencomo, N., and Atapour-Abarghouei, A. Cross-architecture LLM ensembles, feature-based reranking and retrieval-augmented prompting for legal information processing. In *Proceedings of the Workshop on the Thirteenth International Competition on Legal Information Extraction and Entailment (COLIEE 2026)* (Singapore, 2026), pp. 1–10.

[6] Althammer, S., Askari, A., Verberne, S., and Hanbury, A. Dossier@coliee 2021: Leveraging dense retrieval and summarization-based re-ranking for case law retrieval, 2021.

[7] Altman, D. G., and Bland, J. M. Diagnostic tests 1: Sensitivity and specificity. *BMJ 308*, 6943 (1994), 1552.

[8] Anand, A., Awekar, U., et al. Explainable information retrieval: A survey. *arXiv preprint arXiv:2012.14758* (2020).

[9] Arellano Law Foundation. The LawPhil project: Philippine laws and jurisprudence databank, 2000. Accessed: 2026-04-05.

[10] Ariai, F., Mackenzie, J., and Demartini, G. Natural language processing for the legal domain: A survey of tasks, datasets, models, and challenges. *ACM Comput. Surv. 58*, 6 (Dec. 2025).

[11] Artstein, R., and Poesio, M. Inter-coder agreement for computational linguistics. *Computational Linguistics 34* (01 2008), 555–596.

[12] Ashley, K. *Artificial Intelligence and Legal Analytics: New Tools for Law Practice in the Digital Age*. Cambridge University Press, 2017.

[13] Athan, T., Boley, H., Governatori, G., Palmirani, M., Paschke, A., and Wyner, A. Legalruleml: From metamodel to use cases - (a tutorial). vol. 8035, pp. 13–18.

[14] Baird, H. S. Document image defect models. In *Structured Document Image Analysis*, H. S. Baird, H. Bunke, and K. Yamamoto, Eds. Springer, 1993, pp. 546–556.

[15] Banarescu, L., Bonial, C., Cai, S., Georgescu, M., Griffitt, K., Hermjakob, U., Knight, K., Koehn, P., Palmer, M., and Schneider, N. Abstract meaning representation for sembanking. In *Proceedings of the 7th Linguistic Annotation Workshop and Interoperability with Discourse* (Sofia, Bulgaria, Aug. 2013), Association for Computational Linguistics, pp. 178–186.

[16] Bangor, A., Kortum, P. T., and Miller, J. T. An empirical evaluation of the system usability scale. *International Journal of Human-Computer Interaction 24*, 6 (2008), 574–594.

[17] Bangor, A., Kortum, P. T., and Miller, J. T. Determining what individual SUS scores mean: Adding an adjective rating scale. *Journal of Usability Studies 4*, 3 (2009), 114–123.

[18] Beltagy, I., Peters, M. E., and Cohan, A. Longformer: The long-document transformer, 2020.

[19] Bench-Capon, T. J. M., and Sartor, G. A model of legal reasoning with cases incorporating theories and values. *Artificial Intelligence 150*, 1–2 (2003), 97–143.

[20] Bernardo, A. S., and Albaña-Garrido, A. P. Disambiguating philippine republic acts: The case of RA 10913. *International Journal of Legal Discourse 8*, 2 (2023), 221–248.

[21] Bhattacharya, P., Hiware, K., Rajgaria, S., Pochhi, N., Ghosh, K., and Ghosh, S. *A Comparative Study of Summarization Algorithms Applied to Legal Case Judgments*. 04 2019, pp. 413–428.

[22] Bisong, E. *Building Machine Learning and Deep Learning Models on Google Cloud Platform: A Comprehensive Guide for Beginners*. Apress, Berkeley, CA, 2019.

[23] Blair, D. C., and Maron, M. E. An evaluation of retrieval effectiveness for a full-text document-retrieval system. *Commun. ACM 28*, 3 (Mar. 1985), 289–299.

[24] Bommarito, M. J., Katz, D. M., and Bommarito, J. Kl3m tokenizers: A family of domain-specific and character-level tokenizers for legal, financial, and preprocessing applications, 2025.

[25] Bowman, S. R., Angeli, G., Potts, C., and Manning, C. D. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015*

*Conference on Empirical Methods in Natural Language Processing* (2015), pp. 632–642.

[26] Brooke, J. SUS: A quick and dirty usability scale. *Usability Evaluation in Industry 189*, 194 (1996), 4–7.

[27] Campos, R., Evans, J. P., Isidro, J. M., Marques, M., Cunha, L. F., Jorge, A., Nunes, S., and Guimarães, N. Nlp for local governance meeting records: A focus article on tasks, datasets, metrics and benchmark, 2026.

[28] Campos, R., Pacheco, A. F., Fernandes, A. L., Cantante, I., Rebouças, R., Cunha, L. F., Isidro, J., Evans, J., Marques, M., Batista, R., Amorim, E., Jorge, A., Guimarães, N., Nunes, S., Leal, A., and Silvano, P. *CitiLink-Minutes: A Multilayer Annotated Dataset of Municipal Meeting Minutes*. Springer Nature Switzerland, 2026, p. 511–527.

[29] Card, D., Henderson, P., Khandelwal, U., Jia, R., Mahowald, K., and Jurafsky, D. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (Online, Nov. 2020), B. Webber, T. Cohn, Y. He, and Y. Liu, Eds., Association for Computational Linguistics, pp. 9263–9274.

[30] Caruana, R., Niculescu-Mizil, A., Crew, G., and Ksikes, A. Ensemble selection from libraries of models. In *Proceedings of the Twenty-First International Conference on Machine Learning (ICML)* (2004), p. 18.

[31] Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., and Androutsopoulos, I. LEGAL-BERT: The muppets straight out of law school. In *Findings of the Association for Computational Linguistics: EMNLP 2020* (Online, Nov. 2020), T. Cohn, Y. He, and Y. Liu, Eds., Association for Computational Linguistics, pp. 2898–2904.

[32] Chalkidis, I., Fergadiotis, M., Malakasiotis, P., and Androutsopoulos, I. Large-scale multi-label text classification on eu legislation, 2019.

[33] Chalkidis, I., Jana, A., Hartung, D., Bommarito, M., Androutsopoulos, I., Katz, D., and Aletras, N. LexGLUE: A benchmark dataset for legal language understanding in English. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Dublin, Ireland, May 2022), S. Muresan, P. Nakov, and A. Villavicencio, Eds., Association for Computational Linguistics, pp. 4310–4330.

[34] Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research 16* (2002), 321–357.

[35] Chen, J., et al. BGE M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216* (2024).

[36] Chinchor, N. MUC-4 evaluation metrics. In *Proceedings of the 4th Conference on Message Understanding* (1992), Association for Computational Linguistics, pp. 22–29.

[37] City Government of Davao. City Upgrades Legislative Information Support System, August 2025. Accessed: February 13, 2026.

[38] City Government of Davao. Sangguniang panlungsod - department of public administration, 2026. Accessed: 2026-04-06.

[39] Clark, K., Luong, M.-T., Le, Q. V., and Manning, C. D. ELECTRA: Pre-training text encoders as discriminators rather than generators. In *International Conference on Learning Representations (ICLR 2020)* (2020).

[40] Cohen, J. A coefficient of agreement for nominal scales. *Educational and psychological measurement 20*, 1 (1960), 37–46.

[41] Cohen, J. *Statistical Power Analysis for the Behavioral Sciences*, second ed. Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.

[42] Commission on Audit. COA Circular No. 2022-004: Guidelines on the Implementation of the Fifty Thousand Pesos (P50,000.00) Capitalization Threshold for Tangible Items. Tech. rep., Republic of the Philippines Commission on Audit, May 2022. Prescribing a 5-year estimated useful life for government information and communication technology equipment.

[43] Conchas, C. A. J., and Salapa, A. C. Assessing digital governance implementation issues in local government units: A systematic literature review. *International Journal of Engineering Technology Research and Management 9*, 12 (December 2025), 198–208.

[44] Conneau, A., et al. Unsupervised cross-lingual representation learning at scale. *arXiv preprint arXiv:1911.02116* (2020).

[45] Constitutional Commission of 1986. The 1987 Constitution of the Republic of the Philippines. Official Gazette of the Republic of the Philippines, 1987. Article VI, Section 26(2).

[46] Cormack, G. V., Clarke, C. L. A., and Büttcher, S. Reciprocal rank fusion outperforms Condorcet and individual machine learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (2009), pp. 758–759.

[47] Cortez, K., and Davao Today. Leadership and city problems await newly-elected Davao councilors. Rappler, May 2025. Republished with permission from DavaoToday.com for the 2025 Philippine local elections.

[48] Court of Tax Appeals of the Philippines. Rock Steel Resources, Inc. v. City of Davao and Hon. Rodrigo S. Riola (CTA AC No. 139, affirmed by CTA EB No. 1567 and G.R. No. 243823). Court of Tax Appeals Second Division Decision, 2016.

[49] Cronbach, L. J. Coefficient alpha and the internal structure of tests. *Psychometrika 16*, 3 (1951), 297–334.

[50] Cruz, I. A. *Philippine Political Law*. Central Lawbook Publishing, Quezon City, 2007.

[51] Cruz, J. C. B., and Cheng, C. Establishing baselines for text classification and question answering in Tagalog. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL 2020)* (2020), Association for Computational Linguistics, pp. 40–45.

[52] Dagan, I., Glickman, O., and Magnini, B. The PASCAL recognising textual entailment challenge. In *Machine Learning Challenges. Evaluating Predictive Uncertainty, Visual Object Classification, and Recognising Textual Entailment (MLCW 2005)* (2006), vol. 3944 of *Lecture Notes in Computer Science*, Springer, pp. 177–190.

[53] Davies, D. L., and Bouldin, D. W. A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence PAMI-1*, 2 (1979), 224–227.

[54] Davis, F. D. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly 13*, 3 (1989), 319–340.

[55] Deerwester, S., Dumais, S. T., Furnas, G. W., Landauer, T. K., and Harshman, R. Indexing by latent semantic analysis. *Journal of the American Society for Information Science 41*, 6 (1990), 391–407.

[56] Demšar, J. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research 7* (2006), 1–30.

[57] Department of Information and Communications Technology. Government Technical Specifications for Information and Communications Technology Equipment: Standard Office Desktop and Laptop Computers. Tech. rep., Republic of the Philippines DICT, 2020. Technical specifications standardizing 4-to-6-core processors, 8 GB system RAM, and integrated graphics for clerical and administrative workstations.

[58] Department of the Interior and Local Government. Local legislative drafting manual: A technical guide for philippine sangguniang panlungsod and sangguniang bayan members. Tech. rep., Bureau of Local Government Development, DILG, Quezon City, Philippines, 2022.

[59] Deroy, A., Ghosh, K., and Ghosh, S. How ready are pre-trained abstractive models and llms for legal case judgement summarization?, 2023.

[60] Dettmers, T., Lewis, M., Belkada, Y., and Zettlemoyer, L. Llm.int8(): 8-bit matrix multiplication for transformers at scale, 2022.

[61] Dettmers, T., Pagnoni, A., Holtzman, A., and Zettlemoyer, L. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS 2023)* (2024), vol. 36, pp. 10088–10115.

[62] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (Minneapolis, Minnesota, jun 2019), J. Burstein, C. Doran, and T. Solorio, Eds., Association for Computational Linguistics, pp. 4171–4186.

[63] Dietterich, T. G. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation 10*, 7 (1998), 1895–1923.

[64] Dodge, J., Gururangan, S., Card, D., Schwartz, R., and Smith, N. A. Show your work: Improved reporting of experimental results. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (2019), pp. 2185–2194.

[65] Doshi-Velez, F., and Kim, B. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608* (2017).

[66] Dreisbach, J. L., and Demeterio, F. Language use and preference in the multilingual context of davao city, philippines. *Studies in English Language and Education 8* (01 2021), 313–327.

[67] Dwork, C., Feldman, V., Hardt, M., Pitassi, T., Reingold, O., and Roth, A. The reusable holdout: Preserving validity in adaptive data analysis. *Science 349*, 6248 (2015), 636–638.

[68] Edwards, A. L. Note on the "correction for continuity" in testing the significance of the difference between correlated proportions. *Psychometrika 13*, 3 (1948), 185–187.

[69] Elkan, C. The foundations of cost-sensitive learning. In *Proceedings of the 17th International Joint Conference on Artificial Intelligence* (2001), vol. 17, Morgan Kaufmann Publishers Inc., pp. 973–978.

[70] Faul, F., Erdfelder, E., Lang, A.-G., and Buchner, A. G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods 39*, 2 (2007), 175–191.

[71] Fawcett, T. An introduction to ROC analysis. *Pattern Recognition Letters 27*, 8 (2006), 861–874.

[72] Fleiss, J. L. Measuring nominal scale agreement among many raters. *Psychological bulletin 76*, 5 (1971), 378.

[73] Fried, B. H. Ex ante and ex post. *Stanford Law Review* (2015).

[74] Friedman, M. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association 32*, 200 (1937), 675–701.

[75] Fuller, L. L. *The Morality of Law*, revised ed. Yale University Press, New Haven, 1964.

[76] Gabuya, R. J. B. Towards digital legislation: Readiness of the Sangguniang Bayan of LGU motiong for paperless sessions. *International Journal of Research and Innovation in Social Science 9*, 9 (September 2025), 2980–2991.

[77] Gao, T., Fisch, A., and Chen, D. Making pre-trained language models better few-shot learners, 2021.

[78] Gao, Y., Dligach, D., Miller, T., Tesch, S., Laffin, R., Churpek, M. M., and Afshar, M. Hierarchical annotation for building a suite of clinical natural language processing tasks: Progress note understanding, 2022.

[79] Goebel, R., Kano, Y., Kim, M.-Y., Kwan, C., Satoh, K., Yamada, H., and Yoshioka, M. An overview of the coliee 2025 competition: Legal case law and statute law information retrieval and entailment. In *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law* (New York, NY, USA, 2026), ICAIL '25, Association for Computing Machinery, p. 506–515.

[80] Goebel, R., Kano, Y., Kuribayashi, T., Martinez-Gil, J., Rabelo, J., Satoh, K., Silveira, L., and Yoshioka, M., Eds. *Proceedings of the Twelfth International Competition on Legal Information Extraction and Entailment (COLIEE 2025)* (San Luis, Argentina, 2025).

[81] Goebel, R., Kano, Y., Kuribayashi, T., Martinez-Gil, J., Rabelo, J., Satoh, K., Silveira, L., and Yoshioka, M., Eds. *Proceedings of the Thirteenth International Competition on Legal Information Extraction and Entailment (COLIEE 2026)* (Rome, Italy, 2026).

[82] Goodfellow, I., Bengio, Y., and Courville, A. *Deep Learning*. MIT Press, 2016. `http://www.deeplearningbook.org`.

[83] Goodhue, D. L., and Thompson, R. L. Task-technology fit and individual performance. *MIS Quarterly 19*, 2 (1995), 213–236.

[84] Google Cloud. Document AI and cloud vision API architecture overview. `https://cloud.google.com/document-ai/docs/overview`, 2024.

[85] Government of the Philippines. Republic Act No. 7160: The Local Government Code of 1991. Official Gazette of the Republic of the Philippines, 1991.

[86] Government of the Philippines. Republic act no. 11032: Ease of doing business and efficient government service delivery act of 2018. Tech. rep., Official Gazette of the Republic of the Philippines, 2018.

[87] Government of the Philippines. Official gazette of the republic of the philippines, 2026. Accessed: 2026-04-05.

[88] Greco, C. M., and Tagarelli, A. Bringing order into the realm of transformer-based language models for artificial intelligence and law. *Artificial Intelligence and Law 32*, 4 (Nov. 2023), 863–1010.

[89] Grootendorst, M. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794* (2022).

[90] Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning* (2017), PMLR, pp. 1321–1330.

[91] Gururangan, S., Swayamdipta, S., Levy, O., Schwartz, R., Bowman, S., and Smith, N. A. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)* (New Orleans, Louisiana, June 2018), M. Walker, H. Ji, and A. Stent, Eds., Association for Computational Linguistics, pp. 107–112.

[92] Hannes, W., Šavelka Jaromír, R., W. V., D., A. K., and Karim, B. *Computer-Assisted Creation of Boolean Search Rules for Text Classification in the Legal Domain*. IOS Press, 2019.

[93] Harde, P., Jain, B., and Jain, S. LeCNet: A legal citation network benchmark dataset. In *Proceedings of the 1st Workshop on NLP for Empowering Justice (JUST-NLP 2025)* (Mumbai, India, Dec. 2025), A. Modi, S. Ghosh, A. Ekbal, P. Goyal, S. Jain, A. Joshi, S. Mishra, D. Datta, S. Paul, K. B. Singh, and S. Kumar, Eds., Association for Computational Linguistics, pp. 18–28.

[94] Hart, H. L. A. *The Concept of Law*. Clarendon Press, Oxford University Press, Oxford, UK, 1961.

[95] He, P., Gao, J., and Chen, W. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543* (2023).

[96] Henderson, P., Krass, M., Zheng, L., Guha, N., Manning, C. D., Jurafsky, D., and Ho, D. E. Pile of Law: Learning responsible data filtering from the law and a 256GB open-source legal dataset. *arXiv preprint arXiv:2207.00220* (2022).

[97] Holistic SEO. Sliding window technique and algorithm in NLP, 2025.

[98] Holzenberger, N., Blair-Stanek, A., and Durme, B. V. A dataset for statutory reasoning in tax law entailment and question answering, 2020.

[99] House of Representatives of the Philippines. Rules of the House of Representatives (19th Congress). Tech. rep., House of Representatives, Republic of the Philippines, 2022.

[100] House of Representatives of the Philippines. Legislative information system (LEGIS) open data API. Official Website of the House of Representatives, 2023. Accessed: 2026-04-05.

[101] Huang, H., Zhang, H., Wang, Y., Liu, H., Chen, X., Chen, Y., and Liang, Y. Integrated robust optimization for lightweight transformer models in low-resource scenarios. *Symmetry 17*, 7 (2025).

[102] Humeau, S., Shuster, K., Vecchiato, M. A., and Weston, J. Poly-encoders: Architectures and pre-training strategies for fast and accurate multi-sentence scoring. *arXiv preprint arXiv:1905.01969* (2019).

[103] Humphreys, L., Boella, G., Di Caro, L., Robaldo, L., van der Torre, L., Ghanavati, S., and Muthuri, R. Populating legal ontologies using semantic role labeling. In *Proceedings of the Twelfth Language Resources and Evaluation Conference (LREC 2020)* (2020), European Language Resources Association, pp. 2157–2166.

[104] Ignat, O., Maillard, J., Chaudhary, V., and Guzmán, F. OCR improves machine translation for low-resource languages. In *Findings of the Association for Computational Linguistics: ACL 2022* (2022), pp. 1164–1174.

[105] Italiani, P., Moro, G., and Ragazzi, L. Clash-of-leges: A bilingual dataset for conflict detection and explanation in statutory law. *Expert Systems with Applications 300* (2026), 130182.

[106] Jacovi, A., and Goldberg, Y. Towards faithfully interpretable nlp systems: How should we define and evaluate faithfulness?, 2020.

[107] Jain, S., and Wallace, B. C. Attention is not Explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (Minneapolis, Minnesota, June 2019), J. Burstein, C. Doran, and T. Solorio, Eds., Association for Computational Linguistics, pp. 3543–3556.

[108] Jurafsky, D., and Martin, J. H. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*, 3rd ed. Pearson, 2024. Draft available online; accessed: 2026-03-21.

[109] Kadowaki, K., and Kano, Y. A ModernBERT-based system by team KIS for the COLIEE 2025 pilot task: Toward robust evaluation in legal judgment prediction. In *Proceedings of the Workshop on the Twelfth International Competition on Legal Information Extraction and Entailment (COLIEE 2025)* (Chicago, USA, 2025), ACM, pp. 14–22.

[110] Kaggle Inc. Kaggle: Machine learning and data science cloud infrastructure, 2026. Accessed October 2026.

[111] Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., and tau Yih, W. Dense passage retrieval for open-domain question answering, 2020.

[112] Katz, D. M., Bommarito, M. J., Gao, S., and Arredondo, P. Gpt-4 passes the bar exam. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences 382*, 2270 (02 2024), 20230254.

[113] Katz, D. M., Hartung, D., Gerlach, L., Jana, A., and Bommarito, M. J. Natural language processing in the legal domain. *arXiv preprint arXiv:2302.12039* (2023).

[114] Kelsen, H. *Pure Theory of Law.* University of California Press, Berkeley, CA, 1967.

[115] Khattab, O., and Zaharia, M. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (2020), pp. 39–48.

[116] Kim, m.-y., Rabelo, J., Babiker, H., Rahman, M. A., and Goebel, R. Legal information retrieval and entailment using transformer-based approaches. *The Review of Socionetwork Strategies 18* (01 2024), 1–21.

[117] Klebanov, B., and Beigman, E. From annotator agreement to noise models. *Computational Linguistics 35* (12 2009), 495–503.

[118] Koreeda, Y., and Manning, C. D. ContractNLI: A dataset for document-level natural language inference for contracts. In *Findings of the Association for Computational Linguistics: EMNLP 2021* (2021), Association for Computational Linguistics, pp. 1907–1919.

[119] Kumar, I. E., Venkatasubramanian, S., Scheidegger, C., and Friedler, S. Problems with shapley-value-based explanations as feature importance measures. In *International Conference on Machine Learning* (2020), PMLR, pp. 5491–5500.

[120] Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., and Soricut, R. ALBERT: A lite BERT for self-supervised learning of language representations. In *International Conference on Learning Representations (ICLR 2020)* (2020).

[121] Landauer, T. K., Foltz, P. W., and Laham, D. An introduction to latent semantic analysis. *Discourse Processes 25*, 2-3 (1998), 259–284.

[122] Landis, J. R., and Koch, G. G. The measurement of observer agreement for categorical data. *Biometrics 33*, 1 (1977), 159–174.

[123] Largey, N., Fitzgerald, E., and Mansouri, B. AIIR lab at COLIEE 2026: Fusion of instruction-tuned LLMs and cross-encoders for legal case entailment. In *Proceedings of the Workshop on the Thirteenth International Competition on Legal Information Extraction and Entailment (COLIEE 2026)* (Singapore, 2026), pp. 89–99.

[124] Laurer, M., van Atteveldt, W., Salloum, A., and Welbers, K. Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and multi-task NLI pre-alignment. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2024)* (2024), Association for Computational Linguistics, pp. 2123–2139.

[125] Le-Hong, P., and Cambria, E. A semantics-aware approach for multilingual natural language inference. *Lang. Resour. Eval. 57*, 2 (Feb. 2023), 611–639.

[126] Leonen, M. M. SAJ leonen: Despite risks, legal system should keep abreast with AI developments. Keynote Speech at Manila Tech Summit 2024, August 2024.

[127] Levenshtein, V. I. Binary codes capable of correcting deletions, insertions, and reversals. *Soviet Physics Doklady 10*, 8 (1966), 707–710.

[128] Li, J., Qian, L., Liu, P., and Liu, T. Construction of legal knowledge graph based on knowledge-enhanced large language models. *Information 15* (10 2024), 666.

[129] Li, T., Balke, T., De Vos, M., Satoh, K., and Padget, J. Detecting conflicts in legal systems. pp. 174–189.

[130] Liddy, E. D. Natural language processing. In *Encyclopedia of Library and Information Science*, 2nd ed. Marcel Dekker, Inc., 2001.

[131] Linna, E., and Linna, T. Challenges for generative ai in legal reasoning, 2026.

[132] Liu, A., Swamy, V., Ravichander, A., Wang, Y., Bhagavatula, C., and Le Bras, R. WANLI: Worker and AI Collaborative for Natural Language Inference Dataset Creation. In *Findings of the Association for Computational Linguistics: EMNLP 2022* (2022), Association for Computational Linguistics, pp. 6826–6841.

[133] Lopresti, D. Optical character recognition errors and their effect on natural language processing. *International Journal on Document Analysis and Recognition (IJDAR) 10*, 3–4 (2008), 141–151.

[134] Loshchilov, I., and Hutter, F. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)* (2019).

[135] Louis, A., and Spanakis, G. A statutory article retrieval dataset in French. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Dublin, Ireland, May 2022), S. Muresan, P. Nakov, and A. Villavicencio, Eds., Association for Computational Linguistics, pp. 6789–6803.

[136] Lu, K., Hsu, I.-H., Zhou, W., Ma, M. D., and Chen, M. Multi-hop evidence retrieval for cross-document relation extraction. In *Findings of the Association for Computational Linguistics: ACL 2023* (Toronto, Canada, July 2023), A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds., Association for Computational Linguistics, pp. 10336–10351.

[137] Lucos, R., L P, S., P R, V., Girish, D., M, R., Kurup, P. S., M S, C., and Saleela, D. Neuro-Symbolic Artificial Intelligence: Foundations, Advances, and Future Directions. *SSRN Electronic Journal* (December 2025).

[138] Lundberg, S. M., and Lee, S.-I. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, vol. 30. 2017.

[139] Ma, H. Gnn-ere: A graph neural network framework for entity-relation extraction and intelligent legal case reasoning. *Informatica 49* (12 2025).

[140] Manning, C. D., Raghavan, P., and Schütze, H. *Introduction to Information Retrieval*. Cambridge University Press, Cambridge, UK, 2008.

[141] Marzal, A., and Vidal, E. Computation of normalized edit distance and applications. *IEEE Transactions on Pattern Analysis and Machine Intelligence 15*, 9 (1993), 926–932.

[142] Maxilom, R. M. R. Semantic change of the selected Cebuano words. In *Proceedings of the 22nd Pacific Asia Conference on Language, Information and Computation* (2008), pp. 433–438.

[143] McCoy, T., Pavlick, E., and Linzen, T. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (2019), Association for Computational Linguistics, pp. 3428–3448.

[144] McHugh, M. L. Interrater reliability: The kappa statistic. *Biochemia Medica 22*, 3 (2012), 276–282.

[145] McNemar, Q. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika 12*, 2 (1947), 153–157.

[146] Morrison, D. G. On the interpretation of discriminant analysis. *Journal of Marketing Research 6*, 2 (1969), 156–163.

[147] Muennighoff, N., et al. MTEB: Massive text embedding benchmark. *arXiv preprint arXiv:2210.07316* (2023).

[148] Naguio, A. A. A., and Roxas, R. E. O. Hybrid neural-rule based architectures for Filipino stemming with fine-tuned BERT variants. In *Proceedings of the 38th Pacific Asia Conference on Language, Information and Computation* (Tokyo, Japan, Dec. 2024), N. Oco, S. N. Dita, A. M. Borlongan, and J.-B. Kim, Eds., Tokyo University of Foreign Studies, pp. 675–685.

[149] Naik, A., Ravichander, A., Sadeh, N., Rose, C., and Neubig, G. Stress test evaluation for natural language inference. In *Proceedings of the 27th International Conference on Computational Linguistics* (2018), Association for Computational Linguistics, pp. 2340–2353.

[150] National Assembly of the Philippines. Commonwealth act no. 638: An act to provide for the uniform publication and distribution of the official gazette. Supreme Court E-Library, June 1941.

[151] National Research Council of the Philippines. PH local governance gets reviewed after three decades, 2024.

[152] Ngo, T.-H., Nguyen, H.-T., Nguyen, H.-D., Le, X.-B., Nguyen, L.-D., Tran, Q.-T., Nguyen, H.-T., and Vuong, T.-H.-Y. NOWJ@COLIEE 2026: Adaptive pipelines for legal retrieval and reasoning. In *Proceedings of the Workshop on the Thirteenth International Competition on Legal Information Extraction and Entailment (COLIEE 2026)* (Singapore, 2026), pp. 119–128.

[153] Nie, Y., Williams, A., Dinan, E., Bansal, M., Weston, J., and Kiela, D. Adversarial NLI: A new benchmark for natural language understanding. In *Proceed-*

*ings of the 58th Annual Meeting of the Association for Computational Linguistics* (2020), Association for Computational Linguistics, pp. 4885–4901.

[154] Nogueira, R., and Cho, K. Passage re-ranking with bert, 2020.

[155] Nuñez, K. B. Q., Abcede, M. P. G., Salazar, L. R. R., Chua, Y. T., Romero II, V. M., Gabud, R. S., and Regonia, P. R. R. POMI: A corpus to support a natural language inference-based approach for detecting misinformation in Philippine online news. *Datenbank-Spektrum 26* (2026), 25–37.

[156] Nussbaum, Z., et al. Nomic embed: Training a reproducible long context text embedder. *arXiv preprint arXiv:2402.01613* (2024).

[157] Oard, D. W., et al. Desperately seeking Cebuano. In *Proceedings of the ACM SIGIR Workshop on Analytics for Noisy Unstructured Text Data* (2003).

[158] Palmirani, M., Governatori, G., Athan, T., Boley, H., Paschke, A., and Wyner, A. LegalRuleML Core Specification Version 1.0. Oasis standard, OASIS Open, August 2021. Approved 30 August 2021.

[159] Paruchuri, V. Surya: Multilingual document OCR, layout analysis, and line detection. `https://github.com/VikParuchuri/surya`, 2024.

[160] Patumbon, R. G. G. Davao City Council Passes Landmark Measures. SunStar Davao, February 2025. SunStar Davao Special Report on 20th and 21st Sangguniang Panlungsod legislative output.

[161] Patumbon, R. G. G. SP implements less paper with LISSP. SunStar Davao, February 2025. Reporting on official announcement by Davao City Councilor Bonz Andre Militar.

[162] Paul, S., Mandal, A., Goyal, P., and Ghosh, S. Pre-trained Language Models for the Legal Domain: A Case Study on Indian Law. In *Proceedings of the 19th International Conference on Natural Language Processing (ICON)* (2022), Association for Computational Linguistics, pp. 253–262.

[163] Philippine Commission. Act no. 453: An act providing for the publication by the insular government of an official gazette under the general direction of the department of public instruction. Supreme Court E-Library, September 1902.

[164] Poliak, A., Naradowsky, J., Aparicio, A., Duh, K., and Van Durme, B. Hypothesis only baselines in natural language inference. In *Proceedings of the 7th Joint Conference on Lexical and Computational Semantics* (2018), Association for Computational Linguistics, pp. 180–191.

[165] Prechelt, L. Early stopping—but when? In *Neural Networks: Tricks of the Trade*, vol. 1524 of *Lecture Notes in Computer Science*. Springer, 1998, pp. 55–69.

[166] Quevedo, E., Cerny, T., Rodriguez, A., Rivas, P., Yero, J., Sooksatra, K., Zhakubayev, A., and Taibi, D. Legal natural language processing from 2015 to 2022: A comprehensive systematic mapping study of advances and applications. *IEEE Access 12* (2024), 145286–145317.

[167] Rabelo, J., Goebel, R., Kim, M.-Y., Kano, Y., Yoshioka, M., and Satoh, K. Overview and discussion of the competition on legal information extraction/entailment (coliee) 2021. *The Review of Socionetwork Strategies 16*, 1 (April 2022), 111–133.

[168] Rajpurkar, P., Jia, R., and Liang, P. Know what you don't know: Unanswerable questions for SQuAD. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics* (2018), Association for Computational Linguistics, pp. 784–789.

[169] Ranathunga, S., and de Silva, N. Some languages are more equal than others: Probing deeper into the linguistic disparity in the NLP world. In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (Online only, Nov. 2022), Y. He, H. Ji, S. Li, Y. Liu, and C.-H. Chang, Eds., Association for Computational Linguistics, pp. 823–848.

[170] Reganit, J. C. Diokno files bill to break language barriers in PH laws. Philippine News Agency, August 2025. Report on House Bill No. 3863 (Batas sa Sariling Wika Act).

[171] Reimers, N., and Gurevych, I. Sentence-bert: Sentence embeddings using siamese bert-networks, 2019.

[172] Republic of the Philippines. Executive order no. 292: Instituting the Administrative Code of 1987. Official Gazette of the Republic of the Philippines, July 1987. Book I, Chapter 5, Section 20.

[173] Republic of the Philippines. Republic act no. 10173: An act protecting individual personal information in information and communications systems in the government and the private sector (Data Privacy Act of 2012). Official Gazette of the Republic of the Philippines, Vol. 108, No. 35, p. 4398, 2012.

[174] Ribeiro, M. T., Singh, S., and Guestrin, C. "why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2016), pp. 1135–1144.

[175] Ribeiro, M. T., Wu, T., Guestrin, C., and Singh, S. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (2020), Association for Computational Linguistics, pp. 4902–4912.

[176] Robertson, S., and Zaragoza, H. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval 3*, 4 (2009), 333–389.

[177] Rocchio, J. J. Relevance feedback in information retrieval. In *The SMART Retrieval System: Experiments in Automatic Document Processing*, G. Salton, Ed. Prentice-Hall, Englewood Cliffs, NJ, 1971, pp. 313–323.

[178] Roque, M., Abejuela, N., Chu, S., Cabatuan, M., and Sybingco, E. Legal information retrieval through embedding models and synthetic question generation: Insights from the Philippine tax code. In *Proceedings of the 38th Pacific Asia Conference on Language, Information and Computation* (Tokyo, Japan, Dec. 2024), N. Oco, S. N. Dita, A. M. Borlongan, and J.-B. Kim, Eds., Tokyo University of Foreign Studies, pp. 354–362.

[179] Rousseeuw, P. J. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics 20* (1987), 53–65.

[180] Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence 1*, 5 (2019), 206–215.

[181] Ruhl, J. B., and Katz, D. M. Measuring, monitoring, and managing legal complexity. *Iowa Law Review 101*, 1 (2015), 191–244.

[182] Rust, P., Pfeiffer, J., Vulić, I., Ruder, S., and Gurevych, I. How good is your tokenizer? on the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (Online, Aug. 2021), C. Zong, F. Xia, W. Li, and R. Navigli, Eds., Association for Computational Linguistics, pp. 3118–3135.

[183] Sagum, R. A., Clacio, P. A. C., Cayetano, R. E. R., and Lobrio, A. D. F. Philippine court case summarizer using latent semantic analysis. *Procedia Comput. Sci. 227*, C (Jan. 2023), 474–481.

[184] Sangguniang Panlungsod of Davao City. Davao City Ordinance No. 0310-07: The Watershed Protection, Conservation and Management Ordinance of Davao City. Official Records of the Sangguniang Panlungsod, City Government of Davao, 2007.

[185] Sangguniang Panlungsod of Davao City. Davao City Ordinance No. 0367-12: The New Comprehensive Anti-Smoking Ordinance of Davao City. Official Records of the Sangguniang Panlungsod, City Government of Davao, 2012.

[186] Sangguniang Panlungsod of Davao City. Resolution No. 001-22: Internal Rules of Procedure of the 20th Sangguniang Panlungsod. Tech. rep., Davao City Government, 2022.

[187] Sangguniang Panlungsod of Davao City. Davao City Ordinance No. 0270-23: The Comprehensive Speed Limit Ordinance of Davao City. Official Records of the Sangguniang Panlungsod, City Government of Davao, 2023.

[188] Sangguniang Panlungsod of Davao City. Official website of the Sangguniang Panlungsod of Davao City, 2026. Accessed: 2026-04-05.

[189] Sanh, V., Debut, L., Chaumond, J., and Wolf, T. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108* (2019).

[190] Saracevic, T. Relevance: A review of the literature and a framework for thinking on the notion in information science. part ii: Nature and manifestations of relevance. *Journal of the American Society for Information Science and Technology 58* (11 2007), 1915–1933.

[191] Sartor, G. *Legal Reasoning: A Cognitive Approach to the Law*, vol. 5 of *A Treatise of Legal Philosophy and General Jurisprudence*. Springer, Dordrecht, 2005.

[192] Sauro, J., and Lewis, J. R. *Quantifying the User Experience: Practical Statistics for User Research*. Morgan Kaufmann, Waltham, MA, 2012.

[193] Savelka, J., and Ashley, K. Discovering explanatory sentences in legal case decisions using pre-trained language models. In *Findings of the Association for Computational Linguistics: EMNLP 2021* (Punta Cana, Dominican Republic, Nov. 2021), M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds., Association for Computational Linguistics, pp. 4273–4283.

[194] Schwartz, R., Dodge, J., Smith, N. A., and Etzioni, O. Green AI. *Communications of the ACM 63*, 12 (2020), 54–63.

[195] Sculley, D. Web-scale k-means clustering. In *Proceedings of the 19th International Conference on World Wide Web (WWW '10)* (2010), Association for Computing Machinery, pp. 1177–1178.

[196] Shao, Y., Mao, J., Liu, Y., Ma, W., Satoh, K., Zhang, M., and Ma, S. Bert-pli: modeling paragraph-level interactions for legal case retrieval. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence* (2021), IJCAI'20.

[197] Shen, Z., Liu, J., Pang, Y., and Rao, Y. Hopweaver: Cross-document synthesis of high-quality and authentic multi-hop questions, 2025.

[198] Smith, R. An overview of the Tesseract OCR engine. In *Proceedings of the Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)* (2007), IEEE Computer Society, pp. 629–633.

[199] Snow, R., O'Connor, B., Jurafsky, D., and Ng, A. Y. Cheap and fast—but is it good? evaluating non-expert annotations for natural language tasks. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing* (Honolulu, Hawaii, October 2008), Association for Computational Linguistics, pp. 254–263.

[200] Song, K., Tan, X., Qin, T., Lu, J., and Liu, T.-Y. Mpnet: Masked and permuted pre-training for language understanding. In *Advances in Neural Information Processing Systems (NeurIPS)* (2020), vol. 33, pp. 16857–16867.

[201] Sovrano, F., Vitali, F., and Palmirani, M. *Making Things Explainable vs Explaining: Requirements and Challenges Under the GDPR*. Springer International Publishing, 2021, p. 169–182.

[202] Sperber, D., and Wilson, D. *Relevance: Communication and Cognition*, 2nd ed. Blackwell Publishers, Oxford, 1996.

[203] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research 15*, 56 (2014), 1929–1958.

[204] Strubell, E., Ganesh, A., and McCallum, A. Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)* (2019), pp. 3645–3650.

[205] Supreme Court of the Philippines. Tañada v. Tuvera, G.R. no. L-63915. Philippine Reports, Vol. 136, pp. 27–45, Lawphil Project, April 1985. En Banc Decision on the Mandatory Publication of Laws and Presidential Decrees.

[206] Supreme Court of the Philippines. Villacorta v. Bernardo (G.R. No. L-31249). Supreme Court Reports Annotated (143 SCRA 480), 1986.

[207] Supreme Court of the Philippines. Mayor Pablo Magtajas vs. Pryce Properties Corporation, Inc. (G.R. No. 111097). Supreme Court Reports Annotated (234 SCRA 255), 1994.

[208] Supreme Court of the Philippines. City of Davao v. Court of Appeals and Government Service Insurance System (G.R. No. 127383). Supreme Court Reports Annotated (467 SCRA 280), 2005.

[209] Supreme Court of the Philippines. City of Manila vs. Hon. Caridad H. Laguio, Jr. (G.R. No. 118127). Supreme Court Reports Annotated (455 SCRA 308), 2005.

[210] Supreme Court of the Philippines. Mosqueda v. Pilipino Banana Growers & Exporters Association, Inc. (G.R. Nos. 189185 & 189305). Supreme Court Reports Annotated (800 SCRA 313), 2016.

[211] Supreme Court of the Philippines. Mindanao Shopping Destination Corporation v. Duterte (G.R. No. 211093). Supreme Court Reports Annotated (826 SCRA 202), 2017.

[212] Supreme Court of the Philippines. Evasco, Jr. v. Montañez (G.R. No. 199172). Supreme Court Reports Annotated (856 SCRA 320), 2018.

[213] Supreme Court of the Philippines. City of Davao and Tanjili v. ARC Investors, Inc. (G.R. No. 249668). Supreme Court of the Philippines Third Division Resolution, 2022.

[214] Supreme Court of the Philippines. City of Davao v. First Meridian Development, Inc. (G.R. No. 240078, affirming CTA EB No. 1607). Supreme Court of the Philippines Second Division Decision; Court of Tax Appeals En Banc Decision (CTA EB No. 1607), 2022.

[215] Sweller, J. Cognitive load during problem solving: Effects on learning. *Cognitive Science 12*, 2 (1988), 257–285.

[216] Team, Q., Bai, J., Bai, S., Chu, Y., Cui, Z., Dang, K., Deng, X., Fan, Y., Gao, L., Ge, R., et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923* (2025).

[217] Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., and Gurevych, I. Beir: A heterogenous benchmark for zero-shot evaluation of information retrieval models, 2021.

[218] Thorne, J., Vlachos, A., Christodoulopoulos, C., and Mittal, A. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (2018), Association for Computational Linguistics, pp. 809–819.

[219] Tohidi, N., Dadkhah, C., and Gelbukh, A. Abstract meaning representation: A state-of-the-art review. 13–43.

[220] Tran, V.-K., and Nguyen, K.-L. Enhancing inferential consistency in LLM-based legal entailment. In *Proceedings of the Workshop on the Thirteenth International Competition on Legal Information Extraction and Entailment (COLIEE 2026)* (Singapore, 2026), pp. 170–177.

[221] Trong, S. V., and Le, M. N. An empirical evaluation of amr parsing for legal documents, 2018.

[222] Vabalas, A., Gowen, E., Poliakoff, E., and Casson, A. J. Machine learning algorithm validation with a limited sample size. *PLOS ONE 14*, 11 (2019), e0224365.

[223] Van Gestel, R., and Wiener, J. B. Future-oriented research into the expected effects of potential new legislation. *Tilburg University Working Paper* (2011).

[224] van Noord, R., and Bos, J. The role of data resources and NLP preprocessing in robust AMR parsing. In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)* (2017), pp. 36–47.

[225] Van Opijnen, M., and Santos, C. On the concept of relevance in legal information retrieval. *Artif. Intell. Law 25*, 1 (Mar. 2017), 65–87.

[226] van Rijsbergen, C. J. *Information Retrieval*, 2nd ed. Butterworth-Heinemann, London, 1979.

[227] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems* (2017), pp. 5998–6008.

[228] Vellino, A. Artificial intelligence: The very idea: J. haugeland, (mit press, cambridge, ma, 1985); 287 pp. *Artificial Intelligence 29* (09 1986), 349–353.

[229] Vig, J. A multiscale visualization of attention in the transformer model. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations* (2019), pp. 37–42.

[230] Virtucio, M. B., Aborot, J., Abonita, J., Aviñante, R., Co, R., Neverida, M., Osiana, V., Peramo, E., Syjuco, J., and Tan, G. Predicting decisions of the philippine supreme court using natural language processing and machine learning. pp. 130–135.

[231] Vitali, F., and Zeni, F. Towards a country-independent data format: the akoma ntoso experience.

[232] Vu, S. T., Le Nguyen, M., and Satoh, K. Abstract meaning representation for legal documents: an empirical research on a human-annotated dataset. *Artif. Intell. Law 30*, 2 (June 2022), 221–243.

[233] Wang, A., Pruksachatkun, Y., Nangia, N., Singh, A., Michael, J., Hill, F., Levy, O., and Bowman, S. R. Superglue: A stickier benchmark for general-purpose language understanding systems, 2020.

[234] Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., and Bowman, S. R. Glue: A multi-task benchmark and analysis platform for natural language understanding, 2019.

[235] Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., and Wei, F. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533* (2022).

[236] Wang, X., Han, Y., Leung, V. C., Niyato, D., Yan, X., and Chen, X. Convergence of edge computing and deep learning: A comprehensive survey. *IEEE Communications Surveys & Tutorials 22*, 2 (2020), 869–904.

[237] Warner, B., et al. ModernBERT: Bringing BERT into the modern era. *arXiv preprint arXiv:2412.13663* (2024).

[238] Wiegreffe, S., and Pinter, Y. Attention is not not explanation, 2019.

[239] Wu, D., Lawrence, S., and Mansouri, B. AIIR lab at COLIEE 2025: Exploring applications of large language models for legal text retrieval and entailment. In *Proceedings of the Workshop on the Twelfth International Competition on Legal Information Extraction and Entailment (COLIEE 2025)* (Chicago, USA, 2025), ACM, pp. 112–118.

[240] Xiao, C., Zhong, H., Guo, Z., Tu, C., Liu, Z., Sun, M., and Shen, T. Lawformer: A Pre-trained Language Model for Chinese Legal Long Documents. *AI Open 2* (2021), 79–84.

[241] Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., and Nie, J.-Y. C-Pack: Packaged resources to advance general chinese and multilingual information retrieval, 2024.

[242] Yang, J., Ma, W., Zhang, M., Zhou, X., Liu, Y., and Ma, S. Legalgnn: Legal information enhanced graph neural network for recommendation. *ACM Trans. Inf. Syst. 40*, 2 (Sept. 2021).

[243] Yang, Z. Improving the natural language inference robustness to hard dataset by data augmentation and preprocessing, 2024.

[244] Yin, W., Radev, D., and Xiong, C. DocNLI: A large-scale dataset for document-level natural language inference. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021* (Online, Aug. 2021), C. Zong, F. Xia, W. Li, and R. Navigli, Eds., Association for Computational Linguistics, pp. 4913–4922.

[245] Zaheer, M., Guruprasad, G., Kulkarni, A., Perot, D., Alberti, C., Izsak, P., Karchevsky, P., Lepikhin, D., Pfitzner, B., et al. Big Bird: Transformers for Longer Sequences. *Advances in Neural Information Processing Systems (NeurIPS 2020) 33* (2020), 17283–17297.

[246] Zhang, T., Wu, F., Katiyar, A., Weinberger, K. Q., and Artzi, Y. Revisiting few-sample bert fine-tuning, 2021.

[247] Zheng, L., Guha, N., Anderson, B. R., Henderson, P., and Ho, D. E. When does pretraining help? assessing self-supervised learning for law and the casehold dataset, 2021.

[248] Zheng, X., Li, Y., Chu, H., Feng, Y., Ma, X., Luo, J., Guo, J., Qin, H., Magno, M., and Liu, X. An empirical study of qwen3 quantization, 2025.

[249] Zhong, H., Xiao, C., Tu, C., Zhang, T., Liu, Z., and Sun, M. How does NLP benefit legal system: A summary of legal artificial intelligence. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (Online, July 2020), D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds., Association for Computational Linguistics, pp. 5218–5230.

[250] Zhong, X., ShafieiBavani, E., and Yepes, A. J. Image-based table recognition: Data, model, and evaluation. 564–580.

# Appendix A

# Dependencies and Scraping Code

## A.1 Software Dependencies and Imports

Table A.1: Python Package Dependencies and Import Manifest Across the Project Codebase

| Package | Verified | Primary Module / Function | Architectural Role in Pipeline |
|---|---|---|---|
| torch | $\geq$ 2.1.0 | torch.nn, torch.cuda | PyTorch tensor execution, AVX2 CPU operations, neural inference. |
| transformers | $\geq$ 4.36.0 | AutoModelFor SequenceClassification | HuggingFace transformer backbones (DeBERTa-v3, BERT). |
| sentence-transformers | $\geq$ 2.2.0 | SentenceTransformer | Dense bi-encoder provision embeddings (all-MiniLM-L6-v2). |
| accelerate | $\geq$ 0.25.0 | Accelerator | Multi-threaded CPU execution and FP16 tensor batching. |
| rank-bm25 | $\geq$ 0.2.2 | BM25Okapi | Lexical sparse inverted index and candidate provision scoring. |
| scikit-learn | $\geq$ 1.3.0 | sklearn.metrics, KMeans | 5-fold cross-validation, topic modeling, and McNemar tests. |
| scipy | $\geq$ 1.10.0 | scipy.stats. friedmanchisquare | Non-parametric statistical significance hypothesis testing. |
| pandas | $\geq$ 2.0.0 | pandas.DataFrame | Tabular metric aggregation, corpus census, and CSV reporting. |
| numpy | $\geq$ 1.24.0 | numpy.ndarray | Vectorized cosine similarity arrays and matrix arithmetic. |
| pymupdf | $\geq$ 1.23.0 | fitz.open | High-fidelity Folio PDF ingestion and bounding-box rendering. |
| pypdf | $\geq$ 4.0.0 | pypdf.PdfReader | Lightweight municipal ordinance text extraction and page parsing. |
| pillow | $\geq$ 10.0.0 | PIL.Image | Archival scan image preprocessing, deskewing, and contrast scaling. |
| opencv-python | $\geq$ 4.8.0 | cv2.threshold, cv2.bilateral | Adaptive document binarization and typewriter noise filtering. |
| beautifulsoup4 | $\geq$ 4.12.0 | bs4.BeautifulSoup | Autonomous Lawphil HTML DOM parsing and tag stripping. |
| requests | $\geq$ 2.31.0 | requests.Session | Resilient HTTP crawler with exponential backoff and retries. |
| matplotlib | $\geq$ 3.7.0 | matplotlib.pyplot | Publication figure generation and validation loss plotting. |
| plotly | $\geq$ 5.18.0 | plotly.graph_objects | Interactive multi-dimensional statutory cluster visualizations. |
| flask | $\geq$ 3.0.0 | flask.Flask, flask.Response | Local offline HTTP decision-support prototype backend server. |
| streamlit | $\geq$ 1.29.0 | streamlit | Interactive data exploration and cluster visualization prototype. |
| reportlab | $\geq$ 4.0.0 | reportlab.platypus | Authentic Philippine Folio legal PDF document compilation. |
| python-docx | $\geq$ 1.1.0 | docx.Document | Administrative letter and evaluator codebook formatting. |
| openpyxl | $\geq$ 3.1.0 | openpyxl.load_workbook | Excel-based multi-rater annotation workbook compilation. |
| bertopic | $\geq$ 0.16.0 | bertopic.BERTopic | Unsupervised legal domain discovery and semantic modeling. |
| hdbscan | $\geq$ 0.8.33 | hdbscan.HDBSCAN | Density-based provision clustering across latent embeddings. |
| umap-learn | $\geq$ 0.5.5 | umap.UMAP | Non-linear dimensionality reduction for provision visualization. |
| qwen-vl-utils | $\geq$ 0.0.8 | process_vision_info | High-resolution vision token processing for Qwen2.5-VL inference. |

## A.2 Statutory Scraper and Boundary Parser

```python
import os
import re
import json
import time
import requests
from bs4 import BeautifulSoup
from urllib.parse import urljoin

class LawphilStatutoryScraper:
    """Autonomous scraper for national statutes from the Lawphil repository."""
    BASE_URL = "https://lawphil.net/"
    USER_AGENT = "AcademicResearch-LegalNLP-AdDU/1.0"

    def __init__(self, output_dir="data/raw_statutes"):
        self.output_dir = output_dir
        self.session = requests.Session()
        self.session.headers.update({"User-Agent": self.USER_AGENT})
        os.makedirs(output_dir, exist_ok=True)

    def extract_provisions(self, text, statute_type, statute_num):
        """Extracts substantive sections while maintaining structural hierarchy."""
        lines = [line.strip() for line in text.split("\n") if line.strip()]
        title = lines[0] if lines else f"{statute_type} {statute_num}"

        # Section boundary regex pattern
        sec_pattern = re.compile(r'^(Section|Sec\.)\s+(\d+[a-zA-Z]?)\.?\s*(.*)', re.
    IGNORECASE)
        provisions = []
        current_sec = None
        current_body = []

        for line in lines:
            match = sec_pattern.match(line)
            if match:
                if current_sec and current_body:
                    provisions.append({
                        "section_id": f"{statute_type}_{statute_num}_S{current_sec}",
                        "section_num": current_sec,
                        "text": " ".join(current_body)
                    })
                current_sec = match.group(2)
                current_body = [match.group(3)] if match.group(3) else []
            elif current_sec:
                current_body.append(line)

        # Append final boundary
        if current_sec and current_body:
            provisions.append({
                "section_id": f"{statute_type}_{statute_num}_S{current_sec}",
                "section_num": current_sec,
                "text": " ".join(current_body)
            })
        return provisions
```

Listing A.1: Lawphil Statutory Scraper and Provision Boundary Serializer

# Appendix B

# Archival OCR and VLM Code

## B.1 Image Enhancement and Deskewing

```python
import cv2
import numpy as np
from PIL import Image

def preprocess_archival_scan(image_path: str) -> np.ndarray:
    """Preprocesses degraded historical scans to eliminate bleed-through and shadow noise."""
    image = cv2.imread(image_path)
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Bilateral filter removes paper grain while preserving sharp typewriter edges
    denoised = cv2.bilateralFilter(gray, d=9, sigmaColor=75, sigmaSpace=75)

    # Adaptive Gaussian binarization dynamically handles uneven ink fading
    binary = cv2.adaptiveThreshold(
        denoised, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY, blockSize=31, C=15
    )

    # Deskewing based on minimum area bounding box of detected text contours
    coords = np.column_stack(np.where(binary < 255))
    angle = cv2.minAreaRect(coords)[-1]
    if angle < -45:
        angle = -(90 + angle)
    else:
        angle = -angle

    (h, w) = binary.shape[:2]
    center = (w // 2, h // 2)
    m = cv2.getRotationMatrix2D(center, angle, 1.0)
    rotated = cv2.warpAffine(
        binary, m, (w, h), flags=cv2.INTER_CUBIC, borderMode=cv2.BORDER_REPLICATE
    )

    return rotated
```

Listing B.1: Archival Scan Preprocessing and Contrast Enhancement

## B.2 Vision-Language Model Extraction Engine

```python
import torch
```

```python
from transformers import AutoProcessor, Qwen2VLForConditionalGeneration
from PIL import Image

class ArchivalVLMReader:
    """Local 4-bit vision-language inference engine for historical ordinance digitization."""
    MODEL_ID = "Qwen/Qwen2.5-VL-3B-Instruct"

    def __init__(self, device="cuda" if torch.cuda.is_available() else "cpu"):
        self.device = device
        self.model = Qwen2VLForConditionalGeneration.from_pretrained(
            self.MODEL_ID,
            torch_dtype=torch.float16 if self.device == "cuda" else torch.float32,
            load_in_4bit=True if self.device == "cuda" else False,
            device_map="auto"
        )
        self.processor = AutoProcessor.from_pretrained(self.MODEL_ID)

    def extract_verbatim_text(self, image_path: str) -> str:
        """Extracts authentic legislative text with verbatim preservation of legal numerals."""
        image = Image.open(image_path).convert("RGB")
        prompt = (
            "You are an archival legal document transcriber. Transcribe all text on this page "
            "verbatim. Preserve exact legal numbers, monetary penalties, section headers, and "
            "punctuation. Do not correct typos, summarize, or omit clauses."
        )
        messages = [
            {"role": "user", "content": [
                {"type": "image", "image": image},
                {"type": "text", "text": prompt}
            ]}
        ]

        text_input = self.processor.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)
        inputs = self.processor(text=[text_input], images=[image], padding=True, return_tensors="pt").to(self.device)

        with torch.no_grad():
            output_ids = self.model.generate(**inputs, max_new_tokens=2048, temperature=0.01)

        generated_ids = [out[len(inp):] for inp, out in zip(inputs.input_ids, output_ids)]
        return self.processor.batch_decode(generated_ids, skip_special_tokens=True)[0].strip()
```

Listing B.2: Local 4-Bit VLM Inference Engine for Archival Extraction

## B.3 Cloud-Based Batch Archival Transcription Engine (Google Kaggle)

```python
# ==============================================================================
# DAVAO CITY ORDINANCES ARCHIVAL VLM BATCH INGESTION (GOOGLE KAGGLE RUNTIME)
# Hardware: NVIDIA Tesla T4 (16 GB GDDR6) | Model: Qwen/Qwen2.5-VL-3B-Instruct
# ==============================================================================

import os
import gc
import sys
import time
import zipfile
```

```python
from pathlib import Path
import torch
import pymupdf
from PIL import Image
from tqdm.auto import tqdm
from transformers import Qwen2_5_VLForConditionalGeneration, AutoProcessor
from qwen_vl_utils import process_vision_info

# --- 1. ENVIRONMENT VERIFICATION ---
assert torch.cuda.is_available(), "GPU runtime required (Tesla T4 or P100)!"
device_name = torch.cuda.get_device_name(0)
vram_gb = torch.cuda.get_device_properties(0).total_memory / (1024 ** 3)
print(f"[*] Cloud GPU Active: {device_name} ({vram_gb:.2f} GB VRAM)")

# --- 2. INPUT DATASET AND PATH RESOLUTION ---
input_pdfs = list(Path("/kaggle/input").rglob("*.pdf"))
if not input_pdfs:
    zips = list(Path("/kaggle/input").rglob("*.zip"))
    if zips:
        with zipfile.ZipFile(zips[0], "r") as z:
            z.extractall("/kaggle/working/input_pdfs")
        input_pdfs = list(Path("/kaggle/working/input_pdfs").rglob("*.pdf"))

if not input_pdfs:
    raise FileNotFoundError("No PDF files discovered in /kaggle/input directory.")

sample = input_pdfs[0]
root_candidate = None
for parent in [sample.parent, sample.parent.parent, sample.parent.parent.parent]:
    if parent.name.isdigit() and len(parent.name) == 4:
        root_candidate = parent.parent
        break

INPUT_ROOT = root_candidate if root_candidate else sample.parent
OUTPUT_DIR = Path("/kaggle/working/vlm_transcriptions")
OUTPUT_ZIP = Path("/kaggle/working/vlm_transcriptions_batch.zip")
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

# --- 3. MODEL AND PROCESSOR INITIALIZATION (NATIVE FP16) ---
MODEL_ID = "Qwen/Qwen2.5-VL-3B-Instruct"
print(f"[*] Loading {MODEL_ID} in native FP16 (sdpa attention)...")
model = Qwen2_5_VLForConditionalGeneration.from_pretrained(
    MODEL_ID,
    torch_dtype=torch.float16,
    device_map="auto",
    low_cpu_mem_usage=True,
    attn_implementation="sdpa",
)
model.eval()

processor = AutoProcessor.from_pretrained(
    MODEL_ID,
    min_pixels=256 * 28 * 28,
    max_pixels=448 * 28 * 28,
)
print("[OK] Model and processor initialized successfully in FP16.")

# --- 4. VERBATIM PAGE TRANSCRIPTION ENGINE ---
def transcribe_page(pil_image: Image.Image) -> str:
    """Extracts verbatim statutory text using deterministic greedy decoding."""
    messages = [
        {
            "role": "user",
            "content": [
                {"type": "image", "image": pil_image},
                {
                    "type": "text",
                    "text": (
                        "Transcribe the legal text on this page completely and "
```

```
                        "verbatim. Preserve exact section titles, ordinals, "
                        "numbers, dates, names, and currency amounts. "
                        "Do not summarize, omit provisions, or add commentary."
                    ),
                },
            ],
        }
    ]

    text_prompt = processor.apply_chat_template(
        messages, tokenize=False, add_generation_prompt=True
    )
    image_inputs, video_inputs = process_vision_info(messages)

    inputs = processor(
        text=[text_prompt],
        images=image_inputs,
        videos=video_inputs,
        padding=True,
        return_tensors="pt",
    ).to("cuda")

    with torch.inference_mode():
        generated_ids = model.generate(
            **inputs,
            max_new_tokens=1000,
            do_sample=False,
            use_cache=True,
        )

    trimmed_ids = [
        out[len(inp):] for inp, out in zip(inputs.input_ids, generated_ids)
    ]
    transcription = processor.batch_decode(
        trimmed_ids,
        skip_special_tokens=True,
        clean_up_tokenization_spaces=False,
    )[0]
    return transcription.strip()

# --- 5. PER-DOCUMENT INGESTION AND BOUNDARY CROPBOX ---
def process_pdf_document(pdf_path: Path, output_path: Path) -> int:
    """Processes document page-by-page with 3% inset border artifact trimming."""
    doc = pymupdf.open(pdf_path)
    total_pages = len(doc)
    records = []
    try:
        for page_idx in range(total_pages):
            page = doc.load_page(page_idx)
            rect = page.rect
            crop_box = pymupdf.Rect(
                rect.width * 0.03,
                rect.height * 0.03,
                rect.width * 0.97,
                rect.height * 0.97,
            )
            page.set_cropbox(crop_box)
            pix = page.get_pixmap(dpi=105)
            img = Image.frombytes("RGB", [pix.width, pix.height], pix.samples)
            txt = transcribe_page(img)
            records.append(f"--- [Page {page_idx + 1}] ---\n{txt}")
            del pix, img, page
    finally:
        doc.close()

    output_path.parent.mkdir(parents=True, exist_ok=True)
    with output_path.open("w", encoding="utf-8") as f:
        f.write(f"=== ORDINANCE: {pdf_path.stem} ===\n\n")
        f.write("\n\n".join(records))
```

```
149
150     gc.collect()
151     torch.cuda.empty_cache()
152     return total_pages
153
154 def create_checkpoint_zip():
155     """Generates a downloadable ZIP archive of all completed transcripts."""
156     txts = list(OUTPUT_DIR.rglob("*.txt"))
157     if not txts:
158         return
159     with zipfile.ZipFile(OUTPUT_ZIP, "w", compression=zipfile.ZIP_DEFLATED) as zf:
160         for txt in txts:
161             zf.write(txt, arcname=str(txt.relative_to(OUTPUT_DIR)))
162
163 # --- 6. FAULT-TOLERANT BATCH QUEUE EXECUTION ---
164 def run_batch_queue(target_years=None):
165     """Executes full batch queue with skip-on-existing fault tolerance."""
166     all_pdfs = sorted(p for p in INPUT_ROOT.rglob("*.pdf") if p.is_file())
167     if target_years:
168         all_pdfs = [p for p in all_pdfs if any(y in p.parts for y in target_years)]
169
170     print(f"[*] Total PDF Documents Queued: {len(all_pdfs)}")
171     processed, skipped, failed, total_pages = 0, 0, 0, 0
172     start_time = time.time()
173
174     for idx, pdf_path in enumerate(all_pdfs, 1):
175         rel_path = pdf_path.relative_to(INPUT_ROOT)
176         out_path = OUTPUT_DIR / rel_path.with_suffix(".txt")
177
178         # Skip already completed enactments
179         if out_path.exists() and out_path.stat().st_size > 100:
180             skipped += 1
181             continue
182
183         try:
184             pages = process_pdf_document(pdf_path, out_path)
185             processed += 1
186             total_pages += pages
187         except Exception as e:
188             failed += 1
189             print(f"[ERROR] Failed processing {rel_path}: {e}")
190
191         # Checkpoint archive packaging every 25 files
192         if processed > 0 and processed % 25 == 0:
193             create_checkpoint_zip()
194
195     create_checkpoint_zip()
196     elapsed = time.time() - start_time
197     print(f"[COMPLETE] Processed: {processed} files, {total_pages} pages in {elapsed
      /60:.1f}m")
```

Listing B.3: Headless Cloud VLM Batch Extraction Notebook on Google Kaggle (Tesla T4 Runtime)

# Appendix C

# Ordinance Preprocessing Code

## C.1 Text Cleaning and Boundary State Machine

```
import re
from typing import List, Dict

class OrdinanceCleaner:
    """Isolates substantive legislative provisions from raw municipal transcriptions."
    ""

    BOILERPLATE_PATTERNS = [
        re.compile(r'^WHEREAS\b', re.IGNORECASE),
        re.compile(r'^BE IT ORDAINED\b', re.IGNORECASE),
        re.compile(r'^ENACTED by the Sangguniang Panlungsod\b', re.IGNORECASE),
        re.compile(r'^APPROVED,\s*this\b', re.IGNORECASE)
    ]

    SECTION_START = re.compile(
        r'^(SECTION|SEC\.)\s+(\d+[a-zA-Z]?)\.?\s*([A-Z\s\(\)]+)?[\.\-\-]?\s*(.*)',
        re.IGNORECASE
    )

    def clean_and_segment(self, raw_text: str, ordinance_num: str) -> List[Dict]:
        """Parses raw ordinance transcription into substantive operative sections."""
        lines = [line.strip() for line in raw_text.split('\n') if line.strip()]
        sections = []
        current_sec_num = None
        current_title = ""
        current_text = []

        in_operative_body = False

        for line in lines:
            # Check for transition into substantive operative text
            sec_match = self.SECTION_START.match(line)
            if sec_match:
                in_operative_body = True
                if current_sec_num and current_text:
                    sections.append({
                        "ordinance_num": ordinance_num,
                        "section_num": current_sec_num,
                        "section_title": current_title.strip(),
                        "provision_text": " ".join(current_text).strip()
                    })
                current_sec_num = sec_match.group(2)
                current_title = sec_match.group(3) or ""
                current_text = [sec_match.group(4)] if sec_match.group(4) else []
```

```
                continue

            if in_operative_body and current_sec_num:
                current_text.append(line)
            elif not in_operative_body:
                # Discard non-operative preambles
                continue

        # Append terminal section
        if current_sec_num and current_text:
            sections.append({
                "ordinance_num": ordinance_num,
                "section_num": current_sec_num,
                "section_title": current_title.strip(),
                "provision_text": " ".join(current_text).strip()
            })

        return sections
```

Listing C.1: Ordinance Segmentation and Boilerplate Filtering

# Appendix D

# Candidate Retrieval Code

## D.1 Hybrid Retrieval and Rank Fusion

```python
import numpy as np
from typing import List, Dict
from rank_bm25 import BM25Okapi
from sentence_transformers import SentenceTransformer

class DualStreamRetrievalEngine:
    """Stage 1 hybrid coarse retrieval combining Okapi BM25 and dense embeddings."""

    def __init__(self, corpus: List[Dict], dense_model_name: str = "all-MiniLM-L6-v2"):
        self.corpus = corpus
        self.dense_model = SentenceTransformer(dense_model_name)

        # Build Okapi BM25 tokenized index
        tokenized_corpus = [doc["text"].lower().split() for doc in self.corpus]
        self.bm25 = BM25Okapi(tokenized_corpus, k1=1.5, b=0.75)

        # Precompute dense embeddings
        texts = [doc["text"] for doc in self.corpus]
        self.dense_embeddings = self.dense_model.encode(
            texts, batch_size=64, show_progress_bar=False, normalize_embeddings=True
        )

    def retrieve_candidates(self, query: str, top_k: int = 50, rrf_k: int = 60, beta: float = 1.20) -> List[Dict]:
        """Executes hybrid retrieval and merges candidate lists via Reciprocal Rank Fusion."""
        # 1. Sparse BM25 search
        query_tokens = query.lower().split()
        bm25_scores = self.bm25.get_scores(query_tokens)
        bm25_rankings = np.argsort(bm25_scores)[::-1][:top_k * 2]

        # 2. Dense semantic search
        query_emb = self.dense_model.encode([query], normalize_embeddings=True)[0]
        dense_scores = np.dot(self.dense_embeddings, query_emb)
        dense_rankings = np.argsort(dense_scores)[::-1][:top_k * 2]

        # 3. Reciprocal Rank Fusion (RRF) with statutory hierarchy safeguard
        rrf_scores = {}
        for rank, idx in enumerate(bm25_rankings):
            rrf_scores[idx] = rrf_scores.get(idx, 0.0) + (1.0 / (rrf_k + rank + 1))

        for rank, idx in enumerate(dense_rankings):
            rrf_scores[idx] = rrf_scores.get(idx, 0.0) + (1.0 / (rrf_k + rank + 1))
```

```
42
43          # Apply statutory hierarchy boost (prioritizing national codes over local acts
        )
44          for idx in rrf_scores:
45              if self.corpus[idx].get("is_national_statute", False):
46                  rrf_scores[idx] *= beta  # 20% hierarchical priority safeguard (\beta
        = 1.20)
47
48          # Sort and return top-k shortlisted provisions
49          sorted_indices = sorted(rrf_scores.keys(), key=lambda i: rrf_scores[i],
        reverse=True)[:top_k]
50
51          results = []
52          for idx in sorted_indices:
53              candidate = dict(self.corpus[idx])
54              candidate["retrieval_score"] = float(rrf_scores[idx])
55              results.append(candidate)
56
57          return results
```

Listing D.1: Stage 1 Hybrid Retrieval Engine and RRF Fusion

# Appendix E

# Conflict Inference Code

## E.1 Cross-Encoder Inference Engine

```python
import torch
import torch.nn.functional as F
from typing import List, Dict, Tuple
from transformers import AutoTokenizer, AutoModelForSequenceClassification

class LegalConflictInferenceEngine:
    """Stage 2 fine cross-encoder inference with calibrated decision thresholds."""

    def __init__(self, model_path: str = "models/deberta-v3-legal-nli", threshold: float = 0.45):
        self.device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
        self.tokenizer = AutoTokenizer.from_pretrained(model_path)
        self.model = AutoModelForSequenceClassification.from_pretrained(model_path)
        self.model.to(self.device)
        self.model.eval()
        self.threshold = threshold  # Calibrated tau* = 0.45 (C(FN)/C(FP) = 5.0)

    def classify_pair(self, statute_premise: str, ordinance_hypothesis: str) -> Dict:
        """Evaluates an asymmetric premise-hypothesis pair and returns calibrated triage."""
        inputs = self.tokenizer(
            statute_premise,
            ordinance_hypothesis,
            truncation=True,
            max_length=512,
            padding=True,
            return_tensors="pt"
        ).to(self.device)

        with torch.no_grad():
            outputs = self.model(**inputs)
            probs = F.softmax(outputs.logits, dim=-1).squeeze().cpu().numpy()

        # Class indexing: 0: Entailment, 1: Neutral, 2: Contradiction
        p_entail, p_neutral, p_contra = float(probs[0]), float(probs[1]), float(probs[2])

        # Apply cost-sensitive binary triage boundary
        is_conflict = p_contra >= self.threshold

        predicted_label = "Contradiction" if is_conflict else ("Entailment" if p_entail > p_neutral else "Neutral")

        return {
```

```
            "predicted_label": predicted_label,
            "conflict_risk": p_contra,
            "entailment_prob": p_entail,
            "neutral_prob": p_neutral,
            "triage_flag": is_conflict,
            "calibrated_threshold": self.threshold
        }

    def batch_audit(self, pairs: List[Tuple[str, str]], batch_size: int = 16) -> List[
    Dict]:
        """Executes batched cross-encoder inference using dynamic token padding."""
        results = []
        for i in range(0, len(pairs), batch_size):
            batch = pairs[i:i + batch_size]
            premises = [p[0] for p in batch]
            hypotheses = [p[1] for p in batch]

            inputs = self.tokenizer(
                premises,
                hypotheses,
                truncation=True,
                max_length=512,
                padding=True,  # Dynamic padding to longest sequence in batch
                return_tensors="pt"
            ).to(self.device)

            with torch.no_grad():
                logits = self.model(**inputs).logits
                probs = F.softmax(logits, dim=-1).cpu().numpy()

            for prob in probs:
                p_contra = float(prob[2])
                results.append({
                    "conflict_risk": p_contra,
                    "triage_flag": p_contra >= self.threshold
                })
        return results
```

Listing E.1: Stage 2 Cross-Encoder Inference and Calibrated Triage

# Appendix F

# Stop Words and Text Patterns

## F.1 Legal Stop Words

Table F.1: Alphabetical Roster of the 49 Domain-Specific Philippine Legal Stop Words

| Terms 1–13 | Terms 14–25 | Terms 26–37 | Terms 38–49 |
|---|---|---|---|
| act | force | order | secretary |
| amended | further | ordinance | section |
| approved | government | penalties | shall |
| article | hereby | philippines | thereof |
| board | herein | president | thereto |
| city | hereunder | provided | treasurer |
| commission | inconsistent | provisions | violation |
| council | law | purpose | whereas |
| davao | may | pursuant | |
| department | mayor | regulations | |
| duly | office | repealed | |
| effect | official | republic | |
| effective | operative | rules | |

## F.2 Deontic Modality Markers

Table F.2: Deontic Modality Statutory Markers Monitored in Cross-Encoder Attention

| Deontic Class | Statutory Marker Terms and Phrases |
|---|---|
| **Mandatory / Obligatory** | shall, must, is required to, shall be the duty of, directed to, mandatory, compulsory, shall enforce. |
| **Prohibitory / Restrictive** | shall not, cannot, prohibited, unlawful, no person shall, in no case shall, banned, disallowed, forbidden. |
| **Permissive / Discretionary** | may, is authorized to, at the discretion of, optional, allowed, permitted, can, empowered to. |

# Appendix G

# Experimental Notebooks

## G.1 Corpus Preparation and Topic Modeling

```python
# Corpus Preparation and Topic Modeling
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import MiniBatchKMeans

# Load cleaned statutory and municipal provisions
df = pd.read_json("data/unified_dual_statutory_provisions.jsonl", lines=True)

# Compute c-TF-IDF vector representations across substantive provisions
vectorizer = TfidfVectorizer(max_features=10000, stop_words="english", ngram_range=(1, 2))
X_tfidf = vectorizer.fit_transform(df["provision_text"])

# Cluster provisions into 28 sub-clusters across 8 Macro Legal Domains
kmeans = MiniBatchKMeans(n_clusters=28, random_state=42, batch_size=1024)
df["cluster_id"] = kmeans.fit_predict(X_tfidf)

print(f"Successfully clustered {len(df):,} provisions into {kmeans.n_clusters} clusters.")
```

Listing G.1: Corpus Preparation and Topic Modeling

## G.2 Stage 1 Hybrid Retrieval Benchmarking

```python
# Stage 1 Hybrid Retrieval Evaluation
import numpy as np

def evaluate_retrieval_recalls(retrieval_engine, benchmark_queries, k_values=[5, 10, 20, 50]):
    """Evaluates candidate retrieval recall across multiple top-k thresholds."""
    recall_results = {k: 0 for k in k_values}

    for query in benchmark_queries:
        gold_id = query["gold_statute_id"]
        retrieved = retrieval_engine.retrieve_candidates(query["ordinance_text"], top_k=max(k_values))
        retrieved_ids = [doc["section_id"] for doc in retrieved]

        for k in k_values:
            if gold_id in retrieved_ids[:k]:
                recall_results[k] += 1
```

```
    total = len(benchmark_queries)
    return {f"Recall@{k}": f"{(hits / total) * 100:.2f}%" for k, hits in
    recall_results.items()}
```

Listing G.2: Stage 1 Retrieval Sweep and Recall Evaluation

## G.3 Stage 2 Cross-Encoder Fine-Tuning

```
# Stage 2 Cross-Encoder Fine-Tuning Loop
from transformers import AutoModelForSequenceClassification, AdamW,
    get_linear_schedule_with_warmup
from torch.utils.data import DataLoader

model = AutoModelForSequenceClassification.from_pretrained("microsoft/deberta-v3-base"
    , num_labels=3)
optimizer = AdamW(model.parameters(), lr=2e-5, weight_decay=0.01)

# Training loop with validation checkpointing
for epoch in range(5):
    model.train()
    for batch in train_dataloader:
        optimizer.zero_grad()
        outputs = model(**batch)
        loss = outputs.loss
        loss.backward()
        optimizer.step()
        scheduler.step()

    # Validation evaluation for Macro F1 and Conflict F1
    val_macro_f1, val_conflict_f1 = evaluate_nli_fold(model, val_dataloader)
    print(f"Epoch {epoch+1}: Val Macro F1 = {val_macro_f1:.4f}, Conflict F1 = {
    val_conflict_f1:.4f}")
```

Listing G.3: Cross-Encoder Fine-Tuning and Validation

## G.4 Full-Document System Stress Testing

```
# System-Level Full Document Evaluation
def audit_complete_draft_ordinance(draft_pdf_path, pipeline):
    """Executes end-to-end audit: ingestion -> boilerplate filter -> retrieval -> NLI.
    """
    raw_sections = extract_folio_sections(draft_pdf_path)
    operative_sections = [s for s in raw_sections if not is_boilerplate(s)]

    audit_report = []
    for section in operative_sections:
        candidates = pipeline.retrieval.retrieve_candidates(section["text"], top_k=50)
        inference_results = pipeline.inference.classify_candidates(section["text"],
    candidates)
        audit_report.append({
            "section_num": section["num"],
            "max_conflict_risk": max(r["conflict_risk"] for r in inference_results),
            "flagged": any(r["triage_flag"] for r in inference_results)
        })
    return audit_report
```

Listing G.4: End-to-End System Audit and Verification

## G.5 Archival VLM Batch Digitization Notebook (Google Kaggle)

```python
# Archival VLM Ingestion Loop on Google Kaggle GPU (Tesla T4)
def run_archival_batch_ingestion(input_root, output_dir, output_zip, target_years=None):
    """Executes high-throughput headless VLM batch transcription with fault tolerance."""
    all_pdfs = sorted(p for p in input_root.rglob("*.pdf") if p.is_file())
    if target_years:
        all_pdfs = [p for p in all_pdfs if any(y in p.parts for y in target_years)]

    processed, skipped, total_pages = 0, 0, 0
    for idx, pdf_path in enumerate(all_pdfs, 1):
        out_path = output_dir / pdf_path.relative_to(input_root).with_suffix(".txt")
        if out_path.exists() and out_path.stat().st_size > 100:
            skipped += 1
            continue

        pages = process_pdf_document(pdf_path, out_path)
        processed += 1
        total_pages += pages

        # Periodic checkpoint archive packaging every 25 enactments
        if processed > 0 and processed % 25 == 0:
            create_checkpoint_zip(output_dir, output_zip)

    create_checkpoint_zip(output_dir, output_zip)
    return {"processed": processed, "skipped": skipped, "pages": total_pages}
```

Listing G.5: Headless Kaggle Batch Transcription and Checkpoint Packaging

# Appendix H

# User Interface Wireframes

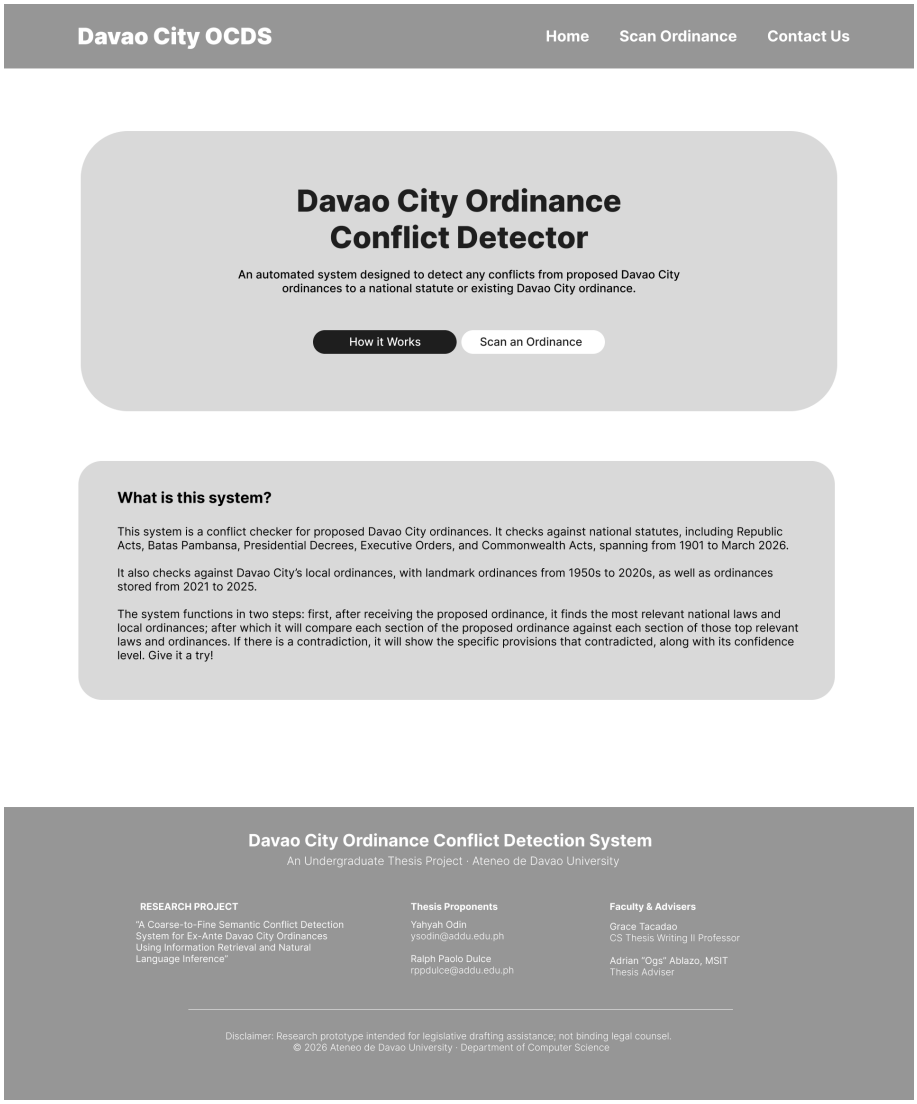



Figure H.1: User Interface Wireframe: Homepage and Draft Ingestion Workspace.

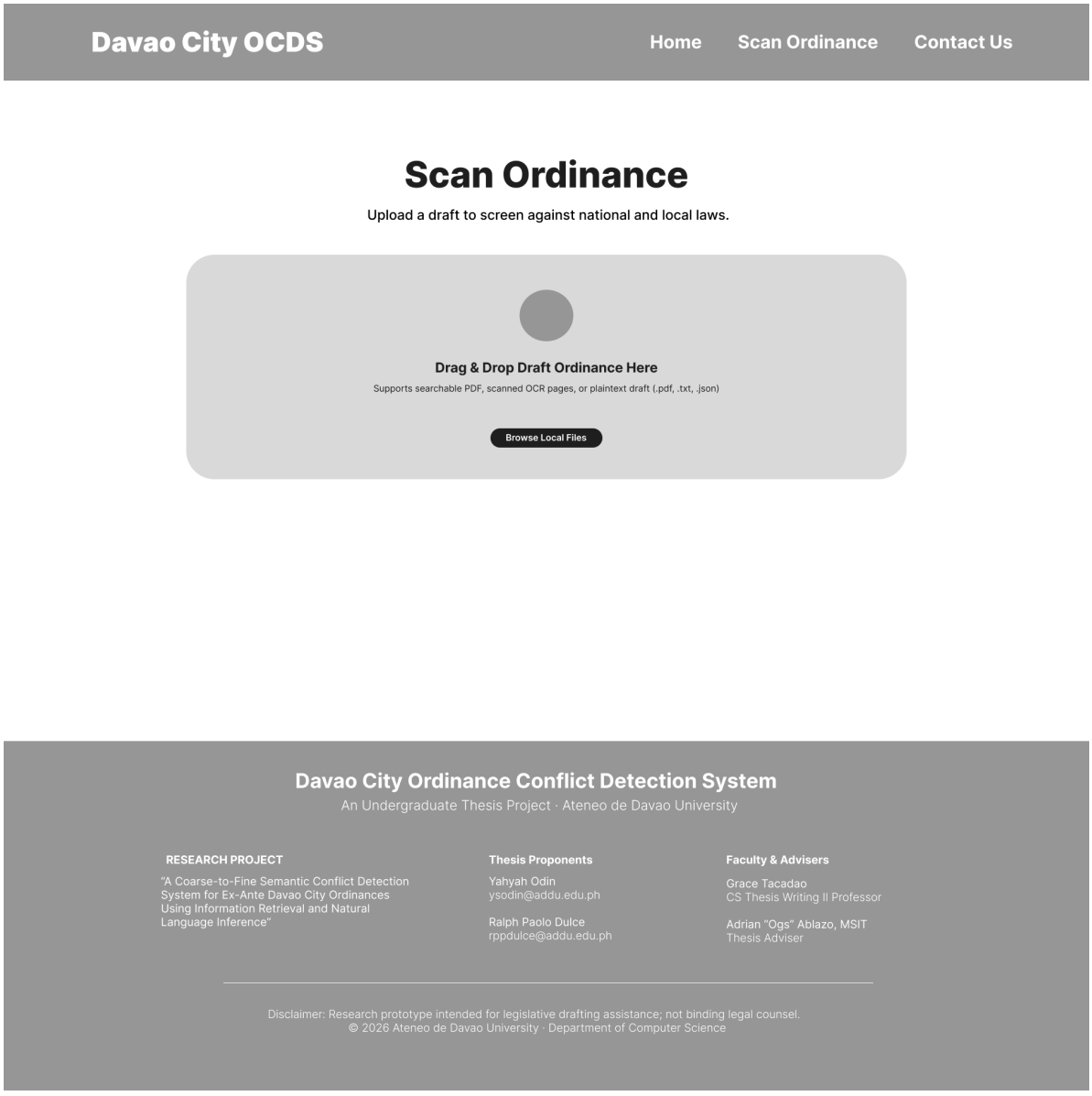



Figure H.2: User Interface Wireframe: Real-Time Legislative Audit Execution View.

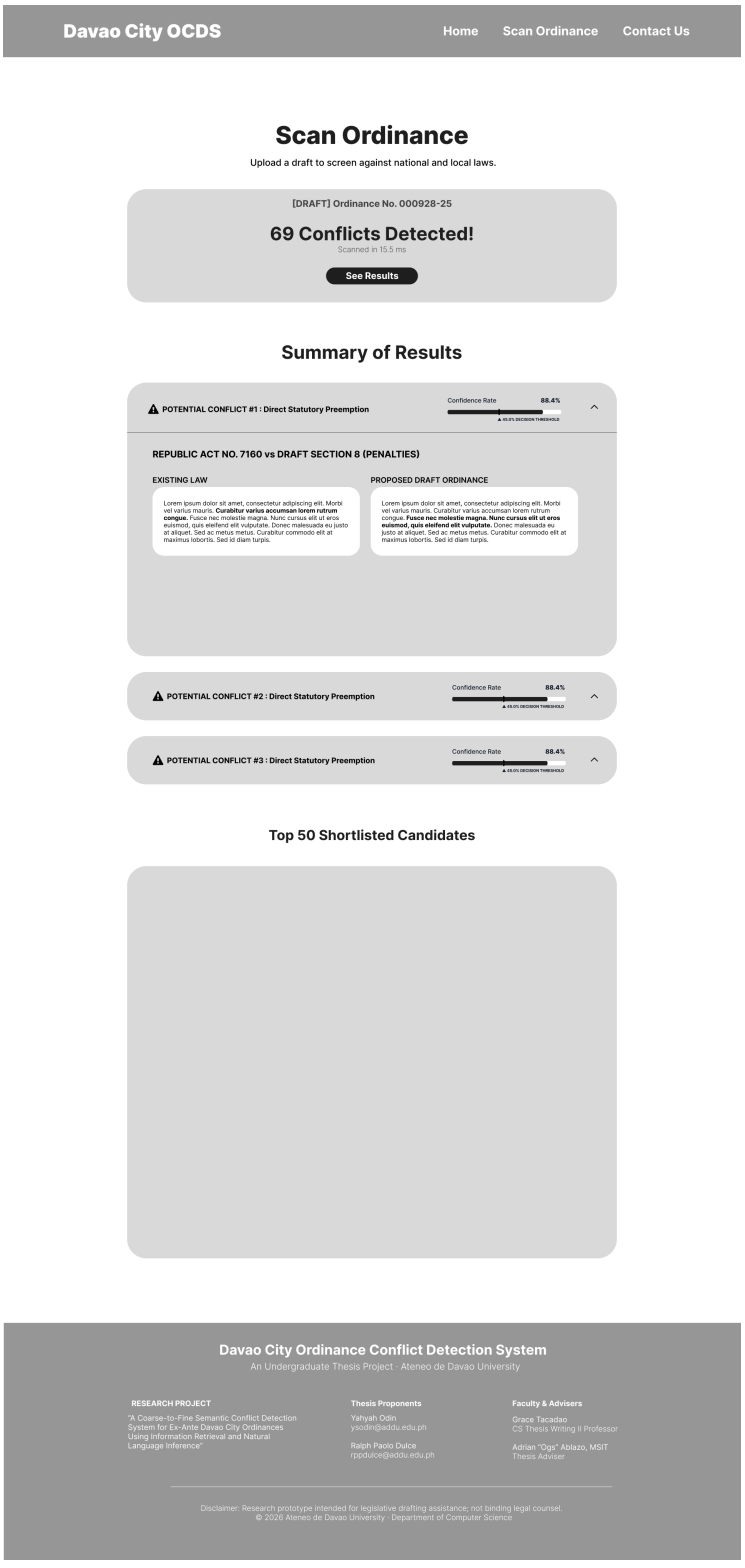



Figure H.3: User Interface Wireframe: Conflict Detection Diagnostic Results and Provision Comparison View.

# Appendix I

# Prototype Dashboard Architecture

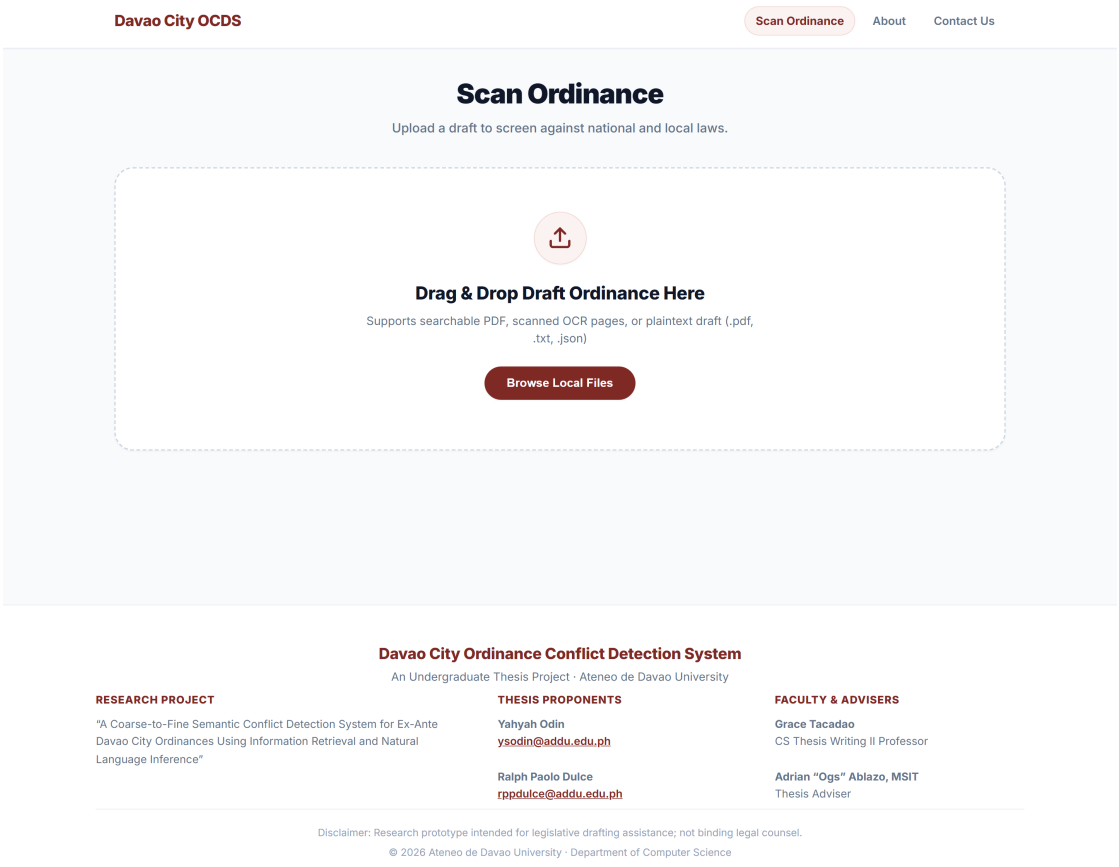



Figure I.1: Draft Ordinance Ingestion Workspace (Scan Ordinance View)

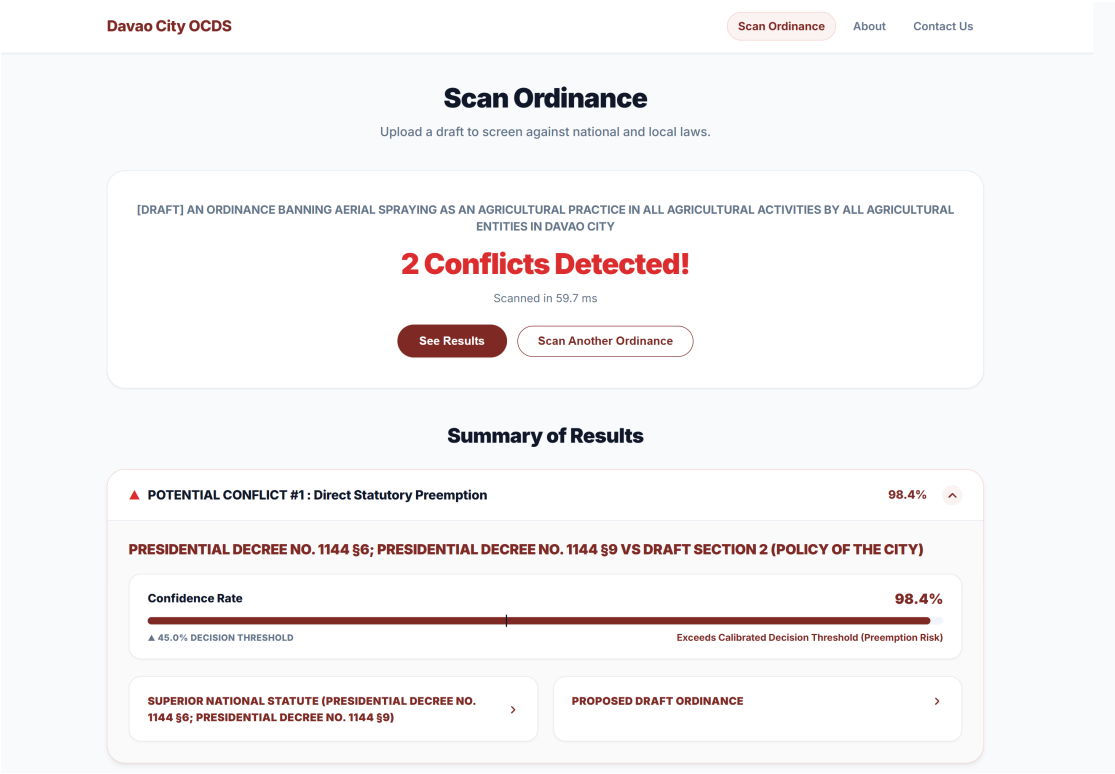



Figure I.2: Audit Results and Conflict Detection Summary View

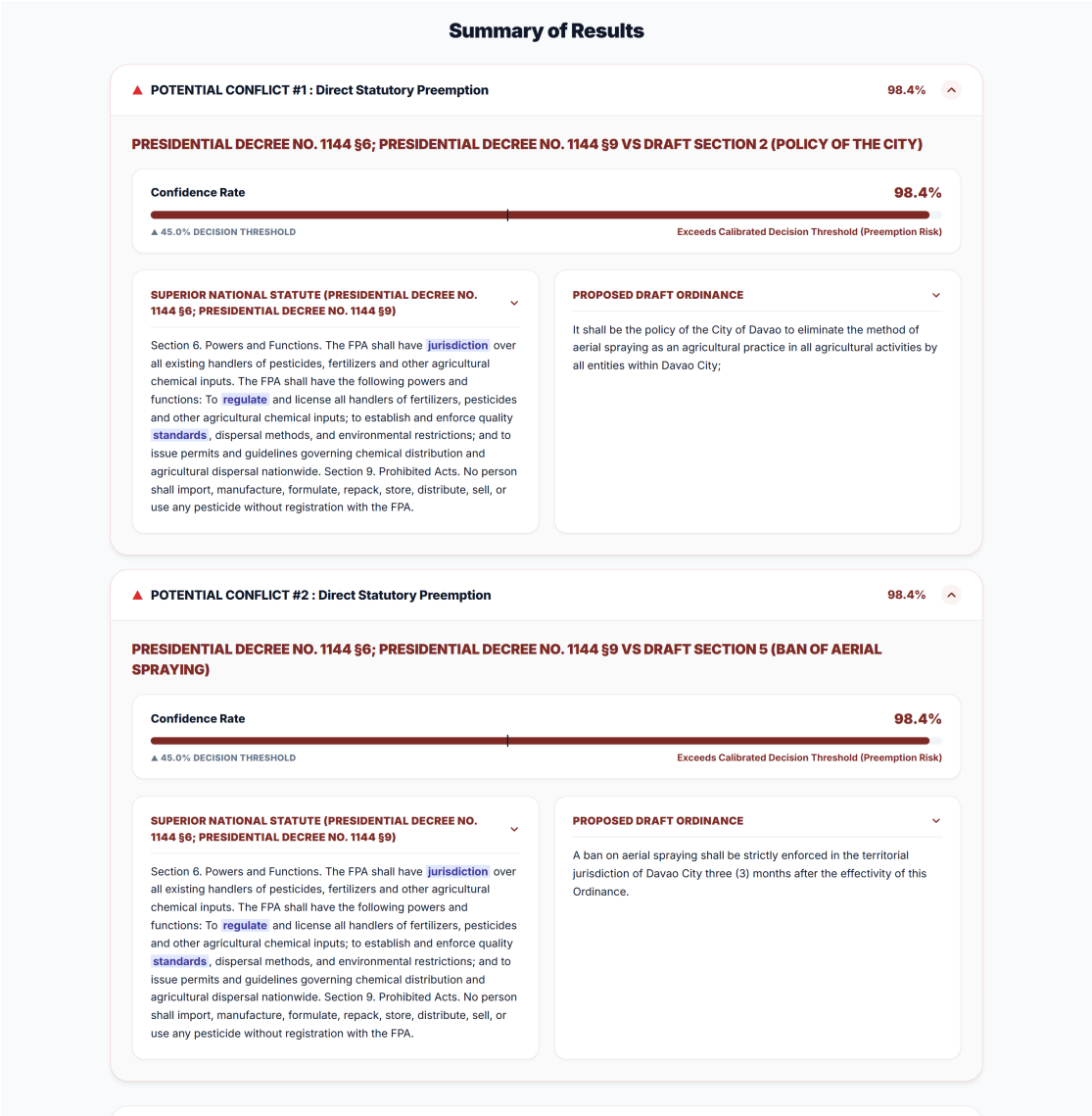



Figure I.3: Expanded Conflict Accordion with Verbatim Statutory Comparison

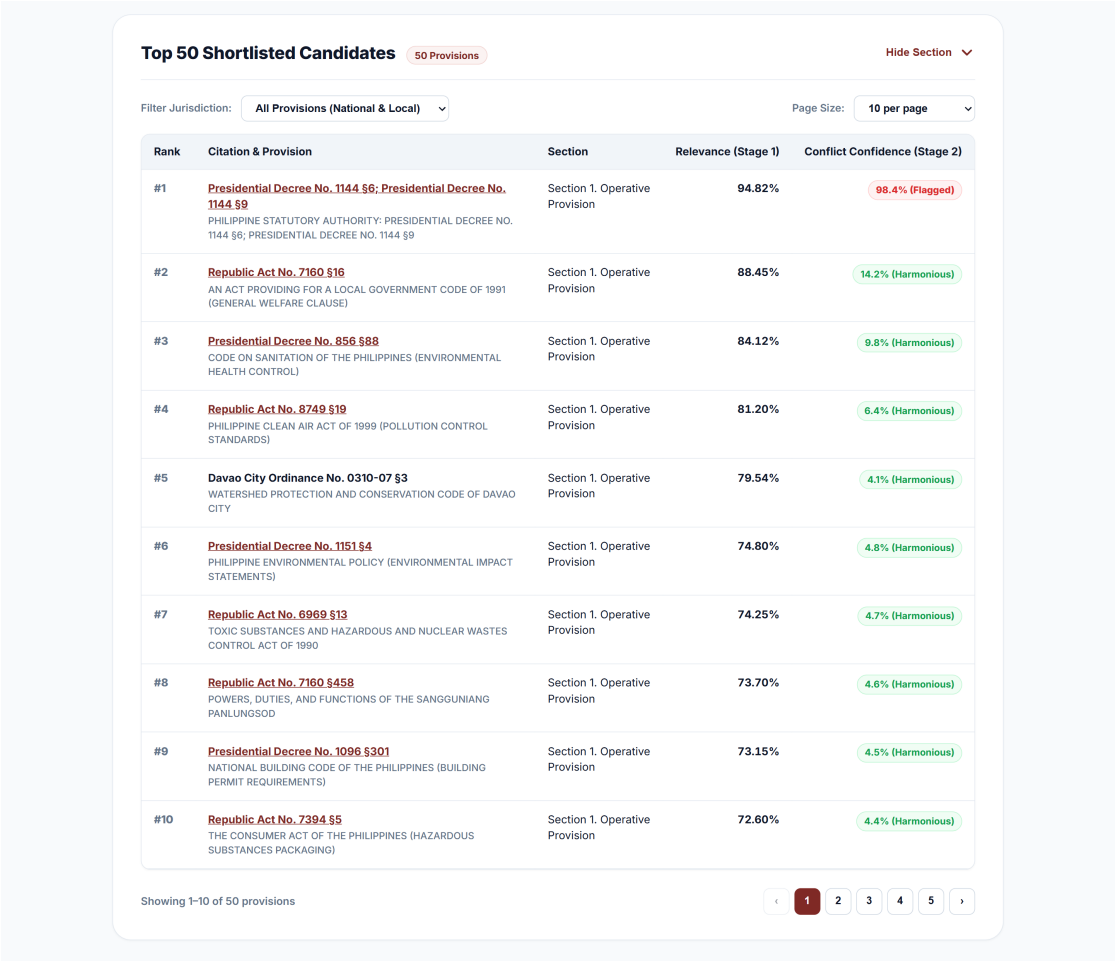



Figure I.4: Top-50 Statutory Candidate Shortlist Table with Pagination and Jurisdiction Filter

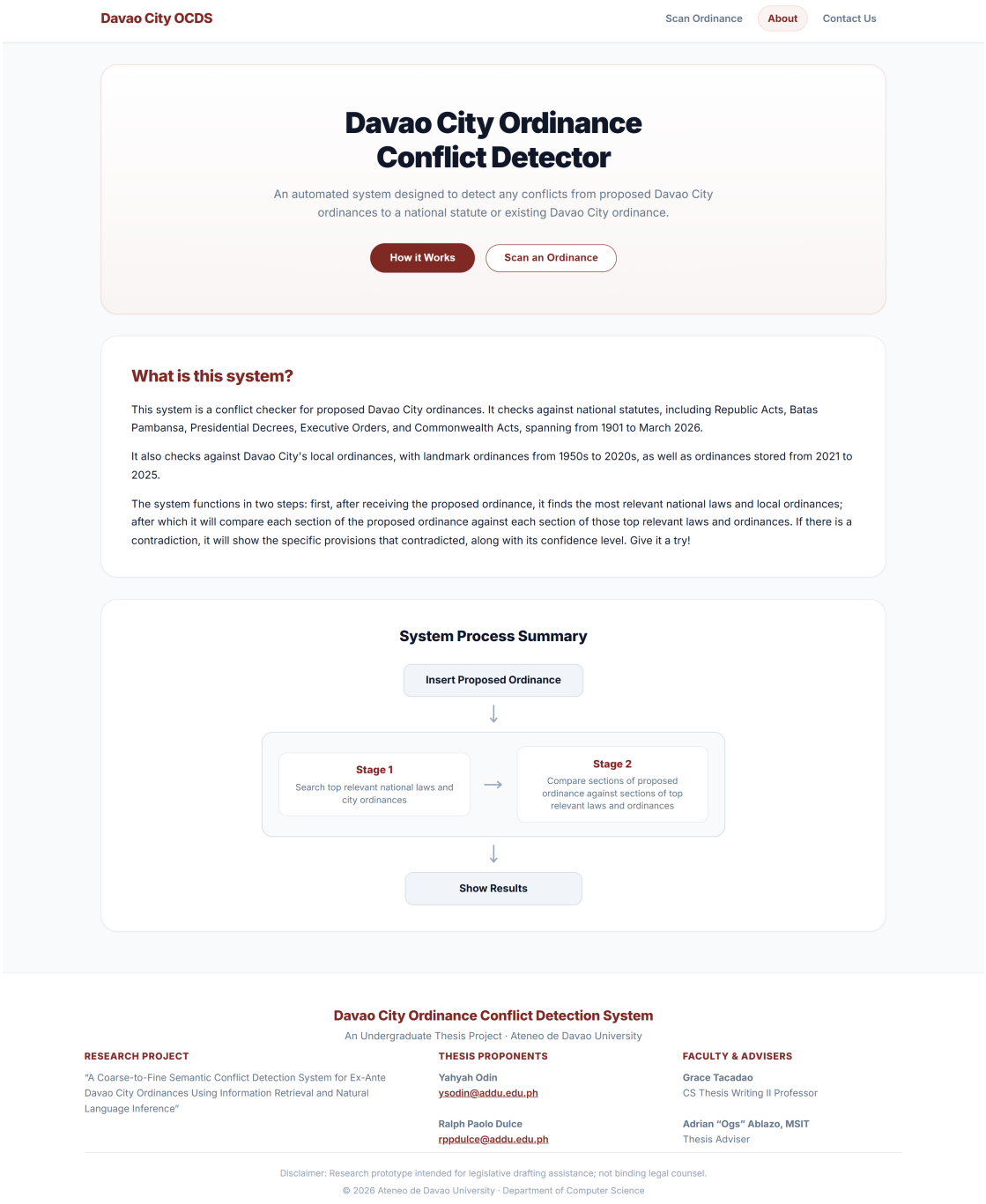



Figure I.5: System Background and Two-Stage Pipeline Overview (About View)

# Appendix J

# Legal Annotation Guide

**SANGGUNIANG PANLUNGSOD LEGAL ANNOTATION GUIDE**

*Evaluating National Statute vs. Local Ordinance Legal Consistency*

Ateneo de Davao University — Computer Studies Cluster

**Thesis Title:** *A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference*
**Thesis Proponents:** Ralph Paolo Dulce & Yahyah Odin
**Advisory & Faculty:** Adrian Ablazo (Thesis Adviser) | Grace Tacadao (Thesis Professor)
**Annotators:** 15 Legal Researchers / Legislative Staff, Sangguniang Panlungsod of Davao City

## J.1 Project Purpose and Your Role

Welcome! As a legal researcher or legislative staff member of the Sangguniang Panlungsod, your everyday work ensures that Davao City ordinances are well-crafted, effective, and legally sound.

For our undergraduate thesis, we are building an automated tool to assist local governments in screening draft ordinance provisions before they are enacted, flagging potential legal clashes against national laws early during the committee review process. To evaluate and train this system, we need Domain Expert Ground Truth—authoritative, real-world consistency judgments from legal professionals like you.

We are truly grateful for your time, dedication, and generosity in sharing your valuable domain expertise to support our undergraduate thesis. Your contributions play a vital role in advancing this research and shaping a meaningful, impactful tool for legislative work.

## J.2 The Core Task: Comparing Text Pairs

For each assigned item, you will compare two legal texts:

1. **National Statute (The Standard):** A specific section from a Philippine Republic Act or national law.
2. **Proposed Local Ordinance (The Draft Clause):** A drafted section of a proposed Davao City ordinance.

When evaluating each item, your key guiding question is:

> *"Based strictly on this specific national statute provision, does the local ordinance draft conflict with it, comply with it, or regulate a completely separate matter?"*

**Evaluation Rule:** Judge each pair only against the specific National Statute section provided in that item. Do not bring in other unlisted statutes from memory.

## J.3 The Three Choices Explained

### J.3.1 Choice 1: Contradiction (Legal Conflict / Overreach)

**Meaning:** The draft local ordinance violates, weakens, or exceeds what the national statute establishes (*ultra vires* under the *Magtajas* doctrine).

- **Exceeding Penalty Caps:** National law (RA 7160 §458) strictly limits city ordinance fines to ₱5,000 and imprisonment to 1 year. A local draft setting a ₱25,000 fine or real property title forfeiture is a contradiction.
- **Exceeding Time Limits:** National law (RA 7581 §6) limits automatic calamity price freezes to 60 days. A draft setting a 180-day freeze is a contradiction.
- **Withholding Guaranteed Rights/Discounts:** National law (RA 11314) guarantees a 20% student fare discount all year. A draft cutting it to 10% or suspending it during summer breaks/holidays is a contradiction.
- **Encroaching on National Agency Powers:** National law gives the Energy Regulatory Commission (ERC) exclusive power over power rates and Congress/NTC exclusive power over telecom franchises. A draft attempting to set local electric rate caps or issue telecom franchises is a contradiction.
- **Euphemisms Hiding Violations:** National law (RA 9344) prohibits detaining minors for curfew violations. A draft that holds minors overnight in lock-up facilities (even if politely termed "overnight reflection in a youth sanctuary") is a contradiction.

### J.3.2 Choice 2: Entailment (Compliant / Lawful Execution)

**Meaning:** The draft local ordinance directly adopts, enforces, or lawfully executes what the national law requires or authorizes.

- **Direct Adoption:** The local draft adopts national highway speed limits (e.g., 80 km/h for passenger cars under RA 4136).
- **Statutory Budget Formulas:** The draft appropriates exactly 20% of its National Tax Allotment to local development (RA 7160 §287) or 5% of regular revenues to disaster funds (RA 10121).
- **Respecting Required Clearance Steps:** The draft requires sand-and-gravel quarry operators to present a national DENR Environmental Compliance Certificate (ECC) before receiving a Mayor's Permit.
- **Non-Custodial Youth Care:** The draft directs apprehended youth to community-based counseling rather than detention, adhering to RA 9344.

### J.3.3 Choice 3: Neutral (Independent / Unrelated Subject Matter)

**Meaning:** Both laws may belong to the same broad field (e.g., health, transportation, taxation), but they govern completely separate obligations and do not conflict or depend on each other.

- **Health:** National law mandates reporting notifiable infectious diseases to DOH, while the local draft sets hygiene rules for wiping waiting room benches or banning mercury thermometers.
- **Local Governance:** National law sets the juvenile criminal exemption age (15 years old), while the local draft sets minimum floor dimensions for daycare centers (30 sq.m.).
- **Public Utilities:** National law sets electric utility rate jurisdiction, while the local draft specifies reflective tape widths on street sweeper vests or median shrub trimming heights.

## J.4 Quick 3-Step Decision Checklist

1. **Step 1:** Does the draft clash with, exceed, or remove what national law provides? → **YES:** Choose **CONTRADICTION**
2. **Step 2:** Does the draft directly enforce, execute, or stay within the national rule? → **YES:** Choose **ENTAILMENT**
3. **Step 3:** Are the two laws talking about completely separate administrative matters? → **YES:** Choose **NEUTRAL**

## J.5 Desk Cheat Sheet: Key Legal Ceilings and Thresholds

Table J.1: Key Statutory Ceilings and Preemption Red Flags Under Philippine Law

| Subject Area | National Law Baseline | Red Flag in Local Draft (Contradiction) |
|---|---|---|
| **Ordinance Penalties** | Max ₱5,000 fine / 1 year jail (RA 7160 §458). | Drafts imposing ₱10,000–₱25,000 fines or real property title confiscation. |
| **Calamity Price Freeze** | Max 60 days duration (RA 7581 §6). | Drafts imposing 90-day or 180-day local price freezes. |
| **Processing Time (EODB)** | Max 3 days (Simple), 7 days (Complex) (RA 11032). | Drafts taking 15 days for simple permits or indefinite holds. |
| **Student Fare Discount** | 20% discount, year-round on all transit (RA 11314). | Drafts cutting discounts to 10% or suspending during vacations. |
| **Juvenile Justice** | Age 15 and below exempt from criminal liability (RA 9344). | Drafts locking up minors or charging 12-year-olds in court. |
| **Real Property Tax** | Basic tax capped at max 2% for cities (RA 7160 §233). | Drafts setting 3% to 5%, or hiking rates without council ordinance. |
| **Amusement Tax** | Capped at max 30%; school plays/events exempt (RA 7160 §140). | Drafts charging 35%, or levying tax on school dramatic/athletic events. |
| **Local Dev. Fund** | At least 20% of Annual Tax Allotment (RA 7160 §287). | Drafts allocating less than 20% (e.g., 7%) or using funds for travel. |
| **Disaster Fund (LDRRMF)** | At least 5% of regular revenue; 30% Quick Response (RA 10121). | Drafts setting 2%, using QRF for tourism, or reverting unspent funds. |

## J.6 Submission and Transmittal

Evaluations are submitted through the standardized digital assessment instrument hosted on Google Forms. In accordance with Republic Act No. 10173 and ethical research approvals, each evaluator evaluates exactly one panel of 70 short legal pairs. Upon conclusion of full-scale deployment, participating legal officers receive the official Certificate of Contribution and Commendation endorsed for their Civil Service 201 records.

# Appendix K

# Consistency Evaluation Questionnaire

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### SP Davao City – Legal Consistency Ground Truth Evaluation

**Introductory Statement**

Magis Greetings!

We are Ralph Paolo Dulce and Yahyah Odin, senior Computer Science students from Ateneo de Davao University, under the advisory of Mr. Adrian "Ogs" Ablazo and course guidance of Ma'am Grace Tacadao. In institutional coordination with the Sangguniang Panlungsod of Davao City, we are conducting our undergraduate thesis entitled: *"A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference."*

**Purpose and Your Role:** Under RA 7160 and the *Magtajas v. Pryce* doctrine, local ordinances cannot contradict national statutes. To aid Second Reading legislative review, our study develops an automated screening tool to flag potential conflicts in proposed Davao City Ordinances against national laws and enacted city ordinances before enactment. As SP legal researchers, your independent legal judgments serve as the expert ground truth benchmark, or essentially the reference standard, to evaluate our system's accuracy.

**Workload:** You are reviewing ONE batch of 70 short provision pairs (taking approximately 60 minutes). You will NOT be asked to review the full 350-item dataset. We deeply appreciate your valuable time, legal expertise, and service in supporting this research. Your insights will directly contribute to advancing data-driven decision support tools tailored to the legislative review workflow of the Davao City Council.

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**Data Privacy Notice & Informed Consent (R.A. 10173)**

Please read the ethical guidelines and data privacy terms governing this research evaluation:

1. **Statutory Compliance & Institutional Oversight:** In strict compliance with Republic Act No. 10173 (The Data Privacy Act of 2012), its Implementing Rules and Regulations, and the ethical standards of the Ateneo de Davao University Research Ethics Committee, this study guarantees the full confidentiality, privacy, and intellectual integrity of your responses.

2. **Purpose & Scope of Data Collection:** All collected data, including your professional background and legal ratings (Contradiction, Entailment, Neutral), will be utilized exclusively for academic research within this undergraduate thesis to benchmark the detection accuracy and inter-rater reliability of an automated statutory conflict screening system.

3. **Strict Confidentiality & Evaluator Anonymization:**

   - *Anonymized Analysis:* All evaluations will be analyzed and presented strictly in aggregate form (e.g., domain accuracy rates, confusion matrices). No individual response will ever be singled out or publicly attributed to any researcher.

   - *Coded Identifiers:* For statistical tracking across the 350-pair benchmark (comprising 198 unique instances), your responses will be processed using an anonymized alphanumeric identifier (e.g., Evaluator-01 to Evaluator-15).

   - *Voluntary Name Provision:* Providing your name in the subsequent demographic section is purely OPTIONAL. If provided, your name will be used strictly by the proponents and faculty to generate a Certificate of Contribution and Commendation endorsed for inclusion in your Civil Service 201 personnel records. If you choose to leave the name field blank, your evaluation will remain completely anonymous.

4. **Voluntary Participation & Right to Withdraw:** Your participation in this evaluation is entirely voluntary. You may pause, skip any question, or withdraw from this study at any stage without incurring any administrative, professional, or employment-related consequences, and without prejudice to your standing at the Sangguniang Panlungsod of Davao City.

5. **Data Retention & Security Safeguards:** All raw submission data will be securely stored within password-protected, encrypted institutional storage accessible solely to the thesis proponents and their academic adviser. Data will be retained only for the duration required for thesis defense and institutional archiving, after which any identifying administrative logs will be permanently deleted.

**Informed Consent Checkbox:**

☐ *I agree and consent to the statement above.*

By selecting the checkbox, you confirm that: (1) you are an active legal researcher, committee staff member, or legislative officer at the Sangguniang Panlungsod of Davao City; (2) you have read and understood the terms of this Data Privacy and Ethics Disclosure; and (3) you freely and voluntarily consent to participate as a domain expert evaluator for this research study.

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**Evaluator Details and Assignment**

1. **Full Name (Optional):** _____

2. **Highest Educational Attainment (Required):**

   ◯ Member of the Philippine Bar (Admitted / Roll of Attorneys)

   ◯ Juris Doctor (J.D.) / Bachelor of Laws (LL.B.)

   ◯ Postgraduate / Master's Degree (e.g., Master in Public Administration, LL.M.)

   ◯ Bachelor's Degree (e.g., Political Science, Legal Management, Philosophy, etc.)

3. **Years of Legislative / Legal Experience (Required):** _____
   *Number of years in legislative drafting, statutory research, ordinance codification, committee review, etc.*

4. **Automated Workload Assignment (Pre-Allocated 70-Item Workload):**

   ◯ SP-ANN-01 to SP-ANN-03 (Panel A: Items GT-001 to GT-070)

   ◯ SP-ANN-04 to SP-ANN-06 (Panel B: Items GT-071 to GT-140)

   ◯ SP-ANN-07 to SP-ANN-09 (Panel C: Items GT-141 to GT-210)

   ◯ SP-ANN-10 to SP-ANN-12 (Panel D: Items GT-211 to GT-280)

   ◯ SP-ANN-13 to SP-ANN-15 (Panel E: Items GT-281 to GT-350)

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**Pairwise Consistency Evaluation Item Structure**

For each assigned item, evaluators read the Controlling National Statute provision and the corresponding Davao City Ordinance provision, then answer the following two standard questions:

1. **Question 1: Legal Consistency Decision (Required):**
   *Mark only one oval.*

   ◯ Contradiction (Legal Conflict / Exceeds Authority)

   ◯ Entailment (Complies With / Directly Executes)

◯ Neutral (Independent / Unrelated Subject Matter)

2. **Question 2: Confidence Rating (Required):**
   *Mark only one oval.*

   ◯ 1 - Low

   ◯ 2

   ◯ 3 - High

**Cross-Reference to Evaluated Text Pairs:**

The complete inventory of all 350 evaluated statutory-ordinance text pairs (Items GT-001 to GT-350) is presented in Appendix M, accompanied by the master ground truth answer key.

# Appendix L

# Annotator Certificate of Contribution

Figure L.1: Sangguniang Panlungsod Legal Annotator Certificate of Contribution Layout.

# Appendix M

# Ground Truth Evaluation Pairs

## M.1 Corpus Stratification and Master Answer Key

Table M.1: Master Ground Truth Corpus Stratification across Eight Macro Legal Domains ($N = 350$)

| Domain ID | Consolidated Macro Legal Domain | Contradiction | Entailment | Neutral | Total Pairs |
|---|---|---|---|---|---|
| 00 | Executive Issuances & Policy Reorganization | 16 | 17 | 17 | 50 |
| 01 | Education & Academic Institutions | 14 | 17 | 14 | 45 |
| 02 | Local Government & Territorial Boundaries | 16 | 17 | 17 | 50 |
| 03 | Public Utilities & Telecom Franchises | 14 | 17 | 14 | 45 |
| 04 | Public Health, Hospitals & Medical Services | 14 | 17 | 14 | 45 |
| 05 | Statutory Codes & General Legal Amendments | 14 | 17 | 14 | 45 |
| 06 | Public Finance & General Appropriations | 12 | 13 | 10 | 35 |
| 07 | Taxation, Tariffs & Revenue Administration | 12 | 13 | 10 | 35 |
| **Total** | **All 8 Consolidated Functional Domains** | **112** | **128** | **110** | **350** |

Table M.2: Master Ground Truth Item Answer Key ($N = 350$)

| Items 001–070 | | Items 071–140 | | Items 141–210 | | Items 211–280 | | Items 281–350 | |
|---|---|---|---|---|---|---|---|---|---|
| ID | Label | ID | Label | ID | Label | ID | Label | ID | Label |
| GT-001 | Contra | GT-071 | Entail | GT-141 | Neutral | GT-211 | Entail | GT-281 | Contra |
| GT-002 | Contra | GT-072 | Entail | GT-142 | Neutral | GT-212 | Entail | GT-282 | Contra |

| Items 001–070 | | Items 071–140 | | Items 141–210 | | Items 211–280 | | Items 281–350 | |
|---|---|---|---|---|---|---|---|---|---|
| **ID** | **Label** | **ID** | **Label** | **ID** | **Label** | **ID** | **Label** | **ID** | **Label** |
| GT-003 | Contra | GT-073 | Entail | GT-143 | Neutral | GT-213 | Entail | GT-283 | Contra |
| GT-004 | Contra | GT-074 | Entail | GT-144 | Neutral | GT-214 | Entail | GT-284 | Contra |
| GT-005 | Contra | GT-075 | Entail | GT-145 | Neutral | GT-215 | Entail | GT-285 | Entail |
| GT-006 | Entail | GT-076 | Entail | GT-146 | Contra | GT-216 | Entail | GT-286 | Entail |
| GT-007 | Entail | GT-077 | Neutral | GT-147 | Contra | GT-217 | Neutral | GT-287 | Entail |
| GT-008 | Entail | GT-078 | Neutral | GT-148 | Contra | GT-218 | Neutral | GT-288 | Entail |
| GT-009 | Entail | GT-079 | Neutral | GT-149 | Contra | GT-219 | Neutral | GT-289 | Neutral |
| GT-010 | Entail | GT-080 | Neutral | GT-150 | Entail | GT-220 | Neutral | GT-290 | Neutral |
| GT-011 | Neutral | GT-081 | Neutral | GT-151 | Entail | GT-221 | Neutral | GT-291 | Neutral |
| GT-012 | Neutral | GT-082 | Neutral | GT-152 | Entail | GT-222 | Neutral | GT-292 | Contra |
| GT-013 | Neutral | GT-083 | Contra | GT-153 | Entail | GT-223 | Contra | GT-293 | Contra |
| GT-014 | Neutral | GT-084 | Contra | GT-154 | Entail | GT-224 | Contra | GT-294 | Contra |
| GT-015 | Neutral | GT-085 | Contra | GT-155 | Neutral | GT-225 | Contra | GT-295 | Contra |
| GT-016 | Contra | GT-086 | Contra | GT-156 | Neutral | GT-226 | Contra | GT-296 | Entail |
| GT-017 | Contra | GT-087 | Entail | GT-157 | Neutral | GT-227 | Entail | GT-297 | Entail |
| GT-018 | Contra | GT-088 | Entail | GT-158 | Neutral | GT-228 | Entail | GT-298 | Entail |
| GT-019 | Contra | GT-089 | Entail | GT-159 | Contra | GT-229 | Entail | GT-299 | Entail |
| GT-020 | Contra | GT-090 | Entail | GT-160 | Contra | GT-230 | Entail | GT-300 | Entail |
| GT-021 | Contra | GT-091 | Entail | GT-161 | Contra | GT-231 | Entail | GT-301 | Neutral |
| GT-022 | Entail | GT-092 | Neutral | GT-162 | Contra | GT-232 | Neutral | GT-302 | Neutral |
| GT-023 | Entail | GT-093 | Neutral | GT-163 | Contra | GT-233 | Neutral | GT-303 | Neutral |
| GT-024 | Entail | GT-094 | Neutral | GT-164 | Contra | GT-234 | Neutral | GT-304 | Neutral |
| GT-025 | Entail | GT-095 | Neutral | GT-165 | Entail | GT-235 | Neutral | GT-305 | Contra |
| GT-026 | Entail | GT-096 | Contra | GT-166 | Entail | GT-236 | Contra | GT-306 | Contra |
| GT-027 | Entail | GT-097 | Contra | GT-167 | Entail | GT-237 | Contra | GT-307 | Contra |
| GT-028 | Entail | GT-098 | Contra | GT-168 | Entail | GT-238 | Contra | GT-308 | Contra |
| GT-029 | Neutral | GT-099 | Contra | GT-169 | Entail | GT-239 | Contra | GT-309 | Entail |
| GT-030 | Neutral | GT-100 | Contra | GT-170 | Entail | GT-240 | Entail | GT-310 | Entail |
| GT-031 | Neutral | GT-101 | Entail | GT-171 | Entail | GT-241 | Entail | GT-311 | Entail |
| GT-032 | Neutral | GT-102 | Entail | GT-172 | Neutral | GT-242 | Entail | GT-312 | Entail |
| GT-033 | Neutral | GT-103 | Entail | GT-173 | Neutral | GT-243 | Entail | GT-313 | Neutral |
| GT-034 | Neutral | GT-104 | Entail | GT-174 | Neutral | GT-244 | Entail | GT-314 | Neutral |
| GT-035 | Neutral | GT-105 | Entail | GT-175 | Neutral | GT-245 | Neutral | GT-315 | Neutral |
| GT-036 | Contra | GT-106 | Neutral | GT-176 | Neutral | GT-246 | Neutral | GT-316 | Contra |
| GT-037 | Contra | GT-107 | Neutral | GT-177 | Neutral | GT-247 | Neutral | GT-317 | Contra |
| GT-038 | Contra | GT-108 | Neutral | GT-178 | Contra | GT-248 | Neutral | GT-318 | Contra |
| GT-039 | Contra | GT-109 | Neutral | GT-179 | Contra | GT-249 | Contra | GT-319 | Contra |
| GT-040 | Contra | GT-110 | Neutral | GT-180 | Contra | GT-250 | Contra | GT-320 | Entail |
| GT-041 | Entail | GT-111 | Contra | GT-181 | Contra | GT-251 | Contra | GT-321 | Entail |
| GT-042 | Entail | GT-112 | Contra | GT-182 | Entail | GT-252 | Contra | GT-322 | Entail |
| GT-043 | Entail | GT-113 | Contra | GT-183 | Entail | GT-253 | Contra | GT-323 | Entail |
| GT-044 | Entail | GT-114 | Contra | GT-184 | Entail | GT-254 | Contra | GT-324 | Neutral |
| GT-045 | Entail | GT-115 | Contra | GT-185 | Entail | GT-255 | Entail | GT-325 | Neutral |
| GT-046 | Neutral | GT-116 | Contra | GT-186 | Entail | GT-256 | Entail | GT-326 | Neutral |
| GT-047 | Neutral | GT-117 | Entail | GT-187 | Neutral | GT-257 | Entail | GT-327 | Contra |
| GT-048 | Neutral | GT-118 | Entail | GT-188 | Neutral | GT-258 | Entail | GT-328 | Contra |
| GT-049 | Neutral | GT-119 | Entail | GT-189 | Neutral | GT-259 | Entail | GT-329 | Contra |
| GT-050 | Neutral | GT-120 | Entail | GT-190 | Neutral | GT-260 | Entail | GT-330 | Contra |
| GT-051 | Contra | GT-121 | Entail | GT-191 | Contra | GT-261 | Entail | GT-331 | Entail |
| GT-052 | Contra | GT-122 | Entail | GT-192 | Contra | GT-262 | Neutral | GT-332 | Entail |
| GT-053 | Contra | GT-123 | Entail | GT-193 | Contra | GT-263 | Neutral | GT-333 | Entail |
| GT-054 | Contra | GT-124 | Neutral | GT-194 | Contra | GT-264 | Neutral | GT-334 | Entail |
| GT-055 | Entail | GT-125 | Neutral | GT-195 | Entail | GT-265 | Neutral | GT-335 | Entail |
| GT-056 | Entail | GT-126 | Neutral | GT-196 | Entail | GT-266 | Neutral | GT-336 | Neutral |
| GT-057 | Entail | GT-127 | Neutral | GT-197 | Entail | GT-267 | Neutral | GT-337 | Neutral |
| GT-058 | Entail | GT-128 | Neutral | GT-198 | Entail | GT-268 | Contra | GT-338 | Neutral |
| GT-059 | Entail | GT-129 | Neutral | GT-199 | Entail | GT-269 | Contra | GT-339 | Neutral |
| GT-060 | Neutral | GT-130 | Neutral | GT-200 | Neutral | GT-270 | Contra | GT-340 | Contra |
| GT-061 | Neutral | GT-131 | Contra | GT-201 | Neutral | GT-271 | Contra | GT-341 | Contra |
| GT-062 | Neutral | GT-132 | Contra | GT-202 | Neutral | GT-272 | Entail | GT-342 | Contra |
| GT-063 | Neutral | GT-133 | Contra | GT-203 | Neutral | GT-273 | Entail | GT-343 | Contra |
| GT-064 | Contra | GT-134 | Contra | GT-204 | Contra | GT-274 | Entail | GT-344 | Entail |
| GT-065 | Contra | GT-135 | Contra | GT-205 | Contra | GT-275 | Entail | GT-345 | Entail |
| GT-066 | Contra | GT-136 | Entail | GT-206 | Contra | GT-276 | Entail | GT-346 | Entail |
| GT-067 | Contra | GT-137 | Entail | GT-207 | Contra | GT-277 | Neutral | GT-347 | Entail |
| GT-068 | Contra | GT-138 | Entail | GT-208 | Contra | GT-278 | Neutral | GT-348 | Neutral |
| GT-069 | Contra | GT-139 | Entail | GT-209 | Contra | GT-279 | Neutral | GT-349 | Neutral |
| GT-070 | Entail | GT-140 | Entail | GT-210 | Entail | GT-280 | Neutral | GT-350 | Neutral |

## M.2 Domain 00: Executive Issuances and Policy Reorganization (Items GT-001 to GT-050)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-001]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 4* | **[Gold: Contradiction]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 4. MANDATORY LOCAL PRICE CEILINGS UPON CALAMITY. — In accordance with local civil defense standards and the emergency management powers of the City Government of Davao, upon official declaration of a state of local calamity by the Sangguniang Panlungsod, the City Mayor, through the City Economic Enterprise Office and the Business Bureau, shall enforce mandatory price ceilings across all grocery and commercial retail establishments within Davao City, which price freeze shall remain in continuous legal effect for a mandatory period of one hundred eighty (180) calendar days from issuance. The City Disaster Risk Reduction and Management Office (CDRRMO) shall maintain continuous operational oversight and issue the corresponding administrative guidelines."* |
| **[GT-002]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 3* | **[Gold: Contradiction]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 3. REORGANIZATION OF DISASTER OPERATIONS. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, the Davao City Disaster Risk Reduction and Management Office (CDRRMO) is hereby dissolved and reconstituted as a private non-governmental foundation operating outside the administrative control of the Office of the City Mayor, governed by an independent civilian board. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-003]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 8* | **[Gold: Contradiction]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 8. PROCESSING PERIOD FOR SIMPLE MAYOR'S PERMITS. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the Business Bureau of Davao City is authorized to process and issue simple business registration renewal clearances within an administrative evaluation timeframe of fifteen (15) working days from the date of complete application submission. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-004]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 4* | **[Gold: Contradiction]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 4. MANDATORY LOCAL PRICE CEILINGS UPON CALAMITY. — In accordance with local civil defense standards and the emergency management powers of the City Government of Davao, upon official declaration of a state of local calamity by the Sangguniang Panlungsod, the City Mayor, through the City Economic Enterprise Office and the Business Bureau, shall enforce mandatory price ceilings across all grocery and commercial retail establishments within Davao City, which price freeze shall remain in continuous legal effect for a mandatory period of one hundred eighty (180) calendar days from issuance. The City Disaster Risk Reduction and Management Office (CDRRMO) shall maintain continuous operational oversight and issue the corresponding administrative guidelines."* |
| **[GT-005]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 3* | **[Gold: Contradiction]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 3. REORGANIZATION OF DISASTER OPERATIONS. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, the Davao City Disaster Risk Reduction and Management Office (CDRRMO) is hereby dissolved and reconstituted as a private non-governmental foundation operating outside the administrative control of the Office of the City Mayor, governed by an independent civilian board. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-006]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 3. LOCAL MONITORING OF PRICE FREEZE DURATION. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, in strict observance of Section 6 of Republic Act No. 7581, the Davao City Local Price Coordinating Council shall monitor retail establishments upon the official declaration of a state of calamity to guarantee that prevailing retail prices of basic necessities remain frozen for a duration not exceeding sixty (60) calendar days. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-007]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 2* | **[Gold: Entailment]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 2. INSTITUTIONAL STRENGTHENING OF CDRRMO. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, in compliance with Section 12 of Republic Act No. 10121, the City Disaster Risk Reduction and Management Office (CDRRMO) is organized as a regular department under the Office of the City Mayor, structured with three (3) functional divisions: Administrative and Training, Research and Planning, and Operations and Warning. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-008]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 4* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 4. MANDATORY ADHERENCE TO PROCESSING TIMELINES. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, in strict adherence to Section 9 of Republic Act No. 11032, the Business Bureau shall process and approve applications for simple business permits within three (3) working days, and complex commercial applications within seven (7) working days, from initial receipt of complete documents. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-009]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 3. LOCAL MONITORING OF PRICE FREEZE DURATION. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, in strict observance of Section 6 of Republic Act No. 7581, the Davao City Local Price Coordinating Council shall monitor retail establishments upon the official declaration of a state of calamity to guarantee that prevailing retail prices of basic necessities remain frozen for a duration not exceeding sixty (60) calendar days. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-010]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 2* | **[Gold: Entailment]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 2. INSTITUTIONAL STRENGTHENING OF CDRRMO. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, in compliance with Section 12 of Republic Act No. 10121, the City Disaster Risk Reduction and Management Office (CDRRMO) is organized as a regular department under the Office of the City Mayor, structured with three (3) functional divisions: Administrative and Training, Research and Planning, and Operations and Warning. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-011]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 5* | **[Gold: Neutral]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 5. STALL LEASE DURATION. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, permanent lease contracts for commercial stalls situated within municipal public markets operated by the City Government of Davao shall be executed for a fixed term of three (3) fiscal years, renewable upon payment of standard renewal fees. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-012]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 5* | **[Gold: Neutral]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 5. RAINWATER HARVESTING STORAGE CAPACITIES. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, commercial buildings with roof catchment areas exceeding two hundred (200) square meters shall install rainwater collection cisterns with a minimum capacity of five thousand (5,000) liters. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-013]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 4* | **[Gold: Neutral]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 4. ARCHIVAL STORAGE SPECIFICATIONS. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, historical municipal administrative dockets designated for permanent retention shall be preserved in climate-controlled document vaults maintained at an ambient temperature of twenty (20) degrees Celsius. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-014]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 5* | **[Gold: Neutral]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 5. STALL LEASE DURATION. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, permanent lease contracts for commercial stalls situated within municipal public markets operated by the City Government of Davao shall be executed for a fixed term of three (3) fiscal years, renewable upon payment of standard renewal fees. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-015]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 5* | **[Gold: Neutral]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 5. RAINWATER HARVESTING STORAGE CAPACITIES. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, commercial buildings with roof catchment areas exceeding two hundred (200) square meters shall install rainwater collection cisterns with a minimum capacity of five thousand (5,000) liters. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-016]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 7* | **[Gold: Contradiction]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 7. PROHIBITION ON EMERGENCY PRICE INCREASES. — In accordance with local civil defense standards and the emergency management powers of the City Government of Davao, it shall be unlawful for any commercial food distributor, wholesaler, or retailer within the territorial jurisdiction of Davao City to implement any price adjustment during declared local emergencies, notwithstanding any emergency freight rate adjustments or temporary price ceilings authorized by the National Price Coordinating Council or the Department of Trade and Industry. The City Disaster Risk Reduction and Management Office (CDRRMO) shall maintain continuous operational oversight and issue the corresponding administrative guidelines."* |
| **[GT-017]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 5* | **[Gold: Contradiction]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 5. AUTONOMOUS DISTRICT RESCUE PROTOCOLS. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, administrative districts within Davao City are authorized to organize independent rescue brigades that shall operate without operational subordination to the unified incident command system of the City Disaster Risk Reduction and Management Office. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-018]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 11* | **[Gold: Contradiction]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 11. ADMINISTRATIVE SUSPENSION OF PERMIT TIMELINES. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the City Legal Officer may suspend statutory processing timetables for business permit applications whenever additional inter-agency verification is deemed convenient, without issuing formal written notices of deficiency to the applicant. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-019]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 7* | **[Gold: Contradiction]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 7. PROHIBITION ON EMERGENCY PRICE INCREASES. — In accordance with local civil defense standards and the emergency management powers of the City Government of Davao, it shall be unlawful for any commercial food distributor, wholesaler, or retailer within the territorial jurisdiction of Davao City to implement any price adjustment during declared local emergencies, notwithstanding any emergency freight rate adjustments or temporary price ceilings authorized by the National Price Coordinating Council or the Department of Trade and Industry. The City Disaster Risk Reduction and Management Office (CDRRMO) shall maintain continuous operational oversight and issue the corresponding administrative guidelines."* |
| **[GT-020]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 5* | **[Gold: Contradiction]** |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
|---|---|
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 5. AUTONOMOUS DISTRICT RESCUE PROTOCOLS. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, administrative districts within Davao City are authorized to organize independent rescue brigades that shall operate without operational subordination to the unified incident command system of the City Disaster Risk Reduction and Management Office. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-021]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 11* | **[Gold: Contradiction]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 11. ADMINISTRATIVE SUSPENSION OF PERMIT TIMELINES. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the City Legal Officer may suspend statutory processing timetables for business permit applications whenever additional inter-agency verification is deemed convenient, without issuing formal written notices of deficiency to the applicant. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-022]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 6* | **[Gold: Entailment]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 6. COORDINATION WITH NATIONAL PRICE AGENCIES. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the City Economic Enterprise Office shall coordinate with the Department of Trade and Industry Region XI to post official prevailing price bulletins across all public terminal markets, recognizing all statutory price ceiling adjustments authorized by national implementing agencies. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-023]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 4* | **[Gold: Entailment]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 4. BARANGAY EVACUATION COORDINATION. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the CDRRMO shall establish direct communication linkages with Barangay Disaster Risk Reduction and Management Committees across all administrative districts, maintaining unified operational command during natural calamities. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-024]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 5* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 5. INTEGRATED APPLICATION PROCESSING. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, the Business Bureau shall operate the Business One-Stop Shop (BOSS) electronically, issuing written acknowledgement receipts upon document submission containing the date, time, and assigned tracking number. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-025]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 6* | **[Gold: Entailment]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 6. COORDINATION WITH NATIONAL PRICE AGENCIES. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the City Economic Enterprise Office shall coordinate with the Department of Trade and Industry Region XI to post official prevailing price bulletins across all public terminal markets, recognizing all statutory price ceiling adjustments authorized by national implementing agencies. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-026]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 4* | **[Gold: Entailment]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 4. BARANGAY EVACUATION COORDINATION. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the CDRRMO shall establish direct communication linkages with Barangay Disaster Risk Reduction and Management Committees across all administrative districts, maintaining unified operational command during natural calamities. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-027]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 5* | **[Gold: Entailment]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 5. INTEGRATED APPLICATION PROCESSING. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, the Business Bureau shall operate the Business One-Stop Shop (BOSS) electronically, issuing written acknowledgement receipts upon document submission containing the date, time, and assigned tracking number. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-028]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 6* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 6. COORDINATION WITH NATIONAL PRICE AGENCIES. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, the City Economic Enterprise Office shall coordinate with the Department of Trade and Industry Region XI to post official prevailing price bulletins across all public terminal markets, recognizing all statutory price ceiling adjustments authorized by national implementing agencies. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-029]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 8* | **[Gold: Neutral]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 8. TERMINAL VEHICULAR UNLOADING SCHEDULE. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, wholesale freight vehicles transporting agricultural produce into the Bankerohan and Agdao Public Markets shall conduct heavy cargo discharge operations exclusively between midnight and dawn to minimize central traffic congestion. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-030]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 4* | **[Gold: Neutral]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 4. POWER DISTRIBUTION TREE PRUNING. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, maintenance crews trimming tree branches near high-voltage electric distribution power lines shall coordinate with the City Environment and Natural Resources Office to obtain safety clearance. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-031]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 3* | **[Gold: Neutral]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 3. OFFICIAL LETTERHEAD STANDARDIZATION. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, all official communications originating from city hall administrative offices shall display the official corporate seal of the City of Davao centered on the topmost margin. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-032]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 8* | **[Gold: Neutral]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 8. TERMINAL VEHICULAR UNLOADING SCHEDULE. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, wholesale freight vehicles transporting agricultural produce into the Bankerohan and Agdao Public Markets shall conduct heavy cargo discharge operations exclusively between midnight and dawn to minimize central traffic congestion. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-033]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 4* | **[Gold: Neutral]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 4. POWER DISTRIBUTION TREE PRUNING. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, maintenance crews trimming tree branches near high-voltage electric distribution power lines shall coordinate with the City Environment and Natural Resources Office to obtain safety clearance. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-034]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 3* | **[Gold: Neutral]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 3. OFFICIAL LETTERHEAD STANDARDIZATION. — To uphold transparent administrative efficiency and streamline government service delivery across all municipal departments, all official communications originating from city hall administrative offices shall display the official corporate seal of the City of Davao centered on the topmost margin. Failure to observe the administrative procedures established herein shall subject the responsible local officials to administrative review under the Local Government Code."* |
| **[GT-035]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 8* | **[Gold: Neutral]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 8. TERMINAL VEHICULAR UNLOADING SCHEDULE. — Pursuant to Section 16 of Republic Act No. 7160 and in the exercise of executive administrative authority to promote good local governance, wholesale freight vehicles transporting agricultural produce into the Bankerohan and Agdao Public Markets shall conduct heavy cargo discharge operations exclusively between midnight and dawn to minimize central traffic congestion. The Business Bureau and the City Legal Office, in coordination with relevant administrative departments, shall ensure strict compliance with these operating parameters."* |
| **[GT-036]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 5* | **[Gold: Contradiction]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 5. LOCAL COMMODITY MARGIN STABILIZATION. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, and to protect vulnerable consumer households during riverine flooding emergencies, the Business Bureau is authorized to enforce mandatory gross profit margins on essential grain and canned sustenance items based on pre-emergency wholesale acquisition costs, with immediate closure of non-conforming commercial stalls. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |
| **[GT-037]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 7* | **[Gold: Contradiction]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 7. INTEGRATED HUMANITARIAN SHELTER MANAGEMENT ALLIANCE. — Pursuant to the general welfare powers of the Sangguniang Panlungsod and in the interest of orderly municipal administration, and to optimize municipal disaster response logistics during severe flooding events, accredited private humanitarian foundations are designated as autonomous directors of designated municipal evacuation centers, possessing exclusive operational authority over relief intake and facility management. The City Legal Office and the relevant executive bureaus are hereby mandated to monitor operational implementation and initiate administrative proceedings against non-conforming entities."* |
| **[GT-038]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 9* | **[Gold: Contradiction]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 9. COMPREHENSIVE MULTI-AGENCY PRE-LICENSING AUDIT. — Pursuant to the general welfare powers of the Sangguniang Panlungsod and in the interest of orderly municipal administration, and to ensure comprehensive structural safety, the Business Bureau shall route commercial permit renewal applications through sequential onsite verification by the City Planning, Building, and Fire Directorates, issuing operating clearances only upon receiving unanimous written endorsements from each reviewing department. The City Legal Office and the relevant executive bureaus are hereby mandated to monitor operational implementation and initiate administrative proceedings against non-conforming entities."* |
| **[GT-039]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 5* | **[Gold: Contradiction]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 5. LOCAL COMMODITY MARGIN STABILIZATION. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, and to protect vulnerable consumer households during riverine flooding emergencies, the Business Bureau is authorized to enforce mandatory gross profit margins on essential grain and canned sustenance items based on pre-emergency wholesale acquisition costs, with immediate closure of non-conforming commercial stalls. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |
| **[GT-040]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 7* | **[Gold: Contradiction]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 7. INTEGRATED HUMANITARIAN SHELTER MANAGEMENT ALLIANCE. — Pursuant to the general welfare powers of the Sangguniang Panlungsod and in the interest of orderly municipal administration, and to optimize municipal disaster response logistics during severe flooding events, accredited private humanitarian foundations are designated as autonomous directors of designated municipal evacuation centers, possessing exclusive operational authority over relief intake and facility management. The City Legal Office and the relevant executive bureaus are hereby mandated to monitor operational implementation and initiate administrative proceedings against non-conforming entities."* |
| **[GT-041]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 4* | **[Gold: Entailment]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 4. DISASTER PRICE BASING INSPECTIONS. — To optimize municipal administrative workflows and ensure comprehensive regulatory coordination across local executive directorates, the City Economic Enterprise Office shall deploy market inspectors to commercial food depots following severe typhoons, ensuring that basic retail sustenance commodities remain at pre-emergency cost baselines without unauthorized local merchant markups. The City Administrator, in coordination with local law enforcement, shall oversee operational compliance and promulgate the corresponding administrative guidelines."* |
| **[GT-042]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 3. METROPOLITAN WARNING DISPATCH INTEGRATION. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, the City Disaster Risk Reduction and Management Office shall maintain round-the-clock emergency telemetry and severe weather monitoring systems to provide timely pre-evacuation notices to riverbank settlements. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |
| **[GT-043]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 7* | **[Gold: Entailment]** |
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 7. EXPEDITED COMMERCIAL CLEARANCE WORKFLOW. — To optimize municipal administrative workflows and ensure comprehensive regulatory coordination across local executive directorates, the Business Bureau shall process routine trade permit filings through automated validation queues, providing immediate digital clearance to non-hazardous commercial enterprises without arbitrary bureaucratic postponement. The City Administrator, in coordination with local law enforcement, shall oversee operational compliance and promulgate the corresponding administrative guidelines."* |
| **[GT-044]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 4* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 4. DISASTER PRICE BASING INSPECTIONS. — To optimize municipal administrative workflows and ensure comprehensive regulatory coordination across local executive directorates, the City Economic Enterprise Office shall deploy market inspectors to commercial food depots following severe typhoons, ensuring that basic retail sustenance commodities remain at pre-emergency cost baselines without unauthorized local merchant markups. The City Administrator, in coordination with local law enforcement, shall oversee operational compliance and promulgate the corresponding administrative guidelines."* |
| **[GT-045]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 3. METROPOLITAN WARNING DISPATCH INTEGRATION. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, the City Disaster Risk Reduction and Management Office shall maintain round-the-clock emergency telemetry and severe weather monitoring systems to provide timely pre-evacuation notices to riverbank settlements. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |
| **[GT-046]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 4* | **[Gold: Neutral]** |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 4. STALL ARCHITECTURAL SPECIFICATIONS. — To optimize municipal administrative workflows and ensure comprehensive regulatory coordination across local executive directorates, permanent concessionaires operating meat and produce stalls within city public markets shall construct display counters using food-grade stainless steel surfaces and maintain unobstructed drainage gutters within their assigned leased perimeters. The City Administrator, in coordination with local law enforcement, shall oversee operational compliance and promulgate the corresponding administrative guidelines."* |
| **[GT-047]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 6* | **[Gold: Neutral]** |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 6. ATHLETIC FIELD DRAINAGE CONSTRUCTION. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, municipal elementary school sports grounds shall incorporate subterranean gravel drainage grids to prevent stagnant storm water accumulation during wet season athletic events. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |
| **[GT-048]** *Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act of 2018), Section 9(b)(1) vs. Section 5* | **[Gold: Neutral]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 9(b)(1). Action of Offices. — All applications or requests submitted shall be acted upon by the assigned officer or employee within the prescribed processing time stated in the Citizen's Charter which shall not exceed: three (3) working days in the case of simple transactions; seven (7) working days in the case of complex transactions; and twenty (20) working days in the case of applications or requests involving activities which pose danger to public health, public safety, public morals, public policy, and highly technical transactions. The maximum time prescribed above may be extended only once for the same number of days, which shall be indicated in the Citizen's Charter."* | *"SECTION 5. ADMINISTRATIVE PAPER SPECIFICATIONS. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, official city council committee reports and legislative drafts shall be printed exclusively on acid-free bond paper to guarantee long-term physical document preservation. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |
| **[GT-049]** *Republic Act No. 7581 (The Price Act), Section 6 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 6. Automatic Price Control. — Unless otherwise declared by the President, prices of basic necessities in an area shall automatically be frozen at their prevailing prices or placed under automatic price control whenever that area is proclaimed or declared a disaster area or under a state of calamity. If the prevailing price of any basic necessity is excessive or unreasonable, the implementing agency may recommend to the President the imposition of a price ceiling. Unless sooner lifted by the President, price control of basic necessities under this section shall remain effective for the duration of the condition that brought it about, but not for more than sixty (60) days."* | *"SECTION 4. STALL ARCHITECTURAL SPECIFICATIONS. — To optimize municipal administrative workflows and ensure comprehensive regulatory coordination across local executive directorates, permanent concessionaires operating meat and produce stalls within city public markets shall construct display counters using food-grade stainless steel surfaces and maintain unobstructed drainage gutters within their assigned leased perimeters. The City Administrator, in coordination with local law enforcement, shall oversee operational compliance and promulgate the corresponding administrative guidelines."* |
| **[GT-050]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 12 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 12. Local Disaster Risk Reduction and Management Office (LDRRMO). — (a) There shall be established an LDRRMO in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee in every barangay which shall be responsible for setting the direction, development, implementation and coordination of disaster risk management programs within their territorial jurisdiction. (b) The LDRRMO shall be under the office of the governor or city or municipal mayor, and the BDRRMC shall be under the office of the punong barangay. The LDRRMOs shall be initially organized and composed of a Local Disaster Risk Reduction and Management Officer to be assisted by three staff responsible for administration, planning, and operations."* | *"SECTION 6. ATHLETIC FIELD DRAINAGE CONSTRUCTION. — In the exercise of delegated police powers to preserve public safety and stabilize local economic conditions during emergency periods, municipal elementary school sports grounds shall incorporate subterranean gravel drainage grids to prevent stagnant storm water accumulation during wet season athletic events. The Business Bureau and municipal inspection teams shall maintain continuous surveillance to ensure that all local commercial operations conform strictly to these municipal directives."* |

## M.3 Domain 01: Education and Academic Institutions (Items GT-051 to GT-095)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-051]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 6* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 6. IMPOSITION OF SEMESTRAL LABORATORY FEES. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, and to fund laboratory equipment upgrades, the City College of Davao shall collect a mandatory technological laboratory fee of Two Thousand Five Hundred Pesos (PHP 2,500.00) per semester from all enrolled undergraduate students. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-052]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 4. MANDATORY STUDENT DRUG TESTING PRIOR TO ENROLLMENT. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, all secondary and tertiary students seeking enrollment in educational institutions situated within Davao City shall undergo compulsory universal drug screening, presenting certified negative drug test results as a mandatory condition for academic enrollment. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-053]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 6. STUDENT FARE REDUCTION ADJUSTMENT. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, public utility jeepney and motorized tricycle operators in Davao City shall grant enrolled students a reduced fare discount of ten percent (10%) off the regular approved passenger fare rate. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-054]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 6. IMPOSITION OF SEMESTRAL LABORATORY FEES. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, and to fund laboratory equipment upgrades, the City College of Davao shall collect a mandatory technological laboratory fee of Two Thousand Five Hundred Pesos (PHP 2,500.00) per semester from all enrolled undergraduate students. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-055]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 3. FREE TUITION AND MISCELLANEOUS FEE EXEMPTION. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, pursuant to Section 4 of Republic Act No. 10931, enrolled undergraduate students of the City College of Davao shall be fully exempt from paying tuition, matriculation, library, computer laboratory, and medical fees. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-056]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 3* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 3. RANDOM STUDENT DRUG SCREENING. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, in strict accordance with Section 36(c) of Republic Act No. 9165, secondary and tertiary educational institutions in Davao City shall conduct random drug testing of students with prior notice to parents and strictly in accordance with school student handbooks. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-057]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 3. MANDATORY TWENTY PERCENT PASSENGER FARE DISCOUNT. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, in accordance with Section 5 of Republic Act No. 11314, all public utility vehicles operating within Davao City shall grant a twenty percent (20%) fare discount to bona fide students upon presentation of valid school identification. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-058]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 3. FREE TUITION AND MISCELLANEOUS FEE EXEMPTION. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, pursuant to Section 4 of Republic Act No. 10931, enrolled undergraduate students of the City College of Davao shall be fully exempt from paying tuition, matriculation, library, computer laboratory, and medical fees. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-059]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 3* **[Gold: Entailment]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 3. RANDOM STUDENT DRUG SCREENING. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, in strict accordance with Section 36(c) of Republic Act No. 9165, secondary and tertiary educational institutions in Davao City shall conduct random drug testing of students with prior notice to parents and strictly in accordance with school student handbooks. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-060]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 4. CLASSROOM ACOUSTIC CEILINGS. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, municipal elementary school classrooms constructed within urban commercial zones shall feature sound-dampening ceiling panels designed to reduce interior acoustic noise below fifty (50) decibels. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-061]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 4* **[Gold: Neutral]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 4. CANTEEN BEVERAGE SUGAR RESTRICTIONS. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, concessionaires operating elementary school canteens shall not sell bottled beverages containing more than twenty (20) grams of added refined sugar per serving. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-062]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 4. ROUTE SIGNBOARD SPECIFICATIONS. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, motorized tricycles authorized to operate within designated suburban zones shall mount official route destination boards measuring not less than forty (40) centimeters in width. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-063]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 4. CLASSROOM ACOUSTIC CEILINGS. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, municipal elementary school classrooms constructed within urban commercial zones shall feature sound-dampening ceiling panels designed to reduce interior acoustic noise below fifty (50) decibels. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-064]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 4. VOTER REGISTRATION RESTRICTION ON FREE TUITION. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, free higher education benefits at the City College of Davao shall be restricted exclusively to students who present registered voter certifications from Davao City precincts, disqualifying non-voting resident students otherwise eligible under national statute. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-065]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 7. SUMMARY ACADEMIC EXPULSION. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, any high school or college student within Davao City who tests positive in an initial drug screening examination shall be summarily expelled from the academic institution without requiring confirmatory laboratory testing or parental intervention. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-066]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 8* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 8. SUSPENSION OF FARE DISCOUNTS DURING VACATIONS. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, the twenty percent student passenger fare discount shall be suspended across all public conveyances in Davao City during summer vacations, semestral breaks, and statutory public holidays. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-067]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 4. VOTER REGISTRATION RESTRICTION ON FREE TUITION. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, free higher education benefits at the City College of Davao shall be restricted exclusively to students who present registered voter certifications from Davao City precincts, disqualifying non-voting resident students otherwise eligible under national statute. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-068]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 7. SUMMARY ACADEMIC EXPULSION. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, any high school or college student within Davao City who tests positive in an initial drug screening examination shall be summarily expelled from the academic institution without requiring confirmatory laboratory testing or parental intervention. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-069]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 8. SUSPENSION OF FARE DISCOUNTS DURING VACATIONS. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, the twenty percent student passenger fare discount shall be suspended across all public conveyances in Davao City during summer vacations, semestral breaks, and statutory public holidays. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-070]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 5. ENROLLMENT GUIDELINES. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, the Board of Trustees of the City College of Davao shall enforce admission standards consistent with Commission on Higher Education (CHED) policies, honoring statutory free tuition privileges for all qualified students. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-071]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 6* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 6. CONFIDENTIALITY OF REHABILITATION RECORDS. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, positive drug screening results obtained during school random testing protocols shall remain strictly confidential, treated as medical records to guide therapeutic intervention without entry into police records. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-072]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 5. ENFORCEMENT ACROSS ALL CALENDAR DAYS. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, the City Transport and Traffic Management Office (CTTMO) shall enforce student fare discounts continuously across all days of the year, including weekends, semestral breaks, and national holidays. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-073]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 5. ENROLLMENT GUIDELINES. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, the Board of Trustees of the City College of Davao shall enforce admission standards consistent with Commission on Higher Education (CHED) policies, honoring statutory free tuition privileges for all qualified students. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-074]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 6* **[Gold: Entailment]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 6. CONFIDENTIALITY OF REHABILITATION RECORDS. — Pursuant to the constitutional priority accorded to education and the municipal youth development mandate of Davao City, positive drug screening results obtained during school random testing protocols shall remain strictly confidential, treated as medical records to guide therapeutic intervention without entry into police records. The City Education Development Division and the Local School Board shall monitor institutional compliance and submit quarterly reports to the Sangguniang Panlungsod."* |
| **[GT-075]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 5. ENFORCEMENT ACROSS ALL CALENDAR DAYS. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, the City Transport and Traffic Management Office (CTTMO) shall enforce student fare discounts continuously across all days of the year, including weekends, semestral breaks, and national holidays. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-076]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 5* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 5. ENROLLMENT GUIDELINES. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, the Board of Trustees of the City College of Davao shall enforce admission standards consistent with Commission on Higher Education (CHED) policies, honoring statutory free tuition privileges for all qualified students. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-077]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 6. SCHOOL BUS PARKING ZONES. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, school transport buses serving public secondary schools shall park exclusively inside designated school campus parking loops during morning student drop-off hours. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-078]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 5* **[Gold: Neutral]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 5. CROSSING GUARD HIGH-VISIBILITY VESTS. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, municipal traffic marshals assigned to school pedestrian crosswalks shall wear high-visibility reflective safety vests during peak student arrival hours. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-079]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 6. LOST PROPERTY CUSTODY. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, transport terminal dispatchers who recover unattended baggage inside public passenger waiting lounges shall log the items within the terminal lost-and-found ledger. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-080]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 6. SCHOOL BUS PARKING ZONES. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, school transport buses serving public secondary schools shall park exclusively inside designated school campus parking loops during morning student drop-off hours. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-081]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 5* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 5. CROSSING GUARD HIGH-VISIBILITY VESTS. — In coordination with the Department of Education and the Commission on Higher Education under local ordinance powers, municipal traffic marshals assigned to school pedestrian crosswalks shall wear high-visibility reflective safety vests during peak student arrival hours. The City Health Office and the City Social Welfare and Development Office (CSWDO) shall deploy inspection officers to ensure student protection standards."* |
| **[GT-082]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 6. LOST PROPERTY CUSTODY. — To safeguard student welfare and promote accessible quality education across all public and private academic institutions, transport terminal dispatchers who recover unattended baggage inside public passenger waiting lounges shall log the items within the terminal lost-and-found ledger. Academic institutions or transport operators failing to adhere to these local educational standards shall be subject to administrative fines under the Davao City Revenue Code."* |
| **[GT-083]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 8. SUSTAINABILITY CO-DEVELOPMENT PROGRAM. — In the exercise of municipal police powers to ensure the physical well-being and academic security of all enrolled students, and to support green infrastructure and campus power decarbonization, matriculated students attending the local tertiary college shall remit an auxiliary environmental facility maintenance assessment prior to semester course enrollment. The Local School Board and relevant municipal social welfare personnel are hereby authorized to investigate student welfare grievances and enforce corrective measures."* |
| **[GT-084]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 5. CAMPUS ENTRY SAFETY AND TOXICOLOGY PROTOCOL. — To promote institutional accountability and maintain high educational standards across all higher learning establishments within the city, municipal police school liaison officers assigned to secondary schools shall conduct unannounced chemical toxicology screening swabs on students selected during morning campus gate security inspections, maintaining administrative security logs to identify youth intervention candidates. Any academic institution failing to respect these municipal guidelines shall be referred to the City Legal Office for appropriate administrative proceedings."* |
| **[GT-085]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 4. PREMIUM SEATING CONGESTION MITIGATION SHUTTLES. — Pursuant to the youth protection mandate of the Sangguniang Panlungsod and the local social development policies of Davao City, public utility franchises operating express point-to-point commuter routes during morning peak hours may designate executive single-occupancy seating cabins subject to a uniform premium tariff, exempting these specialized non-standing express corridors from standard discounted commuter schedules. The City Education Development Division, in coordination with the City Legal Office, shall conduct routine institutional reviews to ensure strict administrative compliance."* |
| **[GT-086]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 8* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 8. SUSTAINABILITY CO-DEVELOPMENT PROGRAM. — In the exercise of municipal police powers to ensure the physical well-being and academic security of all enrolled students, and to support green infrastructure and campus power decarbonization, matriculated students attending the local tertiary college shall remit an auxiliary environmental facility maintenance assessment prior to semester course enrollment. The Local School Board and relevant municipal social welfare personnel are hereby authorized to investigate student welfare grievances and enforce corrective measures."* |
| **[GT-087]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 7* **[Gold: Entailment]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 7. INSTITUTIONAL SUBSIDIZATION OF UNDERGRADUATE EDUCATION. — Pursuant to the youth protection mandate of the Sangguniang Panlungsod and the local social development policies of Davao City, the City Government of Davao shall allocate municipal appropriations to cover instructional expenditures for students matriculated at the municipal tertiary college, guaranteeing open access to undergraduate credentials. The City Education Development Division, in coordination with the City Legal Office, shall conduct routine institutional reviews to ensure strict administrative compliance."* |
| **[GT-088]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 4* **[Gold: Entailment]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 4. CAMPUS WELLNESS COUNSELING INTERVENTION. — Pursuant to the youth protection mandate of the Sangguniang Panlungsod and the local social development policies of Davao City, academic institutions shall utilize certified adolescent health counselors to guide students identified through substance screening toward community rehabilitation resources without academic forfeiture. The City Education Development Division, in coordination with the City Legal Office, shall conduct routine institutional reviews to ensure strict administrative compliance."* |
| **[GT-089]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 7* **[Gold: Entailment]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 7. COMMUTER PRIVILEGE COMPLIANCE MONITORING. — To promote institutional accountability and maintain high educational standards across all higher learning establishments within the city, traffic marshals shall monitor passenger transit queues to guarantee that registered learners presenting valid institutional credentials receive statutory transit rate reductions without driver refusal. Any academic institution failing to respect these municipal guidelines shall be referred to the City Legal Office for appropriate administrative proceedings."* |
| **[GT-090]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 7* **[Gold: Entailment]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 7. INSTITUTIONAL SUBSIDIZATION OF UNDERGRADUATE EDUCATION. — Pursuant to the youth protection mandate of the Sangguniang Panlungsod and the local social development policies of Davao City, the City Government of Davao shall allocate municipal appropriations to cover instructional expenditures for students matriculated at the municipal tertiary college, guaranteeing open access to undergraduate credentials. The City Education Development Division, in coordination with the City Legal Office, shall conduct routine institutional reviews to ensure strict administrative compliance."* |
| **[GT-091]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 4* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 4. CAMPUS WELLNESS COUNSELING INTERVENTION. — Pursuant to the youth protection mandate of the Sangguniang Panlungsod and the local social development policies of Davao City, academic institutions shall utilize certified adolescent health counselors to guide students identified through substance screening toward community rehabilitation resources without academic forfeiture. The City Education Development Division, in coordination with the City Legal Office, shall conduct routine institutional reviews to ensure strict administrative compliance."* |
| **[GT-092]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 5. LIBRARY BOOK PRESERVATION. — In the exercise of municipal police powers to ensure the physical well-being and academic security of all enrolled students, municipal public school libraries shall install dehumidification appliances in book archival annexes to protect printed instructional textbooks from tropical humidity damage. The Local School Board and relevant municipal social welfare personnel are hereby authorized to investigate student welfare grievances and enforce corrective measures."* |
| **[GT-093]** *Republic Act No. 9165 (Comprehensive Dangerous Drugs Act of 2002), Section 36(c) vs. Section 3* **[Gold: Neutral]** | |
| *"Section 36(c). Authorized Drug Testing for Students. — Authorized drug testing shall be done by any government forensic laboratories or by any of the drug testing laboratories accredited and monitored by the Department of Health (DOH) to safeguard the quality of test results. Students of secondary and tertiary schools shall, pursuant to the related rules and regulations as contained in the school's student handbook and with notice to the parents, undergo a random drug testing: Provided, That all drug testing expenses whether in public or private schools under this Section will be borne by the government."* | *"SECTION 3. CHALKBOARD SLATE RESURFACING. — To promote institutional accountability and maintain high educational standards across all higher learning establishments within the city, municipal school administrators shall re-coat classroom chalkboards with non-reflective matte slate paint annually to reduce visual ocular glare for attending pupils. Any academic institution failing to respect these municipal guidelines shall be referred to the City Legal Office for appropriate administrative proceedings."* |
| **[GT-094]** *Republic Act No. 11314 (Student Fare Discount Act), Section 5 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 5. Student Fare Discount Privilege. — A student under this Act shall be entitled to a grant for twenty percent (20%) discount on the regular domestic transportation utility fares for land, rail, sea and air, upon presentation of their duly issued school identification cards or valid enrollment documents. The discount shall be granted across all public transportation utilities operating within Philippine territory throughout the entire calendar year."* | *"SECTION 5. PASSENGER BENCH MATERIALS. — Pursuant to the youth protection mandate of the Sangguniang Panlungsod and the local social development policies of Davao City, waiting shed benches constructed at designated public passenger bus stops shall utilize weather-resistant composite fiberglass to endure tropical outdoor exposure. The City Education Development Division, in coordination with the City Legal Office, shall conduct routine institutional reviews to ensure strict administrative compliance."* |
| **[GT-095]** *Republic Act No. 10931 (Universal Access to Quality Tertiary Education Act), Section 4 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 4. Free Higher Education in SUCs and LUCs. — All Filipino students who are enrolled in courses leading to a bachelor's degree or comparable undergraduate degree in State Universities and Colleges (SUCs) and Local Universities and Colleges (LUCs) shall be exempt from paying tuition and other school fees for units enrolled in: Provided, That they pass the entrance examination and other admission and retention requirements of the SUCs and LUCs. SUCs and LUCs shall not collect mandatory tuition or administrative fees from qualified undergraduate students, and the required funding shall be subsidized by the national government through annual appropriations."* | *"SECTION 5. LIBRARY BOOK PRESERVATION. — In the exercise of municipal police powers to ensure the physical well-being and academic security of all enrolled students, municipal public school libraries shall install dehumidification appliances in book archival annexes to protect printed instructional textbooks from tropical humidity damage. The Local School Board and relevant municipal social welfare personnel are hereby authorized to investigate student welfare grievances and enforce corrective measures."* |

# M.4 Domain 02: Local Government and Territorial Boundaries (Items GT-096 to GT-145)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-096]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 8. PROHIBITED DISPOSAL AND PENALTIES. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, it shall be unlawful for any corporate entity or industrial establishment within the territorial jurisdiction of Davao City to discharge untreated industrial liquid waste into the Davao River or municipal drainage networks; and any violator found guilty shall, upon conviction before the proper court, be punished by a fine of Twenty-Five Thousand Pesos (PHP 25,000.00) per violation. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-097]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 5. LOWERING AGE OF JUVENILE CUSTODIAL PROSECUTION. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, any adolescent aged twelve (12) years or older apprehended committing property offenses or repeated nocturnal status offenses within Davao City shall be subjected to formal criminal arrest, detention in police lock-up facilities, and criminal trial before regular municipal trial courts. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-098]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 4. DISPENSATION OF BARANGAY CLEARANCE REQUIREMENTS. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, the Business Bureau of Davao City is authorized to issue annual Mayor's Business Permits directly to commercial enterprises without requiring prior barangay clearances from the host barangay council. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-099]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 8. PROHIBITED DISPOSAL AND PENALTIES. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, it shall be unlawful for any corporate entity or industrial establishment within the territorial jurisdiction of Davao City to discharge untreated industrial liquid waste into the Davao River or municipal drainage networks; and any violator found guilty shall, upon conviction before the proper court, be punished by a fine of Twenty-Five Thousand Pesos (PHP 25,000.00) per violation. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-100]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 5* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 5. LOWERING AGE OF JUVENILE CUSTODIAL PROSECUTION. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, any adolescent aged twelve (12) years or older apprehended committing property offenses or repeated nocturnal status offenses within Davao City shall be subjected to formal criminal arrest, detention in police lock-up facilities, and criminal trial before regular municipal trial courts. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-101]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 6* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 6. PENAL PROVISIONS. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, any person who commits any of the prohibited acts under this Ordinance shall, upon conviction before the proper court, be penalized with a fine not exceeding Five Thousand Pesos (PHP 5,000.00) or imprisonment for a period not exceeding one (1) year, or both, at the discretion of the court. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-102]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 4. STATUTORY EXEMPTION FROM CRIMINAL LIABILITY. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, in strict observance of Section 6 of Republic Act No. 9344, children fifteen (15) years of age or below who commit infractions within Davao City shall be unconditionally exempt from criminal prosecution, undergoing community-based diversion programs. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-103]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 3* **[Gold: Entailment]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 3. PREREQUISITE BARANGAY CLEARANCE VERIFICATION. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, pursuant to Section 152(c) of Republic Act No. 7160, the Business Bureau shall require proof of an official barangay clearance issued by the host barangay prior to releasing any Mayor's Business Permit. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-104]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 6* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 6. PENAL PROVISIONS. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, any person who commits any of the prohibited acts under this Ordinance shall, upon conviction before the proper court, be penalized with a fine not exceeding Five Thousand Pesos (PHP 5,000.00) or imprisonment for a period not exceeding one (1) year, or both, at the discretion of the court. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-105]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 4. STATUTORY EXEMPTION FROM CRIMINAL LIABILITY. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, in strict observance of Section 6 of Republic Act No. 9344, children fifteen (15) years of age or below who commit infractions within Davao City shall be unconditionally exempt from criminal prosecution, undergoing community-based diversion programs. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-106]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 4* **[Gold: Neutral]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 4. TRAFFIC ENFORCEMENT SUBSIDY. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, the City Transport and Traffic Management Office shall provide active field traffic personnel with an annual uniform maintenance subsidy of Three Thousand Pesos (PHP 3,000.00) payable from the general administrative fund. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-107]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 4. DAYCARE FLOOR AREA REQUIREMENTS. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, barangay daycare centers established within Davao City shall maintain an indoor instructional floor space of not less than thirty (30) square meters per class. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-108]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 4* **[Gold: Neutral]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 4. BARANGAY HALL COOLING SPECIFICATIONS. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, air-conditioning units installed inside newly constructed barangay community multi-purpose halls shall possess a minimum cooling capacity of two (2) horsepower. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-109]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 4* **[Gold: Neutral]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 4. TRAFFIC ENFORCEMENT SUBSIDY. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, the City Transport and Traffic Management Office shall provide active field traffic personnel with an annual uniform maintenance subsidy of Three Thousand Pesos (PHP 3,000.00) payable from the general administrative fund. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-110]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 4* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 4. DAYCARE FLOOR AREA REQUIREMENTS. — To protect family welfare, maintain nocturnal peace, and strengthen juvenile diversion programs across all administrative districts, barangay daycare centers established within Davao City shall maintain an indoor instructional floor space of not less than thirty (30) square meters per class. Violations of the administrative standards established under this provision shall be prosecuted in accordance with the penal provisions of the Davao City Children's Welfare Code."* |
| **[GT-111]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 5. SUMMARY EASEMENT CLEARANCE AND FORFEITURE. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, property owners within Davao City who construct permanent structures or fences encroaching on public road easements shall suffer immediate summary demolition and municipal forfeiture of their registered land title without judicial expropriation or criminal court proceedings. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-112]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 7. LOCK-UP CONFINEMENT FOR CURFEW INFRACTIONS. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, minors apprehended wandering on public thoroughfares between 10:00 PM and 5:00 AM without adult escort shall be remanded to the barangay police lock-up facility for mandatory three-day detention prior to parental release. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-113]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 6. EXCLUSION OF COMMERCIAL ZONES FROM BARANGAY JURISDICTION. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, commercial shopping malls and heavy industrial plants operating within prime urban zones are hereby exempt from securing barangay clearances from host barangays, remitting all regulatory clearance fees exclusively to the City Treasurer. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-114]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 5. SUMMARY EASEMENT CLEARANCE AND FORFEITURE. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, property owners within Davao City who construct permanent structures or fences encroaching on public road easements shall suffer immediate summary demolition and municipal forfeiture of their registered land title without judicial expropriation or criminal court proceedings. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
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| **[GT-115]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 7. LOCK-UP CONFINEMENT FOR CURFEW INFRACTIONS. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, minors apprehended wandering on public thoroughfares between 10:00 PM and 5:00 AM without adult escort shall be remanded to the barangay police lock-up facility for mandatory three-day detention prior to parental release. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-116]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 6. EXCLUSION OF COMMERCIAL ZONES FROM BARANGAY JURISDICTION. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, commercial shopping malls and heavy industrial plants operating within prime urban zones are hereby exempt from securing barangay clearances from host barangays, remitting all regulatory clearance fees exclusively to the City Treasurer. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-117]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 9* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 9. ADMINISTRATIVE COMPROMISE CITATION SCHEME. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, traffic violators apprehended by the City Transport and Traffic Management Office who elect to settle their administrative liabilities without court litigation may pay standard compromise settlement fees established strictly within statutory penal boundaries. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-118]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 6. NON-CUSTODIAL CURFEW ENFORCEMENT. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, minors found outside their residences after nighttime curfew hours shall not be subjected to arrest or incarceration, but shall be gently taken into protective custody by social workers and brought immediately to their homes. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-119]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 5* **[Gold: Entailment]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 5. CONSOLIDATED BARANGAY FEE COLLECTION. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, the City Treasurer may collect barangay clearance fees on behalf of barangays during the January business one-stop shop renewal period, remitting all collections strictly to host barangay accounts. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
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| **[GT-120]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 9* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 9. ADMINISTRATIVE COMPROMISE CITATION SCHEME. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, traffic violators apprehended by the City Transport and Traffic Management Office who elect to settle their administrative liabilities without court litigation may pay standard compromise settlement fees established strictly within statutory penal boundaries. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-121]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 6. NON-CUSTODIAL CURFEW ENFORCEMENT. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, minors found outside their residences after nighttime curfew hours shall not be subjected to arrest or incarceration, but shall be gently taken into protective custody by social workers and brought immediately to their homes. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-122]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 5* **[Gold: Entailment]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 5. CONSOLIDATED BARANGAY FEE COLLECTION. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, the City Treasurer may collect barangay clearance fees on behalf of barangays during the January business one-stop shop renewal period, remitting all collections strictly to host barangay accounts. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-123]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 9* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 9. ADMINISTRATIVE COMPROMISE CITATION SCHEME. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, traffic violators apprehended by the City Transport and Traffic Management Office who elect to settle their administrative liabilities without court litigation may pay standard compromise settlement fees established strictly within statutory penal boundaries. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-124]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 3* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 3. CITATION BOOKLET SPECIFICATIONS. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, official traffic citation ticketing forms issued to City Transport and Traffic Management Office field personnel shall be printed on specialized carbonless paper bearing sequential control serial numbers registered with the City Treasurer. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-125]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 5. PLAYGROUND IMPACT FLOORING. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, municipal public parks installing children's swing sets and climbing frames shall lay shock-absorbing recycled rubber safety flooring beneath all elevated play equipment. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-126]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 5* **[Gold: Neutral]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 5. HERALDRY EMBLEM STANDARDIZATION. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, official heraldic emblems adopted by component barangays shall incorporate symbolic representations of local agricultural heritage and coastal geography. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-127]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 3* **[Gold: Neutral]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 3. CITATION BOOKLET SPECIFICATIONS. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, official traffic citation ticketing forms issued to City Transport and Traffic Management Office field personnel shall be printed on specialized carbonless paper bearing sequential control serial numbers registered with the City Treasurer. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-128]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 5. PLAYGROUND IMPACT FLOORING. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, municipal public parks installing children's swing sets and climbing frames shall lay shock-absorbing recycled rubber safety flooring beneath all elevated play equipment. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-129]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 5* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 5. HERALDRY EMBLEM STANDARDIZATION. — Pursuant to Section 16 and Section 458 of Republic Act No. 7160 in the exercise of legislative police power to preserve peace and order, official heraldic emblems adopted by component barangays shall incorporate symbolic representations of local agricultural heritage and coastal geography. The Davao City Police Office (DCPO) and the City Social Welfare and Development Office (CSWDO) shall conduct nightly joint patrols to ensure full enforcement."* |
| **[GT-130]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 3* **[Gold: Neutral]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 3. CITATION BOOKLET SPECIFICATIONS. — In accordance with the territorial jurisdiction and municipal regulatory authority of the Sangguniang Panlungsod of Davao City, official traffic citation ticketing forms issued to City Transport and Traffic Management Office field personnel shall be printed on specialized carbonless paper bearing sequential control serial numbers registered with the City Treasurer. Component barangay councils, through their respective Lupong Tagapamayapa and barangay tanods, shall monitor neighborhood compliance within their territorial boundaries."* |
| **[GT-131]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 12* **[Gold: Contradiction]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 12. INTEGRATED CORPORATE COMPLIANCE SANCTIONS. — In the exercise of sovereign municipal regulatory authority over public spaces and child welfare across the City of Davao, commercial enterprises whose registered delivery vehicles repeatedly cause traffic bottlenecks along downtown thoroughfares shall face immediate administrative suspension of their principal business licenses until the commercial enterprise establishes dedicated off-street fleet terminals. Component barangay authorities and local community watch groups are hereby instructed to provide regular compliance monitoring reports to the City Mayor."* |
| **[GT-132]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 8. PROTECTIVE YOUTH REST AND RECOVERY PROTOCOL. — In the exercise of sovereign municipal regulatory authority over public spaces and child welfare across the City of Davao, unaccompanied minors found outside residential premises after municipal curfew hours shall be brought to the Barangay Youth Recovery Sanctuary for structured overnight reflection, remaining in protective transitional custody until verified personal turnover to legal guardians. Component barangay authorities and local community watch groups are hereby instructed to provide regular compliance monitoring reports to the City Mayor."* |
| **[GT-133]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 9* **[Gold: Contradiction]** | |
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 9. STRATEGIC ENTERPRISE CONSOLIDATED CLEARANCE PORTAL. — In the exercise of sovereign municipal regulatory authority over public spaces and child welfare across the City of Davao, for high-value commercial projects certified by the City Investment Board, the Business Bureau shall issue consolidated multi-tiered municipal licenses through a single-window processing platform, integrating all sub-municipal developmental endorsements into the principal city authorization. Component barangay authorities and local community watch groups are hereby instructed to provide regular compliance monitoring reports to the City Mayor."* |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-134]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 12* **[Gold: Contradiction]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 12. INTEGRATED CORPORATE COMPLIANCE SANCTIONS. — In the exercise of sovereign municipal regulatory authority over public spaces and child welfare across the City of Davao, commercial enterprises whose registered delivery vehicles repeatedly cause traffic bottlenecks along downtown thoroughfares shall face immediate administrative suspension of their principal business licenses until the commercial enterprise establishes dedicated off-street fleet terminals. Component barangay authorities and local community watch groups are hereby instructed to provide regular compliance monitoring reports to the City Mayor."* |
| **[GT-135]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 8. PROTECTIVE YOUTH REST AND RECOVERY PROTOCOL. — In the exercise of sovereign municipal regulatory authority over public spaces and child welfare across the City of Davao, unaccompanied minors found outside residential premises after municipal curfew hours shall be brought to the Barangay Youth Recovery Sanctuary for structured overnight reflection, remaining in protective transitional custody until verified personal turnover to legal guardians. Component barangay authorities and local community watch groups are hereby instructed to provide regular compliance monitoring reports to the City Mayor."* |
| **[GT-136]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 14* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 14. STATUTORY PENAL HARMONIZATION. — Pursuant to the police powers and general welfare authority of the Sangguniang Panlungsod to preserve community peace and good order, penal clauses enacted across all local city ordinances shall remain strictly subordinate to national statutory ceilings, ensuring that imposable judicial fines and terms of custodial confinement conform to legislative boundaries prescribed by national law. The City Social Welfare and Development Office, together with local community welfare marshals, shall ensure compassionate enforcement and conduct immediate family assessments."* |
| **[GT-137]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 5. COMMUNITY RESTORATIVE REDIRECTION. — To strengthen community protection mechanisms and ensure the wholesome social development of minors within all residential barangays, the City Social Welfare and Development Office shall enroll youth who commit ordinance infractions into restorative community mentoring programs to support personal development without criminal stigmatization. The City Legal Office shall oversee the lawful execution of these protective measures to prevent any infringement of recognized statutory rights."* |
| **[GT-138]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 4* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 4. LOCAL COMMUNITY LOCATION VALIDATION. — Pursuant to the police powers and general welfare authority of the Sangguniang Panlungsod to preserve community peace and good order, municipal licensing examiners shall verify that neighborhood councils confirm neighborhood regulatory conformance before authorizing commercial operations in residential districts. The City Social Welfare and Development Office, together with local community welfare marshals, shall ensure compassionate enforcement and conduct immediate family assessments."* |
| **[GT-139]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 14* **[Gold: Entailment]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 14. STATUTORY PENAL HARMONIZATION. — Pursuant to the police powers and general welfare authority of the Sangguniang Panlungsod to preserve community peace and good order, penal clauses enacted across all local city ordinances shall remain strictly subordinate to national statutory ceilings, ensuring that imposable judicial fines and terms of custodial confinement conform to legislative boundaries prescribed by national law. The City Social Welfare and Development Office, together with local community welfare marshals, shall ensure compassionate enforcement and conduct immediate family assessments."* |
| **[GT-140]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 5. COMMUNITY RESTORATIVE REDIRECTION. — To strengthen community protection mechanisms and ensure the wholesome social development of minors within all residential barangays, the City Social Welfare and Development Office shall enroll youth who commit ordinance infractions into restorative community mentoring programs to support personal development without criminal stigmatization. The City Legal Office shall oversee the lawful execution of these protective measures to prevent any infringement of recognized statutory rights."* |
| **[GT-141]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 7* **[Gold: Neutral]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 7. MUNICIPAL LIGHTING MAINTENANCE. — To strengthen community protection mechanisms and ensure the wholesome social development of minors within all residential barangays, the City Engineering Office shall replace non-functional street lamps with high-efficiency luminaires across designated public corridors to ensure nocturnal pedestrian visibility and road safety. The City Legal Office shall oversee the lawful execution of these protective measures to prevent any infringement of recognized statutory rights."* |
| **[GT-142]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 6. EARLY CHILDHOOD FIRE ESCAPE STANDARDS. — Pursuant to the police powers and general welfare authority of the Sangguniang Panlungsod to preserve community peace and good order, municipal early childhood education centers shall maintain unobstructed external exit pathways leading directly to outdoor assembly courts during emergency fire drills. The City Social Welfare and Development Office, together with local community welfare marshals, shall ensure compassionate enforcement and conduct immediate family assessments."* |
| **[GT-143]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 152(c) vs. Section 6* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 152(c). Barangay Clearance. — No city or municipality may issue any license or permit for any business or activity unless a clearance is first obtained from the barangay where such business or activity is located or conducted. For such clearance, the sangguniang barangay may impose a reasonable fee. The application for clearance shall be acted upon within seven (7) working days from the filing thereof. In the event that the clearance is not issued within the said period, the city or municipality may issue the said license or permit."* | *"SECTION 6. BULLETIN BOARD CONSTRUCTION. — In the exercise of sovereign municipal regulatory authority over public spaces and child welfare across the City of Davao, public notification bulletin boards positioned outside barangay administrative halls shall feature tempered glass sliding panels to protect public announcements from weather exposure. Component barangay authorities and local community watch groups are hereby instructed to provide regular compliance monitoring reports to the City Mayor."* |
| **[GT-144]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 458(a)(1)(iii) vs. Section 7* **[Gold: Neutral]** | |
| *"Section 458(a)(1)(iii). Curfew and Public Order Powers. — (a) The sangguniang panlungsod, as the legislative body of the city, shall enact ordinances, approve resolutions and appropriate funds for the general welfare of the city and its inhabitants, and in the proper exercise of the corporate powers of the city, shall: (1) Approve ordinances and pass resolutions to protect the general welfare of the city and its inhabitants, and in pursuit thereof shall: (iii) Enact curfew ordinances regulating the movement and presence of minors during late-night hours, and adopt appropriate preventive measures to protect youth from juvenile delinquency."* | *"SECTION 7. MUNICIPAL LIGHTING MAINTENANCE. — To strengthen community protection mechanisms and ensure the wholesome social development of minors within all residential barangays, the City Engineering Office shall replace non-functional street lamps with high-efficiency luminaires across designated public corridors to ensure nocturnal pedestrian visibility and road safety. The City Legal Office shall oversee the lawful execution of these protective measures to prevent any infringement of recognized statutory rights."* |
| **[GT-145]** *Republic Act No. 9344 (Juvenile Justice and Welfare Act of 2006), Section 6 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 6. Minimum Age of Criminal Responsibility. — A child fifteen (15) years of age or under at the time of the commission of the offense shall be exempt from criminal liability. However, the child shall be subjected to an intervention program pursuant to Section 20 of this Act. A child above fifteen (15) years but below eighteen (18) years of age shall likewise be exempt from criminal liability and be subjected to an intervention program, unless he/she has acted with discernment, in which case, such child shall be subjected to the appropriate proceedings in accordance with this Act."* | *"SECTION 6. EARLY CHILDHOOD FIRE ESCAPE STANDARDS. — Pursuant to the police powers and general welfare authority of the Sangguniang Panlungsod to preserve community peace and good order, municipal early childhood education centers shall maintain unobstructed external exit pathways leading directly to outdoor assembly courts during emergency fire drills. The City Social Welfare and Development Office, together with local community welfare marshals, shall ensure compassionate enforcement and conduct immediate family assessments."* |

## M.5 Domain 03: Public Utilities and Telecom Franchises (Items GT-146 to GT-190)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-146]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. REVISED SPEED LIMITS ON NATIONAL ARTERIAL HIGHWAYS. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, the maximum allowable vehicular speed for private passenger cars traversing the national highway stretches within Davao City is hereby established at one hundred twenty (120) kilometers per hour. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-147]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 4* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 4. COMPULSORY MUNICIPAL TELECOM FRANCHISING. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, no national telecommunications carrier shall construct, operate, or maintain cellular base stations or fiber transmission grids within Davao City without first securing a local legislative franchise enacted by the Sangguniang Panlungsod. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-148]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 4. LOCAL ELECTRIC DISTRIBUTION TARIFF CAPS. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, the Sangguniang Panlungsod shall review and approve all retail power distribution tariff adjustments proposed by local electric utilities, capping distribution wheeling charges at ten percent (10%) above operational generation costs. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-149]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. REVISED SPEED LIMITS ON NATIONAL ARTERIAL HIGHWAYS. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, the maximum allowable vehicular speed for private passenger cars traversing the national highway stretches within Davao City is hereby established at one hundred twenty (120) kilometers per hour. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-150]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 3. SPEED LIMIT HARMONIZATION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, in accordance with Section 35 of Republic Act No. 4136, maximum vehicular speeds along national highways within city territorial limits shall not exceed eighty (80) kilometers per hour for passenger automobiles. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-151]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 3* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 3. LOCAL BUILDING PERMIT HARMONIZATION. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, in recognition of Section 4 of Republic Act No. 7925, telecommunications public utilities possessing valid national franchises from Congress shall secure standard structural building permits from the City Building Official prior to erecting cellular towers. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-152]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 3* **[Gold: Entailment]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 3. RECOGNITION OF NATIONAL ERC JURISDICTION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, recognizing the exclusive jurisdiction of the Energy Regulatory Commission under Section 43 of Republic Act No. 9136, local electric distribution utilities shall maintain public electric distribution networks along municipal road rights-of-way in accordance with safety standards. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-153]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 3. SPEED LIMIT HARMONIZATION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, in accordance with Section 35 of Republic Act No. 4136, maximum vehicular speeds along national highways within city territorial limits shall not exceed eighty (80) kilometers per hour for passenger automobiles. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-154]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 3. LOCAL BUILDING PERMIT HARMONIZATION. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, in recognition of Section 4 of Republic Act No. 7925, telecommunications public utilities possessing valid national franchises from Congress shall secure standard structural building permits from the City Building Official prior to erecting cellular towers. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-155]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 4* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 4. STREET SIGN ELEVATION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, overhead municipal street name signboards installed above secondary intersections shall be positioned at a vertical clearance of not less than five (5) meters above road asphalt. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |

**[GT-156]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 4* **[Gold: Neutral]**

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| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 4. BATTERY RECYCLING PROCEDURES. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, expired lithium-ion rechargeable battery packs utilized in municipal handheld radio transceivers shall be turned over to the City General Services Office every two (2) years for hazardous waste recycling. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |

**[GT-157]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 4* **[Gold: Neutral]**

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| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 4. REFLECTIVE TAPE SPECIFICATIONS. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, high-visibility safety vests issued to municipal roadway street sweepers shall feature reflective retro-prismatic bands measuring not less than five (5) centimeters in width. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |

**[GT-158]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 4* **[Gold: Neutral]**

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|---|---|
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 4. STREET SIGN ELEVATION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, overhead municipal street name signboards installed above secondary intersections shall be positioned at a vertical clearance of not less than five (5) meters above road asphalt. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |

**[GT-159]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 7* **[Gold: Contradiction]**

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 7. FREIGHT SPEED DEREGULATION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, heavy freight trucks carrying export fruits along the Davao-Bukidnon National Highway are exempted from national speed limit restrictions, permitting vehicular operating speeds up to ninety (90) kilometers per hour. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-160]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 6. LOCAL INTERCONNECTION ROUTING RESTRICTIONS. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, telecommunications carriers operating within Davao City are prohibited from routing voice or data transmission across competing cellular networks within city limits without prior municipal council authorization. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-161]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 6. MUNICIPAL ADJUDICATION OF POWER DISPUTES. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, the City Legal Officer is authorized to adjudicate billing disputes, overcharging complaints, and rate contested cases between residential consumers and power distribution utilities within Davao City. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-162]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 7. FREIGHT SPEED DEREGULATION. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, heavy freight trucks carrying export fruits along the Davao-Bukidnon National Highway are exempted from national speed limit restrictions, permitting vehicular operating speeds up to ninety (90) kilometers per hour. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-163]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 6* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 6. LOCAL INTERCONNECTION ROUTING RESTRICTIONS. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, telecommunications carriers operating within Davao City are prohibited from routing voice or data transmission across competing cellular networks within city limits without prior municipal council authorization. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-164]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 6. MUNICIPAL ADJUDICATION OF POWER DISPUTES. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, the City Legal Officer is authorized to adjudicate billing disputes, overcharging complaints, and rate contested cases between residential consumers and power distribution utilities within Davao City. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-165]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. CTTMO SPEED ENFORCEMENT PROTOCOL. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, the City Transport and Traffic Management Office (CTTMO) shall calibrate electronic radar speed guns to enforce statutory speed restrictions across crowded city streets and highway corridors. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-166]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 5. UNDERGROUND UTILITY CONDUIT COORDINATION. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, telecommunications public utilities operating under valid NTC certificates shall coordinate with the City Engineer when excavating underground conduit corridors along public streets. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-167]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 5* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
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| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 5. UTILITY POLE RELOCATION PROCEDURES. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, electric distribution utilities shall coordinate with the City Engineer when relocating utility poles affected by municipal road widening projects, following ERC distribution guidelines. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-168]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. CTTMO SPEED ENFORCEMENT PROTOCOL. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, the City Transport and Traffic Management Office (CTTMO) shall calibrate electronic radar speed guns to enforce statutory speed restrictions across crowded city streets and highway corridors. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-169]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 5. UNDERGROUND UTILITY CONDUIT COORDINATION. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, telecommunications public utilities operating under valid NTC certificates shall coordinate with the City Engineer when excavating underground conduit corridors along public streets. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-170]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 5* **[Gold: Entailment]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 5. UTILITY POLE RELOCATION PROCEDURES. — Pursuant to the urban mobility and traffic management authority of the City Government of Davao under the Local Government Code, electric distribution utilities shall coordinate with the City Engineer when relocating utility poles affected by municipal road widening projects, following ERC distribution guidelines. The City Transport and Traffic Management Office (CTTMO) shall deploy traffic enforcers and install appropriate directional signages to ensure full operational compliance."* |
| **[GT-171]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. CTTMO SPEED ENFORCEMENT PROTOCOL. — In the exercise of municipal powers to regulate the use of public thoroughfares and utility corridors across Davao City, the City Transport and Traffic Management Office (CTTMO) shall calibrate electronic radar speed guns to enforce statutory speed restrictions across crowded city streets and highway corridors. The City Engineers Office, in coordination with the Energy Regulatory Commission regional desk, shall conduct periodic technical inspections of utility installations."* |
| **[GT-172]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. ASPHALT WEARING COURSE SPECIFICATIONS. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, hot asphalt pavement mixtures utilized for municipal road resurfacing projects shall incorporate crushed basalt aggregates to ensure surface skid resistance. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-173]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 5. ANTENNA MAST COLOR SCHEME. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, emergency municipal broadcast antenna towers erected on public buildings shall feature alternating bands of aviation orange and white enamel paint. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-174]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 3* **[Gold: Neutral]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 3. WORKSHOP POWER TOOL INSPECTIONS. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, portable electric drills and saws utilized in the municipal carpentry workshop shall undergo bi-monthly insulation testing by certified city electricians. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-175]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 5* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 5. ASPHALT WEARING COURSE SPECIFICATIONS. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, hot asphalt pavement mixtures utilized for municipal road resurfacing projects shall incorporate crushed basalt aggregates to ensure surface skid resistance. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-176]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 5. ANTENNA MAST COLOR SCHEME. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, emergency municipal broadcast antenna towers erected on public buildings shall feature alternating bands of aviation orange and white enamel paint. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-177]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 3* **[Gold: Neutral]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 3. WORKSHOP POWER TOOL INSPECTIONS. — To protect commuter safety, reduce vehicular congestion, and ensure equitable access to essential infrastructure networks, portable electric drills and saws utilized in the municipal carpentry workshop shall undergo bi-monthly insulation testing by certified city electricians. Transport operators and utility service providers failing to observe these municipal regulations shall face immediate citation and administrative impoundment."* |
| **[GT-178]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 9* **[Gold: Contradiction]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 9. DYNAMIC VELOCITY CORRIDOR HARMONIZATION. — To enhance public convenience, alleviate vehicular gridlock, and preserve the physical integrity of city road networks, and to maximize traffic throughput along suburban bypass routes, the City Transport and Traffic Management Office may establish progressive green-wave transit corridors, authorizing motor vehicles maintaining synchronized formation to proceed at prevailing corridor design velocity. Municipal transport marshals shall conduct continuous monitoring along designated corridors to guarantee that vehicular movements strictly observe these guidelines."* |
| **[GT-179]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 8* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 8. PRIORITY CIVIL DEFENSE TELECOMMUNICATIONS QUEUING. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, telecommunications carriers operating base transceiver stations within municipal jurisdiction must configure network routing protocols to allocate dedicated bandwidth channels for municipal civil defense communications during severe weather advisories, conditioning cellular mast operating renewals on annual compliance certification. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-180]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 8. MUNICIPAL GRID RESILIENCY INFRASTRUCTURE SURCHARGE. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, and to finance the conversion of overhead distribution cables into subterranean conduits, the City Council shall establish an auxiliary urban grid hardening fee to be unbundled on consumer electricity invoices, mandating the franchised distribution utility to collect the adjusted schedule. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-181]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 9* **[Gold: Contradiction]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 9. DYNAMIC VELOCITY CORRIDOR HARMONIZATION. — To enhance public convenience, alleviate vehicular gridlock, and preserve the physical integrity of city road networks, and to maximize traffic throughput along suburban bypass routes, the City Transport and Traffic Management Office may establish progressive green-wave transit corridors, authorizing motor vehicles maintaining synchronized formation to proceed at prevailing corridor design velocity. Municipal transport marshals shall conduct continuous monitoring along designated corridors to guarantee that vehicular movements strictly observe these guidelines."* |
| **[GT-182]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 6. CORRIDOR SPEED CALIBRATION. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, the City Transport and Traffic Management Office shall position digital velocity detection monitors along major thoroughfares to deter dangerous rapid transit and enforce statutory vehicular speed ceilings. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-183]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 7* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 7. EXPEDITED NETWORK INFRASTRUCTURE CLEARANCE. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, municipal zoning administrators shall expedite local building clearances for cellular transmission facilities to promote broadband connectivity for underserved rural communities in harmony with national telecommunications development goals. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-184]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 7* **[Gold: Entailment]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 7. CLEARANCE OF POWER DISTRIBUTION CORRIDORS. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, municipal building inspectors shall coordinate with certified electric distribution utilities to ensure that private structures do not encroach upon high-voltage aerial distribution corridors. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-185]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 6. CORRIDOR SPEED CALIBRATION. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, the City Transport and Traffic Management Office shall position digital velocity detection monitors along major thoroughfares to deter dangerous rapid transit and enforce statutory vehicular speed ceilings. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-186]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 7* **[Gold: Entailment]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 7. EXPEDITED NETWORK INFRASTRUCTURE CLEARANCE. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, municipal zoning administrators shall expedite local building clearances for cellular transmission facilities to promote broadband connectivity for underserved rural communities in harmony with national telecommunications development goals. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-187]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 4* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 4. MEDIAN HEDGE PRUNING. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, flowering shrubs planted along central roadway medians shall be trimmed regularly by municipal arborists to prevent visual obstruction of oncoming vehicular headlights. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-188]** *Republic Act No. 7925 (Public Telecommunications Policy Act of 1995), Section 4 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 4. Declaration of National Policy. — Telecommunications is essential to the economic development, integrity and security of the Philippines, and as such shall be developed and administered as to safeguard, enrich and strengthen the economic, cultural, social and political fabric of the Philippines. A fundamental objective of government is to develop and maintain a viable, efficient, reliable and universally accessible telecommunications infrastructure. The expansion of telecommunications services shall be undertaken primarily by the private sector, and the entry and operation of telecommunications public utilities shall be regulated and supervised by the National Telecommunications Commission (NTC) as an independent administrative body pursuant to national policy."* | *"SECTION 4. AUDIO EQUIPMENT CABLE MAINTENANCE. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, sound technicians operating audio amplification consoles in public auditoriums shall coil microphone cables loosely around wooden storage spools to prevent internal copper wire fractures. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-189]** *Republic Act No. 9136 (Electric Power Industry Reform Act of 2001), Section 43(u) vs. Section 5* **[Gold: Neutral]** | |
| *"Section 43(u). Functions of the Energy Regulatory Commission (ERC). — The ERC shall have the original and exclusive jurisdiction over all cases contesting rates, fees, fines and penalties imposed in the exercise of its powers, functions and responsibilities, and over all cases involving disputes between and among participants or players in the energy sector. All actions shall be within the exclusive jurisdiction of the Commission, and local government units shall not modify, alter, or superimpose additional local retail tariff rates upon distribution utilities licensed by the Commission."* | *"SECTION 5. CIRCUIT BREAKER LABELLING. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, electrical distribution panels installed within city government repair depots shall display permanent thermal-printed labels identifying each connected workshop machine. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |
| **[GT-190]** *Republic Act No. 4136 (Land Transportation and Traffic Code), Section 35 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 35. Restriction as to speed. — (a) Any person driving a motor vehicle on a highway shall drive the same at a careful and prudent speed, not greater nor less than is reasonable and proper. (b) Subject to the provisions of the preceding paragraph, the rate of speed of any motor vehicle shall not exceed the following: on open country roads with no blind corners, eighty (80) km. per hour for passenger cars and motorcycles, and fifty (50) km. per hour for motor trucks and buses; on through streets or boulevards, forty (40) km. per hour; on city and municipal streets with light traffic, thirty (30) km. per hour; and through crowded streets, twenty (20) km. per hour. (c) The rates of speed herein prescribed shall not apply to emergency medical vehicles, fire department vehicles, or law enforcement officers pursuing traffic violators."* | *"SECTION 4. MEDIAN HEDGE PRUNING. — Pursuant to the police power of the Sangguniang Panlungsod to optimize urban transit flow and protect public safety along municipal roadways, flowering shrubs planted along central roadway medians shall be trimmed regularly by municipal arborists to prevent visual obstruction of oncoming vehicular headlights. The City Transport and Traffic Management Office, in coordination with local traffic enforcers, shall oversee field operations and issue administrative citation notices."* |

## M.6 Domain 04: Public Health, Hospitals and Medical Services (Items GT-191 to GT-235)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-191]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 4* | **[Gold: Contradiction]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 4. EMERGENCY ADMISSION CASH DEPOSITS. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, city-operated public health facilities and hospital emergency rooms in Davao City are authorized to demand an advance cash deposit of Five Thousand Pesos (PHP 5,000.00) before admitting indigent patients for emergency medical stabilization. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-192]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 5* | **[Gold: Contradiction]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 5. PUBLIC DISCLOSURE OF COMMUNICABLE DISEASE PATIENTS. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, and to alert neighborhood residents during epidemic outbreaks, the City Health Office is authorized to publish the full names, residential addresses, and specific medical diagnoses of individuals infected with notifiable diseases on official barangay bulletin boards. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-193]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 6* | **[Gold: Contradiction]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 6. INDOOR SMOKING IN ENTERTAINMENT ESTABLISHMENTS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, licensed bars, karaoke lounges, and nightlife establishments operating within Davao City are permitted to allow indoor tobacco smoking inside customer seating lounges occupying up to fifty percent (50%) of their total enclosed floor space. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-194]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 4* | **[Gold: Contradiction]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 4. EMERGENCY ADMISSION CASH DEPOSITS. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, city-operated public health facilities and hospital emergency rooms in Davao City are authorized to demand an advance cash deposit of Five Thousand Pesos (PHP 5,000.00) before admitting indigent patients for emergency medical stabilization. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-195]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 3* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 3. COMPREHENSIVE ACCESS TO PRIMARY MEDICAL CARE. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, in compliance with Section 6 of Republic Act No. 11223, the City Health Office shall ensure that all residents of Davao City enjoy immediate access to essential primary and preventive health care services through district health centers. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-196]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 3. STATUTORY REPORTING OF NOTIFIABLE DISEASES. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, pursuant to Section 9 of Republic Act No. 11332, all public and private medical facilities in Davao City shall transmit epidemiological reports of notifiable diseases to the Department of Health while strictly maintaining patient confidentiality. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-197]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 4* | **[Gold: Entailment]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 4. ABSOLUTE PROHIBITION OF PUBLIC SMOKING. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, in strict observance of Section 5 of Republic Act No. 9211, smoking and vaping are absolutely prohibited in all public conveyances, government offices, schools, medical facilities, and enclosed public spaces throughout Davao City. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-198]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 3. COMPREHENSIVE ACCESS TO PRIMARY MEDICAL CARE. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, in compliance with Section 6 of Republic Act No. 11223, the City Health Office shall ensure that all residents of Davao City enjoy immediate access to essential primary and preventive health care services through district health centers. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-199]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 3* | **[Gold: Entailment]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 3. STATUTORY REPORTING OF NOTIFIABLE DISEASES. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, pursuant to Section 9 of Republic Act No. 11332, all public and private medical facilities in Davao City shall transmit epidemiological reports of notifiable diseases to the Department of Health while strictly maintaining patient confidentiality. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-200]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 3* | **[Gold: Neutral]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 3. MERCURY THERMOMETER PROHIBITION. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, public and private clinical laboratories operating within Davao City shall replace all mercury-containing glass thermometers with certified digital sensor thermometers within ninety (90) calendar days. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-201]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 4* | **[Gold: Neutral]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 4. CLINICAL EXAMINATION GLOVE STANDARDS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, disposable nitrile examination gloves procured for city health stations shall have a minimum cuff thickness of zero point zero eight (0.08) millimeters. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-202]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 4* | **[Gold: Neutral]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 4. PARK WASTE BIN SPECIFICATIONS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, waste receptacles installed in public parks shall have a minimum internal volume of fifty (50) liters and display standardized labels for biodegradable and recyclable waste. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-203]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 3* | **[Gold: Neutral]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 3. MERCURY THERMOMETER PROHIBITION. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, public and private clinical laboratories operating within Davao City shall replace all mercury-containing glass thermometers with certified digital sensor thermometers within ninety (90) calendar days. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-204]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 6* | **[Gold: Contradiction]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 6. REAL PROPERTY REQUIREMENT FOR HEALTH ENROLLMENT. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, free primary and emergency healthcare consultations at municipal health centers shall be restricted to residents who present proof of real property ownership within Davao City. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-205]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 7* | **[Gold: Contradiction]** |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
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| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 7. MANDATORY LOCAL FILTERING OF DISEASE REPORTS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, private clinical laboratories detecting notifiable pathogens within Davao City shall transmit laboratory reports exclusively to the host barangay council, withholding statutory epidemiological notices from the Department of Health pending local clearance. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-206]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 4* | **[Gold: Contradiction]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 4. EXEMPTION OF VAPOR PRODUCTS IN WORKPLACES. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, the use of electronic nicotine delivery systems and vaporized tobacco devices is permitted inside enclosed commercial workplaces and municipal government offices. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-207]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 6* | **[Gold: Contradiction]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 6. REAL PROPERTY REQUIREMENT FOR HEALTH ENROLLMENT. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, free primary and emergency healthcare consultations at municipal health centers shall be restricted to residents who present proof of real property ownership within Davao City. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-208]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 7* | **[Gold: Contradiction]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 7. MANDATORY LOCAL FILTERING OF DISEASE REPORTS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, private clinical laboratories detecting notifiable pathogens within Davao City shall transmit laboratory reports exclusively to the host barangay council, withholding statutory epidemiological notices from the Department of Health pending local clearance. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-209]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 4* | **[Gold: Contradiction]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 4. EXEMPTION OF VAPOR PRODUCTS IN WORKPLACES. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, the use of electronic nicotine delivery systems and vaporized tobacco devices is permitted inside enclosed commercial workplaces and municipal government offices. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-210]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 5* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 5. PROMPT EMERGENCY PATIENT STABILIZATION. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, municipal health stations shall stabilize walk-in emergency patients immediately without financial pre-conditions, coordinating transfer to specialized tertiary facilities in accordance with PhilHealth referral protocols. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-211]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 5* | **[Gold: Entailment]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 5. SECURE EPIDEMIOLOGICAL DATA TRANSMISSION. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, the City Health Office Epidemiology and Surveillance Unit shall manage contact tracing data through encrypted digital platforms accessible only to authorized health personnel in compliance with DOH standards. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-212]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 6* | **[Gold: Entailment]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 6. DESIGNATED OUTDOOR SMOKING AREAS. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, designated smoking areas in commercial complexes shall be situated strictly in open outdoor spaces, positioned at least ten (10) meters away from building entrances, operable windows, and public access pathways. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-213]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 5* | **[Gold: Entailment]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 5. PROMPT EMERGENCY PATIENT STABILIZATION. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, municipal health stations shall stabilize walk-in emergency patients immediately without financial pre-conditions, coordinating transfer to specialized tertiary facilities in accordance with PhilHealth referral protocols. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-214]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 5* | **[Gold: Entailment]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 5. SECURE EPIDEMIOLOGICAL DATA TRANSMISSION. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, the City Health Office Epidemiology and Surveillance Unit shall manage contact tracing data through encrypted digital platforms accessible only to authorized health personnel in compliance with DOH standards. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-215]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 6* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 6. DESIGNATED OUTDOOR SMOKING AREAS. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, designated smoking areas in commercial complexes shall be situated strictly in open outdoor spaces, positioned at least ten (10) meters away from building entrances, operable windows, and public access pathways. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-216]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 5* | **[Gold: Entailment]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 5. PROMPT EMERGENCY PATIENT STABILIZATION. — Pursuant to the local health autonomy mandate under Republic Act No. 7160 and in pursuit of the Universal Health Care framework, municipal health stations shall stabilize walk-in emergency patients immediately without financial pre-conditions, coordinating transfer to specialized tertiary facilities in accordance with PhilHealth referral protocols. The City Health Office (CHO) and the City Epidemiological Surveillance Unit shall deploy health sanitary inspectors to ensure strict protocol enforcement."* |
| **[GT-217]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 5* | **[Gold: Neutral]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 5. INFECTIOUS WASTE BIN SPECIFICATIONS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, clinical sharps disposal containers utilized in municipal health stations shall be constructed of rigid, puncture-proof red polypropylene bearing biohazard insignias. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-218]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 3* | **[Gold: Neutral]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 3. VACCINE COLD-CHAIN TEMPERATURE LOGS. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, clinic nurses shall verify and record digital temperatures of vaccine storage refrigerators twice daily to ensure uninterrupted cold-chain preservation. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-219]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 3* | **[Gold: Neutral]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 3. ROTATIONAL LAWN IRRIGATION SCHEDULE. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, automated sprinkler systems installed within urban plazas shall operate exclusively during early morning hours between four o'clock and six o'clock to conserve municipal treated water. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-220]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 5* | **[Gold: Neutral]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 5. INFECTIOUS WASTE BIN SPECIFICATIONS. — In the exercise of municipal police power to preserve public hygiene and prevent the transmission of communicable illnesses, clinical sharps disposal containers utilized in municipal health stations shall be constructed of rigid, puncture-proof red polypropylene bearing biohazard insignias. The Davao City Anti-Smoking Task Force, in coordination with the Davao City Police Office, shall conduct random inspections of regulated public spaces."* |
| **[GT-221]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 3* | **[Gold: Neutral]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 3. VACCINE COLD-CHAIN TEMPERATURE LOGS. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, clinic nurses shall verify and record digital temperatures of vaccine storage refrigerators twice daily to ensure uninterrupted cold-chain preservation. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-222]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 3* | **[Gold: Neutral]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 3. ROTATIONAL LAWN IRRIGATION SCHEDULE. — To protect public health, promote clinical safety standards, and safeguard patient welfare across all medical facilities in Davao City, automated sprinkler systems installed within urban plazas shall operate exclusively during early morning hours between four o'clock and six o'clock to conserve municipal treated water. Non-compliance with the health and sanitation directives established herein shall warrant the immediate revocation of the establishment's sanitary permit."* |
| **[GT-223]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 8* | **[Gold: Contradiction]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 8. PROTOCOL FOR SPECIALIZED TRAUMA ADMISSION. — In the exercise of municipal regulatory authority to maintain clean air, clinical safety, and hygienic living conditions for all residents, patients arriving at municipal emergency departments requiring intensive critical care intervention shall be routed through the Patient Accounting Bureau to establish verified national health insurance enrollment eligibility or documented corporate coverage prior to admission to specialized surgical suites. Municipal health inspectors are hereby empowered to examine business premises and recommend administrative closure for establishments failing to observe these health rules."* |
| **[GT-224]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 8* | **[Gold: Contradiction]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 8. GRANULAR EPIDEMIOLOGICAL CLUSTER MAPPING. — Pursuant to the police power of the Sangguniang Panlungsod to protect community health and prevent environmental hazards within the city, and to maximize community vigilance during epidemic surges, the City Health Office shall publish digital geospatial directory maps on municipal portals identifying the specific residential addresses and compound locations of active quarantined households to facilitate localized neighborhood tracing. The City Health Office, in coordination with the City Legal Office, shall conduct regular hygienic inspections and issue formal compliance notices to property administrators."* |
| **[GT-225]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 7* | **[Gold: Contradiction]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 7. DESIGNATED INDOOR HERITAGE CIGAR DEGUSTATION ROOMS. — To advance public health protection and ensure the comprehensive monitoring of sanitation standards across commercial and healthcare properties, licensed hospitality establishments and boutique cigar lounges may maintain an enclosed indoor tobacco tasting chamber for adult connoisseurs, provided that the venue operates independent negative-pressure ventilation systems certified by the City Health Office. The City Administrator shall coordinate with local barangay health workers to maintain community health surveillance and ensure adherence to this Section."* |
| **[GT-226]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 8* | **[Gold: Contradiction]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 8. PROTOCOL FOR SPECIALIZED TRAUMA ADMISSION. — In the exercise of municipal regulatory authority to maintain clean air, clinical safety, and hygienic living conditions for all residents, patients arriving at municipal emergency departments requiring intensive critical care intervention shall be routed through the Patient Accounting Bureau to establish verified national health insurance enrollment eligibility or documented corporate coverage prior to admission to specialized surgical suites. Municipal health inspectors are hereby empowered to examine business premises and recommend administrative closure for establishments failing to observe these health rules."* |
| **[GT-227]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 4* | **[Gold: Entailment]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 4. DECENTRALIZED MEDICAL CLINIC ACCESSIBILITY. — Pursuant to the police power of the Sangguniang Panlungsod to protect community health and prevent environmental hazards within the city, the City Health Office shall deploy mobile medical teams to remote agricultural settlements to provide free preventative clinical diagnostic consultations and essential therapeutic medications to vulnerable populations. The City Health Office, in coordination with the City Legal Office, shall conduct regular hygienic inspections and issue formal compliance notices to property administrators."* |
| **[GT-228]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 6* | **[Gold: Entailment]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 6. PROTECTION OF COMMUNICABLE DISEASE DATA. — In the exercise of municipal regulatory authority to maintain clean air, clinical safety, and hygienic living conditions for all residents, health workers conducting epidemiological disease tracing shall maintain strict medical confidentiality over patient identities, transmitting surveillance records through secure encrypted channels exclusively to authorized public health epidemiologists. Municipal health inspectors are hereby empowered to examine business premises and recommend administrative closure for establishments failing to observe these health rules."* |
| **[GT-229]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 5* | **[Gold: Entailment]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 5. SMOKE-FREE PUBLIC THOROUGHFARES. — Pursuant to the police power of the Sangguniang Panlungsod to protect community health and prevent environmental hazards within the city, the Anti-Vices Task Force shall monitor public assembly zones and commercial walkways to ensure that individuals refrain from combustion of tobacco products in areas accessible to the general public. The City Health Office, in coordination with the City Legal Office, shall conduct regular hygienic inspections and issue formal compliance notices to property administrators."* |
| **[GT-230]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 4* | **[Gold: Entailment]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 4. DECENTRALIZED MEDICAL CLINIC ACCESSIBILITY. — Pursuant to the police power of the Sangguniang Panlungsod to protect community health and prevent environmental hazards within the city, the City Health Office shall deploy mobile medical teams to remote agricultural settlements to provide free preventative clinical diagnostic consultations and essential therapeutic medications to vulnerable populations. The City Health Office, in coordination with the City Legal Office, shall conduct regular hygienic inspections and issue formal compliance notices to property administrators."* |
| **[GT-231]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 6* | **[Gold: Entailment]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 6. PROTECTION OF COMMUNICABLE DISEASE DATA. — In the exercise of municipal regulatory authority to maintain clean air, clinical safety, and hygienic living conditions for all residents, health workers conducting epidemiological disease tracing shall maintain strict medical confidentiality over patient identities, transmitting surveillance records through secure encrypted channels exclusively to authorized public health epidemiologists. Municipal health inspectors are hereby empowered to examine business premises and recommend administrative closure for establishments failing to observe these health rules."* |
| **[GT-232]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 4* | **[Gold: Neutral]** |
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 4. AUTOCLAVE TEMPERATURE MONITORING LOGS. — In the exercise of municipal regulatory authority to maintain clean air, clinical safety, and hygienic living conditions for all residents, dental clinic personnel operating steam autoclave sterilizers shall log vacuum pressure indicators following each sterilization batch to confirm complete pathogen neutralization. Municipal health inspectors are hereby empowered to examine business premises and recommend administrative closure for establishments failing to observe these health rules."* |
| **[GT-233]** *Republic Act No. 11332 (Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act), Section 9 vs. Section 5* | **[Gold: Neutral]** |
| *"Section 9. Prohibited Acts. — The following shall be prohibited under this Act: (a) Unauthorized disclosure of private and confidential information pertaining to a patient's medical condition or treatment; (b) Non-cooperation of persons and entities that should report and/or respond to notifiable diseases or health events of public health concern; and (c) Non-cooperation of the person or firms categorized as having the notifiable disease, or affected by the health event of public concern."* | *"SECTION 5. OUTPATIENT SEATING HYGIENE PROTOCOL. — Pursuant to the police power of the Sangguniang Panlungsod to protect community health and prevent environmental hazards within the city, custodial staff in public outpatient clinics shall wipe plastic waiting lounge benches with approved disinfectant solutions between morning and afternoon clinical consultation sessions. The City Health Office, in coordination with the City Legal Office, shall conduct regular hygienic inspections and issue formal compliance notices to property administrators."* |
| **[GT-234]** *Republic Act No. 9211 (Tobacco Regulation Act of 2003), Section 5 vs. Section 6* | **[Gold: Neutral]** |
| *"Section 5. Smoking in Public Places. — Smoking shall be absolutely prohibited in the following public places: (a) Centers of youth activity such as playschools, elementary schools, high schools, colleges and universities, youth hostels and recreational facilities for persons under eighteen (18) years old; (b) Elevators and stairwells; (c) Locations in which fire hazards are present, including gas stations and storage areas for flammable materials; (d) Within the buildings and premises of public and private hospitals, medical, dental, and optical clinics, health centers, nursing homes, dispensaries and laboratories; and (e) Public conveyances and public facilities."* | *"SECTION 6. SLIP-RESISTANT WALKWAY SEALANT. — To advance public health protection and ensure the comprehensive monitoring of sanitation standards across commercial and healthcare properties, pedestrian pathways within municipal recreational gardens shall be treated with silicone water-repellent sealant to minimize moss accumulation and prevent pedestrian slips during rainstorms. The City Administrator shall coordinate with local barangay health workers to maintain community health surveillance and ensure adherence to this Section."* |
| **[GT-235]** *Republic Act No. 11223 (Universal Health Care Act), Section 6 vs. Section 4* | **[Gold: Neutral]** |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 6. Service Coverage. — (a) Every Filipino shall be granted immediate eligibility and access to preventive, promotive, curative, rehabilitative, and palliative care for medical, dental, mental and emergency health services, delivered either as population-based or individual-based health services. (b) PhilHealth and the Department of Health (DOH), in coordination with local government units (LGUs), shall endeavor to provide a primary health care delivery system that will afford every Filipino access to a licensed primary care provider of choice without financial hardship at the point of service."* | *"SECTION 4. AUTOCLAVE TEMPERATURE MONITORING LOGS. — In the exercise of municipal regulatory authority to maintain clean air, clinical safety, and hygienic living conditions for all residents, dental clinic personnel operating steam autoclave sterilizers shall log vacuum pressure indicators following each sterilization batch to confirm complete pathogen neutralization. Municipal health inspectors are hereby empowered to examine business premises and recommend administrative closure for establishments failing to observe these health rules."* |

# M.7 Domain 05: Statutory Codes and General Legal Amendments (Items GT-236 to GT-280)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-236]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 4. MORATORIUM ON MINERAL AGREEMENTS. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, the Sangguniang Panlungsod hereby declares an absolute ten-year ban on the issuance and execution of all national Mineral Production Sharing Agreements (MPSA) and Financial or Technical Assistance Agreements (FTAA) across all marine territorial waters off Davao City. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-237]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 5. COMMERCIAL FISHING AUTHORIZATION IN MUNICIPAL WATERS. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, commercial fishing vessels measuring between twenty (20) and one hundred fifty (150) gross tons are hereby authorized to conduct commercial purse-seine and trawl fishing within municipal waters extending three (3) kilometers from the shoreline of Davao City. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-238]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 4. LOCAL MUNICIPAL PERMIT TO CARRY FIREARMS. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, licensed firearm holders residing within Davao City must apply for and secure a City Mayor's Permit to Carry Firearms Outside of Residence, paying an annual municipal fee of Five Thousand Pesos (PHP 5,000.00), prior to transporting registered firearms across city streets. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-239]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 4* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 4. MORATORIUM ON MINERAL AGREEMENTS. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, the Sangguniang Panlungsod hereby declares an absolute ten-year ban on the issuance and execution of all national Mineral Production Sharing Agreements (MPSA) and Financial or Technical Assistance Agreements (FTAA) across all marine territorial waters off Davao City. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-240]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 3. ENVIRONMENTAL COMPLIANCE REQUISITE. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, in accordance with Section 70 of Republic Act No. 7942, commercial quarry operators extracting sand and gravel within Davao City shall secure an Environmental Compliance Certificate from the DENR prior to the issuance of a City Mayor's Quarry Permit. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-241]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 3. ARTISANAL MUNICIPAL FISHER REGISTRATION. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, pursuant to Section 18 of Republic Act No. 8550, the City Agriculture Office shall maintain a municipal fisherfolk registry, granting priority exploitation rights within the fifteen (15) kilometer municipal waters to resident subsistence fisherfolk. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-242]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 3. RECOGNITION OF NATIONAL PNP PTCFOR LICENSES. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, in accordance with Section 31 of Republic Act No. 10591, the Davao City Police Office shall verify that any civilian individual carrying a concealed registered firearm in public places holds a valid Permit to Carry Firearms Outside of Residence issued by the Chief of the PNP. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-243]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 3. ENVIRONMENTAL COMPLIANCE REQUISITE. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, in accordance with Section 70 of Republic Act No. 7942, commercial quarry operators extracting sand and gravel within Davao City shall secure an Environmental Compliance Certificate from the DENR prior to the issuance of a City Mayor's Quarry Permit. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-244]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 3. ARTISANAL MUNICIPAL FISHER REGISTRATION. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, pursuant to Section 18 of Republic Act No. 8550, the City Agriculture Office shall maintain a municipal fisherfolk registry, granting priority exploitation rights within the fifteen (15) kilometer municipal waters to resident subsistence fisherfolk. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-245]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 4. POTTING SOIL COMPOSITION. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, municipal botanical nurseries propagating indigenous hardwood saplings shall blend garden topsoil with river sand in a ratio of two (2) parts topsoil to one (1) part river sand. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-246]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 4. MOORING BOLLARD SPECIFICATIONS. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, cast-iron mooring bollards installed along municipal fish landing wharves shall have a rated holding tensile capacity of not less than thirty (30) metric tons. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-247]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 4. PYROTECHNIC TEMPORARY STORAGE DISTANCES. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, temporary commercial storage warehouses stocking authorized pyrotechnic display materials shall maintain a separation clearance of not less than fifty (50) meters from fuel storage tanks. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-248]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 4. POTTING SOIL COMPOSITION. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, municipal botanical nurseries propagating indigenous hardwood saplings shall blend garden topsoil with river sand in a ratio of two (2) parts topsoil to one (1) part river sand. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-249]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 6* **[Gold: Contradiction]** | |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
|---|---|
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 6. REVOCATION OF NATIONAL ENVIRONMENTAL COMPLIANCE CERTIFICATES. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, the City Environment and Natural Resources Office is authorized to revoke Environmental Compliance Certificates (ECC) issued by the national Department of Environment and Natural Resources for large-scale mining operations situated within regional boundaries. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-250]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 7. EXCLUSION OF MUNICIPAL FISHERS FROM ADJACENT PROVINCES. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, marginal municipal fisherfolk registered in neighboring municipalities across the Davao Gulf are prohibited from casting non-motorized fishing nets in Davao City municipal waters, reserving all fishing grounds exclusively to voters of Davao City. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-251]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 6. PROHIBITION OF NATIONAL PTCFOR PERMITS IN BUSINESS PARKS. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, duly issued national Permits to Carry Firearms Outside Residence (PTCFOR) issued by the Chief of the PNP are hereby rendered null and void within all commercial business districts and shopping malls situated in Davao City. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-252]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 6. REVOCATION OF NATIONAL ENVIRONMENTAL COMPLIANCE CERTIFICATES. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, the City Environment and Natural Resources Office is authorized to revoke Environmental Compliance Certificates (ECC) issued by the national Department of Environment and Natural Resources for large-scale mining operations situated within regional boundaries. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-253]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 7* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 7. EXCLUSION OF MUNICIPAL FISHERS FROM ADJACENT PROVINCES. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, marginal municipal fisherfolk registered in neighboring municipalities across the Davao Gulf are prohibited from casting non-motorized fishing nets in Davao City municipal waters, reserving all fishing grounds exclusively to voters of Davao City. The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-254]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 6. PROHIBITION OF NATIONAL PTCFOR PERMITS IN BUSINESS PARKS. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, duly issued national Permits to Carry Firearms Outside Residence (PTCFOR) issued by the Chief of the PNP are hereby rendered null and void within all commercial business districts and shopping malls situated in Davao City. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-255]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 5. MULTI-PARTITE MONITORING TEAMS. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, the City Environment and Natural Resources Office shall participate in multi-partite monitoring teams organized by the Mines and Geosciences Bureau (MGB) to inspect commercial extraction sites for ecological rehabilitation compliance. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-256]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 6. COMMERCIAL FISHING BOAT CLEARANCE IN DEEP WATERS. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, small commercial fishing vessels between 3.1 and 20 gross tons may operate within municipal waters between 10.1 and 15 kilometers, provided the depth exceeds seven fathoms and upon approval of the Fisheries and Aquatic Resources Management Council (FARMC). The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-257]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 5. WEAPONS RESTRICTION IN GOVERNMENT HALLS. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, civilians holding valid PTCFOR licenses shall deposit their firearms in secure lockboxes at building security entrances before entering City Hall administrative offices, in harmony with national police facility security guidelines. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-258]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 5. MULTI-PARTITE MONITORING TEAMS. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, the City Environment and Natural Resources Office shall participate in multi-partite monitoring teams organized by the Mines and Geosciences Bureau (MGB) to inspect commercial extraction sites for ecological rehabilitation compliance. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-259]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 6. COMMERCIAL FISHING BOAT CLEARANCE IN DEEP WATERS. — In the exercise of municipal regulatory jurisdiction over local water bodies, coastal resources, and municipal public lands, small commercial fishing vessels between 3.1 and 20 gross tons may operate within municipal waters between 10.1 and 15 kilometers, provided the depth exceeds seven fathoms and upon approval of the Fisheries and Aquatic Resources Management Council (FARMC). The City Agriculture Office and the Fisheries and Aquatic Resources Management Council (FARMC) shall oversee maritime zone enforcement."* |
| **[GT-260]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 5. WEAPONS RESTRICTION IN GOVERNMENT HALLS. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, civilians holding valid PTCFOR licenses shall deposit their firearms in secure lockboxes at building security entrances before entering City Hall administrative offices, in harmony with national police facility security guidelines. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-261]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 5. MULTI-PARTITE MONITORING TEAMS. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, the City Environment and Natural Resources Office shall participate in multi-partite monitoring teams organized by the Mines and Geosciences Bureau (MGB) to inspect commercial extraction sites for ecological rehabilitation compliance. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-262]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 5. SOLID WASTE COMPOST AERATION. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, operators at the city solid waste composting facility shall turn organic waste windrows twice weekly to accelerate aerobic microbial decomposition. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-263]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 5* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 5. FISH MARKET FLOOR WASHING SCHEDULE. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, municipal fish terminal trading floors shall be hosed down with high-pressure chlorinated water immediately following the conclusion of morning wholesale bidding auctions. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-264]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 5. SECURITY GUARD UNIFORM ACCESSORIES. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, private security guards deployed at municipal transit terminals shall carry standardized hardwood nightsticks measuring sixty centimeters in length and an emergency whistle. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-265]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 5. SOLID WASTE COMPOST AERATION. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, operators at the city solid waste composting facility shall turn organic waste windrows twice weekly to accelerate aerobic microbial decomposition. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-266]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 5. FISH MARKET FLOOR WASHING SCHEDULE. — To protect ecological stability, ensure sustainable natural resource utilization, and preserve public safety across the city, municipal fish terminal trading floors shall be hosed down with high-pressure chlorinated water immediately following the conclusion of morning wholesale bidding auctions. Violations of the environmental conditions and licensing restrictions specified herein shall be penalized under the Davao City Ecological Solid Waste Management Code."* |
| **[GT-267]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 5. SECURITY GUARD UNIFORM ACCESSORIES. — Pursuant to the environmental management and resource conservation powers of Davao City under national environmental statutes, private security guards deployed at municipal transit terminals shall carry standardized hardwood nightsticks measuring sixty centimeters in length and an emergency whistle. The City Environment and Natural Resources Office (CENRO), in coordination with the Department of Environment and Natural Resources, shall conduct field monitoring."* |
| **[GT-268]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 8* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 8. COASTAL BENTHIC ZONE ECOLOGICAL STEWARDSHIP LEASES. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, commercial dredging operators and marine geological surveying entities operating within the city's coastal baseline shelf must execute an ecological stewardship lease approved by the Sangguniang Panlungsod prior to initiating sub-surface coring or mineral sediment sampling. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |
| **[GT-269]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 8. SEASONAL PELAGIC FISHERY RECOVERY LICENSES. — In the exercise of municipal regulatory authority to maintain ecological balance and prevent unauthorized exploitation of local natural resources, and to optimize economic harvest during migratory oceanic fish runs, the City Agriculture Office may issue seasonal joint-venture fishing endorsements authorizing commercial vessels exceeding three gross tons to operate mid-water purse seines within the inner municipal water baseline. The City Legal Office is hereby authorized to institute summary abatement proceedings and seek environmental damages against entities violating these municipal standards."* |
| **[GT-270]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 8. MUNICIPAL TRANSIT FIREARM VERIFICATION CLEARANCE. — In the exercise of municipal regulatory authority to maintain ecological balance and prevent unauthorized exploitation of local natural resources, citizens transporting licensed personal defense firearms across municipal entry boundaries must present a Certificate of Local Transit Endorsement issued by the City Mayor's Security Directorate confirming legitimate civic travel within municipal territory. The City Legal Office is hereby authorized to institute summary abatement proceedings and seek environmental damages against entities violating these municipal standards."* |
| **[GT-271]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 8. COASTAL BENTHIC ZONE ECOLOGICAL STEWARDSHIP LEASES. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, commercial dredging operators and marine geological surveying entities operating within the city's coastal baseline shelf must execute an ecological stewardship lease approved by the Sangguniang Panlungsod prior to initiating sub-surface coring or mineral sediment sampling. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |
| **[GT-272]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 7* **[Gold: Entailment]** | |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
|---|---|
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 7. POST-EXTRACTION ECOLOGICAL RESTORATION VERIFICATION. — In the exercise of municipal regulatory authority to maintain ecological balance and prevent unauthorized exploitation of local natural resources, municipal environmental officers shall inspect decommissioned aggregate borrow pits alongside national mining geologists to confirm that quarry operators complete vegetative slope re-contouring in conformity with approved environmental rehabilitation commitments. The City Legal Office is hereby authorized to institute summary abatement proceedings and seek environmental damages against entities violating these municipal standards."* |
| **[GT-273]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 4. ESTUARINE BREEDING SANCTUARY RESERVATION. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, municipal coastal patrol marshals shall protect inter-tidal mangrove estuaries from motorized fishing vessels, reserving nearshore breeding nurseries exclusively for traditional handline sustenance fishers. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |
| **[GT-274]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 6. WEAPONS RESTRICTIONS AT CIVIC GATHERINGS. — In the exercise of municipal regulatory authority to maintain ecological balance and prevent unauthorized exploitation of local natural resources, security personnel shall maintain perimeter checkpoints at public festival parades, requiring armed citizens to deposit personal firearms with authorized police marshals during festive assemblies. The City Legal Office is hereby authorized to institute summary abatement proceedings and seek environmental damages against entities violating these municipal standards."* |
| **[GT-275]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 7* **[Gold: Entailment]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 7. POST-EXTRACTION ECOLOGICAL RESTORATION VERIFICATION. — In the exercise of municipal regulatory authority to maintain ecological balance and prevent unauthorized exploitation of local natural resources, municipal environmental officers shall inspect decommissioned aggregate borrow pits alongside national mining geologists to confirm that quarry operators complete vegetative slope re-contouring in conformity with approved environmental rehabilitation commitments. The City Legal Office is hereby authorized to institute summary abatement proceedings and seek environmental damages against entities violating these municipal standards."* |
| **[GT-276]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 4* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 4. ESTUARINE BREEDING SANCTUARY RESERVATION. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, municipal coastal patrol marshals shall protect inter-tidal mangrove estuaries from motorized fishing vessels, reserving nearshore breeding nurseries exclusively for traditional handline sustenance fishers. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |
| **[GT-277]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 6. HERBARIUM BOTANICAL MOUNTING. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, botanical plant specimens archived within the municipal botanical herbarium shall be mounted onto acid-free rag paper using neutral archival adhesive paste to preserve plant cell structures for taxonomic classification. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |
| **[GT-278]** *Republic Act No. 8550 (The Philippine Fisheries Code of 1998), Section 18 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 18. Users of Municipal Waters. — All fishery related activities in municipal waters, as defined in this Code, shall be utilized by municipal fisherfolk and their cooperatives who are listed in the registry of municipal fisherfolk. The municipal or city government may, through its local chief executive and acting pursuant to an appropriate ordinance, permit small and medium commercial fishing vessels to operate within the ten point one (10.1) to fifteen (15) kilometer area from the shoreline in municipal waters, provided, that no commercial fishing is allowed in waters with depth less than seven (7) fathoms, and prior public consultation with the FARMC has been conducted."* | *"SECTION 6. COMMERCIAL WEIGHING SCALE CALIBRATION. — In the exercise of municipal regulatory authority to maintain ecological balance and prevent unauthorized exploitation of local natural resources, the City Treasurer shall calibrate counter-balanced platform scales utilized in fish wholesale trading halls using certified brass test weights to guarantee equitable trade transactions. The City Legal Office is hereby authorized to institute summary abatement proceedings and seek environmental damages against entities violating these municipal standards."* |
| **[GT-279]** *Republic Act No. 10591 (Comprehensive Firearms and Ammunition Regulation Act), Section 31 vs. Section 7* **[Gold: Neutral]** | |
| *"Section 31. Absence of Permit to Carry Outside of Residence. — The penalty of prision correccional and a fine of Ten thousand pesos (P10,000.00) shall be imposed upon any person who is licensed to own a firearm but who shall carry the registered firearm outside his/her residence without any legal permit to carry issued by the Chief of the Philippine National Police (PNP)."* | *"SECTION 7. EMERGENCY FIRE DOUSE INFRASTRUCTURE. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, technicians executing authorized public pyrotechnic displays shall position sealed fifty-gallon water drums and sand buckets beside mortar firing racks to extinguish defective smoldering casings. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |
| **[GT-280]** *Republic Act No. 7942 (Philippine Mining Act of 1995), Section 70 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 70. Environmental Impact Assessment (EIA). — Except during the exploration period of a mineral agreement or permit, an environmental clearance certificate shall be required based on an environmental impact assessment and procedures under the Philippine Environmental Impact Statement System including Sections 26 and 27 of the Local Government Code of 1991 which require prior consultation with affected communities and local government units."* | *"SECTION 6. HERBARIUM BOTANICAL MOUNTING. — Pursuant to the environmental stewardship mandate and police powers of the Sangguniang Panlungsod to preserve marine and terrestrial ecosystems, botanical plant specimens archived within the municipal botanical herbarium shall be mounted onto acid-free rag paper using neutral archival adhesive paste to preserve plant cell structures for taxonomic classification. The City Environment and Natural Resources Office, together with local environmental marshals, shall conduct periodic surveys and enforce municipal conservation rules."* |

## M.8 Domain 06: Public Finance and General Appropriations (Items GT-281 to GT-315)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-281]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 5. ALLOCATION TO LOCAL DEVELOPMENT PROJECTS. — To guarantee transparent financial administration, ensure fiscal discipline, and support vital socio-economic development projects, the City Government of Davao shall appropriate seven percent (7%) of its annual internal revenue allotment shares to finance the City Comprehensive Development Investment Program. Disbursing officers and department heads failing to comply with these statutory allocation thresholds shall be subject to audit disallowance and administrative sanction."* |
| **[GT-282]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 4. LOCAL DISASTER RISK FUND APPORTIONMENT. — To guarantee transparent financial administration, ensure fiscal discipline, and support vital socio-economic development projects, the Sangguniang Panlungsod shall set aside two percent (2%) of estimated regular revenue sources to establish the Local Disaster Risk Reduction and Management Fund (LDRRMF). Disbursing officers and department heads failing to comply with these statutory allocation thresholds shall be subject to audit disallowance and administrative sanction."* |
| **[GT-283]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 5. ALLOCATION TO LOCAL DEVELOPMENT PROJECTS. — To guarantee transparent financial administration, ensure fiscal discipline, and support vital socio-economic development projects, the City Government of Davao shall appropriate seven percent (7%) of its annual internal revenue allotment shares to finance the City Comprehensive Development Investment Program. Disbursing officers and department heads failing to comply with these statutory allocation thresholds shall be subject to audit disallowance and administrative sanction."* |
| **[GT-284]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 4. LOCAL DISASTER RISK FUND APPORTIONMENT. — To guarantee transparent financial administration, ensure fiscal discipline, and support vital socio-economic development projects, the Sangguniang Panlungsod shall set aside two percent (2%) of estimated regular revenue sources to establish the Local Disaster Risk Reduction and Management Fund (LDRRMF). Disbursing officers and department heads failing to comply with these statutory allocation thresholds shall be subject to audit disallowance and administrative sanction."* |
| **[GT-285]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 4* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 4. STATUTORY TWENTY PERCENT DEVELOPMENT ALLOCATION. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, in strict compliance with Section 287 of Republic Act No. 7160, exactly twenty percent (20%) of the City's National Tax Allotment is hereby allocated to the 20% Local Development Fund for capital social and economic infrastructure. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-286]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 3. FIVE PERCENT STATUTORY DISASTER FUND ALLOCATION. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, in accordance with Section 21 of Republic Act No. 10121, five percent (5%) of estimated revenue from regular sources is appropriated to the Local Disaster Risk Reduction and Management Fund (LDRRMF), with thirty percent (30%) dedicated to the Quick Response Fund. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-287]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 4. STATUTORY TWENTY PERCENT DEVELOPMENT ALLOCATION. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, in strict compliance with Section 287 of Republic Act No. 7160, exactly twenty percent (20%) of the City's National Tax Allotment is hereby allocated to the 20% Local Development Fund for capital social and economic infrastructure. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-288]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 3. FIVE PERCENT STATUTORY DISASTER FUND ALLOCATION. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, in accordance with Section 21 of Republic Act No. 10121, five percent (5%) of estimated revenue from regular sources is appropriated to the Local Disaster Risk Reduction and Management Fund (LDRRMF), with thirty percent (30%) dedicated to the Quick Response Fund. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-289]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 3* **[Gold: Neutral]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 3. BID DOCUMENT PRICING. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, prospective bidders for municipal supply contracts shall pay a non-refundable reproduction fee of One Thousand Pesos (PHP 1,000.00) for standard bidding tender dossiers. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-290]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 4* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 4. RELIEF WAREHOUSE PALLET STACKING. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, sacks of National Food Authority relief rice stored in calamity response warehouses shall not be stacked higher than ten (10) tiers upon wooden pallets. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-291]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 3* **[Gold: Neutral]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 3. BID DOCUMENT PRICING. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, prospective bidders for municipal supply contracts shall pay a non-refundable reproduction fee of One Thousand Pesos (PHP 1,000.00) for standard bidding tender dossiers. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-292]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 8. RE-ALIGNMENT OF DEVELOPMENT FUNDS TO ADMINISTRATIVE OVERHEAD. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, the City Budget Officer is authorized to re-align thirty percent of the statutory Local Development Fund to pay for municipal administrative personnel overtime allowances and official foreign travel expenses. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-293]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 7. RE-ALIGNMENT OF QUICK RESPONSE CALAMITY RESERVES. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, the City Council is authorized to transfer sixty percent of the Quick Response Fund (QRF) standby reserve to finance municipal tourism advertising and promotional campaigns. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-294]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 8. RE-ALIGNMENT OF DEVELOPMENT FUNDS TO ADMINISTRATIVE OVERHEAD. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, the City Budget Officer is authorized to re-align thirty percent of the statutory Local Development Fund to pay for municipal administrative personnel overtime allowances and official foreign travel expenses. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-295]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 7* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 7. RE-ALIGNMENT OF QUICK RESPONSE CALAMITY RESERVES. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, the City Council is authorized to transfer sixty percent of the Quick Response Fund (QRF) standby reserve to finance municipal tourism advertising and promotional campaigns. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-296]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 6. ELIGIBILITY OF INFRASTRUCTURE PROJECTS. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, programs financed by the Local Development Fund shall adhere strictly to Joint Memorandum Circulars issued by DILG and DBM, funding capital public works and communal water systems. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-297]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 6. FIVE-YEAR DISASTER SPECIAL TRUST FUND. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, unexpended balances of the LDRRMF shall accrue to a special trust fund solely for disaster risk reduction activities for five (5) years, in full compliance with RA 10121. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-298]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 6. ELIGIBILITY OF INFRASTRUCTURE PROJECTS. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, programs financed by the Local Development Fund shall adhere strictly to Joint Memorandum Circulars issued by DILG and DBM, funding capital public works and communal water systems. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-299]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 6* **[Gold: Entailment]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 6. FIVE-YEAR DISASTER SPECIAL TRUST FUND. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, unexpended balances of the LDRRMF shall accrue to a special trust fund solely for disaster risk reduction activities for five (5) years, in full compliance with RA 10121. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-300]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 6* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 6. ELIGIBILITY OF INFRASTRUCTURE PROJECTS. — In the exercise of the corporate and financial management powers of the City Government of Davao over public revenue allocations, programs financed by the Local Development Fund shall adhere strictly to Joint Memorandum Circulars issued by DILG and DBM, funding capital public works and communal water systems. The City Planning and Development Office (CPDO) shall align capital outlay disbursements with the approved Annual Investment Plan."* |
| **[GT-301]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 4. BID OPENING AUDIO-VISUAL RECORDINGS. — To guarantee transparent financial administration, ensure fiscal discipline, and support vital socio-economic development projects, the Bids and Awards Committee Secretariat shall record all competitive public bidding sessions using digital video recording equipment. Disbursing officers and department heads failing to comply with these statutory allocation thresholds shall be subject to audit disallowance and administrative sanction."* |
| **[GT-302]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 5. RELIEF BLANKET TEXTILE STANDARDS. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, emergency thermal blankets procured for calamity evacuation centers shall consist of woven fleece fabric resistant to fraying. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-303]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 4. BID OPENING AUDIO-VISUAL RECORDINGS. — To guarantee transparent financial administration, ensure fiscal discipline, and support vital socio-economic development projects, the Bids and Awards Committee Secretariat shall record all competitive public bidding sessions using digital video recording equipment. Disbursing officers and department heads failing to comply with these statutory allocation thresholds shall be subject to audit disallowance and administrative sanction."* |
| **[GT-304]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 5. RELIEF BLANKET TEXTILE STANDARDS. — Pursuant to the local fiscal autonomy provisions of Republic Act No. 7160 and in accordance with national public expenditure standards, emergency thermal blankets procured for calamity evacuation centers shall consist of woven fleece fabric resistant to fraying. The City Budget Office and the City Treasurer's Office shall verify budgetary appropriations and submit compliance certifications to the Sangguniang Panlungsod."* |
| **[GT-305]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 6* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 6. STRATEGIC REPROGRAMMING OF UNALLOCATED DEVELOPMENT BALANCES. — Pursuant to the financial administration authority of the Sangguniang Panlungsod to optimize municipal capital allocation and disaster preparedness, and to support municipal trade missions and sister-city investment summits, unexpended appropriations under the annual local development fund may be reprogrammed by executive order to finance city protocol hospitality and international civic delegation hosting. The City Treasurer and the City Budget Officer shall maintain strict fiduciary accounting and furnish annual expenditure reports to the City Council."* |
| **[GT-306]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 5. ANNUAL FISCAL SURPLUS RECOVERY PROTOCOL. — Pursuant to the financial administration authority of the Sangguniang Panlungsod to optimize municipal capital allocation and disaster preparedness, at the conclusion of the municipal fiscal year, unutilized appropriations remaining in the localized emergency response allocation shall be transferred to the City General Fund to support municipal public infirmary maintenance and community health center infrastructure. The City Treasurer and the City Budget Officer shall maintain strict fiduciary accounting and furnish annual expenditure reports to the City Council."* |
| **[GT-307]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 6. STRATEGIC REPROGRAMMING OF UNALLOCATED DEVELOPMENT BALANCES. — Pursuant to the financial administration authority of the Sangguniang Panlungsod to optimize municipal capital allocation and disaster preparedness, and to support municipal trade missions and sister-city investment summits, unexpended appropriations under the annual local development fund may be reprogrammed by executive order to finance city protocol hospitality and international civic delegation hosting. The City Treasurer and the City Budget Officer shall maintain strict fiduciary accounting and furnish annual expenditure reports to the City Council."* |
| **[GT-308]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 5* **[Gold: Contradiction]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 5. ANNUAL FISCAL SURPLUS RECOVERY PROTOCOL. — Pursuant to the financial administration authority of the Sangguniang Panlungsod to optimize municipal capital allocation and disaster preparedness, at the conclusion of the municipal fiscal year, unutilized appropriations remaining in the localized emergency response allocation shall be transferred to the City General Fund to support municipal public infirmary maintenance and community health center infrastructure. The City Treasurer and the City Budget Officer shall maintain strict fiduciary accounting and furnish annual expenditure reports to the City Council."* |
| **[GT-309]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 7* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 7. LONG-TERM COMMUNITY ASSET CREATION. — Pursuant to the financial administration authority of the Sangguniang Panlungsod to optimize municipal capital allocation and disaster preparedness, municipal development appropriations shall prioritize durable public capital facilities including rural farm access roadways, barangay maternity clinics, and communal flood control barriers. The City Treasurer and the City Budget Officer shall maintain strict fiduciary accounting and furnish annual expenditure reports to the City Council."* |
| **[GT-310]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 4. STATUTORY DISASTER RISK FINANCING. — In the exercise of sound fiscal stewardship over municipal resources and reserve funds held by the City Government of Davao, the local expenditure budget shall preserve mandatory fiscal reserves dedicated exclusively to disaster prevention, pre-disaster equipment acquisition, and rapid post-calamity relief operations. The City Planning and Development Office shall ensure that municipal development appropriations correspond directly to approved citywide priorities."* |
| **[GT-311]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 7* **[Gold: Entailment]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 7. LONG-TERM COMMUNITY ASSET CREATION. — Pursuant to the financial administration authority of the Sangguniang Panlungsod to optimize municipal capital allocation and disaster preparedness, municipal development appropriations shall prioritize durable public capital facilities including rural farm access roadways, barangay maternity clinics, and communal flood control barriers. The City Treasurer and the City Budget Officer shall maintain strict fiduciary accounting and furnish annual expenditure reports to the City Council."* |
| **[GT-312]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 4. STATUTORY DISASTER RISK FINANCING. — In the exercise of sound fiscal stewardship over municipal resources and reserve funds held by the City Government of Davao, the local expenditure budget shall preserve mandatory fiscal reserves dedicated exclusively to disaster prevention, pre-disaster equipment acquisition, and rapid post-calamity relief operations. The City Planning and Development Office shall ensure that municipal development appropriations correspond directly to approved citywide priorities."* |
| **[GT-313]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 5. TREASURY VAULT KEY PROTOCOL. — To ensure transparent financial management and maintain robust reserve balances for emergency humanitarian operations across the city, combinations and physical safety keys to the municipal treasury currency storage vault shall remain under the personal custody of the City Treasurer and the Assistant Treasurer. The City Legal Office shall conduct periodic administrative reviews to ensure that all financial disbursements adhere strictly to local legislative authorizations."* |
| **[GT-314]** *Republic Act No. 10121 (Philippine Disaster Risk Reduction and Management Act of 2010), Section 21 vs. Section 6* **[Gold: Neutral]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 21. Local Disaster Risk Reduction and Management Fund (LDRRMF). — Not less than five percent (5%) of the estimated revenue from regular sources shall be set aside as the LDRRMF to support disaster risk management activities. Of the amount appropriated for LDRRMF, thirty percent (30%) shall be allocated as Quick Response Fund (QRF) or stand-by fund for relief and recovery programs in order that living conditions in communities stricken by disasters may be normalized as quickly as possible. Unexpended LDRRMF shall accrue to a special trust fund solely for the purpose of supporting disaster risk reduction activities within the next five (5) years."* | *"SECTION 6. EMERGENCY FUEL STORAGE INSPECTIONS. — In the exercise of sound fiscal stewardship over municipal resources and reserve funds held by the City Government of Davao, the City Central Emergency Communications and Dispatch facility shall inspect underground diesel fuel storage tanks quarterly to verify fuel readiness for standby emergency electric generators. The City Planning and Development Office shall ensure that municipal development appropriations correspond directly to approved citywide priorities."* |
| **[GT-315]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 287 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 287. Local Development Projects. — Each local government unit shall appropriate in its annual budget no less than twenty percent (20%) of its annual internal revenue allotment for local development projects. Copies of each local development plan shall be furnished the Department of the Interior and Local Government (DILG)."* | *"SECTION 5. TREASURY VAULT KEY PROTOCOL. — To ensure transparent financial management and maintain robust reserve balances for emergency humanitarian operations across the city, combinations and physical safety keys to the municipal treasury currency storage vault shall remain under the personal custody of the City Treasurer and the Assistant Treasurer. The City Legal Office shall conduct periodic administrative reviews to ensure that all financial disbursements adhere strictly to local legislative authorizations."* |

# M.9 Domain 07: Taxation, Tariffs and Revenue Administration (Items GT-316 to GT-350)

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| **[GT-316]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 4. REAL PROPERTY TAX LEVY RATE ADJUSTMENT. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, the basic real property tax rate on all commercial and residential lands situated within the City of Davao is hereby fixed at five percent (5%) of the assessed property value. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-317]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 4. IMPOSITION OF AMUSEMENT TAX ON ADMISSION. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, there is hereby levied an amusement tax to be collected from the proprietors of movie theaters, concert auditoriums, and entertainment venues at the rate of thirty-five percent (35%) of gross receipts from admission fees. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-318]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 4* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 4. REAL PROPERTY TAX LEVY RATE ADJUSTMENT. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, the basic real property tax rate on all commercial and residential lands situated within the City of Davao is hereby fixed at five percent (5%) of the assessed property value. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-319]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 4* **[Gold: Contradiction]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 4. IMPOSITION OF AMUSEMENT TAX ON ADMISSION. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, there is hereby levied an amusement tax to be collected from the proprietors of movie theaters, concert auditoriums, and entertainment venues at the rate of thirty-five percent (35%) of gross receipts from admission fees. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-320]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 3. FIXING BASIC REAL PROPERTY TAX RATE. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, in accordance with Section 233 of Republic Act No. 7160, the basic real property tax levied upon taxable real property in Davao City is fixed at one point five percent (1.5%) of assessed value. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-321]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 3. STATUTORY TEN PERCENT AMUSEMENT TAX RATE. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, pursuant to Section 140 of Republic Act No. 7160, the amusement tax levied on commercial admission tickets to cinemas, theaters, and concert halls is hereby fixed at ten percent (10%) of gross receipts. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-322]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 3* **[Gold: Entailment]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 3. FIXING BASIC REAL PROPERTY TAX RATE. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, in accordance with Section 233 of Republic Act No. 7160, the basic real property tax levied upon taxable real property in Davao City is fixed at one point five percent (1.5%) of assessed value. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-323]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 3* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 3. STATUTORY TEN PERCENT AMUSEMENT TAX RATE. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, pursuant to Section 140 of Republic Act No. 7160, the amusement tax levied on commercial admission tickets to cinemas, theaters, and concert halls is hereby fixed at ten percent (10%) of gross receipts. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-324]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 4. CADASTRAL MONUMENT SPECIFICATIONS. — Pursuant to the taxing and revenue-raising powers vested in the City Government of Davao under Title Two, Book II of Republic Act No. 7160, cylindrical concrete monuments marking municipal cadastral survey boundaries shall measure fifteen (15) centimeters in diameter by sixty (60) centimeters in length. The City Treasurer's Office and the City Assessor's Office shall implement systematic collection protocols and issue updated tax assessment notices."* |
| **[GT-325]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 4. PROJECTION BOOTH AIR VENTILATION. — Pursuant to the taxing and revenue-raising powers vested in the City Government of Davao under Title Two, Book II of Republic Act No. 7160, movie theater projection booths housing digital cinematic laser projectors shall feature dedicated exhaust ductwork delivering an airflow rate of not less than three hundred (300) cubic feet per minute. The City Treasurer's Office and the City Assessor's Office shall implement systematic collection protocols and issue updated tax assessment notices."* |
| **[GT-326]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 4* **[Gold: Neutral]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 4. CADASTRAL MONUMENT SPECIFICATIONS. — Pursuant to the taxing and revenue-raising powers vested in the City Government of Davao under Title Two, Book II of Republic Act No. 7160, cylindrical concrete monuments marking municipal cadastral survey boundaries shall measure fifteen (15) centimeters in diameter by sixty (60) centimeters in length. The City Treasurer's Office and the City Assessor's Office shall implement systematic collection protocols and issue updated tax assessment notices."* |
| **[GT-327]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 7. ADMINISTRATIVE PROPERTY TAX INCREASES. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, the City Assessor is empowered to adjust residential real property tax rates upward by administrative circular without enacting an amendatory revenue ordinance through the City Council. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-328]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 6* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 6. AMUSEMENT TAXATION ON SCHOOL EVENTS. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, admission fees charged for inter-school athletic competitions and university educational stage plays held in Davao City shall be subject to municipal amusement taxes. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-329]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 7* **[Gold: Contradiction]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 7. ADMINISTRATIVE PROPERTY TAX INCREASES. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, the City Assessor is empowered to adjust residential real property tax rates upward by administrative circular without enacting an amendatory revenue ordinance through the City Council. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-330]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 6. AMUSEMENT TAXATION ON SCHOOL EVENTS. — In the exercise of municipal legislative authority to enact local tax measures and establish equitable assessment schedules, admission fees charged for inter-school athletic competitions and university educational stage plays held in Davao City shall be subject to municipal amusement taxes. The Business Bureau shall require proof of local tax settlement prior to the issuance or renewal of any commercial mayor's permit."* |
| **[GT-331]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 5. ADMINISTRATIVE TAX ASSESSMENT APPEALS. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, property owners dissatisfied with land valuation assessments may appeal to the Local Board of Assessment Appeals (LBAA) pursuant to Section 226 of RA 7160. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-332]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 5. STATUTORY TAX EXEMPTIONS FOR EDUCATIONAL CONCERTS. — Pursuant to the taxing and revenue-raising powers vested in the City Government of Davao under Title Two, Book II of Republic Act No. 7160, in strict observance of Section 140 of RA 7160, amateur athletic exhibitions, school plays, and educational musical concerts are exempt from municipal amusement taxes upon verification by the City Treasurer. The City Treasurer's Office and the City Assessor's Office shall implement systematic collection protocols and issue updated tax assessment notices."* |
| **[GT-333]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 5* **[Gold: Entailment]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 5. ADMINISTRATIVE TAX ASSESSMENT APPEALS. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, property owners dissatisfied with land valuation assessments may appeal to the Local Board of Assessment Appeals (LBAA) pursuant to Section 226 of RA 7160. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-334]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 5. STATUTORY TAX EXEMPTIONS FOR EDUCATIONAL CONCERTS. — Pursuant to the taxing and revenue-raising powers vested in the City Government of Davao under Title Two, Book II of Republic Act No. 7160, in strict observance of Section 140 of RA 7160, amateur athletic exhibitions, school plays, and educational musical concerts are exempt from municipal amusement taxes upon verification by the City Treasurer. The City Treasurer's Office and the City Assessor's Office shall implement systematic collection protocols and issue updated tax assessment notices."* |
| **[GT-335]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 5* **[Gold: Entailment]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 5. ADMINISTRATIVE TAX ASSESSMENT APPEALS. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, property owners dissatisfied with land valuation assessments may appeal to the Local Board of Assessment Appeals (LBAA) pursuant to Section 226 of RA 7160. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-336]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 5. GIS CADASTRAL MAP PLOTTING. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, geographic information system cadastral base maps maintained by the City Assessor shall designate residential land parcels using light yellow shading. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-337]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 5. ADMISSION TICKET SECURITY PAPER. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, physical admission tickets printed for theatrical cinema houses shall utilize security watermark paper to prevent unauthorized counterfeiting. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-338]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 5* **[Gold: Neutral]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 5. GIS CADASTRAL MAP PLOTTING. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, geographic information system cadastral base maps maintained by the City Assessor shall designate residential land parcels using light yellow shading. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-339]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 5* **[Gold: Neutral]** | |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
|---|---|
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 5. ADMISSION TICKET SECURITY PAPER. — To generate necessary local government revenues for basic public services while maintaining fair commercial tax administration, physical admission tickets printed for theatrical cinema houses shall utilize security watermark paper to prevent unauthorized counterfeiting. Taxpayers failing to remit the prescribed municipal taxes within the statutory deadlines shall incur surcharges and interest under the Davao City Revenue Code."* |
| **[GT-340]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 6. SPECIAL URBAN INFILL LAND DEVELOPMENT LEVY. — To promote sound public finance and ensure systematic revenue administration across all commercial enterprises operating within Davao City, an auxiliary municipal infrastructure recovery assessment shall be levied on unimproved commercial acreage situated in designated urban revitalization zones, determined using a graduated multiplier of assessed market valuation approved by the City Assessor to deter land speculation. The City Administrator shall oversee the coordinated implementation of these local revenue measures across all administrative districts."* |
| **[GT-341]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 8* **[Gold: Contradiction]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 8. COMMERCIAL AUDITORIUM PERFORMANCE FACILITY CHARGE. — Pursuant to the revenue administration authority of the Sangguniang Panlungsod to maintain fair local tax assessments and commercial stability, operators of commercial performing arts auditoriums hosting ticketed dramatic plays or musical presentations shall remit a municipal cultural development charge calculated on gross ticket admissions, irrespective of the educational or collegiate character of the performing organization. The City Treasurer and the City Assessor are hereby directed to maintain updated valuation registries and ensure equitable assessment practices."* |
| **[GT-342]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 6* **[Gold: Contradiction]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 6. SPECIAL URBAN INFILL LAND DEVELOPMENT LEVY. — To promote sound public finance and ensure systematic revenue administration across all commercial enterprises operating within Davao City, an auxiliary municipal infrastructure recovery assessment shall be levied on unimproved commercial acreage situated in designated urban revitalization zones, determined using a graduated multiplier of assessed market valuation approved by the City Assessor to deter land speculation. The City Administrator shall oversee the coordinated implementation of these local revenue measures across all administrative districts."* |
| **[GT-343]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 8* **[Gold: Contradiction]** | |

| Controlling National Statute (Premise) | Davao City Ordinance Provision (Hypothesis) |
|---|---|
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 8. COMMERCIAL AUDITORIUM PERFORMANCE FACILITY CHARGE. — Pursuant to the revenue administration authority of the Sangguniang Panlungsod to maintain fair local tax assessments and commercial stability, operators of commercial performing arts auditoriums hosting ticketed dramatic plays or musical presentations shall remit a municipal cultural development charge calculated on gross ticket admissions, irrespective of the educational or collegiate character of the performing organization. The City Treasurer and the City Assessor are hereby directed to maintain updated valuation registries and ensure equitable assessment practices."* |
| **[GT-344]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 4. PROPERTY VALUATION HARMONIZATION. — In the exercise of municipal taxing jurisdiction to generate equitable public revenues for municipal social development programs, the City Assessor shall maintain uniform schedules of fair market value across municipal appraisal zones, ensuring that assessed rates on residential parcels remain strictly aligned with statutory municipal taxing limits. The Business Bureau and the City Legal Office shall institute administrative and legal collection remedies against non-compliant commercial taxpayers."* |
| **[GT-345]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 4. RECOGNITION OF STUDENT ARTISTIC TAX IMMUNITY. — In the exercise of municipal taxing jurisdiction to generate equitable public revenues for municipal social development programs, municipal revenue examiners shall grant automatic exemption clearances to youth athletic competitions and collegiate stage productions, preserving the non-taxable status of educational theatrical performances. The Business Bureau and the City Legal Office shall institute administrative and legal collection remedies against non-compliant commercial taxpayers."* |
| **[GT-346]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 4. PROPERTY VALUATION HARMONIZATION. — In the exercise of municipal taxing jurisdiction to generate equitable public revenues for municipal social development programs, the City Assessor shall maintain uniform schedules of fair market value across municipal appraisal zones, ensuring that assessed rates on residential parcels remain strictly aligned with statutory municipal taxing limits. The Business Bureau and the City Legal Office shall institute administrative and legal collection remedies against non-compliant commercial taxpayers."* |
| **[GT-347]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 4* **[Gold: Entailment]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 4. RECOGNITION OF STUDENT ARTISTIC TAX IMMUNITY. — In the exercise of municipal taxing jurisdiction to generate equitable public revenues for municipal social development programs, municipal revenue examiners shall grant automatic exemption clearances to youth athletic competitions and collegiate stage productions, preserving the non-taxable status of educational theatrical performances. The Business Bureau and the City Legal Office shall institute administrative and legal collection remedies against non-compliant commercial taxpayers."* |
| **[GT-348]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 6* **[Gold: Neutral]** | |

| **Controlling National Statute (Premise)** | **Davao City Ordinance Provision (Hypothesis)** |
|---|---|
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 6. DIGITAL CADASTRAL ARCHIVING. — Pursuant to the revenue administration authority of the Sangguniang Panlungsod to maintain fair local tax assessments and commercial stability, digital cadastral parcel boundaries registered within municipal property databases shall be backed up daily to offsite encrypted storage arrays to preserve municipal land records. The City Treasurer and the City Assessor are hereby directed to maintain updated valuation registries and ensure equitable assessment practices."* |
| **[GT-349]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 140 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 140. Amusement Tax. — (a) The province or city may levy an amusement tax to be collected from the proprietors, lessees, or operators of theaters, cinemas, concert halls, circuses, boxing stadia, and other places of amusement at a rate of not more than thirty percent (30%) of the gross receipts from admission fees. (c) The holding of operas, concerts, dramas, recitals, painting and art exhibitions, flower shows, musical programs, literary and oratorical presentations, except pop, rock, or similar concerts shall be exempt from the payment of the tax hereon imposed."* | *"SECTION 6. AUDITORIUM CHAIR UPHOLSTERY STANDARDS. — In the exercise of municipal taxing jurisdiction to generate equitable public revenues for municipal social development programs, fabric upholstery utilized for cinema auditorium seating shall be treated with chemical fire-retardant sprays to inhibit flame propagation in the event of an electrical fire. The Business Bureau and the City Legal Office shall institute administrative and legal collection remedies against non-compliant commercial taxpayers."* |
| **[GT-350]** *Republic Act No. 7160 (The Local Government Code of 1991), Section 233 vs. Section 6* **[Gold: Neutral]** | |
| *"Section 233. Rates of Levy on Real Property Tax. — A province or city or a municipality within the Metropolitan Manila Area shall fix a uniform rate of basic real property tax applicable to their respective localities as follows: (a) In the case of a province, at the rate not exceeding one percent (1%) of the assessed value of real property; and (b) In the case of a city or a municipality within the Metropolitan Manila Area, at the rate not exceeding two percent (2%) of the assessed value of real property."* | *"SECTION 6. DIGITAL CADASTRAL ARCHIVING. — Pursuant to the revenue administration authority of the Sangguniang Panlungsod to maintain fair local tax assessments and commercial stability, digital cadastral parcel boundaries registered within municipal property databases shall be backed up daily to offsite encrypted storage arrays to preserve municipal land records. The City Treasurer and the City Assessor are hereby directed to maintain updated valuation registries and ensure equitable assessment practices."* |

# Appendix N

# Domain Keyword Dictionaries

## N.1 Macro Legal Domain Keywords

Table N.1: Discriminative Legal Keywords Across Eight Macro Domains ($c$-TF-IDF)

| Macro Legal Domain | Top Discriminative Legal Keywords and Phrasal Tokens |
|---|---|
| **1. Administrative Governance** | sangguniang panlungsod, plantilla, personnel, memorandum, quorum, committee, appointment, tenure, reorganization, civil service, devolved |
| **2. Fiscal and Revenue** | tax, assessment, schedule of market values, surcharge, fee, collector, gross receipts, amusement tax, real property, delinquent, treasury |
| **3. Penal and Public Order** | fine, imprisonment, violation, citation, unlawful, misdemeanor, confiscation, summary revocation, curfew, public intoxication, citation ticket |
| **4. Environmental Resources** | watershed, protected area, ambient air, wastewater, solid waste, segregation, clean water act, timberland, buffer zone, quarry, denr |
| **5. Public Health and Sanitation** | quarantine, communicable disease, sanitary permit, veterinary, abattoir, immunization, potable water, epidemic, medical clinic, meat inspection |
| **6. Labor and Social Welfare** | senior citizen, pwd, discount, minimum wage, occupational safety, maternity, day care, informal settler, livelihood, indigenous cultural community |
| **7. Commercial and Trade** | business permit, mayor's permit, sole proprietorship, corporation, retail trade, price ceiling, weights and measures, fair trade, consumer act |
| **8. Transportation and Infrastructure** | tricycle franchise, route, public utility vehicle, loading zone, building code, occupancy permit, road right-of-way, terminal, motorized |

# Appendix O

# Training Sweeps and Evaluation Logs

## O.1 Hyperparameter Sweep Logs

Table O.1: Stage 2 Cross-Encoder Hyperparameter Sweep Logs (DeBERTa-v3-base)

| **Learning Rate ($\eta$)** | **Batch Size** | **Weight Decay ($\lambda$)** | **Warmup Ratio** | **Val Loss** | **Macro $F_1$** |
|---|---|---|---|---|---|
| $1 \times 10^{-5}$ | 16 | 0.01 | 0.10 | 0.412 | 0.841 |
| $2 \times 10^{-5}$ | 8 | 0.01 | 0.10 | 0.365 | 0.867 |
| $\mathbf{2 \times 10^{-5}}$ | **16** | **0.01** | **0.10** | **0.342** | **0.884** |
| $2 \times 10^{-5}$ | 32 | 0.01 | 0.10 | 0.378 | 0.859 |
| $3 \times 10^{-5}$ | 16 | 0.01 | 0.05 | 0.358 | 0.871 |
| $3 \times 10^{-5}$ | 16 | 0.10 | 0.10 | 0.384 | 0.856 |
| $5 \times 10^{-5}$ | 16 | 0.01 | 0.10 | 0.449 | 0.823 |

## O.2 Threshold Sensitivity and Confusion Matrix

Table O.2: Multi-Threshold Decision Sensitivity on Contradiction Detection ($N = 350$ Full Benchmark In-Sample Operating Profile)

| **Threshold ($\tau^*$)** | **Precision (%)** | **Recall (%)** | **$F_1$-Score** | **False Negatives** | **False Positives** |
|---|---|---|---|---|---|
| 0.20 | 69.0 | 97.3 | 0.807 | 3 | 49 |
| 0.30 | 79.9 | 95.5 | 0.870 | 5 | 27 |
| 0.40 | 89.7 | 93.8 | 0.917 | 7 | 12 |
| 0.42 | 91.3 | 93.8 | 0.925 | 7 | 10 |
| **0.45** | **92.9** | **92.9** | **0.929** | **8** | **8** |
| 0.50 | 94.3 | 88.4 | 0.912 | 13 | 6 |
| 0.60 | 95.7 | 80.4 | 0.874 | 22 | 4 |
| 0.70 | 96.5 | 73.2 | 0.832 | 30 | 3 |
| 0.80 | 97.5 | 68.8 | 0.806 | 35 | 2 |

*Note:* Evaluated across all 350 ground truth pairs (198 unique instances) in-sample to characterize the operational sensitivity profile across varying decision thresholds. The optimal operational threshold $\tau^* = 0.45$ was identified on the 52-pair validation set ($\mathcal{D}_{\text{val}}$, validation Contradiction $F_1 = 0.8947$, zero false negatives) and validated out-of-sample under zero-leakage premise-grouped cross-validation ($N = 198$ unique instances, Table 4.14).

Table O.3: Three-Class Confusion Matrix at Calibrated Threshold $\tau^* = 0.45$ ($N = 350$)

| | **Predicted Class** | | |
|---|---|---|---|
| **Actual Gold Class** | **Contradiction** | **Entailment** | **Neutral** |
| Contradiction ($N = 112$) | **104** | 2 | 6 |
| Entailment ($N = 128$) | 4 | **118** | 6 |
| Neutral ($N = 110$) | 4 | 6 | **100** |

# Appendix P

# Attestation of Primary Authorship and AI Disclosure Statement

**Ateneo de Davao University**
**COMPUTER STUDIES CLUSTER**
**SCHOOL OF ARTS AND SCIENCES**
P.O. Box 8016 Davao City
Philippines

**Attestation of Primary Authorship* and AI Disclosure Statement**

We, **Ralph Paolo Dulce and Yahyah Odin**, the proponents of thesis entitled **"A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference"**, hereby declare that:

- We are the sole authors of this thesis and its technical output.
- We have fully acknowledged and referenced the ideas, techniques and work of others, whether published or unpublished, in our thesis.
- We have prepared the thesis specifically for the degree of **Bachelor of Science in Computer Science** while under supervision at Ateneo de Davao University.
- Our thesis document does not contain work extracted from a thesis, dissertation or research paper previously presented for another degree or diploma at this or any other university.
- Our technical output was not subcontracted to a third party.

**AI Disclosure Statement**

- We hereby declare that any use of artificial intelligence (AI) tools in the preparation of this thesis and its technical output:

| AI Tool Used | Purpose and Description of Use | Sections/Components Where AI Was Used |
|---|---|---|
| **Google Antigravity / Gemini Models** | Assisted as an agentic pair-programmer for Python engineering: scaffolding modular codebase architecture (`src/`), automated execution harnesses, data pipelines for dual-corpus JSONL compilation (27,357 enactments, 178,494 provisions), batch OCR wrappers, and engineering the offline interactive web prototype interface (`demo/`). | Codebase Pipelines (`src/`, `scripts/`) and Interactive Web Prototype (`demo/`) |
| **Google Antigravity / Gemini Models** | Assisted in LaTeX typesetting, multi-pass compilation orchestration (`pdflatex`/`bibtex` in `scripts/build_paper.py`), resolution of build diagnostics, formatting dedicated full-page landscape plates (`pdflscape`) for empirical tables, managing front-matter listings, and synchronizing the automated single-source numbers registry. | Thesis Manuscript (`main.tex`, Chapters 1–5), Appendices A–P, and Numbers Registry |
| **Google Antigravity / Matplotlib / Seaborn** | Assisted in scientific data visualization and publication graphics: scripting 300-DPI figure generation for longitudinal word count evolution, 2D SVD latent semantic topic space with covariance ellipses, 94.75% 512-token compliance curves, Stage 1 Recall@k charts, training loss dynamics, and token attention attribution heatmaps. | Publication Figures across Chapters 3 & 4, Appendices, and Flowcharts (`figs/`) |
| **Gemini Flash 3.8 (Google)** | Assisted in syntactic clause drafting and formal legislative enacting phrase templating of candidate local ordinance provisions under strictly human-specified statutory premises, legal mechanisms, and conflict types (Section 3.5.3), conforming to Davao City Sangguniang Panlungsod and DILG drafting manuals without fabricating legal facts. | Methodology (Section 3.5.3) and Ground Truth Benchmark Formulation (`data/`) |

**Adapted from Faculty of Graduate Research, Statement of Authorship and Thesis Length for PhD Students, Victoria University of Wellington. 2009*

Figure P.1: Institutional AI Disclosure Form (Page 1 of 2).

| AI Tool Used | Purpose and Description of Use | Sections/Components Where AI Was Used |
|---|---|---|
| **Claude 3.5 Sonnet, ChatGPT, & DeepSeek** | Assisted as adversarial simulated peer reviewers and mock defense interrogators: stress-testing scientific claims, identifying narrative overclaiming, and conducting forensic checks on mathematical and statistical consistency (McNemar $2\times contingency tables on = 53$, Friedman ANOVA ranks, paired Wilcoxon tests, Wilson score CIs, and Bayesian low-prevalence screening dynamics). | External Peer Review Simulation and Manuscript Stress-Testing (Chapters 1–5, Abstract, Conclusions) |
| **Claude 3.5 Sonnet / Gemini Models** | Assisted in editorial prose refinement, sentence flow transitions, eliminating repetitive phrasing or conversational filler, and ensuring adherence to academic continuous paragraph formatting in strict accordance with departmental writing guidelines. | Manuscript Prose Polish across Chapters 1–5, Abstract, and Appendices |
| **Google Antigravity / Gemini Models** | Assisted in authoring defense presentation slides strictly following the department 7×7 rule, structuring word-for-word spoken presenter scripts with foundational visual walkthroughs, and developing the interactive HTML presentation slide viewer (`docs/presentation/slides_viewer.html`). | Oral Presentation Slide Decks and Interactive Presentation Viewer (`docs/presentation/`) |
| **Evaluated ML Models (Research Subjects)** | Deep neural cross-encoders (`DeBERTa-v3`, `ModernBERT`, `Legal-BERT`, `all-MiniLM-L6-v2`) and vision-language models (`Qwen2.5-VL-3B-Instruct`) were evaluated strictly as offline empirical machine learning models and experimental research subjects under deterministic evaluation on local LGU hardware, not as generative text authors. | Methodology and Experimental Results (Chapters 3 & 4, Appendices A–P) |

We further declare that:

- We have accurately disclosed the use of AI tools in the preparation of this thesis and its technical output.
- We take full responsibility for the accuracy, originality, integrity, and academic quality of the submitted work, including any content generated or assisted by AI tools.
- We have reviewed and verified all AI-generated or AI-assisted content incorporated into the thesis and its technical output and have ensured that appropriate acknowledgements and citations are provided, where necessary.

By signing below, we affirm that the statements above are true and accurate.

**Signed:** _____ **Date:** _____
**Ralph Paolo Dulce**

**Signed:** _____ **Date:** _____
**Yahyah Odin**

**Adapted from Faculty of Graduate Research, Statement of Authorship and Thesis Length for PhD Students, Victoria University of Wellington. 2009*

Figure P.2: Institutional AI Disclosure Form (Page 2 of 2).